10th Edition

ROSEN'S

Concepts and MCQs

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1 Airway

KEY CONCEPTS

- Anticipating the clinical course of the patient's condition and assessing the likelihood of deterioration are crucial to the decision to intubate, especially if the patient is to leave the emergency department (ED) for a period of time (e.g., interfacility transfer, diagnostic testing).
- Although videolaryngoscopy (VL) has reduced the chance of a failed intubation attempt caused by difficult anatomic features that often thwart direct laryngoscopy (DL), an assessment of the patient for potential difficult intubation, bag-mask ventilation (BMV), ventilation using an extraglottic device (EGD), and cricothyrotomy is an essential step before a neuromuscular blocking agent (NMBA) is administered. The mnemonics *LEMON, ROMAN, RODS,* and *SMART* can serve as useful aids.
- Physiologic derangement can contribute to morbidity and mortality during emergency airway management. Cardiovascular optimization with fluids, blood, and pressor agents should be undertaken, when time allows, to reduce the risk of circulatory collapse and cardiac arrest.
- In the absence of a crash patient (agonal, unresponsive to laryngoscopy) or difficult airway, rapid sequence intubation (RSI) is the airway management method of choice for ED patients.
- Tube placement confirmation using end-tidal carbon dioxide ($ETco_2$) is essential after intubation; failure to detect adequate quantities of exhaled CO_2 is evidence of esophageal intubation until proven otherwise.
- VL increases first-attempt intubation success, even when compared with DL combined with various optimization techniques. First-attempt success (FAS) is associated with fewer adverse events and better outcomes. Emergency airway managers should learn, and adopt, VL as the method of choice for emergency intubation.
- Cricothyrotomy is indicated in the "can't intubate, can't oxygenate" failed airway situation and should be performed once this has been identified. Delays may increase the likelihood or severity of hypoxic injury to the patient.
- Etomidate is used in more than 90% of all RSIs. Either rocuronium or succinylcholine is a reasonable NMBA for use during RSI. Rocuronium has less potential for adverse effects but a longer duration of action.
- EGDs are rarely used in ED airway management but offer additional options for rescue oxygenation of the failed airway and are used in many prehospital systems.

1. Which of the following is considered unreliable for assessing the need to establish an artificial airway?
- **a.** Absence of a gag reflex
- **b.** Absence of swallowing on command
- **c.** Level of consciousness
- **d.** Patient's ability to phonate
- **e.** Pooling of secretions in the oropharynx

Answer: a. The gag reflex can be absent in up to 25% of normal adults. Moreover, there is no evidence that the presence or absence of a gag reflex corresponds to a patient's ability to protect his or her airway. Therefore it should not be used as an indicator of the need for intubation.

2. Which of the following is the most reliable overall method for confirmation of correct tube placement after endotracheal intubation?
- **a.** Chest and gastric auscultation
- **b.** Chest radiography
- **c.** Detection of colorimetric or quantitative end-tidal carbon dioxide ($ETco_2$)
- **d.** Measurement of exhaled volume

Answer: c. Detection of $ETco_2$ after endotracheal intubation is the most reliable of the options listed for the confirmation of tube placement. (A fiberoptic scope passed through the endotracheal tube, with visualization of the tracheal rings, is the gold standard but is not generally required.) Limitations of colorimetric CO_2 detection should be appreciated in cardiac arrest patients. In these situations, a bulb aspiration device may provide helpful information, even though this technique is generally not as reliable as $ETco_2$ detectors. The other listed options, traditional as they may be, are prone to failure and should not be relied on for confirmation of tube placement.

3. During rapid sequence intubation (RSI) of a hypotensive blunt abdominal trauma patient, which of the following will mitigate the risk of circulatory collapse during emergency airway management if performed before medications are administered?
- **a.** Central venous access
- **b.** Choosing propofol instead of ketamine for induction
- **c.** Resuscitating with packed red cells
- **d.** Obtaining an abdominal computed tomography (CT) scan to better characterize the degree of bleeding

Answer: c. Before induction and paralysis, the preintubation optimization step of RSI suggests that volume resuscitation with either isotonic fluids or blood in a hemodynamically unstable patient should be performed, if time allows, to reduce the chance of cardiac arrest during or immediately after intubation.

4. In which of the following conditions is succinylcholine contraindicated?
- **a.** Acute burn <5 days
- **b.** Acute head injury secondary to motor vehicle accident
- **c.** Acute spinal cord injury <5 days
- **d.** Renal failure with a serum potassium level of 4.7 mEq/L
- **e.** Multiple sclerosis

Answer: e. Succinylcholine has been associated with severe fatal hyperkalemia when administered in specific clinical circumstances. The risk of succinylcholine-induced hyperkalemia in patients with denervation syndromes begins with the onset of disease and continues indefinitely. With respect to acute burns, trauma, stroke, spinal cord injury, and intra-abdominal sepsis, the risk of hyperkalemia with succinylcholine use becomes evident 5 days after the onset of injury or disease process. Succinylcholine is not contraindicated in renal failure; however, known elevations in the potassium level may warrant use of another neuromuscular blocking agent.

5. Which of the following provides the highest ETo_2 after 3 minutes of ambient pressure tidal volume breathing?
- **a.** Nonrebreather mask with oxygen flow at 15 L/min
- **b** High-flow nasal cannula
- **c.** Nonrebreather facemask at flush rate oxygen (40 to 90 L/min)
- **d.** Venturi mask

Answer: c. Flush flow rate oxygen using a nonrebreather mask provides an ETo_2 of 85%. This is in comparison with an ETo_2 level in the mid-50s with a nonrebreather mask at 15 L/min flow. The increased inflow rate outcompetes room air entrainment around the margin of the mask and increases the fraction of inspired oxygen (Fio_2) resulting in better nitrogen washout.

6. Until how long after an acute burn is succinylcholine considered safe to use for RSI?
- **a.** 30 minutes
- **b.** 12 hours
- **c.** 24 hours
- **d.** 48 hours
- **e.** 5 days

Answer: e. Succinylcholine can produce severe (and fatal) elevations in serum potassium levels after administration in patients with burns. However, this vulnerability to succinylcholine-induced hyperkalemia is not clinically significant until at least 5 days after the acute burn. As a result, succinylcholine remains the paralytic of choice if rapid sequence intubation occurs less than 5 days after the burn.

2 Mechanical Ventilation and Noninvasive Ventilatory Support

KEY CONCEPTS

- Noninvasive positive-pressure ventilation (NPPV) is often adequate for reversal of impending respiratory failure due to a rapidly reversible cause, and should be considered as the first-line therapy for patients with exacerbations of chronic obstructive pulmonary disease (COPD) and acute cardiogenic pulmonary edema (ACPE) in whom immediate intubation is not required.
- NPPV should generally be avoided for definitive management of patients with pneumonia or acute respiratory distress syndrome (ARDS) unless the patient is clearly improving or has a do-not-intubate status. Otherwise, endotracheal intubation and mechanical ventilation are preferred.
- For patients in need of ventilatory support, bilevel positive airway pressure (BiPAP) and continuous positive airway pressure (CPAP) have clinical equipoise. BiPAP should begin with an inspiratory pressure setting of 10 cm of H_2O and expiratory pressure of 5 cm of H_2O and be evaluated frequently for tolerance and need to titrate up or down.
- Pressure-controlled ventilation (PC) delivers breaths at a predetermined pressure but with variable volume, while volume-controlled ventilation (VC) delivers a predetermined inspiratory volume but variable pressures.
- Assist-control ventilation (A/C) delivers a required number and volume of breaths per minute while synchronized intermittent-mandatory ventilation (SIMV) synchronizes mandatory breaths with spontaneous breaths. Either mode can be PC or VC.
- Invasive mechanical ventilation requires dynamic, ongoing monitoring. After intubation, blood gas analysis should be performed to confirm appropriate ventilation and provide correlation with noninvasive monitoring of oxyhemoglobin saturation and end-tidal CO_2.
- Positive pressure can have adverse hemodynamic consequences.
- Plateau pressure should be maintained below 30 cm H_2O.
- Elevation in ventilation pressures suggests ventilator circuit obstruction, bronchospasm, mainstem intubation, tension pneumothorax or hemothorax, decreased chest wall compliance or increased chest wall rigidity.
- Inappropriately low ventilation pressures, particularly in conjunction with hypoxia, indicates ventilator circuit leak or a faulty connection, endotracheal tube cuff leak, accidental extubation, or esophageal intubation.
- The Richmond Agitation Sedation Score (RASS) or a similar scoring system should be used to manage sedation and analgesia of the mechanically ventilated patient. Avoid unnecessary use of prolonged neuromuscular blockade. When RASS is used, a target score of –2 to 0 avoids both over and under sedation.

1. Which of the following is the most important consideration in the decision to initiate noninvasive positive-pressure ventilation (NPPV)?
- **a.** Degree of acidosis
- **b.** Degree of respiratory distress
- **c.** Hemodynamic profile
- **d.** Level of consciousness

Answer: b. Although NPPV will be helpful for many patients in distress, the need for emergent intubation is an absolute contraindication to NPPV. Both hemodynamic profile and level of consciousness are important determinants in the decision to implement NPPV, but both are relative (rather than absolute) contraindications to its use. Patients with chronic obstructive pulmonary disease (COPD), acute cardiogenic pulmonary edema (ACPE), and asthma are more likely to show benefit with NPPV than patients with pneumonia, but NPPV can be initiated for any patient with respiratory distress. Patients with hypoxia or hypercarbia may be acidemic, but the decision to initiate NPPV will be predicated on mental status and work of breathing rather than on degree of acidosis alone.

2. Which of the following is a potential adverse effect of intubation and positive-pressure ventilation?
- **a.** Decreased mean intrathoracic pressure
- **b.** Decreased ventilation/perfusion ratio
- **c.** Increased cardiac output
- **d.** Increased work of breathing

Answer: d. Positive-pressure ventilation (PPV) is associated with several complications. Some of these can quickly become life-threatening. This reality underscores the importance of familiarization with the common problems that arise as a result of PPV. Most complications result from changes in thoracic physiology when positive pressure is present for part or all of the respiratory cycle. Potential adverse effects of PPV include an increased work of breathing because of asynchrony or improperly set triggers, an increase in intrathoracic pressure, decreased venous return to the heart and decreased cardiac output, an increased ventilation/perfusion ratio, air trapping and intrinsic positive end-expiratory pressure, barotrauma, decreased renal blood flow and glomerular filtration rate with fluid retention, nosocomial infections of the lungs and sinuses, respiratory alkalosis, and agitation and increased respiratory distress.

3. What is the primary physiologic effect of applying positive end-expiratory pressure (PEEP) during mechanical ventilation?
- **a.** Decrease cardiac output
- **b.** Decrease intrapulmonary shunting
- **c.** Decrease ventilation/perfusion mismatch
- **d.** Increase functional residual capacity (FRC)

Answer: d. Although all of these effects can be attributed to the application of PEEP, its primary physiologic effect is to increase FRC by maintaining patency of injured or flooded alveoli that would otherwise collapse at the end of exhalation. Increasing the FRC may improve both oxygenation and lung compliance. PEEP increases Pao_2 at a constant fraction of inspired oxygen (Fio_2) by decreasing intrapulmonary shunting and ventilation/perfusion mismatch. One of the potential adverse effects to PEEP is decreased cardiac output.

4. Regarding a patient who develops acute distress on mechanical ventilation, which of the following is the most accurate?
- **a.** Accidental extubation is a common cause of increased airway pressure.
- **b.** Anaphylaxis would cause immediate increases in both peak inspiratory pressure (PIP) and plateau pressure (Pplat).
- **c.** All patients in distress should be immediately removed from the ventilator and bagged.
- **d.** Compared to patients with restrictive lung disease, patients with obstructive lung disease and asthma are more likely to decompensate without an appropriate expiratory time.

Answer: d. One of the most life-threatening complications of mechanical ventilation is loss of adequate cardiac output because of elevated intrathoracic pressure from intrinsic PEEP (iPEEP). Intrinsic PEEP can be precipitated by a respiratory rate that is too high, which does not allow patients to fully exhale before the delivery of another breath. Patients with obstructive lung disease such as COPD or acute asthma are particularly sensitive to this phenomenon, also known as "breath stacking." Although patients with restrictive lung disease may develop iPEEP with an inappropriate respiratory rate, this is much more likely to occur in patients with obstructive conditions. If patients are not hemodynamically unstable, they do not need to be removed from the ventilator, and in hemodynamically unstable patients, pneumothorax should only be presumptively treated if removing the patient from the ventilator does not improve the situation. Accidental extubation is a common cause of decreased airway pressures, and although anaphylaxis would cause increases in PIP, it would not typically cause increases in Pplat.

3 Shock

KEY CONCEPTS

- Shock can occur with normal arterial blood pressure and not all patients with arterial hypotension have shock.
- A base deficit more negative than –4 mEq/L or a serum lactate level greater than 4.0 mmol/L warrants a presumptive diagnosis of shock.
- Urine output is a reliable index of vital organ perfusion in patients with suspected shock. Normal urine output is 1.0 mL/kg/h. Output less than 0.5 mL/kg/h indicates severe renal hypoperfusion in patients without preexisting disease.
- A combination of a worsening base deficit, increasing lactate level, and low urine output represents persistent or worsening shock.
- Hemorrhagic shock is preferentially treated with blood products, using a balanced transfusion approach of packed red blood cells, fresh frozen plasma, and platelets.
- Early initiation of balanced fluid resuscitation, vasopressor support, and prompt antimicrobial therapy improve outcomes in patients with septic shock.

1. A 72-year-old male presents with crushing substernal chest pain with associated diaphoresis starting 2 hours prior to arrival. Blood pressure is 72/50. ECG shows anterior ST elevation myocardial infarction. Lungs demonstrate diffuse rales, and chest XR demonstrates bilateral infiltrates consistent with pulmonary edema. Bedside ultrasound demonstrates severely impaired left ventricular function. In addition to arranging for rapid reperfusion with either cardiac catheterization or intravenous thrombolytics, which is the most appropriate initial treatment for his cardiogenic shock?

a. Bolus of 30 mL/kg crystalloid fluid.
b. Initiate epinephrine at 0.1 mcg/kg/min.
c. Initiate norepinephrine at 0.1 mcg/kg/min.
d. Transfer patient for placement of an intraaortic balloon catheter.

Answer: c. Norepinephrine (or inotropes such as dobutamine) are first-line agents for cardiogenic shock. Crystalloid fluids should not be administrated to patients with cardiogenic shock and pulmonary edema. Epinephrine carries an increased risk of tachydysrhythmias and no better outcomes than norepinephrine or inotropes in cardiogenic shock. While this patient might benefit from an intraaortic balloon pump, treatment of his shock should not be delayed through transfer.

2. A 56-year-old female presents with fever, dyspnea, and productive cough. Heart rate is 122, respiratory rate is 24, and blood pressure is 82/46. Which of the following is the most appropriate choice for initial fluid resuscitation?

a. Lactated Ringers.
b. Blood products.
c. Hydroxyethyl starch (HES).
d. Normal Saline.

Answer: a. Balanced crystalloids are the preferred choice of fluid for patients with septic shock. Blood products may be indicated for patients with a hemoglobin greater than 7 mg/dL. Synthetic colloids such as HES cause renal failure and should be avoided. Hypertonic saline has not been proven to improve outcomes in sepsis. Normal saline can cause a hyperchloremic metabolic acidosis and acute kidney injury, particularly using large volumes.

3. An 18-year-old unrestrained driver is transported to the emergency department (ED) after being thrown from his vehicle during a motor vehicle collision. He was intubated in the field and received an intravascular bolus of 3 L of normal saline before arrival to the ED. His initial Glasgow Coma Score (GCS) is 7, and his blood pressure on arrival is 80/50 mm Hg. Which of the following would be the most appropriate to initiate immediately on arrival to the ED?

a. Dobutamine.
b. Dopamine.
c. Norepinephrine.
d. Packed red blood cell (PRBC) transfusion.

Answer: d. In patients with signs of hemorrhagic shock, immediate PRBC transfusion should be initiated. This assists with volume expansion and oxygen delivery to the brain. Vasopressors and positive inotropes will be of little benefit before volume replacement, and hetastarch has no proven benefit for initial resuscitation in head injury patients.

4 Brain Resuscitation

KEY CONCEPTS

- Neuronal injury is a dynamic process that continues for hours or days after an ischemic insult to the brain.
- Avoid hypotension and hypoperfusion by maintaining a mean arterial pressure greater than 65 mm Hg and a cerebral perfusion pressure of 60 mm Hg.
- Maintain normal oxygen levels or mild hyperoxemia, with a PaO_2 of 80 to 120 mm Hg and oxyhemoglobin saturations in the high 90s. Avoid hypoxemia and significant hyperoxemia.
- Intracranial pressure (ICP) elevation can further exacerbate ischemic brain injury. Initial management should include optimizing patient positioning while providing adequate analgesia and sedation. Management should then be escalated in a stepwise fashion to include hypertonic therapy, deeper sedation and ultimately barbiturates, hypothermia, and surgery as needed.
- Avoid hyperventilation (target $PaCO_2$ of 35 to 40 mm Hg) as it reduces cerebral blood flow. In the event of life-threatening cerebral herniation or significant ICP elevation, therapeutic hyperventilation is appropriate only as a short-term intervention bridging to more definitive therapy (i.e., craniotomy).
- Promptly treat seizures with intravenous (IV) lorazepam. Prophylactic administration of antiepileptic drugs is not recommended. Initiate continuous electroencephalogram (EEG) monitoring if ongoing seizures are a concern.
- Fever is an important mediator of secondary brain injury. Treat temperatures greater than 38°C with acetaminophen.
- Unresponsive survivors of out-of-hospital cardiac arrest should have rapid initiation of targeted temperature management (TTM) in the emergency department and be maintained at a constant target of 33°C–36°C in an ICU setting for 24 h after resuscitation.
- Because withdrawal of life-sustaining treatment due to perceived poor neurological prognosis is the most common cause of death in cardiac arrest survivors, it is important to provide accurate and evidence-based prognostic information to families. The neurological examination is generally an unreliable predictor of outcome until at least 72 h from normothermia.

1. To maximize cerebral blood flow, a patient with a normal intracranial pressure who is undergoing resuscitation should be ventilated to maintain a partial pressure of carbon dioxide ($Paco_2$) within what range?
   - **a.** 20 to 25 mm Hg.
   - **b.** 25 to 30 mm Hg.
   - **c.** 30 to 35 mm Hg.
   - **d.** 35 to 40 mm Hg.
   - **e.** 40 to 45 mm Hg.

Answer: d. Carbon dioxide is a potent vasoactive agent, and lowering of the $Paco_2$ by hyperventilation results in rapid reduction of cerebral blood flow (CBF). Since reductions in CBF reduce total cerebral blood volume, hyperventilation may transiently abort brainstem herniation in the presence of critically elevated intracranial pressure (ICP) until osmotherapy or ventriculostomy can be initiated. When ICP is not elevated, however, the vasoconstriction and increased cerebrovascular resistance (CVR) caused by hyperventilation can cause potentially dangerous reductions in CBF. In general, ventilation to maintain a $Paco_2$ between 35 and 40 mm Hg is safe and appropriate.

2. Which of the following is a class 1 recommendation from the 2015 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care?
   - **a.** Forty-eight hours from arrest is the earliest time the clinical examination may be used for neuroprognostication.
   - **b.** All comatose patients with return of spontaneous circulation (ROSC) after in- and out-of-hospital cardiac arrest, including both shockable and non-shockable rhythms, should receive targeted temperature management.
   - **c.** EEG monitoring is not necessary or indicated for comatose patients following cardiac arrest.
   - **d.** Infusion of cold saline should be used to induce cooling in the pre-hospital setting following cardiac arrest.

Answer: b. The 2015 guidelines update confirms that all comatose (lack of meaningful response to verbal commands) patients with ROSC following cardiac arrest should receive targeted temperature management (TTM), regardless of location of arrest or rhythm type. Because the neurological examination is unreliable in the first few days following arrest, 72 hours is the **earliest** possible time that one may consider using it to provide prognostic information to surrogate decision makers. Non-convulsive seizures are common following cardiac arrest therefore EEG monitoring, if available, should be performed frequently or continuously in comatose survivors of cardiac arrest. Finally, multiple randomized controlled trials showed that cold intravenous fluids in the pre-hospital setting increased the rate of pulmonary edema and rearrest. We do not recommend this practice.

3. Select the best answer. Which of the following statements is false?
   - **a.** Early magnetic resonance imaging (MRI) and serum biomarkers have a clearly established role in determining the prognosis of patients within 48 hours after cardiac arrest.
   - **b.** Recent data suggest that rate of survival to hospital discharge following cardiac arrest is 12%, with 75% of survivors achieving a favorable neurological outcome.
   - **c.** In cardiac arrest survivors presenting to the emergency department , the initial neurological examination is not a reliable predictor of long-term neurological prognosis.
   - **d.** The vast majority of 1-year survivors of cardiac arrest are able to live independently without disability.

Answer: Statement a is false. The role of early imaging, neurophysiologic testing, and serum biomarkers in predicting outcome has not yet been clearly established at any time point, especially not within 48 hours.

4. Which of the following statements about neuronal damage following ischemic injury is correct?
   - **a.** A patient with return of spontaneous circulation after 15 minutes of cardiac arrest has likely already suffered substantial neuronal cell death.
   - **b.** There is a single biological pathway by which ischemia triggers neuronal injury and eventually results in cell death.
   - **c.** Cell death is delayed following cardiac arrest, suggesting that therapeutic interventions may mitigate long-term neuronal injury.
   - **d.** It is not important to avoid secondary neuronal injury following cardiac arrest.

Answer: c. Although a cascade of cellular pathways will have been triggered, the resulting neuronal cell death is usually delayed by hours or days. This suggests that therapeutic interventions, such as targeted temperature management, can reduce cell death and improve outcomes. Careful avoidance of additional secondary insults is also important to maximize the potential for favorable neurological recovery.

5. Select the best answer. Which of the following is associated with worse neurologic outcomes in comatose survivors of cardiac arrest?
   - **a.** All of these.
   - **b.** Hyperthermia.
   - **c.** Hypotension.
   - **d.** Hypoxia.
   - **e.** Only hypotension and hyperthermia.

Answer: Statement a is the best answer. Hypotension, hypoxia, and hyperthermia (as well as hyperglycemia and seizures) are all associated with worse neurologic outcomes in comatose survivors of cardiac arrest.

KEY CONCEPTS

- Cardiopulmonary resuscitation (CPR) quality is critical to successful resuscitation from cardiac arrest. Important benchmarks of quality CPR include compression rate 100–120 compressions/min, compression depth 5–6 cm, compression fraction at least 80%, full chest recoil between compressions, and a ventilation rate of 10 breaths/min.
- Restoration of adequate cardiac function is the defining factor of return of spontaneous circulation (ROSC). Restoration of good neurologic function is the defining factor of a successful resuscitation.
- Resuscitation of a cardiac arrest victim does not end with ROSC. Rapid diagnosis and proper management of the pathologic condition(s) that precipitated and resulted from the arrest, as well as goal-directed post–cardiac arrest care, can improve outcomes.
- Immediate percutaneous coronary intervention is indicated in patients with ST segment elevation myocardial infarction following ROSC, independent of neurological status.
- Hypothermic targeted temperature management (32°C–36°C [89.6°F–96.8°F] for 24 h) has been shown to improve survival and functional outcome of comatose cardiac arrest survivors.

1. Which of the following statements regarding the epidemiology of out-of-hospital cardiac arrest is true?
   a. Most patients have an automated external defibrillator applied prior to emergency medical services (EMS) arrival.
   b. Most patients receive bystander CPR.
   c. Most surviving to hospital discharge will not have major persistent neurologic deficits.
   d. Only 2% of EMS-treated out-of-hospital cardiac arrests survive to hospital discharge.

Answer: c. It is estimated that 180,000 patients are treated for out-of-hospital cardiac arrest each year in the United States. The number of patients receiving bystander cardiopulmonary resuscitation (CPR) remains low, averaging 40%. The proportion of emergency medical services (EMS)–treated cardiac arrest patients with an initial shockable rhythm has declined over time to 18% in recent US studies. Automated external defibrillators are applied in a minority of cases prior to EMS arrival. Recent epidemiologic data from cardiac arrest registries indicate the survival rate to hospital discharge for EMS-treated, out-of-hospital cardiac arrest is about 10%. Of patients surviving to hospital discharge, independent of neurologic status on presentation, 79% have good neurologic function.

2. A 75-year-old man presents with return of spontaneous circulation (ROSC) after 2 minutes of ventricular fibrillation and successful defibrillation by EMS. The patient is unresponsive to verbal and painful stimuli. Vital signs on arrival are pulse, 120 beats/min; blood pressure, 130/70 mm Hg; respiratory rate, 10 breaths/min; temperature, 36°C (96.8°F); and oxygen saturation, 94%. The patient has intravenous access. The next most appropriate examination or procedure is:
   a. Anteroposterior (AP) chest radiograph
   b. Arterial blood gas (ABG)
   c. Comprehensive neurologic examination
   d. Electrocardiography

Answer: d. Acute coronary syndrome is a common cause of out-of-hospital cardiac arrest. Electrocardiography should be performed as soon as possible after ROSC to evaluate for ST segment elevation. Because it is impossible to determine survival or neurologic status in the immediate post-arrest period, ST segment elevation myocardial infarction (STEMI) should be treated aggressively with percutaneous coronary intervention (PCI) independently of coma or other laboratory values such as those provided on ABG analysis. Oxygen saturations above 94% are adequate for tissue perfusion, and hyperoxia may be harmful. AP chest radiographs may be important to evaluate ventilatory status if the patient is unstable.

3. Which chest compression/ventilation ratio is recommended during adult resuscitation efforts performed by health care professionals before placement of an advanced airway?
   a. 10:1
   b. 20:1
   c. 20:2
   d. 30:2

Answer: d. A 30:2 compression/ventilation ratio is currently recommended for health care professionals in all adult resuscitation scenarios. Although recent evidence has suggested that chest compression-only CPR is effective when performed by bystanders in the out-of-hospital setting, there is inadequate evidence to recommend this as an alternative strategy for health care professionals, except when inadequate personnel are present to provide compressions, ventilation, and other resuscitative activities.

4. Which of the following statements regarding hypothermic targeted temperature management (HTTM) in comatose survivors of cardiac arrest is true?
   a. Gradual rewarming should occur over 4 hours.
   b. Pregnancy is an absolute contraindication.
   c. Prolonged pharmacologically induced paralysis without sedation is often required to control shivering.
   d. Target core body temperature should be 32° to 36°C.

Answer: d. Induction of prolonged HTTM in comatose survivors of cardiac arrest has been shown to improve survival and functional outcome. A target temperature in the range of 32°C to 36°C (89.6°F to 96.8°F) should be selected and maintained. The time to achieve this temperature has not been clearly defined, and it has been suggested there is a broad therapeutic window. In the studies showing a benefit, maintenance of hypothermia occurred for 12 to 24 hours, followed by gradual rewarming over 12 to 24 hours. There are no absolute contraindications to induced hypothermia after arrest. Shivering, which inhibits cooling, can be prevented with sedation and pharmacologic paralysis. However, prolonged paralysis should be avoided because of the risk of unrecognized seizure activity in post–cardiac arrest patients.

5. For end-tidal pressure of carbon dioxide ($PETCO_2$) to be a reliable indicator of cardiac output during cardiac arrest, which of the following must be present?
   a. Mechanical chest compressions must be performed.
   b. The patient must be in asystole.
   c. The patient must be normothermic
   d. The patient must have relatively constant minute ventilation.

Answer: d. Although $PETCO_2$ will change in direct relationship to cardiac output, alterations in minute ventilation will concentrate or dilute the fixed amount of expired CO_2, influencing the $PETCO_2$ measured independently of cardiac output. Therefore, for $PETCO_2$ to be a reliable indicator of cardiac output, minute ventilation must be held relatively constant. The relationship of $PETCO_2$ and cardiac output during CPR is not dependent on rhythm, mechanisms of chest compressions, or temperature. High-dose vasopressor therapy can cause a decrease in cardiac output during CPR, despite increased myocardial blood flow, which results in a decreased $PETCO_2$.

6 Pain Management

KEY CONCEPTS

- Acute pain is an urgent condition for the patient. Pain should be rapidly assessed, treated, and frequently reassessed in tandem with diagnostic evaluations.
- Therapy for acute pain is different than for chronic pain. Chronic pain treatment should be undertaken in consultation with the clinician(s) responsible for the patient's long-term management. In general, opioid analgesic agents should not be administered in the ED or prescribed for outpatient therapy for chronic pain patients unless the plan is agreed to by the responsible outpatient clinician.
- Titrated IV opioid analgesics are the primary therapeutic approach for the treatment of moderate and severe acute pain. When intravenous access is not indicated or attainable, SC administration is preferable to the IM route.
- Opioid therapy should be reserved for pain that has not been or cannot adequately be treated with nonopioid therapies.
- Oral oxycodone, with an onset of action similar to that of IM or SC opioids, can be used for moderate pain when the IV route is not otherwise needed. Oxycodone and other oral opioids should be administered and prescribed as a single-drug preparation, not as part of a combination with acetaminophen.
- Ambulatory treatment with opioids should be confined to the period of acute pain. Most opioid prescriptions from the ED for acute injury should be for 3 to 5 days, after which the patient is transitioned to nonopioid analgesia or reevaluated by an outpatient clinician.
- Acetaminophen and NSAIDs should be added to pain therapy when not contraindicated. Their analgesic effects are additive to those of opioids and each other.
- There is no evidence to support the concept that diagnosis based on physical examination findings will be impaired by administering opioid pain medications to achieve reasonable patient comfort.
- There is no evidence that morphine causes more smooth muscle spasm than other opioids. Morphine is safe and appropriate for patients with acute biliary or renal colic.
- Patients who are known to be diverting or abusing opioids should not be prescribed opioids for use as outpatients. Patients with chronic pain syndromes, or those with chronic conditions that may cause acute pain (e.g., dental caries), should be offered alternative pain management options, and opioids should generally be avoided.
- Topical and local anesthetics can treat pain associated with most ED procedures and should be considered for use in isolated painful conditions.
- Low tissue pH (5 or 6) in infected tissue impairs the effectiveness of local anesthesia.

1. Which of the following statements is true regarding pain transmission?
   a. Cardiac pain is transmitted via the sympathetic system.
   b. Central post-stroke neuropathic pain is associated with parietal infarcts.
   c. Descending modulation of pain is mediated primarily through γ-aminobutyric acid (GABA).
   d. Peripheral neurotransmitters include prostaglandins, histamines, and substance P.

Answer: A. As a general rule, all visceral pain is carried via sympathetic afferents to ganglia and then to the spinal cord. Prostaglandins, substance P, and histamine sensitize peripheral afferents but are not neurotransmitters. The dorsal column tracts can down-modulate ascending pain signals. Central post-stroke pain is clinically seen most often after thalamic strokes. Descending tracts that modulate pain processing at the dorsal horn use norepinephrine and serotonin, with the effect of the former being most important regarding analgesia.

2. Which of the following analgesics is matched with the correct feature?
   a. Fentanyl—prolonged QT interval on electrocardiography
   b. Hydromorphone—active metabolites
   c. Meperidine—muscle rigidity
   d. Oxycodone—serotonin syndrome

Answer: D. Oxycodone has been associated with serotonin syndrome when coadministered with selective serotonin reuptake inhibitor (SSRI) medications. This is due to an active metabolite it shares with morphine, hydromorphone, hydrocodone, and codeine. The following are the other correct associations:

Meperidine—anticholinergic toxicity, active metabolites
Fentanyl—muscle rigidity (chest wall)
Hydromorphone—inactive metabolite

3. A 32-year-old male patient undergoing treatment for an ankle sprain returns to the emergency department (ED) because of inadequate pain relief from the medicines he was prescribed. He is currently taking oxycodone, 10 mg PO every 4 hours, and ibuprofen, 400 mg every 4 hours. What is the next most appropriate medicine to add to his pain treatment regimen?
   a. Add acetaminophen, 650 mg q4h.
   b. Add tramadol, 50 mg PO q4h.
   c. Increase ibuprofen to 800 mg.
   d. Increase oxycodone to 15 mg.

Answer: A. Acetaminophen provides additive analgesia to nonsteroidal antiinflammatory drugs (NSAIDs) and opioids, with few adverse effects at low doses, and it should be incorporated in acute pain treatment when not contraindicated. The pain-relieving effects of ibuprofen have not been shown to be greater when 800 mg is used versus 400 mg, so increasing the dose of ibuprofen is unlikely to improve pain relief and will increase the risk of adverse effects of the NSAIDs. An increased dose of oxycodone would result in improved pain relief but increases the risk of adverse effects; it should be tried after other nonopioid treatments have failed. In this case, acetaminophen should be attempted first before increasing the oxycodone dose. Tramadol would not be indicated before acetaminophen because of its high rate of dizziness and nausea.

7 Procedural Sedation and Analgesia

KEY CONCEPTS

- Safe, effective procedural sedation and analgesia requires high-level experience and sound protocols for patient selection and patient monitoring.
- After procedural sedation, patients should be discharged with and remain in the company of a responsible adult for 4 to 8 h after recovery.
- Propofol is the agent of choice for deep sedation in the emergency department but requires supplementation with an opioid analgesic when a painful procedure is planned.
- The absence of a preprocedure fasting period is not a contraindication to procedural sedation for an emergent or time-sensitive condition.
- Pulse oximetry is mandatory during sedation, and end-tidal CO_2 should be monitored if moderate or deep sedation is planned. Oxygen should be administered to patients undergoing procedural sedation.

1. When do most adverse events associated with emergency department procedural sedation occur?
   a. During the manipulation or intervention
   b. 5 to 20 minutes after the last sedative dose
   c. 20 to 30 minutes after the last sedative dose
   d. 30 to 60 minutes after the last sedative dose

Answer: b. High-risk times are 5 to 20 minutes after the last medication administration and at the completion of the procedure, when there is no longer a painful stimulus but the patient remains sedated.

2. Which of the following statements is most accurate regarding the use of fentanyl for procedural sedation?
   a. Chest wall rigidity is a common occurrence but may be reversed with naloxone.
   b. For deep sedation, a single dose of 1 to 2 mcg/kg IV is often given before sedation to achieve good pain control. A smaller dose of sedative agent may then be added and titrated to effect.
   c. Rapid IV administration is generally safe.
   d. Respiratory depression is less likely with fentanyl than with an equipotent dose of morphine.

Answer: b. For deep sedation, less respiratory depression is generally observed if a single pain-relieving dose of fentanyl is used before sedation, followed by small titrated doses of a sedative agent. Sufficient analgesia is generally attained with fentanyl dosages of 1 to 2 μg/kg IV. Rapid IV administration of large doses is more likely to precipitate respiratory depression or arrest. Chest wall rigidity is very rarely seen and generally only with large, rapid, IV boluses of more than 7 μg/kg IV. If severe chest wall rigidity syndrome is precipitated, it often requires paralysis to ventilate the patient adequately. It is not reversed with naloxone. Equivalent dosages of morphine show more histamine release and hypotension than fentanyl but have similar respiratory depression potential.

3. Which of the following statements regarding the use of ketamine is false?
   a. Benzodiazepine coadministration has not been shown to reduce the incidence of emergence phenomenon in children.
   b. Despite increased secretions, airway reflexes are generally well maintained.
   c. Hypotension is common.
   d. Profound analgesic and sedative effects occur with minimal respiratory depression.

Answer: c. Ketamine increases the release of catecholamines on administration and supports blood pressure well. It also decreases smooth muscle tone in the bronchial tree and may have a benefit in patients with reactive airways disease. Several studies have failed to show benefit with the concurrent administration of low to moderate dosages of benzodiazepines in preventing emergence phenomenon in children. These studies have shown a slightly increased risk of side effects. Their routine use is discouraged and should be reserved for the actual treatment of severe emergency phenomena.

4. Which of the following statements regarding the use of propofol is true?
   a. Propofol has a long duration of action and provides significant analgesia.
   b. Propofol has significant antiemetic properties.
   c. Propofol increases intracranial pressure.
   d. Propofol is well tolerated in volume-depleted patients.

Answer: b. Propofol is a short-acting, sedative-hypnotic, cerebroprotective agent with no analgesic but profound antiemetic properties. Its adverse effects include dose-dependent respiratory depression, apnea, hypotension, and pain on injection. Preload-dependent patients are particularly susceptible to hypotension. Its combined use with ketamine is common. The two agents are thought to have synergistic effects that balance each other's deficits. Their combined use has been shown to improve provider satisfaction, sedation quality, and decrease emesis but has not been shown to be clinically superior to either agent used alone in regard to respiratory depression, airway complications, or improved recovery times.

5. Which of the following statements is true regarding the need for fasting before procedural sedation?
   a. A 6-hour period of fasting is required after the ingestion of liquids or solids before procedural sedation.
   b. Preprocedural fasting is required in all cases.
   c. The recommendation for preprocedural fasting is based on controlled trials involving patients undergoing procedural sedation.
   d. The risk of vomiting and loss of the airway protective reflexes is low during procedural sedation.

Answer: d. The American Society of Anesthesiologists (ASA) currently recommends a period of 2 hours after ingestion of clear liquids, 4 hours after ingestion of breast milk, and 6 hours after the ingestion of other liquids or solids before the performance of procedural sedation. This recommendation is based on expert consensus and has been extrapolated from data on patients receiving general anesthesia and manipulation of the airway during intubation and extubation. There are no published studies showing an increased risk of aspiration after a liquid or solid meal nor benefits of fasting before procedural sedation. Large studies have shown no clinically significant differences with airway complications, emesis, or other adverse effects among groups of patients stratified by their preprocedural fasting status. The risks of procedural sedation should be balanced against the risks of delaying a time-sensitive procedure.

8 Fever in the Adult Patient

KEY CONCEPTS

- Younger adults with fever usually have benign self-limited disease with low mortality. The challenge in this group is to identify rare causes of fever such as meningitis or septic condition when confronted with a predominance of self-limited viral and focal bacterial illness.
- For older patients (greater than 65 years), immune-compromised patients, or those with chronic disease, fever indicates a high risk for serious disease. Temperature elevation may be minimal in these patients, who often are unable to mount a significant febrile response to serious infection. Bacterial infection is the most common cause of fever in these patients. Three body systems—the respiratory tract, urinary tract, and skin and soft tissue—are the target for more than 80% of these infections.
- Atypical symptoms of illness are common in older febrile patients. Altered mental status, difficulty with ambulation, frequent falls, and general functional decline may be the only signs of serious infection in older patients.
- The white blood cell count is not a discriminatory test for patients with fever, may incorrectly indicate serious infection when none is present, or may be normal in the presence of life-threatening infection.
- In febrile patients with serious signs and symptoms, early empirical antibiotic therapy is often indicated as is treatment for severe acute respiratory syndrome (SARS)-2 coronavirus (COVID-19) and as appropriate influenza. The choice of antivirals, antibiotics, and other therapies are based on the likely cause of the fever as well as concomitant conditions, such as absolute neutropenia and end-stage renal disease.

1. Which three body systems are the target of more than 80% of bacterial infections in patients older than 65 years?
   a. Central nervous system, respiratory system, genitourinary system
   b. Central nervous system, respiratory system, skin and soft tissue
   c. Central nervous system, urinary system, skin and soft tissue
   d. Respiratory system, genitourinary system, skin and soft tissue

Answer: d. Patients older than 65 years who present with fever represent a group at high risk for serious disease. Morbidity and mortality in this group are significant. Between 70% and 90% are hospitalized, and 7% to 9% die within 1 month of admission. Infection is the most common cause of fever in these patients, and most of these infections are bacterial in nature. More than 80% of these infections originate from the respiratory system, genitourinary system, and skin and soft tissue.

2. A 65-year-old man presents after "briefly collapsing" at a nearby bus stop. On arrival, he is confused, opens his eyes spontaneously, and follows simple commands. A medical information card found in his wallet reveals a history of hypertension and gout. Vital signs reveal a blood pressure of 95/50 mm Hg, heart rate of 95 beats/min, respiratory rate of 24 breaths/min, temperature of 41.5°C, and arterial oxygen saturation (Sao_2) of 95%. His finger-stick glucose level is normal. Pertinent physical examination findings include pallor, dry mucous membranes, Glasgow Coma Score (GCS) of 14, and ataxia. An intravenous bolus of normal saline is started. Which of the following is the most appropriate next step in the management of this patient?
   a. Administer broad-spectrum antibiotics.
   b. Mist the patient with water and place him in front of a fan.
   c. Obtain a head CT scan.
   d. Perform a lumbar puncture.

Answer: b. In terms of initial resuscitative efforts, the most significant finding in this patient is a temperature of 41.5°C. Almost all cases of temperatures higher than 41.0°C (107°F) are due to hyperthermia rather than to fever. Sustained temperatures higher than 41.0°C (106°F) are rare but can be damaging to neural tissue and require rapid, aggressive, cooling measures, such as misting the patient with tepid water and placing him or her in front of a fan, immersion in cool water, or application of cooling blankets. Each of the other four options could potentially be performed during this patient's stay in the emergency department; however, the immediate focus is on temperature reduction.

3. Of the cases of functional decline in nursing home patients, 75% are due to which of the following?
   a. Congestive heart failure
   b. Coronary artery disease
   c. Dementia
   d. Infection

Answer: d. Of the cases of functional decline (e.g., difficulty ambulating, anorexia, decreased activity, new urinary incontinence) in nursing home patients, 75% are due to infection.

4. An 85-year-old man is brought to the emergency department (ED) by his daughter, who says that he has had a fever and has been eating less than normal for 2 days. The patient has a history of hypertension and severe dementia. He can provide no useful information. His present temperature was recorded by the triage nurse to be 38.5°C (101°F). The remainder of his vital signs are within normal limits. The only abnormal findings on physical examination are slightly dry mucous membranes, confusion, and poor social interaction. The daughter confirms his mental status to be at baseline. Which of the following are the two most important ancillary tests to perform in this patient?
   a. Blood culture and urine culture
   b. Urinalysis and chest radiography
   c. Urinalysis and white blood cell count
   d. White blood cell count and chest radiography

Answer: b. The two most important ancillary tests in the evaluation of fever in the adult patient, and especially in older patients, who frequently have atypical presentations, are urinalysis and chest radiography. Chest radiographs are often helpful in the diagnosis of pulmonary infection but may be difficult to interpret in the patient with chronic obstructive pulmonary disease (COPD), congestive heart failure (CHF), or other chronic lung disease. Urinalysis, although not foolproof, is highly accurate for urinary tract infection, especially in men. Although the white blood cell count is almost universally used in the evaluation of febrile patients, it lacks sufficient sensitivity and specificity to be of discriminatory value. Cultures are ordered in selected patients; however, the delay in obtaining results precludes any influence in emergency evaluation and treatment. Other tests that have relevance in select patients with fever include Gram staining, cerebrospinal fluid (CSF) analysis, thyroid function studies, ultrasonography, and computed tomography (CT) of the abdomen or head.

9 Weakness

KEY CONCEPTS

- Weakness is a common complaint among emergency department (ED) patients, with a preponderance in elders and those with chronic disease, and therefore may require a broad approach to investigating underlying causes.
- Patients may use the term weakness to reflect a variety of vague symptoms including decreased motor strength, fatigue, poor energy, dyspnea, or even depression.
- Global weakness is typically caused by a systemic condition while a focal neurologic deficit or pattern can often be traced to a specific lesion within the central or peripheral nervous systems.
- Although not always detectable in the acute period, lesions to the upper motor neurons (UMNs) tend to produce signs that include spasticity to extension in the upper extremities, spasticity to flexion in the lower extremities, hyperreflexia, pronator drift, and Hoffmann and Babinski signs.
- Lesions to lower motor neurons (LMNs) typically result in flaccidity, decreased reflexes, fasciculations, or muscle cramps.

1. A 65-year-old man with a history of atrial fibrillation, on warfarin, with a supratherapeutic international normalized ratio (INR) of 4, presents with sudden onset right leg weakness and back pain. On examination, he is tachycardic to 108 beats/min and has 3/5 weakness to the right hip flexors and extensors, knee flexors and extensors, as well as ankle dorsiflexion and great toe extension. However, ankle plantar flexion is preserved. His knee reflexes are absent, but Achilles reflexes are normal. He has a normal Babinski reflex and no spasticity. He has sensory deficits throughout the anterior and posterior parts of his proximal leg as well as the anterior lower leg and dorsum of the foot. His posterior lower leg and plantar surface, however, have normal sensation. Rectal tone is normal, and there is no urinary retention. What is the most likely diagnosis?

a. Anterior cord syndrome from epidural hematoma
b. Cauda equina syndrome
c. Guillain-Barré syndrome
d. Hemorrhagic anterior cerebral artery (ACA) distribution stroke
e. Retroperitoneal hematoma with lumbar plexopathy

Answer: e. This patient has a spontaneous retroperitoneal hematoma compressing the lumbar nerve plexus.

2. A 45-year-old man has had gradual onset of progressive weakness to his face and trouble swallowing for 2 days. On examination, he has bilateral ptosis with dilated, poorly reactive pupils, bilateral upper and lower facial muscle weakness, poor soft palate rise, and slurred speech. His oral mucosa is dry. Arms and legs have 5/5 strength. He has no sensory deficits. He has a palpable distended bladder. His symptoms have not abated since onset, and they are getting worse. What is the most likely diagnosis?

a. Brainstem stroke
b. Botulism
c. Muscular dystrophy
d. Myasthenia gravis
e. Organophosphate poisoning

Answer: b. This patient has acute onset of progressive neuromuscular junction weakness. He has autonomic findings of abnormal pupil response to light, dry oral mucosa from decreased salivary production, and a distended bladder. These imply that his problem is with the release of acetylcholine (ACh) rather than the nicotinic receptor. The latter would not have autonomic findings. The most appropriate cause of acute onset, progressive impairment in the release of ACh, as listed in the choices, is botulism.

3. A 23-year-old woman presents with the sudden onset of weakness of her face, arm, and leg 2 hours ago. On examination, she has weakness to her upper and lower face on the left. She cannot abduct her left eye. She has 3/5 strength to her upper and lower extremities on her right side. She has a right pronator drift and an upgoing toe on the right. Sensation is decreased in her right upper and lower extremities as well. There is no aphasia, neglect, or visual field deficit. What is the most likely diagnosis?

a. Midbrain stroke because of cardioembolic stroke
b. Middle cerebral artery (MCA) distribution stroke
c. Multiple sclerosis
d. Myasthenia gravis
e. Pons stroke because of vertebral dissection

Answer: e. This patient has acute onset of crossed face and extremity weakness, with upper motor neuron (UMN) signs in the extremities. Both upper and lower extremities are affected, which makes a corticospinal tract (CST) lesion more likely than a cerebral cortex lesion. Her left-sided facial weakness is representative of a peripheral cranial nerve (CN) VII that is actually due to infarction of the CN VII nucleus in the brainstem. The CN VI deficit on that side is due to proximity of this nucleus as well. The CST runs just anterior to these nuclei within the brainstem. These CN nuclei lie in the pons.

4. A 21-year-old man awoke this morning with weakness to his right hand and right foot. He admits to drinking heavily the night before and falling asleep on the floor. On examination, he appears well, with weakness to right wrist extension and thumb extension, as well as sensory deficits over the dorsal surface of his hand and first and third digits. He also has weakness to ankle dorsiflexion and great toe extension on the right. He has sensory deficits over the anterior lower leg and the dorsum of his foot. Biceps and ankle reflexes are intact, he has no pronator drift, and his toes are downgoing. What is the most likely diagnosis?

a. Brainstem stroke
b. Brown-Séquard syndrome
c. Compressive neuropathy
d. MCA distribution stroke
e. Polyradiculopathy secondary to disk disease

Answer: c. He has radial nerve and peroneal nerve palsies because of compression while lying passed out on the floor for an unspecified time.

5. A 70-year-old man has had trouble swallowing and progressive weakness of his hands over the past 2 months. On examination, his speech is slurred, his voice is nasal, and he has fasciculations to his face, tongue, and over his pectoralis muscles and deltoids bilaterally. He has 4/5 strength to shoulders, biceps, triceps, and hand grip bilaterally. Stiffness to extension is present at both elbows. He has bilateral pronator drift and 3+ biceps reflexes. He tends to smile inappropriately. He has no sensory symptoms. What is the most likely diagnosis?

a. Amyotrophic lateral sclerosis (ALS)
b. Brainstem stroke
c. Chronic demyelinating polyneuropathy
d. Parkinsonism
e. Polymyositis

Answer: a. This patient has lower motor neuron (LMN) signs (fasciculations) and UMN signs (pronator drift and increased reflexes) in similar distribution. The combination of upper and lower motor neuron involvement makes ALS the leading diagnosis among those listed. His dysphagia and inappropriate smiling are due to a release of the medulla from upper motor neuron regulation.

10 Cyanosis

KEY CONCEPTS

- Cyanosis occurs due to an absolute amount of desaturated hemoglobin (~5 g/dL) rather than a percentage; anemic patients exhibit cyanosis at a lower Pa_{O_2} than those with normal hemoglobin levels.
- Cyanosis is an insensitive indicator of tissue oxygenation; its presence suggests hypoxemia, but its absence does not exclude it.
- Central cyanosis is most commonly due to global arterial hypoxemia or abnormal hemoglobin forms; peripheral cyanosis is due to vasoconstriction or reduced flow of normally oxygenated hemoglobin to the peripheral tissues.
- Methemoglobin has a chocolate brown color, even when the blood is exposed to room air; pulse oximetry for patients with methemoglobinemia typically reads 85%, regardless of the Pa_{O_2} or Sa_{O_2}.
- Methemoglobinemia may be due to inherited congenital errors in enzyme function or acquired secondary to exposure to oxidizing agents such as certain drugs and toxins.
- Congenital heart disease is a diagnostic consideration in all infants presenting with cyanosis.
- Sulfhemoglobin is often reported as methemoglobin on CO-oximetry; patients with methemoglobinemia on CO-oximetry who do not respond to methylene blue treatment likely have sulfhemoglobinemia.
- Methylene blue is the treatment of choice for symptomatic methemoglobinemia or at levels of methemoglobin greater than 20%.
- All patients with a first episode of cyanosis or cyanosis of uncertain cause require hospitalization.

1. Which of the following statements regarding methemoglobin is true?

a. Methemoglobin begins to produce cyanosis when its concentration reaches 25% of total hemoglobin.
b. Methemoglobin changes from dark purple to light red when exposed to room air.
c. Methemoglobin normally accounts for 5% of total hemoglobin.
d. Methylene blue accelerates the reduction of methemoglobin to hemoglobin.
e. The primary method of reducing methemoglobin to hemoglobin is glucose-6-phosphate dehydrogenase (G6PD)–dependent.

Answer: d. Methemoglobin normally accounts for less than 1% of total hemoglobin. Cyanosis generally results when more than 10% to 15% of the total hemoglobin is methemoglobin (absolute level of 1.5 g/dL), which has a dark purple-brown color, even when exposed to room air. Methemoglobin is reduced to ferrous hemoglobin primarily by nicotinamide adenine dinucleotide (NADH)–cytochrome b5 reductase, an enzyme system present in red blood cells. A secondary nicotinamide adenine dinucleotide phosphate (NADPH)–dependent system uses glutathione and G6PD to reduce methemoglobin to hemoglobin. This secondary pathway normally plays a minor role but is accelerated by methylene blue.

2. Which of the following conditions produces central cyanosis?

a. Hypothermia
b. Raynaud phenomenon
c. Shock
d. Venous insufficiency
e. Ventricular septal defect

Answer: e. Central cyanosis is caused by decreased arterial oxygen saturation, often secondary to shunting of venous unsaturated hemoglobin into the arterial circulation, or the presence of abnormal hemoglobin. A ventricular septal defect causes shunting of deoxygenated blood from the right side of the heart to the left, with resultant decreased arterial oxygen saturation and central cyanosis. The remaining options would cause peripheral cyanosis.

3. In the cyanotic patient, clinical improvement with supplemental oxygen is most suggestive of which of the following underlying processes?

a. Arterial emboli
b. Congenital heart disease with right-to-left shunting
c. Hyperventilation
d. COPD
e. Methemoglobinemia

Answer: d. Clinical improvement with oxygen suggests diffusion impairment, hypoventilation, or an increased V/Q ratio due to mismatch. Patients who do not respond to oxygen are more likely to have decreased V/Q ratio abnormalities, such as shunting from pulmonary consolidation or congenital heart disease with right-to-left shunting, or abnormal hemoglobin. Arterial emboli will generally not be affected by supplemental oxygen.

4. Soon after receiving topical upper airway anesthesia with benzocaine, a patient becomes tachypneic and complains of chest pain. The patient is noted to be cyanotic. Supportive measures are initiated, and CO-oximetry reveals a methemoglobin level of 40%. Which of the following should be administered?

a. Benadryl
b. Hyperbaric oxygen
c. Methylene blue
d. Nitroglycerin
e. Thrombolytics

Answer: c. Local anesthetics such as benzocaine, lidocaine, and prilocaine can cause an acquired methemoglobinemia. Urgent treatment with oxygen and methylene blue is indicated in such patients, especially when accompanied by symptomatic hypoxia (e.g., dysrhythmias, angina, respiratory distress, seizures, coma) and methemoglobin levels greater than 20%.

5. An otherwise healthy 35-year-old woman presents with headache, mild shortness of breath, and central cyanosis. She is currently taking trimethoprim-sulfamethoxazole and phenazopyridine for a urinary tract infection. She does not improve with supplemental oxygen. Her chest radiograph is normal. CO-oximetry shows a methemoglobin level of 28%. Her medications are withheld. The patient shows no improvement after 2 mg/kg of methylene blue is administered. Ongoing management should include which of the following treatments?

a. Calcium channel blockers
b. Hydroxocobalamin
c. Phlebotomy
d. Sodium thiosulfate
e. Supportive care

Answer: e. This patient likely has sulfhemoglobinemia, a rare complication of the medication phenazopyridine (Pyridium), as well as sulfa-containing drugs such as trimethoprim-sulfamethoxazole. Standard CO-oximetry will often report sulfhemoglobin as methemoglobin. Both sulfamethoxazole and phenazopyridine can also cause methemoglobinemia. Patients with an elevated methemoglobin level and no response to methylene blue likely have sulfhemoglobinemia, which is less severe than methemoglobinemia and only requires supportive care and removal of the offending agent. Phlebotomy is the treatment for polycythemia. Sodium thiosulfate and hydroxocobalamin are both used to treat cyanide toxicity, and calcium channel blockers may be used to treat peripheral cyanosis resulting from vasospasm (Raynaud phenomenon).

KEY CONCEPTS

- Syncope is defined as a sudden, spontaneous loss of consciousness and postural tone with rapid, complete, and spontaneous recovery. This loss of consciousness typically lasts seconds to minutes and is not followed by a persistent alteration in mental status.
- Syncope is caused by a transient global cerebral hypoperfusion. It is the final common pathway for a wide variety of underlying causes, and requires a systematic approach to diagnosis.
- Pre-syncope, or near-syncope, is defined as the sudden onset of a sense of impending or incomplete loss of consciousness.
- Most syncopal episodes have benign causes, but serious underlying conditions are possible. The causes of syncope can be organized into the following three categories: (i) cardiac, (ii) reflex (includes vasovagal syncope), (iii) orthostatic.
- National guidelines recommend a 12-lead electrocardiogram for all patients with syncope but most patients without serious underlying disease will have a normal or nondiagnostic electrocardiogram (ECG).
- National guidelines do not recommend routine laboratory testing or imaging (e.g., chest x-ray, neuroimaging, echocardiogram) and these tests should be obtained only when history or physical examination suggests such evaluation.
- Hospitalization or placement in an observation unit after emergency department (ED) evaluation for syncope is based on the specific, identified cause (e.g., gastrointestinal hemorrhage, malignant arrhythmia, pulmonary embolism).
- Patients with a clear description of reflex syncope (e.g., vasovagal syncope) without serious underlying disease are generally appropriately discharged directly from the ED.
- Several syncope clinical risk scores have been developed for ED use, but none have been definitively shown to outperform unaided clinical gestalt with respect to risk stratification and resource utilization.

1. Which of the following statements regarding the epidemiology of syncope is true?

a. Of athletes who die during exercise, the vast majority have had a prior episode of syncope.
b. People younger than 65 years account for 50% of all patients admitted for syncope from the emergency department.
c. Syncope in young adults is typically secondary to significant pathology.
d. Syncope is responsible for approximately 5% to 6% of all emergency department (ED) visits.
e. There is a slight female predominance among ED patients presenting with syncope.

Answer: e. Syncope accounts for approximately 1% to 2% of ED visits in the United States. There is a slight female predominance among ED patients with syncope. Approximately 32% of these patients are hospitalized, and people aged 65 years or older account for the majority of such hospitalizations. Benign causes of syncope predominate in adolescents and young adults. Approximately 30% of athletes who die during exercise have had a prior episode of syncope as a sentinel event.

2. Syncope resulting from serious pathology is usually caused by which of the following?

a. Transient ischemic attack (TIA)
b. Cardiac dysrhythmias
c. Pulmonary embolism
d. Atrial myxoma
e. Toxic-metabolic abnormalities

Answer: b. The principal serious causes of syncope are cardiac dysrhythmias. TIA is less frequently encountered but equally serious. Stroke generally does not cause isolated syncope. Toxic-metabolic abnormalities may induce syncope through alterations in blood pressure or cardiac rhythm. Structural cardiac lesions, such as atrial myxoma, and sudden interruption of right ventricular outflow by pulmonary embolism, can also cause sudden loss of consciousness but are less common.

3. Which of the following findings most strongly suggest that a patient presenting with syncope can be safely discharged from the emergency department (ED)?

a. Age under 85 years and a normal chemistry panel
b. A normal chest x-ray and CT scan of the head
c. Normal heart rate and temperature in the ED
d. Normal ECG findings and no past medical history
e. Absence of abdominal pain and absence of headache

Answer: d. A normal electrocardiogram in a patient without other significant risk factors (e.g., advanced age, preexisting heart disease) may be considered for outpatient disposition. The chemistry panel, chest x-rays, and CT head scan are commonly ordered but are of low diagnostic and prognostic utility. Abdominal pain, fever, and headache are important associated signs and symptoms to assess for, but their absence has little prognostic utility.

Depressed Consciousness and Coma 12

KEY CONCEPTS

- Coma is a state of depressed consciousness in which a patient is not aware, is not awake, and does not respond to vigorous stimulation. Consciousness consists of arousal (subcortical) and awareness (cortical).
- Damage to the dorsal brainstem, thalamus, axonal projections to the cortex, or extensive injury to bilateral cortices may result in depressed consciousness or coma.
- Toxicologic, metabolic, infectious, and other disorders causing diffuse brain injury cause 65% of coma cases; of these, toxins are the most common. Structural brain diseases account for most of the remaining cases.
- A patient with depressed consciousness is unlikely to provide a reliable history. Historical information should be elicited from other available sources, such as emergency medical services personnel, family members, the patient's belongings, or medical records.
- An abrupt onset of coma suggests a stroke, seizure, or cardiac event.
- The neurologic examination should include an evaluation of level of consciousness, cranial nerves, brainstem reflexes, and motor responses.
- Pinpoint pupils may represent a pontine infarct or intoxication from opioids, clonidine, or cholinergic substances.
- Nonconvulsive status epilepticus should be suspected in cases of coma of undetermined cause and is diagnosed by electroencephalography.
- Hypoglycemia and hypoxia are two easily identified and reversible causes of coma.
- An empiric trial of naloxone will lead to rapid reversal of opioid toxicity and other select medication overdoses.
- Targeted temperature management is recommended in comatose adult patients with return of spontaneous circulation after cardiac arrest from either a shockable or nonshockable rhythm, with a goal temperature range of 32°C to 36°C. In children who are comatose post arrest, it is reasonable to target a temperature range of 36°C to 37.5°C. Hyperthermia should be strictly avoided for all comatose postarrest patients.
- Most patients with coma will require intensive care monitoring. If the cause of coma is not treatable in the initial facility and the patient needs a higher level of specialty care (e.g., structural lesion requiring neurosurgery), then a coordinated transfer should be facilitated.

1. In infants, what is the most common cause of a depressed level of consciousness?
- **a.** Accidental toxic ingestion
- **b.** Inborn error of metabolism
- **c.** Infection
- **d.** Physical abuse

Answer: c. The most common causes of depressed consciousness vary with patient age. In infants, infectious causes are most common; however, trauma secondary to physical abuse and metabolic derangements from inborn errors of metabolism can also be seen. Accidental toxic ingestions are often seen in younger children but are very uncommon in infants. Young adults and adolescents are more likely to present with depressed consciousness after recreational drug use or trauma. Older adults are particularly vulnerable to infectious causes, medication effects, and alterations in their living environments.

2. What Glasgow Coma Scale (GCS) score would be given to an adult patient who opens the eyes to painful stimuli, speaks in an incomprehensible manner, and withdraws from pain?
- **a.** 7
- **b.** 8
- **c.** 9
- **d.** 10

Answer: b. Using the GCS, this adult patient would receive 2 points for eye opening to pain, 2 points for a verbal response consisting of incomprehensible sounds, and 4 points for withdrawing from painful stimuli.

3. What Glasgow Coma Scale (GCS) score would be given to a pediatric patient who opens the eyes and flexes the extremities to painful stimuli and who is irritable and continually cries?
- **a.** 6
- **b.** 7
- **c.** 8
- **d.** 9

Answer: d. Using the GCS, this pediatric patient would receive 2 points for eye opening to pain, 4 points for persistently being irritable, and 3 points for flexion in response to painful stimuli. The pediatric GCS takes into account that younger children might not yet be verbal.

4. Awareness of one's self and surroundings defines which of the following?
- **a.** Cognition
- **b.** Consciousness
- **c.** Judgment
- **d.** Orientation

Answer: b. Consciousness is defined as the awareness of one's self and surroundings; it is made up of both arousal and awareness. Cognition involves knowing and understanding, especially through the perception of what is experienced by the senses; judgment is the ability to process input data to generate more meaningful information; and orientation is the ability to determine the relative position of oneself in relation to one's person, place, time, and situation.

5. In the evaluation of a comatose patient, which of the following responses most clearly indicates brainstem dysfunction?
- **a.** A drop of saline on the cornea of one eye causes both eyes to blink
- **b.** Infusion of cold water into the right ear causes rightward deviation of the eyes
- **c.** Stimulation of the posterior pharynx has no effect on the soft palate of pharyngeal muscles
- **d.** When the head is rotated from side to side in the horizontal plane, the eyes remain in midposition and move in unison with the head.

Answer: d. Brainstem function is evaluated through the various brainstem reflexes. The oculocephalic reflex is tested by rotating the head from side to side in the horizontal plane. In patients with normal brainstem function, the eyes should move in a direction opposite of head movement and appear to maintain visual fixation on a single point in space. The corneal reflex is tested by touching the edge of the cornea with a wisp of cotton or drop of water and observing for blinking of the eyes. The oculovestibular reflex is tested by infusing cold water into the external auditory canal. In patients with an intact brainstem, the stimulus causes sustained conjugate deviation of the eyes toward the ear being irrigated. The gag reflex is tested by stimulating the posterior pharynx and observing for brisk elevation of the soft palate and contraction of the pharyngeal muscles. Because the gag reflex is symmetrically reduced or absent in a subset of normal individuals, the utility of this brainstem reflex is limited.

13 Confusion

KEY CONCEPTS

- Confusion and delirium are symptoms, not a diagnosis.
- Focal cortical dysfunction, such as from tumor or stroke, typically does not cause confusion.
- Any underlying clinical process that disrupts optimal central nervous system (CNS) functioning can result in confusion.
- Emergent causes of confusion that need immediate detection and treatment include hypoglycemia, hypoxemia, hypotension, sepsis, and toxic ingestions.
- Assessment of attention is fundamental for the assessment of patients with confusion as disturbances in attention are consistent with delirium versus psychiatric illness or dementia.
- Recommended tools for identifying patients with delirium in the emergency department are the Delirium Triage Screen (DTS) and brief Confusion Assessment Method (bCAM), if indicated.
- Delirium often goes unrecognized unless a structured assessment tool is used.
- Sedatives, including antipsychotics, are useful for managing undifferentiated agitation while the diagnostic evaluation is in progress.

1. A 70-year-old man with a chief complaint of confusion is brought to the emergency department by his family. Which of the following initial assessments should be included?

a. Blood pressure
b. Pulse oximetry
c. Temperature
d. All of these

Answer: d. Confusion may result from shock states, hypoglycemia, and hypoxia. Evaluation for these conditions is a priority. Confusion is a symptom rather than a medical condition, and reversible causes should be investigated.

2. A variety of brief screening tools may aid in the detection of confusion. All of these tests focus on which cardinal finding of confusion?

a. Attention impairment
b. Decreased level of arousal
c. Disorientation
d. Long-term memory impairment

Answer: a. Inattention is the critical finding for diagnosis of confusion and delirium. All of the screening tools (brief Confusion Assessment Method [bCAM], Mini-Mental State Exam [MMSE], Quick Confusion Scale [QCS]) evaluate this abnormality, although other components of consciousness are tested in some models.

3. A 30-year-old patient is brought to the emergency department for evaluation of odd behavior. Which of the following characteristics might suggest a psychiatric cause for the behavior?

a. Auditory hallucinations
b. Disorientation
c. Fever
d. Visual hallucinations

Answer: a. Auditory hallucinations are common in psychiatric illness. If hallucinations are present in organic causes of delirium, they are usually visual, tactile, or olfactory. Orientation is generally preserved with primary psychiatric disorders unless psychosis or severe impairment is present.

4. Postictal confusion is common in patients with seizures, but if improvement in consciousness does not occur within 20 to 30 minutes after seizure cessation, which of the following conditions should be considered?

a. Electrolyte abnormalities
b. Hypoglycemia
c. Nonconvulsive or subtle status epilepticus
d. All of these

Answer: d. For a patient with a generalized convulsive seizure, termination of the seizure activity is usually **followed** by improvement of mental status within a short period of time. For the patient with persistently altered consciousness or prolonged confusion, consider causes of provoked seizures with prolonged altered mental status or persistence of subtle seizures.

14 Seizures

KEY CONCEPTS

- Lifetime seizure incidence reaches up to 10% in the US population and depends upon multiple factors including age, history of epilepsy, structural brain or neurodegenerative disease, genetic predisposition, acuity and severity of metabolic derangements.
- Seizures are the result of an imbalance in the excitation and inhibitory neuronal synapses in the cerebral cortex or limbic system.
- Epilepsy is a condition associated with a lower seizure threshold and an inherently higher risk of recurrent seizures in a lifetime.
- Acute symptomatic seizures occur at, or in close temporal relationship with, a distinct provoked event, and may occur with any toxic or metabolic, structural, ischemic, inflammatory, or infectious insult.
- The initial evaluation of seizure includes the identification of seizure characteristics, duration, and etiology of possible inciting events. Recognizing the occurrence of similar events in the past is key in determining triggers and distinguishing the management priorities between patients with epilepsy or with acute symptomatic seizures.
- The majority of seizures will cease spontaneously within 5 minutes; however, when physiologic seizure termination pathways are overwhelmed resulting in status epilepticus, characterized by the persistent and exponential refractory response to abortive seizure therapies. The magnitude of treatment refractoriness is time dependent; therefore a prompt stepwise escalation of therapies aiming at seizure termination is warranted.
- Prompt identification and, whenever possible, reversal of seizure triggers are priorities. In particular, assessing for and correcting hypoglycemia should occur as early as possible.
- Initial nonpharmacologic interventions include ensuring physical safety of patients with seizures to prevent traumatic injuries and aspiration. During convulsions, focusing on lateral decubitus positioning and suctioning of oral cavity are important to prevent aspiration; the use of intraoral devices may lead to trauma without additional benefit.
- Pharmacologic interventions center on the timely and dose-appropriate administration of parenteral benzodiazepines, which are first-line therapy (i.e., lorazepam intravenous [IV] up to 0.1 mg/kg in divided doses, or midazolam IV/intramuscular [IM]/intranasal [IN] 5 to 10 mg).
- In pregnant patients, eclampsia should be considered the underlying etiology of de novo seizures after 20 weeks of gestational age and up to 8 weeks postpartum. Clinical evaluation should focus on evaluating for associated symptoms (e.g., headache, visual abnormalities and confusion), as well as focal neurologic deficits. Neuroimaging should be considered if neurologic deficits persist, prolonged loss of consciousness or encephalopathy. IV magnesium 4 to 6 g is first-line therapy in the pregnant patient experiencing new-onset seizures.
- During the postictal state, key priorities in management include ensuring airway patency with optimal positioning in those with alteration of consciousness, evaluation for nonconvulsive seizures and acute brain injuries or inciting factors that warrant further definitive treatment.
- Nonconvulsive status epilepticus can only be diagnosed with electroencephalography (EEG) monitoring and should be considered in any patient with prolonged, unexplained altered mental status. Patients at higher risk include those with poorly controlled epilepsy, acute brain injury, sepsis, and intoxication.
- A patient with new-onset seizures with a normal neurologic examination who returns to baseline mental status and does not have comorbid disease does not require diagnostic testing beyond, serum glucose and sodium levels, a pregnancy test (in women), and a noncontrast head computed tomography (CT); a toxicologic screen is considered in select patients.

1. A 63-year-old woman with end-stage renal disease on hemodialysis, diabetes, and hypertension is brought to the emergency department after losing consciousness during hemodialysis. The episode occurred towards the end of the hemodialysis section approximately 45 minutes ago and was associated with foaming at the mouth, cyanosis, and bilateral tonic-clonic movements of upper extremities lasting nearly 3 minutes. On examination, T 36.6°C, HR 89 bpm, SBP 195 mm Hg, RR 11, pulse oximetry 94%, Glasgow Coma Score (GCS) is 10 with subtle nystagmus, and no meningismus. Computed tomography of the head (CTH) demonstrated hypodensities in bilateral parieto-occipital regions.

What is the leading diagnosis and required work up?

a. Nonconvulsive status epilepticus from hypertensive emergency; electroencephalography (EEG)
b. Cortical blindness from bilateral posterior cerebral artery stroke; computed tomography angiogram
c. Coma from meningoencephalitis; lumbar puncture
d. Coma from uremia; blood urea nitrogen (BUN)

Answer: a. The description of the episode of loss of consciousness with associated cyanosis, and tonic-clonic convulsions is highly suspicious for a generalized tonic-clonic seizure. The patient has not returned to baseline mentation as demonstrated by GCS of 10 and is displaying subtle signs of potentially ongoing seizures (nystagmus); thus the most likely diagnosis is nonconvulsive status epilepticus, which can be diagnosed only with an electroencephalogram. The elevated blood pressure along with hypodensities in the posterior cortical region on CTH are highly suggestive of posterior reversible encephalopathy syndrome, a form of hypertensive emergency. The patient has acute encephalopathy and the stem of the question does not display findings suggestive of cortical blindness (preserved pupillary light reflex with complete binocular vision loss). Lack of prodrome of headache and confusion, fever, and meningismus make meningoencephalitis less likely. The episode happened towards the end of hemodialysis section, so profound uremia is unlikely.

2. You suspected posterior reversible encephalopathy syndrome (PRES) and order a STAT electroencephalography (EEG) while lowering the blood pressure in a stepwise approach. While the patient is waiting to be admitted to the neurology intensive care unit and connected to EEG, the bedside registered nurse reports new left gaze deviation, fine eyelid twitching, and more prominent nystagmus that was refractory to 1 mg of lorazepam. Your index of suspicion for nonconvulsive status epilepticus is high and you decide to treat empirically. The patient weighs 52 kg.

What is the first-line therapy for status epilepticus treatment?

a. Fosphenytoin 1000 mg intravenous (IV)
b. Lorazepam 5 mg IV
c. Levetiracetam 3000 mg IV
d. Lacosamide 300 mg IV

Answer: b. The first-line therapy for seizures and status epilepticus is benzodiazepines. In the hospital setting, lorazepam IV is most commonly used. Although all the other choices are reasonable second-line antiseizure drugs, the patient has received only 1 mg of lorazepam at this point, and a total of 5mg of lorazepam (0.1 mg/kg) is considered appropriate first-line therapy.

3. According to the best evidence available from randomized controlled trials, which second-line antiseizure regimen is recommended for benzodiazepine-resistant convulsive status epilepticus?

a. Fosphenytoin 20 mg phenytoin equivalent (PE)/kg
b. Valproic acid 40 mg/kg
c. Levetiracetam 60 mg/kg
d. All of the above

Answer: d. According to the Established Status Epilepticus Treatment Trial (ESETT) fosphenytoin 20 mg PE/kg (max 1500 mg), valproic acid 40 mg/kg (max 3000 mg), and levetiracetam 60 mg kg (max 4500 mg) were equally effective in terminating convulsive status epilepticus in nearly half of randomized patients (fosphenytoin 45%, valproic acid 46%, levetiracetam 47%). The ESETT trial has provided the best

evidence to date on the second-line treatment of convulsive status epilepticus.

4. A 37-year-old woman with remote history of head trauma from a motor vehicle collision presents to the emergency department with ongoing bilateral convulsions. The patient's significant other witnessed the onset of convulsions 45 minutes ago following an argument. She received a total of 10 mg of intramuscular (IM) midazolam while being transported by the emergency medical services. On examination, T 36.3°C, HR 120 bpm, SBP 115 mm Hg, RR 22, pulse oximetry 100% on a nonrebreather mask; she has her eyes closed and resists passive eye opening of the examiner, does not respond otherwise to the examiner and has ongoing asynchronous, nonstereotypic, and suppressible convulsions of both upper extremities with associated pelvic thrust.

   What are the most important nonpharmacologic initial interventions in patients with suspected generalized convulsive status epilepticus?

   a. Placing an oropharyngeal airway device, lateral decubitus positioning, establishing an intravenous (IV) line
   b. Checking glucose fingerstick, establishing an IV line, lateral decubitus positioning
   c. Suctioning oral cavity, placing oropharyngeal airway device, lateral decubitus positioning patient
   d. Checking glucose fingerstick, placing nasopharyngeal airway device, supine positioning

Answer: b. Initial nonpharmacologic interventions include ensuring physical safety of patients with seizures to prevent traumatic injuries and aspiration. During convulsions, focusing on lateral decubitus positioning and suctioning of oral cavity are important to prevent aspiration; the use of intraoral devices may lead to trauma without additional benefit. Nasopharyngeal airway devices may help with transient obstruction that may occur in the peri-ictal period, but oropharyngeal devices should be avoided as they may induce gagging and vomiting and may damage the teeth or tongue. The serum glucose level should be determined in every seizing or postictal patient as hypoglycemia is a common metabolic cause of provoked seizures. Although some first-line seizure abortive medications can be given IM, intraosseous (IO), per rectum (PR), intranasal (IN), establishing an IV line is a helpful initial step to facilitate timely pharmacologic interventions.

5. An additional 6 mg of lorazepam IV (0.1 mg/kg) slows down the movements but does not abort them completely. What is the most common differential diagnosis for generalized convulsive status epilepticus?

   a. Nonepileptic spells
   b. Neuroleptic malignant syndrome
   c. Complex migraine
   d. Cardiogenic syncope

Answer: a. Although all the answers reflect conditions that can be considered seizure mimics, nonepileptic spells, or pseudoseizures, represent the most common differential diagnosis for generalized convulsive status epilepticus with an incidence of approximately 10% of subjects enrolled in clinical trials. Nonepileptic spells tend to last greater than 5 minutes, thus meeting criteria for convulsive status epilepticus by duration of events, and be resistant to high doses of benzodiazepines. Neuroleptic malignant syndrome is also associated with prolonged spells of abnormal movement, but the type of involuntary movement (typically tonic extension of all extremities, opisthotonos and rigors) and associated hyperthermia, tachycardia, and diaphoresis can distinguish these disorders. Complex migraine typically manifests as lateralizing dysesthesia in association with severe throbbing headache with phono/photophobia, and not with prolonged convulsions. Convulsions associated with cardiogenic syncope usually occur in the setting of hypoperfusion and are brief, unlike this case.

6. Which additional features may assist in the diagnosis of nonepileptic spells?

   a. Disproportionally short postictal confusional state
   b. Normal lactic acid and creatine kinase
   c. Lack of reactive leukocytosis
   d. All of the above

Answer: d. Although the only reliable way to diagnose nonepileptic spells is by characterizing the lack of an underlying seizure rhythm on electroencephalography (EEG) during spells, clinicians should be attentive to clues in the presentation and work-up that suggest a pseudoseizure diagnosis. Prolonged tonic-clonic convulsions (i.e., generalized convulsive status epilepticus) is invariably associated with some degree of rhabdomyolysis and elevation of lactic acid. The complete blood count often displays reactive leukocytosis. Postictal-related confusion or acute encephalopathy is one of the hallmarks of generalized tonic-clonic seizures and carries a direct relationship with the duration of seizures. Short periods of confusion following prolonged bilateral convulsions is suggestive of a nonepileptic spell diagnosis.

7. A 23-year-old woman currently 34 weeks' pregnant with her first baby presents to the emergency department with headache and fatigue. On exam, T 36.2°C, HR 89 bpm, SBP 145 mm Hg, RR 27, pulse oximetry 96%; she is fully oriented, complains of subjective blurry vision and has 2+ pitting edema of lower extremities. You suspect preeclampsia and initiate the work-up.

   What is the typical period for the development of preeclampsia and eclampsia?

   a. 16–37 weeks of gestational age
   b. 20 weeks of gestational age until 8 weeks postpartum
   c. 20 weeks of gestational age until delivery
   d. None of the above

Answer: b. In pregnant patients, the risk of eclampsia is highest from 20 weeks of gestational age up to 8 weeks postpartum. Clinical evaluation should focus on evaluating for associated symptoms (e.g., headache, visual abnormalities and confusion), as well as focal neurologic deficits.

8. What are the red flags that prompt neuroimaging in patients presenting with eclamptic seizures?

   a. No red flags needed, all patients should get a noncontrast computed tomography of the head with abdominal shielding
   b. If persistent neurologic deficits, including prolonged loss of consciousness or encephalopathy
   c. If more than two seizures are observed within 24 hours
   d. If seizures are refractory to benzodiazepines

Answer**: b.** Neuroimaging should be considered if neurologic deficits persist, prolonged loss of consciousness, or encephalopathy. Intravenous magnesium 4 to 6 g is first-line therapy in the pregnant patient experiencing new-onset seizures. Benzodiazepines and nonteratogenic antiseizure medications are reasonable alternatives in magnesium-refractory cases.

9. A 61-year-old previously healthy man presents to the emergency department with a first seizure. The episode consisted of left arm and face clonic movements with secondary generalization and postictal lethargy and left hemiparesis lasting 10 minutes. On examination, T 37.1°C, HR 79 bpm, SBP 105 mm Hg, RR 17, pulse oximetry 96%, GCS is 15 with no focal deficits. Given the focality of seizure onset and the patient age, you obtained a noncontrast computed tomography of the head, which was unremarkable. Which of the following factors should be considered when counseling patients on recurrent seizure risk after a first-time seizure?

a. Presence of focal abnormalities on electroencephalography (EEG)
b. Presence of focal abnormalities on neuroimaging
c. Identification of seizure triggers
d. All of the above

Answer: c. Up to 45% of adults presenting with a first unprovoked seizure will experience another within 2 years, most occurring in the first year 2. The risk is higher with known brain lesions, neuroimaging or EEG abnormalities, or nocturnal seizures 2; the latter are associated with epilepsy syndromes.

10. A 71-year-old man with advanced dementia from multiple prior strokes, atrial fibrillation on warfarin, localization related epilepsy on levetiracetam presents to the emergency department with a breakthrough seizure. The episode consisted of speech arrest and clonic movements of right face and arm. There was no postictal state identified. On examination, T 36.1°C, HR 109 bpm, SBP 115 mm Hg, RR 15, pulse oximetry 97%, GCS is 15 with no focal deficits. Which of the following factors should be considered when deciding on the need for neuroimaging after a breakthrough seizure?
a. No red flags needed, all patients should get a noncontrast CTH
b. Young age, specifically in the pediatric population
c. Use of vitamin K antagonist
d. Lack of identifiable triggers

Answer: c. Neuroimaging should be considered if neurologic deficits persist, prolonged loss of consciousness or coma, advanced age, immunocompromised state, coagulopathy (including anticoagulant drugs) or hypercoagulability, history of malignancy, preceding severe thunderclap headache, status epilepticus, in patients with stigmata of neurocutaneous syndromes, and suspected trauma.

11. What is the most common metabolic cause of seizure activity?
a. Hypercalcemia
b. Hyperglycemia
c. Hypermagnesemia
d. Hypocalcemia
e. Hypoglycemia

Answer: e. Seizure activity secondary to metabolic derangements is most commonly caused by hypoglycemia. The only treatment required in this situation may be intravenous (IV) glucose. Prolonged seizure activity may also cause hypoglycemia so that the cause-and-effect relationship may sometimes be reversed, and further therapy may be required.

12. A 15-year-old girl is brought to the emergency department for evaluation of a recent seizure. While awaiting laboratory results, she begins to have further seizure activity. Which of the following is the optimal first-line agent to terminate her seizure activity?
a. Fosphenytoin
b. Lorazepam
c. Phenobarbital
d. Phenytoin
e. Valproic acid

Answer: b. Benzodiazepines are the optimal first-line agents for stopping seizure activity in patients of all ages. Available agents include lorazepam, diazepam, and midazolam. Phenytoin is recommended as second-line therapy for adults with persistent seizure activity. The prodrug, fosphenytoin, can be administered more quickly, can be given intramuscularly, and has less of a tendency to cause hypotension. Second-line therapy for children is phenobarbital. Third-line therapy is pentobarbital, propofol, or a benzodiazepine infusion. Valproic acid should be considered for patients who are on chronic valproic therapy and whose levels are subtherapeutic.

13. A 24-year-old man is brought to the emergency department by emergency medical services (EMS). The patient's mother reports that she found her son seizing on the floor of her living room approximately 30 minutes before arrival at the hospital. Two months ago, the patient returned from Mexico, where he had been incarcerated for 6 months. The mother reports that during the past 2 months she has seen her son consistently take his seizure medicine and several other pills for a "bad lung infection" he got in Mexico. She cannot remember the names of any of the medications. Several doses of intravenous (IV) lorazepam have been administered, with no effect on the patient's seizure activity. Which of the following medications would be the most effective in aborting his seizure activity?
a. Diazepam
b. Magnesium sulfate
c. Phenytoin
d. Pyridoxine
e. Valproic acid

Answer: d. Several historical clues in this scenario point to tuberculosis being the "bad lung infection" in this patient. In patients with seizures that are refractory to benzodiazepines, isoniazid (a common medication for tuberculosis) overdose is a possibility and should be considered. Pyridoxine is the only fully effective pharmacologic treatment for toxic isoniazid seizures, although benzodiazepines have been shown to suppress seizure activity in some cases.

14. A mother arrives with her 10-year-old daughter (41 kg) who has been seizing for at least 10 minutes. The patient has a history of epilepsy, and a home dose of rectal diazepam has been ineffective. The mother states that the child has been in her usual state of good health until the seizure began, and there has been no history of trauma. Which of the following is the most appropriate initial action?
a. Administer 10 mg midazolam intramuscularly.
b. Consult neurology to obtain a bedside electroencephalogram.
c. Endotracheal intubation with vecuronium and etomidate.
d. Establish vascular access and administer 2 mg of lorazepam.
e. Obtain an immediate computed tomography scan of the head.

Answer: a. Early, aggressive benzodiazepine administration is associated with decreased morbidity and mortality in status epilepticus. Intramuscular midazolam is superior to intravenous lorazepam; in addition, the dose of lorazepam is inadequate. Endotracheal intubation may ultimately be required but is a secondary priority; use of a long-acting neuromuscular blockade agent, such as vecuronium, should be avoided. Cranial computed tomography may or may not be needed in this patient, depending on the response to benzodiazepine therapy. Bedside electroencephalograms are most useful in diagnosing nonconvulsive status epilepticus.

15. Paramedics present with a 24-year-old woman with a history of epilepsy after a seizure. She is somnolent but easily arousable and oriented to self and year. Her vital signs are normal, there are no signs of trauma, and an empty expired bottle of phenytoin is found in her purse. Her prehospital finger stick blood glucose level is 163 mg/dL. Which of the following treatment options is most correct?
a. Administer 20 mg/kg fosphenytoin intramuscularly and observe the patient until she returns to baseline.
b. Establish vascular access, and administer 4 mg lorazepam intravenously (IV).
c. Establish vascular access, obtain a phenytoin level, and administer 1 to 2 mg lorazepam IV if the patient begins to seize.
d. Place the patient in a monitored setting, establish vascular access, and withhold diagnostic tests and treatments unless patient's condition changes.

Answer: c. Medication noncompliance is a frequent cause of seizures in adults with epilepsy. It is recommended to check phenytoin levels before administering additional drug. Because the patient is not in status epilepticus, a low dose of benzodiazepines can be considered for patients who begin to seize while undergoing a period of observation.

16. Which of the following is not part of the routine emergency department evaluation and treatment of a 21-year-old healthy woman with a first seizure?

a. Discharge, with early outpatient neurology follow-up
b. Evaluation of serum electrolyte levels
c. Initiation of antiepileptic drug therapy
d. Performance of cranial computed tomography

Answer: c. Adults presenting with a first seizure should undergo cranial computed tomography and evaluation of serum electrolyte and glucose levels because abnormalities would likely influence disposition while identifying potentially life-threatening conditions. For otherwise healthy adults with normal findings after evaluation, early outpatient follow-up can be considered. In some cases, initiation of antiepileptic drugs can be considered after a first seizure; however, this should be done in consultation with the neurologist responsible for outpatient follow-up.

15 Dizziness and Vertigo

KEY CONCEPTS

- Neurologic complaints, such as imbalance, dysarthria, or numbness in patients with dizziness/vertigo raise the likelihood of transient ischemic attack (TIA) or stroke as the cause.
- Benign paroxysmal positional vertigo (BPPV) requires head movement to elicit symptoms. Consequently, the Hallpike test should *not* be performed if the patient is actively symptomatic during history taking (and the patient's head has not been recently moved) because such a history is inconsistent with BPPV.
- When performing the Hallpike test, the head should be turned to the side 45 degrees prior to laying the patient back into the head-hanging position.
- A positive Hallpike test should elicit upbeating (fast phase towards eyebrows) nystagmus.
- The Epley maneuver is used to treat posterior semicircular canal BPPV, which is the most common subtype of BPPV.
- Central causes of nystagmus are more likely when the pattern of nystagmus is purely vertical, downbeating (fast phase beating toward the nose), nonfatigable, direction changing with gaze, or spontaneous pure torsional.
- The presence of auditory symptoms (e.g., hearing loss, tinnitus) suggests a peripheral cause of the vertigo.
- Acute vestibular syndrome is diagnosed when dizziness develops acutely; is constant; is accompanied by nausea or vomiting, unsteady gait, nystagmus, and intolerance to head motion; and persists for longer than a day.
- Neck injury resulting in a vertebral artery dissection causes posterior circulation ischemia presenting with vertigo.
- Nystagmus is the cardinal sign of inner ear disease and the principal objective evidence of vestibular dysfunction.
- *HINTS* (Head Impulse test, Nystagmus, Test of Skew) is a bedside oculomotor examination test that helps to differentiate central from peripheral vertigo in symptomatic patients with a first ever onset of constant vertigo from acute vestibular syndrome.
- Meclizine (Antivert) has a time of onset of approximately 1 hour.
- Benzodiazepines are not recommended for the treatment of vestibular neuritis or labyrinthitis in patients who are discharged home, because they can interfere with the process of vestibular rehabilitation.

1. A patient presents to the emergency department with dizziness and is diagnosed with horizontal canal benign paroxysmal positional vertigo (BPPV). What maneuver can be used to treat the patient?

a. Barbecue roll
b. Epley maneuver
c. Hallpike test
d. Head impulse test

Answer: a. The supine roll test, in which the patient lies flat on the gurney and the head is turned to each side, is used to diagnose horizontal canal BPPV, whereas the barbecue roll maneuver is used to treat the horizontal variant of BPPV. The Epley maneuver is used to treat posterior canal BPPV. The Hallpike test is used to diagnose posterior canal BPPV. The head impulse test is used to diagnose vestibular neuritis and labyrinthitis.

2. A 70-year-old woman presents with a first ever episode of severe vertigo. Which of the following examination findings should prompt the physician to order imaging tests and/or consultation with a neurologist?

a. Direction changing nystagmus on change in head position
b. Direction changing nystagmus on change in lateral gaze
c. Positive head impulse test
d. Torsional upbeat nystagmus during Hallpike test

Answer: b. Direction changing nystagmus on change in gaze is concerning for a central cause of vertigo and makes up part of the HINTS test. Direction changing nystagmus with changes in head position is expected in a patient with horizontal canal benign paroxysmal positional vertigo (BPPV) and is a benign finding. A positive head impulse test is expected in vestibular neuritis, while torsional upbeat nystagmus is expected in posterior canal BPPV.

3. Internuclear ophthalmoplegia most often suggests a diagnosis of which of the following conditions?

a. Horizontal canal benign paroxysmal positional vertigo (BPPV)
b. Labyrinthitis
c. Multiple sclerosis
d. Vestibular neuritis

Answer: c. Internuclear ophthalmoplegia is diagnosed when, on eye movement, the adducting eye shows little to no movement while the abducting eye moves normally. In a patient presenting with vertigo, this finding is virtually pathognomonic for multiple sclerosis.

4. Which of the following is a central cause of vertigo?

a. Labyrinthitis
b. Ménière disease
c. Vertebrobasilar insufficiency
d. Vestibular neuritis

Answer: c. All the other causes are peripheral.

5. Continuous vertigo of what duration is used to define acute vestibular syndrome?

a. 1 hour
b. 8 hours
c. 24 hours
d. 1 week

Answer**: c.** Acute vestibular syndrome has an arbitrary cutoff of continuous vertigo for at least 1 day in part to help differentiate acute vestibular syndrome from attacks of Ménière disease or prolonged migrainous vertigo.

KEY CONCEPTS

- Patients with a known headache disorder who present with a change in the character of the headache should be evaluated for potential serious causes.
- The physical examination in the headache patient focuses primarily on mental status, funduscopic exam, meningeal signs, and neurologic examination, particularly cranial nerves (CNVs) II, III, IV, and VI, as well as specific cerebral territories.
- Mild to moderate primary and nonspecific headaches are treated with nonsteroidal analgesic drugs (NSAIDs). Severe headaches can be managed parenterally with dopamine antagonists (metoclopramide or prochlorperazine), migraine-specific agents (triptans), or NSAIDS (Ketorolac). Opioid pain medications are not indicated for the treatment of primary headaches.
- Most patients with headache do not require neuroimaging. When obtained, neuroimaging is tailored to the specific elements of concern in the differential diagnosis.
- The differential diagnosis of sudden severe headache includes subarachnoid hemorrhage (SAH) or other intracranial hemorrhage (ICH), cerebral venous thrombosis, and cervical artery dissection (Box 16.1).
- In patients suspected for SAH, a stepwise application of Ottawa SAH Rule, computed tomography (CT) imaging, followed by lumbar puncture (LP) and/or CT angiography (CTA) is appropriate.
- Antibiotics should be administered prior to performing a LP when bacterial meningitis is suspected.

1. Of the following, which is most appropriate in the initial evaluation of a patient with nontraumatic headache?
   a. Computed tomography (CT) scan of brain
   b. Electroencephalogram (EEG)
   c. Magnetic resonance imaging scan of brain
   d. Thorough neurological evaluation

Answer: d. A thorough neurological examination may reveal deficits not seen on gross evaluation, prompting expansion of the differential diagnosis to include more concerning etiologies. Depending on the history and remainder of the physical, a normal neurological examination may be reassuring and obviate need for advanced imaging studies.

2. In the setting of headache, the presence of nausea and vomiting are diagnostic of which of the following underlying causes?
   a. Glaucoma
   b. Increased intracranial pressure
   c. Migraine
   d. None of the above

Answer: d. Nausea and vomiting are completely nonspecific. Migraine headaches, increased intracranial pressure, temporal arteritis, and glaucoma can all be manifested by severe nausea and vomiting, as can some systemic viral infections with headache. Such factors may point toward the intensity of the discomfort but are not specific in establishing the diagnosis.

3. Which of the following is the most appropriate initial diagnostic study for a pregnant woman with normal vital signs and a presentation concerning for a headache emergency?
   a. Computed tomography (CT)
   b. Inflammatory markers (e.g. erythrocyte sedimentation rate [ESR], C-reactive protein [CRP])
   c. Magnetic resonance angiography (MRA)/magnetic resonance venogram (MRV)
   d. No studies are indicated

Answer: a. CT studies with and without contrast are able to identify many vascular causes of headache, for which pregnancy is a risk factor. Iodinated contrast is safe in pregnancy. Magnetic resonance imaging (MRI) without contrast is another option; however, MRA and MRV are relatively contraindicated due to use of gadolinium contrast, which has not been shown to be safe in pregnancy. Inflammatory markers are nonspecific and will not confirm or rule out important emergency headaches in this population. For emergency headaches in pregnancy deferral of additional studies is not appropriate.

4. A 42-year-old male patient presents with the sudden onset of severe headache that worsened over the 30 minutes while at rest 4 hours prior to arriving in the emergency department (ED). His examination reveals only mild distress, no neck stiffness or meningismus, and normal funduscopic and neurological examinations. What is the most appropriate diagnostic test?
   a. Intraocular pressure determination
   b. Sphenopalatine block
   c. Computed tomography
   d. Lumbar puncture

Answer: c. The patient has a sudden onset severe headache and Ottawa Subarachnoid Hemorrhage Rule can be applied. However, his age being greater than 40, the rule is unable to exclude subarachnoid hemorrhage (SAH). Next step is computed tomography (CT) because he is within the 6-hour window. Lumbar puncture and or/ CT angiography can be considered if initial noncontrast CT is normal.

17 Diplopia

KEY CONCEPTS

- The diagnostic approach to diplopia is aimed at determining (1) if the diplopia is monocular, (2) if there is a restrictive or mechanical issue in the orbit or orbital structures, (3) if there is a palsy of one or more of the oculomotor cranial nerves, (4) if there is a neuroaxial process involving the brainstem and related cranial nerves, or (5) if there is a systemic neuromuscular process.
- Monocular diplopia is most commonly caused by an ophthalmologic problem related to refractory distortions or retinal buckling.
- Binocular diplopia resolves when either eye is closed and is the result of a misalignment in the visual axes.
- Important historical features in assessing diplopia includes onset, directionality and orientation of the diplopia; presence of pain, and the presence of other associated symptoms.
- An isolated CN III palsy presents with diplopia in all directions of gaze, except on lateral gaze to the affected side; the eye is deviated down and out; ptosis and a dilated pupil may also be present.
- A CN IV palsy results in vertical or torsional diplopia that worsens on looking down and toward the nose.
- A CN VI palsy will cause the affected eye to turn inward and results in diplopia that worsens on lateral gaze toward the affected side.
- Cavernous sinus or posterior orbit disease (orbital apex syndrome) may present with ipsilateral diplopia involving CN III, IV and or VI; associated symptoms may include pain, exophthalmos, chemosis, and possibly ipsilateral periorbital facial numbness or dysesthesia.
- Multiple sclerosis or localized pathology in the brainstem, such as tumor, lacunar infarct, basilar artery thrombosis, vertebral artery dissection or ophthalmoplegic migraine, may present with focal lesions presenting as diplopia.
- The diplopia caused by myasthenia gravis is associated with ptosis, gets worse as the patient fatigues, and improves with rest or on placing ice over the eye.
- Initial treatment for diplopia is directed toward imminent threats to airway and ventilation (e.g., with botulism and myasthenia gravis) and immediate or rapidly evolving threats to the central nervous system (CNS) (e.g., stroke, meningoencephalitis or Wernicke encephalopathy).

1. A 65-year-old man with a longstanding history of diabetes and hypertension presents with sudden onset of persistent diplopia that began a few hours before arrival. He describes left retro-orbital discomfort, and his examination is notable for a left eye that is deviated laterally and downward, with a palsy of movement medially and upward. He also has a left-sided ptosis but no conjunctival injection, chemosis, or proptosis. His pupils are equal in size at 4 mm, round, and equally reactive to light in both a direct and consensual reflex, and his examination is otherwise unremarkable. What is the most likely cause of the diplopia?

a. Brain tumor
b. Cerebral aneurysm
c. Microvascular ischemia
d. Orbital apex syndrome

Answer: c. Based on examination, this is a patient who has a pupil-sparing complete CN III (third nerve) palsy. Because his pupillary examination is normal, with an otherwise complete CN III palsy, the palsy is very unlikely to be due to external compression from a brain tumor, aneurysm, or orbital apex process. It is a typical presentation of microvascular ischemia, to which the patient is predisposed, given his history of diabetes and hypertension.

2. A 56-year-old woman presents with recurrent episodes of diplopia that have been ongoing for a week. She describes double vision that gradually comes and goes, typically worse at the end of the day, with no particular direction or orientation to the diplopia. The patient's coworker, who is present in the emergency department (ED) with her, states that the patient's eyes "looked droopy" during an animated staff meeting they attended that afternoon but look normal now. The patient also describes waxing and waning general muscular weakness that has also been present this past week but denies any other symptoms and states that when she rests, she feels better. With which entity are her symptoms most consistent?

a. Botulism
b. Hypothyroidism
c. Miller-Fisher syndrome
d. Myasthenia gravis

Answer: d. The patient and coworker are describing what appears to be an activity-related diplopia, with generalized muscle weakness and lack of other focal symptoms, all very suggestive of a possible neuromuscular process (myasthenia gravis). Miller-Fisher syndrome would not be associated with muscle weakness and would not wax and wane. Botulism would typically have a more progressive course, with other associated bulbar symptoms. Diplopia may be associated with hypothyroidism if it is a presentation of or treatment complication of Graves disease but would not change so markedly with activity.

3. A 76-year-old man with hypertension, hypercholesterolemia, and diet-controlled diabetes presents with a sudden onset of diplopia that developed 30 minutes before arrival. Paramedics state that the patient's wife reported that he suddenly began staggering around the room, unable to bear weight on his left side. On examination, the patient has normal vital signs except for mild hypertension and has a right CN III palsy, with left arm and leg weakness. He has no airway complaints and denies any pain. What is the most appropriate initial response?

a. Checking blood gas levels and assessing the patient's negative inspiratory force
b. Emergent treatment with botulinum antitoxin
c. Initiating broad-spectrum antibiotics to cover upper respiratory pathogens
d. Initiating clinical measures to address an acute ischemic stroke

Answer: d. The paroxysmal onset of the patient's symptoms, with focal neurologic symptoms and signs, suggests an ischemic event. His crossed deficits and discrete CN III palsy suggest localization in the brainstem.

4. Which constellation of symptoms is most concerning for foodborne botulism?

a. Double vision, headache, and right leg weakness
b. Double vision, left eye discomfort, and periorbital swelling
c. Double vision, nystagmus, and confusion
d. Double vision, slurred speech, difficulty swallowing, and dry mouth

Answer: d. Double vision, slurred speech, difficulty swallowing, and dry mouth would be present with foodborne botulism.

5. A 45-year-old woman presents with progressively worsening double vision and the inability to adduct the eye on one side in the contralateral direction during lateral gaze that resolves during convergence on physical examination. What disease should be suspected with these findings?

a. Multiple sclerosis
b. Botulism
c. Myasthenia gravis
d. Idiopathic intracranial hypertension

Answer: a. An internuclear ophthalmoplegia (INO), suggested by an inability to adduct the eye on one side in the contralateral direction during lateral gaze that resolves during convergence, implicates a lesion in the medial longitudinal fasciculus (MLF) and is typically found in patients with multiple sclerosis.

KEY CONCEPTS

- Critical diagnoses, such as caustic injury, orbital compartment syndrome, and narrow angle (acute angle closure) glaucoma, require immediate treatment and ophthalmology consultation.
- Prompt and prolonged irrigation is essential for patients who experience caustic injury to the eye.
- Headache and nausea may be prominent symptoms in patients with narrow angle glaucoma.
- Complete abolition of a foreign body sensation after instillation of local anesthetic solution indicates a high likelihood of a superficial corneal lesion.
- Keratitis, inflammation of the cornea, is most commonly caused by a viral infection, but may also be caused by recent ultraviolet light exposure, chemical injury, or hypoxic injury from contact lens use.
- A localized corneal defect with edematous, inflammatory changes may signal corneal ulceration.
- A corneal dendritic pattern may signal a herpetic infection, which can progress to corneal opacification and visual loss.
- Pain, consensual photophobia, perilimbic conjunctival injection, and a miotic pupil could signal iritis or uveitis. The cause may be trauma or an underlying autoimmune disease. The presence of cell and flare in the anterior chamber can help identify these conditions.
- Conjunctivitis is usually self-limited and rarely requires antibiotic treatment.

1. A patient who normally wears contact lenses is diagnosed with bacterial conjunctivitis. Which of the following is the preferred treatment in this patient?
   a. Bacitracin
   b. Chloramphenicol
   c. Erythromycin
   d. Moxifloxacin

Answer: d. Patients who wear contact lenses are at increased risk for infections with *Pseudomonas* and, in the setting of bacterial conjunctivitis, should be prescribed a quinolone, barring any contraindications.

2. A 15-year-old boy presents to the emergency department (ED) after having been shot in the face with a BB gun. He has a solitary penetrating wound just inferior to his left eye. His visual acuity in the left eye is limited to light perception, but he reports having normal vision prior to the injury. He has significant proptosis of his left eye, and his fundus is clearly seen with direct ophthalmoscopy. Intraocular pressure (IOP) of the affected eye is 50 mm Hg. His mental status is normal. What is the most appropriate next step in the management of this patient?
   a. CT scan of the head and face
   b. ED observation with repeated neurologic examinations
   c. Lateral canthotomy and inferior cantholysis
   d. Plain radiography of the face

Answer: c. The described patient likely has a retrobulbar hematoma with visual acuity changes and an elevated IOP. The elevated IOP with a clear fundoscopy are findings consistent with no penetration into the globe. Although CT scan of the head and face is indicated to further delineate specific injuries, lateral canthotomy and inferior cantholysis is emergently necessary for orbital decompression in an attempt to salvage vision. This sight-saving procedure should not be delayed more than 90 minutes after injury when severe findings (decreased visual acuity and significantly increased IOP) are present. Ophthalmology consultation would be indicated emergently.

3. A collection of pus in the anterior chamber of the eye is known as which of the following?
   a. Cotton wool spot
   b. Dacryocystitis
   c. Hyphema
   d. Hypopyon

Answer: c. A collection of layered pus in the dependent portion of the anterior chamber is called a hypopyon.

4. Which of the following results from inflammation of a meibomian gland?
   a. Blepharitis
   b. Chalazion
   c. Dacryocystitis
   d. Hordeolum

Answer: b. Inflammation of a meibomian gland with the subsequent formation of a subcutaneous nodule within the eyelid is known as a chalazion. This condition typically resolves spontaneously over several days. Authorities often recommend warm compress application and gentle massage of the nodule several times a day, although there is no evidence supporting this. If complete resolution does not occur within 2 weeks, the patient should be referred to an ophthalmologist.

5. A neonate presents 7 days after birth with a unilateral red eye. Further examination reveals inflamed conjunctiva, purulent discharge and negative uptake on fluorescein examination. The child was born full term after a pregnancy complicated by poor prenatal care. Which of the following diagnoses is most likely in this patient?
   a. Chemical conjunctivitis
   b. *Chlamydia trachomatis*
   c. Corneal abrasion
   d. *Neisseria gonorrhoeae*

Answer: b. Neonates are at high risk for serious ophthalmologic diseases. Typically, they present with chemical conjunctivitis from antibacterial prophylaxis (i.e., erythromycin ointment) at birth within 2 days. Gonococcal conjunctivitis is seen 2 to 5 days after birth. Chlamydial conjunctivitis is seen 5 to 12 days after birth. Corneal abrasions, while common in this age group, will most likely be evident on fluorescein examination of the cornea.

KEY CONCEPTS

- Sore throat can indicate a range of pathology, from simple pharyngitis to deep space infection, with or without airway compromise.
- Physical examination, supplemented by nasopharyngoscopy when appropriate, is key to determining threats to the airway and establishing a diagnosis.
- If oropharyngeal examination reveals minimal findings, lower or deeper structures may be involved, and endoscopic examination of the upper airway is recommended.
- The modified Centor criteria can help identify adult patients with a higher likelihood of Group A Streptococcus (GAS) infection. Patients with a score of –1 to 1 are unlikely to have GAS infection (10% to 15% likelihood, 95% confidence interval 5% to 25%), while a score of 4 or 5 corresponds to a 40% to 50% likelihood of GAS infection (95% confidence interval 31% to 51%).
- Viral pharyngitis, the most common cause of pharyngitis, is self-limiting and antibiotics are more harmful than helpful in its management.
- For GAS-positive pharyngitis, a single injection of penicillin or a 10-day course of oral penicillin may decrease the duration of symptoms, transmission to close contacts, and prevent rare suppurative and nonsuppurative complications.

1. A 20-year-old female presents to the emergency department complaining of a sore throat. She has a temperature of 39°C, tender cervical lymphadenopathy, exudates on exam and denies any history of cough, rhinorrhea, nausea, vomiting or diarrhea. Her modified Centor score is which of the following?

a. 2
b. 3
c. 4
d. 5

Answer: c. This patient meets four Centor criteria: fever, tender cervical lymphadenopathy, exudates, and absence of a cough. The age criteria is +1 point for age 3 to 14 years, and −1 point for age greater than 45 years.

2. A 39-year-old male presents to the emergency department complaining of a sore throat. He is sitting forward on the stretcher, drooling, with inspiratory stridor. His oropharynx exam shows minimal erythema and no exudate. Which of the following conditions is the likely diagnosis?

a. Epiglottitis
b. Group A streptococcal pharyngitis
c. Periodic fever, aphthous stomatitis, pharyngitis, and cervical adenitis (PFAPA) syndrome
d. Viral pharyngitis

Answer: a. Simple pharyngitis, either bacterial or viral, and PFAPA syndrome do not cause airway compromise. Drooling and inspiratory stridor are symptoms of airway compromise, and epiglottitis is the only diagnosis on this list that results in airway compromise.

3. In adult patients who test positive for Group A streptococcal (GAS) pharyngitis who have no allergies, which of the following is the most appropriate antibiotic choice?

a. Azithromycin 12 mg/kg/day (max dose, 500 mg) × 5 days
b. Clindamycin 7 mg/kg/dose (max dose, 300 mg) TID × 10 days
c. Metronidazole 30 mg/kg/day divided TID × 10 days
d. Penicillin VK 50 mg/kg/day divided QID × 10 days

Answer: d. Penicillin VK is the first-line treatment for GAS pharyngitis in patients without allergies. Clindamycin and azithromycin are options in penicillin allergic patients. Metronidazole is not a treatment of GAS pharyngitis.

4. A 24-year-old female presents for sore throat, difficulty swallowing, and a muffled voice. On examination, she has fullness, swelling and induration of the left posterior peritonsillar area with uvular deviation to the right. What is the next step in the management of this patient?

a. Computed tomography imaging
b. Discharge with antibiotics, clindamycin 300 mg TID × 10 days
c. Drainage either by needle aspiration or incision and drainage
d. Rapid Antigen Detection Test for Group A Streptococcus

Answer: c. This patient has a peritonsillar abscess. In a clear-cut case such as this, no imaging is needed. Definitive treatment is with either needle aspiration or incision and drainage.

5. In which age group is Group A streptococcal infection most common?

a. Age <3 years
b. Age 5–15 years
c. Age 15–45 years
d. Age >45 years

Answer: b. GAS pharyngitis is most common in children aged 5 to 15 years. It is less common in young adults, and rare in children less than three, or adults over the age of 45.

Hemoptysis 20

KEY CONCEPTS

- Hemoptysis is caused by infection, trauma, cancer, or coagulopathy or as a complication of invasive pulmonary procedures.
- "Massive hemoptysis" is defined as greater than 100 mL of blood loss or approximately ½ cup of blood in a 24-hour period or a bleeding rate ≥100 mL/h.
- Plain radiographs are the initial screening test in most cases of massive hemoptysis, although high-resolution chest computed tomography scans are more sensitive and can supplant plain chest x-rays as the initial diagnostic test.
- Bronchial artery embolization is highly effective, with hemostasis rates ranging from 85% to 95%.
- With massive hemoptysis, hypoxia is the more immediate concern rather than volume resuscitation, and early intubation to ensure adequate oxygenation is paramount. Most causes of mortality are a result of asphyxiation.
- If a tracheo-innominate artery fistula (TIF) is suspected, then overinflation of the tracheostomy balloon or digital pressure at the site of bleeding should be performed for immediate hemorrhage control.

1. What is the most common cause of trace hemoptysis (blood-tinged sputum)?
   - **a.** Bronchiectasis
   - **b.** Bronchitis
   - **c.** Cancer
   - **d.** Congestive heart failure
   - **e.** Pulmonary embolism

Answer: b. The most common cause of small-volume hemoptysis is bronchitis.

2. Disruption of which of the following vessels is responsible for the vast majority of cases of massive hemoptysis?
   - **a.** Aorta
   - **b.** Bronchial arteries
   - **c.** Pulmonary arteries
   - **d.** Pulmonary veins
   - **e.** Tracheobronchial capillaries

Answer: b. Massive hemoptysis almost exclusively involves one of the two sets of vessels that constitute the lung's dual blood supply. Bronchial arteries, direct branches from the thoracic aorta, are responsible for supplying oxygenated blood to the lung parenchyma. Disruption of these vessels can result in sudden and profound hemorrhage. Although small in caliber, the bronchial circulation is a high-pressure system and the cause in nearly 90% of the cases of massive hemoptysis requiring embolization. Although they transmit large volumes of blood, pulmonary arteries are at much lower pressure and, unless affected at a very central location, are less likely to cause massive hemoptysis. Trace hemoptysis typically originates from tracheobronchial capillaries that become disrupted with vigorous coughing or minor bronchial infections.

3. Which of the following statements regarding the evaluation of hemoptysis is true?
   - **a.** Chest x-ray localizes the site of bleeding in nearly 90% of cases of patients with large volume hemoptysis.
   - **b.** Conventional angiography is the preferred diagnostic test to detect both bronchial and nonbronchial arterial causes of massive hemoptysis.
   - **c.** High-resolution CT, even with recent advances in technology, remains diagnostically inferior to angiography.
   - **d.** Patients with fever, cough, and blood-tinged sputum should receive a chest x- ray even if the vital signs are normal.

Answer: b. In patients with massive hemoptysis, plain films may localize the site of hemorrhage in approximately 50% of patients. However, high-resolution multidetector CT of the chest is the principal diagnostic test to detect both bronchial and nonbronchial arterial causes of massive hemoptysis. CT is diagnostically comparable with, but less invasive than, conventional angiography, which currently is done as a combined diagnostic/therapeutic modality. A chest CT scan should be obtained in high-risk patients (smokers and oncology patients) or in any patient with moderate-to-severe bleeding even if the initial chest radiograph is normal. CT localization of hemorrhage can expedite bronchoscopic evaluation or guide subsequent interventional procedures. Patients with "garden variety" bronchitis, normal oxygenation, and a normal exam do not require imaging in the emergency department.

4. A 50-year-old man presents after an episode of hemoptysis. He describes coughing up several large clots of dark blood. During his evaluation, he coughs and expectorates approximately 5 mL of clotted blood. The patient's vital signs are normal, and no abnormalities are noted on physical examination. His chest radiograph is normal. Which of the following is the most appropriate next step in the management of this patient?
   - **a.** Admission to an observation unit
   - **b.** Consultation for bronchoscopy
   - **c.** Consultation for percutaneous embolization
   - **d.** Discharge home with follow-up in 24 hours
   - **e.** Obtain chest high-resolution computed tomography (HRCT) scan

Answer: e. Since the advent of HRCT, radiologic evaluation has had an integral role in the evaluation and management of patients with hemoptysis. Unless the initial chest radiograph is diagnostic or the patient is hemodynamically unstable, an HRCT scan should be obtained in most cases. Further management strategy should occur in conjunction with pulmonary and thoracic surgery consultants, guided by the HRCT results.

5. A 58-year-old man with a single lung transplant presents to the emergency department (ED) with what appears to be large-volume hemoptysis. He was just discharged from the endoscopy suite, where he had a number of surveillance biopsies performed. He looks pale and diaphoretic with an initial oxygen saturation of 71%. After placement of an intravenous line and supplemental oxygen, the next most appropriate step is:
   - **a.** Blood transfusion
   - **b.** High-resolution computed tomography scan of the chest
   - **c.** Intubation
   - **d.** Thoracic surgery consultation

Answer: c. This patient is profoundly hypoxic and will need imaging outside of the ED and invasive procedures. The primary cause of mortality in patients with hemoptysis is asphyxiation. All resuscitative and procedural efforts will be futile without intubation and adequate oxygenation.

21 Dyspnea

KEY CONCEPTS

- Dyspnea results from a variety of conditions, ranging from nonurgent to life-threatening. Neither the clinical severity nor the patient's perception correlates well with the seriousness of underlying pathology.
- Dyspnea is subjective and the differential diagnosis can be divided into acute, acute on chronic, and chronic causes, of which the majority are cardiopulmonary. Other causes include metabolic, infectious, neuromuscular, traumatic, psychiatric, and hematologic conditions.
- Chronic or progressive dyspnea usually denotes primary cardiac or pulmonary disease. Acute dyspneic spells may result from asthma exacerbation, infection, pulmonary embolism, cardiac dysfunction, or psychogenic causes.
- All patients experiencing dyspnea, regardless of etiology, should be promptly evaluated. Bedside pulse oximetry readings should be obtained, and the patient placed on a cardiac monitor.
- If the pulse oximetry is less than 92% on room air, the patient should be placed on supplemental oxygen either by nasal cannula or mask, depending on the degree of desaturation.
- If altered level of consciousness, or hypoxia cannot be improved with suctioning, supplemental oxygen, or airway adjuncts then breathing should be assisted with manual or mechanical ventilation, either noninvasively for the short term, or with intubation for prolonged ventilation.
- Unstable patients or patients with critical diagnoses must be stabilized and may require an emergent procedure and admission to an intensive care unit. Urgent patients who improve with emergency department management may be admitted to an intermediate care unit. Patients diagnosed with urgent conditions in danger of deterioration without proper treatment or patients with severe comorbidities, such as diabetes, immunosuppression, or cancer, may also require admission for observation and treatment.

1. Orthopnea is best defined by which of the following?
   a. A respiratory rate greater than normal
   b. A tidal volume that exceeds metabolic demands
   c. Decreased end-tidal carbon dioxide levels
   d. Dyspnea in a recumbent position

Answer: d. Dyspnea in a recumbent position, commonly described as the number of pillows the patient uses to rest comfortably in bed (e.g., two-pillow orthopnea).

2. Stridor is most likely due to which of the following conditions?
   a. Bronchospasm
   b. Guillain-Barré syndrome
   c. Laryngeal edema
   d. Malignancy

Answer: c. Stridor is an upper airway noise caused by airway narrowing. Of the given options, stridor is most likely due to laryngeal edema.

3. A 34-year-old male was struck repeatedly with a pipe in the right chest. He becomes acutely more dyspneic during emergency medical services (EMS) transport and becomes clammy, hypotensive, and more tachycardic on arrival to the emergency department (ED). Examination reveals tachypnea, right chest crepitus, and decreased breath sounds. Which of the following actions should be done next?
   a. Needle chest decompression of the right chest
   b. Perform portable chest radiograph
   c. Provide supplemental oxygen by non-rebreather mask
   d. Rapid sequence intubation (RSI) and endotracheal intubation

Answer: a. Needle chest decompression is indicated for management of a likely tension pneumothorax. If ultrasound is immediately available, it can be used to confirm pneumothorax, but in this patient, who is in cardiovascular collapse, immediate intervention is necessary.

4. Which of the following causes of acute respiratory distress is most likely to be associated with decreased respiratory effort?
   a. Pleural effusion
   b. Tension pneumothorax
   c. Cyanide poisoning
   d. Multiple sclerosis

Answer: d. multiple sclerosis. Several different neuromuscular diseases are associated with decreased respiratory effort in acute distress due largely to inadequate ability to create sufficient negative inspiratory force and generate adequate tidal volume and minute ventilation.

5. When assessing a patient with acute respiratory distress, a shark fin waveform on capnography suggests which of the following conditions?
   a. Cardiac ischemia
   b. Pneumothorax
   c. Pulmonary embolism
   d. Pneumonia
   e. Bronchospasm

Answer: e. Bronchospasm. The classic "shark fin" indicates bronchospasm such as with significant asthma exacerbation. See Fig. 21.1 for assessment paradigm considerations.

22 Chest Pain

KEY CONCEPTS

- Tension pneumothorax is a clinical diagnosis, treated with needle decompression, followed by tube thoracostomy.
- Patients with suspected acute coronary syndrome (ACS) are risk-stratified by history, electrocardiogram (ECG), and troponin levels. Those with ST segment elevation myocardial infarction (STEMI) undergo revascularization via fibrinolysis or percutaneous coronary intervention (PCI). Those with non–ST segment elevation myocardial infarction (NSTEMI) do not require immediate PCI. Those with a nondiagnostic ECG and troponin level are managed with observation.
- Thoracic dissection is diagnosed with CT angiography. The initial management of dissection is with urgent control of heart rate, followed by lowering of blood pressure. Further management, medical or surgical, depends on the location of the dissection.
- Pulmonary embolism is diagnosed using a combination of history, serum D-dimer measurement, and imaging, usually CT angiography. Patients with a low pretest probability and normal D-dimer level do not have pulmonary embolism as the cause of their chest pain presentation.
- Patients with pericardial effusions undergo echocardiography to evaluate for cardiac tamponade. Those with signs of shock may require emergent pericardiocentesis.

1. A 25-year-old man presents with the sudden onset of unilateral chest pain, followed almost immediately by respiratory distress. He is noted to have a blood pressure of 75/45 mm Hg, pulse of 130 beats/min, and decreased breath sounds on the right side of his chest. What is the most appropriate initial step in the management of this patient?

a. Administer intravenous (IV) antibiotics.
b. Infuse a 2-L bolus of normal saline.
c. Obtain a chest radiograph.
d. Perform a tube thoracostomy.

Answer: d. Tension pneumothorax is a critical diagnosis that must be made and remedied, if present, in the first few moments of the rapid stabilization and assessment phase of any patient encounter. If a patient presents with chest pain, respiratory distress, shock, and unilateral reduction or absence of breath sounds, immediate intervention with needle or tube thoracostomy is required.

2. A 65-year-old man with a past medical history of prostate cancer presents with chest pain. His blood pressure is 60/40 mm Hg, and his pulse is 145 beats/min. The ECG shows diffuse ST segment elevation, and cardiomegaly is seen on his chest radiograph. What is the most appropriate first step in the management of this patient?

a. Administration of dobutamine
b. Thrombolysis
c. Cardiac catheterization
d. Cardiac ultrasonography

Answer: d. Prompt bedside cardiac ultrasonography would be the most appropriate next step in the management of this patient, who presents with symptoms and signs of pericardial effusion with tamponade. If confirmed by ultrasonography, immediate pericardiocentesis would logically follow in an effort to reverse the signs of shock in this patient. Other signs that could accompany the presentation are low voltage on the ECG and elevated jugular venous pressure on examination.

3. A "tearing" sensation is classically described for which of the following causes of chest pain?

a. Aortic dissection
b. Esophageal rupture
c. Mallory-Weiss tear
d. Pneumothorax

Answer: a. "Tearing" pain that may migrate from the front to back or back to front is classically described in aortic dissection. Descriptions such as "squeezing," "crushing," or "pressure" lead to the suspicion of a cardiac ischemic syndrome, although cardiac ischemia can be characterized by nonspecific discomfort, such as "bloating" or "indigestion." "Sharp" or "stabbing" pain is seen more in pulmonary and musculoskeletal diagnoses. Patients complaining of a burning- or indigestion-type of pain may initially be suspected of having a gastrointestinal cause; however, because of the visceral nature of chest pain, all causes of pain may present with any of the preceding descriptions.

4. Uremia is a risk factor associated with which of the following causes of chest pain?

a. Acute coronary syndrome
b. Aortic dissection
c. Pericarditis
d. Pneumothorax
e. Pulmonary embolism

Answer: c. Uremia is a risk factor for the development of pericarditis. Other risk factors associated with the development of pericarditis include infection, autoimmune disease, acute rheumatic fever, recent myocardial infarction or cardiac surgery, malignancy, radiation therapy to the mediastinum, and prior pericarditis.

5. A narrow pulse pressure is more closely associated with which of the following diagnoses?

a. Acute myocardial infarction
b. Aortic dissection
c. Pericarditis with effusion
d. Pulmonary embolism

Answer: c. A narrow pulse pressure is a pivotal finding in the diagnosis of pericarditis with associated pericardial effusion. Other characteristic, but less specific, potential findings in the patient with pericarditis include hypotension, tachycardia, fever, and jugular venous distention (JVD). The more specific finding of pericardial rub is also heard in some patients with pericarditis.

6. The finding of Hamman's sign is most consistent with which of the following?

a. Cholecystitis
b. Mediastinitis
c. Pericarditis
d. Pulmonary embolus

Answer: b. Hamman's sign is an audible systolic "crunch" heard on cardiac auscultation that is produced by the heart moving against air in the mediastinum. This can be heard in conditions such as esophageal rupture, mediastinitis, and pneumomediastinum.

7. Right ventricular strain on the ECG of a patient complaining of chest pain would be most consistent with which of the following diagnoses?

a. Acute myocardial infarction
b. Coronary ischemia
c. Coronary spasm
d. Pulmonary embolus

Answer: d. In the setting of chest pain, right ventricular strain as evidenced on the ECG is highly suspicious for pulmonary embolus.

8. The ECG finding of PR segment depression would be more commonly found in which of the following causes of chest pain?

a. Pericarditis
b. Pulmonary embolus
c. ST segment elevation myocardial infarction (STEMI)
d. Subendocardial infarction
e. Unstable angina

Answer: a. The electrocardiographic findings most commonly associated with pericarditis are diffuse ST segment elevation and PR segment depression. The ECG in patients with unstable angina is most often normal or nonspecific. T wave inversion may be seen in these patients. The characteristic ECG findings with subendocardial infarction are T wave inversion and/or ST segment depression in concordant leads. Classic STEMI is manifested electrocardiographically by ST segment elevation (>1 mm) in contiguous leads, a new left bundle branch block (LBBB), or Q waves 0.04 second or more in duration. Many possible electrocardiographic findings are associated with pulmonary embolus, usually manifestations of right ventricular strain.

KEY CONCEPTS

- While the etiology of abdominal pain is frequently benign, requiring little work-up or intervention, abdominal pain can also be the presenting symptom of catastrophic illness, requiring life-saving interventions to be implemented within minutes to hours (see Table 23.1).
- Patients who are pregnant, of advanced age, or immunocompromised are at higher risk for acute pathology and require expeditious evaluation, which generally includes imaging studies.
- Patients with prior surgery, especially bariatric surgery, have unique anatomic features that put them at risk for pathology not seen in patients that have never had abdominal surgery.
- Patients with hemodynamic instability require resuscitation in conjunction with diagnostic evaluation, and often surgical (or gynecologic) consultation. Bedside ultrasonography can be helpful in these patients.
- Pain medication while evaluating stable patients is appropriate and generally should not be withheld. We recommend the administration of IV morphine at a dose of 0.05 to 0.10 mg/kg (usually 2 to 5 mg in adults) every 15 to 20 minutes until pain is controlled (pain score <4).
- Intra-abdominal infections are often polymicrobial and require broad spectrum antibiotic coverage with regimens such as: piperacillin/tazobactam 3.375 g IV, ciprofloxacin 400 mg IV + metronidazole 500 mg IV, or cefepime 2 g IV + metronidazole 500 mg IV.
- Stable patients with unrevealing evaluations whose symptoms improve may be considered for disposition to home with close follow-up. High-risk patients without a clear diagnosis may require in-hospital observation.

1. A 65-year-old man presents with acute abdominal pain. He has a history of hypertension, diabetes, and an abdominal aortic aneurysm. On examination, he is hypotensive and tachycardic with a distended, rigid abdomen. Which of the following is the best method for diagnosis?
- **a.** Computed tomography
- **b.** Exploratory laparotomy
- **c.** Plain radiograph
- **d.** Right upper quadrant ultrasound

Answer: b. Exploratory laparotomy. Patients with known abdominal aortic aneurysm (AAA) and unstable vital signs must be assumed to have ruptured AAA and should be brought to the operating room without delay. If the patient were stable, CT would be indicated. Plain radiograph is not indicated in this patient, nor would right upper quadrant ultrasound be warranted.

2. A 25-year-old woman presents with abdominal pain. She has stable vitals. On pelvic examination, she has tenderness in the right adnexal region. Which of the following is the most appropriate initial imaging study?
- **a.** Computed tomography
- **b.** Magnetic resonance imaging
- **c.** Plain radiograph
- **d.** Pelvic ultrasound

Answer: d. Pelvic ultrasound. Women of child-bearing age with lower abdominal pain and adnexal tenderness on examination are best imaged with pelvic ultrasound. CT may be warranted if the ultrasound is inconclusive or suggestive of, but not diagnostic of, appendicitis. Plain radiographs are not helpful in undifferentiated lower abdominal pain. MRI could be a consideration if the patient were pregnant and there was concern for appendicitis.

3. A 70-year-old man presents with abdominal pain associated with one loose bowel movement and an episode of non-bloody emesis. He has a history of hypertension, diabetes, cigarette smoking, and atrial fibrillation. He rates his pain as "10/10." On examination, he has stable vitals. His abdomen is soft and nontender without signs of peritonitis. Which of the following is the most appropriate imaging study to confirm the suspected diagnosis?
- **a.** Computed tomography angiography
- **b.** Magnetic resonance imaging
- **c.** Plain radiograph
- **d.** Right upper quadrant ultrasound

Answer: a. Computed tomography angiography. The patient presents with "pain out of proportion" to examination, multiple vascular risk factors, and atrial fibrillation, which increases his risk for mesenteric ischemia. Computed tomography angiography would be the best screening test. Magnetic resonance imaging would be too time-consuming, plain radiographs have limited utility, and right upper quadrant ultrasound is not warranted in this patient with diffuse, non-localized symptoms.

4. A 30-year-old man presents with abdominal pain and fever. Which of the following would place him at higher risk for life-threatening pathology?
- **a.** Drinking from a mountain spring
- **b.** Prednisone therapy for systemic juvenile idiopathic arthritis
- **c.** Recent cruise with norovirus outbreak
- **d.** Use of metoprolol for essential hypertension

Answer: b. Prednisone therapy for systemic juvenile idiopathic arthritis. Low-dose steroid therapy places patients at higher risk when they present with abdominal pain. Drinking from a mountain spring increases risk for giardia. Neither this nor exposure to norovirus place the patient at increased risk for life-threatening pathology. Hypertension increases risk for vascular conditions such as abdominal aortic aneurysm, but this would be unlikely in a young patient controlled on metoprolol.

5. What is a characteristic of visceral abdominal pain?
- **a.** Easily described
- **b.** Midline
- **c.** Sharp in nature
- **d.** Well-localized

Answer: b. Midline. Visceral pain is classically midline, poorly localized, and less easily described than somatic pain, which is often referred to as "sharp."

KEY CONCEPTS

- Jaundice is the clinical manifestation of elevated serum bilirubin, which arises through the metabolism of hemoglobin. Elevated bilirubin occurs when: (1) increased bilirubin is produced due to hemolysis, (2) liver dysfunction prevents conjugation of bilirubin, (3) an obstruction prevents secretion of bilirubin into the intestines.
- Jaundice is usually not evident on physical examination until the total serum bilirubin concentration rises above 2.5 mg/dL. Jaundice is first apparent in the conjunctiva, sublingually, or on the hard palate.
- Unconjugated bilirubin that is not bound to albumin can cross the blood-brain barrier, causing adverse neurologic effects; conjugated bilirubin is not neurotoxic.
- New-onset painless jaundice is the classic presentation for a neoplasm involving the head of the pancreas.
- In cases of unexplained hepatocellular injury, a quantitative acetaminophen level may be helpful.
- If the etiology of ascites is unknown, serum ascites albumin gradient (SAAG) determination will aid in distinguishing the cause as well as the presence of portal hypertension.
- Ultrasonography is the preferred initial imaging modality in evaluating for biliary obstruction, whereas CT is preferred if malignant obstruction is suspected or the full abdomen needs to be evaluated.
- Elevated direct bilirubin with transaminase elevation is indicative of hepatocellular inflammation or injury.
- Diagnosis of spontaneous bacterial peritonitis (SBP) is made by a peritoneal fluid neutrophil count of greater than 250. Treatment typically consists of cefotaxime 2 g IV and albumin 1.5 g/kg.

1. A 56-year-old male presents with fever and abdominal distention. Bedside ultrasound reveals ascites and the results of paracentesis indicate possible spontaneous bacterial peritonitis (SBP). What daily medication should be stopped upon admission?

a. Amlodipine
b. Crestor
c. Lactulose
d. Nadolol

Answer: d. Beta blocker use in patients with cirrhosis with SBP should be discontinued as they have been shown to increase mortality.

2. A 43-year-old female presents with 1 month of painless abdominal swelling. On examination she has a diffusely swollen abdomen with a fluid wave. To help determine the etiology, you obtain the serum ascites albumin gradient (SAAG). Which value is consistent with cirrhosis?

a. 0.25 g/dL
b. 0.5 g/dL
c. 1 g/dL
d. 1.5 g/dL

Answer: d. A (SAAG) value of greater than or equal to 1.1 g/dL is found in patients with portal hypertension. There are many causes of portal hypertension, including cirrhosis, liver failure, or heart failure. A SAAG value of less than 1.1 g/dL may be found in patients with conditions such as lupus or pancreatitis.

3. A 48-year-old male with a history of cirrhosis presents with 3 days of abdominal pain and fever. Abdominal paracentesis suggests spontaneous bacterial peritonitis (SBP). Which of the following initial treatments is most appropriate?

a. Cefotaxime, albumin, and discontinue beta blockade
b. Ceftriaxone and dexamethasone
c. Ceftriaxone alone
d. Ciprofloxacin and metronidazole

Answer: a. The empirical antibiotic of choice is a third-generation cephalosporin (e.g., cefotaxime). Albumin administration has been shown to reduce risk for mortality and hepatorenal syndrome. If the patient has a history of cirrhosis and is taking a nonselective beta blocker, it should be discontinued because it has been shown to increase mortality in patients with SBP.

4. A 41-year-old male with a history of cirrhosis presents with fever, abdominal distension, and confusion. A paracentesis is performed in the evaluation of spontaneous bacterial peritonitis (SBP). What are the diagnostic criteria found in the ascitic fluid that confirms SBP?

a. Ascitic fluid neutrophil count >100
b. Ascitic fluid neutrophil count >250
c. Ascitic fluid total WBC count >100
d. Ascitic fluid total WBC >250

Answer: b. The presence of more than 250 polymorphonuclear cells per cubic millimeter of ascitic fluid is diagnostic for SBP.

5. A 55-year-old female presents with 1 month of diffuse abdominal swelling and pain. She reports a long history of alcohol use. In evaluating this patient for jaundice, at what bilirubin level does jaundice typically become clinically apparent?

a. 1 mg/dL
b. 2 mg/dL
c. 2.5 mg/dL
d. 4 mg/dL

Answer: d. Clinical jaundice is usually not evident until the total serum bilirubin concentration rises above 2.5 mg/dL. Jaundice is first apparent sublingually, in the conjunctiva or on the hard palate.

Nausea and Vomiting 25

KEY CONCEPTS

- Nausea and vomiting can result from a primary problem in the gastrointestinal (GI) tract but can also result from problems in the neurological, vestibular, urogenital, cardiac, or other systems.
- In the acutely vomiting patient, associated symptoms and medication history are most helpful in narrowing the differential diagnosis.
- Laboratory studies are not necessary for all patients with vomiting. In patients with severe or protracted vomiting, consider checking electrolytes and renal function, particularly when intravenous (IV) rehydration is initiated.
- In patients with undifferentiated nausea or vomiting or vomiting due to non-obstructive GI disease, ondansetron is the first-line antiemetic due to its low side effect profile. However, no definitive evidence exists for the superiority of one antiemetic agent over others.
- Although evidence is limited, metoclopramide is the antiemetic of choice in hyperemesis gravidarum and vomiting associated with headache, and ondansetron is the drug of choice in chemotherapy-induced vomiting.
- Ondansetron is the first-line agent for children with vomiting due to acute gastroenteritis.
- Consider cannabinoid hyperemesis syndrome (CHS) in all patients with protracted vomiting. Emergency department (ED) treatment of CHS includes the use of capsaicin cream as well as haloperidol or lorazepam.

1. **A** 50-year-old male patient presents with a complaint of vomiting. He states the pancakes he ate 12 hours ago for breakfast came out undigested. He admits to several similar episodes in preceding weeks. He is otherwise asymptomatic and has normal vital signs. His physical examination is normal. Which of the following conditions is most likely?

a. Constipation
b. Gastric outlet obstruction
c. Hepatitis
d. Increased intracranial pressure

Answer: b. Vomiting intact food eaten over 12 hours earlier is considered pathognomonic for gastric outlet obstruction. None of the other choices have this as a feature.

2. Antihistamines would most effectively control the nausea and vomiting caused by which of the following conditions?

a. Chemotherapy administration b. Gastritis
c. Gastroparesis
d. Labyrinthitis

Answer: d. Antihistamines are particularly useful in nausea and vomiting associated with labyrinthitis, motion sickness, and vestibular disorders by directly inhibiting vestibular stimulation and vestibular-cerebellar pathways. The anticholinergic effect may also contribute to their effectiveness in vertigo and motion sickness.

3. Which of the following is the most effective for the treatment of cannabinoid hyperemesis syndrome?

a. Capsaicin cream
b. Cessation of marijuana use
c. Haloperidol
d. Lorazepam

Answer: b. Cessation of marijuana use has been demonstrated to decrease syndrome recurrence. The other choices are frequently utilized in the acute treatment of cannabinoid hyperemesis syndrome, and have only been described in case reports or series.

4. **A** 28-year-old female presents to the emergency department with severe vomiting. She is 7 weeks pregnant. She has experienced nausea during this pregnancy which she had controlled with oral ginger and a pyridoxine/doxylamine combination pill, but for the last 3 days she has been unable to tolerate anything by mouth. Her electrolytes and renal function are normal. Bedside ultrasound reveals a single live intrauterine pregnancy. Her urinalysis reveals large ketones. In addition to intravenous fluids, which of the following medications should be selected as the first line intravenous antiemetic agent?

a. Dexamethasone
b. Haloperidol
c. Metoclopramide
d. Ondansetron

Answer: c. Metoclopramide is a first-line agent due to its safety for the developing fetus. Ondansetron should be reserved for second line as there is some suggestion of the capacity for fetal harm. Olanzapine is indicated for chemotherapy-associated vomiting and haloperidol is best used for gastroparesis or cannabinoid hyperemesis syndrome. Dexamethasone is reserved for patients who fail first- and second-line therapies.

5. What is the most common cause of nausea and vomiting in the pediatric population?

a. Acute gastroenteritis
b. Drug side effects
c. Intussusception
d. Motion sickness

Answer: a. In pediatrics, acute gastroenteritis is the most common cause of vomiting. Studies on management of vomiting in this population focus on children with this condition.

26 Gastrointestinal Bleeding

KEY CONCEPTS

- Routine placement of a nasogastric tube in patients with suspected upper gastrointestinal bleeding (UGIB) is not recommended, because this procedure fails to reliably provide useful data to guide management and is associated with unnecessary patient discomfort and potential complications.
- Patients older than 35 years with UGIB should have an electrocardiogram (ECG) performed early in their evaluation, because patients with UGIB and coronary disease (including occult coronary disease) may develop an acute coronary syndrome that presents with atypical symptoms, such as nausea and abdominal discomfort.
- A blood urea nitrogen (BUN) to creatinine (Cr) ratio of greater than 35 is 90% specific for UGIB. However, the BUN:Cr ratio is poorly sensitive and cannot be used to rule out UGIB.
- Patients with exsanguinating hemorrhage should receive an emergent transfusion of untyped packed red blood cells. For stable patients without known coronary artery disease, transfusion should be initiated when the hemoglobin level is less than 8 g/dL.
- Standard pharmacologic therapy for UGIB includes intravenous (IV) proton pump inhibitors and prokinetic agents. Patients with cirrhosis and suspected esophageal varices should receive octreotide and antibiotics.
- Based on available evidence, we do not recommend the routine use of tranexamic acid for patients with UGIB.
- Platelet transfusion is not indicated for patients on antiplatelet medications unless the platelet count is <50,000 per μL.
- Empirical administration of fresh frozen plasma (FFP) or prothrombin complex concentrate (PCC) to patients with cirrhosis or anticoagulant use is not indicated.
- Stable patients with lower gastrointestinal bleeding (LGIB) who can tolerate bowel preparation are best managed through urgent gastroenterology consultation for colonoscopy, while unstable patients should be seen by an interventional radiologist for possible selective embolization.

1. What is the most common cause of upper gastrointestinal (GI) hemorrhage in adults in the United States?
   - **a.** Esophagitis
   - **b.** Gastritis
   - **c.** Liver cirrhosis
   - **d.** Peptic ulcer disease

Answer: d. Despite a decreasing prevalence with the treatment of *Helicobacter pylori*, peptic ulcer disease (PUD) remains the most common cause of upper GI bleeds in adults in the United States.

2. Management of unstable patients with massive upper gastrointestinal bleeding (UGIB) should include which of the following?
   - **a.** Placement of at least two large-bore IVs.
   - **b.** Avoidance of rapid crystalloid infusion in effort to decrease rebleeding rates
   - **c.** Empiric treatment with fresh frozen plasma (FFP)
   - **d.** Routine intubation for airway protection

Answer: a. Initial resuscitation of unstable patients with massive GI bleeding should include placement of two large bore IV catheters and rapid crystalloid infusion with consideration of early blood transfusion. Intubation should be reserved for patients with altered mental status, respiratory distress, or high concern for aspiration. FFP should be given empirically only for patients receiving a massive transfusion protocol or in select patients with coagulopathy.

3. Which of the following statements is true regarding platelet administration for patients with acute gastrointestinal (GI) bleeding?
   - **a.** Patients requiring massive transfusion should receive a balanced transfusion approach of 1:1:1 packed red blood cells (PRBCs):platelets:FFP.
   - **b.** Platelets should be administered to a target level of >100,000 per μL
   - **c.** All patients with GI bleeding on antiplatelet medications such as aspirin should receive an empiric platelet transfusion
   - **d.** Antiplatelet medications do not increase the risk of acute GI bleeding

Answer: a. Despite a high risk of GI bleeding while taking antiplatelet medications, the routine use of platelet transfusion for patients on antiplatelet medications has not been shown to be beneficial. Platelet transfusion should be reserved for patients who are thrombocytopenic <50,000 per μL or those requiring massive transfusion protocol.

4. Elevations in which of the following serum markers in the setting of an acute gastrointestinal bleed is predictive of increased 30-day mortality?
   - **a.** C-reactive protein
   - **b.** Lactate
   - **c.** Serum osmolality
   - **d.** White blood cell count

Answer: b. An initial lactate greater than 2.5 mmol/L is associated with the development of in-hospital hypotension and an increased rate of 30-day mortality.

5. Appropriate pharmacologic therapy in the emergency department for most acute upper gastrointestinal bleeds (UGIB) includes which of the following agents?
   - **a.** Antibiotics
   - **b.** Erythromycin
   - **c.** Proton pump inhibitor (PPI)
   - **d.** Somatostatin

Answer: c. PPIs decrease rebleeding, surgery, and mortality in patients with an acute UGIB from an ulcer. Because peptic ulcer disease is the most common cause of UGIB, it is appropriate to administer PPIs for most acute UGIB.

KEY CONCEPTS

- Hospital-acquired *Clostridioides difficile* and norovirus infection are the most prevalent causes of fatal illness from diarrhea in the United States.
- Key elements of the history in the patient with diarrhea include recent travel, hospitalization or antibiotic use, dietary factors, immunosuppression, fever, and presence of blood in stool.
- Acute diarrhea is most often viral and treated with supportive therapy.
- Fluids and foods with excess sugar, caffeine, or high fat content should be avoided in patients with diarrhea.
- In selected nontoxic patients, loperamide, initial dose of 4 mg by mouth and subsequent doses of 2 mg by mouth after each loose stool (not to exceed 16 mg in 24 hours), is typically safe and effective in providing symptom control.
- Empiric antibiotics are uncommonly indicated in patients well enough to go home. However, they should be considered in patients with fever or bloody diarrhea who are immunocompromised or in travelers diarrhea. Empiric antibiotics are prudent whenever sepsis is suspected.
- Fluoroquinolones (ciprofloxacin 500 mg by mouth twice daily or levofloxacin 500 mg by mouth daily for 3 to 5 days) or azithromycin (500 mg by mouth daily for 3 days) cover the majority of enteric pathogens if empiric antibiotics are required.
- A single dose of azithromycin, 1000 mg by mouth, is sufficient antibiotic treatment for travelers diarrhea.
- First line treatment for *C. difficile* is vancomycin 125 mg by mouth four times daily for 10 days. Second line treatment is fidaxomicin 200 mg by mouth for 10 days.
- The efficacy of probiotics in reducing stool frequency in acute infectious diarrhea is unclear.

1. Which of the following is not a common cause of diarrhea in a patient with human immunodeficiency virus (HIV) disease?
   a. Antiretroviral therapy
   b. Cytomegalovirus
   c. Giardia
   d. Intestinal lymphocyte proliferation in acute HIV infection
   e. Shigella

Answer: d. Acute HIV infection is associated with depletion (not proliferation) of intestinal lymphocytes from intestinal mucosa, which alters the gut microbiome leading to increased incidence of viral, bacterial, and protozoal causes of diarrhea. Gastrointestinal cytomegalovirus is associated with advanced HIV infection (CD4+ counts less than 50), which can lead to severe hemorrhagic colitis. In addition, all classes of antiretroviral drugs may cause diarrhea, which can affect more than 25% of individuals taking combined antiretroviral therapy.

2. In which of the following patients would *Clostridioides difficile* testing be least useful?
   a. Patient with diarrhea who is taking omeprazole for known peptic ulcer disease.
   b. Patient with diarrhea after recent tonsillitis treated with amoxicillin.
   c. Patient with diarrhea who is a long-term resident of a skilled nursing facility.
   d. Patient with acute vomiting and watery diarrhea following contact with a toddler with similar symptoms.
   e. Patient with uncontrolled HIV and acute watery diarrhea.

Answer: d. Patient in answer D likely has a viral gastroenteritis, and C. difficile testing is more likely to result in false positive result or identify chronic colonization. Risk factors for C. difficile colitis include antibiotic or gastric acid suppressive therapy, skilled nursing facility residence, and chronic medical conditions.

3. Evaluation of stool for ova and parasites would prove least beneficial in which of the following subsets of patients presenting with diarrhea?
   a. Patients concurrently taking clindamycin with diarrhea for 1 week
   b. Patients recently returning from a backpacking trip
   c. Patients returning from a trip to Nepal
   d. Patients with chronic diarrhea
   e. Patients with human immunodeficiency virus (HIV) infection

Answer: a. The assessment of stool for ova and parasites is not routinely recommended in most cases of diarrheal illness. This study is used in patients with chronic diarrhea (Entamoeba histolytica and Cryptosporidium); patients with a history of travel to developing countries, particularly to Nepal or areas of Russia (Cryptosporidium, Giardia, and Cyclospora); patients with exposure to infants in daycare centers (Cryptosporidium and Giardia); and patients with HIV infection (E. histolytica and Giardia). Acute diarrhea can last up to 2 weeks.

4. In which of the following patients is loperamide a safe treatment option?
   a. Adult patient with acute watery diarrhea of unknown etiology
   b. Adult patient with sepsis and bloody diarrhea
   c. Elderly patient with a history of bowel obstruction with watery diarrhea
   d. Pediatric patient with acute watery diarrhea from norovirus
   e. Pediatric patient with travelers diarrhea

Answer: a. Loperamide, a bowel-selective opioid agonist, is an effective tool to reduce diarrhea severity. Although generally safe, it has been associated with toxic megacolon and obstruction in select groups, such as the elderly, pediatric patients (especially younger than age 3), and patients with a severe inflammatory colitis. Adult patients in which the risk of obstruction or severe inflammatory colitis is low are candidates for loperamide treatment.

5. In which of the following patients is empiric antibiotic treatment the most reasonable option?
   a. Adult patient returning from Mexico with watery diarrhea.
   b. Adult patient with bloody diarrhea during outbreak of *Escherichia coli O157:H7*
   c. Pediatric patient returning from Thailand with bloody diarrhea.
   d. Pediatric patient with fever and bloody diarrhea.
   e. Well-appearing adult patient with suspected norovirus based on ill contacts.

Answer: a. Travelers diarrhea generally responds well to a single dose of azithromycin 1000 mg PO which can decrease time to resolution of symptoms with limited side effects. The choice to initiate antibiotics must be tailored to weigh potential benefits and harms of the treatment for the individual patient. Empiric antibiotics are generally contraindicated in children, as well as patients in which E. coli O157:H7 is a possibility, as antibiotics increase the risk of hemolytic uremic syndrome. In cases of suspected viral-mediated diarrhea, such as a clustered norovirus outbreak, antibiotics are generally not indicated.

28 Constipation

KEY CONCEPTS

- Constipation is a variable term used by patients and may refer to straining with defecation, hard or infrequent stools, pain with a bowel movement, a sensation of incomplete evacuation, or abdominal bloating.
- Primary constipation is caused by functional abnormalities of the gastrointestinal tract.
- Secondary constipation is due to diet, related medical or psychiatric disorders, or medication side effects.
- The etiology of constipation can frequently be discovered through a thorough history.
- The physical examination of patients with constipation focuses on the patient's appearance, the presence of pain, and structural abnormalities identified on abdominal or rectal examination.
- Lab work to evaluate constipation is typically not indicated.
- Imaging to evaluate constipation is not typically indicated when the physical examination is normal.
- The treatment of constipation for most patients is empiric and focuses on lifestyle modifications in addition to laxatives.
- Patients with constipation should be referred for outpatient evaluation and testing when they fail empiric treatment.
- Opioid-induced constipation is common and may require peripherally acting μ-opioid receptor antagonists for management.

1. A 48-year-old female with no known past medical history presents to the emergency department with the chief complaint of constipation. She states she has not had a bowel movement in the past 7 days. She has no history of similar symptoms. She denies taking any medications or any past surgical history. On examination, she has no abdominal tenderness with palpation or masses. Her rectal examination is unremarkable for fissures, and the patient does not have any blood or stool in her rectal vault. What is the next step in this patient's management?

a. A plain film abdominal x-ray
b. Discharge home with recommendation for laxatives and plan for lifestyle modification
c. Emergent outpatient gastroenterology follow-up
d. Trial of at home docusate and strict return precautions if plan fails

Answer: b. The patient in this question has acute constipation. Because her history and examination does not have any red flag concerns, this patient can be safely discharged home without any testing. This patient will require laxatives to help her acute constipation, but lifestyle management should help to prevent similar episodes in the future. There is no utility for plain film abdominal x-rays in the emergency department for the evaluation of constipation. Docusate has been shown to be ineffective as first line management for constipation unless the patient has a rectal abnormality or rectal pain that prevents the passage of stool. This patient is well-appearing on examination and therefore, she does not require intravenous (IV) hydration. Because this patient's constipation is acute and she has a benign examination, she does not require emergent gastroenterology follow-up.

2. A 96-year-old male from a nursing home presents with the chief complaint of constipation. He has a past medical history significant for cerebrovascular accident and now receives all of his medications, hydration, and feedings through a gastrostomy tube due to dysphagia. The patient has been on a bowel regimen of twice daily polyethylene glycol through his gastrostomy tube with minimal, watery stool output over the past week. The patient appears to be uncomfortable, but he is afebrile and hemodynamically stable. On examination, the patient's abdomen is soft and there is no focal tenderness. On rectal exam, you feel a large amount of firm stool in the rectal vault. What is the appropriate next step in this patient's management?

a. Admission for colonoscopy
b. Analgesia and disimpaction
c. Glycerin suppository
d. Oral milk of magnesia

Answer: b. This is an elderly male with chronic constipation for a multitude of reasons. Because of his age and possibly due to his cerebrovascular accident, he has denervation of his rectum preventing him from sensing rectal fullness. The large volume of stool in his rectum needs to be removed for his comfort, and disimpaction with analgesia will be necessary. The use of a glycerin suppository to lubricate the stool to be expelled is unlikely to be successful as this patient has lost the ability to expel stool on his own. Because this patient does not receive medications by mouth, milk of magnesia should not be given orally. Because his abdominal examination is benign and he is hemodynamically stable, the likelihood that he has a critical intra-abdominal process requiring imaging is low. Finally, this patient does not require admission to the hospital, nor does he require an emergent colonoscopy at this time.

3. A 32-year-old female with chronic pain followed closely by a pain management specialist presents with constipation. She normally takes a bowel regimen of bisacodyl daily in addition to consuming a high-fiber diet. Her physician recently increased her opioid medication dosage. Since then, she has had difficulty having more than one small, hard stool per week. She complains of abdominal discomfort but has not had any nausea or vomiting. On examination, her abdomen is soft without focal tenderness. Her rectal examination is normal. What is the appropriate next step in this patient's management?

a. A trial of a peripherally acting μ-opioid receptor antagonist and plan for close outpatient follow up with her care team
b. Consultation with her pain management specialist to decrease her medication dosage back to previous levels
c. Lab testing to rule out an electrolyte abnormality causing this patient's symptoms
d. Nonopioid pain medication for her discomfort while in the emergency department and plan for gastroenterology follow-up

Answer: a. This is a patient experiencing opioid-induced constipation. Her previous bowel regimen was sufficient until her medication doses increased. Because this patient has failed her bowel regimen, it is reasonable to try a peripherally acting μ-opioid receptor antagonist to block the effects of the opioids on the colonic μ receptor. Decreasing this patient's pain medications back to previous doses may lead to withdrawal. Lab testing is unlikely to be of benefit. Providing analgesia for discomfort will not relieve constipation.

4. Which of the following is true regarding constipation?
   - **a.** The diagnosis of constipation requires lab testing and imaging in the emergency department.
   - **b.** Most patients with constipation and a normal examination can be safely discharged home with empiric treatment.
   - **c.** Constipation is less common in the elderly due to low fiber, mechanical soft diets.
   - **d.** Chronic constipation occurs when a patient experiences decreased stool output for at least 1 year.

Answer: b. The term constipation refers to a multitude of symptoms, not a medical diagnosis. Patients often have different personal definitions for the term "constipation." Chronic constipation refers constipation symptoms for 3 or more months. Constipation is more common in the elderly due to sedentary lifestyles and low-fiber, mechanically soft diets. In the ED, lab testing and imaging is often unnecessary for patients with a normal physical exam. Most well-appearing patients with constipation and a normal examination can be discharged home with empiric management.

5. Which of the following is true regarding treatment options for constipation?
   - **a.** Bisacodyl works by increasing water penetration and thereby softens the stool.
   - **b.** Docusate sodium should be prescribed to all patients with chronic constipation given its effectiveness as a laxative.
   - **c.** Lifestyle modification is the first line treatment for patients with acute constipation.
   - **d.** Sodium phosphate and magnesium hydroxide should be avoided in patients with renal impairment.

Answer: d. The side effect profiles for many common treatments of constipation should be considered prior to prescribing to special patient populations. Both sodium phosphate and magnesium hydroxide can increase the levels of phosphorus and magnesium in the body, respectively, in patients with renal insufficiency/impairment. Bisacodyl is a laxative and works by stimulating intestinal motility. Docusate sodium is a stool softener, not a laxative, that should not be used as first line or solo therapy due to its marginal efficacy. Patients with acute constipation should be prescribed both a laxative for immediate treatment and lifestyle modification for longer-term management.

Acute Pelvic Pain 29

KEY CONCEPTS

- Acute pelvic pain in women is often from a gynecologic source, but urinary and intra-abdominal sources are also common. Less frequently, the pain may arise from vascular, musculoskeletal, neurologic, or psychiatric disorders.
- Potentially lethal diagnoses associated with acute pelvic pain include ectopic pregnancy, ovarian cyst with significant hemorrhage, and domestic violence; highly morbid conditions presenting with acute pelvic pain include pelvic inflammatory disease and ovarian torsion.
- Nearly all women of childbearing age with pelvic pain should have a pregnancy test performed, and most should have a pelvic ultrasound examination.
- Ectopic pregnancy should be excluded in the pregnant patient with pelvic pain. Bedside ultrasound is an excellent test for confirming an intrauterine pregnancy (IUP); it excludes ectopic pregnancy with a high degree of certainty in patients who are not using assisted reproductive technology.
- Pregnant patients with acute pelvic pain may also have non–pregnancy-related disorders; appendicitis, nephrolithiasis, and ovarian torsion, among others, remain in the differential diagnosis.
- Many patients with acute pelvic pain require imaging as part of their assessment. If a gynecologic source is suspected, begin with an ultrasound and then progress to computed tomography (CT) or magnetic resonance imaging (MRI), if needed. The presence of an ovarian cyst on imaging does not necessarily explain the patient's pain, and further evaluation may be required. Ultrasound after normal CT is unlikely to be informative. Ovarian torsion may be radiographically occult.

1. Which of the following statements regarding the evaluation of patients with pelvic pain is true?
   a. Ancillary testing can be limited to a urine pregnancy test in most patients.
   b. Bimanual examinations have been shown to result in highly reliable findings, with substantial interobserver agreement.
   c. Patients typically localize visceral pain with a high degree of accuracy.
   d. Thorough history taking is adequate to exclude most life-threatening conditions.
   e. None of the above.

Answer: e. It is rare that any particular finding on history or physical examination is reliable enough to make or exclude a particular diagnosis conclusively in patients presenting with pelvic pain, so ancillary testing (beyond a simple pregnancy test) is commonly required in the evaluation of these patients. Ultrasound is the most useful adjunct for most patients. The bimanual examination may, at times, provide important information. Unfortunately, however, findings on pelvic examination are somewhat subjective and unreliable and may serve more to localize the process to one side or the other rather than diagnose it or even limit it to the reproductive organs. Although pelvic pain often originates from the reproductive organs, it may arise from any structures that lie adjacent to or course through the pelvis. Visceral pain afferents supplying the pelvic organs have common innervation with the appendix, ureters, and colon. Their significant overlap makes accurate localization difficult for both patient and emergency clinician.

2. A 26-year-old female presents with right lower quadrant (RLQ) abdominal pain. She states her last menstrual period was 8 weeks ago. Bimanual pelvic examination reveals tenderness in the RLQ and right adnexal area. The patient's vital signs include a regular pulse of 120 beats/min and a blood pressure of 110/65 mm Hg. Urinalysis is unremarkable, and the urine pregnancy test is positive. What is the most appropriate next test?
   a. Pelvic ultrasonography
   b. Cervical cultures
   c. Complete blood count (CBC)
   d. Computed tomography (CT)
   e. Magnetic resonance imaging (MRI)

Answer: a. This follows the algorithms in Figs. 29.1 and 29.2. The most life-threatening pathology requiring urgent or emergent intervention is hemorrhage from a ruptured ectopic pregnancy. Pelvic ultrasound scan is rapid and is the first step in an evaluation of a suspected ruptured ectopic pregnancy. In this case, bedside ultrasonography, if available, is strongly recommended.

3. A 30-year-old female presents with lower abdominal pain. She is not ill appearing and has lower abdominal, uterine, bilateral adnexal, and cervical motion tenderness on pelvic examination. She has a negative urine pregnancy test and urinalysis. What is the most appropriate next step in the patient's management?
   a. Await cervical culture results.
   b. Obtain a complete blood count.
   c. Obtain a CT scan.
   d. Obtain an erythrocyte sedimentation rate (ESR).
   e. Treat with antibiotics.

Answer: e. The constellation of uterine tenderness, bilateral adnexal tenderness, and cervical motion tenderness is classically associated with pelvic inflammatory disease (PID), particularly when the pain onset is during or just after menstruation. However, the diagnosis may be made without the presence of all three signs, and treatment may be initiated with only one sign in an at-risk patient, as given in the 2015 Centers for Disease Control and Prevention (CDC) guidelines. Ultrasound is not strictly necessary in this case but should be performed if there is any concern for a complication, such as tubo-ovarian abscess.

4. A 35-year-old female using assisted reproductive technology presents with severe lower left quadrant (LLQ) abdominal pain and tenderness isolated to the left adnexal area on pelvic examination. The urine pregnancy test is positive, and the urinalysis is unremarkable. Rapid bedside ultrasonography reveals an intrauterine pregnancy with a gestational age of 6 weeks, 5 days, and moderate free pelvic fluid. Which diagnosis should be expediently further investigated?
   a. Heterotopic pregnancy
   b. Round ligament pain
   c. Simple ovarian cyst
   d. Tubo-ovarian abscess
   e. None of the above

Answer: a. Women who are actively undergoing infertility treatment are at increased risk for ectopic pregnancy, heterotopic pregnancy, ovarian torsion, and ovarian hyperstimulation syndrome.

5. A 28-year-old female presents with sudden-onset, severe left adnexal pain that started 1 hour ago. The patient has left adnexal tenderness to palpation, but there is no vaginal discharge or bleeding, and no cervical motion tenderness. Urinalysis and urine pregnancy are negative. Pelvic ultrasound is unremarkable, and you do not suspect intra-abdominal pathology. The patient has required multiple doses of opiates for pain control in the emergency department (ED). What is the most appropriate next step in management?
   a. Admit the patient for serial abdominal examinations.
   b. Consult gynecology.
   c. Discharge the patient with pain control and clinic follow-up.
   d. Discharge the patient with treatment for PID and clinic follow-up.
   e. Obtain a CT scan of the pelvis.

Answer b. Ovarian torsion should be suspected based on this patient's history and examination. Because torsion can be a radiographically occult diagnosis, gynecology should be consulted as laparoscopy may be needed for definitive diagnosis. Less emergent conditions, such as nephrolithiasis, could also be considered.

6. A 26-year-old female who is pregnant at 30-weeks' gestation presents with vaginal bleeding and pelvic cramping. She states the cramping and bleeding started 2 hours ago and she has filled one pad so far. She has not had any prenatal care. Which of the following should be initially avoided during this patient's evaluation?

a. Obstetrics consultation
b. Intravenous (IV) fluids
c. Pelvic examination
d. Type and Screen
e. Ultrasound

Answer c. Pregnant patients beyond 20-weeks' gestation with complaints of vaginal bleeding should undergo transabdominal pelvic ultrasound for placental localization before the pelvic examination. The concern is that placenta previa, which can be worsened by pelvic examination, may be the cause of the bleeding. Obstetrics will need to be involved in this patient's care, as two important potential diagnoses are placenta previa and placental abruption. In any case, the patient will need obstetric observation and fetal monitoring. A type and screen should be performed as Rho(D) immune globulin is indicated if the patient is Rh negative.

7. A 34-year-old female presents with severe abdominal pain. She does not allow for an abdominal examination and keeps pushing your hand away any time you attempt to even mildly palpate her abdomen. Her vital signs are stable. What is the most important and appropriate next step in this patient's care?

a. IV fluids
b. IV opiates
c. Oral (PO) opiates
d. PO acetaminophen
e. Do not give any medications at this time

Answer b. We recommend early administration of analgesia for patients with significant pain. Analgesia greatly improves patient comfort and the reliability of the physical examination, which is otherwise hampered by the patient's extreme pain, tenderness, or both. For severe pain, intravenous opioids such as morphine or hydromorphone are rapid and effective, titratable, and generally considered safe in pregnancy. In a patient who may require surgery, it is recommended to keep the patient nil per os (NPO).

KEY CONCEPTS

- Pregnancy status is the single most important determination to make in evaluating a patient with vaginal bleeding.
- The use of the term *dysfunctional uterine bleeding* is no longer recommended, and the term *abnormal uterine bleeding* is preferred.
- The etiologies of abnormal uterine bleeding can be divided into structural and nonstructural causes using the PALM-COEIN classification. Structural causes include ***p***olyps, ***a***denomyosis, ***l***eiomyomas, and ***m***alignancy (**PALM**). Nonstructural causes include ***c***oagulopathy, ***o***vulatory dysfunction, ***e***ndometrial, ***i***atrogenic, and ***n***ot yet classified causes (**COEIN**).
- For nonpregnant patients, the numerous causes of abnormal uterine bleeding are best categorized by patient age. The possibility of cancer should always be considered in postmenopausal women.
- Hormonal and nonhormonal treatments can be initiated in the emergency department to temporize an acute bleeding episode in a nonpregnant patient until they can follow up with their gynecologist.
- Vaginal bleeding is common in early pregnancy. Most patients will be diagnosed with threatened miscarriage, but ectopic pregnancy should always be considered at any level of serum β-hCG.
- Vaginal bleeding after the 20th week of pregnancy is less common and is often associated with significant morbidity and mortality for the mother and fetus. These patients should be managed in close consultation with an obstetrician.

1. To avoid the potential for misdiagnosis and possible disruption of a desired intrauterine pregnancy, American College of Obstetricians and Gynecologists (ACOG) recommends using which value of β-human chorionic gonadotropin (β-hCG) for the discriminatory zone?

a. 1500 IU/mL
b. 2000 IU/mL
c. 3500 IU/mL
d. 4500 IU/mL

Answer: c. The discriminatory zone has traditionally been accepted to be approximately 2000 IU/mL for transvaginal ultrasound (TVUS). However, viable intrauterine pregnancies may not yet be visualized on TVUS at β-hCG levels greater than 2000 IU/mL. Therefore ACOG currently advises that if the concept of the discriminatory zone is to be used to help diagnose ectopic pregnancy, the value should be as high as 3500 IU/mL to avoid the potential for misdiagnosis and possible disruption of a desired intrauterine pregnancy.

2. What is the most common cause of uterine bleeding in the immediate postpartum period?

a. Uterine rupture
b. Vaginal lacerations
c. Retained products of conception
d. Uterine atony

Answer: d. Postpartum hemorrhage is a leading cause of maternal mortality. Within the first 24 hours after delivery, this is usually caused by uterine atony. After 24 hours, the cause is frequently retained products of conception.

3. A 30-year-old woman with a history of polycystic ovarian syndrome (PCOS) and prior deep vein thrombosis (DVT) presents to the ED with 10 days of irregular, heavy uterine bleeding. Her vital signs are normal. What is the first step in her ED management?

a. Check complete blood count (CBC).
b. Order urine pregnancy test.
c. Administer 25 mg intravenous conjugated estrogen.
d. Prescribe a taper of oral contraceptive pills.

Answer: b. Pregnancy status is the single most important determination to make when evaluating a patient with vaginal bleeding and should be determined after assessing stability.

4. A 30-year-old woman with a history of polycystic ovarian syndrome (PCOS) and prior deep vein thrombosis (DVT) presents to the emergency department (ED) with 10 days of irregular, heavy uterine bleeding. Her vital signs are normal, and a urine pregnancy test is negative. She currently takes no medication. Which of the following is the most appropriate treatment?

a. Oral contraceptive taper
b. Medroxyprogesterone acetate
c. Oral tranexamic acid
d. Ibuprofen

Answer: d. Patients with PCOS may have anovulatory cycles and frequently benefit from hormonal treatment such as oral contraceptive pills. However, as the patient has a history of prior DVT she has a contraindication to hormonal treatments as well as oral tranexamic acid. Ibuprofen would therefore be the most appropriate treatment.

5. A 32-year-old woman who is 26 weeks pregnant describes a 1-hour history of painless vaginal bleeding after tripping and falling from a standing height. Her vital signs are normal, and physical examination of the abdomen reveals a nontender uterus with a fundal height 1 cm above the umbilicus. Of the following, which would be the most appropriate next step in evaluating this patient?

a. Abdominal magnetic resonance imaging (MRI)
b. Bimanual examination
c. Sterile speculum examination
d. Ultrasound

Answer: d. Painless vaginal bleeding after the 20th week of pregnancy is suggestive of placenta previa. Pelvic examination (speculum or bimanual) should be deferred until ultrasound has excluded placenta previa as the cause of bleeding.

31 Back Pain

KEY CONCEPTS

- Acute low back pain is a common, costly, recurring and painful condition that often has no recognizable or dangerous cause.
- While most cases of back pain are non-specific and improve without laboratory evaluation or imaging, the aim of the emergency department (ED) assessment is to identify those serious and emergent pathologies that require urgent treatment.
- A thorough history and physical examination is performed with the goal of uncovering high-risk features predisposing the patient to an emergent or life-threatening etiology.
- The presence of multiple key clinical findings may increase the probability of disease and is often an indication for further investigation of patients with back pain. However, blindly allowing the presence or absence of these individual findings to guide diagnostic treatment will lead to potentially unnecessary, misleading, and costly investigations in most patients.
- Key clinical findings should supplement, not replace, clinical judgement when selecting appropriate laboratory and radiologic testing of patients with back pain. Imaging and laboratory studies are indicated only when there is evidence of neurologic deficit or multiple key clinical findings suggesting a dangerous or systemic pathologic cause.
- Patients who have low back pain emergencies are generally classified into five groups: (1) new onset back pain with neurologic findings in a patient with a malignancy history; (2) back pain and symptoms of epidural compression syndrome; (3) back pain with symptoms suggesting an infectious cause; (4) back pain with gross muscle weakness or paralysis; and (5) back pain with severe or progressive neurologic deficits.
- When a critical or emergent diagnosis is strongly suspected, MRI and spine surgery consultation should be undertaken emergently.
- Adherence to published guidelines will decrease improper laboratory studies and imaging, thereby lowering costs, improving ED throughput and improving overall patient care.
- Conservative therapies are recommended for most patients with nonemergent back pain given the high likelihood for recovery.

1. What is the most likely cause of back pain in a 48-year-old patient with bilateral leg pain and weakness, urinary retention, decreased rectal tone, and saddle anesthesia?

a. Abdominal aortic aneurysm
b. Bone metastasis
c. Epidural abscess
d. Herniated disk
e. Primary bone tumor

Answer: d. The listed symptoms and signs describe cauda equina syndrome, which is usually caused by a large central herniated disk. Other less common causes include tumor and infection.

2. What percentage of asymptomatic persons less than 60 years old will have herniated disc findings on MRI?

a. 10%
b. 25%
c. 50%
d. 75%

Answer: b. Disc disease is a component of normal aging and is a very nonspecific finding. In fact, one in four asymptomatic persons <60 years and one in three >60 years will have MRI findings of a herniated disc.

3. A 35-year-old man presents with severe back pain that radiates down his right leg. He reports that while lifting a heavy box at work 2 weeks ago, he felt a "pop" in his lower back. He has not been able to return to work since the injury occurred. The patient spoke with his lawyer and was told to come directly to the emergency department to get an magnetic resonance imaging (MRI). He denies having any other symptoms and reports no significant past medical history. During the physical examination, the patient is asked to lie on his back, with his knees extended. His right leg is elevated and, at 50 degrees, he reports severe pain running down the lateral aspect of his right leg to his foot. The patient is then asked to sit with his knees flexed and legs hanging over the side of the bed. His legs are passively extended, with no production of pain. The remainder of the physical examination is normal. What is the most appropriate next step in managing this patient?

a. Computed tomography (CT) of the lumbar spine
b. Discharge home
c. Emergent neurosurgical consultation
d. MRI of the lumbar spine
e. Radiography of the lumbar spine

Answer: b. Several aspects of this scenario point to malingering. The most convincing relates to the physical examination findings. The straight leg raise (SLR) is the classic test for sciatic nerve root irritation. The absence of a positive result generally rules out nerve root irritation. To perform the SLR, the patient is positioned supine, knee extended, and leg elevated until pain is elicited. A positive result is pain radiating down the leg below the knee in a dermatomal distribution when the leg is elevated between 30 and 70 degrees. In a patient who may be malingering, the SLR can be performed with the patient sitting on the side of the bed with knees flexed. Passively straightening the legs in this position should produce equally positive results if true nerve root irritation exists.

4. Disk herniation with involvement of the L5 nerve root will present with which of the following findings?

a. Decreased or absent ankle jerk
b. Decreased patellar reflex
c. Diminished sensation of the lateral small toe
d. Impaired plantar flexion
e. Weakness with extension of the great toe

Answer: e. Involvement of the L5 nerve root presents with weakness, with extension of the great toe, decreased sensation in the first web space, and normal reflexes. An S1 radiculopathy is characterized by diminished sensation of the lateral small toe, impaired plantar flexion, and decreased or absent ankle jerk. The patellar reflex is associated with L2–4.

5. A history of IV drug use increases the risk for which of the following causes of acute back pain?

a. Abdominal aortic aneurysm
b. Epidural hematoma
c. Malignancy
d. Transverse myelitis
e. Vertebral osteomyelitis

Answer: e. Vertebral osteomyelitis and spinal epidural abscess are diagnosed most frequently in an at-risk population that includes patients with a history of diabetes, chronic renal failure, IV drug use, alcoholism, cancer, AND recent surgery or trauma.

6. Spinal epidural abscess is most commonly caused by which of the following pathogens?

a. *Mycobacterium tuberculosis*
b. *Pseudomonas aeruginosa*
c. *Staphylococcus aureus*
d. *Staphylococcus epidermidis*
e. *Streptococcus pyogenes*

Answer: c. *S. aureus* causes 70% of spinal epidural abscesses.

32 Multiple Trauma

KEY CONCEPTS

- Immediately after a trauma patient arrives the emergency department (ED), the primary survey should be performed in a standardized fashion. The goal of the primary survey is to rapidly identify and initiate the treatment of critical, and life-threatening injuries.
- The extended Focused Assessment with Sonography for Trauma (eFAST) examination should be performed early in the evaluation of the trauma patient (ideally as part of the primary survey). Thoracic examination of the trauma patient by ultrasound is more accurate than plain radiography.
- Any patient with potentially life-threatening injuries should have blood typing and screening performed. When transfusion is indicated, blood products should be transfused in a 1:1:1 or 1:1:2 ratio of plasma to platelets to packed red blood cells.
- Tranexamic acid (TXA) is indicated for patients with evidence of significant hemorrhage or shock and is given as a 1-g intravenous bolus followed by a 1-g infusion over 8 h. Results are best if started within an hour of injury but benefit may occur when it is given within 3 h.
- Special consideration should be given to the elderly who demonstrate increased morbidity and mortality in settings where traumatic injury initially appear less concerning. Patients over the age 65 with one or more of the following comorbid conditions (coagulopathy, cirrhosis, COPD, CAD, or DM) have more than twice the mortality risk than younger patients.

1. A 33-year-old mother and her 2-year-old-son are brought in by paramedics after they were both hit by a car moving at 15 mph. Although mother and child had an identical mechanism of injury, the son would be at greater risk for all the following injuries, with the following exception:
   a. Head injury
   b. Hypothermia
   c. Intra-abdominal injury
   d. Multisystem injury
   e. Posttraumatic stress disorder

Answer: e. Injury patterns can differ significantly between adults and children subjected to similar mechanisms of trauma. The major anatomic distinctions relate to the smaller size and surface area, larger head-to-body ratio, and less protected abdominal cavity of the child. As a result, children are more vulnerable to multisystem injury in blunt trauma, including significant head and intra-abdominal injuries, as well as being at greater risk for hypothermia.

2. Which of the following is the goal of the primary survey?
   a. Determine which consultations should be obtained.
   b. Do an AMPLE (allergies, medications, past medical history, last meal, environments and events) history.
   c. Obtain pertinent historical data from the paramedics.
   d. Perform a radiographic evaluation.
   e. Rapidly identify critical life-threatening diagnoses and begin treatment at the time of the diagnosis.

Answer: e. The emergency clinician should use a standardized approach to the initial evaluation of these patients. Following the Advanced Trauma Life Support (ATLS) algorithm of ABCs in the primary survey allows the timely identification of critical diagnoses and intervention without delay. The primary survey should be performed in a standardized fashion immediately after the patient arrives in the emergency department. The goal of the primary survey is to identify critical, life-threatening diagnoses rapidly and begin treatment at the time of diagnosis. The goals of the secondary survey are to obtain pertinent historical data about the patient and injury as well as evaluate and treat injuries not found on the primary survey. An AMPLE history should be obtained.

3. An 89-year-old man who was a restrained front seat passenger with a history of hypertension, anxiety disorder, and dementia is being evaluated after a head-on collision. His home medications include an angiotensin-converting enzyme (ACE) inhibitor for the hypertension, lorazepam for the anxiety, and olanzapine (Zyprexa) for the dementia. The patient does not have any complaints but is noted to have a blood pressure of 80/50 mm Hg and heart rate of 100 beats/min. In evaluating the patient, you should suspect that the asymptomatic hypotension is most likely due to which of the following?
   a. Antihypertensive medication use
   b. Antipsychotic medication use
   c. Benzodiazepine use
   d. Blood loss

Answer: d. Lower extremity weakness, gait disturbances, decreased visual acuity, and the use of psychotropics, antihypertensives, and sedatives have been associated with falls in older adults, resulting in major injury. The use of these medications, particularly antihypertensives, should not be considered causative in trauma patients with hypotension until acute hemorrhage has been ruled out. In addition, anticoagulants, antiplatelet drugs, and aspirin are commonly prescribed, and their effects should be suspected and reversed, if warranted.

4. A severely injured, hypotensive trauma patient is being considered for permissive hypotension because she has a contained retroperitoneal hematoma and is not actively hemorrhaging. In permissive hypotension, the mean arterial pressure is restored to a goal of 50 mm Hg. Which of the following should help you decide against using permissive hypotension?
   a. Age >80 years
   b. Age <10 years
   c. Associated traumatic brain injury
   d. Hemoglobin of 10 g/dL
   e. Intoxication

Answer: c. In the severely injured, hypotensive trauma patient, restoration of normal blood pressure may be undesirable. The concept of permissive hypotension is based on the concern that resuscitation to normal blood pressures may increase bleeding from a site that is contained and not actively hemorrhaging. In permissive hypotension, the mean arterial pressure (MAP) is restored to a goal of about 50 mm Hg. Studies have shown that this strategy leads to less blood product use, less bleeding, and lower incidence of coagulopathy. However, the provider should be aware that permissive hypotension is contraindicated in the management of traumatic brain injury because of the risk of hypoperfusion.

5. An 87-year-old male with history of atrial fibrillation on anticoagulation, COPD, hypertension, and hyperlipidemia fell from a standing height sustaining a pelvic fracture requiring blood transfusion. Which of the following should raise concern for increased mortality in geriatric trauma patients?
   a. Anticoagulation
   b. Age >85
   c. Isolated pelvic fracture
   d. COPD
   e. All of the above

Answer: e. All of the above can contribute to both morbidity and mortality of geriatric trauma patients. Often times with less severe injury patterns, this population will have increased transfusion needs, complications due to hospitalizations, immobilization, and prolonged rehabilitation times. Patients with one or more of: coagulopathy, cirrhosis, COPD, CAD, or DM have shown twice the risk of death in the setting of trauma. Additionally, elderly patients who sustain a pelvic ring injury are prone to increased mortality with those >85 years at highest risk.

6. A 52-year-old female involved in a motor vehicle collision (MVC) at 7 pm is transferred to a Level 1 trauma center. She arrives at 11 pm and despite appropriate isotonic resuscitation and remains hypotensive. Transfusion protocol is initiated. What is the next best intervention?
   a. Administer tranexamic acid (TXA) 1 g over 10 minutes
   b. Administer TXA 1 g over next 8 hours
   c. Repeat primary survey
   d. None of the above
   e. All of the above

Answer: c. As with any transferred trauma patient any non-response or change in condition should prompt rapid re-evaluation.

33 Head Trauma

KEY CONCEPTS

- *Head trauma* is a broad term describing an external trauma to the craniofacial area of the body from blunt, penetrating, blast, rotational, or acceleration-deceleration forces. The term *head injury* refers to a clinically evident injury on physical examination, and the term *brain injury* indicates an injury to the brain itself.
- Traumatic brain injury (TBI) is often categorized into mild (Glasgow Coma Scale score, 13–15), moderate (GCS score, 9–12), and severe (GCS score, 3–8), but this actually represents a spectrum of injury. Patients with a presentation GCS score of 13–15 who are stable or improving are unlikely to have CT scan findings that warrant intervention.
- The motor component of the GCS is the strongest predictor of outcome following TBI.
- Secondary systemic insults such as hypoxia and hypotension worsen neurologic outcome and should be corrected as soon as detected.
- A noncontrast head CT scan is the imaging modality of choice when TBI is suspected.
- Anticonvulsant prophylaxis with phenytoin or levetiracetam and broad-spectrum antibiotics should be given to patients with penetrating brain injuries for 7 days postinjury.
- Patients can deteriorate from an expanding intracranial hematoma after a mild traumatic brain injury (MTBI), and should undergo serial evaluations, including GCS scoring.
- An MTBI can be easily overlooked when an alert patient presents with distracting injuries. Specifically, assess for disorientation, confusion, amnesia, or disordered awareness (with or without loss of consciousness).
- Imaging of patients with MTBI should follow a validated guideline, such as the Canadian CT Head Rule and New Orleans Criteria. Emergency clinicians should select the system most applicable for their setting and patient population.
- Intoxicated individuals are high-risk patients. Alcohol and drug use affect the GCS score and may significantly obscure the neurologic examination.
- Most patients with MTBI can be discharged from the emergency department (ED) with a normal examination and after a reasonable period of observation (4–6 h) or following a negative CT scan of the head.
- Patients should be discharged with instructions describing the signs and symptoms of acute and delayed complications of MTBI. All discharge instructions should be relayed to a responsible third party.
- Athletes with a concussive head injury should be immediately removed from play and not return until they have been evaluated by a health care provider with expertise in concussion management. There should be a gradual stepwise increase in physical activity.
- Older adults (age greater than 60 years) may have significant intracranial injuries and not show signs of deterioration, especially if their baseline cognitive functioning is impaired; clinicians should therefore have a low threshold for obtaining CT scans with these patients.
- Falls in older adults, including low-mechanism falls, should prompt emergency clinicians to consider the possibility of brain injury.

1. The primary goal in the emergency care of a head trauma patient is prevention or reduction of secondary insults that are known to worsen outcome after traumatic brain injury (TBI). This includes acutely preventing:
   a. Hypotension
   b. Hypoxia
   c. Malnutrition
   d. All of the above
   e. Hypotension and hypoxia

Answer: e. Avoiding or reversing hypotension and hypoxia is essential in the acute management of TBI patients in the emergency department. Hypotension and hypoxia each double the risk of mortality following TBI.

2. Which of the following statements is true regarding cerebral blood flow (CBF), cerebral perfusion pressure (CPP), and intracranial pressure (ICP)?
   a. CBF autoregulation is lost below a CPP of 60 mm Hg.
   b. CPP closely parallels diastolic blood pressure.
   c. Normal ICP is 65 to 195 mm Hg.
   d. CPP = mean arterial pressure (MAP) – ICP.
   e. The only resistance to CBF is ICP.

Answer: d. CBF depends on CPP, which is the blood flow pressure gradient. CBF resistance is provided by mean systemic venous pressures and ICP, predominantly by the latter. CPP closely parallels MAP offset by ICP; thus, the formula CPP = MAP – ICP. ICP is estimated clinically by the central venous pressure unless a ventricular catheter is in place and ICP can be directly determined. CBF autoregulation is lost below a CPP of 40 mm Hg. Normal ICP is 5 to 15 mm Hg or 65 to 195 mm H_2O.

3. Which of the following parameters are associated with a worsened outcome after traumatic brain injury (TBI)?
   a. Hematocrit (Hct) <30%
   b. Temperature >38.5°C (101.3°F)
   c. Partial pressure of oxygen (Po_2) <60 mm Hg
   d. A and C
   e. All of the above

Answer: e. All of the above. The following are associated with worsened outcomes after TBI:
Hematocrit (Hct) <30%
Temperature >38.5°C (101.3°F)
Systemic blood pressure (SBP) <90 mm Hg
Po_2 <60 mm Hg

4. A 27-year-old man presents after a motor vehicle collision (MVC) with a severe closed head trauma. On examination, you calculate a Glasgow Coma Scale (GCS) score of 5 and a left dilated pupil, with a sluggish pupillary reflex compared with the right. What other finding will your examination likely reveal?
   a. Left carotid bruit
   b. Left foot weakness
   c. Loss of controlled pain/temperature sensation
   d. Right carotid bruit
   e. Right-sided hemiparesis

Answer: e. Uncal herniation is the most common posttraumatic herniation syndrome. The initial pressure compresses the third cranial nerve (CN III), causing ipsilateral pupillary sluggishness, ptosis, anisocoria, and impaired extraocular movements. Contralateral hemiparesis can develop early after an initial normal motor examination. In some cases, the contralateral uncus is compressed, resulting in ipsilateral weakness (Kernohan notch syndrome).

5. A 24-year-old man presents with a closed head injury after a motor vehicle collision (MVC). The physical examination is remarkable for a sluggish left pupil, right-sided hemiparesis, and a Glasgow Coma Scale (GCS) score of 12. What should be the most appropriate next step in this patient's management?
   a. 3% hypertonic saline IV
   b. Intubation and moderate hyperventilation
   c. Mannitol, 1 g/kg IV
   d. Methylprednisolone IV
   e. Pentobarbital IV

Answer: b. The most rapid effect on intracranial pressure (ICP) reduction is achieved via intubation and moderate hyperventilation to a P_{CO_2} of 30 to 35 mm Hg. The effect peaks within minutes and should be considered a short-term intervention, with an expected ICP reduction of 25%. Prolonged hyperventilation, however, is dangerous. Steroids may worsen outcome after traumatic brain injury (TBI). Mannitol is generally efficacious and exerts an effect within minutes and lasts hours. Other neuroprotective effects include volume expansion, viscosity reduction, cerebral blood flow (CBF) improvement, and free radical scavenging. Hypertonic saline data are encouraging but inconclusive; stronger data exist in the pediatric literature. Barbiturates exert a modest ICP-lowering effect because of their lowering of the cerebral metabolic rate and oxygen demand.

6. What is the minimum time after becoming asymptomatic that an individual should refrain from playing sports after a concussion if no loss of consciousness (LOC) or prolonged posttraumatic amnesia occurred at the time of injury?
   a. 24 hours
   b. 48 hours
   c. 1 week
   d. 2 weeks
   e. 1 month

Answer: c. All current recommendations for return to play after a sports-related concussion state that players with concussion should not return to play for at least 1 week after they have become asymptomatic. This is usually increased to at least a symptom-free month for an extended LOC or prolonged posttraumatic amnesia occurring at the time of the concussion.

7. A 15-year-old boy presents after being hit in the head with a baseball. He has a Glasgow Coma Scale (GCS) score of 7 and a large hematoma of his scalp, anterior and superior to his right ear. In addition, he is noted to have unequal pupils and a sluggish papillary light reflex of the right eye. Which of the following is most likely in this patient?
   a. Central transtentorial herniation
   b. Cerebellotonsillar herniation
   c. Downward transtentorial herniation
   d. Uncal herniation
   e. Upward transtentorial herniation

Answer: d. Uncal herniation is often associated with traumatic extra-axial hematomas in the lateral middle fossa of the temporal lobe. The classic signs and symptoms are caused by compression of the ipsilateral uncus of the temporal lobe on the U-shaped edge of the tentorium cerebelli as the brain is forced through the tentorial hiatus. As compression of the uncus begins, CN III is compressed; anisocoria, ptosis, impaired extraocular movements, and a sluggish pupillary light reflex develop on the side ipsilateral to the expanding mass lesion. This phase may last for minutes to hours, depending on how rapidly the expanding lesion is changing. As the herniation progresses, compression of the ipsilateral oculomotor nerve eventually causes ipsilateral pupillary dilation and nonreactivity.

Initially, in the uncal herniation process, the motor examination can be normal, but contralateral Babinski responses develop early. Contralateral hemiparesis develops as the ipsilateral peduncle is compressed against the tentorium. With continued progression of the herniation, bilateral decerebrate posturing eventually occurs; decorticate posturing is not always seen with the uncal herniation syndrome. In some patients, the contralateral cerebral peduncle is forced against the opposite edge of the tentorial hiatus. Hemiparesis is then detected ipsilateral to the dilated pupil and mass lesion, termed *Kernohan notch syndrome,* and causes false-localizing motor findings. As uncal herniation progresses, direct brainstem compression causes additional alterations in the LOC, respiratory pattern, and cardiovascular system. Mental status changes may initially be subtle, such as agitation, restlessness, or confusion, but soon lethargy occurs, with progression to frank coma. The patient's respiratory pattern may initially be normal, followed by sustained hyperventilation. With continued brainstem compression, an ataxic respiratory pattern develops. The patient's hemodynamic status may change, with rapid fluctuations in blood pressure and cardiac conduction. Herniation that is uncontrolled progresses rapidly to brainstem failure, cardiovascular collapse, and death.

8. Central pontine myelinolysis is a potential adverse event associated with the administration of which of the following medications?
   a. Etomidate
   b. Hypertonic saline
   c. Mannitol
   d. Methylprednisolone
   e. Pentobarbital

Answer: b. Central pontine myelinolysis is a potentially adverse event associated with hypertonic saline administration.

9. What percentage of mild traumatic brain injury (MTBI) patients discharged from the emergency department (ED) will continue to have post concussive symptoms at 3 months and 1 year after injury?
   a. 5% at 3 months and 10% at 1 year
   b. 40% at 3 months and 50% at 1 year
   c. 30% at 3 months and 15% at 1 year
   d. 5% at 3 months and 5% at 1 year
   e. None of the above

Answer: c. Studies of ED patients have indicated that as many as 30% of patients with a discharge diagnosis of MTBI will have symptoms at 3 months postinjury, and up to 15% will continue to be symptomatic at 1-year postinjury.

10. Per the Canadian Computed Tomography Head Rule (CCHR) which criteria are NOT considered "High-Risk" for intracranial injury and possible neurologic intervention?
    a. A Glasgow Coma Scale (GCS) score that is persistently <15 after 2 hours
    b. Suspected open or depressed skull fracture
    c. Vomiting 2 or more times
    d. Age over 50 years
    e. Any sign of basal skull fracture

Answer: d. All are high risk except D. High-risk are patients age ≥65 years. Other criteria that suggest important brain injury on CT include amnesia before impact ≥30 min and dangerous mechanism (pedestrian struck by vehicle, occupant ejected from vehicle, fall from elevation >3 feet [five stairs]).

Facial Trauma 34

KEY CONCEPTS

- The face is central to the patient's ability to breathe, eat, and communicate. Injuries to the face can have serious psychological and psychosocial consequences.
- Facial injuries may be prevented by the appropriate use of safety devices including motor vehicular seat belts, child restraints, air bags, helmets, and mouth and face guards.
- The epidemiology of facial injury is changing, with an increasing proportion of injuries occurring as a result of interpersonal violence. A careful history is required, and the possibility of abuse should be considered for every patient.
- Shock from facial trauma is rare and results only from obvious external bleeding. Facial injuries should not distract the emergency clinician from searching for other causes of shock.
- Assertive management of the airway is indicated in a patient with severe facial injuries. Surgical management (cricothyroidotomy) may be required, particularly with penetrating gunshot wounds.
- Directed facial CT scanning is the optimal imaging technique in patients with obvious injuries.
- Definitive facial treatment may be delayed, if necessary, to address other serious injuries.

1. A nasal fracture complication that requires urgent treatment is:
- **A.** Periorbital swelling
- **B.** Septal hematoma
- **C.** Laceration
- **D.** Epistaxis
- **E.** Subcutaneous edema

Answer: B. A septal hematoma is a collection of blood within the nasal septum that can occur following trauma. A septal hematoma should be drained promptly to avoid the risk of developing a septal abscess or ischemia of the septal cartilage leading to a saddle nose deformity. With the use of an anesthetic, the hematoma can be aspirated or incised, drained, and packed to avoid reoccurrence.

2. A patient with a mandibular fracture may complain of:
- **A.** Trismus
- **B.** Epistaxis
- **C.** Paresthesias of the inferior eyelid, nose, and upper lip
- **D.** Ocular muscle entrapment
- **E.** Postnasal drip and sweet taste in the mouth

Answer: A. Individuals with mandibular fractures may report pain with chewing, abnormal bite, jaw malalignment, paresthesias of the lip and chin area, trismus, oral bleeding, and dental injuries. An orthopantomogram (Panorex) has 92% sensitivity in the diagnosis of a mandibular fracture; however, when there is high clinical suspicion or concern for associated traumatic injuries a facial CT scan should be obtained. Epistaxis can be seen with isolated nasal fractures as well as maxillary fractures. Postnasal drip and a sweet taste in the mouth could be related to a CSF leak, which can be seen in frontal and nasal bone fractures. Ocular muscle entrapment can be related to orbital and midface fractures.

3. A patient presents following a physical altercation complaining of eye pain, decreased vision, and swelling. On physical examination he is noted to have periorbital ecchymosis, swelling, and tenderness, an irregular, tear-shaped pupil, and a collection of blood in the anterior chamber of the eye. Which of the following diagnostic or treatment options should be avoided?
- **A.** CT imaging of the face
- **B.** Ophthalmologic consultation
- **C.** Prophylactic antibiotics
- **D.** Thorough eye examination including tonometry
- **E.** Medications for pain and nausea and place patient in semirecumbent position

Answer: D. The patient is presenting with a history of blunt trauma and examination findings consistent for a globe rupture. A thorough history including mechanism and anticoagulant use should be obtained. CT imaging should be performed to look for additional traumatic injuries. The patient should be medicated for pain and nausea to avoid any increase in intraocular pressures resulting in possible aqueous fluid loss. Prophylactic antibiotics are initiated to prevent secondary endophthalmitis. The affected eye should not be manipulated, a rigid shield should be used for protection, and ophthalmology should be emergently consulted.

4. Cribriform plate fractures can be associated with:
- **A.** LeFort I fracture
- **B.** CSF rhinorrhea
- **C.** Dental alveolar fractures
- **D.** Ageusia
- **E.** Tripod fractures

Answer: B. Cribriform plate fractures are commonly associated with severe traumatic injuries including TBI and CSI. Symptoms can include tenderness, swelling, bleeding, CSF rhinorrhea, and anosmia. More complex midface fractures, LeFort III, or significant nasal trauma affecting the NOE complex are associated with fractures of the cribriform plate. ATLS guidelines should be followed when evaluating this patient. CT imaging including head, C-spine, and face are obtained and a CTA to evaluate for blunt cerebrovascular injury is often performed. Appropriate specialists (neurosurgery) should be consulted and involved in the patient's care.

35 Spinal Trauma

KEY CONCEPTS

- NEXUS or CCR decision rules may be used to determine the need for radiographic imaging in the awake, evaluable trauma patient.
- CT scanning is preferred over plain radiography in the evaluation of the trauma patient with potential spinal injury, especially if axial imaging is needed for evaluation of other injuries.
- Suspicion for an anterior cord syndrome warrants prompt neurosurgical consultation because it is a potentially surgically correctable lesion.
- Neurogenic hypotension is a diagnosis of exclusion in the trauma victim and should not be considered the cause of hypotension unless the presence of coexisting hemorrhagic shock, cardiac tamponade, or tension pneumothorax has been eliminated.
- Because neurogenic hypotension can lead to hypoperfusion and secondary spinal cord ischemia, prolonged, severe hypotension (systolic blood pressure < 70 mm Hg) should be prevented and treated.
- Methylprednisolone or any other corticosteroid is not beneficial in the treatment of acute spinal cord injury (SCI) and should not be used.
- Emergency department management of SCI includes care to prevent further cord injury, pressure ulcers, bladder distention, and gastric distention.

1. Which of the following is a stable cervical spine fracture?
- **A.** Anterior atlantoaxial dislocation without fracture
- **B.** Bilateral facet dislocation
- **C.** Flexion teardrop
- **D.** Jefferson fracture
- **E.** Unilateral facet dislocation

Answer: E. See Table 35.1 for a classification of spinal injuries according to stability.

2. A 28-year-old man is brought to the emergency department (ED) after a rollover motor vehicle collision (MVC). He is moderately hypotensive, unable to flex his elbows, and has diffuse lower extremity paralysis. What is the likely site of the lesion?
- **A.** C4
- **B.** C5
- **C.** C6
- **D.** C7
- **E.** C8

Answer: C. See Table 35.2

3. Which of the following statements regarding high-dose methylprednisolone after spinal cord injury is true?
- **A.** Dexamethasone is superior to methylprednisolone if a concurrent closed head injury is present.
- **B.** High-dose methylprednisolone should not be used to treat spinal cord injury.
- **C.** It is efficacious after penetrating injury if given within 4 hours.
- **D.** It is efficacious in cases of spinal shock.
- **E.** It is more efficacious after thoracic than after lumbar injuries.

Answer: B. Evidence that high-dose methylprednisolone is a clinically efficacious intervention in the management of acute, blunt, partial spinal cord injury is lacking and, because of severe side effects, it should not be used.

4. A 27-year-old man presents after a high-speed rollover MVC. The physical examination is remarkable for T8 motor-sensory deficit and a moderately distended abdomen that is nontender. Vital signs are heart rate, 108 beats/min, blood pressure, 88/40 mm Hg, respiratory rate, 22 breaths/min, temperature, 35°C (95°F), and oxygen (O_2) saturation, 96%. Which of the following tests or treatments is indicated?
- **A.** Baseline laboratory tests and observation
- **B.** Computed tomography (CT) scan of the abdomen
- **C.** Intravenous phenylephrine infusion
- **D.** Packed red blood cell transfusion
- **E.** Thoracolumbar spine films

Answer: B. Spinal cord injury often renders the abdominal examination unremarkable. CT, ultrasonography, diagnostic peritoneal lavage, or some combination is necessary to rule out intra-abdominal injury.

5. A 23-year-old man presents after a rollover MVC. He is brought by emergency medical services (EMS) from the scene fully restrained on a backboard, with a cervical collar. The physical examination is remarkable for moderate symmetric numbness below the neck, symmetric arm and leg weakness, intact reflexes, and diminished rectal tone. Vital signs are heart rate, 94 beats/min, blood pressure, 80/46 mm Hg, respiratory rate, 24 breaths/min, and O_2 saturation, 96%. Which of the following treatment sequences is most indicated?
- **A.** Crystalloid to CT scan to phenylephrine
- **B.** Crystalloid to focused assessment with sonography for trauma (FAST) examination to transfusion
- **C.** Crystalloid to phenylephrine to transfusion
- **D.** Dopamine to crystalloid to CT scan
- **E.** Transfusion to phenylephrine

Answer: B. Spinal shock should not be considered the cause of hypotension unless the patient is flaccid and areflexic. Crystalloid is the first step regardless of the traumatic hypotensive cause. The possibilities of coexisting hemorrhagic shock, cardiac tamponade, tension pneumothorax, or other life-threatening injuries should first be addressed. The absence of vasomotor activity in patients with neurogenic hypotension may mask the usual presentation of these life-threatening injuries. In this case, the lack of flaccidity and presence of reflexes argues for a nonneurogenic cause for the hypotension.

6. Which of the following is not a characteristic of flexion-distraction injuries of the lumbar spine?
- **A.** Compression of the anterior vertebral body
- **B.** Associated intrabdominal injuries are common
- **C.** Axis of rotation anterior to the vertebral body
- **D.** Unstable fracture pattern

Answer: C. Flexion distraction injuries are characterized by an axis of rotation posterior to the anterior longitudinal ligament causing compression of the anterior vertebral body and distraction of the posterior elements with failure of all three columns. It is an unstable fracture pattern with a high rate of associated intra-abdominal injuries.

7. Presence of which of the following findings warrants imaging to evaluate for blunt cerebrovascular injury?
- **A.** Isolated seat belt abrasion
- **B.** LeFort I fracture
- **C.** Frontal scalp hematoma
- **D.** Cervical bruit in 40-year-old extricated after MVA

Answer: D. A cervical bruit auscultated in a patient under the age of 50 is a sign of possible blunt cerebrovascular injury and requires dedicated arterial imaging. A seat belt sign with significant swelling, pain or altered mental status is indication for screening for BCVI, but not an isolated seat belt abrasion. Midface fractures with instability, LeFort grades II and III, are associated with BCVI. Isolated scalp hematoma is not associated with an increased risk of BCVI.

36 Neck Trauma

KEY CONCEPTS

- Significant injuries to the cerebral vessels can occur as a result of blunt or penetrating trauma to the neck.
- Blunt trauma to the neck can lead to immediate or delayed ischemic strokes, and a low threshold for imaging vessels when neurologic findings or risk factors for cervical vessel injury exist is recommended.
- A no-zone approach to penetrating neck trauma is becoming widely accepted as opposed to a zone-based selective management algorithm.
- Multidetector CT angiogram is the initial imaging modality of choice in essentially all symptomatic cases of blunt or penetrating neck trauma.
- Both penetrating and blunt neck injuries can present with minimal to no initial signs of injury; a high index of suspicion is therefore required.
- Because of the paucity of neck trauma in children, many adult algorithms have been adopted for use in the pediatric population

1. The presence of which of the following signs after penetrating neck trauma would indicate a likely benefit from surgical intervention?
   a. Decreased or absent radial pulse
   b. Small degree of hemoptysis
   c. Horner syndrome
   d. Muffled voice
   e. Stable hematoma

Answer: a. "Hard" signs of penetrating neck trauma are the presence of an expanding hematoma, severe active bleeding, shock not responding to fluids, decreased/absent radial pulse, vascular bruit/thrill, cerebral ischemia, and airway obstruction. Most patients with hard signs benefit from surgical intervention.

2. Which of the following is true regarding pharyngoesophageal injuries?
   a. Esophageal barium swallow is the best initial x-ray modality for evaluating esophageal injuries.
   b. Gastrografin has greater sensitivity than barium and is therefore the preferred contrast agent in the initial evaluation of esophageal injuries.
   c. Confirmatory barium study is not necessary after a negative Gastrografin swallow study.
   d. Computed tomography angiogram (CTA) is the initial study of choice in evaluating pharyngoesophageal injuries and never requires additional confirmatory studies
   e. None of the above is true.

Answer: e. CTA is a reasonable exam with which to begin evaluation of a patient suspected to have PE injury but confirmatory studies in the setting of an equivocal CTA are often required. Esophageal barium swallow has greater sensitivity than Gastrografin but is associated with risk of delayed mediastinal fibrosis in the event of contrast extravasation. Most literature supports a confirmatory barium study in the setting of negative water-soluble imaging.

3. The pathophysiology and sequelae of hanging include all of the following except
   a. Pulmonary edema
   b. Hypoxic –ischemic brain injury
   c. Cardiogenic pulmonary edema as a result of Takotsubo cardiomyopathy
   d. Neurogenic pulmonary edema from massive parasympathetic discharge
   e. Presence of Tardieu spots

Answer: d. Pulmonary edema can occur from several mechanisms: neurogenic pulmonary edema from centrally mediated, massive *sympathetic* discharge; post-obstructive from relief of the marked negative intrapleural pressure generated by forceful inspiratory effort against an extrathoracic obstruction; and cardiogenic pulmonary edema, which is increasingly recognized as a result of hanging-associated Takotsubo cardiomyopathy.

4. Which of the following statements is true regarding zonal injuries to the neck?
   a. Zone 2 is the most commonly injured, has the highest mortality and is most amenable to surgical exploration
   b. Zone 1 extends from the sternal notch to the cricoid cartilage
   c. Penetrating trauma injuries to the neck often span multiple zones, and the most significant injury may lie in a different zone than the point of entry
   d. Current consensus supports a "no-zone" approach which uses clinical examination coupled with computed tomography angiogram (CTA) to arrive at the best diagnostic and therapeutic pathways
   e. All of the above are true.

Answer: e. Despite decades of relying on a zone-based selective algorithm, current consensus supports a "no-zone" approach which uses clinical examination coupled with CTA, not the zone of injury, to arrive at the best diagnostic and therapeutic pathways

5. Which of the following statements regarding airway management after penetrating neck trauma is true?
   a. Awake fiberoptic intubation is the first-line technique.
   b. Bag-valve-mask ventilation should be high tidal volume, low rate.
   c. Cervical spine immobilization is typically unnecessary.
   d. Nasotracheal intubation is relatively contraindicated in neck trauma.
   e. Preintubation nasogastric tube (NGT) placement may be lifesaving.

Answer: c. Unless there is concomitant blunt injury or evidence of spinal cord injury, cervical immobilization is not needed. Oral intubation after rapid sequence intubation (RSI) is the technique of choice and is successful in almost all cases. Although rarely a first line choice, nasotracheal intubation has been used successfully in these trauma patients. Gentle bag-valve-mask technique with low pressure is indicated to avoid venous air embolism (VAE) and subcutaneous emphysema. NGT placement, if done at all, should be gentle and placed after intubation.

37 Thoracic Trauma

KEY CONCEPTS

- Even relatively minor chest wall injuries, such as rib fractures, may result in serious complications in elderly patients and patients with preexisting pulmonary disease if adequate analgesia and close follow-up care are not provided.
- Unless there are abnormalities on the electrocardiogram (ECG) or an elevated serum troponin level, there is no need to pursue the diagnosis of myocardial contusion with more sophisticated testing.
- Many patients with myocardial rupture or traumatic aortic rupture survive to reach the hospital and can be salvaged with rapid diagnosis and intervention.
- Pericardial tamponade can be diagnosed accurately before hemodynamic decompensation occurs by standard cardiac ultrasound performed by emergency clinicians.
- Chest computed tomography (CT) scan is the diagnostic test of choice for blunt aortic injury even in the presence of normal chest radiographs.
- The NEXUS-Chest CT criteria can be used to determine the need for chest CT in patients with blunt trauma.

1. An 18-year-old man presents after a motor vehicle collision (MVC) in which he was ejected from the vehicle. The paramedics have been administering bag-valve-mask ventilation en route because of respiratory distress and now report increased resistance with ventilations. The patient has decreased breath sounds on the left. His blood pressure is 80/40 mm Hg, and his pulse is 145 beats per minute. His respirations are agonal, with a rate of 5 breaths per minute. Which of the following is the most appropriate next step in the management of this patient?

a. Anteroposterior chest radiograph
b. Emergency department thoracotomy (EDT)
c. Endotracheal intubation
d. Needle decompression
e. Tube thoracostomy

Answer: d. This clinical scenario depicts a patient with a tension pneumothorax. He has decreased blood pressure, decreased breath sounds, and, most important, an increased resistance to ventilation, which is the earliest sign of the development of a tension pneumothorax. Immediate decompression with a large-bore needle is the correct initial management in this condition.

2. Which of the following is the most sensitive electrocardiogram (ECG) manifestation of myocardial contusion?

a. Biphasic T wave
b. Left bundle branch block
c. Right bundle branch block
d. Sinus tachycardia
e. Transient ST segment elevation

Answer: d. Sinus tachycardia is present in approximately 70% of patients with documented myocardial contusion and is the most sensitive sign for this condition. It is, however, also the least specific.

3. A patient presenting with blunt thoracic trauma complains of shortness of breath and chest pain. On physical examination, he is tachypneic with chest wall bruising and moist rales on the right side on auscultation. Which of the following is the *least* likely finding?

a. Consolidation within 6 hours of injury
b. Diffuse patchy alveolar infiltrates on chest radiograph in 24 hours
c. Low partial arterial pressure of oxygen (Pao_2) on arterial blood gas sampling
d. Patchy alveolar infiltrate on chest radiograph within minutes of injury
e. Rib fractures

Answer: b. This patient has physical findings consistent with pulmonary contusion. All answers are correct except B. Delayed onset of diffuse alveolar infiltrates is more consistent with acute respiratory disease syndrome (ARDS). The development of ARDS is diffuse and usually delayed, with onset typically between 24 and 72 hours after injury.

4. A 55-year-old man complains of chest wall pain after a high-speed motor vehicle collision (MVC). He has ecchymosis of the left lateral chest wall. You notice that there is outward movement of the left lateral chest wall on expiration. Which of the following statements regarding this patient's problem is *not true?*

a. Chest radiograph likely demonstrates parenchymal contusions.
b. Intubation splints the chest wall internally.
c. Multiple rib fractures are likely.
d. Positioning of patient with injured side down improves symptoms.
e. The cornerstone of treatment is pulmonary physiotherapy.

Answer: d. This patient has a flail chest. Out-of-hospital or emergency department (ED) stabilization of the flail segment, by positioning the person with the injured side down or placing a sandbag on the affected segments, has been abandoned. Endotracheal intubation and positive-pressure ventilation will internally splint the chest wall, making the flail segment difficult to detect on physical examination. The cornerstones of therapy include pulmonary physiotherapy, effective analgesia, selective use of endotracheal intubation and mechanical ventilation, and close observation for respiratory compromise.

5. A 50-year-old woman is brought in by emergency medical services on a backboard after a motor vehicle collision (MVC), complaining of shortness of breath. She has decreased breath sounds on the right side of the chest. A chest tube is placed, with a return of 200 mL of blood in the first hour, 200 mL in the second hour, and 350 mL in the third hour. What is the next step in the management of this patient?

a. Check coagulation profile
b. Conservative management and transfusion as needed
c. Emergency thoracotomy
d. External fixation of rib fractures
e. Insertion of a second thoracostomy tube

Answer: c. Immediate drainage of more than 1500 mL of blood from the pleural cavity is usually considered an indication for urgent thoracotomy. Perhaps even more predictive of the need for thoracotomy is a continued output of at least 200 mL/h for 3 hours.

6. A 37-year-old man presents with chest pain after a motor vehicle collision (MVC). He states that his chest hit the steering wheel. On initial evaluation, the patient is without fractures of the ribs or sternum, but there is a small chest wall contusion. The initial chest radiograph is negative, and the electrocardiogram (ECG) shows nonspecific ST-T wave changes. What is the next step in the emergency management of this patient?

a. Admit to telemetry for 23-hour observation
b. Discharge home after repeat ECG and troponin at 6 hours
c. Discharge home after repeat ECG in 1 hour
d. Discharge home if echocardiogram is negative
e. Discharge home if initial troponin is negative

Answer: b. In patients who have minor injuries and are otherwise asymptomatic, elevated troponin levels and minor ECG abnormalities do not necessarily indicate a clinically significant myocardial contusion. Very few of these patients will develop complications. However, normal troponin level (4 to 6 hours after injury), along with normal (or unchanged) ECGs, correlate with minimal risk of cardiac complications. Echocardiography is rarely required in this low-risk subset of patients who have minor injuries and are otherwise asymptomatic.

7. A 30-year-old woman presents intubated by emergency medical services on a backboard with C-spine immobilization. She was found unresponsive and hypotensive after a front-end collision. She was the driver of the vehicle and unbelted. Despite fluid resuscitation, the patient continues to be tachycardic and hypotensive. On physical examination, you note jugular venous distention (JVD) and a harsh murmur that sounds like a splashing mill wheel. An electrocardiogram (ECG) reveals electrical alternans. Which of the following statements is true regarding the patient's underlying problem?
   a. Echocardiogram will reveal diastolic collapse of the right ventricle and fluid in the pericardium.
   b. Focused assessment with sonography in trauma (FAST) will demonstrate echogenicity in Morrison's pouch.
   c. Patchy consolidation will be seen on chest radiograph.
   d. Pericardiocentesis is not indicated.
   e. She will have a negative focused abdominal sonogram.

Answer: A. Early use of emergency department (ED) ultrasonography may facilitate the early diagnosis of cardiac rupture and pericardial tamponade. The combination of shock and JVD (or an elevated central venous pressure [CVP]) in a patient with blunt chest trauma should immediately suggest pericardial tamponade.

38 Abdominal Trauma

KEY CONCEPTS

- The accuracy of a physical examination is limited in cases of abdominal trauma. It is rendered less reliable by distracting injury, altered sensorium (e.g., head trauma, alcohol or drug intoxication, developmental delay, psychiatric illness), and spinal cord injury.
- Stab and gunshot wounds frequently violate the lung parenchyma, diaphragm, mediastinum, intraperitoneal cavity, and retroperitoneum in some combination.
- Physical examination with extended focused assessment with sonography for trauma (e-FAST), followed by computed tomography (CT) scan when indicated, provides accurate diagnosis for the majority of blunt and penetrating abdominal trauma patients.
- Emergent laparotomy is indicated for patients sustaining a stab wound in the setting of hemodynamic compromise, peritoneal signs, evisceration, or left-sided diaphragmatic injury. Patients not meeting these criteria should undergo a combination of e-FAST, local wound exploration (LWE), CT scan, and serial examination, depending on the location of the wound
- Emergent laparotomy is indicated for patients sustaining a gunshot wound in the setting of hemodynamic compromise, peritoneal signs or evisceration. Patients not meeting these criteria should undergo a combination of e-FAST, CT scan, and serial examination, then proceed to laparotomy or admission for observation pending those results.
- The critical determinant in hemodynamically unstable patients with pelvic fracture is the existence of active intraperitoneal hemorrhage. Diagnosis of this by e-FAST, CT scan, or peritoneal aspiration is an indication for laparotomy, whereas its absence prompts diagnostic and potentially therapeutic angiography.

1. The extended focused assessment with sonography for trauma (e-FAST) scan of a patient with blunt abdominal trauma shows a hypoechoic stripe in the pouch of Douglas. Which of the following is correct?
   - **a.** In the presence of hemodynamic instability, this indicates a need for laparotomy.
   - **b.** The patient needs to go for emergent laparotomy immediately.
   - **c.** The patient requires repeat abdominal examinations and FAST examinations in the emergency department (ED).
   - **d.** There is at least 500 mL free fluid in the abdomen.
   - **e.** This indicates a ruptured bladder.

Answer: a. The pouch of Douglas is one of the areas of ultrasound inspection for an e-FAST examination. If free fluid is present and the patient is hemodynamically unstable, the patient should forego computed tomography (CT) scanning for the operating room. FAST examinations are effective in detecting as little as 100 mL of free fluid in the abdominal cavity.

2. An 18-year-old man presents after a high-velocity front-end vehicle collision. He has a blood pressure of 90/70 mm Hg, heart rate of 120 beats/min, respiratory rate of 17 breaths/min, and a Glasgow Coma Score (GCS) of 13. On physical examination, he has a tender abdomen and an unstable pelvis. An extended focused assessment with sonography for trauma (e-FAST) examination is positive for free fluid in the abdomen. What should be the patient's disposition?
   - **a.** Admission to the trauma service for observation
   - **b.** Emergency department (ED) observation for 12 hours with repeat FAST examinations
   - **c.** Interventional radiology for pelvic angiography and embolization
   - **d.** Operating room for emergency laparotomy

Answer: d. The e-FAST examination is an excellent bedside tool in guiding care in unstable patients with pelvic fractures. If the e-FAST exam shows intraperitoneal free fluid, the patient should be taken to the operating room emergently for laparotomy. If the e-FAST exam is negative and no other readily identifiable source of hypotension is present, emergent pelvic angiography and embolization in interventional radiology (IR) is appropriate.

3. A 27-year-old male presents 4 hours after an isolated stab wound to the anterior abdomen. His vital signs are heart rate 84 beats/min and blood pressure 115/64, and the lactate level is 0.9 mg/dL. His extended focused assessment with sonography for trauma (e-FAST) examination is negative for free fluid. He denies alcohol and drug use and appears clinically sober. Which of the following statements regarding this patient's subsequent management is true?
   - **a.** A negative computed tomography (CT) scan rules out the need for further evaluation.
   - **b.** If local wound exploration (LWE) definitively demonstrates that the wound does not violate the peritoneum, the patient can be discharged from the emergency department (ED).
   - **c.** The negative e-FAST examination rules out intra-abdominal injury requiring operative intervention.
   - **d.** The patient meets criteria for emergent laparotomy.

Answer: b. Simple anterior abdominal stab wounds that do not violate the peritoneum can be discharged from the ED after appropriate wound care (see Fig. 38.6). CT scans poorly visualize both the small bowel and diaphragm and cannot be used in isolation to rule out injury in penetrating abdominal trauma. A negative e-FAST examination does not rule out significant intra-abdominal injury or even small-volume hemoperitoneum in either penetrating or blunt abdominal trauma. Not all anterior stab wounds meet indication for laparotomy, even in the presence of peritoneal penetration.

4. A 67-year-old female who is taking warfarin for atrial fibrillation presents after a high-mechanism motor vehicle collision. Her heart rate is 142 beats/min and blood pressure is 84/40 after 1 L of normal saline. Her Glasgow Coma Score (GCS) is 6, and her left pupil is 6 mm versus 3 mm on the right. Her physical examination is notable for a seat belt sign on the abdomen. Which of the following is *not* an acceptable approach to her initial assessment and treatment?
   - **a.** Perform an extended focused assessment with sonography for trauma (e-FAST) examination to evaluate for the presence of intra-abdominal fluid.
   - **b.** Perform chest and pelvic radiographs in the resuscitation bay.
   - **c.** Perform empirical craniotomy concurrently with laparotomy in the operating room after a positive diagnostic peritoneal aspiration.
   - **d.** Perform endotracheal intubation and begin mild hyperventilation.
   - **e.** Proceed to radiology for an emergent abdominal computed tomography (CT) scan.

Answer: e. The patient is hemodynamically unstable, with suspected intra-abdominal injuries in conjunction with signs of herniation. CT scanning of the abdomen would be inappropriate in this patient. Several concurrent management options to stabilize the patient and determine the source of her hypotension are desirable (see Fig. 38.10). Endotracheal intubation allows airway control and possible hyperventilation to delay impending herniation. Chest and pelvic radiographs rule out other sources of ongoing hemorrhage and may support emergent laparotomy. An e-FAST examination can confirm the presence of intra-abdominal free fluid, which in the setting of hemodynamic instability is an indication for laparotomy. Finally, after confirmation of intra-abdominal blood by e-FAST, proceeding to the operating room, with or without emergent CT scanning of the head, are management options. If the head CT is foregone because of instability, empirical craniotomy is an acceptable management option.

5. Which of the following statements regarding splenic injuries in blunt abdominal trauma is true?
   - **a.** A computed tomography (CT) scan with a grade IV splenic laceration indicates the need for laparotomy.
   - **b.** Angiographic embolization has no role in high-grade splenic lacerations.
   - **c.** Bedside extended focused assessment with sonography for trauma (e-FAST) can accurately rule-out splenic injury.
   - **d.** CT scanning followed by serial abdominal examinations and hematocrits is a reasonable management option for certain splenic injuries at experienced centers.
   - **e.** Mononucleosis increases the risk of splenic laceration from seemingly minor blunt abdominal trauma.

Answer: d. Serial abdominal examinations, laboratory tests, or repeat e-FAST examinations are reasonable management options at experienced centers. High-grade splenic lacerations, although having a higher rate of failed nonoperative or angiographic embolization management, do not represent a definitive indication for laparotomy. Angiographic embolization may preserve some of the immune function of the spleen, even in the setting of high-grade splenic lacerations. Bedside e-FAST, although sensitive for intra-abdominal fluid, is a relatively poor test for the evaluation of solid organ injury. Mononucleosis does increase the risk of splenic laceration. Trauma can be so minor that the patient may have little recollection of the remote trauma responsible.

39 Genitourinary Trauma

KEY CONCEPTS

- Microscopic or gross hematuria is suggestive of genitourinary trauma; however, the degree of hematuria does not correlate well with either the severity or anatomy location of injury.
- The kidney is the most frequently injured genitourinary organ, and imaging should be considered in patients with gross hematuria or microscopic hematuria with hemodynamic instability.
- Most noniatrogenic ureteral injuries are due to penetrating mechanisms and are unilateral; blunt ureteral trauma is rare. Delayed computed tomography (CT) images after intravenous (IV) contrast should be obtained in patients with mechanism or findings suggestive of ureteral trauma.
- CT scan with IV contrast is not sensitive for diagnosing bladder injury, and retrograde cystography should be obtained to allow proper distention of the bladder to allow for urinary extravasation.
- Pelvic fractures associated with hematuria strongly suggest bladder or urethral injury.
- Retrograde urethrography (RUG) should be performed in patients with pelvic fractures and hematuria, those with perineal ecchymosis or swelling, or those with blood at the urethral meatus, prior to attempting bladder catheterization, because blind passage of a Foley catheter in these scenarios can worsen a preexisting urethral injury.
- Genital injury is rarely life threatening but prompt diagnosis and evaluation are necessary to decrease the likelihood of future morbidity in these patients.

1. Which of the following scenarios associated with clinically significant blunt renal injuries in adult patients would the *least likely* to warrant further evaluation?

a. A sudden decelerating mechanism of injury in a patient without microhematuria
b. A sudden decelerating mechanism of injury in a patient without shock
c. Gross hematuria
d. Microscopic hematuria
e. Microscopic hematuria in a patient with shock

Answer: d. Renal injury requiring intervention is rare in the absence of gross hematuria or shock. Gross hematuria warrants careful evaluation for significant genitourinary injury, although the degree of hematuria does not necessarily correlate with the degree or grade of injury and significant genitourinary trauma can occur without hematuria. Microscopic hematuria in a blunt trauma patient without shock is not an indication for renal imaging, even if there is evidence of local trauma (e.g., costovertebral angle tenderness or localized ecchymosis). Although renal injury may uncommonly be identified on imaging for these patients, the injuries are mild and do not require intervention. Hemodynamic instability with evidence of intraperitoneal injury on abdominal examination, presence of pelvic fracture, a penetrating trauma mechanism, or presence of lower rib fractures, with or without hematuria, are indications for further investigation. In addition, imaging is advisable for patients with gross hematuria, targeted to the entire urinary tract or localized to the lower tract (bladder and urethra), depending on the mechanism and location of the trauma.

2. A 27-year-old male presents after a motor vehicle collision complaining of abdominal pain. He was the restrained driver of a car struck on the driver's side by a delivery truck. His vital signs are blood pressure 118/72, heart rate 70 beats/min, and respiratory rate 16 breaths/min. Physical examination reveals left upper quadrant abdominal tenderness without guarding or rebound tenderness. There is no blood at the urethral meatus or scrotal hematoma. A Foley catheter is placed without difficulty and drains gross blood. A radiograph of the pelvis reveals fractures of the left superior and inferior pubic rami, and a focused assessment with sonography in trauma (FAST) examination reveals free fluid in the splenorenal pouch. Which of the following diagnostic strategies is most appropriate?

a. Intravenous (IV) contrast-enhanced computed tomography (CT) of the abdomen and pelvis with delayed images of the bladder after clamping the Foley catheter to allow the IV contrast to collect in the bladder
b. IV contrast-enhanced CT of the abdomen and pelvis with delayed images of the renal collecting system
c. IV contrast-enhanced CT scan of the abdomen and pelvis and retrograde CT cystogram
d. IV contrast-enhanced CT of the abdomen and pelvis and retrograde urethrogram
e. Retrograde cystogram followed by IV pyelogram

Answer: c. This patient has clinical features concerning for several possible injuries, including a splenic laceration, renal injury, and bladder rupture. Significant urethral injury is less likely, given the examination findings and ease of Foley catheter placement. Because the patient is stable without apparent indication for laparotomy, the most appropriate diagnostic evaluation would be IV contrast-enhanced CT of the abdomen and pelvis and retrograde CT cystogram. The former will evaluate for solid organ injury and the latter for bladder rupture. It is essential that cystography not be obtained in an antegrade fashion, because such studies (e.g., injecting IV contrast material, clamping the Foley catheter, and allowing the examination to depend on antegrade filling of the bladder from renal excretion of progressively dilute contrast material) may produce incomplete and spurious findings because of inadequate distention of the bladder.

3. A 72-year-old male presents with flank and pelvis pain after he slipped and fell on an icy sidewalk. His examination reveals normal vital signs and abrasions over the left flank and left iliac crest and is otherwise unremarkable. He has grossly clear urine, but a urinalysis reveals 25 red blood cells per high-power field. Radiographs of the pelvis and hips reveal no fracture, and the patient is able to ambulate without difficulty. What is the most appropriate next step?

a. Obtain a renal ultrasound scan.
b. Obtain an IV contrast-enhanced CT of the abdomen and pelvis.
c. Perform a retrograde cystogram.
d. Perform a retrograde urethrogram.
e. Treat the patient's pain and discharge him home with outpatient urology follow-up in 1 week.

Answer: e. A significant genitourinary injury is unlikely, given the patient's history, physical examination, and urine findings. However, outpatient urology follow-up until microhematuria has cleared is advisable to be certain that it does not represent another more serious underlying (nontraumatic) condition.

4. A 35-year-old female presents after being stabbed with an ice pick during a robbery. Her examination is normal except for a 0.5-cm, hemostatic wound to the right flank at the level of the second lumbar vertebrae. Bedside ultrasonography reveals no free intra-abdominal fluid. Her urinalysis does not contain blood. What is the most appropriate next step?
   a. Obtain a renal ultrasound scan.
   b. Obtain an IV contrast-enhanced CT scan of the abdomen and pelvis, with additional images of the renal collecting system 10 minutes after contrast injection.
   c. Perform a retrograde cystogram.
   d. Perform a retrograde urethrogram.
   e. Treat the patient's pain, counsel her on appropriate wound care, and discharge her home with outpatient urology follow-up in 1 week.

Answer: b. In cases of penetrating renal trauma, the presence or absence of hematuria is not a reliable predictor of upper urinary tract injury. The location of the penetrating injury in relation to the urinary tract is the most important determining factor in deciding the need for radiographic investigation. Therefore the absence of hematuria in a patient with a gunshot or stab wound in proximity to the urinary tract does not eliminate the need for IV contrast-enhanced CT as the initial diagnostic examination. Significant injuries to the kidney and ureter may occur in penetrating trauma without hematuria. Additional images obtained at 10 minutes after contrast injection are indicated to evaluate for delayed contrast extravasation and to maximize the sensitivity of the study.

5. A 25-year-old male presents after an unfortunate incident resulting in a scrotal injury and a left testicle that was traumatically amputated (the patient brings the amputated testicle with him in a towel). Which of the following would be *least* indicated?
   a. Analgesics for the patient
   b. Emergent urologic consultation
   c. Place the amputated testicle directly on ice
   d. Prepare the patient for possible operative exploration by the surgeon
   e. Wrap the amputated testicle in saline soaked gauze

Answer: c. Amputations of either the testicles or penis require immediate surgical evaluation. In the interim, the amputated part should be wrapped in saline-soaked gauze, and then placed in a sealed bag, which can then be placed in another bag that is filled with ice. The amputated part should never be placed directly on ice. Direct pressure is usually adequate to achieve hemostasis.

6. A 30-year-old female presents after blunt abdominal trauma after a motor vehicle collision, and an ultrasound demonstrates free fluid. You suspect a bladder rupture. A true statement about this entity includes which of the following:
   a. A bladder contusion may be successfully managed with Foley catheterization drainage.
   b. Extraperitoneal bladder injuries are typically managed with surgical intervention.
   c. It is less important to distinguish an intraperitoneal from an extraperitoneal bladder rupture (EBR) because the management options are essentially the same.
   d. Most patients with an intraperitoneal bladder rupture (IBR) will resolve with Foley catheter drainage.
   e. Most penetrating bladder injuries will resolve with Foley catheter drainage.

Answer: a. The distinction between EBR and IBR is important because the management differs. Contusions and extraperitoneal injuries due to blunt trauma are typically managed conservatively with Foley catheterization, unless they are complicated by other intra-abdominal injuries, bladder neck injuries, or bone fragments in the bladder wall or if open reduction is performed on an associated pelvic fracture. In contrast, given the extremely low likelihood of IBR and penetrating injuries healing with conservative therapy, almost all patients with these types of injuries are taken to the operating room for exploration and repair. Without surgery, there is an extremely high likelihood of complications, which include infections and fistula formation.

Peripheral Vascular Trauma 40

KEY CONCEPTS

- The overall condition of the patient with an acute peripheral vascular injury determines the extent of emergency department (ED) evaluation and stabilization. Critically injured patients may require immediate surgery, which should not be delayed for confirmatory studies of obvious vascular injury.
- Arterial injury may be readily apparent or clinically occult. In patients with high-energy blunt mechanisms, computed tomography angiography (CTA) should be the initial diagnostic modality of choice. In patients with lower-energy mechanisms, serial physical examinations may be performed instead.
- Symptoms of arterial injury may be delayed by hours to months after the initial injury. Late onset of symptoms suggests delayed thrombosis, pseudoaneurysm or arteriovenous fistula (AVF) formation, compartment syndrome, or intermittent claudication, resulting from stenosis or reliance on small-caliber collateral vessels for arterial perfusion.
- Compartment syndrome frequently develops in extremities with arterial injury, particularly injuries of the lower leg, and fasciotomy is often required.
- Many vascular injuries are amenable to endovascular treatment with self-expanding stents. This results in fewer complications, lower cost, and earlier discharge from the hospital.

1. An 18-year-old man presents complaining of left lower leg pain. He was involved in a motor vehicle collision (MVC) 1 month earlier and required surgical repair of a left femur fracture. Which of the following is the *least* likely cause of the patient's symptoms?
   a. Compartment syndrome
   b. Intimal flap
   c. Pseudoaneurysm
   d. Thrombosis
   e. Vessel stricture formation

Answer: a. Delayed thrombosis can occur months to years after injury if the injured vessel heals with stricture formation and decreased blood flow distally, followed by stasis and clot formation. Although flow is not altered by small intimal flaps and the associated soft tissue wounds often appear benign initially, they may become a nidus for thrombosis that can occur hours to months after the initial injury. The cavity of a pseudoaneurysm is in direct communication with the lumen of the vessel, so embolization of mural clots may produce distal arterial occlusion. Patients with pseudoaneurysm are commonly seen months to years later with symptoms of compression neuropathy or peripheral arterial embolism. The delayed presentation of this patient makes compartment syndrome the least likely cause of his leg pain.

2. A 45-year-old man complains of right leg pain and edema shortly after surgery for a midshaft femur fracture. Physical examination reveals decreased distal pulses in the extremity. What is the most important diagnosis to consider in this patient?
   a. Anemia causing poor tissue oxygenation
   b. Compartment syndrome
   c. Intimal flap
   d. Pseudoaneurysm
   e. Vessel stricture

Answer: b. After restoration of arterial flow to a previously ischemic limb, a cascade of reperfusion injury has been identified that results from release of oxygen free radicals, lipid peroxidation, and influx of intracellular calcium. These mediators give rise to progressive cellular damage, edema, and necrosis, thereby propagating the vicious cycle that increases compartment pressure.

3. After vascular injury to an extremity, how many hours are generally required before irreversible damage occurs?
   a. 2 hours
   b. 6 hours
   c. 10 hours
   d. 12 hours
   e. 24 hours

Answer: b. Although individuals may vary, 6 hours of complete warm ischemia is generally considered the point at which irreversible nerve and muscle damage begins to occur. After 6 hours of warm ischemia, 10% of patients will have irreversible damage; by 12 hours, 90% will have irreversible damage. Artificially cooling the limb to just higher than freezing temperature will reduce the metabolic demands of ischemic tissues and greatly prolong the tissue's tolerance of ischemia to 24 hours or more.

4. A 25-year-old man who sustained a gunshot wound to the right leg has an ankle-brachial index (ABI) of 0.9. What is the most appropriate next step in this patient's management?
   a. Admit for observation and repeat examinations.
   b. Discharge the patient home with surgical follow-up.
   c. Obtain an emergent computed tomography (CT) angiogram.
   d. Obtain immediate vascular consultation.
   e. Perform a Doppler ultrasound scan.

Answer: a. Patients with an ABI of 0.90 to 0.99 merit observation for 12 to 24 hours for repeat examinations and ABI measurements to detect evolving injury. In general, a ratio of less than 0.90 is considered abnormal and is an indication for further investigation, such as computed tomography angiography (CTA).

5. Which of the following is *least* likely to cause vascular injury to an extremity?
   a. Close-range shotgun wound
   b. Crush injury
   c. Electrical injury
   d. Gunshot wound from a long distance
   e. Massive tissue avulsion

Answer: d. Close-range shotgun wounds can cause significant blunt trauma to vessels, as well as a higher rate of bone and nerve damage than gunshot wounds. An open avulsion injury to a limb is particularly severe because the skin is the final structure to be disrupted, and there is inevitable injury to deeper vessels and nerves. Vascular injury must be suspected in patients with massive soft tissue avulsion or crush injury, displaced long bone fractures, and electrical or lightning injuries.

41 General Principles of Orthopedic Injuries

KEY CONCEPTS

- Many orthopedic injuries can be stabilized and treated definitively by the emergency clinician. Consultation with an orthopedist should be sought for the treatment of some long bone fractures, open fractures, injuries with joint violation, tendon injuries, and injuries with neurovascular compromise.
- A careful history and physical examination can predict radiographic findings in orthopedic injuries with a high degree of accuracy.
- Open fracture management should focus on the early administration of antibiotics, tetanus prophylaxis, dressing coverage of the wound, and splinting. Suggested therapy for open fractures includes a first-generation cephalosporin, with the addition of an aminoglycoside for type II or III fractures.
- Compartment syndrome is associated most commonly with a closed long bone fracture of the tibia but also is well described in the thigh, forearm, arm, hand, and foot and can occur with isolated soft tissue trauma. Clinical examination remains the diagnostic cornerstone of acute compartment syndrome, which may then be confirmed by compartment pressure measurement.
- Because of the anatomic location and blood supply distribution, certain bones may undergo avascular necrosis after fracture, especially if fractures are comminuted and go untreated for any length of time. The femoral head, talus, scaphoid, and capitate are particularly prone to this complication.
- Fat embolism syndrome is a serious consequence of fractures, occurring most commonly after long bone leg fractures in young adults and after hip fractures in older patients. Acute respiratory distress syndrome is the most serious manifestation. Neurologic involvement, manifesting as restlessness, confusion, and deteriorating mental status, as well as thrombocytopenia resulting in a petechial rash are early signs.
- Patients should be evaluated for the ability to ambulate safely prior to discharge. The use of a walker or other device, if it can be used safely, is acceptable. It also must be assured that the patient or a caregiver can carry out essential activities of daily living.

1. A 40-year-old diabetic woman is brought to the emergency department (ED) after crashing her car into a tree at moderate to high speed. She has a minimally displaced trimalleolar fracture and no other injuries after full body scanning and evaluation by the trauma team. While awaiting admission to the floor she develops blood-filled fracture blisters over the anterior ankle. What is the next step?
   a. Debride blisters and apply silver sulfadiazine cream
   b. Arrange for urgent surgery within 8 hours
   c. Cover with sterile dressing and elevate foot and ankle
   d. Measure the compartment pressure in the ankle

Answer: c. Fracture blisters sometimes follow high-energy injuries, especially when the overlying skin is thin and there are few hair follicles or sweat glands. Debridement and application of silver sulfadiazine is sometimes used when blood is not present. It is contraindicated in the presence of diabetes. Conservative treatment with a sterile dressing is recommended and surgery through the involved area should be delayed.

2. A patient presents with severe pain in the leg despite the removal of a cast 3 days after suffering a closed tibia fracture. Which of the following requires an emergent fasciotomy?
   a. A high clinical suspicion based on extreme pain and tenseness in the compartment
   b. The absence of pulses detected with a hand-held Doppler stethoscope
   c. Tissue pressure less than the diastolic pressure
   d. Tissue pressure of 20 mm Hg

Answer: a. Compartment syndrome cannot be diagnosed on the basis of specific tissue pressure. The tolerance to tissue ischemia varies among individuals because of shock, compensatory hypertension, altered tone in resistance vessels, and other unknown factors. Inadequate perfusion and relative ischemia begin when the tissue pressure within a closed compartment increases to within 20 mm Hg of a patient's diastolic pressure or, more accurately, within 30 mm Hg of the mean arterial pressure. When tissue pressure equals or exceeds the patient's diastolic pressure, tissue perfusion effectively ceases. Clinical judgment is paramount. Delayed diagnosis and fasciotomy frequently result in high damage claim awards.

3. An intoxicated alcoholic man is brought into the ED after being found on a bench and is found to have a wrist drop. Which is the likely cause of the wrist drop?
   A. Axonotmesis
   B. Neurotmesis
   C. Neuropraxia
   D. Increased compartment pressure

Answer: c. The patient likely has a contusion of the radial nerve; axonotmesis and neurotmesis involve more severe injury such as crushing or severing. Isolated neurologic dysfunction is not a feature of compartment syndrome.

41.4.)A 60-year-old woman presents to the emergency department (ED) with a closed deformity to the wrist as shown in the figure (show old figure 42.4 in text). In discussing this injury to a consultant, which is the best description?
   a. Compound fracture of the wrist
   b. Severely angulated fracture of the wrist
   c. Fracture of the distal radius with dorsal displacement and volar angulation
   d. Fracture of the distal radius with severe anterior displacement

Answer: c. Fractures should be described using the most precise language possible. The injury is a closed injury, so it is not compound. Fracture displacement in the forearm or wrist should be described as volar and dorsal, and occasionally as radial or ulnar (as the forearm is viewed in the anatomic position). Descriptions should convey as much information as possible.

5. A 50-year-old nonprofessional weightlifter presents after sudden pain in the upper arm described as a "pop." There is bunching up of the biceps muscle and bruising at its insertion. Which drug class is commonly associated with this condition?
   a. Nonsteroidal antiinflammatory drugs (NSAIDs)
   b. Angiotensin converting enzyme inhibitors
   c. Fluoroquinolones
   d. Statins

Answer: c. The patient appears to have rupture of the biceps brachialis tendon. Tendinopathy is associated with use of fluoroquinolones but not NSAID use, hypertension, or hyperlipidemia.

42 Hand Injuries

KEY CONCEPTS

- For general testing of the motor nerve function of the hand, have the patient make an "OK" sign with their thumb and index finger (testing the median nerve), while spreading/abducting the third, fourth, and fifth fingers (ulnar nerve) and dorsiflexing the wrist (radial nerve).
- For general sensory exam, test two-point discrimination on the finger tufts of the index finger (median nerve), the little finger (ulnar nerve), and the dorsum of the first webspace (radial nerve).
- The most important treatment for the majority of hand injuries is applying an appropriate splint. The neutral position for general fractures of the hand or fingers is achieved by placing a volar splint ideally with the wrist at 20 to 30 degrees extension, MCP at 70 to 90 degrees flexion, and the PIP and DIP kept in extension. This is also known as intrinsic plus position.
- The most common extensor tendon injury is a terminal tendon disruption from sudden flexion of the extended DIP joint resulting in a *mallet injury.*
- A flexor tendon injury should be considered when an injured finger does not assume its naturally flexed position on *cascade sign* testing.
- Traditionally, patients with acute open wounds on the hand have been treated prophylactically with antibiotics, but there is little scientific data supporting the clinical efficacy of this practice.
- For a stable finger tuft with a subungual hematoma without external disruption of the nail plate, trephination without antibiotics is the only treatment needed.
- Small tuft avulsions defined as less than 1 cm^3 can be treated conservatively, while larger amputations should have emergent consultation with a hand surgeon for consideration for flap coverage or reimplantation. The presence of exposed bone or tendon indicates the need for surgical intervention.
- Most hand infections are based on skin flora; treatment for outpatient injuries are oral first-generation cephalosporins. With the growing emergence of community-acquired MRSA, antibiotics such as doxycycline, TMP-SMX, or clindamycin should be considered. For inpatient injuries, IV first-generation cephalosporins or vancomycin is recommended.
- The four classic Kanavel findings for infectious flexor tenosynovitis are fusiform swelling of the digit, tenderness along the tendon sheath, digit held in flexion at rest, and pain with passive extension of the digit.

1. Which of the following demonstrates the motor function of the median nerve?
   a. Extending the hand at the wrist
   b. Making an "OK" sign with the thumb and first finger
   c. Pulling the thumb across the palm to touch the little finger
   d. Spreading out the fingers of the hand

Answer: B. Making an "OK" sign with the thumb and first finger. The median nerve controls the flexors of the thumb and index finger that allow the hand to make the "OK" sign. The ulnar nerve innervates the adductor pollicis that pulls the thumb across the palm so it can touch the little finger and innervates the interosseous muscles to allow the hand to spread out or abduct the fingers. The radial nerve innervates the extensor carpi radialis longus and the other wrist extensors.

2. A 25-year old woman comes to the emergency department complaining of hitting the tip of her finger on a basketball. She has difficulty extending the tip of her ring finger. What is the most appropriate splinting technique for this injury?
   a. Aluminium volar splint with finger in position of function
   b. Dorsal blocking splint with DIP at 15 degrees of flexion
   c. Stack splint with extension of distal phalanx
   d. Ulnar gutter splint to include ring finger to the tip

Answer: C. Stack splint with extension of distal phalanx. The description of the injury is consistent with a mallet finger injury, where the distal extensor tendon has pulled off the proximal end of the distal phalanx. The premolded Stack splint keeps the distal phalanx in extension and can remain in place for 6 weeks for healing. The position of comfort for the finger in an aluminium volar splint is with the PIP and DIP at approximately 10 to 15 degrees of flexion, not complete extension as needed to allow the tendinous insertion to reattach. A straight aluminium splint could be used on the dorsal surface of the finger to keep the DIP joint straight. A dorsal blocking splint is used for flexor tendon lacerations or ruptures, preventing any extension in the digit and hand to decrease the contraction of the flexor tendon, which will increase the extensor tendon strain. The ulnar gutter splint provides much more extensive immobilization than is necessary for this injury.

3. After cutting her hand while washing a drinking glass, a patient complains of a laceration to her finger. In which location of the palmar hand surface is this laceration most likely to cause a lack of flexion to the index finger PIP joint?
   a. Base of the thenar eminence
   b. Distal end of the middle phalanx
   c. Proximal end of the proximal phalanx
   d. Volar crease of the MCP joint

Answer: D. Volar crease of the MCP joint. The flexor digitorum superficialis (FDS) functions to flex the finger at the PIP joint. It attaches at the proximal end of the middle phalanx as well as the distal end of the proximal phalanx. A laceration along the volar crease of the MCP joint will cut the FDS tendon when the fingers are in extension and reduce the flexion of the PIP joint. The flexor digitorum profundus (FDS) inserts at the proximal end of the distal phalanx. Cutting the finger from the mid aspect of the middle phalanx distally results in an injury to the FDP and an inability to flex at the DIP joint. There are no tendons running within the thenar webspace.

4. Which of the following is the most disease-distinguishing finding of a patient with pyogenic flexor tenosynovitis?
   a. Involved finger is held in flexion
   b. Pain on extension of the finger
   c. Tenderness over the tendon sheath
   d. Uniform swelling of the involved finger

Answer: B. Pain on extension of the finger. All of these findings are components of Kanavel's findings for infectious tenosynovitis. Even though uniform involved swelling has been listed as the most common sign in some studies, this is misleading as swelling is the most common sign in many hand problems and is very nonspecific. What most commonly separates purulent flexor tenosynovitis from other finger infections is pain with passive extension. Many patients with even just cellulitis have pain with palpation over the area of infection on the finger, so this does not distinguish it well. Only half of the patients diagnosed with pyogenic flexor tenosynovitis have all of Kanavel's findings.

5. A patient presents after a fist fight with a puncture wound to his hand. Which description of the presentation has the best prognosis?
   a. Fingertip penetration
   b. Delay in treatment
   c. Depth of the wound
   d. Involvement of joint

Answer: A. Fingertip penetration. The three factors that have been found to worsen the prognosis for a "fight bite" injury are delay in identification and treatment of the wound, an increase depth of the wound as it is more likely to include the tendon, and intrusion into the joint of the tooth. A puncture wound of the fingertip is much less likely to create a significant infection due to the close septae that decrease the likelihood of spreading bacteria in an open wound.

43 Wrist and Forearm Injuries

KEY CONCEPTS

- On plain radiographs of the wrist, three distinct arcs, known as *Gilula's lines*, and equal spacing between carpus bones (1–2 mm), known as *parallelism*, assist in the radiographic diagnosis of carpal injuries.
- In the setting of trauma, there is a high incidence of occult fractures and soft tissue injuries of the wrist. Because of the associated risk of malunion, nonunion, posttraumatic arthritis, and avascular necrosis (AVN), splint immobilization is recommended if pain persists despite normal appearing radiographs.
- Routine wrist radiographs which include anteroposterior (AP), lateral, and oblique projections, may fail to detect scaphoid fractures.
- Thumb spica immobilization is recommended for suspected scaphoid and other carpal fractures. Expedited orthopedic follow-up for repeat assessment, radiographs, or advanced imaging (e.g., MRI, CT, or bone scan) is indicated.
- Triquetral dorsal chip fractures are best seen on the standard lateral view of the wrist as a small avulsion fracture fragment, although a more oblique pronated lateral view may be necessary to visualize these types of fractures.
- Hamate and pisiform fractures are best visualized with a carpal tunnel or reverse supinated oblique radiograph.
- Lunate dislocations result in a characteristic triangular appearance of the lunate on the posteroanterior (PA) view (commonly referred to as the "piece of pie" sign) owing to rotation of the lunate in a volar direction. This rotation also is visible on the lateral view of the wrist, where the lunate appears like a cup tipped forward, spilling its contents into the palm (referred to as the "spilled teacup" sign).
- A Colles fracture is a transverse fracture of the distal radial metaphysis, which is dorsally displaced and angulated. The Smith fracture is a transverse fracture of the metaphysis of the distal radius, with associated volar displacement and angulation.
- Ulna fractures associated with radial head dislocation are commonly known as Monteggia fractures. Galeazzi fractures refer to fractures of the middle to distal third of the radius associated with injury to and dislocation of the distal radial ulnar joint (DRUJ).

1. In the setting of acute trauma and negative radiographs of the wrist, which clinical examination method(s) is (are) used to detect occult scaphoid fractures?
- **a.** Anatomic snuffbox tenderness
- **b.** Scaphoid tubercle tenderness
- **c.** Thumb metacarpal compression tenderness
- **d.** A positive Watson's scaphoid shift test
- **e.** All of the above

Answer: e. In the setting of acute trauma and negative radiographs, tenderness to palpation within the anatomic snuffbox, on the scaphoid tubercle, with thumb metacarpal compression, or a positive Watson's scaphoid shift test are all suggestive of occult scaphoid fracture.

2. A 45-year-old man complains of wrist pain after falling on an outstretched hand. What injury is shown in Figure 43.21 of the patient's wrist?
- **a.** Barton's fracture
- **b.** Lunate dislocation
- **c.** Perilunate dislocation
- **d.** Scaphoid fracture
- **e.** Scapholunate dissociation

Answer: e. The radiograph shows the signet ring, or cortical ring sign, which refers to the rotary subluxation of the scaphoid and oval appearance of the tubercle in the anteroposterior (AP) view of the wrist. On a properly positioned radiograph, this sign is typically associated with scapholunate widening, suggesting ligamentous laxity or dissociation. The signet ring sign is also used to describe pulmonary computed tomography (CT) imaging of bronchiectasis in relation to a dilated bronchus and associated pulmonary artery.

3. A 55-year-old man complains of wrist pain after a fall onto an outstretched arm. There is pain and swelling along the carpal bones. Also noted is decreased two-point sensation distally on the tips of the index and middle digits. Which structure is most likely injured?
- **a.** Median nerve
- **b.** Radial artery
- **c.** Radial nerve
- **d.** Ulnar artery
- **e.** Ulnar nerve

Answer: a. The median nerve courses through the carpal tunnel on the volar aspect of the wrist. It provides sensation to most of the palm and thumb, half of the ring finger and, specifically, the tips of the index and middle digits. The median nerve is the most common neurapraxia associated with Colles fractures.

4. Which nerve is commonly associated with a Monteggia fracture?
- **a.** Muscular branch of the radial nerve
- **b.** Posterior interosseous nerve
- **c.** Deep branch of the ulnar nerve
- **d.** Median nerve
- **e.** Ulnar nerve

Answer: b. Injury to the posterior interosseous nerve (PIN), a deep branch of the radial nerve, is commonly associated with a Monteggia fracture. Because the PIN innervates the finger extensors along with the supinator, associated injury is often manifested by weakness or paralysis of the thumb or finger extension.

5. A karate player presents with pain over the ulnar aspect of his hand. On physical examination, there is tenderness over the ulnar aspect of the wrist, distal to the volar crease. Paresthesias in the distribution of the ulnar nerve are present and the patient also complains of hand clumsiness. Standard X-rays are negative. What additional view might reveal the diagnosis?
- **a.** Clenched fist view
- **b.** Carpal tunnel view
- **c.** Scaphoid view
- **d.** External Oblique view
- **e.** Metacarpal view

Answer: b. Pisiform or hook of hamate fractures may cause the constellation of findings in the above scenario and are better appreciated on the carpal tunnel view. The clenched fist view is best for identifying scapholunate widening while the scaphoid view is optimal for identifying scaphoid fractures. The external oblique view would not optimize imaging of these two bones and the metacarpal view is not an actual view.

Humerus and Elbow Injuries 44

KEY CONCEPTS

- Clinical decision rules for the elbow joint have not been validated. Radiographs should be obtained when there is limitation in range of motion, moderate to severe pain, obvious deformity, joint effusion, or significant tenderness or crepitus over any of the bony prominences or the radial head.
- The threshold for radiographic imaging should be lower in pediatric patients (with the exception of radial head subluxations), owing to the presence of open growth plates and limitations to the physical examination.
- Injuries that result in neurovascular compromise necessitate prompt intervention and consultation with an orthopedic specialist for reduction and potential operative intervention.
- In children with a traumatic wrist injury, normal radiographs should prompt consideration of an elbow injury causing referred pain to the wrist.
- On lateral elbow x-ray, a small anterior fat pad, parallel to the anterior surface of the humerus, can be a normal finding. Any convex ("sail sign") anterior fat pad and all posterior fat pads are pathological and indicate the presence of joint effusion.
- In the setting of trauma, patients with a radiological posterior fat pad sign of the elbow are assumed to have an intra-articular skeletal injury. In adults, a posterior fat pad sign is indicative of a radial head fracture, whereas in children, a supracondylar fracture is more likely. In the absence of trauma, inflammation and infection also cause effusions with positive fat pad signs.
- Radial nerve injury is the most common complication of humeral fractures. This is most often a benign neurapraxia that resolves spontaneously. Radial nerve injuries associated with penetrating trauma or open fractures are likely to represent anatomical disruption requiring operative exploration.
- The radius and ulna, bound together firmly by the annular ligament and interosseous membrane, typically displace as a unit and dislocate posteriorly following a traumatic injury.
- Biceps tendon rupture is more common in men, between ages 40 to 60, resulting from an unexpected extension force applied to the arm flexed at 90 degrees. Smoking, diabetes, chronic renal failure, systemic lupus erythematosus, rheumatoid arthritis, and steroid or fluoroquinolone therapy may predispose to this injury.

1. A 4-year-old boy is brought to the emergency department (ED) by his mother after falling from a swing set. He complains of right wrist pain and resists detailed examination of the arm. Radiographs are normal. What is the most appropriate next step in this patient's management?

a. Computed tomography (CT) scan of the wrist
b. Elbow radiograph
c. Reassurance
d. Splinting for 3 to 6 weeks
e. Triple-phase bone scan after 7 to 10 days

Answer: b. In children with wrist pain and traumatic mechanism of injury, the absence of a clear-cut explanation for the pain (e.g., normal radiograph) should prompt suspicion for an elbow injury producing referred pain.

2. A 3-year-old girl comes in from home with arm pain after being playfully swung by her parents. The child appears uncomfortable and refuses the move her arm. After manipulation and a palpable click, the child appears more comfortable however after 30 minutes will not range her arm for the provider. What is the most appropriate next step for this patient?

a. Obtain an XR to evaluate for fracture
b. Continue to observe the patient with frequent re-evaluations
c. Observe for neurovascular check
d. Discharge home with parents with strict return precautions
e. Consult Orthopedics

Answer: d. In children with radial head subluxation, the child, fearful for eliciting a pain response, may continue to hold their arm in protection while appearing more comfortable. Appropriate comfort and discharge may be appropriate with reliable caregivers and with instructions to return within 24 hours if the child does not return to normal.

3. A 67-year-old right-hand dominant man fell onto his right arm and suffered an injury to his upper arm, with crepitus, gross deformity, and is diagnosed with a midshaft humerus fracture. 30 minutes after placement in a sling begins having difficulty extending the thumb and first finger. Repeat radiography shows improvement in alignment of prior fracture. What is the appropriate action?

a. Adjust splint and discharge to home
b. Do not adjust splint and discharge to home
c. Adjust splint and admit for surgery
d. Do not adjust splint and admit for surgery
e. Admit for compartment checks

Answer: c. The patient is now presenting with a radial nerve injury and palsy, likely after movement of the fracture fragments. The offending manipulation should be undone to potentially improve nerve function, and the patient should then be admitted for operative repair.

4. A 57-year-old man presents with acute left upper extremity pain while suffering an injury doing pullups. The patient has visible deformity of his proximal upper extremity and is splinting his arm and unable to supinate and flex his forearm. The patient is neurovascularly intact. X-rays of the shoulder and humerus are unremarkable. An ultrasound diagnoses a torn biceps tendon. What is the most appropriate treatment plan?

a. Orthopedics consultation for emergent operative repair of the ruptured tendon
b. Splinting and urgent orthopedic follow-up within 48 to 72 hours for operative repair
c. Splinting and follow-up with PCP in 1 week
d. Splinting and discharge with instructions for physical therapy

Answer: b. Patients with a biceps tendon rupture should be evaluated for fractures with plain radiography. If negative, consultation with orthopedics for urgent follow-up with possible operative repair is the most appropriate follow-up plan.

5. A 46-year-old construction worker falls onto an extended arm and has a sudden injury to their elbow. The patient arrives with a gross deformity to the distal humerus/elbow. Radiography reveals a posterior elbow dislocation with a wide humerus-olecranon space. The patient as 1+ radial pulses on the affected arm but is otherwise neurovascularly intact. What is the most appropriate next step?

a. Manual reduction of the elbow dislocation and discharge to follow up with orthopedics
b. Manual reduction of the elbow and a CT angiography to evaluate for brachial artery injury
c. Orthopedics consultation for emergent operative reduction
d. Vascular surgery consultation for management of potential brachial artery injury

Answer: b. The patient likely has a brachial artery injury. These are more common in elbow dislocations with a wide separation between the olecranon and humerus. While there may be persistent pulses distally, the presence of pulses does not eliminate the risk for brachial artery injury. The injury should be reduced, and the patient should undergo CT angiography for evaluation of potential vascular injury.

45 Shoulder Injuries

KEY CONCEPTS

- Axillary nerve function is best evaluated by testing the motor function of the deltoid muscle.
- The three-view trauma series of the shoulder (true anteroposterior, scapular Y, and axillary views) leads to an accurate radiographic diagnosis of most fractures and dislocations, although specialized views are sometimes necessary. Consider the presence of unfused epiphyses in adolescents and young adults.
- Most shoulder girdle fractures can be treated with simple immobilization with good functional outcomes.
- The most important aspect of scapular fractures, scapulothoracic dissociation, and posterior sternoclavicular joint (SCJ) dislocations is the high incidence of associated injuries to the ipsilateral lung, chest wall, mediastinum, or shoulder girdle complex.
- Type III acromioclavicular joint (ACJ) dislocations can be treated conservatively (immobilization, range of motion exercises, and strengthening).
- Recurrence is a common complication after anterior dislocation, especially in male patients younger than 30 years old, and such patients likely benefit from arthroscopic surgical repair.
- Posterior dislocation should be included in the differential diagnosis of any shoulder injury, particularly in patients who report shoulder pain after a seizure.
- Ultrasound can be a useful technique for diagnosing tendon tears, selected soft tissue conditions, fractures, and dislocation, as well as for confirming successful reduction.
- Early initiation of passive shoulder range-of-motion exercises reduces the risk of adhesive capsulitis when the shoulder is immobilized for any reason.

1. A 27-year-old male unrestrained driver is involved in a high speed motor vehicle collision (MVC) with direct impact of the steering wheel into his anterior chest wall. Which shoulder girdle injury represents a true orthopedic emergency?
   a. Anterior glenohumeral dislocation
   b. Anterior sternoclavicular dislocation
   c. Posterior glenohumeral dislocation
   d. Posterior sternoclavicular dislocation

Answer: d. Posterior sternoclavicular dislocations can be associated with life-threatening injuries within the superior mediastinum and intrathoracic cavity; therefore, they are considered true orthopedic emergencies and should be reduced expeditiously. Reported complications associated with posterior sternoclavicular dislocations include compression or lacerations of the great vessels, tracheal compression, pneumothorax, thoracic outlet syndrome, tracheoesophageal fistula, and injury to the brachial plexus.

2. A 30-year-old patient presents complaining of right shoulder pain and limited range of motion. On physical examination, you note that the shoulder has a "squared off" appearance and is held in abduction and external rotation. The patient is unable to adduct the arm or internally rotate without severe pain. What is the best way to manage this patient without the aid of other emergency department (ED) personnel?
   a. Hippocratic method
   b. Kocher maneuver
   c. Milch technique
   d. Traction/countertraction method

Answer: c. The ideal method for reduction of an anterior shoulder dislocation should be simple, quick, and effective; require little assistance; and cause no additional injury to the shoulder. It is wise to be familiar with several techniques of reduction because none is uniformly successful. The Milch technique (see Table 45.3) allows for reduction by a single practitioner, and it can be attempted without procedural sedation. Older techniques, such as the Hippocratic method (traction with the foot in the axilla) and the Kocher maneuver (leverage, adduction, and internal rotation), are no longer recommended because of a high incidence of associated complications (axillary nerve injury, humeral shaft and neck fractures, and capsular damage). Traction/countertraction requires two people to reduce the shoulder (see Table 45.3).

3. A 19-year-old skateboarder presents to the emergency department (ED) complaining of left shoulder pain with his left arm in adduction and supporting the elbow with his right hand. An obvious clavicular defect is palpable on exam. Urgent orthopedic consultation is recommended for which type of clavicle fracture?
   a. Minimally displaced midclavicular fracture
   b. Type I lateral clavicle fracture
   c. Type II lateral clavicle fracture
   d. Type III lateral clavicle fracture

Answer: c. Type I lateral clavicle fractures are stable and minimally displaced because the coracoclavicular ligaments are intact. Type II fractures are associated with a torn coracoclavicular ligament and have a tendency to displace because the proximal fragment lacks any stabilizing forces. Type III injuries involve the articular surface. More urgent orthopedic consultation (before 72 hours) is recommended for type II lateral clavicle fractures, because these fractures have up to a 30% incidence of nonunion and may benefit from surgical repair.

4. A 75-year-old right-handed woman presents to the emergency department (ED) after sustaining a ground level fall onto an outstretched right arm. She is holding her right arm in adduction and supporting the weight of the arm with her left hand. On examination she does not have an obvious palpable defect but there is significant pain in her shoulder with any attempt to move her arm. She has intact lateral deltoid sensation and full symmetric radial pulses. The x-ray demonstrates a proximal humerus fracture with minimal displacement. Which of the following is the most appropriate next step in care?
   a. Emergent orthopedic consultation for operative repair
   b. Pain control, immobilization, and discharge with outpatient follow-up with orthopedics
   c. Orthopedics consultation for outpatient surgical repair after resolution of swelling
   d. Emergent vascular surgery consult due to high risk of associated vascular injury

Answer: b. Proximal humerus fractures with minimal displacement respond well to conservative care. Patients may begin physical therapy as soon as one week after the fracture. Surgical repair is rarely indicated for minimally displaced fractures, hence answers (a) and (c) are incorrect. She has good radial pulses and these fractures are not commonly associated with secondary vascular injury.

5. A 30-year-old recreational basketball player presents after an awkward fall onto an outstretched left arm during a pickup basketball game with severe left shoulder pain. He is supporting his left elbow with his right hand and refuses to move his arm due to pain. There is a dimple inferior to the acromion and diffuse pain with palpation throughout his shoulder. There is no evidence of neurovascular injury. True and standard AP radiographs do not show any fracture or obvious dislocation. The technician was unable to obtain an axillary lateral due to significant pain with abduction. Which of the following is the next best step?
   a. Obtain modified axillary lateral or trans-scapular Y view radiograph
   b. Upper extremity CT with intravenous contrast
   c. Discharge with outpatient follow-up with primary care
   d. Refer to orthopedics for likely complete supraspinatus tear

Answer: a. Posterior dislocations glenohumeral dislocations are uncommon but cannot be excluded with a true orthogonal view. Options to consider in this patient include a modified axillary lateral, apical oblique, or transscapular Y view. A missed posterior dislocation can result in a locked posterior dislocation requiring orthopedic intervention under general anesthesia. If there is a high clinical suspicion, a non-contrast shoulder CT scan can reveal the diagnosis; however, the most appropriate next step would be to obtain orthogonal views. Contrast is only necessary if there is suspicion for an associated vascular injury which is uncommon with posterior GH dislocations. Although an acute rotator cuff tear can present similarly, this patient has had incomplete radiographic imaging and therefore a diagnosis of rotator cuff tear represents premature closure.

46 Pelvic Injuries

KEY CONCEPTS

- The most serious pelvic ring injuries caused by high-energy impact are (1) anteroposterior compression fractures ("open-book" fracture), (2) vertical shear fractures, and (3) fractures involving significant displacement. These injuries are associated with major blood loss and transfusion requirements.
- Pelvic fractures are a marker for serious injury to other organ systems. The vast majority of patients who die after sustaining a pelvic fracture have multiple trauma.
- Careful examination of the perineum and buttocks, as well as digital rectal and vaginal examinations, are necessary to diagnose open fractures and expanding hematomas.
- Computed tomography (CT) is the imaging test of choice to diagnose pelvic fracture and concurrent intra-abdominal injuries for patients stable enough to undergo imaging. CT aids in establishing surgical priorities and planning of definitive orthopedic care.
- There is an increasing incidence of low-energy (e.g., fall from standing) pelvic fractures in older patients which often are managed medically due to multiple co-morbidities and poor bone quality that would make an anatomic reduction difficult to obtain.
- Fragility pelvic fractures in older patients often involve the anterior column of the acetabulum, are more comminuted, and result in severe impaction of the femur head.
- In the hemodynamically unstable patient who cannot undergo CT imaging, the anteroposterior radiograph usually reveals serious pelvic fractures that cause major pelvic bleeding, which is sufficient information to undertake pelvic stabilization.
- The combination of posterior arch fracture plus hypotension is associated with a mortality rate of approximately 50%.
- Early fluid resuscitation with blood products in a 1:1:1 ratio (packed red blood cells: platelets: fresh frozen plasma) is recommended for unstable patients suspected of having active pelvic bleeding.
- Trauma hospitals should have institutional guidelines and mechanisms to facilitate early decisions regarding treatment for pelvic hemorrhage. Treatment options include angiography and embolization, pelvic packing, invasive fixation, or a combination of these.
- Unstable patients with a positive focused assessment with sonography in trauma (FAST) and a pelvic fracture should be treated with laparotomy with pelvic stabilization and possible pelvic packing, followed by angiography.
- Unstable patients with a negative FAST and a pelvic fracture should be treated with pelvic stabilization (such as a pelvic binder) and angiography, and then a repeat FAST and laparotomy if they remain unstable.
- "Open-book" pelvis fractures should be internally compressed with a pelvic binder or sheet to reduce the size of the pelvis, unless the fracture forces have already internally rotated the hemi-pelvis. In this case, further internal compression may cause an increase in the pelvic diameter.
- Patients with pelvic fractures are at high risk for subsequent deep venous thrombosis (DVT), despite the use of mechanical and chemical thromboprophylaxis.

1. While examining a patient with blunt abdominal trauma, you note blood at the urethral meatus. What is the most appropriate next step in the patient's management?
- **a.** Order a cystography
- **b.** Order a retrograde urethrogram
- **c.** Order a urinalysis
- **d.** Order an anterograde urethrogram
- **e.** Order an intravenous pyelography

Answer: b. Blood at the urethral meatus necessitates a retrograde urethrogram followed by a cystogram. Gross hematuria is investigated by a combination of urethrography, intravenous pyelography, cystography, and computed tomography (CT).

2. An avulsion fracture caused by the hamstring muscles most likely involves which of the following structures?
- **a.** Anterior inferior iliac spine
- **b.** Anterior superior iliac spine
- **c.** Iliac crest
- **d.** Ischial tuberosity
- **e.** Sacrum

Answer: d. Forceful contraction of the hamstrings can result in an injury of ischial tuberosity. The anterior inferior iliac spine may be avulsed during strenuous contraction of the rectus femoris muscle. A portion of the iliac crest epiphysis may be avulsed by contraction of the abdominal muscles. The anterior superior iliac spine may be avulsed by forcible contraction of the sartorius muscle.

3. A 25-year-old patient involved in a motor vehicle collision (MVC) presents with a suspected unstable pelvic fracture. The vital signs are blood pressure, 90/50 mm Hg; heart rate, 120 beats/min; and respiratory rate, 22 breaths/min. What is the most appropriate next step in this patient's management?
- **a.** Call trauma service for immediate laparotomy
- **b.** Call orthopedics for fixation
- **c.** Chest and pelvis radiographs
- **d.** Computed tomography (CT) scan of the abdomen and pelvis
- **e.** Perform serial abdominal and focused assessment with sonography in trauma (FAST) examinations

Answer: c. When patients are too unstable to undergo CT investigation, the anterior/posterior portable radiograph is useful in screening for pelvic injuries that are most associated with major blood loss. Findings with this technique that have been reported to predict the need for transfusion include "open-book" fracture, displacement of 0.5 cm or more at any fracture site in the pelvic ring, and displaced symphysis pubis or obturator ring fracture. Chest radiographs should be performed to rule out associated injuries that might be the cause of the patient's hypotension and tachycardia (e.g., hemothorax, tension pneumothorax, widened mediastinum).

4. Which of the following treatments can provide definitive control of hemorrhage from severe pelvic fracture?
- **a.** Blood transfusion
- **b.** Commercial circumferential pelvic compression devices
- **c.** External fixation
- **d.** Transfusion of concentrated clotting factors
- **e.** Wrapping pelvis with sheet

Answer: c. External fixation is an invasive strategy aimed at cessation of pelvic hemorrhage. Options A, B, and E are important for the emergency clinician to quickly stabilize a mechanically unstable pelvic fracture and hopefully minimize bleeding, but these options, per se, are not definitive treatments. Transfusion of blood products is important in overall resuscitation of the patient but does not stop bleeding from pelvic fractures.

5. Which of the following radiographic findings necessitates further evaluation of the pelvic ring?
- **a.** Asymmetry of the pelvis caused by rotation
- **b.** Duverney fracture
- **c.** Sacroiliac joint 4 mm wide on pelvic radiograph
- **d.** Symphysis pubis 7 mm wide
- **e.** Transverse fracture of the sacrum

Answer: d. On the anteroposterior radiograph, the symphysis pubis is normally no more than 5 mm wide, and a small (1 or 2 mm) vertical offset of the left and right pubic rami may be normal. Normally, the sacroiliac joint is approximately 2 to 4 mm wide. On the anteroposterior view, the physician may judge the degree of pelvic rotation caused by technique and positioning by the presence of asymmetry.

47 Femur and Hip Injuries

KEY CONCEPTS

- A hip dislocation is an orthopedic emergency. The likelihood of avascular necrosis (AVN) is related to both the initial degree of trauma and the amount of time the femoral head remains out of joint. Reduction of the hip within 6 h after dislocation significantly decreases the incidence of AVN.
- When a painful hip makes ambulation difficult and plain radiographs do not reveal a fracture, computed tomography (CT) or magnetic resonance imaging (MRI) should be performed. MRI is the gold standard for diagnosis—although this does not need to be completed in the emergency department (ED).
- In patients with intertrochanteric fractures, hemodynamic instability can result from dehydration and blood loss. Up to 70% of patients with these injuries are under-resuscitated.
- It is important to identify acetabular fractures before closed reduction is attempted, because intra-articular bone fragments can interfere with effective reduction.
- In elderly patients, the use of femoral nerve blocks should be considered, due to the potential adverse effects of parenteral opioids.

1. Which of the following fractures has the best outcome and lowest rate of complications?
   a. Femoral neck fracture
   b. Insufficiency fracture
   c. Intertrochanteric fracture
   d. Lesser trochanteric fracture
   e. Subtrochanteric fracture

Answer: d. Treatment of a lesser or greater trochanteric fracture consists of pain control and early mobilization with crutches; weight bearing is allowed as tolerated by pain. Outpatient management of this injury is possible with a satisfactory social situation. The prognosis is good, and healing is generally excellent. The mortality rate during the first year after a femoral neck (also known as an *insufficiency fracture*) fracture is 15%. Intertrochanteric fractures have an associated mortality rate of 10% to 30% in the first year. The reported mortality rate from subtrochanteric fractures ranges from 10% to 15%.

2. A 40-year-old man complains of persistent thigh pain that worsens over 2 weeks after an assault with a baseball bat. His pain is worse with knee extension. On physical examination, you note that there is a small palpable mass at the mid-anterior thigh. What should be your next confirmatory test?
   a. Bone scan for fracture
   b. Bone scan for stress fracture
   c. Computed tomography (CT) scan for tumor
   d. Magnetic resonance imaging (MRI) for evaluation of torn muscle
   e. Radiograph for heterotropic calcification

Answer: e. Myositis ossificans (heterotrophic ossification) is pathologic bone formation at a site where bone is not normally found. Traumatic myositis ossificans results most commonly from a direct blow to muscle. It should be suspected when symptoms persist past 10 to 14 days or if symptoms intensify several weeks after the trauma. The ossific mass is often palpable and may limit motion, depending on its location.

3. An 80-year-old woman presents complaining of pain in her right leg after a motor vehicle collision (MVC). Emergency medical services (EMS) noted a swelling and a deformity to her thigh, so they placed a traction device on her leg. Her blood pressure is 80/40 mm Hg; heart rate, 110 beats/min; and respiratory rate, 18 breaths/min. What is the most appropriate next step in the management of the patient's fracture?
   a. Femoral nerve block and leave the traction device in place
   b. Femoral nerve block and remove the traction device
   c. Intravenous morphine and leave the traction device in place
   d. Intravenous narcotic pain medication and remove the traction device
   e. Immediate orthopedic consult to take the patient to the operating room for an open reduction and internal fixation (ORIF)

Answer: b. Although the patient may ultimately need to go to the operating room, the patient is currently unstable and requires resuscitation and evaluation for other injuries. A femoral nerve block is a valuable adjunct or alternative to systemic analgesics in a patient at risk for hypotension and has been underused by emergency clinicians. Prolonged traction during the assessment and management of other injuries can cause or worsen serious neurologic injury in the thigh by producing potentially damaging tension on the sciatic or femoral nerves.

4. What is the most common complication of a proximal femur fracture?
   a. Avascular necrosis (AVN)
   b. Myositis ossificans
   c. Osteomyelitis
   d. Pulmonary embolism
   e. Septic arthritis

Answer: a. AVN is the most common complication of proximal femur fractures (despite optimal treatment) because of the complex arterial anatomy. Deep infection in the form of osteomyelitis or septic arthritis is more common with femoral neck fractures because the fracture line extends into the joint. Pulmonary embolism is another significant complication and is the leading cause of death 7 days after fracture in orthopedic patients.

5. A 60-year-old woman presents complaining of right hip pain after a trip and fall at home. The patient denies loss of consciousness or other symptoms. You note that the patient's right leg is internally rotated, and the thigh is adducted and flexed at the hip joint so that the ipsilateral knee is resting on the opposite thigh. Which is the correct maneuver?
   a. Apply traction and splint the leg in full extension.
   b. Apply traction to an extended knee and flexed hip at 90 degrees with a gentle rotational component.
   c. Consult orthopedics immediately to take the patient to the operating room.
   d. Provide analgesia and Holter monitor for discharge when ambulatory.
   e. Use the Allis maneuver and place the patient in a knee immobilizer.

Answer: e. A patient with a posterior dislocation typically holds the hip flexed, adducted, and internally rotated. The knee of the affected extremity rests on the opposite thigh. The Allis technique is usually effective for both posterior and obturator dislocations. With the knee flexed, the operator applies steady traction in line with the deformity. The hip is slowly brought to 90 degrees of flexion while steady upward traction and gentle rotation are applied.

6. A mother brings her 11-year-old boy for evaluation of left knee pain that is worse after physical activity. Radiographs of the knee were negative. What is the most appropriate next step in the patient's management?
   a. Hip radiograph with frog-leg views
   b. Joint aspiration for evaluation of transient synovitis
   c. Place a knee immobilizer and ensure follow-up in 1 week
   d. Radiograph of right knee for comparison
   e. Rest, ice, compression, elevation, and orthopedic follow-up urgently

Answer: a. This patient may have a slipped capital femoral epiphysis (SCFE), which most commonly develops in boys 10 to 17 years old during their period of rapid growth. Referred pain to the knee is a classic manifestation, and patients frequently present with groin, thigh, or knee pain rather than hip pain. Initially, anteroposterior, lateral, and frog-leg lateral radiographs of the hip should be obtained. The frog-leg lateral projection shows the hip in a plane midway between the anteroposterior and lateral views.

7. A 75-year-old woman presents after a fall from standing. She has right hip pain and tenderness to palpation but no obvious deformity. Right knee and ankle examinations are normal, without tenderness, deformity, or external signs of trauma. Hip and pelvis radiographs are negative for fracture, but the patient is unable to bear weight on her right leg. The next appropriate step in management is:

- **a.** Admit to the hospital for bed rest.
- **b.** Discharge the patient home with analgesia and a walker.
- **c.** Obtain magnetic resonance imaging (MRI) of the hip to assess for fracture not identified by radiographs.
- **d.** Obtain radiographs of the rest of her right leg to ensure no occult fracture is present.

Answer: c. With hip injury, if radiographs do not show a fracture or suggestion of injury and the patient is unable to ambulate, further imaging studies should be obtained to evaluate for occult fracture. Two percent to 10% of all hip fractures are radiographically occult. Failure to detect these fractures results in increased morbidity and mortality.

8. Which of the following injuries is appropriate for traction splinting by prehospital providers?

- **a.** Femoral fracture with bone protrusion through the skin
- **b.** Posterior hip dislocation
- **c.** Severe crush injury of leg with obvious deformity of the knee
- **d.** Suspected closed mid shaft femoral fracture

Answer: d. Traction splints can provide pain relief, immobilization, and limit blood loss when applied correctly to a femoral fracture. However, contraindications to the use of traction splints include pelvic fractures, patellar fractures, ligamentous knee injuries, and tibia or fibula fractures. Traction in the prehospital setting should not be applied to any open fracture that has exposed bone. Such reduction pulls grossly contaminated bone fragments back into the wound before adequate débridement can be undertaken in the operating room.

9. A 15-year-old female gymnast presents after experiencing the sudden onset of severe groin pain during a dismount when she landed in a flexed-hip position. The pain radiates into her abdomen, and flexion of the hip produces pain, but there is no deformity noted. What is the most likely radiographic finding?

- **a.** Avascular necrosis (AVN) of the femoral head noted on magnetic resonance imaging (MRI) scan
- **b.** Diastasis of the pubic symphysis on anteroposterior pelvis radiographs
- **c.** Femoral neck fracture on dedicated anteroposterior hip radiograph
- **d.** Iliopsoas muscle with some associated hemorrhage on computed tomography (CT) scan

Answer: d. Gymnasts and dancers are the group of athletes most likely to experience an injury to the iliopsoas as a result of sudden forceful hip flexion against resistance. Severe pain often is experienced in the groin, thigh, or low back region. Severe intra-abdominal pain is common at the muscle origin and may dominate the clinical picture. Examination reveals groin tenderness and pain with active hip flexion. Radiographs of the femur should be obtained to identify an avulsion fracture of the lesser trochanter. CT scan frequently will demonstrate a large hematoma. Bed rest with partial flexion at the knee and hip generally is required for 7 to 10 days. With severe strains, symptoms may persist for 2 to 3 months. Referral to a sports medicine specialist is appropriate.

10. A 45-year-old male presents with a posterior hip dislocation after a motor vehicle crash (MVC) noted on radiographs. After sedation of the patient and reduction of the dislocated hip, what is the most appropriate next step in the patient's management?

- **a.** Have the patient ambulate to assess the stability of the joint.
- **b.** Measure the femoral compartment pressure.
- **c.** Obtain post-reduction hip radiographs to assess for additional injuries and adequate reduction.
- **d.** Place a traction splint.

Answer: c. Obtaining post-reduction radiographs to ensure adequate reduction and evaluate for associated injuries is essential. After closed reduction, the hip should be tested for stability, which is accomplished by gently taking it through a full range of motion to see whether it will re-dislocate. After testing has ensured stability, the injured extremity should be placed in a knee immobilizer, and an abduction pillow should be applied to prevent repeat dislocation.

48 Knee and Lower Leg Injuries

KEY CONCEPTS

- Knee dislocation often causes vascular injury to the popliteal artery. Early revascularization is crucial. Hard signs of vascular injury include absent pedal pulses, cool mottled foot, expanding popliteal hematoma, or popliteal hemorrhage. When any of these are present, angiography or emergent surgical exploration is indicated.
- Soft signs of popliteal artery injury include asymmetric pedal pulses and foot or leg paresthesias. Computed tomography (CT) angiography or duplex ultrasound study is indicated when these are present.
- In the absence of signs of popliteal injury, the knee dislocation patient can be observed for 24 h, with measurement of the ankle-brachial index (ABI) every 3–4 h. An ABI >0.9 over a 24-h period effectively excludes significant popliteal artery injury.
- Tibial plateau fractures should be considered in patients with a traumatic knee injury and inability to bear weight. Fractures may be radiographically occult, so CT imaging should be considered if clinical suspicion is high. In general, if a patient is unable to bear weight after a knee injury but no acute fracture is identified in the emergency department (ED), advanced imaging should be considered.
- Extensor mechanism injuries such as patellar and quadriceps tendon injuries are important to recognize, as delayed diagnosis is associated with substantial morbidity. A palpable defect in the quadriceps tendon or patellar tendon or over the patella and an inability to perform a straight leg raise should raise suspicion for such injuries. Knee immobilization and urgent orthopedic follow-up are advised.
- A dislocated patella will be fixed in a position superolateral to its normal position. Reduction is performed by gentle knee extension with medially directed or valgus directed pressure on the displaced patella.
- Injuries to the cruciate and collateral ligaments may not be detectable on initial examination because of effusion and splinting. Emergent diagnosis is not necessary, but the patient should undergo a follow-up examination by an orthopedist within a week.
- Lipohemarthrosis is an uncommon and subtle sign of occult fracture but should be sought on radiographs of traumatized knees.
- Compartment syndrome is a serious complication of tibial shaft fracture, usually occurring 24–48 h after injury. Orthopedic consultation or measurement of compartment pressure should be obtained when the patient's pain is increasing despite immobilization and support of the fracture.

1. A 45-year-old woman presents after a high-impact motor vehicle collision with pain in the left knee. She thinks her knee may have gone out of place but isn't sure. The distal neurovascular examination is intact. What is the most appropriate management?
   a. Compression wrap and early range of motion exercises
   b. Discharge if radiographs are negative with partial weight bearing
   c. Knee immobilizer with orthopedic follow-up
   d. Measure ankle-brachial index and perform duplex ultrasonography if less than 0.9
   e. Orthopedic consultation

Answer: d. This patient had a knee dislocation. Knee dislocation is uncommon but should be considered in the setting of an appropriate injury mechanism because 50% of all knee dislocations are reduced spontaneously before emergency department (ED) arrival. Reduction before ED arrival does not lessen the likelihood of vascular injury, and vascular injury should be considered in patients with severe ligamentous injuries and injuries caused by high-energy mechanisms. Vascular injury to the popliteal artery is the most severe complication and is the major cause of morbidity and limb loss. Management of suspected knee dislocation involves an algorithm designed to be noninvasive but sensitive for arterial injury.

2. A 42-year-old man was at the gym performing squatting exercises. As he was coming to a stand, he had an immediate onset of pain in his right thigh. He presents with an inability to extend his knee. What is the most likely physical finding?
   a. High-riding patella
   b. Mid-quadriceps muscle tenderness and deformity
   c. Popliteal swelling and tenderness
   d. Positive Lachman test
   e. Suprapatellar tenderness with patella baja

Answer: e. This is a quadriceps tendon rupture. Patients with extensor disruption may have signs and symptoms that include an acute onset of pain, swelling, ecchymoses over the anterior aspect of the knee, a palpable defect in the patella, quadriceps tendon, or patella tendon loss or limited ability for active leg extension; extension lag usually is seen when the last 10 degrees of extension are performed haltingly or with difficulty. With quadriceps rupture, a low-riding patella (patella baja) and inferior retraction may be seen.

3. A college football lineman presents complaining of left knee pain and inability to bear weight. He was on the scrimmage line when the ball was placed in motion and, as he lunged forward, he twisted his leg and heard an audible pop in his knee. He was immediately unable to bear weight. On physical examination, you find the left knee swollen, with a positive Lachman test. What is the most sensitive way to diagnose the injury accurately?
   a. Arthroscopy of the knee
   b. Computed tomography (CT) scan of the knee
   c. History and physical examination
   d. Magnetic resonance imaging (MRI) of the knee
   e. Plain radiography of the knee

Answer: a. Clinical evaluation is moderately sensitive for anterior cruciate ligament (ACL) tears but, in the acute phase, is often inaccurate because of swelling and splinting. Acutely, only plain films are indicated as long as dislocation is not suspected. Tibial plateau fractures detected on plain film may require CT scanning to determine the need for admission for early operative repair. Arthroscopy is the gold standard for diagnosis of soft tissue injuries of the knee. MRI is useful but may miss small tears and anatomic abnormalities, and a normal study may still lead to arthroscopy if symptoms persist. MRI scanning of the knee is rarely indicated in the emergency setting.

4. A man presents with severe pain of his right lower leg. He was in the ED the previous night for splint placement for a tibial fracture. After removal of the splint, you see a lower leg with mild ecchymosis, healing abrasions, and palpable posterior tibial and dorsalis pedis pulses. He has severe pain on passive movement of his first toe and decreased sensation of the first toe web space. Which of the following should be the next step?
   a. Admit for intravenous (IV) antibiotics and pain control.
   b. Obtain venous Doppler scans.
   c. Resplint the patient.
   d. Test the anterior compartment pressures.
   e. Test the superficial posterior compartment pressures.

Answer: d. The lower leg is divided into the following compartments by deep partitions of the investing crural fascia—anterior, lateral, superficial posterior, and deep posterior. The anterior compartment contains the tibialis anterior, long toe extensor muscles, anterior tibial artery, and deep peroneal nerve, which supplies sensation to the first web space of the foot.

5. A severe medial ankle sprain with no ankle fracture necessitates evaluation of which of the following?
   a. Cervical spine
   b. Femoral nerve
   c. Lumbar spine
   d. Popliteal artery
   e. Proximal fibula

Answer: e. An important exception to fibular fractures being stable is a Maisonneuve fracture (see Fig. 48.13). This involves a medial ankle disruption (deltoid ligament tear or medial malleolar fracture), with complete tearing of the syndesmosis joining the tibia and fibula and fracture of the proximal fibula. This results in an unstable ankle mortise because the fibula now floats free relative to the tibia, and surgical fixation is required.

KEY CONCEPTS

- The Ottawa Ankle and Foot Rules should be used to evaluate the need for x-rays in ankle and foot injuries.
- CT scans are indicated for negative x-rays of the ankle and foot when a high clinical concern of fracture exists.
- Ankle dislocations with obvious vascular compromise should be reduced promptly, before radiographs are obtained.
- The entire fibula should be examined if a medial malleolar fracture is present to rule out a proximal fibular (*Danis-Weber type C or Maisonneuve*) fracture.
- Osteochondral lesions of the talar dome, nondisplaced fractures of the lateral posterior process of the talus, and fracture of the anterior process of the calcaneus are important to include in the differential diagnoses of presumed ankle sprains.
- Patients discharged with a diagnosis of ankle sprain should be encouraged to seek outpatient follow-up should ankle pain and swelling persist for longer than 2 weeks.
- Patients with Achilles tendon rupture are still capable of weak plantar flexion. Ultrasound may be used to confirm the diagnosis if clinical uncertainty exists.
- A Lisfranc injury should be considered with any fracture or dislocation in the tarsometatarsal region, particularly with fractures of the second metatarsal base. When suspicion for a Lisfranc injury is high, even if plain radiographs and CT scans are negative, immobilization and orthopedic referral are indicated.
- Fifth metatarsal tuberosity (zone 1) fractures should be carefully differentiated from diaphyseal fractures. Orthopedic referral is indicated for zones 2 and 3 fractures.
- Stress fracture should be considered in patients with long-standing foot pain, particularly if symptoms are in the metatarsal region. This may be diagnosed by plain radiography, CT scanning, or radionuclide bone scanning.
- Compartment syndrome may occur in any of the four major compartments of the foot. Diagnosis is made by high clinical suspicion and measurement of the intracompartmental pressures.

1. A middle-aged farmer presents after an accident with a plow with a crushed open distal tibia fracture. What is the appropriate antibiotic choice?

a. Cefazolin
b. Cefazolin and gentamycin
c. Cefazolin, gentamycin, and penicillin G
d. Ciprofloxacin
e. Pen VK and cephalexin

Answer: C. Because the patient has a probable soil contaminant in his open fracture, he will need the addition of penicillin G to cover *Clostridium perfringens*. For low-energy injuries with mild to moderate contamination, a broad-spectrum cephalosporin is usually sufficient. Heavily contaminated wounds require the addition of gram-negative bacterial coverage, typically with an aminoglycoside. Adding penicillin G as a third antibiotic is necessary for farm- or soil-related crush injuries.

2. A 30-year-old woman complains of pain in the ankle and difficulty walking after attempting a jump shot while playing basketball. Her past medical history includes a recent urinary tract infection, for which she is taking ciprofloxacin. On physical examination, her ankle joint is not swollen, there is a palpable defect 4 cm proximal to the posterior ankle, and she has weakened plantar flexion. Which of the following is true concerning her condition?

a. The Boehler angle should be less than 20 degrees.
b. In a seated position, squeezing the calf causes foot dorsiflexion.
c. It requires a stress radiograph view.
d. On the radiograph, there may be an opacification at the Kager triangle.
e. Talofibular ligament laxity is present.

Answer: D. The patient has a ruptured Achilles tendon. On examination, a visible and palpable tendon defect 2 to 6 cm proximal to the calcaneal insertion may be present. In some cases of complete Achilles tendon rupture, weak plantar flexion may still be present because of the actions of the tibialis posterior, toe flexors, and peroneal muscles. When the Achilles tendon is intact, squeezing the calf muscles should cause passive plantar flexion of the foot (Thompson test). Lateral radiographs of the ankle may confirm rupture by showing opacification of the triangular fatty tissue–filled space anterior to the Achilles tendon (Kager triangle) or an irregular contour and thickening of the tendon.

3. A 24-year-old patient presents with pain in the left lateral ankle after a twisting injury. On physical examination, you find no point tenderness on the medial or lateral distal 6 cm of the malleoli, no proximal fibular tenderness, and no navicular tenderness. Which of the following physical findings would necessitate radiographs being taken in the emergency department?

a. Inability to ambulate to the car
b. Laxity on anterior drawer testing
c. Moderate swelling with ecchymosis
d. Tenderness at the base of the fifth metatarsal
e. Tenderness over the deltoid ligament

Answer: D. The Ottawa Ankle Rules advise ankle radiographs when any of the following are present: tenderness at the posterior edge of the distal 6 cm or tip of the lateral malleolus, tenderness at the posterior edge of the distal 6 cm or tip of the medial malleolus, or inability to bear weight for at least four steps immediately after the injury and at the time of evaluation. Foot radiographs in the setting of blunt ankle trauma are advised when any of the following are present: tenderness over the navicular, tenderness at the base of the fifth metatarsal, or inability to bear weight for at least four steps immediately after the injury and at the time of evaluation.

4. A 40-year-old female falls down some stairs and her ankle is forced into sudden dorsiflexion and eversion. She reports pain over the lateral aspect of the ankle and reports a "snapping" sensation with active ankle eversion, otherwise the rest of the examination is unremarkable. The plain radiograph of the ankle is negative with the exception of what appears to be a large avulsed fragment over the lateral malleolar region. The optimal management of this patient is:

a. Reassuring the patient that this is an avulsion fracture and make her weightbearing as tolerated.
b. Order a CT scan to evaluate the origin of the avulsed fragment.
c. Orthopedic evaluation in the emergency department.
d. Outpatient dynamic ultrasound and referral to orthopedics.
e. Splinting, nonweightbearing, and outpatient referral to orthopedics.

Answer: D. The patient's mechanism of injury, symptoms, and x-ray findings are consistent with a rupture of the superior peroneal retinaculum with peroneal tendon dislocation. If the diagnosis is in doubt, dynamic ultrasound can be used to evaluate the peroneal tendon because it will show dislocation of the tendon when the foot is dorsiflexed and everted. Rupture of the superior peroneal retinaculum invariably requires operative repair.

5. A 32-year-old male presents after a fall in which his foot was forced into dorsiflexion. He has severe pain and swelling on his dorsal foot just anterior to the ankle. In the ED, he is diagnosed with a nondisplaced talar neck fracture (Hawkins type 1). His pain is controlled, and he has no associated injuries. What is the disposition for this patient?
   - **a.** Place in a walking boot or a cast, weightbearing as tolerated, and follow up with orthopedics in 1 to 2 weeks.
   - **b.** Orthopedics consult in the emergency department is required and the patient should be admitted.
   - **c.** Immobilize in a padded, noncircumferential splint and arrange outpatient orthopedics within 48 hours.
   - **d.** Immobilize in a cast and nonweightbearing for 6 weeks, with outpatient orthopedics follow-up in 1 to 2 weeks.

Answer: C. Hawkins type 1 fractures are nondisplaced. These fractures are treated conservatively and are the only talar neck fractures that are appropriate for outpatient orthopedic follow-up. Follow-up, however, should be urgent in 24 to 48 hours, and the patient should certainly be immobilized in a well-padded and noncircumferential splint. Depending on local practice, orthopedic consultation in the ED is reasonable.

6. A 44-year-old male presents after a fall from a ladder in which he landed on his right foot. He has severe pain in his heel and is diagnosed with an intraarticular calcaneus fracture on plain x-rays. Which of the following is incorrect in the emergency management of this patient?
   - **a.** Assessment for other traumatic injuries, particularly spinal injury.
   - **b.** Once pain and swelling are controlled, the patient may be immobilized in a padded splint with orthopedic follow-up arranged for 48 hours.
   - **c.** The patient should get a CT scan of the foot, if available, in the emergency department.
   - **d.** Orthopedic surgery should be consulted to assess the patient in the emergency department.
   - **e.** Place the patient in a well-padded splint with the foot elevated.

Answer: B. Patients with intraarticular calcaneus fractures should have an orthopedics consultation in the ED. Treatment may be conservative, but even when operative repair is chosen, the repair may be done electively at a time when the swelling has decreased. It is important to assess these patients for other associated injuries and for patients to be placed in a well-padded splint with the foot elevated to the level of the heart, to decrease swelling. CT imaging aids in classification and decisions about management.

7. A 22-year-old male football player presents with midfoot pain and difficulty weightbearing after being tackled in a college game. You are concerned about a Lisfranc injury, but his plain radiographs appear normal. What is the recommended next step in his management?
   - **a.** MRI in the emergency department
   - **b.** Attempt weightbearing x-rays or a CT scan, to assess for ligamentous disruption.
   - **c.** Immobilize in a nonweightbearing splint and refer for follow-up with orthopedic surgery.
   - **d.** Ultrasound to assess for disruption of the Lisfranc ligament
   - **e.** Orthopedics consultation for stress fluoroscopy

Answer: B. Diagnosing low-energy, ligamentous Lisfranc injuries is a challenge. The American College of Radiology guidelines recommend an attempt at weightbearing stress views to look for diastasis of greater than 2 mm between base of the first and second metatarsals, or a difference greater than 1 mm between the base of the first and second metatarsals comparing both feet. If weightbearing x-rays are not possible, a CT scan is recommended. If clinical concern persists, more advanced imaging with MRI or fluoroscopic stress views may be necessary, but these can be arranged at the time of specialist follow-up. Ultrasound has not yet demonstrated utility.

8. A 28-year-old female suffers a twisting injury to her foot during a fall. She is tender at the base of her fifth metatarsal. Plain radiographs reveal an acute, nondisplaced fracture at the level of the intermetatarsal articulation. Which of the following management options is incorrect?
   - **a.** She can be treated functionally in a supportive shoe and can bear weight as tolerated. Orthopedic follow-up should be arranged.
   - **b.** This type of injury may be managed surgically. She should be referred for orthopedic assessment as an outpatient.
   - **c.** She should be placed in a weightbearing cast or walker boot for 3 months with outpatient follow-up.
   - **d.** She can be provided with a walker boot but instructed to remove it when able. Orthopedic follow-up should be arranged.
   - **e.** She can be treated functionally in a supportive shoe if necessary. No follow-up is needed

Answer: E. This is a zone 2 fracture. The treatment of these fractures is evolving and varies according to patient factors. In high-level athletes, primary surgical management may be of benefit. Historically, these injuries were thought to have a high rate of nonunion and prolonged immobilization has been the standard approach in the average patient. More recent data, however, suggest that the rate of symptomatic nonunion is actually quite low, and there is evidence that treating these patients functionally in supportive footwear, perhaps with initial use of a protective boot, is efficacious and more economical. However, as there remains a risk of nonunion, orthopedic follow-up is recommended for these injuries. Fractures in zone 1 are the only base of fifth fractures that do not require orthopedic follow-up.

50 Wound Management Principles

KEY CONCEPTS

- Risk factors for wound infection include crush mechanism; long (>5 cm) deep penetrating wounds; high-velocity missiles; diabetes; and contamination with saliva, feces, soil, or other foreign matter.
- Soaking wounds in povidone-iodine (Betadine) is toxic to healthy tissue. Prepare the skin with a chlorhexidine-alcohol solution.
- The most effective intervention to decrease wound bacterial counts and infection is thorough cleansing, with use of saline or tap water irrigation at approximately 8 psi. Attaching an 18-gauge needle to a 30-mL syringe creates an irrigant force of 7 or 8 psi.
- Bupivacaine is the preferred local anesthetic agent for the care of most wounds. In adults, the maximal reported safe dose is approximately 2 to 2.5 mg/kg without epinephrine and 3 to 3.5 mg/kg with epinephrine.
- The discomfort associated with the injection of bupivacaine into wounds can be reduced by warming the anesthetic and by adding sodium bicarbonate (0.1 mL of sodium bicarbonate to 10 mL of bupivacaine).
- High-risk wounds should not be sutured primarily but may be repaired in 4 or 5 days (i.e., delayed primary closure).
- Tissue adhesives offer several advantages compared to the suturing of simple wounds. However, the use of adhesives is not recommended for lacerations longer than 4 cm, for high-tension wounds, or in areas subject to frequent repetitive movements, such as joints and hands.
- Skin tears in elders are best treated with Steri-Strips, augmented with topical skin adhesives when tension exists.
- Antibiotics are indicated for through-and-through intraoral lacerations, cat bites, some dog and human bites, puncture injuries to the foot in high-risk individuals, open fractures, and wounds involving exposed tendons or joints.
- Tetanus immunization should be provided soon after injury but can be given days or weeks later as the incubation period for tetanus is 7 to 21 days (range, 3 to 56 days).
- Tdap is recommended for patients 65 years old or older requiring tetanus prophylaxis.

1. Which of the following is associated with an increased risk of infection?
   a. Avoidance of epinephrine use in wound anesthesia
   b. High-pressure irrigation
   c. Use of clippers instead of razors for hair management near wound
   d. Use of silk suture material
   e. Use of tape over sutures for wound closure

Answer: d. Silk yields the highest infection rates, whereas monofilament synthetic substances have the lowest risk of infection. Other risk factors are listed in eBox 50.1

EBOX 50.1 Risk Factors for Wound Infection

1. Location: Leg and thigh, then arms, then feet, then chest, then back, then face, then scalp
2. Contamination with devitalized tissue, foreign matter, saliva, or stool
3. Blunt (crush) mechanism
4. Presence of subcutaneous sutures
5. Type of repair: Risk greatest with sutures > staples > tape
6. Anesthesia with epinephrine
7. High-velocity missile injuries
8. Diabetes

2. A 32-year-old man presents with 20 stab wounds to his arms and legs sustained in an assault. He weighs 80 kg. Which of the following is an appropriate dosage of wound anesthesia?
   a. 240 mg of 1% lidocaine
   b. 320 mg of 0.5% bupivacaine
   c. 400 mg of 0.5% bupivacaine with epinephrine
   d. 700 mg of 1% lidocaine with epinephrine

Answer: a. In adults, the maximal reported safe dose of bupivacaine is approximately 2.5 mg/kg without epinephrine and 3.5 mg/kg with epinephrine, assuming the injection is a wound infiltration technique and not one in a highly vascular area. General dose guidelines for lidocaine are 3 to 5 mg/kg without and 5 to 7 mg/kg with epinephrine, respectively. A 1% lidocaine solution contains 10 mg/mL. The actual percent solution does not matter; the total milligram dose does.

3. A 23-year-old female presents with a laceration to her thigh. As you begin to apply anesthetic, she tells you that when she went to the dentist she had an allergic reaction to procaine. Which of the following should you use?
   a. Benzocaine
   b. Benzocaine with epinephrine
   c. Bupivacaine
   d. Lidocaine from a multidose vial
   e. Tetracaine

Answer: c. Allergy to local anesthetics is uncommon. Two distinct groups of "caine" anesthetics exist. The esters include procaine, tetracaine, and benzocaine. The second group, including lidocaine and bupivacaine, belongs to the amide family. Allergy to the esters is uncommon. True allergy to agents in the amide family is rare. No cross-reactivity occurs between the amide and ester families, so an agent from a different group may be chosen. A "preservative-free" preparation should ideally be used because the para-aminobenzoic acid in multidose vials may cause an amine-like reaction that may be confused with an allergy to the primary agent.

4. Which of the following factors is most likely to require no antibiotic prophylaxis?
   a. Cat bite
   b. Diabetic with contaminated wound
   c. Dog bite
   d. Human bite
   e. Puncture wound through rubber sole

Answer: c. Antibiotic prophylaxis is often provided for patients with wounds with gross contamination, patients with severe crush injuries, and immunocompromised patients. Prophylaxis is also required for patients with cat bites. Seven of eight randomized trials of dog bite wounds showed no benefit with antibiotics. Human bites or lacerations to the metacarpophalangeal (MCP) joint are termed "fight bites," and these wounds have a high incidence of infection. Thus, of the choices given, dog bites have the lowest incidence of infection and therefore are the least likely to require prophylactic antibiotics.

5. Antibiotic coverage of a cat bite must target which of the following pathogens?
   - **a.** Bacteroides species
   - **b.** *Clostridium perfringens*
   - **c.** *Eikenella corrodens*
   - **d.** *Pasteurella multocida*
   - **d.** Pseudomonas species

Answer: e. The organisms found in cat bites include *Staphylococcus* species, *Streptococcus* species, and, most often, *P. multocida. P. multocida* is usually found in infected cat bite wounds and is present in the normal oral flora of 70% of all cats. *P. multocida* is sensitive to penicillin, but the infection is often polymicrobial. *P. multocida* is resistant to dicloxacillin, cephalexin, and clindamycin, and there are many erythromycin-resistant strains. Amoxicillin with clavulanate is the current recommendation for antibiotic prophylaxis for cat bites.

6. A 74-year-old patient presents with a gaping wound from a dog bite, complaining of pain at the site. The dog belongs to a friend, has reportedly had "all his shots," and is in custody. The patient is from Central America but has lived in the US for more than 50 years; he does not recall his immunization history. Besides copious irrigation and wound dressing, which of the following should be included in the treatment of this patient?
   - **a.** Diphtheria, pertussis, tetanus toxoids (Tdap) and tetanus immunoglobulin (TIG)
   - **b.** Rabies immunization
   - **c.** Tetanus toxoid
   - **d.** Tetanus toxoid and immunoglobulin
   - **e.** B and D

Answer: a. Studies show that many people are inadequately immunized, especially patients older than 70 years old, immigrants, and people with no education beyond grade school. Patient immunization histories are often unreliable. Given the inability to predict which wounds are high risk, all wounds should be approached with suspicion. Inadequately immunized patients need a dose of Tdap and TIG. Because studies suggest that 10% to 40% of the population in the United States is inadequately immunized against diphtheria, diphtheria vaccination should be given along with tetanus toxoid. Many adults are not immunized against pertussis, and the incidence of disease has been rising since the 1980s. In 2005, a new acellular form of the pertussis vaccine became available and is recommended for all adults.

51 Foreign Bodies

KEY CONCEPTS

- If the history and mechanism of injury are compatible with ocular penetration or if a small puncture wound of the globe is noted, anteroposterior and lateral radiographs of the orbit are an appropriate initial step when the foreign body is thought to be radiopaque. Computed tomography (CT) and ultrasound are complementary diagnostic studies.
- Although most otic and nasal foreign bodies are amenable to emergency department (ED) removal, instrumentation of these anatomical areas must be undertaken with great care because removal attempts can cause more injury than the foreign body itself.
- Most airway foreign bodies are seen in pediatric patients and may not be visible on plain films. A normal radiograph does not rule out an aspirated foreign body.
- The patient with critical airway obstruction and impending or actual respiratory arrest requires one of three options: (1) forced expulsion of the foreign body; (2) direct laryngoscopy with attempted manual removal with Magill forceps; or (3) cricothyroidotomy to bypass an obstruction, or intubation to push the foreign body distally into the right mainstem bronchus.
- Esophageal foreign bodies are typically found at one of the three constriction locations: (1) proximal esophagus at the level of the cricopharyngeal muscle and thoracic inlet—radiographically, the clavicular level; (2) midesophagus at the level of the aortic arch and carina; and (3) distal esophagus just proximal to the esophageal-gastric junction—radiographically, a level two to four vertebral bodies cephalad to the gastric bubble.
- Esophageal foreign bodies (e.g., coins) usually are oriented in the coronal plane, and airway objects usually are oriented in the sagittal plane.
- Unless caustic or sharp, foreign bodies in the stomach and bowel are increasingly managed with a conservative approach with watchful waiting and tracking of the object's progression through the gastrointestinal tract.
- In the perineal region, foreign body removal tends to be more difficult than anticipated by emergency clinicians and be physically and psychologically traumatic for patients. Consultation with appropriate specialists (e.g., urology or general surgery) is indicated.
- The most important determinant of successful soft tissue foreign body removal is an understanding of the object's precise location.

1. A 33-year-old construction worker presents with left eye pain. On gross inspection, you note a watery discharge and moderate erythema. The patient tells you she was working with her coworker when she felt something hit her eye. She admits to not wearing protective goggles. Fluorescein examination of the eye reveals rivulets of dye tracking from a corneal defect. What is the next appropriate step in the patient's management?

a. Attempt ultrasound to find a foreign body
b. Check intraocular pressures
c. Obtain orbital computed tomography (CT) scan and consult ophthalmology
d. Obtain plain radiographs of the orbits
e. Topical antibiotics and ophthalmology follow-up in 24 hours

Answer: c. Rivulets of fluorescein tracking from the puncture (i.e., positive Seidel test) are helpful in identifying intraocular penetration. Ultrasound is, as always, operator dependent, and the pressure of the probe on the orbit may risk further injury to an open globe. Checking for intraocular pressures with an open globe is contraindicated. Compared with plain radiographs, CT delivers less radiation to the lens. Multiplanar reconstruction minimizes streak artifacts, affording better localization of intraorbital objects.

2. A 2-year-old presents with purulent drainage from the right naris and was brought to the ED by a parent concerned about sinusitis. The patient's vital signs are heart rate 110 beats/min, respiratory rate 15 breaths/min, and temperature 37.7°C. On physical examination, you see a small plastic ball in the naris surrounded by swelling and purulent discharge. What should be the next step?

a. Attempt to displace the foreign body posteriorly
b. Blow air into the contralateral naris to help dislodge the foreign body
c. Consult the otolaryngologist for removal under anesthesia
d. Make sure you have a right-angle probe, suction, and alligator forceps
e. Treatment with topical vasodilators to facilitate removal

Answer: d. The emergency clinician can remove most nasal foreign bodies. Posterior movement of a nasal foreign body risks aspiration; objects should be removed anteriorly via suction or traction. In some circumstances, it may be prudent to place patients in lateral decubitus, perhaps with additional Trendelenburg positioning, to help prevent the aspiration of objects. Foreign bodies can sometimes be easily removed via positive pressure applied to the patient's mouth (not the contralateral naris, which should instead be clamped closed to increase pressure on the involved-side naris). Pretreatment with vasoconstrictive spray may improve chances of success. If positive-pressure techniques are not indicated or do not work, it is important to have necessary instruments close at hand to proceed with foreign body removal attempts. These instruments include a blunt-tipped right-angle probe (to maneuver posterior to the foreign body), suction equipment, and alligator forceps.

3. A 14-month-old girl presents in respiratory distress after eating a hot dog for lunch. Her mother states that she had stepped out of the kitchen, and when she came back, her daughter was sitting forward with noisy breathing and obvious distress. Which of the following management interventions is contraindicated?

a. Back blows
b. Blind finger sweep as the first step
c. Chest thrusts
d. Direct laryngoscopic visualization
e. Intubate for respiratory distress

Answer: b. Blind finger sweeping has resulted in the conversion of partial to complete airway obstruction when objects are displaced into the subglottic space. For this reason, the technique has lost favor as an initial maneuver in pediatric and adult patients. It is recommended that up to five back blows be delivered (with the patient in a head-down position), followed by chest thrusts. Intubation or needle cricothyroidotomy may be performed if other maneuvers fail and circumstances dictate their need.

4. A 45-year-old man was at dinner when he had a choking episode. When he recovered, he experienced new-onset wheezing and presents with the following vital signs: blood pressure 140/90 mm Hg, heart rate 110 beats/min, respiratory rate 20 breaths/min, and

arterial oxygen saturation (Sao_2) 92%. On chest radiograph, you see a flat, fixed diaphragm on the right with a mediastinal shift to the left and inadequate left-sided expansion. What is the next step in the management of this patient?

a. Albuterol nebulizers and steroids intravenously
b. Consult pulmonary for bronchoscopy
c. CT scan of the chest with intravenous contrast
d. Needle decompression of the left side
e. Needle decompression of the right side

Answer: b. Air trapping and atelectasis are the most common early clues to airway foreign body presence, with bronchiectasis and bronchial stenosis developing later. In air trapping, a comparison of inspiratory and expiratory films shows a flat, fixed diaphragm on the involved side, and the heart and mediastinum shift to the uninvolved side during expiration. When the foreign object is distal to the oropharynx, however, subspecialty consultation is the safest and most expeditious means for foreign body removal. As a general rule, early bronchoscopy in any patient with a suspected foreign body is key to reducing morbidity and mortality.

5. A concerned father brings in his toddler, who has swallowed a button battery. Which of the following management strategies is most appropriate for this patient?

a. Endoscopy is indicated if it is above the lower esophageal sphincter.
b. Endoscopy is not indicated if it is found in the esophagus.
c. Expectant management is appropriate.
d. If it is in the small bowel, further surveillance is not indicated.

Answer: a. If a disk battery has been ingested, its location must be ascertained with immediate removal if it has lodged in the esophagus. If the button battery has passed distal to the esophagus, the patient can be observed, with follow-up radiography to confirm spontaneous passage through the gastrointestinal tract. Nifedipine is occasionally effective in managing food boluses but should not be used to manage nonorganic foreign bodies.

6. Which of the following is an initial management option for esophageal food bolus impactions in the ED?

a. Glucagon
b. Nifedipine
c. Papain
d. Sublingual nitroglycerine

Answer: a. Although none of the medical therapies are as good as endoscopy, Glucagon is likely more efficacious than nitroglycerin and nifedipine. Enzymatic degradation of an impacted meat bolus using the proteolytic enzyme papain, should not be performed because of risks of esophageal perforation. The gold standard intervention strategy for esophageal foreign body removal is endoscopy.

7. What do you expect to see on the chest radiograph of a child who has swallowed a coin?

a. Foreign body anterior to tracheal air column
b. Foreign body causing air trapping on the left side
c. Visualized flat foreign body in the coronal plane
d. Visualized round foreign body in the coronal plane
e. Visualized round foreign body in the transverse plane

Answer: d. Esophageal foreign objects usually align themselves in the coronal plane and are posterior to the tracheal air column on lateral view. Coins in the esophagus lie in the coronal position in virtually all cases because the opening into the esophagus is much wider in this orientation.

52 Mammalian Bites

KEY CONCEPTS

- Mammalian bites require an evaluation for both trauma and their risk of infection.
- Cat and human bites are at higher risk for infection than dog bites.
- Most mammalian bite wound infections are polymicrobial. *Pasteurella* species are the most common pathogens in dog and cat bites.
- Preventing infection from mammalian bites relies more on vigilant cleaning, debridement, and irrigation than prophylactic antibiotics.
- Prophylactic antibiotics are most effective when started within 3 hours of the bite. The recommended duration of treatment is 5 days.
- Amoxicillin-clavulanate (Augmentin) is the prophylactic antibiotic of choice for dog, cat, and human bites.
- The decision to close mammalian bite wounds must weigh the benefit of improved cosmesis against the risk of wound infection.
- Given the risk of infection, mammalian bite wounds to the hand should not be closed primarily. Most facial bites can safely be closed if done so within 24 hours of the bite.
- Clenched-fist injuries or "fight bites" have high rates of damage to deep structures and infection.
- Mammalian bites to the hand should receive prophylactic antibiotics. Infected human bites to the hand are treated on an inpatient basis with intravenous antibiotics, operative debridement, and irrigation.

1. A 75-year-old woman presents with a puncture mark on her hand. She reports being bitten by her cat the previous night. Which of the following statements regarding this patient's injury is *true?*

a. *Capnocytophaga canimorsus* is the organism of concern.
b. Cats are very clean animals and do not carry any virulent strains.
c. Cats produce superficial infections because their teeth are not long enough to inoculate past the dermis.
d. Concern exists for a virulent gram-negative bacterium, which can produce a rapid cellulitis.
e. Irrigation and topical antibiotics are indicated.

Answer: d. Cats have long, slender, pointed teeth that can penetrate tendons, joints, and bone, inoculating bacteria deep into these tissues. Cat bites have a substantially higher risk of infection than dog bites do. Another important factor in the development of wound infection after cat bites involves the presence of *Pasteurella multocida,* a highly virulent, gram-negative, facultatively anaerobic rod found in the oral cavity or nasopharynx of 70% to 90% of healthy cats.

2. What is the most appropriate antibiotic prophylaxis for a cat bite?

a. Amoxicillin-clavulanate
b. Clindamycin
c. Erythromycin
d. First-generation cephalosporins
e. Vancomycin

Answer: a. In vitro, *P. multocida* is sensitive to penicillin, ampicillin, tetracycline, fluoroquinolones, amoxicillin-clavulanate, second- and third-generation cephalosporins, and trimethoprim-sulfamethoxazole.

3. A 19-year-old man presents 10 hours after sustaining a dog attack. He has multiple lacerations on his head and right hand. What is the appropriate management of this patient?

a. Repair the hand and the head lacerations
b. Repair the hand laceration but not the head laceration
c. Repair the head laceration but not the hand laceration
d. Use Steri-Strips on all and give him a prescription for amoxicillin-clavulanate (Augmentin)
e. Wound cleansing and bandaging

Answer: c. For dog bites, the infection rate of hand wounds is as high as 30%, regardless of suturing, whereas the infection rate of dog bites elsewhere averages 9%. Similarly, dog bites of the face and neck (including punctures) have lower infection rates of 0% to 5% even when they are sutured. Bite wounds of the face and scalp from any species that are less than 12 hours old may be cleansed well and sutured. Most other bite wounds that are going to be closed should have this done within 6 hours.

4. Selected monkey bites require postexposure prophylaxis with which of the following medications?

a. Amoxicillin clavulanate
b. Clindamycin
c. Flagyl
d. Tetracycline
e. Valacyclovir

Answer: e. Valacyclovir to prevent herpes B virus infection. Other terms for this virus include *herpesvirus simiae, herpesvirus B,* and *monkey B virus*. This virus has serologic cross-reactivity with herpes simplex virus (HSV) type 1 and type 2, which cause herpetic lesions in humans. Seventy-five to 100% of monkeys of the genus *Macaca* (macaques) used for biomedical research are seropositive for the B virus. Other antibacterials may be needed if cellulitis infection ensues.

53 Venomous Animal Injuries

KEY CONCEPTS

- Snake venom causes neurotoxicity and hematotoxicity, but one usually predominates, depending on the species of snake.
- Pit vipers have triangular or arrow-shaped heads, elliptical pupils, and characteristic pits found bilaterally midway between the eye and the nostril.
- The amount of crotalid antivenom given depends the severity of the bite. Currently, Crotalidae polyvalent immune Fab (CroFab) and F(ab′)2 (Anavip) are the antivenoms of choice for crotalid bites. Children require the same amount of antivenom as adults.
- Envenomation from exotic international snakes such as cobras, kraits, and mambas often require antivenom for life-threatening neurologic and hematologic toxicity.
- Arthropods such as hymenoptera account for more deaths from envenomation than snakes, usually as a result of allergic or anaphylactic reactions.
- Black widow spider bites are neurotoxic, causing severe pain and muscle spasms; an antivenom is available but used only in severe cases. Brown recluse spider bites cause necrotizing skin wounds.
- Nematocyst (jellyfish) stings should be neutralized with vinegar or hot water, and fish stings with hot water. Stings from venomous fish (e.g., lionfish and sting rays) are treated with circulating hot water to denature the toxin and cause vasodilation.
- Marine antivenoms are available for severe box jelly fish, stonefish, and sea snake envenomation.

1. Which of the following venomous snakes in the United States is a member of the neurotoxic Elapidae family?

a. Copperheads
b. Coral snakes
c. Rattlesnakes
d. Water moccasins

Answer: b. Pit vipers from the family Viperidae are the most prevalent venomous snakes in the United States. They are native to every state except Maine, Alaska, and Hawaii. They are classified into three main groups: true rattlesnakes (genus *Crotalus*), copperheads and water moccasins (genus *Agkistrodon*), and pygmy and Massasauga rattlesnakes (genus *Sistrurus*). Pit vipers account for 98% of all venomous snakebites in the United States. Other families include Colubridae, Hydrophiidae, Elapidae (to which the neurotoxic coral snake belongs), and Crotalidae.

2. A 30-year-old man complains of pain, burning, and swelling of his hand after gardening. Physical examination shows two puncture marks on his thenar eminence, as well as moderate localized swelling. When you inquire about it, he said that he did have some sudden burning pain while digging around in the ground by a bush, but he dismissed it as abrasions from sharp sticks. That was 2 hours ago. What should you do next?

a. Apply a constricting band to impede venous and arterial flow.
b. Discharge with instructions for rest, ice, immobilization, and use of oral antibiotics.
c. Give tetanus and clindamycin for prophylaxis.
d. Obtain baseline laboratory studies and observe the patient for increasing symptoms.
e. Test compartment pressure and prepare for fasciotomy.

Answer: d. The symptoms are typical for a pit viper snakebite. The most consistent symptom associated with pit viper bites is immediate burning pain in the area of the bite, whereas pain may be minimal with bites of Elapidae and other exotic snakes. Tetanus prophylaxis may be indicated, but antibiotics are not. Fasciotomy is rarely if ever indicated for snakebite. A constricting band may be useful for first aid in the field with certain neurotoxic snakes, but not in North America. Immobilization may be helpful, but ice and antibiotics are not.

3. A 43-year-old woman sustained a snakebite while in her backyard in Florida. She was not able to secure the snake, but she remembered it to be colorful. She presents with ptosis, slurred speech, and nausea. What is true about the type of snake most likely involved in this case?

a. Death usually occurs from coagulopathies.
b. Eastern species are the most venomous.
c. The snake has a heat-sensitive organ between eyes and nostrils on both sides of the head.
d. The snake has a triangle-shaped head.
e. The snake has elliptical pupils.

Answer: b. This woman sustained a bite from a coral snake. All of the above except B are related to pit vipers. The coral snake can be differentiated from the king or milk snake by two characteristics: the nose of the coral snake is black, and the red and yellow bands are adjacent on the coral snake. Eastern coral snake is considered more deadly as opposed to its western relative. There are no records of fatalities caused by the western species. Ptosis is common and often the first outward sign of envenomation. Other signs and symptoms include vertigo, paresthesias, fasciculations, slurred speech, drowsiness, dysphagia, increased salivation, nausea, and proximal muscle weakness.

4. Which of the following is true about snake antivenom?

a. Antivenom may reverse symptoms of envenomation including laboratory abnormalities.
b. Antivenom should be administered around the wound.
c. Even mild envenomations require antivenom.
d. Exotic snakebites require only antivenom if neurologic symptoms are present.
e. Pregnancy is a contraindication to receive antivenom.

Answer: a. Pregnancy is not a contraindication to antivenom therapy. Any victim of a venomous snakebite with moderate or severe envenomation is a candidate for antivenom. Antivenom may reverse the local signs, systemic symptoms, and laboratory abnormalities associated with envenomation. As with coral snakes, many patients do not show any early signs after envenomation by exotic snakes. The antivenom should be administered before neurologic changes develop. All antivenom should be administered intravenously.

5. A 28-year-old man presents with a copperhead snakebite that occurred 1 hour ago. He complains of pain and moderate swelling of his right hand and wrist. He has no systemic symptoms and has a pulse of 80 bpm and a normal blood pressure. His initial coagulation studies are normal. What should be done?

a. Admit for observation and antivenom
b. Discharge with elevation, ice, and pain medications
c. Give antivenom if swelling spreads to forearm
d. Immediate antivenom, four vials intravenously
e. Observe 12 hours for signs of increasing envenomation

Answer: e. The majority of copperhead bites do not require antivenom. Some toxicologists will administer antivenom if extremity swelling is progressive and may cause long-term disability. Copperhead bites generally cause a moderate amount of swelling that may peak 24 to 36 hours after the bite. All venomous snakebites should be observed for at least 12 hours if signs of envenomation are present. If systemic signs develop or the patient develops a coagulopathy, antivenom may be indicated.

6. Which of the following patients is most likely to be the first discharged home safely?

a. A pregnant woman with black widow envenomation
b. A 3-year-old child with a scorpion sting 1 hour before arrival
c. A 55-year-old man with hypertension and coronary artery disease with black widow envenomation
d. An 8-year-old with a coral snake bite
e. An 18-year-old with an unknown snakebite 8 hours before arrival

Answer: e. A scorpion sting in a child should be observed for at least 6 hours. Symptomatic children with stings should be admitted. Most venomous snakes will show signs of envenomation with 6 hours. If this patient is asymptomatic, it is likely that the snake was nonvenomous or this was a "dry bite." All children with envenomation and coral snake bites should be admitted for observation. Pregnant patients and those with coexisting medical problems should be admitted after black widow envenomation.

7. While walking in shallow water, a patient accidentally steps on a stonefish. Which of the following is the least effective diagnostic and treatment intervention?

a. Irrigating the wound with vinegar
b. Obtaining radiographs for retained foreign body
c. Observing for cardiovascular and respiratory symptoms
d. Removing the spine with forceps
e. Using hot water to relieve the pain

Answer: a. Stonefish, a type of bony fish, may cause serious cardiac and respiratory symptoms, which can be prevented by early administration of the appropriate antivenom. The fish spines of bony fish should be removed with forceps because they are thick and less likely to break off at the skin (like a bee stinger). In all cases, the wound should be copiously irrigated. Vinegar has been shown to be useful for some types of nematocyst injuries from jellyfish. Significant analgesia is achieved by submersion of the wound in hot water for 30 to 90 minutes or until improvement. There is a stonefish antivenom in limited supply for severe envenomation.

8. A patient presents with a necrotic lesion on his midthigh that started as a bleb while he was working in his woodshed 3 days ago. The wound has grown gravitationally. Her blood pressure is 110/80 mm Hg, respiratory rate 16 rpm, heart rate 110 bpm, and temperature 38.3°C. You suspect an envenomation of which of the following?

a. *Centruroides exilicauda*
b. *Hapalochlaena maculosa*
c. *Haplopelma lividum*
d. *Latrodectus mactans*
e. *Loxosceles reclusa*

Answer: e. The cobalt blue tarantula, *Haplopelma lividum,* is an aggressive spider with toxic venom. The black widow spider is *Latrodectus mactans*. *Centruroides exilicauda,* which is found in Arizona, is a particularly dangerous kind of scorpion. The blue-ringed octopus is *Hapalochlaena maculosa*. The brown recluse spider, *Loxosceles reclusa,* causes an initial white area of vasoconstriction at the site of the bite within 3 or 4 hours. A bleb then forms in the center of this area, and an erythematous ring arises on the periphery. The lesion at this stage resembles a bull's-eye. The bleb darkens, necroses during the next several hours to days, and continues to spread slowly and gravitationally.

KEY CONCEPTS

- After removing the patient from the source of injury, burns should be cooled with room temperature water while avoiding hypothermia in patients with large burns.
- Clinical signs such as facial burns, hoarseness, drooling, carbonaceous sputum, and singed nasal hairs indicate inhalation injury; however, they are poor predictors of injury severity.
- Confirmation of inhalation injury is best accomplished by direct visualization of the glottic inlet via flexible or rigid video laryngoscopy augmented by topical anesthesia and judicious sedation, when necessary.
- Worsening hoarseness, edema, or soot in the supraglottic region necessitates immediate intubation.
- Supplemental oxygen should be given in patients with suspected inhalation injury and determination of carbon monoxide levels should be performed.
- Crystalloid resuscitation is required to support vital organ perfusion and is guided by urine output. Care should be taken to avoid over-resuscitation.
- Pain relief is required for larger burns and is accomplished through both pharmacological and non-pharmacological methods. Pain levels should be assessed frequently.
- Patients with large body surface area burns (>20% for adults and >10% for children and the elderly) or deep burns will require admission to a burn center. Most other patients presenting to the ED will have small and superficial burns that can be managed with over-the-counter or commercially available topical agents.

1. A 30-year-old man presents with a burn to both anterior aspects of his forearms after being burned by a radiator. He has severe pain and has deep (reticular) dermis extension of the burn. You call the consultant and describe the burn as which of the following?
   a. Superficial burn of 4.5% body surface area
   b. Superficial partial thickness burn of 9% body surface area
   c. Deep partial thickness burn of 4.5% body surface area
   d. Full thickness burn of 2.5% body surface area
   e. Deep full thickness burn of 9% body surface area

Answer: c. Deep partial thickness burns extend through the epidermis into the deep (reticular) dermis. Body surface area is determined by the rule of nines. Nine percent for each upper extremity means that the forearm is approximately one fourth of 9%. Two forearms burns are half of 9%. Partial thickness burns are often more painful than full thickness burns, in which all of the nerve endings are destroyed.

2. A 3-year-old boy presents with circumferential burns involving both upper extremities, including his hands, from pulling a boiling pot of water off the stove. The burns are mixed partial thickness and full thickness burns. Which of the following best describes the body surface area burned and the most appropriate disposition?
   a. 9% and admit to pediatrics with surgery consultation
   b. 9% and transfer to burn unit
   c. 18% and transfer to burn unit
   d. 18% and admit to pediatric surgery
   e. 20% and admit to pediatric intensive care unit

Answer: c. Children with burns over 10% total body surface area (TBSA) should be transferred to a burn unit. In addition, hand burns should be treated at a burn unit. Circumferential burns are not consistent with a splash injury, and child abuse should be suspected and reported.

3. A 55-year-old, 80-kg man presents with partial thickness burns of both his legs, front torso, and groin. What is the initial fluid resuscitation according to the Parkland formula?
   a. 500 mL LR solution in the first 4 hours
   b. 1100 mL normal saline in the first hour
   c. 1100 mL of LR solution in the first hour
   d. 2400 mL normal saline in the first hour
   e. 4200 mL LR solution in the first 4 hours

Answer: c. The amount of LR solution required for the first hour can be rapidly estimated with the Parkland formula by multiplying the estimated total body surface area (TBSA) of the partial and full thickness burn (55%) by body weight in kilograms (80 kg) and dividing by 4.

4. A 25-year-old woman presents with a partial thickness burn to her right forearm after a grilling accident. You note areas of gray discoloration with decreased blanching in the erythematous region of the burn. In addition to irrigation, debridement, and dressing with a nonadherent dressing, which of the following would be the most appropriate treatment?
   a. Apply silver sulfadiazine and follow-up with plastic surgery in 1 week
   b. Calculate the Parkland formula and administer fluids before discharge
   c. Educate the patient about daily dressing changes and have her follow-up with plastic surgery in 24 to 48 hours
   d. Immerse her forearm in ice water for pain control
   e. Unroof soft blisters and have the patient follow-up with her primary care physician in 1 week

Answer: c. The distinction between superficial and deep partial thickness burns is important in that deep partial thickness burns often do not heal within 2 or 3 weeks and may result in severe scarring and contractures, especially in children. As a result, deep partial thickness burns that do not heal within 21 days may require excision and skin grafting to minimize scarring. Deep partial thickness burns may also progress to full thickness burns during the course of several days after injury. Burns less than 20% TBSA can be treated with oral hydration. Although there are little data to guide treatment, blisters are generally left intact initially if possible unless tense or over a joint. They may later require debridement.

5. Which of the following is an indication for intubating a patient who was found in a burning house and requires transfer 2 hours away to a burn center?
   a. Soot in the airway and singed nasal hair
   b. Dyspnea and hypoxia
   c. Patient unable to handle own secretions
   d. Altered mental status
   e. All of the above

Answer: e. Traditionally, inhalation injury was diagnosed on the basis of clinical findings, such as facial burns, singed nasal vibrissae, carbonaceous sputum, and a history of injury within a closed space. However, these findings are neither highly sensitive nor highly specific. If the patient is at a burn center and symptoms are mild, then an approach involving flexible laryngoscopy and close observation in an ICU is reasonable. However, if the patient requires prolonged transport to a tertiary care center then intubation is recommended if signs or symptoms or airway involvement are present.

6. What is the most appropriate management for a superficial partial thickness burn on the forearm?
   a. A clean dry dressing, such as gauze
   b. Petroleum gauze
   c. Systemic antibiotics and silver sulfadiazine
   d. Topical antibiotic ointment
   e. B and D

Answer: e. Superficial partial thickness burns may be treated with a topical antibiotic ointment, petroleum gauze or one of several commercially available burn dressings. Silver sulfadiazine, as well as dry dressings, will slow re-epithelialization. Silver sulfadiazine is appropriate for infected or heavily contaminated burns. Systemic antibiotics are *not* indicated for non-infected burns.

55 Chemical Injuries

KEY CONCEPTS

- For chemical injury, the degree of skin destruction is determined mainly by the properties of the toxic agent, its concentration, and the duration of its contact.
- Chemical injuries are commonly encountered after exposures to acids and alkalis.
- Hazardous materials (HazMat) are substances that can cause physical injury and damage the environment if improperly handled.
- In dealing with HazMat incidents, two distinct goals need to be achieved: (1) Containing the HazMat, extinguishing the fire and explosions, and eventually cleaning the site, and (2) evacuating victims exposed to the HazMat from the scene, decontaminating, and rapidly evaluating for treatment.
- Decontamination consists of removing contaminated clothing and hydrotherapy (i.e., washing of the skin) for the majority of exposures. For lithium, potassium, and sodium exposure, hydrotherapy is contraindicated because of the exothermic reaction upon contact with water.
- Alkali burns tend to penetrate deeper into tissue layers than acidic burns; and as a result, are associated with greater morbidity.
- Hydrofluoric acid burns can be associated with significant systemic hypocalcemia.
- Exposure to various toxic gases can occur from common industrial settings, and knowledge of these agents is necessary for proper treatment by the emergency clinician.
- Unconventional chemical weapons may be categorized into four major classifications: nerve agents, vesicants, choking agents, and cyanide agents.

1. A 34-year-old man presents with burning, erythema, and blurred vision after cleaning liquid is splashed in his eye. Which of the following should you do?

a. Be concerned about acidic burn.
b. Be concerned about alkali burn.
c. Postpone irrigation until confirmation of the type of cleaning liquid.
d. Use a miotic agent.
e. Use copious irrigation until a pH of 7.4.

Answer: e. When a chemical injury to the eye is suspected, regardless of the offending agent, copious irrigation should be started immediately. Irrigation with water or normal saline should continue until the pH is at a physiologic level (approximate pH of 7.4). All but the mildest burns should be treated with a long-acting cycloplegic and mydriatic.

2. A patient presents to the emergency department (ED) complaining of severe burning of both hands since leaving her building this morning. In her history, you find out that the metal doors of her housing complex had just been cleaned of all rust. She has small blisters on her palms and a white appearance to the skin. What is the next step?

a. Consult plastics for an alkali burn.
b. Leave blisters, cover with bacitracin and gauze.
c. Mix calcium gluconate with KY jelly, apply to hands, and cover with gloves.
d. Obtain pain control and prescribe oral clindamycin.
e. Administer intravenous magnesium.

Answer: C. The rust cleaning and physical findings are descriptive of a hydrofluoric acid burn. It is also commonly used in the production of microelectronics, etching glass, for removing rust, and for cleaning cement and bricks. Calcium gluconate (2.5%) gel is the preferred topical agent. However, this gel is often not available in hospital pharmacies, but it can be made by mixing 3.5 g calcium gluconate powder in 150 mL of a water-soluble lubricant. The gel should be secured by an occlusive cover, such as a latex glove.

3. In a hydrofluoric burn, what electrolyte abnormality causes systemic manifestations contributing to this chemical's significant morbidity and mortality?

a. Hypercalcemia
b. Hypermagnesemia
c. Hypocalcemia
d. Hypokalemia
e. Hyponatremia

Answer: C. Hydrofluoric acid binds calcium and magnesium ions with strong affinity. Systemic manifestations of fluoride toxicity are at least partly related to hypocalcemia and include abdominal pain, muscle fasciculations, nausea, seizures, ventricular dysrhythmias, and cardiovascular collapse. Of note, hyperkalemia is often a terminal finding in fatal cases.

4. A 30-year-old female presents by emergency medical services confused and hypoxic. She has a history of depression, and was found with an empty bottle of fluoxetine, which she takes for her depression, and cefalexin and phenazopyridine, which she is currently taking for a urinary tract infection. She is noted to be hypoxic with a saturation of 83%, which does not improve with oxygen administration. She is speaking full sentences but is confused. The nurse informs you the arterial blood sample blood has a "chocolate brown" appearance. Which of the following represents the best management strategy?

a. Obtain a carboxyhemoglobin level and consider hyperbaric oxygen
b. Obtain a methemoglobin level, and consider administration of methylene blue
c. Determine if acidosis is present, and if present, administer sodium thiosulfate
d. Determine if the urine fluoresces with a Wood's lamp
e. Examine the urine for presence of crystals.

Answer: B. The patient is suffering from methemoglobinemia. For those symptomatic patients without glucose-6-phosphate dehydrogenase (G6PD) deficiency, 2 mL/kg of 1% methylene blue can be administered over 3 to 5 minutes. Symptoms typically improve within 20 minutes, although repeat doses may be needed. Severe cases can be treated with exchange transfusion. Candidates for exchange transfusion include those with G6PD with significant toxicity from methemoglobinemia or those patients who fail to respond to methylene blue.

5. A 21-year-old man presents to the emergency department (ED) with chronic shortness of breath and a "huffer's rash." The patient has dry, cracked skin with perioral pyoderma. The cause of these symptoms is most likely which of the following?

a. Anhydrous ammonia
b. Cyanide
c. Hydrocarbons
d. Nitrites
e. White phosphorous

Answer: C. The toxicity from hydrocarbons can affect many different organs, but the lungs are the most commonly affected organ. The toxicity is directly related to the volatility and inversely related to the viscosity and surface tension. Chronic dermal exposure to hydrocarbons can result in perioral or perinasal dermatitis with pyoderma. This so-called "huffer's rash" is primarily seen with recreational abuse.

6. Nerve agent poisoning may be rapidly fatal. Besides appropriate decontamination and atropinization, which of the following should be administered?

- **a.** British anti-Lewisite 2 mg/kg intramuscularly
- **b.** Methylene blue 2 mg/kg intravenously
- **c.** *N*-acetylcysteine 140 mg/kg orally
- **d.** Pralidoxime 600 mg intramuscularly
- **e.** Pyridostigmine 1 mg intravenously

Answer: D. Nerve agents work by affecting acetylcholine (ACh) levels via inhibition of acetylcholinesterase. ACh receptors are found on the postsynaptic receptor of cholinergic synapses. These receptors can be either nicotinic or muscarinic. The effects at the muscarinic receptors include excess secretions and smooth muscle contractions. The initial recommended treatment is 2 mg of atropine for adults, although much larger doses will likely be required. Pralidoxime should also be administered to patients with suspected or known ingestion with significant symptoms. This agent potentially helps prevent irreversible inhibition of the acetylcholinesterase enzyme.

7. A 31-year-old man was found in his apartment after an apparent suicide attempt. He smells of bitter almonds. He is obtunded with hypotension, tachypnea, and normal oxygen saturation. After decontamination and resuscitation, what is the next step?

- **a.** Atropine
- **b.** Hydroxocobalamin
- **c.** Methylene blue
- **d.** Naloxone
- **e.** Sodium thiosulfate

Answer: B. This patient likely has cyanide poisoning. One older method of treatment involves the administration of amyl nitrite, sodium nitrite, and sodium thiosulfate. The U.S. Food and Drug Administration (FDA) approved, and we recommend hydroxocobalamin (Cyanokit) to treat cyanide intoxication. Hydroxocobalamin binds to cyanide to form cyanocobalamin, which subsequently undergoes renal excretion.

KEY CONCEPTS

- Antibiotics do not have a role in the management of pulpitis, whether reversible or irreversible.
- Missing teeth must be accounted for. They may be hidden, left at the scene, aspirated, swallowed, or traumatically impacted.
- Avulsed permanent teeth should be stored in milk, saline, or a commercially available solution and reimplanted at the earliest possible opportunity. Primary teeth are not reimplanted.
- A localized periapical abscess that is successfully drained in an immunocompetent patient does not require antibiotics.
- Cover dental fractures that involve the dentin or pulp with a calcium hydroxide paste then promptly refer to a dentist.
- Periodontal splinting material can stabilize reimplanted teeth or luxated teeth once they have been put in the appropriate position.
- Advanced imaging for localized periapical abscess is unnecessary; however, contrast-enhanced CT of the face is necessary if signs or symptoms of a **deep neck infection** are present.
- Supraperiosteal nerve blocks can anesthetize individual teeth but are less effective for adult mandibular teeth since the surrounding bone is thicker.

1. A 27-year-old previously healthy man presents with dental pain, facial swelling, and a muffled voice. Physical examination is remarkable for trismus, poor dentition with diffuse periodontal disease, and an elevated floor of the mouth. You determine endotracheal intubation is necessary. Which of the following statements regarding the patient's management is *true?*

a. An awake intubation technique is indicated.
b. Pre-induction muscle relaxants will decrease the trismus.
c. Rapid sequence induction with rocuronium is indicated.
d. Succinylcholine will worsen the trismus.
e. Transtracheal retrograde wire intubation is indicated.

Answer: a. The trismus is mechanical (presence of a mass, abscess of phlegmon). Neuromuscular blockade, regardless of the choice between depolarizing or non-depolarizing agents, is not likely to help. Patients with anatomically difficult airway and symptoms of airway obstruction are optimally managed with an awake intubation, using a either a videolaryngoscope or flexible endoscope is indicated. Patients with limited oral access will be best managed with nasotracheal intubation over a flexible intubating bronchoscope. Transtracheal retrograde wire intubation is rarely, if ever, indicated in modern emergency practice. Traditional "muscle relaxants" are sedatives and have no direct muscle-relaxing properties.

2. A 21-year-old man presents with tooth pain. He underwent a left mandibular premolar extraction three days ago. Examination is remarkable for a painful but nonbleeding tooth socket. The face and mandible are not swollen. Which of the following statements regarding this patient's condition is *true?*

a. Analgesia is the main goal of emergency department (ED) therapy.
b. Antibiotics are indicated.
c. Dental consultation in the ED is indicated.
d. Opioids are the treatment of choice.
e. Provocation of socket bleeding is encouraged to form new clot.

Answer: a. A dry socket is an exquisitely painful post-extraction syndrome that typically occurs 3 or 4 days later. The pathophysiology is loss of the healing blood clot and localized bone infection. Treatment consists of pain control. Optionally, gentle irrigation, and packing with gauze soaked in eugenol can be performed. Opioids may be used, but this would be in addition to nonsteroidal antiinflammatory drugs (NSAIDs), which are excellent analgesics for dental pain and are preferred due to the inflammatory component.

3. A 2-year-old previously healthy toddler presents with an avulsed tooth after hitting his mouth on the ground after a trip and fall. Physical examination is remarkable for a non-toxic appearing child who is crying and has an open slightly oozing pocket at the maxillary central incisor. The mother has brought the tooth stored in milk and the event occurred one hour ago. Which of the following statements regarding the patient's management is *true?*

a. Set-up for conscious sedation should be arranged, because the replacement of the avulsed tooth may be painful.
b. The mother should store the tooth in the patient's mouth.
c. The tooth should be placed in Hank's balanced salt solution.
d. The tooth should be replaced in the socket as soon as possible.
e. The tooth should not be reimplanted.

Answer: e. Management of recovered avulsed teeth depends on the age of the patient and the length of time for which the tooth has been displaced. Avulsed primary teeth are not reimplanted. Reimplanted primary teeth ankylose or fuse to the bone, so although the dentofacial complex grows downward and forward, the reimplantation site does not. There also may be interference with the eruption of the permanent tooth. Cosmetic deformity results in either case. Thus, this patient should be referred to a specialist for consideration of a space maintainer or cosmetic appliance.

4. A 25-year-old previously healthy woman presents with recurrent pain from her temporomandibular joint disorder (TMD). The pain is described as being dull, and it worsens during the course of the day. Physical examination is remarkable for mild tenderness to palpation over the temporomandibular joint (TMJ) and some spasm noted over the masseter and internal pterygoid. Which of the following statements regarding the patient's management is *true?*

a. A computed tomography (CT) scan of the face would be helpful.
b. A Panorex should be obtained.
c. Patient should be prescribed narcotic agents.
d. Treatment consists of external application of heat for 15 minutes four to six times per day, soft diet, analgesics including nonsteroidal antiinflammatory drugs (NSAIDs).
e. Ultrasound of the joint should be obtained.

Answer: d. TMJ radiographs are not helpful. Treatment consists of the external application of heat for 15 minutes four to six times per day, soft diet, analgesics including NSAIDs, and consideration of a muscle relaxant, such as diazepam (2 to 10 mg up to four times per day). Patients should be referred to a dentist specializing in TMD, such as a periodontist or a periodontal prosthodontist.

5. A 32-year-old previously healthy man presents with a fractured maxillary lateral incisor after a mechanical trip and fall. Physical examination is remarkable for a portion of the tooth that has an ivory-yellow appearance. You determine that he has a fractured tooth. Which of the following statements regarding the patient's management is *true?*

a. A calcium hydroxide paste should be used to cover the exposed dentin.
b. A dressing can be placed for comfort.
c. An urgent pulpotomy is indicated.
d. If this were a pediatric patient or an adolescent, it would be considered less serious, because children have more dentin than adults.
e. The patient will likely need an emergent root canal.

Answer: a. This patient has a fracture involving the dentin, which has an ivory-yellow appearance. The pulp continually lays down dentin throughout the life of the tooth. In a child, the pulp is relatively large in size, and there is less dentin; the inverse is true in the adult. Because dentin is a microtubular tissue capable of preventing bacteria to percolate into the pulp chamber, fractures involving dentin are more serious in children and adolescents. In younger patients, the management of dentin fractures involves the immediate placement of a dressing of calcium hydroxide paste over the exposed dentin. Early intervention may prevent contamination of the pulp and avoid the need for subsequent root canal. In an adult, who has a greater thickness of dentin compared with pulpal tissue, there is less need for urgent referral to a dentist. A dressing can be placed on the tooth for comfort. Referral should be made to a dentist for the next working day.

57 Ophthalmology

KEY CONCEPTS

- Prophylactic topical antibiotics are not indicated for the treatment of corneal abrasions, and eye patches are not recommended because they can mask a worsening infection.
- Eyelid lacerations that may require referral to a plastic or ophthalmic surgeon include those with lid margin lacerations, a canalicular laceration, or levator or canthal tendon injuries.
- Alkaline burns to the cornea and conjunctiva need to be irrigated until a neutral pH is attained because they produce a liquefactive necrosis that penetrates and dissolves tissue.
- Admit hyphema patients with sickle cell trait, uncontrolled elevations in intraocular pressures (IOPs), hyphema of greater than 50%, and when there is concern for rebleeding.
- Avoid manipulation, palpation, or tonometry on a suspected globe rupture pending an ophthalmology consult.
- Scleritis, an autoimmune inflammatory process involving the sclera, can be confused with episcleritis caused by inflammation in the more superficial episcleral layer of the eye.
- Episcleritis, unlike scleritis, is associated with much less discomfort, a pinker and more pronounced peri-limbal injection from superficial episcleral vessels that—unlike the deeper injected scleral vessels in scleritis—will vasoconstrict and blanch with 10% phenylephrine. Treatment of both involves topical corticosteroid drops.
- Endophthalmitis is an infection of the eye itself, typically following intraocular surgery, that is treated with intravitreal antibiotics.
- Herpes zoster keratoconjunctivitis can complicate herpes zoster ophthalmicus. Emergent ophthalmologic consultation and treatment with systemic antiviral agents are required.
- Treatment of acute angle-closure glaucoma requires both a reduction in the production and an increase in the outflow of aqueous humor (see Box 57.1).
- With anisocoria, the following considerations help in determining which pupil—the larger or the smaller—is the pathological one: (1) parasympathetic innervation constricts a pupil in bright light, whereas sympathetic stimulation helps dilate a pupil in the dark; (2) an abnormally small pupil may therefore be due to either a decrease in sympathetic stimulation or an augmentation of parasympathetic stimulation—but likely the former (e.g., Horner syndrome); (3) an abnormally large pupil may therefore be due to either a decrease in parasympathetic stimulation or an augmentation of sympathetic stimulation—but likely the former (e.g., partial third-nerve palsy from compression, Adie's pupil, pharmacological mydriasis); or (4) the abnormally small pupil will usually look worse in the dark, whereas the abnormally large pupil will usually look worse in the light.

1. A 23-year-old male presents with left periorbital pain after being struck with a fist. On examination, there are no globe injuries but marked periorbital swelling is noted. Computed tomography (CT) of the face reveals an orbital floor fracture. Which of the following would be the most likely physical findings?

a. Cheek anesthesia, enophthalmos, and limitation of upward gaze
b. Cheek anesthesia, ptosis, and limitation of inferior gaze
c. Forehead anesthesia and afferent papillary defect
d. Forehead anesthesia, diplopia, and limitation of lateral gaze
e. Ptosis, miosis, and ipsilateral anhydrosis

Answer: a. An orbital floor fracture may entrap the inferior rectus and inferior oblique muscles, resulting in diminished upward gaze. Other findings may include ptosis, enophthalmos, ipsilateral cheek/lip anesthesia, and orbital emphysema. As many as 25% of such patients have associated globe injuries. Option E describes Horner syndrome, which is not a typical finding.

2. A 20-year-old male presents with periorbital pain and swelling after a blow to the eye by a softball. Physical examination reveals proptosis with significant decline in vision and limitation of ocular motion in all planes. Tonometry reveals an intraocular pressure (IOP) of 45 mm Hg. Which of the following should be the first indicated maneuver?

a. Acetazolamide 500 mg IV, mannitol 20 g IV, and topical timolol
b. Computed tomography (CT) scan of the head and face
c. Endotracheal intubation and hyperventilation
d. Immediate lateral canthotomy and cantholysis
e. Ophthalmologic consultation

Answer: d. These findings should make one suspect retrobulbar hemorrhage. All of these interventions are likely indicated. Intraocular hypertension may compromise central retinal artery flow. Although immediate ophthalmologic consultation and pressure-lowering maneuvers are indicated, lateral canthotomy and cantholysis will provide the most rapid temporizing measure to preserve vision.

3. A 43-year-old male presents with acute ocular pain after a splash injury from drain cleaner. What should be the sequence of interventions?

a. Copious irrigation for 10 minutes, pH testing, cyclopentolate cycloplegia, topical antibiotics/intraocular pressure (IOP) measurement
b. Intravenous (IV) analgesia, cyclopentolate cycloplegia, IOP measurement, isotonic irrigation
c. IOP measurement, analgesia, head-up position, cycloplegia
d. Phenylephrine cycloplegia, isotonic irrigation for 10 minutes, pH testing, slit-lamp examination for foreign bodies
e. Phenylephrine cycloplegia, slit-lamp examination for foreign bodies, isotonic irrigation for 10 minutes, pH testing

Answer: a. Copious irrigation, ideally beginning at the scene, is the cornerstone of management. Nitrazine pH testing after 10 minutes should guide the need for continued irrigation. Cycloplegia, IOP measurement, and topical antibiotics come after pH normalization. Phenylephrine is contraindicated for cycloplegia in these cases because of its vasoconstrictive properties.

4. A 17-year-old girl who wears contact lenses presents with a 24-hour history of right eye pain. Physical examination reveals a right corneal abrasion at the 6-o'clock position of the limbus. Appropriate treatment consists of which of the following?

a. Cessation of contact lens wear, eye irrigation (qid) with isotonic saline solution, followed by instillation of undiluted topical tetracaine for 5 days
b. Emergent ophthalmology consultation
c. Tetanus prophylaxis, eye patching for 48 hours, antibiotic ointment, and a 24-hour recheck
d. Tetanus prophylaxis, topical nonsteroidal antiinflammatory drugs (NSAIDs), cessation of contact lens wear, and a 24-hour recheck
e. Topical nonsteroidal medications, topical anti-pseudomonal antibiotic, and a 24-hour recheck

Answer: e. Tetanus prophylaxis is not indicated for corneal abrasion unless there is corneal perforation or contamination with organic material. Topical NSAIDs reduce corneal abrasion pain. Antipseudomonas coverage with cessation of contact lens wear is appropriate. Eye patching is not indicated. Administration of undiluted topical anesthetics for more than 24 hours is untested and may be dangerous. Oral analgesics may be needed.

5. How do patients with subconjunctival hemorrhage most commonly present?
   - **a.** Asymptomatic blood in the eye, noticed in the mirror or by a friend
   - **b.** Decreased visual acuity
   - **c.** Foreign body sensation
   - **d.** Modest pain
   - **e.** Photophobia

Answer: a. Any significant symptoms, such as pain, decreased vision, foreign body sensation, or photophobia, should spark the search for more serious pathology. Bilateral hemorrhage in the absence of a clear cause (e.g., severe vomiting) should raise suspicion for coagulation issues.

6. A 38-year-old man presents with unilateral left-sided visual loss after a motor vehicle collision (MVC). The only clinical finding is a left-sided hyphema rising to 50% of the height of the anterior chamber. Intraocular pressure (IOP) is 17 mm Hg in the unaffected eye and 29 mm Hg in the affected eye. Appropriate management should include which of the following?
   - **a.** Cycloplegia, intravenous (IV) mannitol, ophthalmology consultation
   - **b.** IV analgesia and antibiotic, immediate ophthalmologic consultation for decompression resulting from intraocular hypertension
   - **c.** Oral acetazolamide, shield, antiemetics, 24-hour recheck
   - **d.** Topical beta-blocker, shield, modest analgesia, admission
   - **e.** Topical beta-blocker, topical nonsteroidal antiinflammatory drugs (NSAIDs) for pain, shield, 24-hour recheck

Answer: d. Significant hyphema is an indication for admission. The presence of elevated IOP requires urgent treatment (which might also include topical alpha agonists or IV acetazolamide, and so on), shield, elevation of the head, and cautious use of systemic analgesics. Any form of platelet inhibition would be contraindicated.

7. What is the major complication of hyphema?
   - **a.** Detached retina
   - **b.** Glaucoma
   - **c.** Horner syndrome
   - **d.** Rebleeding
   - **e.** Vitreous hemorrhage

Answer: d. Rebleeding typically occurs 2 to 5 days later as the clot retracts. It is most common in patients with elevated intraocular pressures (IOPs), hyphema greater than 30% of the anterior chamber, and with delayed presentation. Rebleeding may lead to glaucoma and synechia formation.

8. A 48-year-old woman presents with right eye pain, photophobia, and decreased vision after a motor vehicle collision (MVC). Physical examination reveals an irregularly shaped pupil and a small hyphema. Photophobia, decreased acuity, minimal pupil reactivity, and bloody chemosis are seen on examination. What is the most likely diagnosis?
   - **a.** Acute angle-closure glaucoma
   - **b.** Blunt ciliary injury
   - **c.** Iridodialysis
   - **d.** Scleral rupture
   - **e.** Traumatic miosis

Answer: d. Scleral rupture occurs either at the insertion of the extraocular muscles or at the limbus, where the sclera is the thinnest. A "teardrop" pupil is often seen and may be accompanied by bloody chemosis or severe subconjunctival hemorrhage. Brownish-black pigment prolapse may also be seen. Intraocular pressure (IOP) may be low, but tonometry is generally contraindicated in cases of suspected globe injury.

9. A 26-year-old man presents with a 3-day history of right eye pain, decreased vision, and photophobia. He reports a history of left eye trauma 6 weeks prior, with hyphema, traumatic iritis, and persistent decreased vision. He is otherwise healthy. Physical examination reveals photophobia in the right eye with bilateral decreased vision. Before the past 3 days, the vision in the right eye had been perfect. What is the most likely explanation for his right eye symptom?
   - **a.** Collagen vascular disease
   - **b.** Post-traumatic conjunctivitis
   - **c.** Post-traumatic retinal tear
   - **d.** Spontaneous vitreal hemorrhage
   - **e.** Sympathetic ophthalmia

Answer: e. Sympathetic ophthalmia is an autoimmune inflammatory response in the unaffected eye, days to months after uveal trauma in the opposite eye. Pain, photophobia, and decreased vision are common. This patient had no findings consistent with conjunctivitis or collagen vascular disease, and a retinal tear would not typically be painful.

10. Oral antibiotics are indicated for which of the following?
    - **a.** Blepharitis
    - **b.** Chalazion
    - **c.** Dacryocystitis
    - **d.** Endophthalmitis
    - **e.** Hordeolum

Answer: c. Dacryocystitis is an infection of the lacrimal sac from nasociliary duct obstruction. Warm compresses are also recommended and may be helpful, although evidence is lacking. Warm compresses and topical antibiotics are appropriate for the other conditions. Intravitreal antibiotics are indicated for endophthalmitis.

11. Bedside ocular ultrasonography can provide useful information for which of the following conditions?
    - **a.** Lens dislocation
    - **b.** Retinal detachment
    - **c.** Vitreous hemorrhage
    - **d.** All of the above

Answer: d. A displaced lens can be seen in the relatively hypoechoic vitreous. Vitreous hemorrhage and retinal detachment can both be diagnosed with emergency department bedside ultrasonography.

KEY CONCEPTS

Otitis Media

- Otitis media is caused by eustachian tube inflammation and dysfunction leading to fluid accumulation in the middle ear and proliferation of nasopharyngeal bacteria, most commonly *Streptococcus pneumoniae, Haemophilus influenzae,* and *Moraxella catarrhalis.*
- Diagnosis of acute otitis media (AOM) is made based on the presence of a middle ear effusion and signs of inflammation, including ear pain, tympanic membrane erythema and bulging, and fever.
- Bullae may be seen in some cases of AOM. This is known as bullous myringitis and treatment for this condition does not differ from non-bullous AOM.
- Mastoiditis is the most common suppurative complication of AOM though is rare in the modern era.
- Most cases of AOM resolve spontaneously, and observation for 2 to 3 days may be offered instead of immediate antibiotics in appropriately selected pediatric patients. Observation strategies are not validated in adult patients.
- Amoxicillin is the recommended initial antibiotic treatment of choice for most patients with AOM. Recurrent otitis media or treatment failures warrant selection of an alternate antibiotic regimen.
- Antibiotics alone do not provide improvement in pain in the first 24 hours of AOM, and pain should be addressed with the use of over-the-counter analgesics, such as acetaminophen or ibuprofen.
- Otitis media with otorrhea from an acute perforation is treated in the same manner as AOM without perforation. These perforations are typically small and resolve spontaneously.
- In patients with tympanostomy tubes presenting with acutely increased otorrhea, treatment with ototopical fluoroquinolone drops is recommended.

Otitis Externa

- Otitis externa (OE) is most commonly seen in warm climates and summer months. *Pseudomonas aeruginosa* and *Staphylococcus aureus* are the most common causative pathogens.
- Patients with immunocompromised states (diabetes) and persistent OE despite treatment should be evaluated for necrotizing OE, a severe, life-threatening infection of the outer ear and surrounding structures.
- Topical antibiotic drops are effective for most cases of OE. Systemic antibiotics are not indicated in immunocompetent patients with infection limited to the external ear canal.
- Fluoroquinolone drops are effective in most cases of OE, and they may be used in patients with a non-intact tympanic membrane. Aminoglycoside ear drops are ototoxic and should not be used in patients with a tympanic membrane (TM) perforation.

Necrotizing (Malignant) External Otitis

- Necrotizing external otitis is an aggressive from of OE with high associated mortality. Patients with diabetes mellitus, immunocompromise, and advanced age are at particular risk.
- Necrotizing otitis media should be considered in at-risk patients with progressive OE unresponsive to treatment.
- CT scanning can aid in diagnosis of necrotizing otitis media by demonstrating bony erosion and characteristic soft-tissue abnormalities.
- Necrotizing otitis media is treated with systemic antibiotics with activity against *Pseudomonas aeruginosa.* Ciprofloxacin is an appropriate initial antibiotic treatment.
- Patients with necrotizing otitis media should be managed in coordination with an otolaryngologist. Surgical intervention may be needed in refractory cases.

Mastoiditis

- Mastoiditis is the most common suppurative complication of otitis media, though it is a relatively rare condition. It may occur in patients without known preceding otitis media.
- The most common physical findings with mastoiditis are postauricular erythema and tenderness, protrusion of the auricle, and an abnormal TM.
- CT scanning is indicated in patients with suspected intracranial involvement from mastoiditis.
- Mastoiditis is treated with vancomycin (15 mg/kg IV), and an agent active against *Pseudomonas aeruginosa* is added in those who have previously received antibiotics.
- Consultation with an otolaryngologist is advised to assist in management of mastoiditis.

Sudden Hearing Loss

- Sudden sensorineural hearing loss (SSNHL) is defined as acute hearing loss over a period less than 3 days and is often misdiagnosed.
- SSNHL is idiopathic in 70% of cases. It may also be secondary to infection, otologic disease, trauma, vascular disease, hematologic disorders, or neoplasm.
- Imaging is not warranted in the ED for SSNHL unless a space-occupying lesion is suspected.
- SSNHL is treated with steroids (prednisone 1 mg/kg/day up to 60 mg, tapered over 10 to 14 days), and prompt surgical follow up for further evaluation.

Epistaxis

- Epistaxis is distressing to patients but rarely life-threatening. Most nosebleeds arise from an anterior locus.
- A causative link between hypertension and epistaxis has not been clearly established; however, patients are often hypertensive upon presentation, which may be associated with persistent bleeding.
- Most cases of anterior epistaxis can be managed in the ED with a combination of direct pressure, topical vasoconstrictive drugs, chemical cautery and packing, as needed. Tranexamic acid is an effective adjunct to these treatments that decreases immediate and delayed bleeding.
- Posterior epistaxis requires specialized packing techniques and monitoring in an inpatient setting.
- Refractory cases of epistaxis may require embolization or surgical treatment
- Routine use of prophylactic antibiotics in patients treated with anterior packing for epistaxis is not recommended.

Sialolithiasis

- Sialolithiasis is most commonly seen in the submandibular salivary gland. Risk factors include dehydration, anticholinergic and diuretic medications and smoking.
- Sialolithiasis may be treated with manual massage of the gland, sialogogues and anti-inflammatory pain medications.
- CT scan is the preferred imaging modality for suspected sialolithiasis and can identify stones and evidence of infection.
- Sialadenitis is an associated infection of the salivary glands, and antibiotics such as amoxicillin/clavulanate or clindamycin should be added to the treatment regimen.

Neck Masses

- Neck masses in children are likely inflammatory or benign congenital abnormalities. In adults, neck masses commonly represent malignancies.
- Physical examination findings such as stridor, dysphagia, voice changes and referred pain can help localize symptoms and guide diagnostic approach.
- CT scan with contrast is the preferred initial imaging modality to characterize the neck mass and involvement of surrounding structures.
- Patients with airway compromise secondary to a neck mass warrant emergent airway management and ENT evaluation.
- Patients with hoarseness lasting greater than 2 weeks and patients with neck masses unresolved within 2 weeks should be evaluated in follow-up by an otolaryngologist.

1. *Bullous myringitis* is most commonly caused by which of the following bacterial pathogens?
   a. *Streptococcus pneumoniae*
   b. *Pseudomonas aeruginosa*
   c. *Mycoplasma pneumoniae*
   d. *Staphylococcus aureus*

Answer: a. Bullae may be seen on the TM in up to 5% of cases of AOM in young children. While this finding has been historically associated with *M. Pneumoniae*, the bacterial pathogenesis of bullous myringitis is similar to that of non-bullous AOM and is treated similarly.

2. Which of the following patients with acute otitis media would be appropriate candidates for observation in lieu of immediate antibiotic treatment?
   a. A 6-month-old female with bilateral middle ear effusions with red bulging tympanic membranes and fever to 103.5°F
   b. A 6-year-old male with three days of ear pain, fever to 101.5°F and an erythematous, bulging TM with purulent material seen in the external ear canal
   c. A 2-year-old male without significant medical history presenting with 1 day of right ear pain and fever to 100.5°F. His exam reveals an erythematous TM with a middle ear effusion. There is no otorrhea or TM perforation noted.
   d. A 5-month-old female with fever to 101.2°F with an erythematous, bulging left TM seen on otoscopy.

Answer: c. Observation for 48 to 72 hours can be considered in lieu of immediate antibiotics in pediatric patients aged 6 months to 2 years with unilateral AOM of <48 hours duration, in the absence of severe symptoms (including temperature <102.2°F) or otorrhea. Patients older than 2 years with unilateral or bilateral AOM without otorrhea or severe symptoms can be appropriate for an observation strategy. Patients with otorrhea and those younger than 6 months should be treated with antibiotic therapy and are not appropriate for an observation approach. This decision should be made in conjunction with caregivers, and reliable follow-up must be in place. This approach is not validated in adult patients.

3. A 72-year-old female with a history of poorly controlled diabetes mellitus presents to the ED with severe left ear pain of 2 weeks duration. On ear examination, granulation tissue is noted in the external auditory canal. The patient has facial asymmetry, which her family states developed 2 days ago. What is the most appropriate imaging modality for this patient in the ED?
   a. MRI with contrast
   b. CT with contrast
   c. Skull x-ray
   d. Transcranial doppler ultrasound

Answer: b. This patient is presenting with findings consistent with necrotizing external otitis: granulation tissue in the EAC and facial paralysis. CT scan is the most appropriate choice for imaging in the ED and will demonstrate bony erosion and associated soft-tissue abnormalities. MRI and nuclear medicine studies may be helpful later in the patient's course.

4. Routine use of prophylactic antibiotics in patients treated with anterior packing for epistaxis has clearly demonstrated which of the following benefits?
   a. Prevention of toxic shock syndrome
   b. Reduction in incidence of acute otitis media
   c. Prevention of acute sinusitis
   d. None of the above

Answer: d. Administration of prophylactic antibiotics in patients discharged with anterior nasal packing has not been shown to prevent the listed infectious complications and is not recommended in patients without immunocompromise, valvular disorders, or in those whose packing will be left in place for less than 48 hours.

5. Neck masses in pediatric patients are more likely to represent an _____ process, while neck masses in adult patients are most likely secondary to a _____ process.
   a. inflammatory, inflammatory
   b. neoplastic, inflammatory
   c. inflammatory, neoplastic
   d. neoplastic, neoplastic

Answer: **c.** In adults, neck masses should raise concern for malignancy, as 80% of non-thyroid neck masses in adults are neoplastic in etiology. In children, the majority of neck masses are inflammatory in nature or secondary to benign congenital abnormalities.

KEY CONCEPTS

- Inhaled and systemic steroid medications are effective in controlling airway inflammation and have important roles in the management of asthma exacerbations.
- Inhaled bronchodilators and systemic corticosteroids remain the mainstays of management for most acute asthma exacerbations.
- Acute severe asthma requires rapid identification. Treatment may use strategies not used in mild-to-moderate exacerbations, such as infusion of magnesium sulfate, use of non-invasive ventilation, and endotracheal intubation.
- Ventilator management in the intubated asthmatic is critical and includes lower tidal volumes (6 to 8 mL/kg) and low respiratory rates, often less than 10 per minute.
- A normal partial pressure of carbon dioxide (P_{CO_2}) in a pregnant patient represents hypercarbia.
- Elevated lactic acid levels are common in critically ill asthmatics and do not reflect deterioration or a poor prognosis.
- ED management of acute asthma is expanding (up to 24 hours) as more non-critically ill asthmatics are treated in observation units.
- Integration of discharged patients with acute asthma into chronic management strategies to prevent relapse requires physicians be familiar with controlling medications, such as inhaled corticosteroids and long-acting beta agonists.

1. Which of the following is a risk factor for sudden death from asthma?
   a. A hospitalization for asthma in the past year but not within the past 30 days
   b. An emergency department (ED) visit for asthma in the past year but not within the past 30 days
   c. Current use of systemic corticosteroids
   d. Patient perception that the current exacerbation is very severe

Answer: c. Risk factors for sudden death are current or recent corticosteroid use, ED or hospitalization within the past 30 days, more than two hospitalizations for asthma in the past year, more than three ED visits for asthma in the past year, using more than two beta-agonist canisters per month, previous intubation or intensive care unit (ICU) visit, and difficulty perceiving symptoms or their severity.

2. A 23-year-old man with known severe asthma presents with an acute asthma flare over 2 hours. Physical examination reveals a well-developed man in marked respiratory distress. Heart rate is 120 beats/min, oxygen saturation is 90%, respiratory rate is 26 breaths/min, blood pressure is 140/92 mm Hg, and oral temperature is 98.7°F (37.2°C). Current medications are albuterol metered-dose inhaler (MDI) and fluticasone inhaler 500 µg twice daily. What therapy is recommended for this acute flare?
   a. Albuterol 2.5 mg nebulized, Ipratropium 0.5 mg, methylprednisolone 125 mg intravenously, and magnesium sulfate 2 g intravenously
   b. Epinephrine infusion at 5 µg/min
   c. Ipratropium 500 µg nebulized × three doses with methylprednisolone 125 mg intravenously
   d. Methylprednisolone 125 mg intravenously with salmeterol nebulized via continuous nebulization

Answer: a. Short-acting inhaled beta-agonists with ipratropium are the cornerstone of acute asthma management. Corticosteroids are indicated in any moderate or severe flare. Oral steroids are as efficacious as intravenous (IV) steroids if the patient can take oral medications. Magnesium sulfate may obviate the need for intubation or decrease hospital length of stay in severe cases. Salmeterol is a slow-onset, long-acting beta-2 agonist (LABA) not indicated in acute asthma management.

3. What is the medication combination of choice for the rapid sequence induction of an asthmatic?
   a. Etomidate/succinylcholine
   b. Ketamine/succinylcholine
   c. Midazolam/pancuronium
   d. Propofol/rocuronium

Answer: b. No choice is contraindicated. Ketamine is the sedative of choice because of its bronchodilatory effect. Propofol may have the same benefit, although less profoundly. Thiopental releases histamine, and etomidate does not bronchodilate. Succinylcholine releases trace amounts of histamine, but this is not known to cause any adverse effect. Rocuronium has an onset time similar to that of succinylcholine, no histamine release, and a prolonged duration of action. Either succinylcholine or rocuronium is acceptable for rapid sequence intubation (RSI) in acute asthma.

4. Which of the following is a risk factor for death in patients presenting with an asthma attack?
   a. Currently taking theophylline
   b. Family history of asthma
   c. Presence of symptoms for 1 week
   d. Use of three albuterol metered-dose inhalers (MDIs) per month

Answer: d. The following are risk factors for death from asthma:

- Past history of sudden severe exacerbations
- Prior intubation for asthma
- Prior asthma admission to an intensive care unit (ICU)
- Two or more hospitalizations for asthma in the past year
- Three or more emergency department (ED) care visits for asthma in the past year
- Hospitalization or an ED care visit for asthma within the past month
- Use of more than two MDI short-acting beta-2 agonist canisters per month
- Current use of or recent withdrawal from systemic corticosteroids
- Difficulty perceiving severity of airflow obstruction
- Comorbidities such as cardiovascular diseases or other systemic problems
- Serious psychiatric disease or psychosocial problems
- Illicit drug use, especially inhaled cocaine and heroin

5. A 25-year-old woman presents with wheezing and shortness of breath from asthma. She was recently exposed to cigarette smoke. She denies cough and fever. You cannot get much more of a history from her at this time because she finds it difficult to speak in complete sentences. Her vital signs are blood pressure, 136/85 mm Hg; heart rate, 110 beats/min; respiratory rate, 32 breaths/min; and temperature, 99°F (37.2°C). Her oxygen saturation is 92%. Her PEF is 50% of predicted. On physical examination, you note bilateral wheezing, regular tachycardia, and accessory muscle use. The remainder of her examination is normal. Over the course of 1 hour, she receives supplemental oxygen, three doses of nebulized albuterol (5 mg) mixed with ipratropium (0.5 mg), and oral prednisone 50 mg. She now reports feeling somewhat better. She speaks in longer sentences but still cannot speak in complete sentences. A repeat peak flow measurement is now 60% of predicted. Otherwise, there are no changes on a repeat physical examination. You plan to admit her to your ED observation unit. What is an appropriate next step in the management of this patient?
   a. Additional nebulized albuterol
   b. Intravenous (IV) magnesium sulfate
   c. IV methylprednisolone (Solu-Medrol)
   d. Oral montelukast

Answer: a. This patient presents with a moderate-to-severe asthma exacerbation. She has responded to initial therapy but continues to have moderate symptoms. Additional adrenergic medications are indicated. Because she is tolerating nebulized medications and is responding, there is no need for IV or subcutaneous adrenergic agents, such as terbutaline. IV magnesium sulfate is a smooth muscle relaxant, but it is generally reserved for more severe asthma exacerbations. Oral and IV steroids have the same efficacy, and regardless of the route, only need to be administered every 6 to 8 hours. Montelukast is a leukotriene-modifying drug that is used in chronic management.

60 Chronic Obstructive Pulmonary Disease

KEY CONCEPTS

- Chronic obstructive pulmonary disease (COPD) is "a common, preventable and treatable disease that is characterized by persistent respiratory symptoms and airflow limitation that is due to airway or alveolar abnormalities usually caused by significant exposure to noxious particles or gases."
- COPD exacerbation is a worsening of symptoms from baseline beyond day-to-day variation and requiring additional treatment. It is most commonly caused by viral infections, bacterial infections, environmental pollutants, particulate matter, thrombotic disease, or temperature changes.
- Dyspnea, cough, increased sputum production, and sputum purulence are the most common symptoms of COPD exacerbations. They are a result of airway inflammation, increased mucus production, and air trapping.
- Treatment includes nebulized short-acting beta-agonists such as albuterol, short-acting anticholinergics such as ipratropium, and glucocorticoids such as prednisone.
- Antibiotics should be given to patients with COPD exacerbations who have signs of lower respiratory tract infection, increased purulence of their sputum, or who have respiratory failure. Antibiotics should be provided empirically based on local resistance patterns to *Streptococcus pneumoniae, Moraxella catarrhalis,* and *Haemophilus influenzae* with common regimens including amoxicillin/clavulanate, macrolides, or tetracyclines.
- Patients with COPD are at increased risk for thromboembolic disease due to a sedentary lifestyle and chronic inflammation. Patients admitted for COPD exacerbation presenting a pleuritic chest pain or signs of heart failure without an infectious precipitant should be screened for pulmonary embolism.
- Patients with COPD exacerbations commonly present with tachyarrhythmias including atrial fibrillation, atrial flutter, and multifocal atrial tachycardia.
- Oxygen supplementation should be administered with a saturation goal of 88% to 92%.
- In severe exacerbations, patients develop a rapid, shallow breathing pattern which decreases the exhalation time, causes hyperinflation, increases the proportion of dead space ventilation, and causes respiratory muscle fatigue.
- Bi-level noninvasive ventilation is the first-line therapy for patients with hypercapnic respiratory failure and acute COPD exacerbations without hemodynamic instability, severe agitation, or respiratory arrest. Implementing noninvasive ventilation decreases the mortality, intubation rates, and hospital length of stay.
- If intubation is required, ventilation priorities are decreasing the patient work of breathing and limiting dynamic hyperinflation primarily using a low respiratory rate of 10 to 14 breaths/min and tidal volume of ≤8 mL/kg predicted by body weight. Respiratory acidosis with a pH greater than 7.2 should be tolerated without manipulating the minute ventilation.
- All patients being discharged from the emergency department should be counseled on smoking cessation, adequate inhaler techniques, and offered pneumococcal and influenza vaccination if not current.

1. A patient presents to the emergency department with 8 months of progressive dyspnea. There is no recent change in the patient's symptoms. On exam, there is a prolonged expiratory phase without wheezing, no rales, no pedal edema. The electrocardiogram is normal. The chest x-ray revealed hyperinflation with a flattened diaphragm. The diagnosis of chronic obstructive pulmonary disease (COPD):

a. Can be made clinically in this patient
b. Can be made radiographically with demonstration flattening of the diaphragms and hyperinflation on chest x-ray
c. Requires demonstration of spirometry-proven airflow limitation with the ratio of forced expiratory volume in one second (FEV_1) to forced vital capacity (FVC) after bronchodilators less than or equal to 0.7
d. Requires genetic testing and historical features including increased cough, sputum production, noxious stimuli, and dyspnea

Answer: c. The diagnosis of COPD requires spirometry-proven airflow limitation defined as a ratio of FEV_1 divided by FVC after bronchodilators of less than 0.7. Patients who have normal spirometry with chronic symptoms and possible emphysema on CT scan are not considered to have COPD.

2. In a patient with a diagnosis of chronic obstructive pulmonary disease (COPD), increased dyspnea, sputum production, and sputum purulence worse than baseline day-to-day variations for the past three days requiring evaluation in the emergency department is defined as:

a. Chronic bronchitis
b. COPD exacerbation
c. Emphysema
d. Progression of underlying disease

Answer: b. COPD exacerbation can be defined as a worsening of symptoms from baseline day-to-day variations that requires treatment. The most common symptoms include increased dyspnea, cough, increased sputum production and increased sputum purulence.

3. A 64-year old male with a history of severe chronic obstructive pulmonary disease (COPD) with baseline 3 L/min oxygen requirement presents to the emergency department with complaint of dyspnea. On physical exam, he is taking rapid shallow breaths with a respiratory rate of 50, requiring 8 L/min by nasal cannula to maintain a SpO_2 of 88%, but is awake and following commands. A blood gas reveals a pH of 7.25 and a pCO_2 of 70 mm Hg. Chest x-ray reveals hyperinflated lungs without evidence of pneumothorax or infiltrate. Antibiotics, nebulized short-acting beta-agonists, and glucocorticoids are initiated. The next step in management is:

a. Continue medical management
b. Intubate the patient for anticipated clinical decline
c. Start noninvasive bi-level positive pressure ventilation
d. Trial high flow nasal cannula

Answer: c. This patient is presenting with respiratory failure without clear contraindications to noninvasive ventilation. The patient should be trialed on noninvasive ventilation and receive inhaled bronchodilators including albuterol and ipratropium, glucocorticoids, and antibiotics. Glucocorticoids have similar bioavailability and efficacy when given PO or IV, however in patients with respiratory distress we recommend avoiding oral medications.

4. The rapid, shallow breathing pattern that develops in a chronic obstructive pulmonary disease (COPD) exacerbation _____ exhalation time causing hyperinflation, and _____ the proportion of dead space ventilation causing hypercapnia and respiratory muscle fatigue.

a. Decreases, decreases
b. Decreases, increases
c. Increases, decreases
d. Increases, Increases

Answer: b. The rapid, shallow breathing pattern that develops in a COPD exacerbation decreases exhalation time causing hyperinflation and increases the proportion of dead space ventilation causing hypercapnia and respiratory muscle fatigue. If untreated, this will progress to respiratory failure.

5. When treating a patient with a chronic obstructive pulmonary disease (COPD) exacerbation who requires invasive mechanical ventilation, setting the initial tidal volume to 8 mL/kg predicted body weight and the respiratory rate to 10 to 14 breaths/min helps prevent the development of:

a. Acidosis
b. Dynamic hyperinflation
c. Hypercapnia
d. Expiratory flow limitation

Answer: b. Decreasing the respiratory rate in an intubated patient with COPD allows for complete emptying and return to FRC, decreasing the development of dynamic hyperinflation and auto-positive end expiratory pressure (PEEP).

61 Upper Respiratory Tract Infections

KEY CONCEPTS

- Viral infections cause most cases of pharyngitis. Patients should not be treated with antibiotics based on symptoms and exam alone. Patients with a Centor criteria score of 0 or 1 do not require further testing or treatment. Those with a score of 2 or greater should undergo rapid antigen testing with treatment decisions based on results. Throat cultures are recommended in children but are not necessary in adults.
- Antibiotics for group A *Streptococcus* pharyngitis are aimed at symptom reduction, decreasing transmission, and decreasing suppurative complications. Rheumatic fever is very rare and likely from a shift in streptococcal M proteins rather than antibiotic use.
- Infectious pharyngitis presents with an acute onset and resolves within days. In subacute and chronic cases of pharyngitis, consider abscess formation, neoplastic causes, HIV, and autoimmune disease.
- In streptococcal pharyngitis, a single high dose of corticosteroids, such as 10 mg dexamethasone, is safe and reduces symptom severity.
- Consider deep space infection and epiglottitis in patients who present with neck pain, hoarseness, and have a benign oropharyngeal exam.
- Airway edema from palatine or lingual tonsillitis, epiglottitis, or deep space infection is rare, but life-threatening.
- Deep space infections such as retropharyngeal and parapharyngeal abscesses are difficult to diagnose clinically, and contrast-enhanced CT is recommended.
- Transcervical or intraoral ultrasound is useful in the diagnosis and treatment of peritonsillar abscess.
- Acute rhinosinusitis is likely viral and will resolve with supportive care, including nasal irrigation with hypertonic saline.
- Although rarely needed, first-line antibiotic treatment for acute rhinosinusitis is amoxicillin or amoxicillin-clavulanate for 5 days.

1. A previously well 18-year-old male presents with throat pain and fever to 39 °C for the past 2 days. He denies vomiting, diarrhea, cough, or rhinorrhea. He is not sexually active. Physical exam is notable for tender anterior cervical lymphadenopathy and symmetrically swollen and erythematous palatine tonsils with a gray-white tonsillar exudate. The remainder of his neck and physical exam is benign. What is the most appropriate next step?
   a. Empirically prescribe amoxicillin for 10 days
   b. Obtain a complete blood count with differential
   c. Obtain an intraoral tonsillar ultrasound
   d. Perform a rapid strep antigen test and prescribe antibiotics only if there is a positive result

Answer: D. This patient is presenting with pharyngitis, with sore throat, fever, exudative pharyngitis, and cervical lymphadenopathy without associated cough or rhinorrhea. A majority of cases are viral and caused by common cold viruses. Bacterial infection is only responsible for approximately 5% to 10% of adult and 20% to 30% of pediatric cases. Rapid antigen testing is recommended as a first-line diagnostic test for GAS pharyngitis, because it is highly sensitive and specific when performed correctly. A positive test indicates the presence of GAS and does not require follow up testing.

2. A 19-year-old female patient presents to the emergency department with vision changes and headaches. She reports a sore throat 10 days ago started to improve, but never completely resolved. For the past 3 days she noted increasing left-sided neck pain and fevers. On exam she appears uncomfortable with a fever and mild tachycardia. She is warm and diaphoretic with a dry forehead on the left. Her left eyelid is slightly lower than the right and her left pupil is 3 mm smaller than the right. Her extraocular movements are intact, and her intraoral exam shows symmetric tonsils with mild erythema and a midline uvula. She has mild trismus. What is most likely causing her symptoms?
   a. Parapharyngeal abscess with invasion of the carotid sheath
   b. Peritonsillar abscess with compression of the carotid sheath
   c. Spread of infection into the cavernous sinus causing thrombosis and intracranial abscess
   d. Submandibular space infection and spread into retropharynx

Answer: A. Complications from retropharyngeal or parapharyngeal abscess include airway compromise, abscess rupture leading to aspiration of pus, and involvement of the carotid sheath. The carotid sheath is surrounded by deep fascia and includes the carotid artery, internal jugular vein, and vagus nerve, and thus infection can result in compromise of any of these structures including arterial erosion, aneurysm, Lemierre syndrome, palsy of cranial nerves IX–XII, and mediastinitis.

3. A 34-year-old male presents with complaints of high fever, left-sided face pain, and purulent nasal discharge for 10 days. He has been attempting symptom management with nasal saline irrigation and intranasal corticosteroids, but after initial improvement reports worsening symptoms. What is the next step in treatment?
   a. CT scan of his face to evaluate for complications of rhinosinusitis
   b. Obtain bacterial culture via sinus puncture
   c. Prescribe a 5-day course of amoxicillin
   d. Supportive care with antiinflammatory medications, rest, and continued nasal saline irrigation

Answer: C. The diagnosis of acute rhinosinusitis is clinical. The diagnosis of bacterial rhinosinusitis requires one of the three criteria: at least 10 days of persistent symptoms without improvement, three to four days of severe symptoms including a fever greater than 39 degrees Celsius with nasal discharge or facial pain without improvement, or onset of progressive symptoms with worsening symptoms after initial improvement. Given the risks and lack of benefit of routine antibiotic use in rhinosinusitis, antibiotics should only be used for patients who meet the definition of bacterial rhinosinusitis with these criteria.

4. Patients presenting with symptoms concerning for a retropharyngeal abscess are best evaluated using what modality?
   a. Contrast-enhanced CT
   b. Intraoral ultrasound
   c. Lateral neck x-ray
   d. Visualization with a fiberoptic scope

Answer: A. Contrast-enhanced CT scan is the preferred diagnostic test for the identification of retropharyngeal abscess. CT findings are characterized by a fluid collection with central hypodensity and complete ring enhancement with scalloping. Fat stranding and edema characterized by low-density thickening without peripheral enhancement may represent an early phase of infection before an abscess develops.

5. This bacterium is frequently implicated in non-GAS pharyngitis and is also most commonly implicated in suppurative jugular thrombophlebitis:
   a. *Arcanobacterium haemolyticum*
   b. *Fusobacterium necrophorum*
   c. *Haemophilus influenzae* type B
   d. Methicillin-resistant *Staphylococcus aureus*

Answer: B. *Fusobacterium necrophorum* is an anaerobic gram-negative rod that is part of the normal oral flora and causes pharyngitis in patients 15 to 45 years old with a similar presentation to GAS. *Fusobacterium* is the primary causative agent in septic jugular vein thrombophlebitis (Lemierre syndrome). *Fusobacterium* should be considered in young adults with ongoing severe symptoms.

KEY CONCEPTS

- Empirical antimicrobial therapy should be started in the emergency department (ED) for patients admitted with pneumonia.
- *Streptococcus pneumoniae* is the most commonly encountered pathogen in hospitalized patients, especially those requiring the intensive care unit.
- No characteristic radiographic pattern is pathognomonic for a specific pneumonia pathogen.
- *Legionella* should be suspected in patients with gastrointestinal or neurologic symptoms presenting with pneumonia.
- As part of the evaluation of patients with pneumonia, the patient's immune status should be considered. Patients with HIV and other immunosuppressive conditions are at risk for opportunistic infections, such as *Pneumocystis jiroveci.*
- Empirical therapy should treat the most likely pathogens for the clinical situation, such as *S. pneumoniae, Haemophilus influenzae, Mycoplasma pneumoniae,* and *Chlamydia pneumoniae,* and should be consistent with current national treatment guidelines, such as those from the American Thoracic Society/Infectious Disease Society of America (ATS/IDSA).
- Community-acquired methicillin-resistant *Staphylococcus aureus* (MRSA) is an uncommon cause of community-acquired pneumonia (CAP), but empirical coverage of MRSA should be strongly considered for patients with severe pneumonia and sepsis, with concomitant influenza, contact with someone infected with MRSA, or radiographic evidence of necrotizing pneumonia.
- Patients with prior use of intravenous antibiotics, neutropenia, or underlying bronchiectasis are at increased risk of infection with *Pseudomonas aeruginosa.* Empirical therapy for high-risk or critically ill patients should cover for *P. aeruginosa.*
- Disposition is dictated by the patient's underlying medical conditions, severity of illness, likelihood of clinical deterioration, and feasibility of home care and outpatient follow-up.

1. Which of the following statements is true regarding the treatment of community-acquired pneumonia (CAP)?
   a. Amoxicillin monotherapy is not an option as it does not effectively cover atypical pneumonia.
   b. Macrolide resistance almost always prohibits azithromycin monotherapy for CAP.
   c. The most common atypical pneumonia in CAP is *Chlamydophila.*
   d. Clinical history and prodrome often lead to the identification of the causative organism.

Answer: b. The 2020 American Thoracic Society/Infectious Disease Society of America (ATS/IDSA) guidelines no longer recommend macrolide monotherapy as *Streptococcus pneumoniae* resistance is greater than 30% in most settings in the United States. Amoxicillin is now recommended as a first-line agent in the treatment of CAP in healthy patients without comorbidities.

2. A 92-year-old male nursing home resident presents to the emergency department (ED) with a history of hypertension and nonproductive cough. He has a SaO_2 at triage of 82% but is speaking full sentences and with supplemental oxygen his saturation increases to 95%. Chest x-ray reveals diffuse infiltrates. Which of the following viral causes of pneumonia is this patient likely to have?
   a. COVID-19
   b. Respiratory syncytial virus (RSV)
   c. Parainfluenza virus
   d. Metapneumovirus

Answer: a. The novel coronavirus causing a worldwide pandemic is more likely to cause severe disease in patients with comorbidities and those living in nursing homes. SARS-CoV-2 often causes profound hypoxia that corrects with supplemental oxygen and causes diffuse infiltrates.

3. Which of the following causative organisms would be an indication for respiratory isolation?
   a. *Histoplasma capsulatum*
   b. *Herpes simplex virus* (HSV)
   c. *Mycobacterium tuberculosis*
   d. *Pneumocystis jiroveci*

Answer: c. Any patient with suspected tuberculosis (TB) infection should be placed in respiratory isolation and all staff should take appropriate personal protective precautions. None of the other causative agents require isolation.

4. A 60-year-old man with a past medical history of diabetes controlled with insulin and diagnosis of influenza 7 days prior presents with cough, weakness, and purulent sputum production. Vital signs are temperature 38.3°C (101°F) oral, heart rate, 130 beats/min, blood pressure, 80/50 mm Hg, respiratory rate, 30 breaths/min, and oxygen saturation, 92%. The chest radiograph reveals consolidation in the left lower lung (LLL). What is the most appropriate antibiotic therapy?
   a. Ceftriaxone plus levofloxacin plus vancomycin
   b. Ceftriaxone with a macrolide
   c. Fluoroquinolone only
   d. Trimethoprim-sulfamethoxazole (TMP-SMX)
   e. Vancomycin only

Answer: a. In patients requiring hospitalization for community-acquired pneumonia (CAP), coverage for CAP would typically be with a combination of a macrolide with a β-lactam. Because this patient has signs of septic shock, consideration should be given to the addition of an agent for methicillin-resistant *Staphylococcus aureus* (MRSA; vancomycin) in addition to CAP coverage.

5. Which of the following is true regarding the use of procalcitonin in emergency department (ED) patients with pneumonia?
   a. Procalcitonin is not useful in determining if a pneumonia is bacterial or viral.
   b. Procalcitonin predicts mortality in ED patients.
   c. It is helpful is predicting who can be discharged from the ED.
   d. Procalcitonin is useful in guiding antimicrobial therapy.

Answer: a. While sometimes requested by hospitalist colleagues, the ED use of procalcitonin has not been shown to be useful in determining mortality, viral versus bacterial etiology, mortality, or prognosis.

KEY CONCEPTS

- Point of care thoracic ultrasound can be used to rule out pneumothorax with high sensitivity.
- For healthy young patients with a small primary spontaneous pneumothorax, observation with supplemental oxygen is an appropriate treatment option. For larger symptomatic primary spontaneous pneumothorax, simple aspiration with a catheter is often successful.
- In most cases of secondary pneumothorax, tube thoracostomy should be considered because less invasive approaches are associated with treatment failure.
- The most common causes of pleural effusion in the United States are congestive heart failure, malignancy, and bacterial pneumonia. Pulmonary embolism is an uncommon cause of pleural effusion.
- Thoracentesis should be performed under ultrasound guidance to minimize the risk of complications.
- On a frontal (anteroposterior or posteroanterior) projection, a volume of at least 200 mL of pleural fluid is required to detect a pleural effusion. Ultrasound can detect as little as 50 mL of pleural fluid and can be easily preformed at bedside.

1. What is the most common condition associated with secondary spontaneous pneumothorax in adults in the United States?

a. Chronic obstructive pulmonary disease
b. Collagen vascular disease
c. *Pneumocystis* pneumonia
d. Pulmonary malignancy

Answer: A. Chronic obstructive pulmonary disease is the most common condition associated with secondary spontaneous pneumothorax, although all the conditions listed may also be causes.

2. Which of the following are indications of a pneumothorax on bedside ultrasound?

a. Identification of a lung point
b. Lack of lung sliding identified
c. When in M-mode, you see a "barcode" appearance
d. All of the above

Answer: D. Pneumothorax may be difficult to identify on bedside ultrasound and there are several ways to help identify a pneumothorax. Identifying an area of lung that lacks lung sliding and has a "barcode" appearance on M-mode indicates that the parietal and visceral pleura are not sliding over each other and is highly suggestive of a pneumothorax. Identifying a lung point, the point where a pneumothorax transitions to a normal pleural interface, can also help identify a pneumothorax.

3. A 32-year-old male with no significant past medical history presents with acute onset of right pleuritic chest pain, cough, and shortness of breath. The symptoms occurred at rest. Physical examination is remarkable for a tachycardia of 110 beats/min and respiratory rate of 32 and use of accessory muscles to breathe. Chest radiography reveals an estimated 5-cm right pneumothorax. Which of the following would be suitable management?

a. Admission for 100% face mask oxygen and repeat radiography in 1 day
b. One-time air aspiration and repeat radiography in 6 hours
c. Reassurance and observation
d. Tube thoracostomy

Answer: D. Large (2 cm) primary spontaneous cases and patients who are showing signs of respiratory distress should be treated with tube thoracostomy. If the primary spontaneous pneumothorax is small (<2 cm), then observation alone is appropriate. For a larger primary spontaneous pneumothorax (> 2cm) without significant symptoms, either observation alone or intervention with either simple aspiration or tube thoracostomy may be appropriate and a shared decision-making discussion with your patient is appropriate to determine best course of action.

4. Which of the following statements is true regarding the routine application of suction after tube thoracostomy?

a. It improves the rate of lung expansion.
b. It increases the risk of reexpansion pulmonary edema.
c. It is associated with increased rates of empyema.
d. It is not routinely indicated.

Answer: D. Suction neither accelerates lung reexpansion nor improves outcomes. It is indicated when reexpansion with a Heimlich or water seal device does not occur after 24 to 48 hours. A pressure of at least −20 cm H_2O should be used.

5. A 68-year-old man with a history of esophageal cancer presents with progressive fever, chest pain, and shortness of breath over 24 hours. Chest radiography demonstrates a possible left lower lobe pneumonia and large left pleural effusion. Pleural fluid analysis reveals pH of 6.95, glucose level of 47 mg/dL, 11,500 white blood cells (WBCs)/mm^3 (82% neutrophils), and protein level 75% of plasma levels. What are the indicated next steps in management?

a. Antibiotics and fluid resuscitation
b. Antibiotics and tube thoracostomy
c. Antibiotics, tube thoracostomy, and esophageal oral contrast study
d. One-time pleural aspiration for fluid analysis

Answer: C. This is an exudative pleural effusion as defined by Light's criteria (see Box 63.3). A pH less than 7.0 suggests emphysema or esophageal rupture. This patient is at risk for both; hence, the need to assess esophageal integrity. A pH less than 7.0 with glucose level less than 50 mg/dL are indications for tube thoracostomy. Normal pleural fluid has a WBC count of less than 1,000/mm^3.

Acute Coronary Syndromes 64

KEY CONCEPTS

- Symptoms of acute coronary syndrome (ACS) are variable. Clinicians must consider ACS in a broad range of presentations, especially in patients prone to subtle or non-chest pain presentations (for example, women, older patients, or those with comorbidities such as diabetes).
- The ECG is crucial in the initial evaluation of ACS. Limitations of the 12-lead ECG in suspected ACS include initial nondiagnostic findings, evolving fluctuations with ongoing symptoms, anatomic myocardial blind spots, and confounding or obscuring patterns, such as left bundle branch block (LBBB).
- It is crucial to recognize ECG patterns that may indicate ischemia or impending infarction but do not meet traditional STEMI criteria (e.g., Wellens syndrome, significant aVR elevation, de Winter syndrome, Sgarbossa criteria). Patients with these findings may benefit from prompt catheterization.
- Patients with ECG findings of STEMI should undergo emergent reperfusion therapy. Percutaneous coronary intervention (PCI) is recommended with a goal door-to-balloon time of 90 minutes. If PCI is not feasible, patients should be administered fibrinolytic therapy within 30 minutes.
- Clinical decision tools, such as EDACS or the HEART score, may assist in the risk stratification of patients with suspected ACS.
- Serial troponin testing, combined with an accelerated diagnostic protocol, allows for identifying patients at low risk for ACS. High-sensitivity troponin (hs-T) testing may further reduce the time needed for evaluation. Validation of hs-T testing protocols is ongoing. Clinicians should be aware that testing characteristics vary by assay.
- Patients with suspected ACS who are low-risk by diagnostic protocols are unlikely to benefit from further diagnostic testing in the ED or admission and may be discharged with outpatient follow-up.

1. A 44-year old male presents after an episode of left-sided chest pain without radiation that lasted for approximately 5 minutes. He described the pain as a "pressure" without radiation or other significant symptoms. Onset was at rest and the symptoms are now resolved. The patient is a non-smoker, and medical history is significant only for hypertension. There is no significant family history of CAD. ECG is significant for LVH without other findings. Initial troponin testing is negative (less than the normal limit). What is the most appropriate next step in management?

a. Admission for serial troponin testing
b. Coronary CT angiography
c. Discharge to outpatient follow-up
d. Stress echocardiography

Answer: C. Utilizing the HEART score can assist in the risk stratification of this patient. Ten minutes of chest "pressure" without other typical features of ACS may be considered "moderately suspicious." Given this, the patient's heart score is 3 (1 point each for history, nonspecific repolarization disturbance on ECG, and the presence of 1 to 2 risk factors). This patient is at low risk for ACS and may be appropriately discharged from the emergency department.

2. A 48-year-old man with history of hypertension and hypercholesteremia presents with chest pain and hyperacute T waves in an anterior distribution on the initial ECG. During your initial history and physical examination the patient experiences ventricular fibrillation that responds to cardiopulmonary resuscitation (CPR) and defibrillation after being pulseless for a period of 3 minutes. Following cardiac arrest, the patient is comatose, with the following vital signs: HR 110 beats/min, BP 160/98 mm Hg, RR 12 breaths/min (intubated), temperature 36.5°C, and O_2 saturation 96%. A repeat ECG demonstrates a large, evolving anterior STEMI. Which of the following treatment plans is most appropriate?

a. Administer aspirin, PSY_{12} inhibitor, IV UFH, IV fibrinolysis, and admission to intensive care unit (ICU)
b. Administer aspirin, PSY_{12} inhibitor, IV UFH, IV fibrinolysis, initiation of therapeutic hypothermia, and admission to ICU
c. Neurologic examination for brain death and admission to palliative care because outcome is almost universally fatal
d. Rapid revascularization with percutaneous coronary intervention (PCI), initiation of therapeutic hypothermia, and admission to ICU for comprehensive postresuscitation care
e. Supportive care, and admission to ICU

Answer: D. A neurologic examination immediately following cardiac arrest is poorly prognostic of a favorable neurologic outcome with modern postresuscitation care. In the absence of multiple "unfavorable resuscitation features" (prolonged arrest, unwitnessed arrest, nonshockable rhythm, advanced age, etc.), revascularization with percutaneous coronary intervention (PCI) and immediate application of therapeutic hypothermia is warranted. Although not contraindicated, fibrinolysis is inferior to PCI following cardiac arrest and should only be used when a patient is not a candidate for PCI.

3. Which of the following is an absolute contraindication to fibrinolytic therapy?

a. Age older than 75 years
b. Appendectomy performed 2 months ago
c. Previous coronary artery bypass grafting (CABG)
d. Previous hemorrhagic stroke
e. Systolic blood pressure of 175/90 mm Hg following administration of vasoactive agents

Answer: D. Although patients older than 75 years have a higher risk of intercerebral hemorrhage, age should not be considered a contraindication to fibrinolysis. Recent major surgery or trauma is a relative contraindication for fibrinolysis; however, the term *recent* is variably defined in the fibrinolytic literature and never as more than 6 weeks. Although prior CABG patients should be preferentially considered for PCI, there is no contraindication to fibrinolytic use in these patients if PCI is not available. Systolic blood pressure above 150 mm Hg is a risk factor for intracerebral hemorrhage. Only hypertension persistently above 200/120 mm Hg, despite reasonable efforts, should be considered an absolute contraindication.

4. Which of the following drugs provides mortality benefit in the setting of AMI?

a. Aspirin
b. Intravenous beta blocker
c. Intravenous morphine
d. Nitroglycerin
e. Oxygen

Answer: A. The ISIS-2 trial has demonstrated that aspirin independently reduces mortality by 23% in the setting of AMI. Intravenous morphine has not been shown to improve mortality and has been associated with mortality. Although nitroglycerin does improve symptoms and cause vasodilation, it has never been proven to improve mortality. Oxygen beyond that needed to maintain an oxygen saturation of 94% has been associated with additional mortality. The use of intravenous beta blockers does not offer significant benefit and is associated with an increased rate of adverse events.

5. A 42-year-old male patient presents with 45 minutes of chest pain. The ECG is depicted below (Fig. Q64.5). You are working at a noninvasive (i.e., no PCI capability) hospital; transfer time to the closest major medical center with PCI capability is 4.5 hours considering weather and logistics. The patient has no contraindications for fibrinolysis. Which of the following statements is most appropriate?

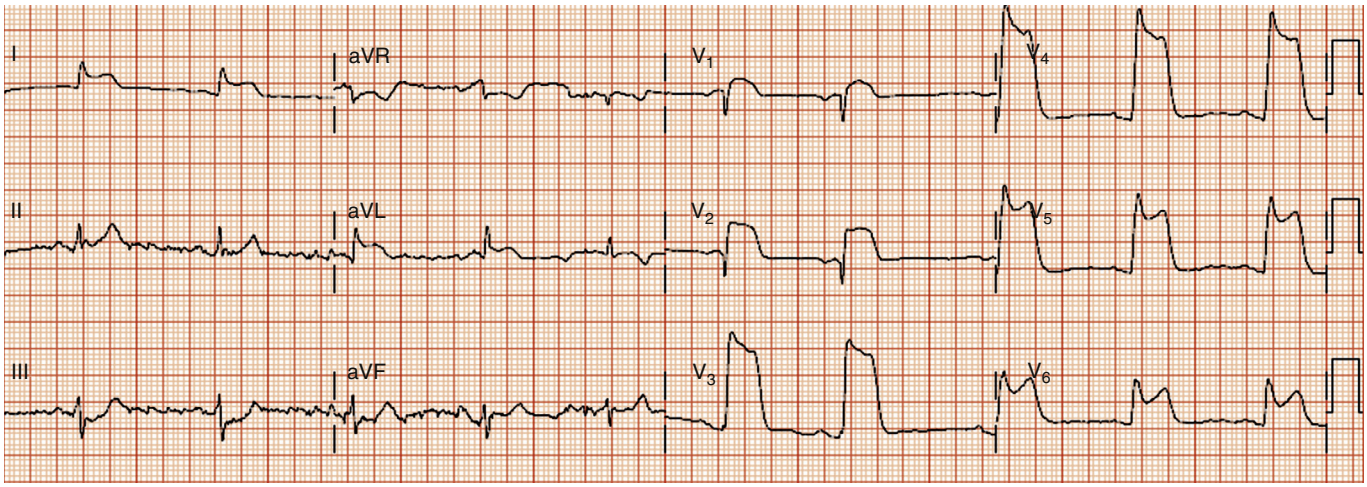
a. The patient must be transferred rapidly to the closest PCI center, with initiation of appropriate β-adrenergic blocking agents and antiplatelet and anticoagulant therapies before transfer.

b. The patient should receive a fibrinolytic agent followed by appropriate antiplatelet and anticoagulant therapies with admission to your hospital's ICU.

c. The patient should receive a fibrinolytic agent followed by appropriate antiplatelet and anticoagulant therapies, with transfer to the closest PCI center for immediate PCI.

d. The patient should receive a fibrinolytic agent followed by appropriate antiplatelet and anticoagulant therapies, with transfer to the closest PCI center within 24 hours for reevaluation and consideration of immediate PCI.

Answer: D. The ECG demonstrates an extensive anterolateral STEMI. The patient is young and has presented early in the STEMI evolution. This patient is at extreme risk due to the extensive nature of the STEMI and yet can benefit significantly from early reperfusion therapies. A delay of more than 60 to 120 minutes in this patient is not appropriate for the initiation of reperfusion therapies; furthermore, he is a candidate for a fibrinolytic agent. The early initiation of reperfusion therapy (fibrinolysis or PCI) is vital to reduce morbidity and mortality. Such a significant delay in this case for PCI is not justified, so a fibrinolytic agent is preferred. On arrival at the closest PCI center, the patient can be evaluated for PCI if he has not demonstrated successful reperfusion with resolution of chest discomfort and normalization of the ST segment elevation.



65 Dysrhythmias

KEY CONCEPTS

- Electrical therapy is appropriate for unstable patients in whom a dysrhythmia is the cause of symptoms—pacing if the heart rate is slow, countershock with sedation if fast.
- Any regular new-onset, symptomatic, wide-complex tachycardia should be assumed to be ventricular tachycardia until proven otherwise.
- Type II second-degree AV block is never a normal variant and implies a conduction block below the AV node. When the conduction ratio is 2:1, a type II block should be assumed until proven otherwise. Pacing should be readily accessible.
- Any tachycardia exceeding a rate of 225 to 250 beats/min, regardless of the QRS complex morphology, should be considered an accessory pathway syndrome. Nodal blocking agents should be avoided.
- Irregularity can be difficult to appreciate in tachycardia over 200 beats/min. Atrial fibrillation can be missed if R-R intervals at fast rates are not carefully tracked.

65.1. What is the primary electrochemical difference between pacemaker and nonpacemaker cells?

a. Lack of a plateau phase 3 in nonpacemaker cells
b. Rapid phase 0 upstroke in nonpacemaker cells after stimulus
c. Slow calcium ion influx during phase 2 for pacemaker cells
d. Slow phase 4 spontaneous depolarization in pacemaker cells
e. Transient membrane repolarization by potassium channel closure during phase 1 for pacemaker cells

Answer: d. The spontaneous return to a depolarization threshold during phase 4 (diastole) characterizes pacemaker cells. Both cell types then exhibit a rapid phase 0 upstroke resulting from sodium ion (Na^+) influx, brief repolarization resulting from potassium ion (K^+) efflux (phase 1), plateau phase resulting from balanced calcium ion (Ca^{2+}) entry and K^+ efflux (phase 2), and then repolarization resulting from Ca^{2+} channel closure and K^+ efflux (phase 3).

65.2. For a reentrant tachydysrhythmia to occur, what three conditions exist?

a. Electrolyte disturbance, ischemia, and altered conduction in an endogenous atrioventricular pathway
b. Electrolyte disturbance, two conduction pathways, with one of the pathways being slower
c. Ischemia, two conduction pathways, with one of the pathways being slower
d. Two conduction pathways, one path being slower, and differing responsiveness
e. Two conduction pathways with equal responsiveness

Answer: d. Remember that a conducting pathway is bidirectional. In a typical scenario, the alpha pathway of the atrioventricular (AV) node is the anterograde conducting limb, and the beta pathway is the retrograde conducting limb. Reentrant dysrhythmias are almost always AV nodal and narrow complexes that start and end abruptly.

65.3. Classic antifibrillatory effects are seen with which class of antidysrhythmic?

a. IA
b. IB
c. IC
d. II
e. III

Answer: e. Class III agents, of which amiodarone is the prototype, prolong the action potential and refractory period duration. Class I agents have variable effects on depolarization rate and repolarization duration.

65.4. The most frequent proarrhythmic effects are seen with which class of antidysrhythmic?

a. IA
b. IB
c. IC
d. II
e. III

Answer: c. Class IC agents, such as flecainide, encainide, and propafenone, markedly slow depolarization and conduction and prolong repolarization and action potential duration. Class IB agents generally have the least proarrhythmic effect.

65.5. A 49-year-old woman presents with a sudden onset of palpitations and shortness of breath. This has happened once before. She has no past history and takes no medications. Vital signs are temperature, 36.0°C (96.8°F) oral, blood pressure, 115/69 mm Hg, heart rate 156 beats/min, respiratory rate 24 breaths/min, and oxygen (O_2) saturation, 98%. Her electrocardiogram (ECG) is shown in Fig. 65.28. What is the most appropriate intervention?

a. Adenosine, 6 mg IV
b. Digoxin, 0.25 mg IV
c. Diltiazem, 0.4 mg/kg IV
d. Propranolol, 1 mg IV
e. Synchronized electrical cardioversion after IV sedation with midazolam

Answer: a. Adenosine causes slowing of conduction in the anterograde and retrograde pathways, with no effect on ventricular contractility. It converts a high percentage of narrow-complex tachycardias to sinus rhythm, but with a 25% recurrence rate. Diltiazem would not be unreasonable, but the quoted dose is too high. Calcium channel blockers also exert their effects only on the anterograde pathway, with little direct effect on accessory pathways. Contractility may be diminished. Digoxin use has been largely supplanted by adenosine and class IV agents. Its onset of action after IV use is 1.5 to 2 hours. Cardioversion would not be indicated unless the patient exhibited hemodynamic instability.

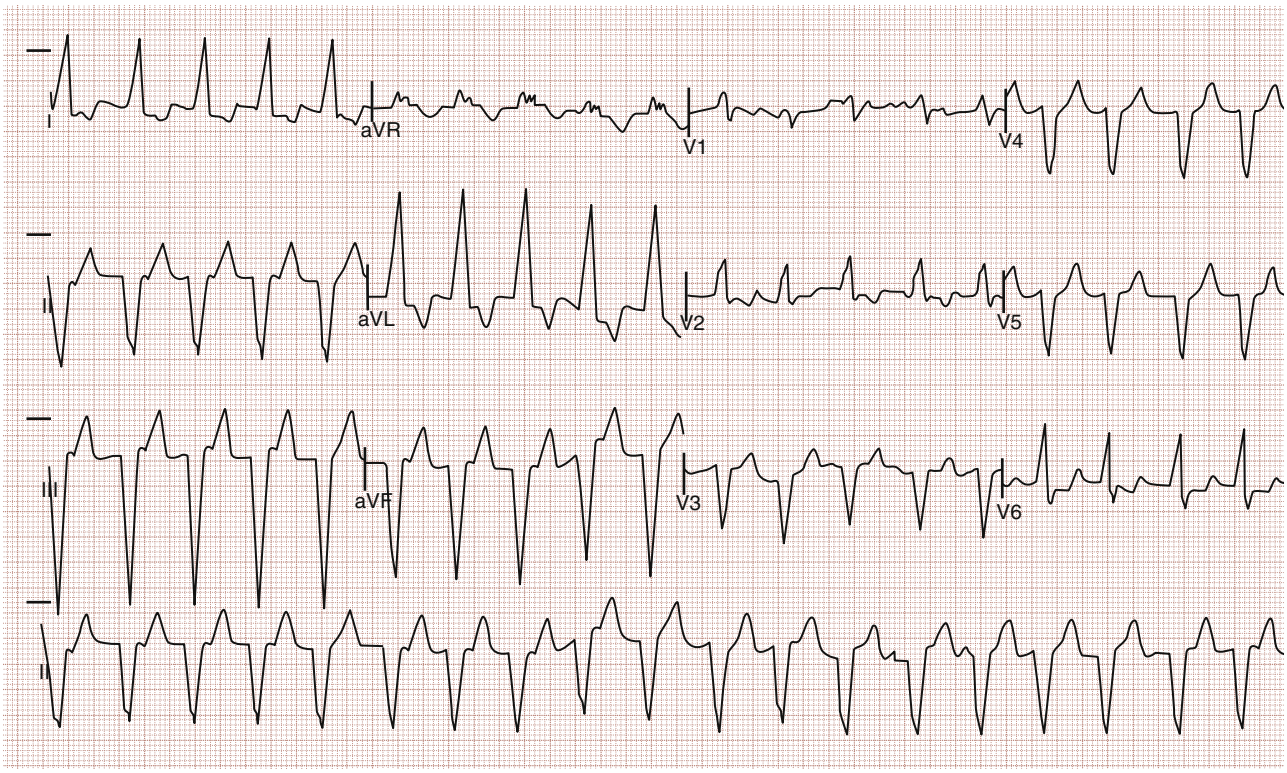
KEY CONCEPTS

- Pacemaker malfunction soon after implantation (within 6 to 8 weeks) is usually a result of a lead problem, such as a lead displacement, or a pacemaker programming issue, such as a pacing rate too slow for the patient's needs.
- Pacemaker malfunction is categorized as failure to pace, oversensing, undersensing, or pacing at an inappropriate rate (too fast or too slow).
- Pacemaker or implantable cardioverter-defibrillator (ICD) interrogation should be performed if a patient's presentation (e.g., shortness of breath, palpitations, syncope, shock, cardiac arrest, etc.) or ECG (electrocardiogram) is suggestive of device malfunction, abnormal intrinsic rhythm, or worsening cardiac function.
- Modern lithium-iodine batteries are highly reliable; battery-related issues are unlikely to be the cause of abrupt pacemaker failure.
- Consider pacemaker lead infection and endocarditis if a patient with a pacemaker has a fever of unclear cause.
- Right ventricular-paced complexes are normally conducted with a left bundle branch block pattern. A right bundle branch pattern is abnormal and suggests lead displacement.
- Magnet application turns off the sensing and inhibition function, causing the pacemaker or ICD to fire at a preprogrammed fixed rate (typically near 60 beats/minute) and inhibits antitachycardia pacing in ICDs. Magnet removal returns the pacemaker to its programmed pacing mode.
- Defibrillation is safe in patients with a pacemaker or ICD. Pads should be placed at least 10 cm from the subcutaneous implant site of the device.
- Most left ventricular assist device (LVADs) do not produce pulsatile flow; therefore, these patients will not have a palpable pulse.
- All modern VADs tolerate external defibrillation and cardioversion.
- Chest compressions can be performed in an unresponsive, nonperfusing patient with a VAD, supported by typical resuscitative measures.

1. Which of the following conditions is an indication for placement of a biventricular pacemaker?
   a. Third-degree AV block
   b. Brugada syndrome
   c. Severely depressed systolic heart failure and a left bundle branch block
   d. Slow atrial fibrillation

   Answer: C. Cardiac resynchronization therapy, or biventricular pacing, is indicated for patients with systolic heart failure with an ejection fraction less than 35% and a left bundle branch block. By ensuring appropriate synchrony between the ventricles, cardiac output is improved, and multiple studies have shown a mortality benefit in these patients. Options A and C have an indication for a permanent pacemaker but would not benefit from biventricular pacing. Option B has an indication for implantable cardioverter-defibrillator.

2. A 67-year-old female presents with complaints of shortness of breath and a fluttering sensation in her chest. She had a dual-chamber pacemaker placed 10 years ago for sinus node dysfunction. Blood pressure is 124/96 mm Hg. On examination, she is anxious with clear lungs and no evidence of jugular venous distention or peripheral edema. There is no tenderness over her pacemaker insertion site. An ECG is provided as follows. Which of the following is the next best step in the management of this patient?
   a. Administration of IV amiodarone
   b. Drawing blood cultures and starting antibiotics
   c. Immediate cardioversion
   d. Placement of a magnet over the pacemaker



Answer: D. This patient has a pacemaker with an inappropriately fast rate due to a pacemaker-mediated tachycardia. Management of this condition involves disruption of the circuit and can be accomplished by placing a magnet over the pacemaker, which sets the pacemaker into an asynchronous mode with no sensing (e.g., DOO). Cardioversion and amiodarone are less appropriate in this situation. Consultation with a cardiologist and interrogation of the pacemaker should be performed, although this can be done after placement of a magnet over the pacemaker. In the absence of potential for pacemaker infection, there is no role for blood cultures or antibiotics

3. A 67-year-old male with a history of an ischemic cardiomyopathy and an implantable cardiac defibrillator (ICD) presents with complaints of multiple shocks from his ICD in the past day. Interrogation of his ICD shows no evidence of any shocks in the past 2 months. What is the most appropriate management plan?
   a. Administration of intravenous amiodarone
   b. Cardiology consultation for reprogramming of ICD
   c. Reassurance and follow-up with his cardiologist
   d. Temporary ICD deactivation with magnet application

Answer: C. This patient has experienced 2 "phantom shocks." The perception of a shock in the absence of the actual ICD firing occurs in approximately 10% of patients with an ICD and is associated with significant anxiety and stress. Appropriate management includes reassurance and discharge with a patient's cardiologist. There is no role for IV amiodarone, reprogramming of the ICD or magnet application.

4. A 55-year-old man presents with chest pain and hypotension. ECG reveals ST-elevation MI in the inferior and lateral leads, and bedside echocardiogram shows poor systolic function of both ventricles. He has elevated jugular venous pressure, and a plethoric IVC on ultrasound. In addition to percutaneous coronary interventions, which of the following therapies would be most appropriate to consider?
   a. Intra-aortic balloon pump
   b. Temporary LVAD (e.g., Impella or Tandem Heart)
   c. Veno-venous ECMO
   d. Veno-arterial ECMO

Answer: D. This patient shows evidence of right heart failure, with poor RV systolic function and elevated jugular venous and IVC pressures. Intra-aortic balloon pumps and temporary left ventricular assist devices will support primarily the left ventricle. Veno-venous ECMO provides effective oxygenation and ventilation, but very little hemodynamic support. Veno-arterial ECMO may be considered as temporary biventricular support, if there is expected recovery of ventricular function, or if the patient is a candidate for another therapy (e.g., heart transplant or intracorporeal VAD).

5. A 65-year-old man with an intracorporeal left ventricular assist device (LVAD) presents to the emergency department and tells you he has felt lethargic for the past two days. A manual blood pressure cuff and Doppler ultrasound show a systolic blood pressure of 75 mm Hg, and a continuous "hum" is heard on auscultation. The cardiac monitor shows a wide complex tachycardia, and no pulse is palpable. Which is the most appropriate next step in management?
   a. Administer amiodarone
   b. Begin chest compressions
   c. Call heart failure specialist and continue diagnostic evaluation
   d. Immediate cardioversion/defibrillation

Answer: C. A low systolic pressure is expected with a continuous-flow LVAD, as there is generally a very low pulse pressure. The fact that the patient is awake suggests that he is adequately perfusing. His wide complex tachycardia may be a result of a ventricular dysrhythmia, or may be SVT or sinus tachycardia with aberrancy, and should be further investigated in a stable patient before starting interventions. He does not require chest compressions or immediate defibrillation/cardioversion with evidence of adequate perfusion, despite the lack of palpable pulse. While amiodarone may be an appropriate therapy, more information regarding his rhythm and current medications is required before giving this therapy.

Heart Failure

KEY CONCEPTS

- Heart failure (HF) is the clinical syndrome defined by signs and symptoms of elevated intracardiac pressures or depressed cardiac output, which in turn are due to either functional or structural cardiovascular (CV) abnormalities. By definition, HF presentations managed in the emergency department (ED) are acute heart failure (AHF) (ie, HF in which the signs and symptoms require unscheduled care).
- AHF is a multiorgan, multifactorial, and multiple phenotype disease state. Abnormalities in renal function, central venous volume and fluid shifts, arterial vascular tone, neuroendocrine overactivity, microvascular dysfunction, respiratory failure, or myocardial ischemia have bidirectional relationships with AHF as both potential causes and effects.
- Central congestion without hypervolemia is typically due to intercompartmental fluid shifts to the central circulation from venous reservoirs with reduced capacitance (increasing preload) or abrupt increases in central arterial tone (ie, increasing afterload).
- Emergency physician (EP) clinical gestalt based on history and physical exam is inaccurate for AHF diagnosis as often as a quarter of the time.
- The most useful diagnostic test for identifying lung congestion due to AHF is an 8-point lung ultrasound B-line scan, which is more sensitive than chest x-ray for pulmonary edema, and has a higher positive predictive value for AHF than natriuretic peptides.
- Focused cardiac ultrasound (FOCUS) for detecting reduced ejection fraction (EF) misses the 40% to 50% of AHF presentations with preserved EF. It adds little information to risk stratification because EF does not typically change with dynamic worsening of cardiac function or improvement after treatment.
- A useful clinical classification of AHF distinguishes between a vascular phenotype and a cardiac phenotype, though overlap can exist. Distinguishing between systolic and diastolic dysfunction is less helpful in the ED, because EF is preserved in about half of AHF cases, and most AHF patients have both systolic and diastolic dysfunction regardless of their EF.
- The "vascular phenotype" describes an AHF presentation where functional (ie, reversible) CV abnormalities such as increased vascular tone and fluid shifts predominate over structural (ie, irreversible) ones. These patients tend to be hypertensive, less likely to be hypervolemic, respond favorably to intravenous (IV) nitroglycerin, and have a better prognosis despite more abrupt symptom onset.
- The "cardiac phenotype" of AHF involves a predominance of structural (ie, irreversible) CV abnormalities such as more severe chronic myocardial disease, myocardial ischemia, and more complex multiorgan interactions (eg, cardiorenal syndrome). These patients are often hypervolemic and require high-dose IV diuresis. They typically present with a more indolent symptomatic progression, but paradoxically are at higher risk for worse outcomes.
- Noninvasive positive-pressure ventilation is the first-line approach for respiratory distress in AHF and may obviate a need for intubation in most cases.
- The initial dose of furosemide in the ED for AHF is 1 to 2.5 times the patient's total daily oral dose (or 40–80 mg if loop-diuretic naïve) and should be given IV. Although creatinine may rise after IV furosemide, it is rarely indicative of iatrogenic acute kidney injury, and patients with acute cardiorenal syndromes typically benefit from aggressive diuresis.
- Inotropes are only indicated in cases of cardiogenic shock (CS), because they may increase mortality as a pharmacologic class.
- Resuscitation and stabilization of CS should be followed by rapid evaluation from interjectional cardiology or cardiac surgery. While STEMI was traditionally the predominant cause of CS, it may only account for 30% of contemporary cases due to improvements in coronary intervention.
- Numerous high-moderate risk features in AHF define which patients likely require admission or observation, however, low-risk factors (ie, for discharge) are less well defined. When high or moderate risk factors are absent, the patient is adherent to the appropriate guideline-directed medical therapy, and can secure close outpatient follow-up (≤1 week), a shared decision-making discussion regarding discharge from the ED may be appropriate.

1. A 59-year-old female with a smoking history who has never been to the doctor before presents to the ED with 2 weeks of dyspnea and no known past medical history. Evaluation is notable for BNP 650, creatinine 1.5, wheezing on lung exam, CXR with cardiomegaly but no pulmonary edema, and LUS demonstrating more than 3 B-lines in 2 of 4 lung zones on each hemithorax. Which of the following is the most likely/appropriate diagnosis?
   a. Advanced chronic heart failure
   b. Acute decompensated heart failure
   c. Acute exacerbation of chronic obstructive pulmonary disease
   d. De novo acute heart failure

Answer: D. The patient presents with 3 or more B-lines in 2 of 4 lung zones on each hemithorax, which is highly specific for acute heart failure (AHF). Because the patient does not have a known past history of chronic heart failure, her more specific diagnosis is "de novo acute heart failure," and not "acute decompensated heart failure." "Advanced chronic heart failure" is a subset of chronic heart failure at the end stages of chronic pump dysfunction. Because the patient presented for unscheduled care due to heart failure (ie, true of essentially all patients presenting to the ED for HF symptoms, as opposed to a scheduled outpatient visit), this case is by definition acute heart failure and not chronic heart failure. Finally, although wheezes on exam may lead the clinician to think of COPD exacerbation, auscultated wheezing is common in AHF and neither sensitive nor specific (likelihood ratios positive and negative = 0.6, 1.8) enough to significantly inform a diagnosis of AHF versus COPD on its own.

2. A 68-year-old ED patient with a history of HFrEF presents with 3 weeks progressive edema and exertional dyspnea, pulmonary edema on portable CXR, respiratory rate 35, BP 128/67, and SpO_2 86% on room air. Last week his cardiologist increased his home dose of furosemide from 40 mg PO daily to 40 mg BID, with unchanged urine output and no improvement in symptoms. Which of the following is the best step in management?
   a. Nasal cannula oxygen, 160 mg IV push furosemide now
   b. Nasal cannula oxygen, 60 mg IV push furosemide only if creatinine is not elevated from baseline
   c. BiPAP, 160 mg IV push furosemide now
   d. BiPAP, 60 mg IV push furosemide only if creatinine is not elevated from baseline

Answer: C. This patient presents with hypoxia, tachypnea, and other signs of respiratory failure, and noninvasive positive-pressure ventilation (NPPV) such as CPAP or BiPAP should be the first-line resuscitative therapy. Although nasal cannula may sufficiently reverse the hypoxia, NPPV has beneficial effects on work of breathing and possibly on AHF hemodynamics. IV furosemide dosing should be 1 to 2.5 times the patient's daily home PO dose (or 40–80 mg IV if they do not take furosemide). Because this patient was recently increased to 80 mg, a dose of 80–200 mg IV would be possible. Additionally, since the change resulted in no change in symptoms at home, this would suggest the patient is experiencing some aspect of acute cardiorenal dysfunction, loop diuretic resistance, or both. Thus, a dose at the higher end of the acceptable range is likely advisable. Finally, elevations in creatinine due to acute cardiorenal syndrome are usually due to congestion, and if creatinine rises further after IV loop diuresis this typically represents a transient response not associated with long-term renal dysfunction. In other words, waiting for a creatinine level is not necessary in most cases, because worsening creatinine after furosemide rarely represents a true acute kidney injury. By contrast, delayed administration of furosemide in AHF is associated with adverse outcomes.

3. If the patient in the previous question instead had a BP of 200/125, and was diuretic naïve, which of the following would be the best management approach?
   a. BiPAP, nitroglycerin 1–2 mg IV push repeated until SBP 150 mm Hg, furosemide 40 mg IV push
   b. BiPAP, morphine 4 mg IV push, furosemide 40 mg IV push
   c. BiPAP, isosorbide mononitrate 60 mg PO, furosemide 40 mg PO
   d. BiPAP, nitroglycerin 1–2 mg IV push repeated until SBP <150, furosemide 40 mg PO

Answer: A. In this variation of question two, the patient is diuretic naïve and has features suspicious for the vascular phenotype of AHF. Furosemide should therefore be dosed at 40–80 mg IVP, and a 25% or less reduction in SBP should be targeted with repeated IV push nitroglycerin, 1–2 mg per dose. In patients with less distress or less severe hypertension who still show the vascular phenotype, sublingual nitroglycerin is a useful alternative because absorption and bioavailability are similar to IV. Transdermal and PO nitrates are not currently thought to be useful in AHF, by contrast, because of less favorable kinetics.

4. A 78-year-old patient presents to the ED with hypotension and acute respiratory failure. His daughter states he complained of severe crushing chest pain 7 days ago, but refused to go to the hospital. His ECG shows deep Q waves and T-wave inversions in the inferior leads without ST elevation, and harsh systolic murmur is heard on exam. What is the best next step in management?
   a. Dobutamine, consult cardiac surgery for mechanical hemodynamic support and repair of ruptured papillary muscle
   b. Dobutamine, consult interventional cardiology for PCI
   c. Norepinephrine, consult cardiac surgery for mechanical hemodynamic support and repair of ruptured papillary muscle
   d. Norepinephrine, consult interventional cardiology for PCI

Answer: C. This patient presents with cardiogenic shock, likely due to papillary muscle rupture given the murmur consistent with mitral regurgitation, and an ECG suggesting that a STEMI may have occurred several days ago. Surgical management is the definitive therapy. While arranging for surgical management, norepinephrine is the first-line vasoactive therapy for cardiogenic shock patients. Other inotropic and/or vasoactive medications such as epinephrine, dobutamine, milrinone, or levosimendan are reasonable second-line options.

5. In a patient with AHF and a history of HFrEF whose symptoms have resolved after a single dose of IV furosemide in the ED, which of the following profiles suggests the lowest risk to be discharged from the ED?
   a. No renal or hepatic dysfunction, takes no home medications
   b. SBP 165 mm Hg, compliant with home meds including carvedilol and sacubitril-valsartan
   c. Patient's PCP can see them for follow-up in 21 days from now
   d. No comorbidities, de novo presentation of AHF

Answer: B. This patient who has resolution of or marked improvement in symptoms after initial ED treatment is a good candidate for discharge from the ED. In option A, a lack of acute renal or hepatic dysfunction would further increase the appropriateness for discharge, but having HFrEF the patient should be on a minimum of a beta-blocker and ACE/ARB/ARNI for guideline-directed medical therapy (GDMT). Discussion with the patient's physician to initiate these therapies and arrange close follow-up could be an appropriate strategy with answer A, but in answer B the patient is already on appropriate GDMT. Elevated BP in option B is also a sign of very low risk for morbidity and mortality. Patients who are able to follow up with their physician within 7 days of ED discharge after AHF therapy have significantly improved outcomes, so answer C is incorrect. Although option D would suggest a similarly low-risk presentation, with de novo AHF it is reasonable to keep a patient for further testing and evaluation for their cardiac dysfunction and its underlying etiology. Of final note, none of the answer options suggest a patient who necessarily requires a full hospital admission. A 24-hour-or-less observation stay would likely be appropriate for A,C, or D.

KEY CONCEPTS

- Pericarditis and myocarditis have chest pain, laboratory results, and ECG findings that can mimic acute myocardial infarction (AMI). When the diagnosis is not clear, early coronary angiography definitively identifies or excludes AMI.
- Acute pericarditis is treated with aspirin or nonsteroidal antiinflammatory drugs. Adding colchicine decreases the rate of recurrence.
- Cardiac tamponade presents many ways. Patients with dyspnea, distended neck veins, hypotension, and muffled heart sounds should have the diagnosis made by bedside ultrasound. Pericardiocentesis is therapeutic and may help establish the etiology.
- Myocarditis should be considered in any patient with viral illness symptoms and chest pain.
- Patients with newly diagnosed hypertrophic cardiomyopathy should avoid strenuous exertion until evaluated by a cardiologist. Beta-blockers are the mainstay of therapy; nitrates should be avoided.
- Dilated cardiomyopathy presents with symptoms of heart failure or rhythm disturbance.
- Many cardiomyopathies have genetic origins. Send patients for both cardiology and genetics follow-up.

1. A 33-year-old man presents with a 4-hour history of left anterior chest pain associated with mild shortness of breath. He is otherwise healthy except for chronic tobacco use. Vital signs are: blood pressure, 142/92 mm Hg; heart rate, 120 beats per minute; respiratory rate, 24 breaths per minute; temperature, 100.4°F (38.0°C) oral; and oxygen saturation, 97%. Physical examination is remarkable for tachycardia and a friction rub. The patient's electrocardiogram (ECG) is shown below. Which of the following would be the most appropriate therapy?

a. Cardiac catheterization
b. Ibuprofen 600 mg QID and colchicine
c. Nitroglycerin, aspirin 324 mg oral, cardiology consultation
d. Oxygen, serial troponin levels

Answer: b. The ECG shows findings consistent with acute pericarditis. Clinically, diffuse ST elevation is seen in leads I, II, II, aV_L, aV_F, and V_2 to V_6. In contrast to finding an acute myocardial infarction (AMI), ST elevations are concave upward in acute pericarditis rather than convex upward as seen with a myocardial infarction (MI). PR depression is also a frequent finding. As the acute pericarditis resolves, ST changes revert to normal, followed by T wave flattening and later deep symmetrical inversion, which may persist. NSAIDs are the mainstay therapy for uncomplicated pericarditis. Colchicine helps decrease recurrence. Oxygen therapy is not helpful in patients with normal oxygen saturation. Because the ECG, examination, and history point to pericarditis, cardiac catheterization is not needed. There are case reports of thrombolytics use in pericarditis leading to pericardial hemorrhage and major complications.

2. Which of the following statements regarding post–myocardial infarction (MI) pericarditis is true?

a. Classic pericarditis electrocardiogram (ECG) findings are reliably seen.
b. It indicates a greater degree of myocardial damage than those without pericarditis.
c. Large pericardial effusions are common.
d. The incidence of congestive heart failure (CHF) is unchanged.

Answer: b. Both dysrhythmias and CHF are more common in patients who experience post-MI pericarditis. Large effusions are uncommon, and classic pericarditis ECG findings are often overshadowed by the changes of the recent or evolving MI.

3. A 15-year-old female presents during summer break with fever, cough, and chest pain. She has no history of illnesses or medication use. No street drug use. Her vital signs are BP 110/70, T 38.4, P 122, RR 16, SaO_2 96% on RA. ECG shows sinus tachycardia with normal intervals and low voltage. You order a troponin test and it is slightly above the normal range. Which other test result would you expect?

a. Cardiomegaly on chest x-ray
b. Elevated C-reactive protein
c. Elevated sodium and creatinine
d. Regional wall motion abnormalities on POCUS

Answer: d. Patients with myocarditis show regional wall motion abnormalities and global hypokinesis on POCUS. Even though there is inflammation, WBC and CRP are neither sensitive nor specific for myocarditis. Electrolyte and renal abnormalities should not occur. Cardiomegaly is a late finding if it occurs at all.

4. A 55-year-old woman presents with progressive dyspnea, chest pain, and cough over 5 days. She has a past history of renal failure and is on dialysis, last 2 days ago. She does not smoke. Vital signs are: temperature, 100.2°F (37.9°C) oral; heart rate, 120 beats per minute; respiratory rate, 26 breaths per minute; blood pressure, 100/60 mm Hg; and oxygen saturation, 96% on room air. Physical examination is remarkable for 3-cm jugular venous distention at 45 degrees, clear lung fields on auscultation, tachycardia without a friction rub, trace pretibial edema, and weak peripheral pulses that disappear during expiration. Chest radiograph shows an enlarged cardiac silhouette and clear lung fields. What would be the most appropriate initial intervention?

a. Endotracheal intubation with rapid sequence induction
b. Enoxaparin 1 mg/kg subcutaneous
c. Computed tomography (CT) scan of the chest
d. Isotonic fluid bolus and point of care cardiac ultrasound

Answer: d. This patient is presenting with cardiac tamponade, presumably secondary to an uremic pericardial effusion. Pulmonary embolus is a consideration but less likely, given the picture of normal oxygen saturation and an enlarged heart. The initial intervention should be fluid loading to maintain preload and cardiac output, followed by ultrasound confirmation and likely pericardiocentesis. Fluid loading in renal failure may easily result in pulmonary edema, so expeditious relief of tamponade is indicated. It is too early for intubation because the patient is oxygenating adequately.

5. A 44-year-old man complains of swollen legs. He just finished two courses of prednisone for wheezing related to asthma. The first course was prescribed 6 weeks ago in the emergency department (ED), where he was diagnosed with new onset asthma and normal chest radiograph. The second course was prescribed by his family physician 2 weeks ago. The patient denies fever and chest pain and is still mildly short of breath, which is worse at night or with exertion. Examination shows bibasilar rales in his lungs, normal heart sounds, and 1+ edema in both legs up to his knees. What is his diagnosis?

a. Asthma exacerbation
b. Idiopathic dilated cardiomyopathy (DCM)
c. Prednisone-induced edema
d. Prednisone-induced liver failure

Answer: b. The patient is unlikely to have a new diagnosis of asthma. He most likely had a viral process leading to reactive airway disease initially and viral myocarditis later. He unfortunately now has a DCM and symptoms of heart failure. Treatment is supportive. Prednisone-related end-organ damage usually does not occur this quickly.

69 Infective Endocarditis and Valvular Heart Disease

KEY CONCEPTS

- Infectious endocarditis (IE) is more often caused by Staphylococcus species than Streptococcus, and increasingly occurs in elderly, recently hospitalized patients, those with prosthetic valves and intra-cardiac devices, and in patients who inject opioids.
- IE should be considered in any patient with a predisposing condition plus fever; IE frequently presents with a complication, including heart failure, embolic stroke, or osteomyelitis.
- At least two sets of blood cultures (obtained before starting antibiotics) and transthoracic echocardiography are the key diagnostic tests to order for patients with suspected IE.
- Empiric antimicrobial treatment for IE should include vancomycin.
- Consultation with a cardiothoracic surgeon is recommended for left-sided vegetations.
- Acute rheumatic fever is a delayed nonsuppurative complication of streptococcal pharyngitis characterized by arthritis, carditis, chorea, subcutaneous nodules, and erythema marginatum.
- In a patient with severe mitral stenosis, hypovolemia and tachycardia are poorly tolerated. "Slow and full" are appropriate goals.
- In patients with critical aortic stenosis, avoid excessive preload reduction with vasodilators and diuretics.
- In patients with acute aortic insufficiency, classic physical findings may be absent. Medical stabilization entails the cautious use of vasodilators and diuretics. Intraaortic balloon counterpulsation is contraindicated.
- Complications of prosthetic heart valves range from structural failure and thrombosis to systemic embolization, hemolysis, and endocarditis.

1. What is the most common manifestation of acute rheumatic fever (ARF)?
   a. Carditis
   b. Chorea
   c. Erythema marginatum
   d. Polyarthritis

Answer: d. Arthritis occurs early in the course of ARF. The knees, ankles, elbows, and wrists are commonly affected, and pain can be out of proportion to physical findings. Cardiac manifestations are subtle and may reflect endocarditis, myocarditis, or pericarditis. Chorea and erythema marginatum are rare. Chorea is typically a late finding.

2. A 49-year-old woman presents with progressive dyspnea on exertion and orthopnea. Vital signs are temperature 36.7°C (98.1°F; oral), heart rate, 110 beats/min, blood pressure, 135/80 mm Hg, respiratory rate, 22 breaths/min, and oxygen (O_2) saturation, 97% on room air. The physical examination is remarkable for clear lung fields and an irregularly irregular rhythm with a 4/6 diastolic murmur in the left anterior axillary line. She has no peripheral edema. Which of the following would be appropriate hemodynamic management of her cardiac pathophysiology?
   a. Aggressive diuresis
   b. β_1-Agonist to increase chronotropy
   c. Beta blocker
   d. Selective arterial vasodilator

Answer: c. This patient has a picture consistent with atrial fibrillation and mitral stenosis. The apical diastolic murmur and left atrial enlargement, along with progressive dyspnea, all support the diagnosis. Tachycardia is poorly tolerated because of the need for higher left atrial pressures and a longer time during diastole to perfuse across the stenotic valve. Slow and full are appropriate goals. Diuresis might decrease venous return. Any agent producing tachycardia would decrease diastole time and left ventricular preload. An arterial vasodilator would have little effect, given the normal blood pressure and the fact that systemic vascular dilation would not be seen at the mitral valve level as long as the aortic valve was competent.

3. A 32-year-old woman with a history of injection drug use presents with persistent fevers, night sweats, and shortness of breath. Vital signs are temperature 38.2°C (100.8°F; oral), heart rate, 123 beats/min, blood pressure, 105/60 mm Hg, respiratory rate, 26 breaths/min, and oxygen (O_2) saturation, 93% on room air. The physical examination is remarkable for scattered rhonchi and a 2/6 systolic murmur in the left lower sternal border. Which of the following is the most appropriate antibiotic to start?
   a. Clindamycin
   b. Vancomycin
   c. Ceftriaxone
   d. Cefepime

Answer: b. In most cases, empiric therapy in the ED should include vancomycin, and vancomycin alone for one or two doses is a reasonable approach for native valve and prosthetic valve infection and in IDUs. Ceftriaxone can be added to vancomycin to cover HACEK and other gram-negative organisms, but these are responsible for less than 6% of infections. There is no role at this time for empiric use of clindamycin or cefepime.

4. A 62-year-old man with presents 3 days after being admitted for an inferior ST-segment elevation myocardial infarction (STEMI). He underwent a successful percutaneous coronary intervention (PCI) and was just discharged earlier in the day. He developed sudden onset shortness of breath at home, and he arrives in extremis. Vital signs are temperature 37.2°C (98.8°F; oral), heart rate, 112 beats/min, blood pressure, 85/68 mm Hg, respiratory rate, 24 breaths/min, and oxygen (O_2) saturation, 90% on room air. The physical examination is remarkable for severe respiratory distress, diffuse crackles, and a 2/6 systolic murmur in the left lower sternal border. Which of the following is the most appropriate intervention?
   a. Give 2L of crystalloid over an hour.
   b. Start dobutamine.
   c. Initiation of dialysis for volume overload.
   d. Initiate continuous positive airway pressure (CPAP) therapy.

Answer: d. This patient presents with likely acute mitral regurgitation (MR) from a papillary muscle rupture after an myocardial infarction (MI). Giving additional fluid will likely make his pulmonary edema worse. Starting dobutamine without a vasopressor while he is hypotensive could make the hypotension worse. When the diagnosis of acute MR is suspected, initial stabilization in the ED includes treatment of pulmonary edema with intravenous nitrates, diuretics, and noninvasive positive pressure ventilation.

5. A 38-year-old man with presents with progressive dyspnea over a day, starting 2 hours before arrival. Vital signs are temperature 38.9°C (102.0°F; oral), heart rate, 115 beats/min, blood pressure, 120/72 mm Hg, respiratory rate, 26 breaths/min, and oxygen (O_2) saturation, 92% on room air. The physical examination is remarkable for diffuse crackles and a 2/6 systolic murmur in the left lower sternal border. His echocardiogram demonstrates a large, mass on the mitral valve with severe mitral regurgitation (MR). In addition to managing his pulmonary edema, which of the following is the most appropriate intervention?
   a. Administer beta-blockade to reduce the risk of embolization.
   b. Consult Cardiac Surgery.
   c. Start a heparin infusion.
   d. Start digoxin for heart rate control.

Answer: b. Patients with suspected infectious endocarditis (IE) and acute heart failure require immediate consultation by a cardiothoracic surgeon. In general, there has been a shift toward earlier surgical treatment in IE, with approximately one-half of patients with left-sided infection undergoing surgery during the index admission. If possible, patients with left-sided IE should be managed by a multidisciplinary team at a center capable of cardiothoracic surgery. There is no role for addition of beta-blockade in this patient in heart failure. Heparin and digoxin are not indicated for IE.

Hypertension 70

KEY CONCEPTS

- Elevated blood pressure with or without associated symptoms is exceedingly common in the emergency department (ED).
- A true hypertensive emergency is defined by the presence of acute target organ damage (TOD) and is distinct from other clinical presentations.
- Commonly encountered hypertensive emergencies include ischemic and hemorrhagic stroke, myocardial infarction, acute heart failure, aortic dissection, and pre-eclampsia.
- Management strategies regarding target blood pressure reduction, rate of reduction, and agent choice vary widely among the discrete conditions defined as hypertensive emergencies. Thus, clinical decision making in this context requires knowledge of the physiology and an evidence-based approach to each condition.
- For patients without acute TOD, immediate antihypertensive therapy is not needed. However, emergency clinicians play an essential role in the care of this group, providing screening, ongoing surveillance, and linkage to care to prevent secondary complications of this disease.
- Initiation of long-term oral antihypertensive therapy in the ED for those with elevated blood pressure but no established history of hypertension is not recommended; however, emergency clinicians are encouraged to refill, reinitiate, or up-titrate antihypertensive medications in patients with elevated blood pressure who are known to have chronic hypertension.

1. An 85-year-old man presents with acute onset of right-sided weakness that began 1 hour before arrival. His family states that he was recently diagnosed with dementia and has refused to take any of his medications for more than 1 week but had otherwise been relatively healthy. Emergency medical services (EMS) report a prehospital blood pressure of 240/110 mm Hg. On examination, the patient has clear evidence of an acute stroke with severe deficit (National Institutes of Health [NIH] Stroke Scale = 24), but his gag reflex is intact. His initial blood pressure is 230/110 mm Hg. A head computed tomography (CT) scan shows diffuse white matter ischemic disease but no evidence of acute infarct or hemorrhage. The most appropriate next step is which of the following?

a. Repeat the blood pressure and, if 185/110 mm Hg or higher, start a nicardipine infusion at 5 mg/hour.
b. Contact neurosurgery for intracranial pressure monitoring.
c. Give a dose of tissue plasminogen activator (tPA) immediately.
d. Give sublingual nifedipine.

Answer: a. Despite his advanced age, this is a relatively high-functioning individual with acute onset of a severe stroke. He is within the time window for administration of tPA, but his blood pressure is excessively elevated. According to the American Heart Association/American Stroke Association guidelines, efforts should be made to reduce his blood pressure to less than 185/110 mm Hg before tPA administration. Other agents, such as labetalol, would also be appropriate to use.

2. A 27-year-old woman presents with lower back pain for 1 week. She recently moved into the community and has been busy unpacking and moving furniture. She denies any neurologic symptoms and has no medical history. Her triage blood pressure is 170/110 mm Hg, but vital signs are otherwise normal. With the exception of appreciable muscle spasm in her lower back, the physical examination is normal. You give her 10 mg of diazepam and observe her for 1 hour. She is feeling better but her blood pressure is still elevated, at 165/90 mm Hg. At this point you should do which of the following?

a. Give her 0.2 mg of clonidine.
b. Take a more extensive history to see if she is taking oral contraceptives
c. Ignore the blood pressure and discharge her without follow-up.
d. Order a 24-hour urine metanephrine and normetanephrine test.

Answer: b. This patient's blood pressure elevation is clearly not a reaction to her pain, and some further questioning is needed to identify a potential cause. In young women, oral contraceptive agents are a treatable cause of hypertension and should be considered when elevated blood pressure is encountered in these patients. Of note, the likelihood of contraceptive-related hypertension increases with the duration of use. It would not be unreasonable to discharge the patient without further exploring the potential cause of her elevated blood pressure, provided some provision of follow-up was sought. The clinical picture is inconsistent with renal artery stenosis or pheochromocytoma, and an evaluation for these conditions would be inappropriate at this point.

3. A 50-year-old man complains of dull chest pain that began 4 hours before arrival. The patient states he awoke this morning feeling well and that the pain began while shoveling his driveway. He has a 20-year history of hypertension and diabetes but has always been compliant with his medications, which include metformin, Norvasc, and Diovan. His initial blood pressure is 215/120 mm Hg, and his heart rate is 100 beats/min. On examination, his lungs are clear, heart sounds are normal, pulses are bounding and symmetric, and there are no neurologic deficits. An electrocardiogram shows nonspecific T wave inversions in the lateral leads, with no evidence of ST segment elevation or depression and normal intervals. The serum troponin level is elevated. Which of the following is the most likely explanation for his clinical presentation?

a. He has a ruptured dissecting aortic aneurysm that is leaking blood into the retroperitoneal space.
b. He is suffering from a massive pulmonary embolism.
c. He strained an intercostal muscle while shoveling.
d. He is suffering from subendocardial ischemia triggered by a cycle of increased afterload that began while he was shoveling.

Answer: d. Increased afterload can be triggered by exertional activation of the sympathetic nervous system. In the presence of long-standing hypertension, ventricular and aortic stiffness are likely to develop, increasing the potential for afterload-mediated effects on the heart. When suddenly faced with increased resistance, the left ventricular pressure may rise, leading to intrinsic compression of subendocardial myocytes and ischemia. Coupled with earlier transmission of the reflected arterial wave form, the diastolic coronary filling time may also be diminished, producing a clinical picture of acute coronary syndrome. He may also have a coronary artery lesion on which this is superimposed but, as described, the likelihood of his clinical presentation being caused by a complete right coronary occlusion is quite low.

4. Which of the following fundoscopic findings would be the earliest indicator of acute hypertensive retinopathy in a comatose patient whose blood pressure is 260/140 mm Hg?
   - **a.** Copper and silver wiring appearance to the retinal arterioles
   - **b.** Focal intraretinal periarteriolar transudates
   - **c.** Cotton wool spots
   - **d.** Diffuse atrioventricular (AV) nicking

Answer: b. Focal intraretinal periarteriolar transudates are the first abnormality to appear with acute hypertensive retinopathy, preceding all other findings, including cotton wool spots and disk edema. Other findings that are listed may also be seen, but they are indicative of chronic, not acute, retinal involvement. It is important to remember that acute retinal abnormalities may be absent in hypertensive emergency patients and, although fundoscopy is an important tool, the diagnosis should be based primarily on the results of the clinical examination.

5. A 65-year-old woman presents to the emergency department (ED) with chief complaint of elevated blood pressure, which was discovered at her dentist's office prior to a routine dental procedure. Her blood pressure upon admission is 187/101 mm Hg. Otherwise, the patient has no acute complaints, however, she does note chronic low grade chronic headaches and intermittent blurry vision. She has had previous readings of mildly elevated blood pressure at church health fairs, although was never formally diagnosed, and does take medication. Which of the following steps would be appropriate in the context of and ED visit?
   - **a.** Obtain a CT of the head to rule out intracranial pathology.
   - **b.** Administer a short-acting antihypertensive to reduce her blood pressure prior to discharge.
   - **c.** Prescribe a diuretic such as chlothalidone or hydrochlorothiazide, and refer to outpatient follow up with a hospital affiliated primary care provider.
   - **d.** Discharge home with instructions to follow up with a primary care provider.

Answer: d. Most patients who are found to have significantly elevated BP on intake vital signs measurement or who come to the ED because BP was found to be elevated in an outpatient setting or by self-measurement do not have an acute hypertensive emergency. While many patients have vague symptoms, such as headache or mild blurry vision, these symptoms do not correlate with the degree of hypertension. For such patients, a head CT or acute reduction of BP is not indicated and offers no tangible outcome benefit. This patient has a physician and is already on medication. As such, there is no indication to start a new medication or make a new referral at this time.

KEY CONCEPTS

- Aortic dissection most commonly presents as abrupt, sharp, severe pain maximally intense at onset in the chest or back. There may be various seemingly unconnected associated symptoms due to altered blood pressure or insufficiency of disparate vascular beds.
- Definitive diagnosis is most commonly made with an imaging study such as computed tomography of the aorta with intravenous contrast. A combination of standardized clinical assessment, chest x-ray, and serum D-dimer testing can effectively rule out the condition in low-risk patients.
- The critical immediate therapeutic actions are medical therapy to decrease aortic sheer force and blood pressure control. Immediate surgical consultation is indicated.
- Surgical treatments, including endovascular stent placement, are advancing, leading to decreased mortality. This increases the importance and benefit of early diagnosis in the emergency department setting.

1. The single most important agent to treat acute aortic dissection is:
- **a** Nitroprusside.
- **b** Aspirin.
- **c** Antiadrenergic.
- **d** Enalaprilat.

Answer: C. The initial primary goal of care for a patient with an aortic dissection in the emergency department setting is to minimize extension of the dissection. This is accomplished by decreasing the force of cardiac contraction and resulting aortic sheer forces, and driving the heart rate down to 60 beats per minute or less. Acceleration of aortic blood flow, dP/dt, is minimized while maintaining blood pressure in the low normal range. Beta-1 adrenergic blockade therapy is the preferred treatment.

2. A 70-year-old man with a history of poorly controlled hypertension presents with sharp central severe chest pain. ECG, chest x-ray, and serum D-dimer are normal. The next appropriate step is:
- **a** Transthoracic echocardiography in the radiology suite.
- **b** CT angiogram of the aorta.
- **c** Check serial troponin, and discharge with close follow up if negative.
- **d** Consult surgical service.

Answer: B. This patient had a presentation very concerning for an aortic dissection. In high-risk cases, chest x-ray and D-dimer cannot not be used to rule out a dissection. These patients must receive an advanced imaging test.

3. You are working in a small rural hospital when you diagnose a type A aortic dissection based on CT angiogram of the aorta. Vital signs are heart rate 85 beats per minute and blood pressure 130/85 mm Hg. The high-volume referral center is 25 minutes away by ambulance ride.
- **a** Arrange for immediate transfer to high-volume center.
- **b** Stabilize with arterial and central lines and intubation, then arrange transfer.
- **c** Call the intensivist to arrange medical ICU admission on labetalol infusion.
- **d** Call the general surgeon on duty to organize emergent surgery at your facility.

Answer: A. Surgery and endovascular interventions for aortic dissection require coordinated teams with expertise and experience to achieve success. Outcomes for type A dissection patients are generally better at high-volume centers. However, patient transfer incurs a delay in time-sensitive care. Recent data suggest regionalization of care with transfer a mean of 50 miles to high-volume centers is associated with lower operative mortality.

4. A 60-year-old woman presents with chest pain radiating to the back with onset while lifting a 50-lb pot in her garden, and her ECG demonstrates an inferior STEMI. The next appropriate step is:
- **a** MR angiogram of the chest.
- **b** Aspirin and TPA.
- **c** Diltiazem infusion.
- **d** Immediate cardiac catheterization with recommendation to interventional cardiologist for initial aortogram.

Answer: D. Aortography is valuable for the patient with STEMI but some presentation characteristics suggestive of dissection. As time is essential for percutaneous transluminal coronary angioplasty (PTCA) or stenting for STEMI patients, the patient can be brought to the cardiac catheterization lab and aortography performed at the initiation of the procedure in lieu of other imaging. The emergency physician must share his or her concerns with the interventional cardiologist to plan this diagnostic approach.

5. The connective tissue disorder responsible for the most cases of acute aortic dissection is:
- **a** Ehlers-Danlos, type I.
- **b** Ehlers-Danlos, type IV.
- **c** Marfan syndrome.
- **d** Loeys-Dietz syndrome.

Answer: C. Marfan syndrome is the connective tissue disease most commonly associated with acute aortic dissection.

6. There are about 150 million annual ED visits in the United States. An emergency physician who sees 3000 patients per year can expect to see an acute aortic dissection about every:
- **a** 6 months.
- **b** 1 year.
- **c** 5 years.
- **d** 15 years.

Answer: C. Aortic dissection is a rare condition, with only about 10,000 cases a year in the United States. If an emergency physician sees 3000 patients, and the rate of aortic dissection is 10,000 per 150,000,000 ED visits, this results in seeing 0.20 per year, or about 1 every 5 years.

KEY CONCEPTS

- A ruptured abdominal aortic aneurysm (AAA) should be considered in any patient with otherwise unexplained abdominal or back pain. The complete triad of pain, hypotension, and a pulsatile mass may not be present.
- In a patient with an AAA and acute symptoms such as severe abdominal or back pain or hypotension, rupture is imminent or has already occurred.
- A patient with a ruptured AAA and initially normal vital signs can suddenly deteriorate at any time.
- The risk of rupture increases substantially with increased aneurysm size, and most ruptured AAAs have diameters greater than 5 cm.
- Bedside ultrasound may be used to document an AAA or free fluid and assist in the rapid diagnosis of a ruptured AAA.
- The abdominal CT scan is the diagnostic test of choice in the evaluation of the stable patient with suspected ruptured AAA; intravenous contrast is not essential in emergencies.
- The patient who has had endovascular repair of an AAA remains at risk for aneurysm rupture.

1. What is the most common complication of abdominal aortic aneurysms (AAAs)?
   a. Renal artery infarction
   b. Rupture
   c. Development of an aortoenteric fistula
   d. Ureteral obstruction

Answer: B. Although all of the options are potential complications of AAAs, rupture is by far the most common. Rupture typically occurs into the retroperitoneum, allowing possible tamponade of the rupture site. Only between 10 and 30% of ruptures are freely into the peritoneum, which is often rapidly fatal.

2. An 82-year-old male presents after a syncopal episode, which occurred 1 hour prior to arrival and was without preceding symptoms. He has a past medical history significant for hypertension and a surgical repair of an abdominal aortic aneurysm (AAA) 15 years ago. Physical examination reveals a pale man with a heart rate of 112 beats per minute, a blood pressure of 86/54 mm Hg, and a respiratory rate of 24 beats per minute. There is melena on rectal examination and there is no palpable mass on abdominal palpation. Along with appropriate volume resuscitation, what is the most appropriate next step in the management of this patient?
   a. CT scan of the abdomen with contrast enhancement
   b. Abdominal x-rays
   c. Immediate vascular surgery consultation
   d. Addition of albumin to crystalloid infusion

Answer: C. The presence of gastrointestinal bleeding with evidence of an AAA or a history of aortic surgery is an aortoenteric fistula until proven otherwise. The most common site of vascular erosion is the third or fourth portion of the duodenum, causing hematemesis or melena. Initial bleeding may be minor, followed days to weeks later by massive bleeding. Surgical intervention is necessary. Unstable patients may need to be taken directly to the OR based on a strong clinical presumption.

3. Which of the following is true regarding the management of a patient with a suspected abdominal aortic aneurysm (AAA)?
   a. Aggressive crystalloid resuscitation to normalize the blood pressure is warranted.
   b. Permissive hypotension with a goal of a systolic blood pressure of 70 to 90 mm Hg is a reasonable target.
   c. Lowering the blood pressure in hypertensive patients has been shown to improve outcomes.
   d. Bedside ultrasound should always be followed by a CT scan to better define the aneurysm.

Answer: B. Permissive hypotension with a goal of a systolic blood pressure of greater than 70 mm Hg but not necessarily more than 90 mm Hg is a reasonable target. Aggressive crystalloid resuscitation may be harmful. Lowering the blood pressure in hypertensive patients has not been shown to improve outcomes in patients with abdominal aortic aneurysms as it has in patients with dissecting thoracic aortic aneurysms. Although a CT scan can be useful to better define the aneurysm prior to surgery in stable patients, bedside ultrasound may be the only imaging necessary in unstable patients.

4. What is the strongest predictor of mortality in patients with a ruptured abdominal aortic aneurysm (AAA)?
   a. Hypotension
   b. Presentation more than 12 hours after symptom onset
   c. Age older than 70
   d. Anemia on arrival

Answer: A. Although all of the options are predictors of higher mortality, preoperative hypotension is the strongest predictor. That said, there is little data guiding volume resuscitation, fluid choices, and target blood pressures in hypotensive patients with a ruptured AAA. Crystalloid resuscitation before surgical repair can cause dilutional coagulopathy and clot dislodgement by raising the intravascular volume. Judicious volume resuscitation with early use of blood products and a target systolic pressure of 70 to 90 mm Hg is reasonable.

5. Which of the following is most common in the patient with an intact (non-ruptured) 6-cm abdominal aortic aneurysm?
   a. Back pain
   b. Abdominal pain
   c. Sensation of abdominal distention
   d. Absence of any symptoms

Answer: D. Most intact AAAs are asymptomatic. They may be discovered incidentally on physical examination or a radiologic study done for other reasons or may be found in an ultrasonographic aneurysm screening program. Pain in the abdomen or back does not usually develop until the aneurysm ruptures.

6. A 67-year-old man presents with a several-day history of intermittent periumbilical abdominal pain. Physical examination and vital signs are unremarkable except for supraumbilical tenderness. Hemoglobin is 12 g/dL. Contrast-enhanced computed tomography (CT) scan of the abdomen details a 4.2-cm infrarenal abdominal aortic aneurysm (AAA) with an apparent old contained rupture with no obvious new or acute blood. All of his discomfort is relieved by a single dose of morphine sulfate. Which of the following would be the most appropriate course of action?
   a. Vascular surgery clinic follow-up in 1 or 2 days
   b. Emergency department (ED) observation for 6 to 8 hours with serial abdominal examinations
   c. Gastrointestinal consultation for early follow-up and upper or lower endoscopy
   d. Surgical consultation

Answer: D. Watchful waiting is indicated only for asymptomatic aneurysms, regardless of the size (up to 5.5 cm). The majority of "stable" AAAs are neither painful nor tender, and the presence of both in this patient suggests imminent rupture. Although the CT scan did not detect acute blood, the safest course of action would be surgical consultation, with early or imminent rupture as the presumed source of this patient's symptoms

73 Peripheral Arteriovascular Disease

KEY CONCEPTS

- Acute arterial occlusion is a limb-threatening emergency requiring early anticoagulation and Fogarty catheter embolectomy. The clinical diagnosis is based on some variant of the five Ps: pain, pallor, pulselessness, paresthesias, and paralysis. Confirmatory tests are unnecessary and increase the limb's ischemic time.
- Atheroembolism (blue toe syndrome) is associated with cool, painful cyanotic toes in the presence of palpable distal pulses. A proximal source should be localized, most often an atherosclerotic aneurysm in the aorta or the iliac, femoral, or popliteal artery.
- Popliteal aneurysms are bilateral in 60% of patients and often coexist with an abdominal aortic aneurysm.
- The classic Raynaud attack is triphasic: the fingers become pale, blue, and then red. Raynaud disease has no detectable underlying cause and usually has a benign course. Conversely, Raynaud phenomenon has an underlying disorder, usually connective tissue disease.
- The only reliable clinical test for the detection of thoracic outlet syndrome is the elevated arm stress test (EAST).
- Partial arterial lacerations continue to bleed, resulting in an expanding hematoma. Complete arterial transections initially have only moderate bleeding but can result in delayed hemorrhage. Blunt arterial injury may produce intimal disruption resulting in dissection, thrombosis, or obstruction. Arterial vasospasm can accompany injuries adjacent to the blood vessel, but spontaneous resolution always occurs in the absence of arterial disruption or intimal injury.
- Aneurysms and arterial stenoses are characterized by a systolic murmur. Pseudoaneurysms, associated with prior surgical or trauma sites, are characterized by a loud systolic and possibly a separate, faint diastolic murmur. Arteriovenous fistulae are characterized by a harsh "to and fro" murmur associated with a palpable thrill.
- Intra-arterial injection of illicit drugs into the brachial or radial artery is associated with immediate onset of severe, burning pain, and emergency department presentation with patchy blue-purple skin discoloration. Identifying the injection site helps confirm this syndrome, which can be associated with persistent ischemia and tissue loss.

1. Where do most arterial emboli originate?
- **a.** Abdominal aorta
- **b.** Femoral artery
- **c.** Left ventricle
- **d.** Left atrium
- **e.** Thoracic aorta

Answer: c. Eighty-five percent of arterial emboli originate in the heart. Of these, left ventricular thrombus formation after myocardial infarction (MI) accounts for 60% to 70%. Atrial thrombi account for only 5% to 10% of all peripheral arterial emboli.

2. What is the most frequent site of acute arterial embolic occlusion?
- **a.** Carotid artery
- **b.** Common femoral artery
- **c.** Mesenteric artery
- **d.** Popliteal artery
- **e.** Renal artery

Answer: b. The bifurcation of the common femoral artery accounts for 35% to 50% of acute arterial occlusion due to arterial embolism. Embolic occlusion most often occurs at major arterial bifurcations because of the sudden change in vessel diameter at these locations.

3. A 63-year-old male presents with acute onset of left leg pain while walking. He describes it as a shock-like sensation that made his knee buckle. Past history is remarkable for hypertension, diabetes (diet controlled), tobacco use, and a recent lateral wall myocardial infarction. Current medications are aspirin, metoprolol, and lisinopril. Vital signs are temperature, 37.0°C oral; heart rate, 98 beats per minute; blood pressure, 160/105 mm Hg; respiratory rate, 20 breaths per minute; and oxygen (O2) saturation, 96%. Physical examination is remarkable for left lower extremity pallor with decreased light touch sensation, nonpalpable left foot pulses, and minimal capillary refill. What would be the most appropriate next step in the diagnosis and management of this patient?
- **a.** Abdominal ultrasonography
- **b.** Arteriogram
- **c.** Serum lactate level
- **d.** Thoracolumbar magnetic resonance imaging (MRI) scan
- **e.** Vascular surgery consultation

Answer: e. This patient has acute limb ischemia from an acute arterial embolus, most likely originating from his left ventricle secondary to a recent myocardial infarction. Loss of light touch sensation on physical examination indicates jeopardized tissue viability, requiring immediate vascular surgery consultation for emergent Fogarty catheter embolectomy. Reliable diagnosis of an acute arterial embolism can almost always be made by history and physical examination alone. Any additional diagnostic evaluation constitutes an unnecessary delay. Serum lactate level, abdominal ultrasonography, and thoracolumbar magnetic resonance imaging (MRI) scan would not provide useful information. An arteriogram before going to the operating room is an unnecessary delay and may further exacerbate limb ischemia.

4. A supine patient is asked to raise his foot 12 inches above the estimated level of the right atrium and dorsiflex the foot five or six times. He is then brought to a sitting position with his feet hanging. In the absence of severe advanced ischemia, venous filling of the foot should return in less than how many seconds?
- **a.** 1
- **b.** 5
- **c.** 10
- **d.** 15
- **e.** 20

Answer: e. This bedside test is Buerger sign and can provide reliable evidence of advanced ischemia. In the absence of severe advanced ischemia, the lower extremity veins should fill within 20 seconds after being placed in the dependent position.

5. A 73-year-old man presents with acute onset of right lower extremity pain. He has a long history of tobacco use, hypertension, and a several-year history of moderate calf claudication at 50 yards walking. Physical examination reveals signs of chronic atherosclerotic occlusive disease of the bilateral lower extremities, including muscular atrophy, loss of hair over the toes and feet, and thickening of the toenails. Examination of the distal right lower extremity reveals pallor, absent popliteal and foot pulses, and decreased sensation to light touch of the right foot. The cardiac examination is unremarkable, and the 12-lead electrocardiogram (ECG) reveals only normal sinus rhythm. Based on the most likely diagnosis, what is the most appropriate definitive therapy?
- **a.** Acute hyperbaric oxygen therapy
- **b.** Arteriogram to determine the presence of embolus versus in situ thrombosis
- **c.** Intra-arterial thrombolysis
- **d.** Surgical referral for Fogarty catheter embolectomy
- **e.** Surgical referral for Fogarty catheter embolectomy with vascular bypass grafting

Answer: e. This patient has a history and physical examination consistent with long-standing peripheral atherosclerotic occlusive disease, no evidence of a proximal source for embolism, but acute onset of ischemic symptoms and loss of light touch in the affected extremity. The most likely diagnosis is a large, in situ thrombosis precipitating acute limb-threatening ischemia. When limb-threatening ischemia is present, emergent surgical referral for Fogarty catheter embolectomy is indicated, whether caused by acute in situ thrombosis or embolus. With limb-threatening ischemia caused by in situ thrombosis, simple Fogarty catheter embolectomy is insufficient and usually requires additional bypass grafting. Acute hyperbaric oxygen therapy has no role in the treatment of limb-threatening ischemia due to in situ thrombosis or embolism. An arteriogram to determine the presence of embolus versus in situ thrombosis is unwarranted, represents an unnecessary delay, and may further exacerbate ischemia. Intra-arterial thrombolysis takes 6 to 72 hours to work and is contraindicated in cases of limb-threatening ischemia.

6. What percentage of patients presenting with arteriosclerosis obliterans are younger than 50 years old?
   - **a.** 1%
   - **b.** 5%
   - **c.** 10%
   - **d.** 20%
   - **e.** 40%

Answer: d. Peripheral arteriovascular disease can occur in younger patients. Nineteen percent of patients presenting with atherosclerosis obliterans are between the ages of 30 and 50 years old. Of all arteriosclerosis patients, 33% have coexistent coronary artery disease, and 70% to 90% are smokers. The non-smokers have other risk factors including significant hypertension and hyperlipidemia.

7. A 49-year-old woman presents with severe left ankle pain. She describes a fairly sudden development of a left lateral malleolus hemorrhagic blister that transitioned to a painful superficial ulcer over 48 hours. She has no prior history of extremity ulcers, and her only significant past medical history is hypertension. She has a long-standing history of noncompliance with her hypertensive medications and smokes two packs of cigarettes per day. She has no history of myalgias, joint pain, fever, or systemic symptoms. Vital signs are: temperature, 36°C oral; heart rate, 90 beats per minute; blood pressure, 210/125 mm Hg; respiratory rate, 20 breaths per minute; and O2 saturation, 96%. Physical examination reveals a thin black female in distress because of pain. Cardiopulmonary examination is unremarkable. Abdominal, neurologic, and extremity examinations are likewise unremarkable except for a well-demarcated, shallow 4 × 3 cm ulcer over the left lateral malleolus. There is mild erythema but no evidence of active infection. Distal pulses and capillary refill are normal. What would be the most appropriate intervention?
   - **a.** Analgesics and admission for vasculitis evaluation
   - **b.** Surgical consultation for possible embolectomy
   - **c.** Wound care and blood pressure control
   - **d.** Wound care and tapering dose of prednisone
   - **e.** Venous Doppler scans and surgical consultation for possible skin grafting

Answer: c. This patient has a hypertensive ulcer, which is the most painful of lower extremity ulcers. They typically occur over the lateral malleolus, as opposed to venous stasis ulcers, which are more common anteriorly and medially. Ischemic arterial ulcers are more common distally over the digits. Although vasculitis or a collagen vascular disease are possible, the lack of any other systemic symptoms or prodrome argues against this.

74 Pulmonary Embolism and Deep Vein Thrombosis

KEY CONCEPTS

- Hallmarks of deep venous thrombosis (DVT) include unilateral limb pain and swelling, though these findings can be subtle and nonspecific.
- Patients with low pretest probability (PTP) can have DVT ruled out in the ED with a negative D-dimer or venous ultrasound (US), patients with high PTP can have DVT ruled out in the ED with a negative D-dimer and venous US.
- Three-point US evaluates the leg veins proximal to the knee and has a sensitivity of 95% and specificity of 95% for proximal DVT when performed by a certified sonographer. A negative three-point ultrasound in a patient with a moderate or high PTP for DVT should have a D-dimer test or repeat venous ultrasound within 7 days. A single, whole-leg ultrasound excludes DVT in all pretest probabilities.
- Anticoagulation for DVT and pulmonary embolism (PE) can be achieved with a direct-acting oral anticoagulant (DOAC), low-molecular-weight heparin (LMWH), unfractionated heparin or, in rare cases, an alternative anticoagulant. For DVT and most PE, DOACs are as effective as and safer than the combination of heparin and warfarin.
- The treatment of isolated calf vein DVT is controversial, and it is reasonable to withhold anticoagulation in favor of a repeat venous ultrasound within 7 days and close follow-up.
- Distal superficial vein thrombophlebitis can adequately be treated with nonsteroidal antiinflammatory drugs and warm compresses, but proximal superficial vein thrombophlebitis should be treated with anticoagulation.
- Most patients with DVT distal to the iliofemoral region can be safely treated as outpatients as long as close follow-up and access to anticoagulant medications is assured.
- PE can present asymptomatically, with dramatic clinical symptoms, or with sudden cardiac death. Even the most common symptom, dyspnea, is absent in a quarter of patients.
- The first step in the evaluation of possible PE is determining whether testing for PE is indicated.
- The patient's pretest probability for PE, as determined by clinical gestalt or a validated score (e.g., the Wells Score) dictates the approach to objective testing for PE.
- Patients with low gestalt pretest probability and a negative PERC rule may incur more harm than benefit if testing for PE (including D-dimer) is performed.
- Patients with non-high pretest probability (e.g., Wells Score ≤ 6) can have PE ruled out with a negative D-dimer.
- The D-dimer threshold can be adjusted according to the patient's age or the pretest probability of PE. The formula for age adjustment is Age × 10 ng/mL. Using an adjusted D-dimer threshold reduces the need for imaging by about 10% to 15%.
- Patients with high pretest probability or a positive D-dimer require imaging. For most patients, including pregnant women, computer tomography pulmonary angiography (CTPA) is the imaging test of choice.
- For pregnant women, the decision to undergo any testing for PE should be shared with the patient. To minimize fetal radiation exposure, diagnostic testing for PE should start with a combination of D-dimer and bilateral lower extremity venous ultrasound.
- For patients diagnosed with PE, typical resuscitative measures can be harmful. Endotracheal intubation and positive-pressure ventilation can decrease preload and precipitate cardiac arrest. Excessive intravenous fluid administration can lead to worsening right ventricular distention and left ventricular compression.
- Patients with PE should be risk stratified using a combination of vital signs, CTPA, echocardiography, and troponin. High-risk PE is defined by hemodynamic instability, intermediate-risk by right ventricular dysfunction, and low-risk by the absence of either.
- Patients with high-risk PE should be treated with thrombolysis or thromboembolectomy unless contraindicated.
- Patients with intermediate-risk PE are usually not candidates for thrombolysis but may be candidates for catheter-directed or low-dose systemic thrombolysis.
- About 25% of patients with low-risk PE can be treated as outpatients.

1. A deep vein thrombosis localized to which of the following veins can safely be treated without anticoagulation and early follow-up with a repeat venous ultrasound?
   a. Popliteal vein
   b. External iliac vein
   c. Anterior tibial vein
   d. Common femoral vein

Answer: C. See Figure 74.2. The popliteal, external iliac, common femoral, and deep femoral veins are all proximal veins. Proximal DVT should be treated with anticoagulation. DVT isolated to a calf vein like the anterior tibial vein can be observed, without anticoagulation, so long as a follow-up ultrasound can be performed within 7 days to look for propagation.

2. In which of the following patients would the D-dimer level likely be elevated (e.g., >500 ng/mL)?
   a. 52-year-old female with a history of breast cancer in remission
   b. 18-year-old male with an acute febrile illness
   c. 79-year-old male with a history of hypertension
   d. 24-year-old female with a history of obesity

Answer: C. See Box 74.2. Active cancer increases D-dimer levels, but inactive cancer (in remission) does not. Febrile illness, obesity, and long-haul travel do not increase D-dimer levels. While hypertension does not increase D-dimer levels, D-dimer levels do increase with advancing age. The specificity of the D-dimer test in a patient 79 years old is about 10% to 15%.

3. A 29-year-old woman, G2P1, 28 weeks pregnant presents with a brief episode of left-sided chest pain and vague cramping of her left leg. She agrees that she would like to pursue a workup for PE. Which combination of tests is the most appropriate first step?
   a. CTPA and bilateral venous ultrasound
   b. Ventilation/perfusion scan and echocardiogram
   c. D-dimer and CTPA
   d. D-dimer and bilateral venous ultrasound

Answer: D. While the radiation doses from both CTPA and ventilation perfusion scanning are lower than the minimum dose known to affect pregnancy outcomes, it is still best to minimize radiation exposure when possible. Therefore, a diagnostic testing strategy that starts with D-dimer, which can rule out PE if negative, and bilateral venous ultrasound, which can rule in VTE when positive, may obviate the need for imaging with ionizing radiation.

4. What is the preferred anticoagulant for a patient with low-risk PE?
   a. A direct acting oral anticoagulants (DOAC)
   b. A nonsteroidal anti-inflammatory drug (NSAIDs)
   c. Unfractionated heparin
   d. Warfarin

Answer: A. See Figure 74.11. For most patients, and especially for patients with low-risk PE, direct-acting oral anticoagulants (DOACs) are the treatment of choice. The DOACs apixaban and rivaroxaban are approved for the initial treatment of DVT and PE and are associated with lower rates of major hemorrhage than the combination of heparin and warfarin. Nonsteroidal antiinflammatory drug (NSAIDs) can be used to treat superficial thrombophlebitis, but not PE.

5. A 46-year-old woman presents to a tertiary-care hospital with syncope and hypotension. Her oxygen saturation is 88% on room air. Her systolic blood pressures are between 75 to 88 mm Hg during her 30 minutes in the ED. Her CTPA shows acute, bilateral main pulmonary artery PE and right ventricular dilatation. Three weeks ago, she had surgery to resect a glioma from the left cerebral hemisphere. The most appropriate next step is:
   a. 2 L normal saline bolus and endotracheal intubation.
   b. Consult cardiac surgery to consider surgical thromboembolectomy.
   c. Low-dose alteplase, 50 mg IV over two hours.
   d. Anticoagulation with rivaroxaban.

Answer: B. See Figure 74.11. This patient has high-risk, or "massive," PE. She has persistent hypotension and a large central clot with evidence of right ventricular dysfunction. She also has an absolute contraindication to thrombolysis having recently undergone resection of a brain tumor. Patients with right ventricular dilatation from PE may clinically decompensate with excessive IV fluid resuscitation and intubation. Fluid boluses, when given, should be small (250–500 mL). Intubation should be avoided if possible. DOACs like rivaroxaban have not been extensively studied in high-risk PE and, because of their long half-life, should not be used in patients who may undergo thrombolysis or invasive treatment for their PE. Because this patient is not a candidate for thrombolysis, the best first step is to consult a cardiac surgeon to consider open thromboembolectomy.

KEY CONCEPTS

Dysphagia

- Dysphagia can be caused by obstructive lesions (e.g., esophageal neoplasm), motility disorders (e.g., achalasia), or neuromuscular disorders that can be vascular (e.g., cerebral vascular accident), immunologic (e.g., myasthenia gravis, multiple sclerosis [MS]), infectious (e.g., botulism), or metabolic in nature.
- The incidence of achalasia increases with age, presenting insidiously with equal frequency for solids or liquids.
- Dysphasia can be the initial presentation of myasthenia gravis.
- Treatment of dysphagia is directed toward the underlying cause (e.g., myasthenia gravis, MS).
- An outpatient barium swallow or upper gastrointestinal endoscopy is indicated for most patients with dysphagia.

Upper Gastrointestinal Foreign Bodies

- Structural abnormalities of the esophagus are a major risk factor for foreign body obstruction; thus patients who obtain relief of foreign body sensation symptoms should be referred to a gastroenterologist for follow-up evaluation.
- Immediate intervention is indicated for button batteries, food boluses causing high-grade obstruction, or patients in significant distress (e.g., vomiting, gagging, choking, stridor, or inability to tolerate oral intake).
- Urgent (<24 h, and ideally <12 h) intervention is indicated for low-grade obstructions caused by sharp objects, coins lodged in the proximal esophagus, or food boluses.
- Urgent intervention is also recommended for gastric foreign bodies wider than 2.5 cm or longer than 5 cm.
- Flexible endoscopy with procedural sedation is the preferred therapeutic intervention to remove most proximal foreign bodies that can be reached by the scope.

Esophageal Perforation

- Iatrogenic causes, such as a complication of endoscopy, remain the most common cause of esophageal perforation.
- Water-soluble contrast should be used for initial diagnostic imaging studies when esophageal perforation is suspected.
- Admission with broad-spectrum antibiotics (e.g., intravenous vancomycin 15 mg/kg plus piperacillin 3.375 g) and early surgical consultation should occur in the majority of cases of esophageal perforation.
- Select stable patients with small, contained esophageal perforation may be managed conservatively by keeping the patient nothing per os (NPO), by administering broad-spectrum antibiotics and parenteral nutrition, and with a surgeon readily available.

Esophagitis

- Gastroesophageal reflux disease (GERD) is a diagnosis of exclusion in patients who present with chest pain. It is critical to first rule out other diagnoses, such as acute coronary syndrome (ACS).
- Empirical treatment of GERD with lifestyle modifications, H2 antihistamines, or proton pump inhibitors (PPIs) is appropriate, but if there is no improvement with these measures, patients should be referred for further evaluation.
- Sucralfate (1 g qid) can be safely used in pregnant patients with GERD.
- Eosinophilic esophagitis commonly presents as solid food dysphagia. Once food impaction is ruled out, a PPI should be initiated and the patient referred to a gastroenterologist.
- Infectious esophagitis primarily occurs in immunocompromised patients, and endoscopy may be helpful in differentiating among causal agents.
- Patients with pill esophagitis typically present with sudden onset retrosternal pain and odynophagia in the setting of taking medications without water; the diagnosis can often be made on the history alone.
- Medications associated with pill esophagitis include doxycycline, aspirin, nonsteroidal antiinflammatory drugs (NSAIDs), and potassium chloride.

Gastritis and Peptic Ulcer Disease

- Although gastritis cannot be definitively diagnosed based on clinical features alone, a clinical history such as recent NSAID use or alcohol ingestion in the setting of classic symptoms supports a presumptive clinical diagnosis of gastritis.
- The most common cause of gastritis is *Helicobacter pylori* infection.
- First-line treatment of *H. pylori* infection is a PPI (e.g., omeprazole 20 mg bid), amoxicillin (1 g bid), and clarithromycin (500 mg bid) for 14 days.
- The most serious complications of peptic ulcer disease (PUD) include hemorrhage, perforation, penetration, and gastric outlet obstruction.

Gastric Volvulus

- Acute gastric volvulus often presents with the combination of severe epigastric pain, distention, and vomiting, followed by violent nonproductive retching.
- Volvulus has a very high morbidity and mortality. Nasogastric tube reduction can be attempted in the emergency department, but ultimately these patients need emergent surgical intervention.

1. Which of the follow drug regimens is appropriate first-line treatment for *Helicobacter pylori* infection?
   a. Bismuth subsalicylate, famotidine, and clarithromycin
   b. Metronidazole and sucralfate
   c. Omeprazole, amoxicillin, and clarithromycin
   d. Omeprazole and bismuth subsalicylate
   e. Famotidine, omeprazole, and amoxicillin

Answer: c. The recommended triple-treatment regimen for *H. pylori* infection is a proton pump inhibitor (PPI) (e.g., omeprazole, 20 mg bid), amoxicillin (1 g bid), and clarithromycin (500 mg bid) for 14 days. Quadruple therapy with Pepto Bismol, metronidazole, tetracycline, and a PPI is an alternative option. See Box 75.5.

2. What percentage of esophageal foreign bodies require a nonoperative intervention to facilitate removal?
   a. <5%
   b. 10% to 20%
   c. 50%
   d. 75%
   e. >90%

Answer: b. Most foreign bodies pass spontaneously. Approximately 10% to 20% require intervention, but less than 1% require surgery for removal.

3. A 5-year-old child is brought to the emergency department (ED) by his mother after a possible ingestion of a plastic Lego piece. He has had no pulmonary symptoms but reports difficulty swallowing and declines to drink liquids that are offered. What would be the intervention of choice?
   a. Contrast-enhanced CT scan of the chest
   b. Endoscopy
   c. Non–water-soluble barium swallow
   d. Posteroanterior and lateral chest radiography
   e. Water-soluble barium swallow

Answer: b. Because the patient is symptomatic, endoscopy is indicated. A CT scan of the chest is useful for organic and inorganic materials. A basic chest radiograph cannot reliably exclude a foreign body. Barium swallow is difficult in pediatric patients, particularly in a child who is refusing oral fluids. Water-soluble media administration risks development of pneumonitis if aspirated. Non–water-soluble materials risk increased inflammation if leakage occurs into the mediastinum. Barium may also obscure subsequent endoscopic visualization.

4. Which of the following statements regarding the use of glucagon in esophageal obstruction from a food bolus is true?
   a. Glucagon should be administered by the oral route.
   b. Glucagon has antiemetic properties.
   c. Glucagon can facilitate passage of a food bolus localized anywhere in the esophagus.
   d. Glucagon is relatively contraindicated with sharp-edged foreign bodies.
   e. The success rate of glucagon approaches 90% when used with a few hours of food bolus impaction.

Answer: d. Glucagon is a smooth muscle relaxant, so it is theoretically useful only for distal esophageal foreign bodies. There are only anecdotal reports of success with glucagon, but a literature review shows no statistical benefit of glucagon in treating esophageal foreign bodies. There are many adverse effects, including flushing, nausea, and vomiting, that can potentially increase the risk of aspiration. It is contraindicated for use with sharp or damaging foreign bodies. There is only low-level evidence to support the use of effervescent agents, and they are relatively contraindicated after 24 hours because of perforation concerns.

5. Which of the following is an indication for urgent endoscopy?
   a. Button battery in the stomach
   b. Chest pain due to foreign body
   c. Coin in the proximal esophagus
   d. Nausea and vomiting
   e. Object failing to pass out of the esophagus after 12 hours

Answer: c. A coin that remains lodged in the proximal esophagus should be removed. Other indications are inability to handle secretions, sharp objects, esophageal button battery (alkaline), and impactions that fail to pass after 24 hours.

6. Ulcers in which portion of the upper GI tract are most likely to perforate?
   a. Hypopharyngeal
   b. Esophageal
   c. Gastric—antrum
   d. Gastric—body
   e. Duodenal

Answer: e. Duodenal ulcers account for 60% of all perforations, followed by antral gastric ulcers (20%) and gastric body ulcers (20%). Various upper gastrointestinal (GI) symptoms may lead to esophageal ulcerations and severe odynophagia, although perforation is infrequent.

7. A 32-year-old otherwise healthy man presents with acute onset of epigastric pain radiating to his chest that woke him from sleep at 2 AM. It was a burning pain associated with water brash. There were no associated pulmonary symptoms. His past medical history is unremarkable except for tobacco use and heartburn. His electrocardiogram and upright chest radiograph are normal. Vital signs and physical examination findings are unremarkable. He is currently pain free. His troponin level is normal. What is the most appropriate intervention?
   a. Cardiology consultation for catheterization
   b. Contrast-enhanced CT scan of the chest
   c. Discharge on aspirin, 325 mg once daily
   d. Serial troponins
   e. Trial of twice-daily proton pump inhibitor therapy

Answer: e. Peak gastric acid secretion occurs during the early morning hours between 1 AM and 3 AM, with a typical scenario of causing awakening from sleep. The shared afferent neural pathway makes the pain of gastroesophageal reflux disease (GERD) similar to that of pain of cardiac origin. Gastric acid secretion is lowest at approximately 6 AM; thus awakening in the morning with pain from GERD is unusual.

8. A 45-year-old woman presents several hours after an upper endoscopy with severe chest pain and neck discomfort. She is awake and alert, but rates pain as "10 of 10." What is the most appropriate test to confirm the diagnosis?
   a. Abdominal x-ray
   b. Barium contrast esophagography
   c. Gastrografin (water-soluble) contrast esophagography
   d. Ultrasound
   e. Upper endoscopy

Answer: c. We recommend an initial attempt with a water-soluble agent in patients who are awake and alert and are not at risk for aspiration. Barium sulfate is superior for identifying small perforations; however, it may incite an inflammatory response in tissue and should be used only if no initial perforation is identified with water-soluble contrast. Endoscopy is generally not recommended except in cases of penetrating trauma because insufflation could potentially enlarge a minimal transmural opening. CT imaging is a reasonable alternative.

KEY CONCEPTS

Viral Hepatitis

- The clinical presentation of viral hepatitis is highly variable, and many cases may be asymptomatic, particularly in children.
- Hepatitis A is transmitted through oral fecal spread, whereas hepatitis B and C are spread through parenteral or intimate contact.
- Highly effective immunizations exist against hepatitis A and B viruses.
- Postexposure, passive immunization exists for hepatitis A and B viruses, though its use is primarily in the setting of hepatitis B exposure in nonimmunized individuals.
- Direct-acting antiviral regimens using nucleoside inhibitors have revolutionized hepatitis treatment.
- Viral hepatitis may be a reportable disease dependent on jurisdiction.

Alcoholic Hepatitis

- Alcohol-induced liver disease may progress from steatosis to steatohepatitis to cirrhosis, and finally to hepatocellular carcinoma. It is estimated that 8% to 20% of patients with steatosis will eventually progress to cirrhosis. The risk for hepatocellular carcinoma (HCC) in decompensated alcohol-induced cirrhosis approaches 1% per year.
- With cessation of alcohol intake, steatosis may reverse within 2 weeks.
- Laboratory tests may help distinguish alcoholic hepatitis from viral hepatitis, with the former associated with milder transaminase level elevations and a relative predominance of AST to ALT.
- Hypoglycemia is common in patients with alcoholic hepatitis secondary to caloric insufficiency, chronically depleted glycogen stores, and suppressed gluconeogenesis.
- Management of alcoholic hepatitis is guided by severity scoring; Maddrey discriminant function (MDF) is the most commonly utilized and incorporates coagulopathy and bilirubin levels.
- Patients with alcoholic hepatitis and a MDF over 32 should be treated with prednisone, 40 mg PO daily, or methylprednisolone, 32 mg IV daily.

Cirrhosis

- Patients with cirrhosis most often present with complications resulting from loss of hepatocytes, including ascites, variceal bleeding, hepatorenal syndrome, or hepatic encephalopathy.
- The targeted platelet count should generally be greater than 50,000/mm^3 prior to the performance of invasive procedures in patients with cirrhosis. Management of coagulopathy may need to precede major diagnostic or therapeutic interventions.
- An elevated INR or thrombocytopenia is not a contraindication to paracentesis in patients with cirrhosis.
- Albumin, 8 g per L of ascitic fluid removed, should be administered when volumes over 5 L are removed by paracentesis.
- Upper GI bleeding in patients with cirrhosis, often from esophageal varices, carries a 10% to 15% mortality.
- When treating liver-associated coagulopathies in a patient with active bleeding, cryoprecipitate, 1 unit/10 kg body weight, is preferred over fresh-frozen plasma.
- Angiotensin-converting enzyme inhibiting drugs or angiotensin receptor blocking drugs should be avoided in patients with decompensated cirrhosis or ascites because plasma drug concentrations may increase, renal clearance may be markedly decreased, leading to impaired renal function.
- In patients with cirrhosis, mean arterial pressure (MAP) is an independent predictor of mortality.
- Hepatorenal syndrome is heralded by an increasing creatinine level in the setting of liver failure, and is associated with a high rate of mortality.

Hepatic Encephalopathy

- Hepatic encephalopathy is a state of cerebral and neuromuscular dysfunction secondary to an increased ammonia level and its effect on cerebral metabolism. However, the severity of hepatic encephalopathy does not directly correlate with the measured serum ammonia level.
- Underlying conditions that can precipitate encephalopathy in patients with cirrhosis include GI bleeding, hypokalemia, infection, and dehydration.
- Management of hepatic encephalopathy includes correction of underlying electrolyte abnormalities, dietary guidance, administration of lactulose (30–60 g/day), and rifaximin (400 mg PO every 8 hours).
- In addition to lactulose and rifaximin, infusion of branched chain amino acids has shown benefit without increased mortality in patients with hepatic encephalopathy.

Spontaneous Bacterial Peritonitis

- Spontaneous bacterial peritonitis (SBP) is an acute infection of ascitic fluid that should be considered in any patient with ascites and abdominal pain, fever, or unexplained clinical deterioration.
- SBP is most commonly caused by *E. coli* and *Klebsiella*.
- The diagnosis of SBP is dependent on obtaining ascitic fluid for analysis.
- An ascitic fluid granulocyte count greater than 250 cells/mm^3 (100 cells/mm^3 in peritoneal dialysis patients) is generally an indication for antibiotic treatment.
- Treatment of SBP includes cefotaxime 2 g IV every 8 hours for 5 days.

Hepatic Abscesses

- Abscesses may be amoebic or pyogenic in nature. Abdominal ultrasound or CT are the diagnostic imaging modalities of choice.
- Chest radiography may reveal a right lower lobe pleural effusion
- Imaging does not distinguish pyogenic from amebic abscesses.
- Treatment is initiated prior to abscess drainage.

1. Which of the following statements regarding hepatitis A is true?
   - **a.** Fecal shedding and highest infectivity coincide with symptomatic disease.
   - **b.** In the US, approximately 20% of urban-dwelling adults are seropositive.
   - **c.** Occult disease is more common in children than in adults.
   - **d.** The incidence is consistent across ethnic groups.
   - **e.** The most common risk factor for children is travel.

Answer: c. Children are more likely to have occult disease (up to 70%). Adult seropositive rates approach 50% among urban-dwelling adults. The incidence varies widely across ethnic groups. In areas of pediatric vaccinations, increasing adult cases are seen among intravenous drug users (IVDUs) and males who have sex with males. The stage of highest infectivity precedes symptoms.

2. Which of the following statements concerning hepatitis D infection is true?
   - **a.** Hepatitis D is spread primarily via the fecal-oral route.
   - **b.** Infection with hepatitis D is an independent event with a course nearly identical to that of hepatitis A.
   - **c.** It is common to see aspartate aminotransferase (AST) level elevations far in excess of alanine aminotransferase (ALT) level elevations.
   - **d.** Many cases are misdiagnosed as acute or reactivated hepatitis B.
   - **e.** Unconjugated bilirubin levels are 2 or 3 times higher than conjugated levels.

Answer: d. Hepatitis D virus infection can only occur with (coinfection) or after (superinfection) hepatitis B infection. It is spread via the parenteral route, such as by IV drug use. Many cases are misdiagnosed as acute or reactivated hepatitis B because HBV markers will be positive. There are no unique biochemical or laboratory patterns for any of the viral hepatitis infections. Hepatitis D appears to have a direct cytotoxic potential compared with other viral causes where the host immunologic response primarily leads to hepatopathy.

3. A 26-year-old man presents with complaints of pruritus and a raised rash for 7 days. The rash has been associated with nausea and painful symmetrical swelling of both wrists and metacarpophalangeal joints. He has no past medical history and takes no medications. He works in a retail store. Vital signs are normal, and the physical examination is remarkable for right upper quadrant tenderness, bilateral mild wrist effusion with minimal warmth and no erythema, and diffuse skin urticaria. The remainder of the examination is negative. Blood count, chemistries, and liver studies are remarkable for WBC of 11,800 cells/mm^3, AST of 212 IU/L, ALT of 395 IU/L, normal alkaline phosphatase level, and total bilirubin of 2.3 mg/dL. Which of the following diagnostic modalities would be most likely to yield the causative diagnosis?
   - **a.** C-reactive protein level
   - **b.** Hepatitis A antigen
   - **c.** Hepatitis B surface antigen
   - **d.** Joint aspiration
   - **e.** Liver ultrasound

Answer: c. A small number of patients with hepatitis B develop a prodrome of arthralgias and arthritis (symmetric small joints) and dermatitis. The dermatitis is typically urticarial but may be macular, popular, or petechial.

4. Scleral icterus becomes clinically apparent at approximately which serum bilirubin level?
   - **a.** 2 mg/dL
   - **b.** 2.5 mg/dL
   - **c.** 3 mg/dL
   - **d.** 3.5 mg/dL
   - **e.** 4 mg/dL

Answer: b. Icterus is often first noted in the sublingual or subungual areas when the serum bilirubin level is above approximately 2.5 mg/dL.

5. Which of the following statements regarding the typical laboratory profile for a patient with acute viral hepatitis is true?
   - **a.** AST is generally elevated in excess of ALT.
   - **b.** Direct and indirect bilirubin levels are elevated in almost equal proportions.
   - **c.** Lactate dehydrogenase (LDH) levels are almost always normal.
   - **d.** The alkaline phosphatase level is generally elevated 5 to 10 times normal.
   - **e.** The WBC always shows a marked polymorphonuclear leukocytosis.

Answer: b. The alkaline phosphatase level is rarely elevated more than 2 or 3 times normal, and LDH levels are modestly elevated. The WBC count may range from low, with lymphocytic predominance, to a polymorphonuclear (PMN)-predominant leukocytosis. ALT is almost always elevated in excess of AST in the setting of acute viral hepatitis.

6. A 26-year-old woman returns for follow-up after initial evaluation for possible acute hepatitis. Her hepatitis panel has returned with the following results:

Hepatitis A IgM	Negative
Hepatitis A IgG	Negative
Hepatitis B surface antigen	Positive
Hepatitis B surface antigen IgG	Negative
Hepatitis B core antigen IgM	Positive
Hepatitis B core antigen IgG	Negative
Hepatitis C antigen	Negative

Which of the following is the most appropriate diagnosis?
   - **a.** Acute hepatitis A
   - **b.** Acute hepatitis B
   - **c.** Immunity to hepatitis B
   - **d.** Previous hepatitis A
   - **e.** Previous hepatitis B

Answer: b. Acute hepatitis A is characterized by IgM to hepatitis A. Prior infection is determined by IgG antibody. Acute hepatitis B is characterized by the presence of surface antigen and IgM antibody to core antigen. Surface antigen alone may be absent late in the course of the disease or may present chronically unrelated to the current episode. IgG to the core antigen indicates previous infection. IgG to the surface antigen is the best marker for immunity.

7. A 39-year-old man presents with a 4-day history of abdominal pain and nausea. He has no significant past history and takes no medications. Vital signs include a temperature of 37.7°C (99.9°F) oral, pulse of 98 beats/min, respiratory rate of 20 breaths/min, and a blood pressure of 119/68 mm Hg. The physical examination reveals scleral icterus, a normal cardiopulmonary examination, moderate right upper quadrant tenderness without rebound, and guaiac-negative stool. Laboratory assessment reveals the following:

Total bilirubin	9.8 mg/dL
Conjugated bilirubin	4.6 mg/dL
Unconjugated bilirubin	5.2 mg/dL
AST	5300 IU/L
ALT	8400 IU/L
Alkaline phosphatase	750 IU/L
Albumin	3.9 mg/dL
INR	1.2
Hematocrit	42%
Platelet count	396,000/mm^3
WBC	9900/mm^3
Blood urea nitrogen (BUN)	53 mg/dL
Creatinine	0.9 mg/dL

Which of the following courses of action is most appropriate?

a. Admission for observation and GI consultation
b. CT scan of the abdomen with contrast
c. Gastrointestinal (GI) referral for interferon therapy
d. Reassurance
e. Tapering course of corticosteroids

Answer: a. Altered sensorium and prolongation of the PT beyond 5 seconds or INR beyond 1.5 suggests fulminant hepatic failure. Similarly, an unexplained elevation of the BUN or creatinine level may portend hepatorenal syndrome, which is associated with significant morbidity. The laboratory derangements warrant admission or transfer for hydration, close observation, and gastroenterology evaluation. Interferon has demonstrated some success in symptomatic hepatitis B patients but does not affect the early course. There is no role for corticosteroids.

8. The risk of liver injury in men increases as daily consumption of alcohol exceeds which of the following?

a. 10 g
b. 20 g
c. 40 g
d. 60 g
e. 80 g

Answer: e. 80 g of alcohol consumption is equivalent to a six-pack of beer, four to six glasses of wine, or three or four mixed drinks. For women, the risk increases with daily consumption of more than 20 g of alcohol.

9. A 63-year-old man with history of alcohol abuse presents with altered mental status. His family reports 3 days of decreasing ambulation and increasingly nonsensical conversation. He has no other known past medical history and takes no medications. Vital signs are unremarkable. Physical examination reveals a thin, unkempt man who is oriented to person only but is cooperative and follows commands. He falls asleep easily. There is no scleral icterus, and cardiopulmonary, abdominal, stool guaiac, and neurologic examinations are otherwise normal. A noncontrast CT scan of the head is negative for acute pathology. Pertinent laboratory findings are as follows:

Hematocrit (HCT)	34%
Hemoglobin	11.4 g/dL
Platelet count	108,000/mm^3
WBC	9300/mm^3
AST	148 IU/L
ALT	86 IU/L
Total bilirubin	2 mg/dL
Albumin	2.2 mg/dL
Alkaline phosphatase	158 IU/L
INR	1.8
BUN	38 mg/dL
Creatinine	2.1 mg/dL
Ethanol	0 mg/dL
Bicarbonate	30 mmol/L
Sodium	133 mEq/L
Potassium	3.6 mEq/L
Chloride	95 mEq/L

What is the most appropriate intervention?

a. Admission for lactulose, 30 to 60 g daily, titrated to modest diarrhea
b. Determination of further therapy, admission, and treatment based on serum ammonia levels
c. Discharge with the family; neomycin, 500 mg every 4 to 6 hours
d. Oral metronidazole, 250 mg PO every 6 hours
e. Oral vitamin K for 2 weeks

Answer: a. Ammonia accumulates in severe liver disease and crosses the blood-brain barrier to eventually form glutamine. Ammonia levels correlate poorly with encephalopathy. Lactulose is an osmotic cathartic that acidifies colonic contents, causing ammonia trapping. Neomycin is a poorly absorbed aminoglycoside that reduces colonic bacteria but is relatively contraindicated in cases of renal insufficiency. Therapies for hepatic encephalopathy that have been under clinical investigation include metronidazole, zinc, flumazenil, and eradication of *Helicobacter pylori*. Vitamin K would have modest benefit due to loss of hepatic synthetic abilities.

10. A 23-year-old G2P1 woman at 35 weeks of gestation presents with 3 days of fatigue, anorexia, nausea, and vomiting. She reports moderate epigastric and right upper quadrant pain. The physical examination is remarkable for icteric sclerae, slightly dry mucous membranes, and moderate tenderness in the right upper quadrant. She is afebrile and her uterus is not tender. Ultrasound shows a viable fetus at 34 weeks estimated gestation, with good cardiac activity, and liver and gallbladder ultrasound reveals no obvious gallstones or ductal dilations but moderate hepatomegaly. Laboratory analysis is remarkable for the following:

AST	1050 IU/L
ALT	1265 IU/L
Total bilirubin	9.9 mg/dL
Conjugated bilirubin	4.6 mg/dL
Unconjugated bilirubin	5.2 mg/dL
Alkaline phosphatase	328 mg/dL
Glucose	62 mg/dL
Hemoglobin	9.3 g/dL
Platelet count	105,000/mm^3
Serum electrolytes	Normal
Creatinine	0.6 mg/dL
Prothrombin time	14.8 sec
Albumin	3.1 g/dL

What is the most appropriate treatment?

- **a.** Clear liquids, antiemetics, and follow-up outpatient ultrasound in 48 hours
- **b.** Contrast CT scan of the abdomen
- **c.** Hydration, antiemetics, and discharge after symptom resolution
- **d.** Intensive care unit admission for monitoring for DIC
- **e.** Stabilization and urgent delivery

Answer: e. Acute liver failure of pregnancy typically presents in the later third trimester. Treatment involves aggressive fluid and electrolyte support, glucose administration, and immediate delivery. Liver disease generally resolves without sequelae. The illness is more common in primigravidas and twin gestations.

11. What is the most sensitive and specific imaging test for acute cholecystitis?

- **a.** Contrast CT scan
- **b.** Hepatobiliary nuclear scintigraphy with iminodiacetic acid (HIDA) scan
- **c.** Serum alkaline phosphatase level
- **d.** Serum bilirubin level
- **e.** Ultrasonography

Answer: b. Iminodiacetic acid administered IV is taken up by hepatocytes and secreted into bile canaliculi. Visualization of the gallbladder and common duct within 1 hour has a negative predictive value of 98%. Sensitivity of HIDA is diminished at bilirubin levels of 5 to 8 mg/dL.

KEY CONCEPTS

- Acute pancreatitis represents a wide spectrum of disease, ranging from mild to severe life-threatening disease with a mortality rate as high as 30%.
- The most common causes of acute pancreatitis are gallstones and chronic alcohol consumption.
- Acute pancreatitis is diagnosed by the presence two of three criteria—characteristic abdominal pain, serum lipase or amylase levels greater than three times the upper limit of normal, and characteristic findings on abdominal imaging.
- Serum lipase level is preferred over the amylase level because of its greater sensitivity and specificity in diagnosing acute pancreatitis.
- Computed tomography (CT) scan is not routinely recommended in the diagnosis of acute pancreatitis. It should be used in cases of diagnostic uncertainty and assessing for complications.
- Abdominal ultrasound should be performed to evaluate for a biliary etiology of pancreatitis.
- Treatment of acute pancreatitis is mainly supportive with fluid resuscitation and pain management. Lactated Ringers is preferred over normal saline because it is more physiologic and may provide antiinflammatory effects. There is no evidence to support one analgesic agent over another.
- Prophylactic antibiotics are not indicated in the management of acute pancreatitis but should be used in cases of infected pancreatic necrosis or other clear evidence suggesting sepsis or infection.
- Endoscopic retrograde cholangiopancreatography (ERCP) is only indicated in cases of cholangitis or biliary obstruction.
- Most patients with pancreatitis require hospitalization for symptomatic control, monitoring of hydration and nutrition status, and management of complications.
- There are several scoring systems to aid in predicting severity and outcomes in pancreatitis, including Ranson criteria, Acute Physiology and Chronic Health Evaluation II (APACHE II), CT severity index (CTSI), and Bedside Index of Severity in Acute Pancreatitis (BISAP). They are similar in their predictive accuracy, and each has different strengths and weaknesses.
- Chronic pancreatitis is a progressive fibroinflammatory syndrome which impairs both exocrine and endocrine pancreatic function.
- Pancreatic cancer is the seventh most common cause of death from cancer globally with a 5-year survival rate of only 7%.
- Surgical treatment may improve survival in patients whose pancreatic cancer is diagnosed early without metastasis. Most patients have advanced disease at diagnosis.

1. Which of the following statements regarding acute pancreatitis is true?
   a. Esophageal ultrasound is the diagnostic imaging of modality of choice for the diagnosis of acute pancreatitis.
   b. Hypertriglyceridemia is the likely etiology of acute pancreatitis in those with serum triglyceride levels greater than 1000 mg/dL.
   c. The most common cause of acute pancreatitis is alcohol abuse.
   d. Acute pancreatitis is the most lethal pancreatic disease globally.
   e. Pancreatic pseudocysts form within the first week of symptom onset.

Answer: b. Serum triglyceride levels greater than 1000 mg/dL with no other clear causes of acute pancreatitis points to hypertriglyceridemia as the etiology. Imaging is not routinely recommended in the diagnosis of acute pancreatitis. CT is the diagnostic modality of choice; endoscopic ultrasound (EUS) is invasive and not recommended as a first line imaging test. Although acute pancreatitis is the most common pancreatic disease globally, pancreatic cancer carries the highest mortality. Pancreatic pseudocysts generally form after 4 weeks from symptom onset.

2. Which of the following statements about serum lipase and amylase levels is true?
   a. Lipase is an enzyme which aids in the digestion of carbohydrates.
   b. In acute pancreatitis, amylase levels rise in 1 or 2 hours and normalize in 2 days.
   c. Higher lipase and amylase levels are correlated with more severe disease.
   d. Amylase is a less specific test than the serum lipase level for the diagnosis of pancreatitis.
   e. Testing for both lipase and amylase enzymes improves the diagnostic sensitivity and specificity for acute pancreatitis.

Answer: d. Lipase is more specific than amylase in the diagnosis of acute pancreatitis as amylase is produced by multiple organs other than the pancreas and lipase is produced predominantly by the pancreas. Lipase aids in the breakdown of dietary triglycerides into free fatty acids while amylase aids in the breakdown of carbohydrates. Levels rise within 6 to 24 hours and normalize in 3 to 7 days. Higher enzyme levels have not been found to correlate with disease severity. Testing for both enzymes has not been found to improve diagnostic sensitivity and specificity in the diagnosis of acute pancreatitis.

3. Which of the following statements regarding the use of radiographic studies for the evaluation of pancreatitis is true?
   a. Computed tomography (CT) is indicated in the evaluation of all cases of pancreatitis.
   b. Oral administration of a contrast agent for abdominal CT may aggravate pancreatitis.
   c. The initial study of choice in suspected gallstone pancreatitis is ultrasonography.
   d. CT and magnetic resonance imaging (MRI) of the abdomen are equally accurate for visualizing the biliary tract.
   e. Ultrasonography may help to differentiate pancreatitis from pancreatic pseudocyst.

Answer: c. Ultrasonography has a sensitivity of 94% for gallstones further imaging may be necessary to rule out biliary tract pathology. CT is not routinely recommended in the diagnosis of acute pancreatitis but is recommended in cases of diagnostic uncertainty, to rule out other diagnoses, and to assess for complications in patients who worsen or fail to improve. Oral contrast material does not aggravate pancreatitis. A non–contrast-enhanced helical scan may also be helpful if an oral contrast agent cannot be tolerated. MRI is superior to CT in visualizing the biliary tract although often more costly and less accessible. Ultrasonography is generally not helpful in assessing for pseudocysts.

4. A 28-year-old woman presents with recurrent pancreatitis. She is otherwise healthy and takes no medications. This episode of pain was preceded by several similar episodes of intermittent epigastric pain that lasted several hours at a time. A previous ultrasound examination of the liver, gallbladder, and pancreas was normal. She does not smoke, drink, or use over-the-counter medications. The physical examination is remarkable for moderate epigastric tenderness without rebound. Vital signs are normal. Laboratory evaluation reveals an elevated lipase level and moderate leukocytosis. Urinalysis and urine pregnancy test results are normal. What would be the most appropriate intervention?
   a. After stabilization, referral to a gastroenterologist for endoscopic retrograde cholangiopancreatography (ERCP)
   b. After stabilization, referral to a gastroenterologist for upper endoscopy
   c. CT scan of the abdomen to rule out pancreatic pseudocyst
   d. Repeated ultrasound examination of the liver, gallbladder, and pancreas
   e. Symptomatic treatment only unless her clinical picture worsens

Answer: a. Many cases of presumed idiopathic pancreatitis may be due to small stones or sludge that cannot be seen by ultrasound examination but may be seen by ERCP. Pancreatic pseudocyst is more likely in alcoholic pancreatitis and typically occurs gradually, several months after a severe episode.

5. A 68-year-old male presents with several months of epigastric pain, decreased appetite, weight loss, and weakness. Which of the following statements is true?
   a. CT scan will rule out the diagnosis of pancreatic cancer.
   b. His pancreatic adenocarcinoma is most likely to be confined to the pancreas.
   c. Following the diagnosis of pancreatic adenocarcinoma, chemoradiation will be curative.
   d. Chronic pancreatitis is a risk factor for developing pancreatic cancer.
   e. Those residing in developing countries are at higher risk for pancreatic cancer.

Answer: d. Those with chronic pancreatitis are at higher risk of developing pancreatic cancer. Abdominal CT is the initial imaging modality of choice for the diagnosis of pancreatic cancer; however, smaller tumors (<10 mm) are better detected on EUS. Less than 20% of patients are diagnosed with localized and resectable disease. Chemoradiation is generally focused on slowing tumor progression and palliative therapy but is unlikely to be curative without resection. The incidence of pancreatic cancer is higher in developed countries.

KEY CONCEPTS

- Small bowel obstruction (SBO) is a common clinical condition accounting for 2% of patients presenting to the emergency department with abdominal pain, leading to an estimated 300,000 hospitalizations annually in the United States.
- The most common cause of SBO is adhesions from prior abdominal surgery, which are found in approximately 60% of cases. Tumors and abdominal hernias account for another 20% and 10% of cases, respectively.
- The most common presenting symptoms of SBO include colicky abdominal pain, abdominal distention, nausea, or vomiting. However, no single component of the history or physical exam can reliably predict SBO.
- Initial evaluation should focus on identifying SBO patients in need of prompt surgical evaluation including findings concerning for peritonitis, bowel strangulation, or ischemia.
- For patients presenting with SBO likely due to adhesions, most can be successfully managed conservatively with bowel rest, pain control, intravenous fluid, and electrolyte replacement.

1. What is the most common cause of small bowel obstruction in the developed world?

a. Gallstone ileus
b. Hernias
c. Intussusception
d. Postoperative adhesions
e. Tumors

Answer: d. In the developed world, postoperative adhesions account for approximately 60% of cases of small bowel obstruction. Patients with a history of intestinal or pelvic surgery are at the highest risk.

2. Which of the following patients is at the highest risk of developing an obturator hernia?

a. 2-year-old boy with no known medical problems
b. 45-year-old woman with a 1-year history of hysterectomy
c. 67-year-old man with a history of metastatic prostate cancer
d. 80-year-old woman with a 3-month history of rapid weight loss

Answer: d. This type of hernia is especially common in older women who have recently lost a significant amount of weight. The female pelvis is wider, and the obturator canal is more oblique in women. This, in combination with a loss of preperitoneal fat, predisposes to its development. Because an external mass is absent, diagnosis is especially challenging and explains why it carries the highest mortality of any abdominal hernia – nearly 70% when incarcerated.

3. A 55-year-old woman with a history of Roux-en-Y gastric bypass surgery presents with one day of worsening colicky abdominal pain and vomiting. A CT scan reveals an internal hernia. What is the most appropriate disposition?

a. Administer broad-spectrum antibiotics and admit to the medicine floor.
b. Arrange for barium swallow with small bowel follow-through.
c. Insert nasogastric tube and admit to the medicine floor.
d. Prompt surgical consultation and preparation for surgery.

Answer: d. Although rare in the general population, internal hernias are a recognized complication of bariatric surgery, especially when a Roux-en-Y type procedure has been performed. Because of the closed loop nature of an internal hernia, they are not suitable for conservative management and require surgical intervention.

4. What is the length of time from acute ischemia of the intestines to completion of transmural necrosis?

a. 15 minutes
b. 60 minutes
c. 2 hours
d. 6 hours
e. 24 hours

Answer: d. Although the mesenteric circulation is able to adapt to variations in circulation, the small bowel is quickly injured after acute ischemia. Within 15 minutes, structural damage to the intestinal villi can be seen histologically. If blood flow is not restored, mucosal sloughing can start to occur within 3 hours and, by 6 hours, transmural necrosis is complete.

5. A 35-year-old woman taking oral contraceptive pills presents with 2 days of progressively worsening diffuse abdominal pain without peritoneal findings on examination. A CT scan reveals mesenteric venous thrombosis. What is the next most appropriate step?

a. Arrange for formal mesenteric venous angiography to confirm and treat.
b. Arrange for immediate exploratory laparotomy regardless of current clinical status, given the high risk of severe complications.
c. Discharge home because this will resolve without intervention.
d. Institute pain control and admit to the floor.
e. Start anticoagulation with therapeutic dosing of heparin.

Answer: e. The treatment of mesenteric venous thrombosis is unique in that in the absence of peritoneal findings, initial treatment with heparin alone may be adequate. In the vast majority of cases (>75%) an underlying inherited or acquired hypercoagulable state can be identified. Oral contraceptive use accounts for 9% to 18% of cases in young women.

KEY CONCEPT

- Appendicitis is a progressive illness caused by appendiceal luminal distention followed by appendiceal wall ischemia, transmural inflammation, and eventual perforation, with resulting peritonitis.
- Clinical history, physical examination, and laboratory findings need to be combined to formulate a comprehensive assessment. No one finding can definitively diagnose or exclude appendicitis.
- The most useful historical features in evaluating appendicitis are right lower quadrant (RLQ) pain, pain preceding vomiting, and migration of pain to the RLQ.
- The most useful physical findings in evaluating appendicitis are RLQ tenderness or rigidity.
- Cervical motion tenderness is not specific for pelvic pathology and is found in up to 28% of females with appendicitis.
- A rectal examination contributes little and should not be routinely performed in the evaluation of appendicitis.
- The white blood cell count alone is neither sensitive nor specific for appendicitis and offers little in the evaluation.
- When clinicians have a low pretest probability for appendicitis, the combination of a WBC count < 10,000/mm^3 and CRP < 8 mg/L support the exclusion of appendicitis as a likely diagnosis.
- A young, healthy adult patient with classic symptoms and signs of appendicitis should receive graded compression ultrasound as the initial imaging of choice, if available. If the ultrasound is negative or inconclusive and concern for appendicitis persists, the patient should receive a CT of the abdomen and pelvis with IV contrast, which is the most sensitive test for appendicitis. There is no need for oral or rectal contrast when ordering a CT for appendicitis.
- When there is a concern for acute appendicitis, pregnant women should first have graded compression ultrasound followed by MRI if the ultrasound is negative or nondiagnostic.
- Nonoperative management of acute appendicitis (IV antibiotics, admission) is gaining support. The patient should not have high-risk features on imaging (e.g., presence of a fecalith, abscess, tumor, fluid collection, or appendiceal diameter > 1.1 cm) and should be made aware of the risks of treatment failure and recurrent appendicitis, both of which generally require surgical removal of the appendix.
- Because of the risk of malignancy, guidelines recommend follow-up colonoscopy for patients > 40 years old treated with nonoperative management.
- Delays of 12 to 24 hours between appendicitis diagnosis and operative management are not associated with adverse outcomes. Many surgeons will perform an appendectomy the following morning if a patient presents in the evening or overnight hours. Early appendectomy should be considered in elderly patients, because up to 41% of patients 65 years or older develop perforation between diagnosis of appendicitis and surgery.

1. What percentage of women with acute appendicitis have accompanying cervical motion tenderness (CMT)?

a. <5%
b. 15%
c. 25%
d. 50%
e. 75%

Answer: c. Prior to the advent of routine imaging of the appendix, as many as 25% of women with acute appendicitis were initially misdiagnosed because of the presence of CMT.

2. Which of the following statements regarding ultrasonographic visualization of the appendix is true?

a. A compressible appendix is a positive finding.
b. An appendiceal diameter greater than 6 or 7 mm is a positive finding.
c. The sensitivity of ultrasound for appendicitis is 94% to 98%.
d. Ultrasonography has good reliability for detecting a retrocecal appendix.
e. Ultrasonography compares favorably with computed tomography (CT) scanning for the detection of appendicitis.

Answer: b. A noncompressible appendix with a diameter greater than 6 or 7 mm in a setting of clinical features of appendicitis is considered a positive finding. Ultrasound sensitivities are 75% to 90%. It is a less useful modality in the obese, those with peritoneal adhesions, or those with a retrocecal appendix. The sensitivity of CT is much higher than ultrasonography.

3. A 27-year-old G3P2 woman at 22 weeks of gestation presents with 2 days of right lower quadrant (RLQ) abdominal pain. It began midline and later became more pronounced in the RLQ. The physical examination is remarkable for RLQ tenderness without rebound. The gynecologic examination is negative except for a nontender gravid uterus, with good fetal movement by transabdominal ultrasound. Urinalysis shows 8 to 10 white blood cells (WBCs)/high-power field (HPF) and occasional bacteria. Complete blood count (CBC) shows a WBC count of 12,700/mm^3 with 77% neutrophils. Hemoglobin level is 11 g/dL. RLQ ultrasound examination is limited, without visualization of the appendix or secondary signs of appendicitis. Transvaginal ultrasound does not show a gynecologic or obstetric problem. Repeat examination shows continued RLQ tenderness. What is the most appropriate intervention at this point?

a. Administer cephalexin for urinary tract infection and schedule a 48-hour clinic recheck
b. Admit for observation and serial examination
c. Obtain surgical consultation for laparotomy
d. Order a CT scan of the abdomen.
e. Order a magnetic resonance imaging (MRI) scan

Answer: e. MRI for appendicitis is helpful in pregnant women, and radiation exposure should be minimized. Exploratory surgery carries significant risk of premature labor and fetal loss.

4. In young patients with classic symptoms and signs of appendicitis, what is the most appropriate initial intervention?

a. Antibiotics and serial abdominal examinations
b. CT scan of the abdomen
c. MRI scan of the abdomen
d. Surgery
e. Ultrasonography

Answer: e. Ultrasound is the most appropriate initial intervention because it uses no radiation and can often visualize and diagnose appendicitis without significant delay. It has become less common for a patient with a history and examination concerning for appendicitis to undergo surgery without imaging, and the use of advanced imaging has dramatically decreased the negative appendectomy rate. Graded compression ultrasound for appendicitis is specific but lacks the sensitivity of CT, so if clinical suspicion persists after a negative or nondiagnostic ultrasound, CT (or MRI in pregnant patients) is often the best next step.

KEY CONCEPTS

- Gastroenteritis is usually self-limited and requires supportive care only. Routine laboratory testing or stool cultures are not indicated for most patients.
- Patients with gastroenteritis associated with fever, dysentery, bloody stools, severe dehydration, sepsis, a suspicion for *Clostridium difficile,* or an immunocompromised state should generally have a complete blood count, electrolytes, and stool cultures evaluated.
- Caution should be used in the care of the very young and elders with gastroenteritis, because these populations have the highest morbidity and mortality.
- Evaluate all individuals with gastroenteritis for dehydration. Use of oral rehydration therapy (ORT) with hypotonic oral rehydration solution (e.g., World Health Organization [WHO]-modified 245 mOsm ORT solution) is recommended.
- The use of antiemetic medications in patients with gastroenteritis may help with ORT. Ondansetron 0.15 mg/kg up to 8 mg given orally (PO) or intravenously is safe and cost effective.
- Early continuation of normal feeding is advised, and continuation of breastfeeding in mothers with gastroenteritis is suggested, if possible.
- *Campylobacter* and *Salmonella* are the top two causes of culture-proven bacterial enteritis in developed countries. Consult health authorities for local or regional outbreaks and pathogen prevalence.
- Norovirus, previously referred to as the Norwalk-like virus, is the most common cause of acute gastroenteritis in children and adults.
- Bacterial pathogens are responsible for approximately 80% of all cases of traveler's diarrhea, which is usually self-limited. Enterotoxigenic *Escherichia coli* (ETEC) is the most common etiology.
- Antibiotic treatments can prolong the shedding of *Salmonella* non-*typhi* organisms and therefore are not indicated in most gastrointestinal infections.
- Avoid antibiotic treatment in patients with gastroenteritis caused by *E. coli* O157:H7 or Shiga toxin 2–producing organisms as treatment increases the incidence of hemolytic uremic syndrome (HUS) or thrombotic thrombocytopenic purpura (TTP).
- In patients with fever or bloody stools, if antimotility agents are to be given, they should be administered in combination with an antibiotic as these agents may increase the contact time of the toxin or invasive infectious organism within the intestinal mucosa.
- Classic food poisoning usually manifests 1–6 h post ingestion of preformed toxins from bacterial organisms such as *Staphylococcus*, *Bacillus cereus*, or *Clostridium perfringens*. Food poisoning is generally short-lived (24 h), and the treatment is typically supportive care only.
- Risk factors for *C. difficile* colitis include recent antibiotic use (1–4 weeks), recent hospitalization, living in a long-term care facility, solid organ transplant recipients, or the use of antacids.
- Evaluation with abdominal computerized tomography (CT) for toxic megacolon and pseudomembranous colitis is recommended for patients with *C. difficile* colitis who are older, appear septic, have a tender distended abdomen, or have a high leukocytosis. Consider empirical antibiotic treatment with vancomycin 125 mg PO every 8 h for 10 days for suspected *C. difficile.*
- Scombroid poisoning results from eating spoiled dark meat fish; scombroid poisoning is not an allergic reaction but results from ingestion of histamine.
- In addition to the regular enteropathic bacterial pathogens, human immunodeficiency virus (HIV)-positive patients, particularly those with a CD4+ count less than 200/mm^3, have increased susceptibility to certain viruses and parasites, such as *Cytomegalovirus*, *Cyclospora, Cryptosporidium, Isospora, Mycobacterium avium-intracellulare* complex, and *Giardia.*
- Consider *Giardia* for diarrhea lasting more than 2 weeks, with foul-smelling stools and symptoms of flatulence, abdominal bloating, cramping, and recent exposure to contaminated river water.
- Hand hygiene with soap and water or hand sanitizers will contain the spread of most infectious agents in gastroenteritis.

1. After eating undercooked ground beef, a patient develops fever, abdominal pain, bloody diarrhea, and rectal tenesmus. Local health authorities are investigating a common source from a local diner, as 10 other patrons have presented with the same symptoms. Which of the following organism is likely to produce this dysentery outbreak?

a. Giardia
b. Norovirus
c. Shiga toxin–producing *Escherichia coli* (O157:H17)
d. *Staphylococcus aureus*

Answer: c. This patient is most likely infected with Shiga toxin–producing *E. coli* (ETEC) which often results from eating undercooked ground beef. Watery diarrhea that becomes bloody with significant abdominal pain is typical. The diarrhea may be grossly bloody and mimic inflammatory bowel disease or intestinal ischemia. Toxin assays are recommended. Endoscopic findings would be identical to those of any other severe colitis. Antibiotics are not effective and may increase the risk of hemolytic uremic syndrome (HUS) in children. Thrombotic thrombocytopenic purpura (TTP) is a risk in older children and adults. Norovirus typically causes watery non-bloody diarrhea without severe abdominal pain. *S. aureus* causes the classic food poisoning with abdominal cramping and vomiting 1 to 6 hours after ingestion the preformed toxin.

2. A patient from a resource-rich country has traveled to a resource-poor developing country. He develops diarrhea, up to four times a day, with yellow watery stools. There is no fever or abdominal pain. Which of the following organism is most commonly the etiology of Traveler's Diarrhea?
   a. Enterotoxigenic *Escherichia coli* (ETEC)
   b. *Giardia*
   c. *Shigella*
   d. *Vibrio* noncholera

Answer: a. ETEC is the most common cause of traveler's diarrhea, especially with travel to resource poor countries. It is usually self-limited, requiring antibiotic treatment only in severe cases. *Giardia* causes persistent diarrhea for more than 14 days. Giardiasis is commonly acquired from the ingestion of contaminated water, classically a camper who has ingested river water. The symptoms include diarrhea, abdominal cramping, and bloating, with foul-smelling stools. Shigellosis usually causes a more severe illness with dysentery.

3. A 5-year-old patient with a likely viral gastroenteritis for 3 days is determined to be dehydrated. What is optimum mode of rehydration therapy in this patient?
   a. High-glucose solution (e.g., apple juice)
   b. IV normal saline (0.9%)
   c. Oral hydration with clean water
   d. Reduced osmolarity (245 mOsmol/kg) oral rehydration solution

Answer: d. Fluids containing glucose and electrolytes provide optimal rehydration due to the cotransport of water across the intestinal lumen. Reduced osmolarity rehydration solution (e.g., WHO ORS 245 mOsm/kg) is associated with a reduced need for unscheduled IV infusions, lower stool volume, and less vomiting. High-glucose solutions have high osmolarity, and pure water has a low osmolarity; thus neither are optimal for water resorption. IV hydration is generally not indicated or cost effective in patients with mild or moderate dehydration.

4. A patient comes to the emergency department (ED) with a complaint of abdominal cramping and vomiting twice; the first time was the food ingested from a buffet 8 hours ago and the second time yellowish vomitus. There is no diarrhea or fever. There is continued retching intermittently in the ED. What is the likely causative organism?
   a. *Clostridium perfringens*
   b. *Escherichia coli* O157:H7—Shiga-producing toxin
   c. Norovirus
   d. *Salmonella enteritidus*

Answer: a. *Clostridium perfringens* is a common source of foodborne illness. Food cooked in advance, cooled, and rewarmed is the typical culprit. This is a toxin-mediated, diarrhea-predominant syndrome that is almost always self-limited. Ingestion of live organisms or spores is required, and antibiotics are not indicated. Symptoms usually appear within 6 to 12 hours, slightly longer than for staphylococcal food poisoning.

5. A patient presents with non-bloody diarrhea for 5 days, crampy abdominal pain, and bloating. The white blood cell (WBC) is 16,000/mL. Which of the following types of patient and scenario is most likely to put the patient at high risk for *Clostridium difficile*?
   a. A 3-year-old toddler in daycare
   b. Elderly nursing home patient
   c. Nurse who works in the emergency department
   d. Traveler on vacation returning from Mexico

Answer: b. Elderly nursing home patients have the highest risk for acquiring *C. difficile*. These patients are in close quarters and have multiple daily contacts with staff members, making person to person transmission more likely. Antibiotics and antacids are frequently utilized in these patients, which are risk factors for *C. difficile*. The *C. difficile* spores are highly resistant to heat, acid, and antibiotics, making them highly contagious for person to person or surface to person infections. The WBC is generally greater than 15,000/mL.

81 Large Intestine

KEY CONCEPTS

Irritable Bowel Syndrome

- Irritable bowel syndrome (IBS) is a chronic disorder that includes both abdominal pain and bloating and is either diarrhea-predominant, constipation-predominant, or a mixed picture.
- Treatment of IBS is challenging and commonly involves a combination of diet, pharmacological, and behavioral therapy, along with reassurance.

Diverticular Disease

- Diverticular disease can consist of diverticulosis, which can cause bleeding but is often asymptomatic, or inflammation of the involved diverticula, termed diverticulitis.
- Uncomplicated diverticulitis can typically be managed on an outpatient basis, with oral antibiotics if indicated. Complicated diverticulitis may require admission and intravenous antibiotics, often with surgery consultation.

Large Bowel Obstruction

- The most common cause of large bowel obstruction in the United States is colorectal malignancy. Treatment consists of endoscopic placement of an intraluminal stent with surgical resection at a later date.
- Acute colonic pseudo-obstruction commonly occurs in hospitalized elders. Treatment involves medical management, with endoscopic decompression if refractory.

Volvulus

- Sigmoid volvulus often affects elders or those that are in long-term care facilities and can usually be treated by endoscopic decompression.
- Cecal volvulus often affects younger patients and requires surgical management.

Intussusception

- Intussusception is the second most common cause of an acute abdomen in the pediatric population after appendicitis, is usually idiopathic, and may be successfully treated with hydrostatic or pneumatic reduction.
- In adults, intussusception is often associated with neoplasm or malignancy, typically requiring surgical management.

Inflammatory Bowel Disease

- Inflammatory bowel disease (IBD) is characterized by chronic, relapsing, and remitting inflammatory disease. Crohn disease's (CD) transmural inflammation can affect any area of the gastrointestinal (GI) tract, whereas ulcerative colitis (UC) inflammation is more superficial and is limited to the colon and rectum.
- Effective management of IBD requires prompt recognition and treatment of acute relapses, and appropriate choice and monitoring of medications for maintenance of remission. Most patients with CD respond to ileal-release budesonide or systemic steroids; aminosalicylates are less effective. In contrast, aminosalicylates are first-line therapies for UC, with steroids reserved for use only when necessary. Refractory exacerbations for both CD and UC may be treated with immunomodulators or biologic agents.
- Common complications of IBD include the formation of fistulae, strictures, or abscesses; less common yet more severe complications include fulminant colitis, toxic megacolon, or intestinal perforation.
- Chronic inflammation can have extraintestinal effects. Thromboembolic events affecting the venous and arterial systems in IBD patients may be underdiagnosed.

Colonic Ischemia

- Colonic ischemia (CI) typically presents in elders and is the most common ischemia disorder of the GI tract. Local hypoperfusion and reperfusion injury cause crampy abdominal pain over the segment of the colon involved, followed by a short course of bloody diarrhea.
- Although abdominal computed tomography (CT) is not diagnostic for CI, it is useful in supporting the clinical suspicion, assessing the extent of colon involvement, diagnosing complications, and excluding other disorders. Colonoscopy within 48 hours of symptom onset is the most accurate diagnostic study.
- CI is usually self-limited, and most patients experience resolution of symptoms with supportive medical management with bowel rest, hydration, and pain management. Those with more severe disease, especially those with right-sided colonic involvement, may develop peritoneal signs from transmural ischemia requiring antimicrobials and possible surgical intervention.

Stercoral Colitis

- Stercoral colitis is a rare complication of chronic constipation that is typically diagnosed on CT and is associated with high mortality. The condition primarily affects the elderly, individuals living in nursing homes, young patients with neurological impairment, and those with chronic opioid use.
- Fecal impaction leads to increased intraluminal and colonic wall pressure, which then causes inflammatory changes and pressure wall necrosis of the colon. It can be complicated by CI, stercoral ulcer formation, and subsequent perforation.

Radiation Proctocolitis

- Radiation proctocolitis occurs commonly in those that have received pelvic radiation. Acute radiation proctocolitis occurs during the course of treatment and is often self-limited. Chronic radiation proctocolitis can occur months after completion of treatment. Management typically varies based on severity and can range from stool softeners to hyperbaric therapy.

Neutropenic Enterocolitis

- Neutropenic enterocolitis, or typhlitis, occurs in those that have hematologic malignancies, are undergoing chemotherapeutic regimens, or are immunosuppressed for other reasons. It is characterized by neutropenia, fever, and abdominal pain. Treatment involves broad-spectrum antibiotics, bowel rest, and interventions to stimulate leukocyte count recovery.

1. A 59-year-old man with a past history of diverticular disease presents with his second episode of left lower quadrant (LLQ) abdominal pain. He is afebrile, and laboratory examination is remarkable for a leukocytosis of 13,800/mm^3. Physical examination reveals moderate LLQ tenderness without masses or rebound. A computed tomography (CT) scan of the abdomen reveals a small (4 cm) abscess adjacent to the sigmoid colon, with moderate diverticulitis. Which of the following would be the most appropriate treatment?
   a. Admission for intravenous antibiotics
   b. Confirmation with double-contrast barium enema
   c. Discharge on oral antibiotics with 2-day follow-up
   d. Radiology consultation for percutaneous drainage
   e. Surgical consultation for laparotomy

Answer: a. Abscesses smaller than 5 cm are typically treated with intravenous antibiotics followed by an outpatient oral regimen. Larger abscesses may be drained surgically or percutaneously.

2. What is the most common cause of large bowel obstruction in the US.?
   a. Adhesions
   b. Colon cancer
   c. Diverticulitis
   d. Intussusception
   e. Volvulus

Answer: b. Colon cancer accounts for 53% of cases. Volvulus and diverticulitis account for 17% and 12% of cases, respectively. Extrinsic compression from malignancy accounts for 6% of cases. Adhesions, unlike for small bowel obstructions, are a rare cause of large bowel obstructions.

3. Which of the following statements regarding intussusception is true?
   a. Bowel obstruction typically occurs.
   b. CT scans have a high sensitivity for detection of intussusception.
   c. Most adult cases involve the large bowel.
   d. Most adult cases require surgery.
   e. Most children have a causative lesion.

Answer: d. In contrast to children, most adult cases have a causative pathologic lesion, usually located within the small bowel. Complete obstruction occurs in less than 20% of cases. CT scans may miss as many as 50% of cases of intussusception. Surgery is usually required for adults with intussusception.

4. A 29-year-old woman presents with a 4-month history of intermittent abdominal pain with bloating and diarrhea. The diarrhea has been watery, nonbloody, and often nocturnal. Physical examination is remarkable for mild diffuse abdominal tenderness and brown, guaiac-positive stool. Rectal examination also demonstrates a small anal fissure at the 3-o'clock position. Laboratory evaluation is remarkable only for a normocytic anemia with a hemoglobin level of 11.5 g/dL. The diagnosis would most likely be confirmed by which of the following?
   a. Colonoscopy
   b. CT scan of the abdomen
   c. Erythrocyte sedimentation rate
   d. Mesenteric angiography
   e. Response to a high-fiber diet

Answer: a. This presentation is typical for inflammatory bowel disease. Nocturnal diarrhea, blood in the stool, and presence of an eccentric (nonposterior midline) anal fissure argue against IBS or benign diarrhea. Ischemic colitis is unlikely in this age group. Endoscopy with biopsy is the diagnostic intervention of choice.

5. Which of the following statements regarding colonic ischemia is true?
   a. CT scanning of the abdomen is diagnostic
   b. It is rarely associated with bloody stool.
   c. It is typically due to nonocclusive disease.
   d. Isolated right-sided ischemic colitis is associated with comparatively lower mortality.
   e. Specific serum biomarkers may be helpful.

Answer: c. Colonic ischemia is typically due to nonocclusive microvascular disease. It occurs due to low-flow conditions related to congestive heart failure (CHF), renal failure, hypovolemia, or recent illness or surgery. Bloody stools are predictable. It may also occur in younger patients with collagen vascular disease, hematologic abnormalities, or cocaine abuse. Isolated right-sided ischemic colitis is associated with comparatively higher mortality. There are no sensitive or specific laboratory tests, and although sometimes suggestive, CT scanning primarily rules out alternative processes.

KEY CONCEPTS

- Anorectal conditions can be approached using an algorithm that addresses the presence or absence of pain, bleeding, swelling, and pruritus, in combination with an assessment of the patient's overall health (see Fig. 82.2).
- Patients who seek treatment for nonspecific anorectal complaints should be evaluated for the presence of underlying systemic disease (e.g., cancer, diabetes mellitus, immunodeficiency) because disorders of the anorectum may be the initial presentation of associated conditions.
- Patients with a sexually transmitted disease (STD) should be evaluated for HIV infection, use of the anus for sexual purposes, and the possibility of domestic violence or abuse.
- Most anorectal conditions can be symptomatically improved by adherence to the WASH regimen (***w***arm water, ***a***nalgesics, ***s***tool softeners, ***h***igh-fiber diet).
- Thrombosed external hemorrhoids are covered by modified anoderm and may be excised and drained within 48 hours of formation. After 48 hours, the clot begins to dissolve, and conservative management with the WASH regimen is indicated.
- Internal hemorrhoids are covered with mucosa and should be referred to a colorectal surgeon for definitive management.
- Acutely thrombosed, gangrenous fourth-degree internal hemorrhoids require urgent surgical consultation.
- Fistulous tracts should not be probed.
- Pilonidal abscesses should be drained with needle aspiration or with a longitudinal incision off the midline.
- Pruritis ani is caused by a variety of conditions, including infection, hemorrhoids, topical irritants, cutaneous conditions, cancer, and hypersensitivity to foods and drugs (see Box 82.5).
- Sensitivity is required when managing patients with anorectal foreign bodies. Health care provider safety is imperative when evaluating foreign bodies with sharp edges.
- Distal anorectal foreign bodies often can often be removed in the ED, whereas proximal or sharp ones should be removed in the operating room.

1. Which of the following modifying factors is most likely to lead to the development of hemorrhoids as a result of portal hypertension?
- **a.** Alcoholism
- **b.** Obesity
- **c.** Pediatric age group
- **d.** Pregnancy
- **e.** HIV infection

Answer: c. Portal hypertension does not predispose to hemorrhoids, except in children. Bleeding in adults is likely from rectal varices. Most hemorrhoidal bleeding is from the superior rectal artery and thus is bright red. Approximately one-third of pregnant patients experience hemorrhoids in the third trimester or postpartum period. Traumatic deliveries can result in hemorrhoid development. Painless bleeding with defecation is the most common symptom (pain usually occurs if hemorrhoids are thrombosed).

2. What is the most common cause of sudden-onset rectal pain?
- **a.** Anal fissure
- **b.** Proctalgia fugax
- **c.** Sacral radiculopathy
- **d.** Thrombosed external hemorrhoid
- **e.** Thrombosed internal hemorrhoid

Answer: a. Anal fissures typically result from superficial tears in the anoderm, usually occurring in the posterior midline. The pain is heightened by secondary spasm of the anal sphincter.

3. Which of the following approaches to performing incision and drainage of a pilonidal cyst is most appropriate?
- **a.** Elliptic incision
- **b.** Horizontal incision at the center of the affected area
- **c.** Horizontal incision at the lower portion of the affected area
- **d.** Longitudinal incision along the midline
- **e.** Longitudinal incision lateral to the midline

Answer: e. Making a longitudinal incision lateral to the midline spares the area that is most vulnerable to reaccumulation of debris and the inflammatory response. Removal of pus, inflammatory tissue, and ingrown hairs relieves the pain of the abscess and promotes healing.

4. A 68-year-old woman presents with hemorrhoids. On examination, you note three protruding masses that are maroon in color and may be reduced manually into the anorectal opening. What is the most appropriate curative therapy?
- **a.** Application of a concentrated sucrose solution to the affected area
- **b.** Emergent surgical intervention (e.g., banding, sclerosing, hemorrhoidectomy)
- **c.** Prescription of a 7-day course of topical corticosteroid cream
- **d.** Referral to a surgeon for an outpatient procedure
- **e.** Removal of the hemorrhoid with an elliptic incision, including overlying tissue

Answer: d. Third-degree internal hemorrhoids may be manually reduced in the emergency department (ED) but are unlikely to heal spontaneously. Referral for operative therapy is curative. Excision of internal hemorrhoids is contraindicated. Acutely thrombosed external hemorrhoids may be excised in the ED. Temporizing measures include using the WASH regimen—using *w*arm water to encourage reduction of the protruded hemorrhoids, maintaining hygiene, *a*nalgesics, *s*tool softeners to ease passage of stool, and a *h*igh-fiber diet. Topical corticosteroids may be used for 1 or 2 days during an acute exacerbation, but their continued use promotes skin breakdown and itching. A sucrose solution may prove helpful in reducing procidentia (rectal prolapse). A thorough history should be obtained and a physical examination performed to learn if an underlying medical condition may be associated with the hemorrhoids.

83 Renal Failure

KEY CONCEPTS

- The causes of acute kidney injury (AKI) can be classified as prerenal, postrenal, and intrinsic renal disorders. Abrupt cessation of glomerular filtration typically results in a rise of the serum creatinine level of 1 to 2 mg/dL per day.
- Management of AKI is directed first at potentially lethal complications such as hyperkalemia or volume overload and then at reversal of the underlying cause of renal dysfunction. It is important to avoid any further hemodynamic or toxic insults to the kidneys.
- Patients with acute or chronic kidney disease have a limited ability to handle fluid and solute loads and have altered metabolism of many drugs. Therefore, the patient's impaired renal function must be considered when fluid is administered or drugs are prescribed.
- The most rapidly lethal complication of acute and chronic kidney disease is hyperkalemia.
- The most common problems with vascular access devices used for hemodialysis are thrombosis, hemorrhage, and infection. Access infection often presents as fever without an obvious source and, if suspected, appropriate IV antibiotics should be administered presumptively while awaiting blood culture results.
- Peritoneal dialysis–associated peritonitis typically presents with cloudiness of the peritoneal dialysis effluent. The diagnosis is made by a positive Gram stain or finding of more than 100 WBC/mm^3 in the effluent, with at least 50% polys. It is generally treated on an outpatient basis with intraperitoneal antibiotics self-administered by the patient.
- Chest pain in the dialysis patient should be presumed initially to be due to acute coronary syndrome, although other potentially serious causes may also be responsible. Serum troponin levels tend to be elevated in patients with poor renal function, but patients with myocardial infarction show the typical temporal pattern of rise and fall of troponin levels.
- Hypotension in patients with chronic kidney disease (CKD) may be caused by infection but may also be the result of rapid fluid removal during dialysis. This often responds readily to fluid administration. Pericardial tamponade is another cause of hypotension that should be considered for these patients.
- Altered mental status is most commonly due to causes similar to those seen in patients without renal disease but is sometimes the result of overrapid shifts in intravascular fluid and solutes during dialysis, termed *disequilibrium syndrome.*

1. Of the following choices, which is the correct match between type of urine cast and pathologic condition?
   a. Fatty casts—glomerulonephritis
   b. Granular casts—renal parenchymal infection
   c. Hyaline casts—dehydration, exercise
   d. Red cell casts—nephrotic syndrome
   e. White cell casts—acute tubular necrosis (ATN)

Answer: c. Hyaline casts are associated with dehydration, exercise, and glomerular proteinuria. The following are the other correct associations:

- Fatty casts—nephrotic syndrome
- Red cell casts—glomerulonephritis
- White cell casts—renal parenchymal infection
- Granular casts—ATN

2. A 34-year-old man with chronic kidney disease has missed his last two hemodialysis treatments and presents with generalized fatigue and dyspnea on exertion. Vital signs include a blood pressure of 165/94 mm Hg but are otherwise normal. Room air Sao_2 is 94% at rest. A portable chest x-ray shows vascular congestion. Which of the following is an indication for immediate dialysis?
   a. Blood urea nitrogen (BUN) level of 87 mg/dL and serum creatinine level of 10.6 mg/dL
   b. Serum calcium level of 6.8 mg/dL and serum albumin level of 3.0 g/dL
   c. Serum potassium level of 7.6 mEq/L (not hemolyzed)
   d. Serum sodium level of 124 mEq/L
   e. Serum troponin level of 0.08 ng/mL

Answer: c. Severe hyperkalemia is an indication for immediate dialysis in this patient. An ECG should be obtained stat to determine the degree of cardiotoxicity, and immediate medical treatment for hyperkalemia should be given while awaiting initiation of dialysis. Abnormalities of the serum sodium or calcium level can generally be treated without hemodialysis. A minimal troponin level elevation may be seen in patients with chronic kidney disease, but does not generally require specific treatment without other signs of cardiac ischemia; dialysis is not required.

3. What is the most common cause of nontraumatic hematuria?
   a. Carcinoma of the kidney or bladder
   b. Cystitis
   c. Kidney stones
   d. Prostatic hyperplasia
   e. Transfusion reaction

Answer: c. In descending frequency, the most common causes are kidney stones, lower urinary tract infection (UTI), benign prostatic hypertrophy (BPH), carcinoma of the kidney or bladder, urethritis, and glomerulonephritis.

4. A 53-year-old man presents with painless gross hematuria. His only past medical history is an aortic valve replacement for which he takes warfarin. The physical examination is nonfocal. Laboratory evaluation is remarkable for a normal chemistry panel and blood count, too numerous to count (TNTC) red blood cells (RBCs) on urinalysis, and international normalized ratio (INR) of 1.8. What is the next indicated step?
   a. Admission and observation
   b. Parenteral vitamin K
   c. Reassurance
   d. Renal imaging
   e. Withholding warfarin (Coumadin) for 3 days

Answer: d. When hematuria is associated with anticoagulant use, underlying disease can be identified in a significant proportion of patients.

5. A 33-year-old woman presents with mild nonfocal abdominal pain and subjective fever. Her past medical history is significant for hypertension-induced renal failure, for which she is on peritoneal dialysis (PD). The physical examination is remarkable for a temperature of 38°C (100.4°F), blood pressure 190/110 mm Hg, and mild nonfocal abdominal pain. Slightly cloudy peritoneal fluid is aspirated from the PD catheter and sent for analysis. Which of the following statements regarding this patient's condition is correct?
   a. A peritoneal fluid white blood cell (WBC) count >50/mm^3 is diagnostic.
   b. Intravenous antibiotics should be started empirically.
   c. Most cases are due to *Staphylococcus*.
   d. No organism is identified in 50% of cases.
   e. Polymicrobial infections suggest sample contamination.

Answer: c. Most PD-associated cases of peritonitis are due to *Staphylococcus aureus* or *Staphylococcus epidermidis*. No organism is identified in 20% of cases. A polymicrobial infection warrants GI evaluation for possible perforation or intra-abdominal abscess. A PD fluid count >100 cells/mm^3, with a neutrophil count >50% or positive Gram stain, is considered confirmatory. Treatment is typically with intraperitoneal antibiotics given for a 10- to 14-day course.

KEY CONCEPTS

- The emergency department (ED) diagnosis of sexually transmitted infections (STIs) is often based on clinical findings. Empirical antibiotic treatment is warranted to cover the most likely infecting organisms based upon history and physical examination findings. Rapidly available diagnostic tests (Gram stain, darkfield microscopy, wet mount microscopy, and other point-of-care tests) increase diagnostic sensitivity and specificity.
- Confirmatory diagnostic studies (PCR, culture, serology, and others) should be considered even when results are not immediately available. A mechanism for follow-up of test results should be established and appropriate patient contact information obtained.
- STIs frequently coexist. Diagnosis of one STI should prompt consideration and screening for others, including HIV.
- Genital herpes, the most common ulcerating STI, is often transmitted by persons who are unaware that they are infected or are asymptomatic at the time of transmission.
- In a patient with a genital ulcer, visualization of spirochetes on darkfield microscopy is highly specific for the diagnosis of syphilis and provides rapid confirmatory results.
- Single-dose antibiotic therapy should be used for treatment of STIs when possible. Directly observed therapy administered in the ED enhances treatment compliance. Single-dose antibiotic therapy is recommended for treatment of uncomplicated gonorrhea, primary and secondary syphilis, chancroid, and trichomoniasis in men.
- HIV, syphilis, gonorrhea, chlamydia, and chancroid are reportable diseases in all 50 of the United States (US).

1. A 30-year-old pregnant female presents for evaluation of a genital ulcer. Darkfield microscopy reveals spirochetes. She is allergic to penicillin. Which of the following statements is false?

a. Azithromycin is an acceptable treatment alternative for primary syphilis during pregnancy in a patient with known penicillin allergy.

b. Nontreponemal serologic tests for syphilis (rapid plasma reagin [RPR], Venereal Disease Research Laboratory [VDRL]) may yield false-negative results in primary syphilis.

c. Primary syphilis facilitates the transmission and acquisition of human immunodeficiency virus (HIV) infection.

d. Syphilis is a reportable disease in all 50 states.

e. The chancre of primary syphilis will heal spontaneously without antibiotic therapy.

Answer: A. Penicillin remains the drug of choice for treatment of syphilis during pregnancy. A pregnant patient with syphilis and known penicillin allergy should be admitted for desensitization and treatment with penicillin. Nontreponemal serologic tests may yield false-negative results in primary syphilis when antibody titers have not yet risen. False-positive serology may be seen with a variety of other medical conditions. Visualization of spirochetes on darkfield microscopy confirms the diagnosis of syphilis. All sexually transmitted infections (STIs) facilitate the transmission and acquisition of HIV. Reportable STIs in all 50 states include gonorrhea, chlamydia, syphilis, HIV, and chancroid.

2. A 17-year-old female presents with complaints of pelvic pain. She reports multiple sexual partners and inconsistent condom use. Pelvic examination reveals yellow cervical discharge, cervical motion tenderness, and bilateral adnexal tenderness. Pregnancy test is negative. Which of the following statements regarding this scenario is true?

a. All adolescents with pelvic inflammatory disease require hospital admission for intravenous antibiotics.

b. A negative nucleic acid amplification test for gonorrhea and chlamydia reliably excludes the diagnosis of pelvic inflammatory disease.

c. In the absence of an identifiable alternative diagnosis, the clinical diagnosis and empirical treatment of pelvic inflammatory disease is warranted.

d. The clinical diagnosis of pelvic inflammatory disease requires the presence of lower abdominal tenderness *and* cervical motion tenderness *and* adnexal tenderness on physical examination.

e. Women treated as outpatients for pelvic inflammatory disease should have a follow-up evaluation in 2 weeks.

Answer: C. The clinical diagnosis of pelvic inflammatory disease is warranted in a sexually active woman at risk for sexually transmitted infections (STIs) if no alternative diagnosis is identified and any one of the following findings is present on examination: (1) cervical motion tenderness, (2) uterine tenderness, and/or (3) adnexal tenderness. Additional diagnostic criteria (mucopurulent cervical discharge, fever, elevated white blood cell count, positive testing for gonorrhea or chlamydia, and others) improve specificity but decrease sensitivity in the diagnosis of pelvic inflammatory disease (PID). Adolescents with PID may be treated as outpatients using the same criteria as adult women. Women receiving outpatient treatment for PID should be advised to seek a follow-up evaluation within 48 to 72 hours.

3. Which of the following antibiotic regimens is acceptable for outpatient treatment of pelvic inflammatory disease in a 70-kg woman?

a. Azithromycin 1 g *per os* (by mouth) (PO) (single dose) *and* metronidazole 500 mg bid for 14 days

b. Ceftriaxone 125 mg IM (single dose) *and* doxycycline 100 mg bid for 7 days

c. Ceftriaxone 250 mg IM (single dose) *and* azithromycin 1 g PO (single dose)

d. Ceftriaxone 500 mg IM (single dose) *and* doxycycline 100 mg PO bid for 14 days *and* metronidazole 500 mg PO bid for 14 days

e. Metronidazole 2 g PO (single dose) *and* doxycycline 100 mg PO bid for 14 days

Answer: D. Pelvic inflammatory disease is typically a polymicrobial infection. *Neisseria gonorrhoeae* and/or *Chlamydia trachomatis* are frequently implicated organisms, but anaerobes, enteric organisms, and normal vaginal flora may also be present. Empirical treatment of PID should include adequate coverage for gonorrhea and chlamydia. A single dose of ceftriaxone 500 mg IM is adequate treatment for upper tract gonococcal infection in a patient weighing <150 kg. A 14-day course of antibiotics is recommended for adequate coverage of chlamydia and anaerobes in PID.

4. A 24-year-old sexually active male presents with painful genital ulcers. Physical examination reveals a cluster of 2- to 3-mm tender superficial ulcers on the penile shaft. He reports a history of similar lesions in the same location sporadically in the past. Which statement regarding this clinical scenario is false?
   a. Both herpes simplex virus (HSV)-1 and HSV-2 can be transmitted through sexual contact.
   b. Genital herpes is a lifelong viral infection.
   c. Prompt initiation of antiviral medication reduces the duration and severity of symptoms.
   d. Topical antiviral therapy is not recommended.
   e. Use of condoms is not necessary to prevent transmission in the absence of clinically apparent lesions.

Answer: E. Genital herpes is a lifelong infection caused by herpes simplex virus. Sexual transmission is more common with HSV-2 but may also occur with HSV-1. Condom use is recommended during asymptomatic periods, because viral shedding and transmission may occur even in the absence of clinically apparent lesions. Antiviral therapy is not curative. Prompt initiation of systemic antiviral medication within 72 hours (acyclovir, famciclovir, or valacyclovir) reduces the duration and severity of symptoms, particularly at the time of primary infection. Topical antiviral therapy is not recommended.

5. A 24-year-old female presents to the emergency department (ED) complaining of vaginal discharge. A copious frothy whitish discharge is noted on speculum examination. Microscopic examination of a saline wet mount reveals motile flagellated organisms. Which of the following statements regarding this patient's treatment is true?
   a. Metronidazole is the drug of choice for treatment of symptomatic trichomoniasis during all stages of pregnancy.
   b. Punctate hemorrhagic lesions are seen on the cervix in most cases of trichomonas vaginitis.
   c. Tinidazole is a safe alternative for treatment of trichomoniasis during pregnancy.
   d. Trichomoniasis is always symptomatic in men and women.
   e. Wet mount microscopy approaches 100% sensitivity in the diagnosis of trichomonas vaginitis.

Answer: A. Metronidazole is the drug of choice for treatment of symptomatic trichomoniasis during all stages of pregnancy. Tinidazole should be avoided in pregnant women due to limited data regarding safety for use in pregnancy. Visualization of flagellated protozoans on wet mount microscopy of vaginal discharge is highly specific, but only 50% to 65% sensitive for the diagnosis of trichomoniasis. Punctate hemorrhagic lesions on the cervix (so called "strawberry cervix") is seen in up to 10% of cases. Nucleic acid amplification tests for trichomonas are highly sensitive and specific. Trichomoniasis may be asymptomatic in men and women.

6. Single-dose antibiotic therapy is recommended for treatment of which of the following STIs?
   a. Nongonococcal urethritis (NGU)
   b. Vaginal trichomoniasis
   c. Primary and secondary syphilis
   d. Pelvic inflammatory disease
   e. Chlamydia proctitis

Answer: C. Primary and secondary syphilis are treated with a single dose of benzathine penicillin. Gonococcal urethritis, cervicitis, pharyngitis, and proctitis are treated with a single dose of ceftriaxone. Single-dose metronidazole is no longer recommended for vaginal trichomoniasis, but is still recommended for trichomoniasis in men. A 7-day course of doxycycline is now recommended for chlamydia urethritis, cervicitis, pharyngitis, and proctitis in men and nonpregnant women. Treatment of NGU should include chlamydia coverage. Treatment of upper genitourinary tract STIs, including pelvic inflammatory disease and epididymoorchits, requires a longer course of antibiotic therapy.

KEY CONCEPTS

- Urinary obstruction should be considered in patients with a urinary tract infection (UTI) with risk factors for obstruction. Urinary obstruction should be ruled out in patients presenting in septic shock.
- Patients presenting with acute ureteral obstruction require a urinalysis to rule out the presence of a superimposed UTI.
- Acute, uncomplicated UTIs should be treated with fosfomycin, nitrofurantoin, or trimethoprim-sulfamethoxazole. Fluoroquinolones are not recommended as first-line therapy for uncomplicated UTI.
- Acute bacterial prostatitis (ABP) generally affects men between the ages of 20 and 40 years old with a second peak in men older than 60 years old. A very tender and swollen prostate gland is present in more than 90% of patients with acute prostatitis.
- The three primary predictors of stone passage without the need for surgical intervention are calculus size, location, and degree of patient pain. The most important factor that relates to passage of a calculus though the genitourinary tract is its size (stone <5 mm has a 90% chance of passing spontaneously in 4 weeks).
- Imaging is not needed in all patients with renal colic. It should be performed when symptoms and signs are atypical and the diagnosis is uncertain, when high-grade obstruction is suspected, or when the patient appears toxic or has a solitary or transplanted kidney.
- Testicular torsion should be suspected in all patients presenting with acute scrotal pain. There is no historical factor or physical finding that reliably differentiates torsion from other causes of testicular pain.
- Approximately 10% of patients with testicular trauma have associated torsion and require prompt identification and detorsion.
- Sexually active males should receive ceftriaxone and doxycycline to treat epididymitis. Patients in whom enteric organisms are likely the cause of epididymitis should receive fluoroquinolones. Most cases of pediatric epididymitis are idiopathic, and antibiotics are not routinely recommended.
- Acute urinary retention (AUR) is usually caused by an obstructive lesion but also can be the presenting manifestation of other pathologic processes. Patients with AUR and concomitant infection, pelvic mass, or neurologic deficits warrant imaging in the emergency department.
- Although generally associated with a benign process, the presence of gross or microscopic hematuria requires consideration of life-threatening causes such as aortic abdominal aneurysms and malignancy.

1. Which of the following medications is not an appropriate first-line choice for empirical treatment of uncomplicated urinary tract infections (UTIs)?
   a. Fosfomycin
   b. Nitrofurantoin
   c. Trimethoprim-sulfamethoxazole
   d. Ciprofloxacin

Answer: d. The options for treating uncomplicated lower UTI include single-dose therapy with fosfomycin, 5 days of nitrofurantoin, or 3 days of trimethoprim-sulfamethoxazole. Fluoroquinolones such as ciprofloxacin or levofloxacin should not be used as first-line agents for empirical treatment of uncomplicated UTIs. Instead, they should be reserved for patients who have failed or have contraindications to first-line antibiotics.

2. Which of the following drugs is not an appropriate first-line choice for treatment of an uncomplicated urinary tract infection (UTI) in women?
   a. Fosfomycin
   b. Ciprofloxacin
   c. Nitrofurantoin
   d. Trimethoprim-sulfamethoxazole
   e. All of the above

Answer: b. According to the Infectious Disease Society of America (IDSA) practice guidelines, fluoroquinolones should not be used as first-line agents for uncomplicated UTI because of increased resistance.

3. Computed tomography should be undertaken in the patient suspected of having renal colic if which of the following is present?
   a. The patient has a solitary or transplanted kidney.
   b. The patient has gross hematuria.
   c. The patient has had a prior history of nephrolithiasis.
   d. The patient presents in severe pain.

Answer: a. Imaging is appropriate for patients who have a history of nephrolithiasis who do not improve with treatment, have a urinalysis showing infection, have a solitary or transplanted kidney, or for whom a diagnosis other than renal colic is suspected.

4. What percentage of cases of acute bacterial prostatitis eventually develop into chronic bacterial prostatitis?
   a. 1%
   b. 5%
   c. 10%
   d. 20%

Answer: c. Chronic bacterial prostatitis will eventually develop in 10% of patients with acute bacterial prostatitis.

5. In testicular torsion, testicular salvage may be predicted by which of the following factors?
   a. Degree of torsion
   b. Duration of the ischemia
   c. Cremasteric reflex
   d. Age of the patient
   e. Both A and B

Answer: e. Testicular torsion results in obstruction of venous outflow, subsequent compromised arterial flow, and testicular ischemia. Testicular salvage hinges on the degree of torsion and duration of the ischemia.

6. To detorse a testicular torsion, you should first rotate the testis medially to laterally. If this does not produce immediate relief, you should do which of the following?
   a. Assume that the testicle is already necrotic and not amenable to further reduction attempts.
   b. Attempt untwisting past 360 degrees because a higher degree of rotation may be present.
   c. Both A and B.
   d. Reverse the direction of reduction.

Answer: d. Most testes torse laterally to medially, but some may torse medially to laterally. If no immediate relief is obtained by rotating medially to laterally, reverse the direction of the reduction attempt.

KEY CONCEPTS

- Adnexal torsion is easily missed on initial presentation and should be considered in any patient with known risk factors, even if symptoms are subtle or atypical.
- Doppler ultrasound is the preferred initial imaging study for suspected adnexal torsion.
- An ultrasound examination may distinguish among the various types of ovarian cysts and identify associated complications, such as torsion, hemorrhage, and malignancy. Most ovarian cysts are simple follicular cysts that resolve without pharmacologic or surgical intervention.
- Abnormal uterine bleeding (AUB) has many structural, hormonal, and coagulopathic causes. Selected imaging and laboratory testing, based on a careful history and physical examination, can often lead to determination of the cause. Combined oral contraceptive pills can help to regulate the cycle and alleviate AUB.
- Emergency contraception is a safe, effective option to prevent an undesired pregnancy. Levonorgestrel and ulipristal are both effective oral medications and are associated with fewer side effects than the traditional combined contraceptive method. Intrauterine devices should also be considered for emergency contraception if a patient desires a long-term contraceptive option.

1. Which of the following statements regarding ovarian torsion is true?
   a. Abdominal tenderness is predictable
   b. Complete arterial obstruction is common
   c. Computed tomography (CT) has a higher sensitivity than ultrasound
   d. Most cases are associated with an ovarian mass
   e. There is a left-sided predominance

Answer: d. Most cases are associated with a benign ovarian tumor or cyst. There is a modest right-sided predominance. Due to the collateral uterine and ovarian arterial supply, complete arterial obstruction is rare. In cases of intermittent or chronic torsion particularly, abdominal tenderness may be absent. CT has a lower sensitivity than ultrasound, which is still only approximately 71%. Interpret negative studies carefully.

2. Which of the following patterns of menses should be considered abnormal?
   a. A 23-day menstrual cycle
   b. A 40-day menstrual cycle
   c. Bleeding 6 months after menopause
   d. Seven days of menstrual flow
   e. Three days of menstrual flow

Answer: b. A menstrual cycle shorter than 21 days or more than 35 days apart, or flow that is less than 2 days or more than 7 days, is considered abnormal. In the postmenopausal woman, any bleeding 12 months after cessation of menses is considered abnormal.

3. A 33-year-old gravida 3 para 3 woman presents with 7 days of heavy, painless vaginal bleeding. Her only other complaint is dizziness. Urine pregnancy test is negative. Vital signs are blood pressure, 85/40 mm Hg, and heart rate, 130 beats/min. The pelvic examination reveals copious vaginal bleeding through a partially open cervical os. The hemoglobin level is 6.8 g/dL. Which of the following interventions is most appropriate?
   a. 20 μg of ethinyl estradiol daily until the bleeding subsides
   b. 35 μg ethinyl estradiol bid until the bleeding subsides
   c. Blood transfusion and urgent gynecologic consultation for dilation and curettage
   d. Premarin, 25 mg intravenously every 6 hours
   e. Saline hydration followed by a 2-day recheck

Answer: c. This patient is symptomatic, hypovolemic, anemic, and exhibiting ongoing bleeding. Oral estrogens are indicated in cases of modest bleeding. Parenteral estrogen may be used as an adjunct to other therapies for patients requiring admission. The degree of anemia in the face of ongoing bleeding in this case warrants gynecologic intervention.

4. To be most effective, the emergency contraceptive ulipristal should be given as soon as possible but is approved to be given within how many hours of intercourse?
   a. 12
   b. 24
   c. 48
   d. 72
   e. 120

Answer: e. The efficacy of all emergency contraceptive pills in preventing pregnancy is greatest when a contraceptive is taken soon after intercourse. Ulipristal is labeled for 120 hours post coitus. Because it is not as effective as preplanned contraception, women should still be aware of the possibility of pregnancy after its use.

87 Stroke

KEY CONCEPTS

- Anterior circulation strokes result in contralateral hemiparesis of the face and body.
- Vertebrobasilar strokes result in ipsilateral cranial nerve (CN) deficits and contralateral hemiparesis.
- Posterior cerebral artery stroke causes ipsilateral CN III palsy and contralateral homonymous hemianopsia.
- *Wallenberg syndrome* (lateral medullary syndrome) causes vertigo, Horner syndrome, ipsilateral facial numbness, loss of corneal reflex, along with contralateral loss of pain and temperature.
- Cervical artery dissection is a common cause of stroke in young patients; trans ischemic attacks (TIAs) preceding stroke in these patients are often misdiagnosed.
- The goal for eligible patients is to receive thrombolytics within 90 min of symptoms onset; the dose of the tissue plasminogen activator Alteplase (t-PA) is 0.9 mg/kg with 10% given as a bolus and the remaining 90% given over 1 h.
- Overly aggressive blood pressure (BP) management should be avoided in patients with acute ischemic stroke. If thrombolytic therapy is indicated, stringent control of BP is indicated to reduce the potential for intracranial hemorrhage.
- Accurate identification noting the last time a patient was known to be at his or her neurologic baseline should be documented in all patients with stroke.
- Acute ischemic stroke patients receiving Alteplase are at risk of developing a spontaneous intracranial hemorrhage; the risk is lowest in patients with a low National Institutes of Health Stroke Scale (NIHSS) score, no hypertension, no diabetes, and age younger than 70 years.
- In acute ischemic stroke, the patient and/or their families should be informed of the risk and benefit of treatment with thrombolytic therapy.
- The incidence of symptomatic intracerebral hemorrhage after intravenous (IV) thrombolysis is 2%–7%. The incidence of asymptomatic intracranial hemorrhage is higher with 30%–45% of patients experiencing >10 cerebral microbleeds.
- Beyond 4.5 h, the benefit of IV thrombolysis is not clear and should be determined after consulting with the neurologist or stroke specialist.
- In patients with mild nondisabling stroke symptoms, IV thrombolysis is not recommended. However, for otherwise eligible patients with mild but disabling stroke symptoms, IV thrombolysis is recommended for patients within 4.5 h of ischemic stroke symptom onset.
- Patients with a hemorrhagic stroke on coumadin should *be reversed* using vitamin K and prothrombin complex concentrate (PCC). Fresh frozen plasma (FFP) can be used if PCC is not available. However, PCC is preferred over FFP due to more rapid correction of international normalized ratio (INR), lower risk of infection, and lower administered volume.
- Prognosis is worse in acute stroke in the setting of fever, hypotension, hypoxia, and hypo/hyperglycemia. Measures should be taken to rapidly correct these.
- Carotid Doppler, magnetic resonance angiography (MRA), or computed tomography angiography (CTA) studies are recommended before discharge of a patient with TIA from the emergency department (ED).
- TIA has moved from time-based to tissue-based definition (defined as a transient episode of neurologic dysfunction caused by focal brain, spinal cord, or retinal ischemia, without evidence of acute infarction).
- MRI with diffusion-weighted imaging (DWI) has a greater sensitivity than computed tomography (CT) for detecting small infarcts in patients with TIA.
- Patients with TIA or nondisabling strokes are at increased risk of recurrent stroke and should be evaluated and treated immediately in order to reduce the risk of subsequent stroke.
- CTA with CT pulmonary angiogram (CTPA) or MRA with DW-MRI is useful for selecting candidates for mechanical thrombectomy between 6 and 24 h after patient was last known to be well.
- Mechanical thrombectomy can be used in addition to treatment with intravenous thrombolytics. However, pretreatment with intravenous thrombolytics (within 4.5 h of symptom onset) is not required prior to thrombectomy.

1. Which of the following statements concerning ischemic stroke is true?
   a. Anterior circulation strokes are more likely than posterior strokes to show evidence of progression at the time of presentation.
   b. Anterior circulation strokes rarely present with complete loss of consciousness.
   c. Posterior cerebral artery strokes are associated with incontinence, leg weakness greater than arm weakness, and gait clumsiness.
   d. The presence of aphasia suggests an anterior cerebral artery (ACA) distribution stroke, typically left sided.
   e. The presence of diplopia suggests an anterior circulation stroke.

Answer: b. Forty percent of posterior and 20% of anterior circulation strokes present with progressive symptoms. It is rare for anterior circulation (carotid, ACA, and middle cerebral artery [MCA]) strokes to significantly alter consciousness, unless there has been a previous contralateral stroke. ACA strokes primarily affect frontal lobe functions and may also present with primitive grasp and such reflexes. In addition to contralateral motor and sensory defects, MCA strokes may present with expressive aphasia, agnosia, and ipsilateral hemianopsia.

2. Which of the following is not associated with posterior circulation strokes?
   a. Diplopia
   b. Homonymous cranial nerve (CN) and extremity motor deficits
   c. Loss of consciousness
   d. Loss of visual object recognition
   e. Nausea, vomiting, and ataxia

Answer: b. Posterior circulation (vertebrobasilar) strokes involve the vertebral, basilar, and posterior cerebral arteries. Because this system supplies the reticular activating system, cerebellum, brainstem, occipital lobe, and brainstem vomiting centers, loss of consciousness with vomiting, visual changes, and cerebral ataxia may be seen. Ipsilateral CN deficits (because these nuclei largely reside in the brainstem) occur with contralateral "body" deficits resulting from motor/sensory fiber decussation.

3. Headache, vomiting, and a decreased level of consciousness are most commonly seen with which of the following disorders?
   a. Ischemic stroke
   b. Intracranial hemorrhage
   c. Migraine headache
   d. Subarachnoid hemorrhage (SAH)
   e. Tic douloureux

Answer: d. The incidence of headache is highest by far in patients with SAH. Vomiting and depressed loss of consciousness are also generally more common in this group. Pure migraine headache rarely, if ever, causes a depressed loss of consciousness. Tic headaches do not cause loss of consciousness.

4. What percentage of patients with hemorrhagic stroke experience clinical deterioration because of growth in hemorrhage volume within the first hours?
   a. 10%
   b. 20%
   c. 30%
   d. 50%
   e. 75%

Answer: c. Thirty percent of patients with intracerebral hemorrhage (ICH) experience early hemorrhage expansion. Progression of neurologic deficits and decreasing mental status suggest the diagnosis.

5. A 69-year-old male presents with headache, vomiting, aphasia, a right lower facial palsy, and right upper greater than right lower extremity weakness. The symptoms began approximately 4 hours before arrival. Vital signs are temperature 99°C, blood pressure (BP) 180/90 mm Hg, respiratory rate 18 breaths/min, heart rate 92 beats/min, and oxygen saturation 96% on room air. Emergent computed tomography (CT) scan shows a left temporal intracerebral hemorrhage (ICH). Soon after presentation, the patient experiences increased vomiting and a diminishing level of consciousness. What is the most likely explanation for this deterioration?
   a. Accompanying subarachnoid hemorrhage (SAH)
   b. Acute brainstem herniation
   c. Hypoxia from neurogenic pulmonary edema
   d. Increase in volume of the ICH
   e. Myocardial infarction with cardiogenic shock

Answer: d. Approximately one-third of patients with ICH experience early hemorrhage volume expansion. Although brainstem herniation is a possibility, this is typically a delayed sequela with a more gradual presentation. Acute myocardial infarction may be associated with intracranial emergencies but would not likely cause an abrupt mental status change. Neurogenic pulmonary edema may accompany any condition with elevated intracranial pressure (ICP) and would not likely cause an abrupt mental status change.

6. After complete occlusion of cerebral vessels, irreversible neurologic deficits are expected to reliably occur within how many hours?
   a. 2
   b. 3
   c. 4
   d. 5
   e. 6

Answer: e. As a result, ischemic stroke trials, using fibrinolytic or antiplatelet agents, have attempted to recanalize occluded arteries and reperfuse ischemic areas of the brain within a 2- to 6-hour therapeutic window.

7. Which of the following areas of the brain is perfused by the posterior circulation?
   a. Internal capsule
   b. Posterior aspect of the temporal lobe
   c. Putamen
   d. Speech areas of the temporal lobe
   e. Thalamus

Answer: e. The thalamus is perfused by the posterior circulation. The other areas are perfused by the anterior circulation.

8. Which of the following statements regarding stroke etiology is true?
   a. Lacunar strokes reliably cause a pure motor deficit.
   b. Less than 1% of strokes occur in the 15- to 45-year-old age group.
   c. One-third of ischemic strokes are thrombotic.
   d. Strokes resulting from atrial fibrillation likely involve small vessels.
   e. Two-thirds of ischemic strokes are cardioembolic.

Answer: c. One-third of ischemic strokes are thrombotic. Lacunar strokes may cause a pure motor, pure sensory, or ataxic/hemiparesis stroke. Vessel occlusion resulting from atrial fibrillation–induced emboli more likely involves the large vessels. Three percent to 4% of ischemic strokes occur in the 15- to 45-year-old age group.

9. A 29-year-old female presents with a left-sided headache after a moderate-speed motor vehicle collision (MVC). She suffered no loss of consciousness and has no other complaints or obvious injuries. Physical examination is remarkable only for drooping of the left eyelid and slight miosis of the left pupil compared with the right. Which of the following would be the diagnostic test of choice?

a. Brain magnetic resonance imaging (MRI) with gadolinium
b. Contrasted computed tomography (CT) scan of the brain
c. CT angiogram of the carotid arteries
d. Uncontrasted CT scan of the brain
e. Urine drug screen

Answer: c. Carotid or vertebral artery dissection can occur after trauma or mild events, such as yoga, twisting, or prolonged static positions looking upward. The hallmark is unilateral neck pain, face pain, or headache, often with accompanying Horner syndrome. Acutely, cerebral ischemic changes would not be seen on brain imaging. Carotid and vertebral dissection is not a contraindication for thrombolytic therapy in the eligible patient.

10. A 28-year-old G3P3 woman who is 2-week postpartum after an uncomplicated vaginal delivery presents with acute onset of mild headache, lethargy, and double vision. Physical examination is remarkable for normal vital signs and a left eye lateral gaze palsy. The most appropriate intervention is likely to be which of the following?

a. Computed tomography (CT) scan of the brain with possible lumbar puncture
b. CT scan of the brain and intravenous (IV) heparin
c. Erythrocyte sedimentation rate (ESR) and IV corticosteroids
d. IV magnesium
e. Lumbar puncture and IV antibiotics

Answer: b. Cerebral venous thrombosis may present with headache, lethargy, cranial nerve (CN) deficits, seizures, or even psychiatric complaints. CT scan and/or magnetic resonance imaging (MRI)/magnetic resonance angiography (MRA) are likely to reveal the diagnosis. Treatment includes heparin. Neurosurgical consultation is not useful. Subarachnoid hemorrhage (SAH) would not be expected to cause a focal neurologic deficit. Eclampsia and meningitis would be expected to give characteristic findings on history and examination.

11. An 82-year-old male presents with an apparent stroke. He is rapidly evaluated and determined to be within the 3- to 4.5-hour window for fibrinolytic therapy. Which of the following exclusion criteria applies only to the 3- to 4.5-hour criteria and not the 0- to 3-hour criteria?

a. Administration of heparin within the 48 hours preceding the stroke onset
b. Age older than 80 years
c. High clinical suspicion for subarachnoid hemorrhage (SAH)
d. Seizure at the onset of the stroke
e. Symptoms rapidly improving

Answer: b. In the 3- to 4.5-hour window, patients cannot exceed 80 years of age. Heparin administration within the 48 hours preceding stroke onset is a contraindication in both the 0- to 3-hour window as well as the 3- to 4.5-hour window. Similarly, high clinical suspicion for SAH, seizure at the onset of the stroke symptoms, and rapidly improving symptoms are contraindications to fibrinolysis in both time windows.

12. A 66-year-old female presents with a possible transient ischemic attack (TIA). Approximately 1 hour before her arrival, she had a 15-minute episode of strictly right arm and right leg weakness, and her symptoms have now resolved. Her blood pressure (BP) is 165/92 mm Hg. Her prior medical history is significant for hypertension, high cholesterol, and diabetes mellitus. What is her ABCD2 score?

a. 4
b. 5
c. 6
d. 7
e. 8

Answer: c. The patient's ABCD2 score is 6 (age >60 years, BP >140/90 mm Hg, unilateral weakness, symptoms lasting 10 to 59 minutes, and history of diabetes). No speech impairment is reported.

13. A 72-year-old male presents with an apparent stroke. Computed tomography (CT) imaging of the brain demonstrates only a hyperdense middle cerebral artery (MCA) sign and no other ischemic changes. He was last seen neurologically normal 4 hours earlier. Which of the following is a contraindication to fibrinolytic therapy in this patient?

a. His blood glucose is 372 mg/dL.
b. His National Institutes of Health (NIH) Stroke Scale score is 24.
c. His platelet count is 110,000/mm^3.
d. His systolic blood pressure (BP) is 180 mm Hg.
e. He takes warfarin daily.

Answer: e. Any oral anticoagulant treatment (regardless of the patient's international normalized ratio [INR]) is a contraindication to fibrinolysis in the 3- to 4.5-hour treatment window. Other contraindications include an NIH stroke scale score greater than 25, platelet count less than 100,000/mm^3, blood glucose greater than 400 mg/dL, and systolic BP greater than 185 mm Hg.

14. A 75-year-old man is brought to the emergency department (ED) for altered mental status. After computed tomography (CT) imaging of the brain is performed, he is found to have a large intracerebral hemorrhage (ICH). Which of the following options is *not* an appropriate strategy for lowering intracranial pressure (ICP)?

a. Barbiturate-induced coma
b. Hyperthermia induction
c. Hypertonic saline administration
d. Hyperventilation
e. Mannitol administration

Answer: b. Hypothermia is an experimental modality for lowering ICP. Hyperventilation can serve as a temporizing measure for reducing ICP. Mannitol and/or hypertonic saline can also be administered. Inducing a barbiturate coma is also an experimental modality.

KEY CONCEPTS

- Epilepsy is a neurologic condition associated with an intrinsically lower seizure threshold and a higher risk of recurrent seizures without a clear trigger.
- The characterization of seizure semiology, duration, and etiology is important for accurate classification of seizures and status epilepticus; these impact definitive treatment choices.
- There is no single test to confirm that a patient seized, and several seizure mimics, including convulsive syncope, exist. A postictal alteration in mental status makes a seizure five times more likely than syncope.
- Key factors in the evaluation of epilepsy patients who present with breakthrough seizures include: changes in anti-seizure regimen, compliance, the addition of new medications that may lower the seizure threshold or levels of antiseizure drugs, presence of common infections or metabolic derangements, and recent sleep habits.
- Although most seizures are self-limited, the management of patients with seizures involves a targeted search for underlying pathology, treatment of complications associated with convulsions, and the prevention of future episodes.
- Serious systemic complications of seizures and status epilepticus include cardiac arrest, arrhythmias, apnea, hypoxia, acute kidney injury, rhabdomyolysis, acidosis, and death. The prognosis of status epilepticus is directly related to the etiology of seizures.
- Primary seizure prophylaxis should only be given for 7 days following traumatic brain injury. The period is shorter, although less well-defined, in unsecured aneurysmal subarachnoid hemorrhage. Prolonged primary prophylaxis is not recommended because it has not been demonstrated to reduce long-term seizure risk. The duration of secondary seizure prophylaxis and the antiseizure regimen in patients with acute brain injury who had a seizure during hospitalization should be individualized.
- Patients with a first-time seizure who have no known structural brain pathology, normal serum glucose and sodium levels, and normal neurologic examination can be discharged from the ED with appropriate outpatient follow-up.
- Emerging evidence supports the consideration of secondary seizure prophylaxis for first-time unprovoked seizures in selected patients. However, data are heterogeneous, and a thorough discussion with the patient and a specialist is advised before the initiation of anti-seizure drugs.
- Alcohol withdrawal syndrome can include seizures resulting from the cessation or reduction of alcohol consumption leading to an unopposed excitatory sympathomimetic response. Benzodiazepines are the drug of choice and should be supplemented by supportive measures, including electrolyte and thiamine supplementation. In pregnant patients, evaluations for new-onset seizures before 20 weeks should be the same as in nonpregnant patients. After 20 weeks, and up to 8 weeks postpartum, eclampsia is a major cause of seizures and should be included in the differential. IV magnesium remains first-line treatment for patients with eclamptic seizures and should not be delayed. Benzodiazepines and non-teratogenic antiseizure medications are reasonable alternatives in magnesium-refractory cases.
- Post-anoxic status epilepticus, including myoclonic status, frequently observed following cardiac arrest, was considered pathognomonic of poor neurologic outcome. However, in patients lacking factors with high predictive value for poor outcome post cardiac arrest, early antiseizure therapy can lead to improved outcomes.
- Therapeutic approaches to nonconvulsive status epilepticus are commonly extrapolated from convulsive generalized status epilepticus guidelines. However, the presumed etiology of seizures, extent of cortical area involved (focal versus diffuse or generalized), comorbidities, and response to therapy should be considered when selecting an antiseizure therapeutic algorithm.
- Immunomodulation remains the cornerstone of therapy for autoimmune epilepsies, in conjunction with antiseizure drugs. Transdisciplinary decision making is warranted before initiating immune-targeted therapy, which may include high-dose methylprednisolone, intravenous immunoglobulin, plasma exchange, rituximab, cyclophosphamide, and more recently, tocilizumab.

1. Which of the following statements regarding the pathophysiology of seizures is *true?*
   - **a.** During a seizure, prompt loss of consciousness implies a cortical focus.
   - **b.** Generalized seizures involve abnormal electrical activity in one hemisphere.
   - **c.** Regarding seizure activity, acetylcholine is inhibitory.
   - **d.** The clinical seizure activity typically reflects the brain focus of initiation.

Answer: D. Clinical seizure activity typically reflects the site of seizure origin. Generalized seizures involve both brain hemispheres. Acetylcholine is an excitatory neurotransmitter, and GABA is inhibitory. Prompt loss of consciousness with a seizure implies a subcortical focus involving the reticular activating system.

2. Which of the following statements regarding epilepsy is *true?*
   - **a.** Electrolyte disturbances are the most common cause of recurrent seizures in epileptic patients.
   - **b.** Epilepsy is a condition of recurrent provoked seizures.
   - **c.** Epileptic seizures are always generalized tonic-clonic.
   - **d.** Epileptic seizures can be triggered by smells or lack of sleep.

Answer: D. Epilepsy is a condition of recurrent, *unprovoked* seizures. Idiopathic epilepsy almost always starts in childhood. Medication noncompliance is the most common cause of recurrent seizures in epileptic patients. Epileptic seizures can be partial seizures. Some patients with epilepsy have seizures triggered by certain specific sensory stimuli, with flashing lights being the most well-known of these.

3. Which of the following factors does not put patients at risk for seizures?
   - **a.** Acute stroke
   - **b.** Chronic stroke-related lesions
   - **c.** Hyperkalemia
   - **d.** Hypomagnesemia

Answer: C. Indwelling intracranial shunts and other lesions, and hypomagnesemia (as seen in malnourished alcohol-dependent patients) are all risk factors for seizures. Stroke is one of the most common causes of seizures in the elderly. Seizures occurring acutely with a stroke are often partial and reflect the affected brain area, whereas those that occur chronically are often generalized. Hyperkalemia does not cause seizures unless it precipitates hypotension and consequent hypoperfusion of the brain.

4. Which of the following statements regarding patients presenting with first-time seizures is *true?*
   - **a.** All patients should have basic serum electrolyte testing.
   - **b.** All patients should receive a loading dose of an antiepileptic drug and a prescription for an oral regimen prior to discharge.
   - **c.** Nondiabetic patients do not need to have serum glucose checked if their mental status returns to normal.
   - **d.** Persistent altered mental status is an indication for a lumbar puncture.

Answer: D. Persistent altered mental status should trigger consideration of a lumbar puncture, serum glucose, and an electroencephalogram (EEG) among other testing. Non-contrast CT of the head is relatively high-yield in first-time seizure patients, and most guidelines recommend performing one. Patients with no comorbidities who return to baseline mental status do not need broad-based electrolyte testing, but hypoglycemia is a common cause of seizures even in nondiabetic patients and should be routinely tested for in first-time seizure patients. Initiation of antiepileptic drugs in the emergency department (ED) for first-time seizure patients without a known structural brain lesion may do more harm than good.

5. A 23-year-old woman is brought to the emergency department (ED) for a prolonged seizure. By emergency medical service (EMS) report, the patient has no past medical history and no history of seizures. Paramedics report tonic-clonic activity for approximately 15 minutes, refractory to diazepam 5 mg intravenously in the ambulance. Upon arrival to the ED, the patient's seizure activity abruptly ceases, and she lucidly responds to the history and physical examination. She is symptom free. What would be the most appropriate intervention?
   - **a.** A trial of oral phenytoin after an intravenous (IV) loading dose in the ED
   - **b.** Confrontation
   - **c.** Neurology referral for electroencephalogram (EEG) and consultation
   - **d.** Psychiatric consultation

Answer: D. No clinical criteria are 100% specific for the diagnosis of nonepileptic activity. Seizures and nonepileptic activity may coexist. For many patients, these episodes may not be deliberate. The most prudent course of action would be to treat them as possible ictus and refer to a neurologist. Initiation of antiepileptic drugs in the ED is not indicated in first-time seizures of healthy patients who return to normal.

KEY CONCEPTS

- The goals of headache evaluation in the ED are (1) to distinguish between benign primary headache disorders and potentially life-threatening secondary causes of headache and (2) to treat the headache pain effectively and rapidly without causing undue side effects.
- Patients with the following headache presentations are at risk for serious underlying disease: sudden explosive headache; new-onset headache after the age of 50 years; headache associated with papilledema, alteration in or loss of consciousness, or focal neurologic symptoms; subacute headache with increasing frequency or severity; headache associated with fever, cancer, or immunosuppression; and headache triggered by exertion, sexual activity, or Valsalva maneuver.
- The need for diagnostic studies is dictated by the suspected secondary cause of headache.
- Antidopaminergic agents, such as metoclopramide or prochlorperazine, are first-line therapy for migraine.
- Opioids are not first-line therapy for the primary headaches.
- Patients with migraine treated in the ED require a discharge "rescue plan" if the headache recurs.
- High-flow oxygen will terminate the majority of cluster headaches.
- The differential diagnosis of sudden severe headache includes subarachnoid hemorrhage, cerebral venous thrombosis, cervical artery dissection, and idiopathic intracranial hypertension.
- Cerebral venous thrombosis should be suspected in women who have a new type of headache and are pregnant or on birth control pills.
- Carotid artery dissection may cause headache, ptosis, and miosis.
- Patients with post-traumatic headache should only be imaged if they have high-risk features on the Canadian CT Head rule.

1. A 29-year-old female presents within 1 hour of the sudden onset of a severe, diffuse headache accompanied by meningismus and vomiting. Emergent computed tomography (CT) scan is negative. Lumbar puncture (LP) reveals 50,000 red blood cells (RBCs) in tube 1 and 30,000 RBCs in tube 4. Opening pressures are normal, and the sample is negative for xanthochromia. What would be the most appropriate next step?
   a. Admission and observation
   b. Cerebrovascular imaging study
   c. Hydration, analgesics
   d. Magnetic resonance imaging (MRI) scan with gadolinium

Answer: B. CT scan provides a sensitivity of approximately 90% for the detection of subarachnoid hemorrhage (SAH). A traumatic LP, even with diminishing RBC counts with sequential tubes, cannot differentiate between a SAH and traumatic tap. The lack of xanthochromia is predictable, given the acute onset of headache and the 12 hours required for cerebrospinal fluid (CSF) xanthochromia to develop. A cerebrovascular imaging study, such as computed tomography angiography (CTA), magnetic resonance angiography (MRA), or standard angiography would be required next to exclude a source of bleeding, such as an aneurysm or arteriovenous malformation (AVM).

2. A 69-year-old male presents with several months of intermittent left-sided headaches that have been worse at night and occasionally on exposure to cold air. On several occasions, he has noted increased pain while eating. He has had no other symptoms other than modest fatigue. Physical examination is unremarkable with normal vital signs and ophthalmologic and neurologic survey. Laboratory evaluation shows only a mild anemia, with a hemoglobin of 11 mg/dL and normocytic indices. What would be the most appropriate next step?
   a. Computed tomography (CT) scan of the brain
   b. Electrocardiogram
   c. Erythrocyte sedimentation rate (ESR)
   d. Neurology consultation

Answer: C. Temporal arteritis may present with intermittent or continuous symptoms, sometimes associated with fatigue, myalgias, jaw claudication, and mild anemia. The ESR is usually, but not always, diagnostic and would confirm the diagnosis if elevated. Steroids and ophthalmology consultation would then follow. The presence of temporal artery tenderness may be variable because the vasculitis may affect any artery. The diagnosis must be suspected in any elderly patient with recurrent or continuous headaches.

3. With cavernous sinus thrombosis, the clinical picture is usually dominated by which of the following?
   a. Facial pain
   b. Lethargy
   c. Nausea and vomiting
   d. Ocular findings such as pain and proptosis

Answer: D. The symptoms may also include paralysis of extraocular movements. Ocular symptoms are most common with thrombosis of the cavernous sinus, rather than one of the other sinuses.

4. A 28-year-old man with a history of migraine and no other medical history presents with a pounding, bilateral 9/10 headache associated with nausea and vomiting that has lasted for 48 hours. The patient reports experiencing a similar headache about once every few months. Physical exam is noteworthy for a blood pressure of 180/110 mm HG in his right arm and 178/111 mm HG in his left arm. Physical exam, including a detailed neurologic exam, is otherwise completely normal. The most appropriate first step in the management of this patient is the following:
   a. Non-contrast CT scan of the head
   b. Metoclopramide 10 mg IV
   c. Labetalol 20 mg IV
   d. CT angiography of the head and neck

Answer: B. For patients who present with elevated blood pressure and an otherwise normal physical exam in the setting of a typical migraine headache, treatment should first focus on relieving the acute pain. Treatment of blood pressure should commence if the blood pressure remains persistently high after adequate control of pain. Head imaging is not indicated in a patient who reports a typical exacerbation of migraine. CT angiography of the head and neck is most appropriate to rule out cervical artery dissection. Sumatriptan is an appropriate treatment of acute migraine but is relatively contraindicated in a patient with markedly elevated blood pressure.

5. An obese 37-year-old female reports worsening headache × 3 months, which first responded to ibuprofen but now no longer does. The headache is bilateral, pulsating, associated with nausea, and improves when she assumes an upright position. She reports 5-second blackouts of her vision, which occur occasionally when she changes position. A neurologic exam is unremarkable. Retinal exam is deferred because of marked photophobia. Non-contrast head CT scan is normal. The most appropriate next step in the diagnostic workup is:
   a. MRV
   b. D-dimer
   c. CT angiography
   d. Lumbar puncture

Answer: D. The provided history is most consistent with idiopathic intracranial hypertension, which can only be confirmed by lumbar puncture. Lumbar puncture will demonstrate elevated cerebrospinal fluid pressure >250 mm H_2O. MRV and D-dimer can be used to exclude the diagnosis of venous sinus thrombosis, which should be considered if the CSF pressure is indeed elevated. ESR is most useful to exclude giant cell arteritis, a diagnosis that does not occur in patients younger than 50 years. CT angiography can be used to diagnose cerebrovascular pathology.

KEY CONCEPTS

- Delirium is an acute condition characterized by an altered level of attention and awareness. It develops during a short time, and symptoms tend to fluctuate throughout the day.
- Delirium is commonly caused by medications, drug intoxication or withdrawal, infections, metabolic disorders, CNS and cardiovascular events, and autonomic nervous system disturbances.
- Dementia is a chronic condition characterized by cognitive impairment. It is slow in onset and progressive in nature. This disorder has many causes, some of which are reversible with treatment. It is essential to search for reversible underlying etiologies that may be worsening a cognitive impairment.
- Patients with either dementia or psychiatric disorders may present with superimposed delirium, often making identification of the underlying cause of their abnormal behavior difficult.
- The clinician should be wary of attributing behavioral disturbances to psychiatric illness in the presence of abnormal vital signs or abnormal sensorium.
- Nonpharmacologic methods of controlling agitation including reassurance, verbal de-escalation, and avoidance of environmental triggers should be considered in the treatment of patients with dementia.
- Antipsychotics and benzodiazepines are used cautiously in the management of acute agitation in delirium and dementia. The choice of agent is determined by side effect profile and etiology of delirium or acute agitation.
- Antipsychotics may cause QTc prolongation and extrapyramidal side effects, especially when given intravenously.
- Lower doses of medications may be appropriate in older adults to decrease risk of adverse events while effectively treating acute agitation.

1. Which of the following characteristics helps distinguish dementia from delirium?

a. Acute onset
b. Memory impairment
c. Disorientation
d. Delusions

Answer: A. Delirium is characterized by an acute onset of impaired cognition, in comparison to the slower onset of delirium. Memory impairment and disorientation can occur in either process. Delusions are more common in delirium, but can occur in the later stages of dementia as well.

2. A 78-year-old woman presents with 3 days of decreasing ability to concentrate, memory and cognition breakdown, sleep cycle disruption, and fluctuating levels of agitation. Her current medications include levofloxacin (Levaquin) 500 mg/day for a bladder infection, tramadol PRN for knee arthritis, and hydrochlorothiazide 25 mg/day for essential hypertension. Her examination is normal except for a baseline tachycardia, moderate agitation and restlessness, and orientation to person only. Laboratory analysis shows glucose 198 mg/dL, sodium 131 mEq/L, potassium 3.8 mEq/L, creatinine 1.4 mg/dL, white blood cell (WBC) count 11,300 cells/mm^3, hemoglobin 12 g/dL, bicarbonate 25 mEq/L, and a normal urinalysis. What is the most likely etiology for her delirium?

a. Early sepsis
b. Hyperglycemia
c. Hyponatremia
d. Medication effect

Answer: D. Medications are the most common cause of delirium in the older adult population. Common inciting medications include antibiotics (quinolones and macrolides), analgesics, sympathomimetics, antiinflammatories, sedatives, and cardiovascular agents. This level of hyponatremia would not be expected to cause delirium. Likewise, a modest hyperglycemia without associated acidosis would be an unlikely culprit.

3. An 82-year-old man presents with acute delirium. On examination, he is alert and mildly agitated. He is oriented to person and place but not time. He is easily distracted and exhibits a mild bilateral upper extremity resting tremor without asterixis. His neurologic examination is nonfocal. His short-term memory is impaired. What is the central component most key to the diagnosis of delirium in this case?

a. Agitation
b. Disorientation
c. Inattention
d. Memory dysfunction

Answer: C. Disturbance in attention is central to the diagnosis of delirium. Disorientation often accompanies this but is not invariably present. The patient is usually disoriented to time and, less often, place. The delirious patient may also experience visual, auditory, tactile, and olfactory hallucinations; may lose the ability to modulate emotional expression; and often exhibits fluctuating symptoms. Short-term memory is usually impaired, but this is also seen in dementia.

4. Which of the following associations is correct?

a. Droperidol: QT prolongation
b. Haloperidol: Dysphoria
c. Lorazepam: Excessive half-life
d. Meperidine: Cholinergic effects

Answer: A. This may also be seen with haloperidol and the phenothiazines. Meperidine causes dysphoria and possibly some anticholinergic effects. Diazepam results in the longest terminal $T\frac{1}{2}$ of the benzodiazepines.

5. A 63-year-old man presents with acute-onset delirium. He is a known alcoholic, and the family reports a cessation of alcohol intake 36 hours before presentation. He has no other known medical problems. Examination is remarkable for an acutely delirious patient who has active visual and auditory hallucinations and a tremor. Neurologic examination is otherwise negative. Finger-stick glucose is normal. Thiamine 100 mg intravenously fails to improve his symptoms. Which of the following is the intervention most likely to prevent further worsening?

a. Dextrose
b. Haloperidol
c. Lorazepam
d. Magnesium

Answer: C. This patient presents with symptoms of alcohol withdrawal. Wernicke encephalopathy should be considered and thiamine supplementation may be necessary, but lorazepam will prevent progression to withdrawal seizures. Magnesium acts as a cofactor for thiamine, and should also be repleted if levels are found to be low.

KEY CONCEPTS

Multiple Sclerosis

- The clinical picture of multiple sclerosis (MS); is one of marked heterogeneity. The classic clinical syndrome consists of recurring episodes of neurologic symptoms that rapidly evolve over days and slowly resolve over weeks.
- Magnetic resonance imaging (MRI) is a high-yield diagnostic test for MS. When the emergency department (ED) MRI is an option, it should be considered as it may expedite follow up and initiation of treatment.
- MS relapse is treated with high-dose methylprednisolone, typically 1000 mg IV over a course of 3 to 5 days. There is emerging evidence supporting oral therapy as efficacious in the treatment of relapse.

Cerebral Venous Thrombosis

- Cerebral venous thrombosis (CVT) should be suspected in patients (particularly female patients under age 50) presenting with stroke symptoms without risk factors, unexplained new seizures or refractory headaches, or signs of intracranial hypertension.
- The combination of MRI and magnetic resonance venography (MRV) is considered the gold-standard for diagnosis of CVT, though contrast-enhanced computed tomography (CT) with venous phase imaging is an alternative.
- Primary treatment of CVT in the ED is anticoagulation with a weak recommendation for low molecular-weight heparin over unfractionated heparin. Neuro-interventional treatment may have a role in severe cases, and hospital transfer may be needed for advanced neurological care.

Trigeminal Neuralgia

- Trigeminal neuralgia (TN) is characterized by intermittent, unilateral, severe and sharp facial pain precipitated by innocuous sensory stimuli and not explained by other local causes.
- Secondary TN can be associated with MS and other cranial nerve disorders and neurologic conditions. A careful history and physical examination can help identify these causes though imaging may be needed.
- Sodium-channel blockers are first line therapy, either carbamazepine (200 to 400 mg/day, titrated up to 1200 mg/day) or oxcarbazepine (300 to 600 mg/day in two divided doses, titrated up to 1800 mg/day); adverse side effects may lead to treatment failure. While secondary therapies are available, treatment failure with sodium-channel blockers is an indication for surgical referral.

Facial Nerve Paralysis

- The diagnostic dilemma related to facial nerve paralysis typically revolves around distinguishing Bell's palsy from other causes. While diagnostic imaging is not needed to make a diagnosis of Bell's palsy, it may be indicated in cases where the etiology of facial weakness is unclear.
- Patients at high risk for stroke and infection or with features atypical for Bell's palsy (particularly intact forehead movement or bilateral facial paralysis) should be considered for an alternate diagnosis and often require additional diagnostic testing.
- Corticosteroids improve the outcome in Bell's palsy and should be started as soon as possible to maximize their benefit. We recommend 5 days of prednisone, 60 mg, followed by a taper over 5 days.
- Adding antiviral therapy to corticosteroids confers little or no benefit and evidence supporting its use is very weak except in cases of Ramsay Hunt Syndrome.

Vestibular Schwannoma

- Vestibular schwannomas are a common cause of sensorineural hearing loss and may lead to other distressing neurological symptoms. The primary role of the emergency clinicians is referral for testing and treatment when the diagnosis is suspected or confirmed.

Diabetic Cranial Mononeuropathy

- The differential diagnosis for diabetic cranial mononeuropathy includes aneurysm and stroke; a careful history and physical examination can identify cases that require neuroimaging.
- Diabetic patients presenting with a CN III palsy with spared pupillary response and no other deficits in general do not require neuroimaging.

1. A 30-year old woman with no past medical history presents complaining of weakness and spasms in both legs. She describes prior episodes of numbness in her feet that lasted for a few weeks at a time before resolving. She also notes right eye pain, and unilateral papilledema is noted on fundoscopic exam. What is the most appropriate next step in this patient's management?

a. Lumbar puncture
b. Ophthalmology consultation
c. Discharge home and referral to neurologist
d. MRI of the brain and spinal cord
e. IV corticosteroids

Answer: d. This patient is presenting with symptoms suggesting a new diagnosis of Multiple Sclerosis. Early diagnosis and initiation of therapy has been shown to improve outcomes. MRI is a sensitive test in identifying lesions consistent with MS. While MRI may not always be an option in the ED, it should be strongly considered and pursued when available. Lumbar puncture is also a useful test in the evaluation of possible MS, and while there are characteristic CSF findings, LP is of less initial utility than MRI. Non-contrast head CT may be useful in investigating other causes, but it will not identify MS lesions. Discharge home with neurology home is appropriate, but MRI in the ED would likely expedite diagnosis and initiation of therapy.

2. A 65-year-old woman with a history of MS presents with worsening pain, paresthesias, and weakness in her right leg over the last 3 days, which she describes as common symptoms during relapse. She has 4/5 motor strength and increased deep tendon reflexes in her right leg. These findings are noted on prior ED notes. She has a temperature of 102.6°F and otherwise normal vital signs: What is the most appropriate next step in this patient's management?

a. Obtain CBC, chest x-ray, and urinalysis
b. CT of brain
c. Rapid initiation of IV antibiotics followed by lumbar puncture
d. Discharge home with NSAIDs for fever and pain
e. Discharge home with oral corticosteroids

Answer: a. This patient is presenting with an exacerbation of MS. A relapsing and remitting pattern is characteristic. Infection is a common precipitating cause of MS relapse, and investigating the source of infection can help determine the need for and selection of antibiotic therapy. While there may be a need for lumbar puncture and imaging of the brain, these are not necessarily indicated in this scenario. In addition to a possible need for antibiotic therapy, IV or oral corticosteroids can be used to manage the symptoms of MS relapse.

3. A 38-year-old woman presents 8 weeks postpartum with a 1-week history of severe headache and progressively altered mental status, which culminated in a seizure several minutes before presentation. On examination, she is normotensive, appears postictal, but has no focal neurologic findings. Ophthalmoscopic examination reveals papilledema, and non–contrast-enhanced head computed tomography (CT) reveals a dense sagittal sinus and a small venous hemorrhage in the occipital region. What is the most appropriate next step in this patient's management?

a. 325 mg aspirin
b. Dexamethasone 10 mg IV push
c. Hypertonic saline 500-mL bolus
d. Mannitol 1 g/kg
e. Systemic anticoagulation with unfractionated heparin or low-molecular-weight heparin (LMWH)

Answer: e. The patient presents with a venous sinus thrombosis. Although large, randomized trial data do not exist, case series and expert consensus strongly suggest improved outcomes with systemic anticoagulation, even in the setting of venous hemorrhage on head CT. Osmotic agents and steroids have no proven benefit in the management of sinus thrombosis and may cause harm. Antiplatelet agents may be considered if absolute contraindications to anticoagulation exist but probably have lower therapeutic efficacy.

4. A 26-year-old man presents with a left facial droop. The symptoms began painlessly 3 days ago without a prodrome. Examination reveals a left facial droop with an inability to wrinkle the forehead without other associated physical examination abnormalities. What is the most appropriate next step in this patient's management?
a. Discharge home with reassurance that symptoms usually resolve in a few weeks
b. Initiation of oral antiviral therapy
c. Initiation of oral corticosteroids
d. Serologic testing for Lyme disease
e. Urgent non–contrast-enhanced computed tomography (CT) scan of the head

Answer: c. The patient has Bell's palsy, which is a painless left facial nerve palsy. The primary symptom is a left peripheral nerve paralysis often with unilateral dysgeusia, hyperacusis, and external canal and pharyngeal numbness. Facial sensation is intact. Corticosteroids have been shown to improve outcomes, and they should be initiated as soon as possible. Antiviral therapy in combination with corticosteroids may be considered (conflicting evidence exists) but they are less efficacious than corticosteroids when given alone. Given the classic picture and lack of other associated neurologic findings, CT scan of the head is not required and is not likely to provide benefit. Testing for Lyme disease should be considered in Lyme endemic areas or in patients with a history of a tick bite.

5. Which of the following clinical features would make a diagnosis of Bell's palsy less likely?
a. Increased sensitivity to sound
b. Decreased tearing on the affected side
c. A prodromal viral-like illness
d. Rash
e. Altered sensation of taste

Answer: d. In addition to the unilateral facial weakness involving the forehead, Bell's palsy may present with increased sound sensitivity (hyperacusis), altered sense of taste (dysgeusia). Decreased tearing may also occur, and viral illness is known to be a precipitating cause. Rash is not characteristic of Bell's palsy, and the presence of a herpetiform vesicular eruption should raise suspicion for Ramsey-Hunt syndrome. Other features that raise suspicion for an alternative diagnosis include vestibular symptoms, bilateral or forehead-sparing distribution, involvement of other cranial nerves, and signs of systemic illness.

6. A 62-year-old woman presents with one day of episodic sharp unilateral facial pain brought on often by combing her hair. She has had several similar episodes over the last year. Examination of the head and neck, as well as cranial nerves, is unremarkable. Which of the following is the most appropriate next step in this patient's management?
a. The patient should be started on carbamazepine and referred to a neurologist
b. Neuroimaging
c. Discharge with short course of oral prednisone.
d. Timely referral to a neurosurgeon
e. NSAIDs and reassurance that condition generally resolves without therapy

Answer: a. This patient has met diagnostic criteria for trigeminal neuralgia (TN). TN can be a debilitating condition and is marked by episodic severe episodic facial pain brought on by innocuous stimuli. TN is a clinical diagnosis, and while some suggest that MRI be part of the diagnostic process to evaluate secondary causes of TN, urgent neuroimaging is not indicated in the absence of neurologic deficit or other concerns. First line therapy are sodium-channel blockers such as carbamazepine. Patients who continue to have symptoms with pharmacologic therapy should be referred to a surgeon for consideration of microvascular decompression.

7. A 58-year-old woman presents with progressive right-sided hearing loss along with a sensation of fullness in the ear and otalgia. Her otic canal and tympanic membrane appear normal and she has no mastoid tenderness. Her cranial nerve exam demonstrates right sensorineural hearing less, and the remainder of her neurologic exam is normal. Which of the following is the most appropriate next step in this patient's management?
a. Admission for IV steroid therapy
b. Discharge home with instructions for outpatient workup
c. Head CT
d. Prompt otolaryngology consultation
e. Discharge home with otic hydrocortisone/neomycin/polymyxin for pain control

Answer: b. This patient likely has a vestibular schwannoma (VS), a slow-growing tumor of the vestibular nerve that is the most common cause of asymmetric sensorineural hearing loss. Patients may also complain of headache, disequilibrium, and tinnitus. Workup of potential VS, which includes MRI and audiometry, can generally be performed as an outpatient, with subsequent referral to an otolaryngologist or neurosurgeon. CT lacks the necessary sensitivity in the posterior cranial fossa to reliably rule out the presence of vestibular schwannoma.

8. A 54-year-old man with a history of poorly controlled diabetes presents with diplopia. Physical exam reveals the inability to move the right eye superiorly and medially, accompanied by ptosis. The right pupil in larger than the left, and poorly responsive to light. Which of the following is the most appropriate next step in this patient's management?
a. Oral corticosteroid therapy
b. Discharge with patient education on importance of blood sugar control to minimize risk of stroke.
c. Prompt referral to an ophthalmologist
d. MRI brain
e. Initiation of systemic anticoagulation therapy

Answer: d. Cranial neuropathies are uncommon and usually a complication of diabetes. They most often affect extraocular muscles, causing ophthalmoplegia. The oculomotor nerve (III) is most commonly involved, leading to the inability to move the affected eye superiorly and medially. The etiology of diabetic mononeuropathy is thought to be microvascular ischemia. Fibers controlling pupillary size and light reflex run superficially, and while they can be affected by microvascular ischemia, they are more suspectable to external compression. Ophthalmoplegia with pupillary involvement should raise concern of a posterior cerebral artery aneurysm causing compression of CN III. MRI or CT angiogram is indicated in this situation. In the absence of pupillary involvement or other cranial nerve abnormalities, there is not a clear requirement for imaging in the ED. Attention to modifiable risk factors is important as there may be an association of diabetic mononeuropathy with subsequent stroke risk.

92 Spinal Cord Disorders

KEY CONCEPTS

- Nontraumatic spinal cord disorders can be intrinsic or extrinsic, some of which require prompt diagnosis, advanced imaging, and specialist intervention to prevent or limit permanent neurologic dysfunction.
- The bulbocavernosus reflex is cord-mediated. Return of this reflex following a spinal injury marks the termination of spinal shock.
- Anterior cord syndrome is marked by symmetrical motor loss but intact proprioception and vibration sense.
- In patients with sudden severe back pain, consider spinal subarachnoid hemorrhage (SSAH) or spinal epidural hematoma (SEH), both of which are diagnosed using magnetic resonance imaging (MRI).
- Transverse myelitis is inflammation of the spinal cord often associated with a prior viral infection resulting in paraplegia and a defined sensory level impairment. MRI with contrast enhancement is the diagnostic modality of choice. Roughly one-third of patients have a good outcome.
- Cauda equina syndrome can be difficult to differentiate from conus medullaris lesions because both can result in bladder retention, fecal incontinence, leg weakness, and sensory loss in the perineum. Conus lesions are more typically bilateral, whereas cauda equina syndrome may be unilateral. Upper motor neuron findings are expected with conus lesions while cauda equina syndrome is associated with hypo- or areflexia.
- Central cord syndrome is often due to a hyperextension injury. Physical findings are represented with the mnemonic **MUD**: **M**otor deficits > sensory, **U**pper extremities > lower, **D**istal extremity findings > proximal.
- Brown-Séquard syndrome is due to a functional hemisection of the spine (frequently traumatic) resulting in ipsilateral loss of motor function, proprioception, and vibration with contralateral loss of pain and temperature sensation below the level of injury.
- The diagnostic imaging of choice in the majority of suspected spinal disorders is MRI with contrast.
- A syrinx is a cavitary lesion in the spinal cord that presents with a sensory disassociation predominately in the upper extremities. With progression, it can lead to upper extremity weakness and wasting. Symptom exacerbation with cough or Valsalva is typical.
- With compressive lesions of the spinal cord, neurologic status at the time of intervention and the duration of symptoms are directly related to outcome.
- Autonomic dysreflexia is a complication of spinal cord injury that can result in life-threatening hypertension. Hypertension may be the result of bladder distention, fecal impaction, pain, or infection. Treatment focuses on blood pressure management and the identification and treatment of inciting noxious stimuli.

1. A 43-year-old man with a history of alcohol abuse and liver cirrhosis presents with a complaint of one week of progressive difficulty ambulating and low thoracic back pain. He is noted to be febrile on examination. Which of the following diagnostic modalities is most appropriate in this patient?

a. Blood cultures
b. CT myelography
c. Lumbar puncture
d. MRI with IV contrast
e. White blood cell (WBC) count

Answer: D. The patient has a presentation concerning for spinal epidural abscess (SEA). MRI is the imaging modality of choice and should be done emergently. Lab tests may be helpful in supporting the diagnosis but are neither sensitive nor specific. CT may show bony abnormalities concerning for SEA but should not replace MRI. Blood cultures are not useful as a diagnostic tool but may reveal the causative organism and help to guide antibiotic therapy. Lumbar puncture is relatively contraindicated in these patients because it may further seed the infection.

2. Which feature most often distinguishes a conus medullaris from a cauda equina lesion?

a. Back pain
b. Bilateral symptoms
c. Distal motor weakness
d. Sacral anesthesia
e. Urinary incontinence

Answer: B. Because the conus medullaris is a small structure with lumbar and sacral segments represented in a minimal area, a lesion will likely result in bilateral symptoms. As cauda equina is not a true cord lesion but rather reflects dysfunction at the level of nerve roots, the symptoms are frequently unilateral. Both can result in urinary retention, fecal incontinence, saddle anesthesia, and distal motor weakness. Conus medullaris may have upper motor neuron signs (hyperreflexia and increased tone), which will be absent in cauda equina.

3. A 27-year-old woman presents with complaints of back pain and difficulty walking. Her symptoms have been progressive for 2 days. She has no significant past medical history. On history it is discovered that she recently recovered from a bout of "the flu" approximately 3 weeks ago. Physical examination is remarkable for lower extremity hyperreflexia, moderate symmetrical lower extremity weakness, moderate increased tone, a T10 level of sensory loss, and a PVR urine volume of 350 mL. What should be the next intervention?

a. Antibiotics
b. Complete blood count, erythrocyte sedimentation rate (ESR), and antinuclear antibody levels
c. MRI scan with gadolinium enhancement
d. Emergent neurosurgical intervention
e. Steroids

Answer: C. Transverse myelitis is postinfectious in 30% of cases and also idiopathic in 30%. Other causes are autoimmune disorders and infections. Symptoms are typically rapid in onset and progress during 1 or 2 days. Symptoms are often sensory loss below the level of insult, extremity weakness, decreased sphincter tone, and urine retention (often exhibited with an elevated PVR). Back pain may accompany it. Emergent MRI is indicated to rule out other causes. There is no proven efficacious treatment, although steroids have been used. There is an association with multiple sclerosis (MS). Prognosis for recovery is only fair.

4. A 23-year-old male patient with no past medical history presents with longstanding, recurrent headache and neck pain that he reports has been gradually worsening over several months. He denies any fevers, weakness, numbness, vision changes, back pain, or bowel/bladder changes. He notes that he is often afraid to cough or sneeze because that makes the symptoms worse. What is another clinical feature of this patient's most likely diagnosis?

a. Normal gait
b. Loss of pain and temperature sensation in the upper extremities
c. Lower extremity weakness
d. Bladder dysfunction
e. Loss of proprioception and light touch

Answer: B. Syringomyelia typically presents with headache, neck pain, and variable upper extremity dissociative anesthesia: symmetrical loss of pain and temperature sensation with preserved posterior column function. With progression, upper extremity weakness or wasting and lower extremity upper motor neuron changes are expected. Exacerbation with cough and the Valsalva maneuver is typical. There is a 90% association with type I Arnold-Chiari malformation (cerebellar tonsils and medulla projecting into the spinal canal—often the cause of the typical occipital headaches). MRI is diagnostic.

5. In the majority of patients with an extrinsic compressive lesion, surgical outcome is most dependent on which of the following factors?
   a. Age
   b. Laterality of symptoms
   c. Location of lesion
   d. Neurologic status at the time of procedure
   e. Primary source of neoplasm

Answer: D. In patients with SEH or SEA, prognosis is intimately linked with neurologic function at time of decompressive laminectomy or operative intervention. Patients who have had symptoms greater than 72 hours with SEH are unlikely to regain lost function. In SEA, neurologic status rarely improves if intervention is delayed longer than 12 to 36 hours from onset of paralysis.

6. A 37-year-old female with a prior history of transverse myelitis at the T4 level one year ago presents with several hours of sweating, nausea, and headache. At baseline, she is paraplegic to the lower extremities, performs intermittent bladder catheterization, and is insensate inferiorly to the T4 level. She has never had these symptoms before. Her vitals are: blood pressure 270/130, heart rate 54, respiratory rate 16, oral temperature of 97.2°F. ECG performed in triage shows no signs of ischemia or infarct. Which of the following actions should be taken next?
   a. Manage blood pressure with nitrates and perform a thorough physical examination including bladder scan and rectal exam.
   b. Administer aspirin and obtain cardiology consult.
   c. Obtain an emergent CTA of chest to assess for aortic dissection.
   d. Administer benzodiazepines.
   e. Order an MRI with IV contrast.

Answer: A. The patient is experiencing autonomic dysreflexia, seen frequently in patients with spinal cord injuries above the T6 level. Her severe hypertension, diaphoresis, nausea, and headache can mimic other diagnoses, but her history of spinal cord lesion must be taken into consideration. Management focuses around aggressive blood pressure management with nitrates and investigation into the possible noxious stimuli that is causing the syndrome. Bladder-related etiologies (particularly distention or detrusor overactivity) cause most episodes of autonomic dysreflexia. Other causes are fecal impaction, pain, tight clothing, and infection.

7. A 34-year-old male presents after a motorcycle versus car collision. The patient was wearing a helmet but thrown from his bike. He reports burning discomfort and significant weakness to both arms. He also reports some mild numbness but no clear bladder symptoms. What additional findings would you expect to find on physical examination?
   a. Decreased rectal tone
   b. Loss of pain and temperature below the level of the injury
   c. Distal muscle groups are affected more than proximal muscle groups
   d. Loss of position, vibration, and light touch sensations
   e. Lower extremity weakness

Answer: C. The patient has likely suffered a cervical hyperextension injury resulting in a central cord syndrome. Remember the pneumonic "MUD" for central cord syndrome: motor > sensory, upper > lower, distal > proximal. Patients with central cord syndrome have variable sensory impairment and bladder dysfunction and these symptoms are not hallmarks of this partial cord syndrome. Brown-Séquard syndrome is characterized by ipsilateral loss of motor function, proprioception, and vibration with contralateral loss of pain and temperature sensation below the level of injury. Anterior cord syndrome is characterized by loss of motor function, pinprick, and light touch below the level of the lesion with preservation of the posterior column modalities.

8. An 11-month-old female presents to your emergency department with low-grade fevers and progressive refusal to crawl or pull up. Mom reports she took the patient to her pediatrician last week but they didn't find any abnormalities. However, her symptoms have continued to worsen. On exam, you see an otherwise healthy infant who now appears somewhat irritable. Vital signs are as follows: HR 110, RR 18, T 100.2°F. The patient is being held by mom and resists movement. Most of her physical exam appears normal, including a nonfocal neurologic exam, but she cries as you palpate the lumbar spine. Which of the following is true of the likely diagnosis?
   a. Blood cultures frequently reveal the causative organism.
   b. WBC is a helpful finding if elevated.
   c. Plain radiographs are sufficient for diagnosis.
   d. IV antibiotics are usually curative.
   e. The lack of associated neurologic deficits is unusual in this case.

Answer: D. The child has a presentation consistent with discitis. These children may present with sepsis and, as this is an unusual diagnosis, the provider should maintain a broad differential. Children are more likely to develop infections in the vertebral disc and usually have developed symptoms over several weeks. A hallmark of the diagnosis is that patients have normal neurologic exams. MRI is the imaging modality of choice as it is diagnostic and helps to rule out other conditions. The white blood cell (WBC) count is usually normal and blood cultures are usually negative. IV antibiotics are effective at treating discitis and surgery or aspiration are rarely needed.

KEY CONCEPTS

- The diagnosis of a specific peripheral neuropathy generally requires confirmatory ancillary testing; approach in the ED should focus on identifying one of seven categorical patterns.
- Diagnostic approach to peripheral neuropathies involves combining three clinical features: (1) right-left symmetry or asymmetry, (2) proximal-distal location, and (3) sensorimotor modalities affected.
- Any patient with symmetrical weakness, distributed both proximally and distally, with loss or diminution of deep tendon reflexes (DTRs) and variable sensory abnormalities should be treated as having Guillain-Barré syndrome (GBS).
- Respiratory compromise is the primary life-threatening event seen in some peripheral neuropathies; GBS is by far the most common peripheral neuropathic cause of respiratory arrest.
- The definitive treatments for GBS are plasma exchange or intravenous immune globulin (IVIG).
- Most polyneuropathies are characterized by a pattern of distal, symmetrical sensorimotor findings, worse in the lower than in the upper extremities, with a stocking-glove distribution of sensory abnormalities that gradually diminishes as one moves proximally.
- High-level evidence supports the use of pregabalin, gabapentin, and the serotonin and norepinephrine reuptake inhibitor duloxetine in the treatment of diabetic distal symmetrical polyneuropathy (DSPN).
- Radial nerve mononeuropathies are characterized by wrist and finger drop and mild numbness over the skin of the first dorsal interosseus muscle.
- Humeral shaft fractures are associated with radial nerve injury, with "wrist drop" being the hallmark clinical finding.
- The ulnar cutaneous innervation to the hand branches from the main trunk proximal to the nerve entering the Guyon canal. Thus, a lesion at the wrist should not produce sensory abnormalities, whereas one at the elbow would be expected to do so.
- The most specific finding for carpal tunnel syndrome (CTS) is splitting of the sensation on the fourth digit (i.e., normal sensation of the ring finger on the ulnar palmar side with abnormal sensation on the median [radial] palmar side of the same finger).
- Lateral femoral cutaneous mononeuropathy (meralgia paresthetica) is caused by injury to this pure sensory nerve as it passes through or over the inguinal ligament, where it may become entrapped or kinked.
- The most striking feature of a complete common peroneal mononeuropathy is footdrop caused by weakness of foot dorsiflexion.
- The most common neurologic abnormality in Lyme disease is unilateral or bilateral facial nerve palsy, usually occurring within a month of exposure.
- ALS is the most common form of motor neuron disease (MND), and diagnosis requires the presence of both upper and lower motor neuron findings.

1. Which category of peripheral neuropathy tends to occur in an asymmetrical, distal distribution?
- **a.** Autonomic neuropathy
- **b.** Large-fiber neuropathy
- **c.** Mixed motor and sensory neuropathy
- **d.** Pure motor neuropathy

Answer: D. Pure motor and pure sensory peripheral neuropathies tend to occur in an asymmetrical distal pattern.

2. A 26-year-old woman presents with a chief complaint of weakness. She notes a 1- or 2-day onset of easy fatigability and diminished ability to navigate stairs. She has no past history and takes no medications. Vital signs are normal. Physical examination reveals absent lower extremity deep tendon reflexes (DTRs); symmetrical weakness of the quadriceps, calf muscles, and foot/toe dorsiflexion; and minimal sensory loss. Cranial nerve and upper extremity examination is normal. Which of the following is likely?
- **a.** An antecedent viral illness
- **b.** Lack of anal sphincter tone
- **c.** Onset of ocular muscle dysfunction
- **d.** Sparing of the autonomic nervous system

Answer: A. Guillain-Barré syndrome (GBS) is characterized by fairly acute onset of ascending weakness, loss of deep tendon reflexes (DTRs), and variable sensory loss. Antecedent infections often trigger, with common organisms being campylobacter, cytomegalovirus, Epstein-Barr virus, and mycoplasma. Rarely, symptoms begin in the upper extremities. Urinary retention is common, but anal tone is preserved. Ocular muscles are usually spared. Autonomic neuropathy is common, with marked variations in heart rate and blood pressure. Patients with predominantly sensory symptoms tend to have less risk of respiratory embarrassment and a more favorable prognosis. Lumbar puncture shows cerebrospinal fluid (CSF) pleocytosis or may be normal early on.

3. A 26-year-old woman presents with lower extremity weakness and difficulty walking. Examination is remarkable for lower extremity symmetrical weakness with mild symmetrical sensory loss and absent lower extremity deep tendon reflexes (DTRs). Symptom onset has been during 2 days. What should be the next step?
- **a.** Emergent magnetic resonance imaging (MRI)
- **b.** Intravenous immune globulin (IVIG)
- **c.** Lumbar puncture and antibiotics
- **d.** Pulmonary function studies

Answer: D. All patients with Guillain-Barré syndrome (GBS) are at risk of respiratory failure. A forced vital capacity (FVC) of less than 20 mL/kg and a negative inspiratory force of less than 30 cm H_2O are associated with impending ventilatory failure and the need for intubation.

4. Among patients with Guillain-Barré syndrome (GBS) who have normal pulmonary function, which of the following can be monitored to predict impending ventilatory failure?
- **a.** Deltoid strength
- **b.** Extensor neck strength
- **c.** Hand grip strength
- **d.** Masseter strength

Answer: B. Extensor muscle strength has been shown to correlate with ventilatory muscle strength.

5. A 53-year-old diabetic presents with a complaint of increasing difficulty walking in the last several months. He has no other past history but has been an insulin-dependent diabetic for 23 years. Current glucose level is 138 mg/dL, and chemistries and complete blood count are otherwise unremarkable. Examination is remarkable for bilateral lower extremity numbness extending symmetrically to above the knees, loss of the Achilles reflex bilaterally with footdrop, and steppage gait. Which of the following is true?
- **a.** Autonomic neuropathy is unlikely.
- **b.** Facial numbness would necessitate MRI.
- **c.** Hand numbness is expected.
- **d.** Erythrocyte sedimentation rate is likely to be elevated.

Answer: C. Diabetic neuropathy is a progressive, ascending mixed polyneuropathy. Hand numbness and upper extremity symptoms usually begin before the lower extremity symptoms ascend to the knees. Extensive motor loss can occur with gait and grip abnormalities. Skull and face numbness can occur. Autonomic dysfunction is expected.

KEY CONCEPTS

- In patients presenting with acute neuromuscular weakness, complaints of difficulty in breathing or swallowing are signs of bulbar nerve or respiratory muscle compromise with potential airway or ventilatory failure. In such patients, forced vital capacity (FVC) of less than 15 mL/kg or maximal negative inspiratory force of less than 15 mm H_2O is a potential indication for mechanical support, either by intubation or by noninvasive ventilation.
- Initiating new medications in patients with myasthenia gravis may precipitate a crisis. Commonly prescribed cardiovascular agents, including calcium channel blockers, beta-blockers and antidysrhythmics; commonly used antibiotics; and other agents, including corticosteroids, require particular vigilance.
- Myasthenic crisis is treated with either plasma exchange or intravenous immune globulin (IVIG).
- Botulism usually arises as a painless descending paralysis, often first affecting the cranial nerves (with diplopia) and bulbar innervated muscles, without sensory deficits or alteration of consciousness. The treatment is airway management with ventilatory support and administration of antitoxin.
- Injection drug use remains an important cause of wound botulism outbreaks.
- Botulism is an important consideration in evaluation a weak and floppy infant.
- Hypokalemic periodic paralysis is caused by intracellular shift of potassium and so requires frequent serum potassium measurement to avoid overcorrection during treatment with potassium infusion.
- Patients with newly diagnosed hypokalemic periodic paralysis should be evaluated for hyperthyroidism.

1. Which of the following is not found in a neuromuscular junction?
   a. Myelin sheaths
   b. Presynaptic receptors
   c. Postsynaptic receptors
   d. The synaptic cleft
   e. Acetylcholine

Answer: a. The neuromuscular junction is an important unit in regulating muscle function. It consists of both presynaptic and postsynaptic receptors that release and receive the neurotransmitter acetylcholine as it travels through the synaptic cleft. Myelin sheaths are important in quickening the transmission of signals along an axon but do not extend all the way to the neuromuscular junction.

2. Which of these features in a patient's history suggest a diagnosis of myasthenia gravis?
   a. History of small cell lung carcinoma
   b. Worsening of symptoms following a large meal
   c. Ascending paralysis
   d. An erythematous periorbital rash
   e. Improvement of symptoms following plasma exchange

Answer: e. Myasthenia gravis, along with several other neuromuscular disorders, can be treated with plasma exchange. History of small cell lung carcinoma is associated with Lambert-Eaton syndrome. Ascending paralysis is seen more commonly in Guillain-Barré syndrome and tick paralysis. An erythematous periorbital, or heliotrope, rash is seen in dermatomyositis.

3. A 40-year-old female patient with a history of myasthenia gravis presents to the emergency department with mild shortness of breath upon exertion. You are concerned for an impending myasthenic crisis. Which of the following patient factors would most suggest this diagnosis?
   a. Bilateral ptosis
   b. Inability to tolerate secretions
   c. A forced vital capacity (FVC) of 65 mL/kg
   d. A negative inspiratory force (NIF) of 40 cm H_2O
   e. Recent initiation of a new medication for hypertension

Answer: b. Myasthenic crisis is an emergent, life-threatening exacerbation of myasthenia gravis that involves acute weakness of the muscles of the respiratory system. Detecting signs of impending myasthenic crisis and differentiating this from a regular exacerbation of myasthenia gravis is crucial for emergency medicine providers. Pulmonary function tests, such as forced vital capacity (FVC) and negative inspiratory force (NIF) can suggest impending crisis, but these tests should not be the sole factors to consider. Physical exam findings of significant muscle weakness or respiratory distress are the most concerning signs for impending myasthenic crisis. Bilateral ptosis is a sign of acute myasthenia but, by itself, likely does not indicate a significant exacerbation. Inability of the patient to tolerate oral secretions, on the other hand, is a much more concerning exam finding and implies significant muscle weakness and impending crisis.

The previous patient continues to have worsening respiratory symptoms. Both neurology and critical care teams are consulted and are at bedside. The patient is started on pyridostigmine. Which of the following findings would indicate worsening myasthenic crises as opposed to overtreatment with pyridostigmine?
   a. An electrocardiogram showing a heart rate of 35
   b. Pupil size of 7 mm bilaterally
   c. Increasing respiratory fatigue
   d. Significant abdominal discomfort
   e. Excessive sweating

Answer: b. When treating myasthenia gravis, worsening of the patient's condition can be caused by worsening myasthenic crisis or by cholinergic crisis from overtreatment of the patient with anticholinergic agents. These two conditions can present similarly but have different causes and thus different treatments. Worsening myasthenic crisis occurs from undertreatment of the initial crisis by anticholinergic agents where cholinergic crisis occurs from overtreatment of the initial crisis. Patients in cholinergic crisis tend to present with bradycardia, abdominal pain, and increased secretions. In both cholinergic crises and myasthenic crises, increased weakness and respiratory distress can be present. However, mydriasis is much more likely to be seen in myasthenic crisis.

An 8-month-old infant presents to the emergency department with increasing weakness and poor feeding. Physical exam reveals an ill-appearing infant with decreased muscle tone. What is the most DEFINITIVE treatment?
   a. Administration of antibiotics against *Clostridium* species
   b. Administration of corticosteroids
   c. Contacting the Centers for Disease Control and Prevention (CDC) for antitoxin
   d. Contacting the California Department of Health for antitoxin
   e. Serum and stool testing for botulinum toxin

Answer: d. This patient likely has infantile botulism and the definitive treatment is obtaining IV human-derived botulism immunoglobulin, or BabyBIG, from the California Department of Health. The CDC has a heptavalent botulinum toxin for noninfantile cases. IV corticosteroids and antibiotics are not recommended for cases of infantile botulism. Serum and stool testing can diagnose botulism but often have a prolonged turnaround time and so are not useful in emergency department management.

6. Which of the following is TRUE about wound botulism?
   a. It is the most common form of botulism.
   b. Antibiotics do not have a role in wound botulism management.
   c. Compared with other forms of botulism, wound botulism patients more often have abdominal complaints.
   d. It is associated with enteroviruses.
   e. It is associated with topical tobramycin treatment.

Answer: c. Wound botulism is caused by infection of a wound with *Clostridium botulinum* which then causes systemic infection via hematogenous spread. It is an infrequent presentation of *C. botulinum* infection; infantile botulism is the most common form of botulism. Antibiotics as well as surgical consultation are first line in treatment. Patients with wound botulism often present with fever and less often present with abdominal symptoms compared with other forms of botulism. Enteroviruses are not associated with botulism but are associated with acute flaccid myelitis (AFM). Tobramycin treatment can be a trigger for acute myasthenia gravis but not for wound botulism.

7. A patient presents with an episode of flaccid lower extremity paralysis. He notes that he frequently gets these episodes and the condition runs in his family. His serum potassium concentration is 5.9 mEq/L. What is the next best treatment step?
   a. Intravenous (IV) immunoglobulin
   b. IV hydration
   c. Thyroid function testing
   d. Magnesium supplementation
   e. High-dose insulin and dextrose

Answer: b. This patient likely is having an episode of hyperkalemic periodic paralysis. Hyperkalemic periodic paralysis is due to an imbalance in the distribution of potassium at a cellular level. IV hydration helps to rebalance the body's potassium stores. Testing of thyroid function and high-dose insulin therapy may be indicated but are not the immediate next steps. IV immunoglobulin and magnesium supplementation are not indicated in this case.

8. An 8-year-old patient presents to the emergency department with diffuse weakness and myalgias. His parents note that he recently had a muscle biopsy which confirmed an unspecified illness. You also note a rash on the patient's chest. What would you likely find on the patient's physical exam?
   a. Difficulty in sitting upright from a supine position
   b. Difficulty in holding a pencil or pen
   c. Fasciculations of proximal muscle groups
   d. Atrophy of proximal muscle groups
   e. Increase responsiveness of the patellar reflex

Answer: a. This patient likely has dermatomyositis. Dermatomyositis is an inflammatory myopathy that most commonly affects the proximal muscle groups and spares the distal muscle groups. This patient likely would have trouble sitting up as it uses the proximal hip flexor muscles. Distal muscles, such as those in the hand, are less commonly affected. Fasciculations are not commonly seen. Atrophy is generally a much later finding in inflammatory myopathies. Patients also tend to present with decreased patellar reflexes.

9. Which of the following pairs is correct regarding a neuromuscular disease and its pathophysiology?
   a. Botulism—concomitant gastroenteritis is almost always present
   b. Myasthenia gravis—autoantibodies attack postsynaptic acetylcholine (ACh) receptors
   c. Periodic paralysis—hyperkalemic form is most common
   d. Polymyositis—distal muscle weakness precedes proximal muscle weakness
   e. Tick paralysis—symptoms resolve quickly after the tick is removed

Answer: b. Myasthenia gravis is caused by autoantibodies that attack postsynaptic ACh receptors at the neuromuscular junction. Botulism usually does not present with gastroenteritis. Periodic paralysis is most commonly associated with hypokalemia. Polymyositis usually presents first in proximal muscles. Tick paralysis may take days to weeks to resolve after removal of an offending tick.

95 Central Nervous System Infections

KEY CONCEPTS

- CNS infection should be considered in all patients with headache, nuchal rigidity, fever, altered sensorium, or diffuse or focal neurologic findings.
- Patients with suspected CNS infection should be asked about history focused on risk factors for CNS disease and infection and receive a full neurologic examination.
- CSF testing is the most reliable method of assessing the presence of meningitis and should be obtained in all patients in which there is a suspicion for CNS infection.
- Early initiation of empirical antimicrobial therapy is recommended in cases of suspected acute CNS infection. This should occur before imaging or lumbar puncture.
- Patients without focal neurologic symptoms, altered level of consciousness or signs of increased intracranial pressure do not require imaging prior to lumbar puncture.
- We recommend steroid treatment before or alongside treatment with antibiotics in all cases of suspected meningoencephalitis.
- First-line treatment for healthy adults with suspected bacterial meningitis is ceftriaxone or cefotaxime plus vancomycin in most countries given high antibiotic resistance.
- Acyclovir is recommended for patients with suspected encephalitis because HSV is a common pathogen.
- Clinical course and risk factors, such as immunosuppression, dictate the need to test and treat for fungal and tuberculous meningoencephalitis.
- In the absence of a clear clinical picture of viral meningitis, patients should generally be empirically covered for bacterial meningitis and admitted to the hospital.
- Antibiotic chemoprophylaxis should be given for close contacts of patients with meningitis resulting from *N. meningitidis* and contacts of patients with *H. influenzae* living with immunocompromised or unvaccinated individuals.
- CT or MRI can be used in patients with a suspected brain abscess.
- MRI with gadolinium contrast of the entire spine is the definitive imaging test for patients with suspected spinal epidural abscess.

1. Which of the following presentations should receive a lumbar puncture to rule out CNS infection:
   a. 23-year-old healthy female with fever and stiff neck but minimal headache.
   b. 32-year-old male with diabetes and unexplained altered mental status.
   c. 67-year-old male with fever and new right arm weakness.
   d. All of the above

Answer: D. All patients with fever, headache, altered mental status or new neurologic deficits should be considered for possible meningitis or encephalitis. Lumbar puncture and CSF analysis is the most reliable method for assessing for CNS infection; imaging and blood tests are inadequate alone.

2. Patients with suspected bacterial meningitis should receive which of the following interventions FIRST?
   a. Non-contrast CT of the head
   b. MRI with contrast of the head
   c. Lumbar puncture to evaluate for bacterial cause
   d. Steroids and antibiotics for presumed bacterial infection

Answer: D. Patients with suspected bacterial meningitis should receive steroids and antibiotics first to cover for the possibility of bacterial infection. Treatment should not be delayed obtaining neuroimaging first (if necessary), or until the lumbar puncture is completed. However, lumbar puncture should still be obtained expeditiously, because culture yields begin to decrease after 1 hour.

3. Which of the following antimicrobial regimens would be appropriate coverage for suspected meningitis in a 40-year-old patient with no comorbidities in a country with high antibiotic use?
   a. Ceftriaxone
   b. Ceftriaxone and vancomycin
   c. Acyclovir and ceftriaxone
   d. Cefotaxime, vancomycin, and ampicillin

Answer: B. Ceftriaxone and vancomycin are the appropriate initial antibiotics for suspected meningitis or encephalitis in an adult patient. Ceftriaxone alone may be appropriate in lower income countries with less antibiotic use and resistance. Ampicillin is added for patients older than 50 years and neonates due to risk of Listeria. Acyclovir should be added for immunocompromised patients with suspected HSV or VZV infections. Cefepime should be used instead of ceftriaxone in patients at his risk for pseudomonas infections (e.g., recent surgical procedures).

4. What is the most appropriate imaging test for a patient with low back pain, low-grade fever, and a history of IV drug use?
   a. CT lumbar spine without contrast
   b. CT myelogram of the lumbar spine
   c. MRI of the lumbar spine with contrast
   d. MRI of the cervical, thoracic and lumbar spine with contrast

Answer: D. Because exam findings can be inaccurate, patients with suspected epidural abscess should receive an MRI with contrast of their entire spine. Presentations can be subtle and may not include all of the classical features such as fever and back pain; patients must be queried for potential risk factors. CT myelogram of the entire spine is an acceptable alternative for patients unable to undergo an MRI.

5. Contacts with exposure to which of the following types of meningoencephalitis should receive prophylactic antibiotics?
   a. Pneumococcal
   b. Tuberculous
   c. Meningococcal
   d. West Nile

Answer: C. Close contacts with exposure to patients diagnosed with meningococcal meningitis or encephalitis should receive chemoprophylaxis, most commonly with oral rifampin. Chemoprophylaxis is also recommended for immunocompromised close contacts of patients with H. influenzae CNS meningitis.

Thought Disorders 96

KEY CONCEPTS

- Thought disorder symptoms can be precipitated by psychiatric, underlying medical, and toxicologic etiologies.
- Diagnostic testing should be patient specific and based on the particular medical processes that the clinician feels may be causing or exacerbating the thought disorder, rather than panels of routine tests.
- Consider nonphysical interventions first when appropriate, but chemical sedation or physical restraints may become immediately necessary for patients who demonstrate life-threatening aggressive and dangerous behaviors.
- Appropriate disposition depends on the etiology of the underlying psychosis and response to treatment while addressing patient and community safety considerations. Psychiatric consultation is often required.

1. Which of the following pharmacologic agents have been implicated in causing acute psychosis?
- **a.** Aripiprazole, hydralazine, nitroglycerin
- **b.** Diazepam, rifampin, captopril
- **c.** Hydrochlorothiazide, acetaminophen, albuterol
- **d.** Lorazepam, salsalate, rocuronium
- **e.** Penicillin, ceftriaxone, risperidone

Answer: B. Box 96.2 provides an extensive list of other agents that may cause psychosis.

2. Rapid tranquilization using a neuroleptic agent would be indicated in which of the following cases?
- **a.** An intoxicated schizophrenic
- **b.** Anticholinergic psychosis
- **c.** A lactating schizophrenic
- **d.** A phencyclidine overdose
- **e.** A pregnant schizophrenic

Answer: A. Neuroleptics are contraindicated in choices B to E. They should not be the sole agent for alcohol withdrawal but would be useful for acute psychotic agitation.

3. A 45-year-old woman presents to the ED complaining of severe anxiety and unrest. She has a history of moderate schizophrenia, for which she was placed on olanzapine 2 months prior. She has been compliant. Physical examination is remarkable for the presence of anxiety, clear sensorium and orientation, and normal speech. She is restlessly pacing the room and reports being compelled to keep moving. Urine drug screen is negative. What would be the most appropriate therapy?
- **a.** Benztropine orally
- **b.** Lorazepam orally
- **c.** Olanzapine intravenously
- **d.** Psychiatry consultation
- **e.** Ziprasidone intravenously

Answer: A. Akathisia is a state of motor restlessness characterized by a physical need to be constantly moving. The patient does not want to do so but feels compelled. It is most commonly seen in middle-aged patients within the first few months of starting treatment. It may be mistaken for an acute deterioration, but psychotic features are not increased. Treatment is with oral beta-blockers and anticholinergics (benztropine).

4. What is the most common adverse effect seen with neuroleptic agents?
- **a.** Akinesia
- **b.** Dystonia
- **c.** Orthostatic hypotension
- **d.** Pseudoparkinsonism
- **e.** Tardive dyskinesia

Answer: B. Dystonia occurs in 1% to 5% of this patient population. The reaction occurs because of a dopaminergic pathway disruption with a resulting cholinergic predominance. Anticholinergics should be administered parenterally (Benadryl 25 to 50 mg intravenous [IV] or Cogentin 1 or 2 mg IV), followed by 48 to 72 hours of oral follow-up treatment to prevent recurrence. Patients may experience tongue protrusion (buccolingual crisis), upward eye deviation (oculogyric crisis), back arching (opisthotonus), and, rarely, laryngospasm. Symptoms may lessen with voluntary muscle action and increase with stress.

5. A 27-year-old known schizophrenic is brought to the ED for altered mental status. His only known medication is clozapine, which he started 4 weeks ago with subsequent dose increases. He has no other past history. Physical examination reveals a muscular man who is somnolent and diaphoretic. He withdraws all extremities stiffly and grimaces in pain. Vital signs are temperature, 40.5°C; heart rate, 146 beats per minute; blood pressure, 205/125 mm Hg; and respiratory rate 28 breaths per minute. Rectal examination is guaiac positive. Foley placement shows brown urine. What should be the next diagnostic maneuver?
- **a.** Creatine kinase level
- **b.** Head computed tomography (CT) scan
- **c.** Lumbar puncture
- **d.** Thyroid hormone levels
- **e.** Urine drug screen

Answer: A. Neuroleptic malignant syndrome is an idiopathic condition clinically similar to serotonin syndrome and malignant hyperthermia. Milder cases may be confused with serotonin syndrome. Severe cases, related to possible hypothalamic dysfunction, present with fever, rigidity, altered mental status, autonomic instability, elevated creatine phosphokinase (CPK), and possibly rhabdomyolysis. It is seen with both typical and atypical antipsychotics and generally occurs in the first few weeks of treatment. Complications may include hepatic/renal failure, gastrointestinal (GI) hemorrhage, and respiratory failure. Severe cases may require intravenous dantrolene or dopamine agonists (e.g., bromocriptine).

97 Mood Disorders

KEY CONCEPTS

- Patients with apparent mood disorders should be evaluated for medical disorders, medication effects, or substance abuse or withdrawal because these conditions can mimic both depression and mania.
- Mood disorders should be suspected in patients with multiple, vague, nonspecific complaints and in patients who are frequent, heavy users of medical care.
- Patients with mood disorders should be assessed for their suicide potential.
- Pharmacologic treatment and linkage to care after discharge are important parts of managing mood disorders in the emergency setting.

1. What is the lifetime suicide risk for people with major untreated depression?
   - **a.** 5%
   - **b.** 10%
   - **c.** 15%
   - **d.** 20%
   - **e.** 25%

Answer: c. Patients with major depression have a high lifetime suicide risk, and although episodes of acute decompensation with even higher risk can be identified and treated, a certain number of patients succeed in committing suicide.

2. Which of the following abnormalities of central nervous system neurotransmitters is associated with clinical depression?
   - **a.** Decreased hypothalamic-pituitary-adrenal (HPA) activity
   - **b.** Depressed serotonin levels
   - **c.** Elevated gamma-aminobutyric acid (GABA) levels
   - **d.** Elevated norepinephrine levels
   - **e.** Unchanged dopamine levels

Answer: b. The central biochemical features toward which pharmacologic management is directed are depressed levels of norepinephrine and serotonin. Data are also emerging that suggest decreased dopamine levels. The HPA axis may also be altered with elevated cortisol levels.

3. Which of the following statements regarding depression in children and elders is *true*?
   - **a.** Children with depression rarely present with somatic complaints.
   - **b.** Depression in children may be misdiagnosed as attention deficit disorder (ADD).
   - **c.** Depression presents differently from dementia in elders.
   - **d.** Diagnostic criteria for depression in children are different.
   - **e.** Serious depression in elders is generally refractory to treatment.

Answer: b. Depression in children and adolescents can be misdiagnosed as ADD. Somatic complaints are a common feature of children and adolescents presenting with depression, but the diagnostic criteria are not different. Geriatric depression may be manifested in a manner similar to dementia (pseudodementia), but unlike dementia, the depression is highly treatable and reversible once it is recognized.

4. A 31-year-old attorney is brought to the emergency department (ED) by his family for a chief complaint of agitation and a behavioral change. He has no past medical history and takes no medications. The family reports decreased sleep, increased talkativeness, marked increased time and involvement at work, and an uncharacteristic buying spree. Your examination is remarkable for distractibility, gregarious and pressured speech, flight of ideas, and mild psychomotor agitation. Laboratory examination and urine drug screen results are negative. The patient is adamant that he has important things to do and needs to leave. Which of the following statements regarding this patient's most likely diagnosis is true?
   - **a.** Antipsychotic agents are not effective.
   - **b.** Hallucinations would be atypical.
   - **c.** If treated, intravenous valproic acid is indicated.
   - **d.** Initiating treatment in the ED is not indicated.
   - **e.** Multiple antibiotics can cause this clinical picture.

Answer: e. This patient has a fairly classic presentation for acute mania with pressured speech, distractibility, grandiosity, increased involvement (in this case with work), and decreased need for sleep. Multiple drugs may precipitate this, including acyclovir, isoniazid, sulfonamides, the floxins, and chloroquine. An acute manic episode may be manifested with hallucinations and mimic an acute psychosis. ED treatment is usually indicated for this disorder. Acute stabilization is generally effective with major tranquilizers, such as haloperidol.

KEY CONCEPTS

- Patients who present with predominant symptoms of anxiety may be suffering from medical disorders, medication effects, or substance abuse or withdrawal.
- Anxiety may accompany the onset of serious medical disease, cause significant metabolic demands, and stress a marginally compensated organ system.
- Anxiety caused by non-psychiatric illness is usually suggested by the patient's physical examination findings but may require testing to further delineate the cause.
- Oral, intravenous, or intramuscular medication may be necessary for patients who are a significant threat to themselves or others and for anxious patients with significant medical illness.
- Limited benzodiazepine therapy may be helpful for select patients.

1. Which of the following is the most common mental health disorder?
   a. Anxiety
   b. Bipolar
   c. Depression
   d. Schizophrenia
   e. Substance abuse

Answer: A. Many of these patients never receive appropriate care, in part because they choose to present with a physical complaint and disguise their anxiety. Patients with chronic illnesses have higher rates of anxiety and depression than the rest of the population.

2. What is the most common cause of organic anxiety, anxiety that results from a physiologic origin?
   a. Adrenal disorders
   b. Alcohol and drug use
   c. Cardiac disease
   d. Hyperthyroidism
   e. Pulmonary embolus

Answer: B. This may be from intoxication or withdrawal states.

3. A 52-year-old woman presents with 2 months of recurrent episodes of anxiety, mild chest pain, subjective palpitations, hand paresthesias, and occasional muscle spasms. They have occurred weekly in the past but are now increasing in frequency. Her only past history is a thyroidectomy 4 months prior. She is taking levothyroxine (Synthroid) and had normal thyroid levels 2 weeks ago. Her vital signs, physical examination, and electrocardiogram are normal. Laboratory evaluation shows sodium 141 mEq/L, potassium 4.1 mEq/L, creatinine 1.0 mg/dL, bicarbonate 26 mEq/L, chloride 100 mEq/L, and calcium 7.1 mg/dL; a complete blood count is normal. Which of the following should be the next step in her management?
   a. Outpatient clonazepam
   b. Parathyroid hormone level
   c. Psychiatry consultation
   d. Thyroid hormone levels
   e. Urine drug screen

Answer: B. Anxiety is the predominant symptom in 20% of patients with hypoparathyroidism. Other symptoms include paresthesia, muscle cramps, and spasms. Most cases are idiopathic or due to inadvertent parathyroid gland harvest during thyroidectomy. The diagnosis is suggested by a low serum calcium and an elevated phosphate and is confirmed by a depressed parathyroid level.

4. Which of the following statements regarding anxiety and endocrine disorders is true?
   a. Anxiety can often be traced to reactive hypoglycemia.
   b. Anxiety is not a manifestation of hypothyroidism.
   c. Diabetics treated with antianxiety agents have improved hemoglobin A1c levels.
   d. Less than 5% of diabetics experience anxiety.
   e. Patterns of diaphoresis in pheochromocytoma mimic those of a panic attack.

Answer: C. Approximately 15% of diabetics have an anxiety disorder. Treatment improves hemoglobin A1c levels. Anxiety due to reactive hypoglycemia is rare despite the common perception among patients. Pheochromocytoma causes whole-body diaphoresis, whereas panic disorders primarily cause sweaty palms. Hyperthyroidism or hypothyroidism can cause significant anxiety manifestations. It is more related to the rate of change than the level of thyroid hormones.

5. A 23-year-old woman with a history of asthma presents with increasingly frequent episodes of panic attacks. Her medications are an inhaled beta-agonist and an intermittent steroid inhaler. She reports subjective increasing asthma severity as her panic episodes have worsened. When counseling the patient, which of the following statements is most correct?
   a. An anxiety disorder in an asthmatic patient does not increase morbidity.
   b. Anxiety does not precipitate asthma attacks.
   c. Anxiety does not worsen airflow.
   d. Asthmatics are more likely to have an anxiety disorder.
   e. It is difficult to differentiate dyspnea related to asthma from anxiety.

Answer: D. Anxiety can precipitate and prolong an asthma attack. Morbidity and mortality are increased in asthmatic patients who have a coexisting anxiety disorder. Patients who have asthma are twice as likely to have an anxiety disorder and five times as likely to have a phobia. Acute dyspnea from "panic" dyspnea can be differentiated from asthma by clear lungs on auscultation.

6. Which of the following syndromes is not associated with anxiety?
   a. Left hemispheric strokes
   b. Multiple sclerosis
   c. Right hemispheric strokes
   d. Transient ischemia attack
   e. All of the above can be associated with anxiety.

Answer: E. Anxiety may be a component of seizures, tumors, arteriovenous malformations, and ischemic events. It may be the only manifestation of some disorders (e.g., right hemispheric strokes and transient ischemic attacks [TIAs]). The coexistence of anxiety plays an important role in the prognosis and impairment of stroke patients.

7. A 38-year-old woman with a long history of anxiety and panic disorder presents with anhedonia, melancholy, sleep disruption, crying episodes, and some hostility feelings. She has no current anxiety symptoms. Her only medication is clonazepam. She has no known medical illness. Which of the following statements regarding this patient's symptoms is true?

- **a.** Approximately 50% of patients with panic disorder develop major depression.
- **b.** Depression with anxiety and hostility is usually refractory to treatment.
- **c.** The first diagnostic step should be a thyroid panel.
- **d.** The majority of patients with depression have panic attacks.
- **e.** This is likely a drug-induced depression.

Answer: A. Approximately 50% of patients with a primary panic disorder will later develop major depression. Twenty percent of patients with depression have panic attacks. Depression with panic attacks is less responsive to treatment, but depression with anxiety and hostility responds well to antidepressants. Although benzodiazepines can exacerbate symptoms of depression, there is already a high spontaneous rate of depression with anxiety disorders.

8. Which of the following statements regarding benzodiazepine use and anxiety is true?

- **a.** Benzodiazepines are first-line agents for anxiety disorders.
- **b.** Several weeks of treatment are indicated after initial diagnosis.
- **c.** Short-acting benzodiazepines produce a more severe abstinence syndrome.
- **d.** They are particularly useful in patients with alcohol abuse.
- **e.** Withdrawal rebound is less common than with selective serotonin reuptake inhibitors (SSRIs).

Answer: C. SSRIs are the first-line agents for anxiety and panic disorders, but the primary disadvantage is the several-week lag needed for maximal clinical benefit. Benzodiazepines work best for motivated, dependable patients when an immediate reduction of symptoms is indicated or a short-term treatment is necessary. Patients who do not benefit from benzodiazepines within a week are unlikely to do so. Patients with a history of alcoholism or drug abuse, who are excessively/emotionally dependent, or who become anxious from normal stress are at greater risk for dependency. Rebound withdrawal is more likely after short-acting agents.

9. A 29-year-old Caucasian female presents with excessive daytime somnolence. She states that she had been suffering from anxiety associated with her paralegal occupation, and 1 week ago her psychiatrist had started her on a 2-week course of once-daily benzodiazepine therapy, which she takes in the morning. Her anxiety symptoms are well controlled. She asks if you can change her to a new medication because the somnolence is significantly affecting her job performance. What would be the most appropriate course of action?

- **a.** Counsel the patient that she should continue the medication as prescribed because she will soon adapt and the somnolence will likely subside.
- **b.** Discontinue the benzodiazepine and refer her back to her psychiatrist.
- **c.** Have her try dosing the benzodiazepine at bedtime, because this will likely continue to control her anxiety and limit daytime somnolence.
- **d.** Switch the patient to a selective serotonin reuptake inhibitor (SSRI) and refer her back to her psychiatrist.
- **e.** Switch the patient to a shorter-acting benzodiazepine.

Answer: C. Instituting an SSRI should be reserved for primary care physicians or psychiatrists who can monitor the patient more closely, because the response will be delayed. Some patients do adapt to the sedative effects of benzodiazepines but usually only after long-term use. Stopping the benzodiazepine may ultimately be necessary but at the risk of recurrent anxiety. Dosing benzodiazepines at bedtime may minimize daytime sedation and still provide an anxiolytic effect. Shorter-acting benzodiazepines produce a more severe abstinence syndrome when stopped abruptly, and thus most prescribers prefer longer-acting agents.

10. A 52-year-old male construction worker presents with chest pain. He states his symptoms began early this morning and have progressively worsened throughout the day. His symptoms include nervousness, tremors, chest pain, shortness of breath, and palpitations. He states that he has had anxiety for 30 years but has controlled it with the consumption of alcohol. He became unemployed 1 week ago, and his daily alcohol use has diminished significantly. His vital signs are blood pressure (BP) 185/95 mm Hg, heart rate 123 beats per minute, respiratory rate of 20 breaths per minute, and temperature of 98.9°F. His physical examination is remarkable for diaphoresis, tongue fasciculation, both resting and intention tremors, and mild psychomotor agitation while maintaining orientation with a congruent anxious mood and affect. What is the most likely etiology of this patient's symptoms?

- **a.** Acute alcohol withdrawal syndrome
- **b.** Exacerbation of endogenous anxiety secondary to diminished alcohol intake
- **c.** Exacerbation of exogenous anxiety secondary to change in employment status
- **d.** Hypertensive emergency with acute coronary syndrome
- **e.** Reactive anxiety secondary to the onset of chest pain

Answer: A. Hypertensive emergency is unlikely given the level of this patient's BP. On the basis of the history alone, it may be difficult to differentiate organic versus functional anxiety or identify an exogenous trigger, but the abnormal vital signs and physical examination associated with a recent cessation of long-term alcohol consumption makes acute alcohol withdrawal the most likely cause. Given the significant morbidity associated with withdrawal states, this must be addressed acutely. Appropriate diagnosis and management of underlying psychiatric disease will be a secondary concern after the patient's withdrawal is managed.

99 Somatic Symptoms and Related Disorders

KEY CONCEPTS

- Several conditions previously classified in the fourth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) as somatoform disorders are now classified under DSM-V as *Somatic Symptom and Related Disorders* (SSRD). These include somatic symptom disorder (SSD), illness anxiety disorder (IAD), and conversion disorder. They share a common feature of patients' maladaptive and inappropriate psychological response to somatic (bodily) symptoms.
- SSRD patients have approximately twice the rate of medical disease seen in the general population. It is unclear whether this is the consequence of more frequent health care use or whether an increased disease burden prompts these patients to have a greater concern for bodily sensations.
- *SSD* is characterized by disproportionate or persistent health-related thoughts, anxiety, and time and energy devoted to somatic (bodily) symptoms, resulting in disruption of daily life.
- Patients with *IAD,* formerly known as hypochondriasis, have excessive anxiety regarding the possibly having or acquiring a serious medical illness in the presence of minimal or absent somatic symptoms.
- *Conversion disorder*, also known as *functional neurologic symptom disorder*, is characterized by abnormal sensory or voluntary motor function that is found to be incompatible with known neurologic or medical conditions, and that causes significant distress or life impairment.
- The differential diagnoses for SSRD may be broadly divided between (a) psychiatric disorders that manifest somatic symptoms and (b) medical conditions with signs or symptoms that might be attributed to psychiatric disorders.
- A "positive review of systems" in an emergency department (ED) evaluation is similar to a high score on SSRD symptoms severity scales and thus may serve as an inadvertent screen for SSRD. Further research is needed to ascertain whether this is clinically reliable.
- ED care goals for patients with SSRD include establishment of rapport, building a therapeutic alliance, legitimizing the patient's distress, and enhancing the patient's ability to function despite the symptoms.
- Multiple care modalities are available for SSRD treatment. These are typically managed by the patient's primary care physician or psychiatrist.

1. Which of the following characterizes patients with somatic symptom disorder (SSD)?
- **a.** Patients have an inappropriate lack of concern for profound disability.
- **b.** Patients intentionally and deceptively falsify symptoms.
- **c.** Patients strongly believe they have a particular serious medical illness.
- **d.** Patients have a maladaptive psychological response to bodily symptoms.

Answer: d. The defining feature of SSD is the psychological response to somatic symptoms.

2. A patient presents with the inability to move the left side of his body, but he is easily able to turn his head to the right. He also appears relatively unworried about his symptom severity. This presentation suggests which somatic symptom and related disorder (SSRD)?
- **a.** Somatic symptom disorder.
- **b.** Illness anxiety disorder.
- **c.** Conversion disorder.
- **d.** Psychological factors affecting medical illness.

Answer: c. Normal sternocleidomastoid function ipsilateral to the otherwise-paralyzed side and presence of *la belle indifference* suggest conversion disorder.

3. A patient presents multiple, nonspecific symptoms and believes that she has cancer despite multiple previous evaluations in emergency, primary care, and specialist settings. She says she is certain of this diagnosis, that it causes a significant amount of anxiety and that she spends several hours each day researching the subject. Your standard work-up for her presenting symptoms is negative. What should you do?
- **a.** Perform a few more tests to reassure the patient.
- **b.** Tell the patient that this is "all in your head" and promptly discharge.
- **c.** Place the patient under an involuntary psychiatric hold for self-harm.
- **d.** Seek therapeutic alliance, provide reassurance, and refer the patient to primary care or psychiatry for management.

Answer: d. This patient's presentation is consistent with illness anxiety disorder, formerly known as hypochondriasis. Note that patients with this disorder may have few or no somatic symptoms.

4. A predominance of pain is a subtype of which of the somatic symptom and related disorders?
- **a.** Somatic symptom disorder.
- **b.** Illness anxiety disorder.
- **c.** Conversion disorder.
- **d.** Psychological factors affecting medical illness.

Answer: a. Predominant pain is a subtype of somatic symptom disorder in which pain is the defining symptom. This may be due to prior neural sensitization or genetic predisposition.

5. Which of the following is considered the gold standard for diagnosing somatic symptom and related disorders (SSRD)?
- **a.** Elimination of other possible medical conditions via extensive medical testing.
- **b.** Structured clinical interview.
- **c.** A positive review of symptoms.
- **d.** Emergency physician gestalt.

Answer: b. SSRD diagnoses are suggested by overall clinical picture and may be supported by self-report questionnaires. However, final diagnosis is typically based upon formal structured clinical interview in psychiatric or primary care settings. Note that a "positive ROS" is similar to high scores on SSRD symptom severity questionnaires.

KEY CONCEPTS

- Patients who have consciously synthesized symptoms and signs may be divided into two broad diagnostic categories: (1) those with obvious secondary gain (malingering), who control their actions, and (2) those with a motivation of achieving the sick role (factitious disorders), who cannot control their actions.
- The initial management of patients suspected of fabricating disease should include a caring, nonjudgmental attitude and a search for objective clinical evidence of treatable medical or psychiatric illness. Review of old medical records and interviews of family members are often helpful.
- Unnecessary tests, medications, and hospitalizations should be avoided in the absence of objective evidence of a medical or psychiatric disease, and patients should be referred for ongoing primary care.
- In cases of suspected factitious disorder imposed on another (FDIA) involving children or elders, protection of the victim takes first priority.

1. Which of the following statements regarding factitious disorder is true?
- **a.** It involves voluntary and controllable symptom production.
- **b.** Patients are generally well educated and otherwise responsible.
- **c.** Presentations are not related to an identifiable event.
- **d.** The symptoms produced are always physical ones.
- **e.** The underlying motivation is a conscious one.

Answer: B. Many such patients are actually employed in the health care industry. The act of producing symptoms is voluntary but not controllable and derives from a subconscious motivation. Presentations are very often related to a "traumatic" event, such as a breakup. Produced symptoms may be physical (e.g., hematuria) or psychological. The typical patient is an unmarried female younger than 40 years. Despite undergoing invasive procedures and associated hardships, these patients seek more medical care and hospitalization.

2. Which of the following statements concerning Munchausen syndrome by proxy is true?
- **a.** A known physical illness in the child is common.
- **b.** Most perpetrators are demanding, uncooperative, and socially inept.
- **c.** Most maternal perpetrators are not a biologic parent.
- **d.** Psychosis is common in the maternal perpetrator.
- **e.** The mean age of victim diagnosis is 7 to 9 years.

Answer: A. Victim children often have a legitimate illness. Mean age at diagnosis is 40 months. Most have a history of failure to thrive and multiple hospitalizations. The perpetrator receives some personal fulfillment from the care and attention of the hospital staff, which is often admiration for her persistence, willingness to sacrifice, and patience; and she is typically pleasant, medically savvy, and socially skilled. Invasive procedures on the child are often welcomed. Although psychosis is very unusual in the parent, depression, anxiety, and somatization are typical in the perpetrator.

3. A 2-year-old female presents with new-onset seizures. Her past medical history is unremarkable. Laboratory evaluation reveals blood glucose of 20 mg/dL. The patient's mother denies a family history of diabetes or having medications the child might have ingested at home. She works as a nurse at a local hospital and has been with the child all day. The child's symptoms improve with glucose administration and a meal. Your colleague remembers evaluating the child recently for hematuria with a negative evaluation. If you suspect Munchausen syndrome by proxy, which of the following tests would be most helpful in establishing the diagnosis?
- **a.** Basic metabolic panel
- **b.** Computed tomography (CT) scan of head
- **c.** C-peptide and insulin level
- **d.** Electroencephalography (EEG)

Answer: C. The diagnostic criteria for factitious hypoglycemia include high-serum insulin levels along with the absence of serum C-peptide. The C-peptide is removed during the purification of commercial insulin, and so its absence suggests the presence of endogenously administered insulin. In patients with insulinoma, both C-peptide and insulin levels are elevated and detectable.

4. A prison inmate presents after falling from the top bunk in his cell. He is complaining of lower lumbar pain and states he is unable to move or feel his lower extremities from his waist down. On physical examination, lower extremity reflexes are present but the patient denies feeling pain or light touch sensation below the waist. Lumbar spine CT and MRI are negative. Which of the following conditions is most likely?
- **a.** Cord contusion
- **b.** Factitious disorder
- **c.** Malingering
- **d.** Munchausen syndrome

Answer: C. Malingering is the intentional symptom production for secondary gain. There is a marked discrepancy between claimed disability and the actual objective findings. Confessions and proof are rare.

101 Suicidal Behavior

KEY CONCEPTS

- Many suicidal individuals see a physician shortly before their death. An ED visit for suicidal thoughts or behaviors represents an opportunity for a critical intervention that may prevent a subsequent suicide.
- Suicidal thoughts or behaviors are often triggered by short-term crisis, and most survivors are grateful to be alive.
- An empathetic, patient-centered, collaborative approach that incorporates information from collateral sources (e.g., family) can optimize care.
- Suicide precautions in the ED include appropriate use of staff to monitor the patient to prevent attempts of self-harm while in the ED.
- Routine laboratory tests provide little value for most ED patients with self-harm behaviors. Evaluation should be directed to specific concerning signs or symptoms.
- Suicide risk changes over time, and estimation of imminent risk is not currently evidence-based.
- Brief risk assessment by the clinician can identify patients in need of a comprehensive evaluation and consultation with a mental health specialist.
- Patients at low risk of suicide may be discharged to a safe and supportive environment where access to firearms or risk of overdose or poisoning is minimized.
- Discharged patients should receive education and safety planning in the ED and be referred for early mental health follow-up.

1. Which of the following statements regarding suicide is true?
- **a.** Many suicide attempts occur during an acute crisis.
- **b.** Suicide rates are highest in older men.
- **c.** Suicidal behavior may be chronic.
- **d.** Suicidal patients are usually ambivalent about dying.
- **e.** All of the above are true.

Answer: E. Many suicide attempts occur in response to a crisis that may be time-limited or resolvable. Suicide rates are particularly high in older white men. With the exception of psychotic patients, suicidal patients are usually ambivalent about dying. An example of chronic suicidal behavior is drinking in the face of liver disease.

2. Which of the following statements regarding suicide is true?
- **a.** Blacks attempt suicide more than whites.
- **b.** Men attempt suicide more than women.
- **c.** Marriage decreases the likelihood of suicide.
- **d.** Pregnancy increases the risk of suicide.
- **e.** Suicide rates are highest among female teens.

Answer: C. Marriage and pregnancy/motherhood decrease suicide risk. Whites attempt suicide more than blacks, with the highest rate among older white men. Women attempt suicide far more often than men but do not choose lethal means and therefore have a lower success rate.

3. Which of the following is a risk factor for suicide?
- **a.** Prior suicide attempt
- **b.** Access to firearms
- **c.** Alcohol abuse
- **d.** Veteran status
- **e.** All of the above

Answer: E. Additional suicide risk factors include, but are not limited to: adolescence and older age; male gender; certain races/ethnicities (White, American Indian, Alaskan Native); mental disorders; substance abuse; prior suicide attempt; psychosocial stressors (e.g., recent psychiatric hospital discharge, history of trauma or abuse, terminal illness, chronic pain, hopelessness, impulsiveness); environmental stressors (e.g., job loss, bereavement); and sociocultural factors (e.g., isolation, poor access to mental health care, stigma against seeking help, or media exposure to suicide).

4. Most completed suicides involve which of the following?
- **a.** Falls
- **b.** Firearms
- **c.** Piercing
- **d.** Poisonings
- **e.** Suffocation

Answer: B. Fifty percent of completed suicides involve firearms. Seventy percent of attempted suicides involve poisoning.

5. Discharge planning for suicidal patients should include which of these elements?
- **a.** Counseling about reducing access to guns and toxic medications.
- **b.** Involvement of family or friends.
- **c.** Rapid referral to outpatient mental health.
- **d.** Written materials with warning signs and hotline numbers.
- **e.** All of the above.

Answer: E. All of the listed elements are recommended components of ED care and discharge planning for patients evaluated for suicidal thoughts or behaviors but deemed safe for discharge home.

6. A 33-year-old Caucasian man presents with agitation and suicidal ideation. He has a long history of schizophrenia and is currently taking olanzapine and occasional clonazepam. He was hospitalized 3 weeks prior for an accelerated psychotic episode and released on an increased dose of olanzapine. His family brought him here today after visiting him in his apartment and finding him in a room with all the lights off. They note he has been unable to work for more than 2 years. He was formerly employed as an engineer. Your examination is remarkable for a blunted affect, moderate pressured speech, and a depressed mood. What is the most appropriate intervention?
- **a.** Addition of sertraline to olanzapine, 2-day follow-up
- **b.** Admission to psychiatry unit
- **c.** Increase olanzapine, release with family
- **d.** Overnight emergency department (ED) observation, 2-day psychiatry follow-up
- **e.** Parenteral ziprasidone, release with family

Answer: B. Approximately 10% of schizophrenic patients will kill themselves. Psychotic patients who kill themselves are often unmarried Caucasians of high intelligence. A recent psychiatric hospitalization is a suicide risk factor, particularly during the first month post discharge. This patient is high risk and needs admission.

7. Which of the following statements concerning risk assessment of suicidal patients is true?
- **a.** An empathetic approach will reinforce malingering behavior and subsequent ED visits.
- **b.** For patients at low risk of imminent suicide, providers can consider discharge without formal consultation with a mental health professional.
- **c.** Intoxicated patients who, once sober, disavow prior suicidal statements do not need a suicide risk assessment.
- **d.** Routine screening labs should include serum chemistries and urine toxicologic panels.
- **e.** Suicidal patients can be permitted to leave the ED prior to a risk assessment as long as they sign "Against Medical Advice" paperwork.

Answer: B. In a step-wise manner, the ED provider can complete a brief risk assessment to identify which patients do (or do not) require a comprehensive evaluation with a mental health professional. Diagnostic testing should be targeted to individual patients as clinically indicated. An empathetic approach can enhance patient evaluation and care and should be used with all suicidal patients. Acute and chronic alcohol use are both suicide risk factors; even if a patient denies suicidality once sober, a risk assessment may be prudent. No suicidal patient should be allowed to leave the ED before the risk assessment is complete.

8. Which of the following statements regarding involuntary commitment is true?

- **a.** It is not associated with adverse psychiatric consequences.
- **b.** It lowers the rate of future suicides.
- **c.** Most states mandate attempts at involuntary commitment if there is imminent self-harm.
- **d.** Patients who volunteer for admission may still need commitment papers.
- **e.** Statutes are consistent among the states.

Answer: D. Statutes vary widely among the states. Some states require commitment papers even in cases of voluntary admission. Only two states mandate commitment in the face of suspected imminent self-harm. Involuntary commitment does not lower the rate of future suicides and is associated with adverse psychiatric consequences.

102 Arthritis

KEY CONCEPTS

1. Arthritis can be considered in three broad classes: degenerative, infectious, and inflammatory. The number of joints involved, time course, and presence of other systemic symptoms can help in classification.
2. Septic (acute infectious) arthritis should be considered in any patient presenting with acute monoarticular arthritis.
3. No singular physical examination finding, serum test, or synovial fluid test in isolation can diagnose or exclude the diagnosis of a septic joint.
4. Arthrocentesis can aid in diagnosis and should be considered if the diagnosis is unclear or any time infection is suspected.
5. Crystal arthritis and septic arthritis can and do frequently coexist. The presence of crystals in joint aspirate should not be used to exclude infection.
6. Gram stain of synovial fluid is positive only in 50% of patients with confirmed infection and should not be used in isolation to exclude infection.
7. Many common arthritis conditions, including gout, rheumatoid arthritis, and osteoarthritis, can be managed in the outpatient setting. Analgesics, specifically antiinflammatory medications, are the mainstay of initial therapy.
8. A systematic history and physical examination should be performed to evaluate for multiple joint involvement or signs of systemic disease.
9. Intra-articular corticosteroid injections can be considered in select patient populations where systemic therapy is either contraindicated or presents a significant risk to the patient.
10. Consider ultrasound evaluation of acutely painful joints to better evaluate for the presence of joint effusion requiring aspiration.

1. A 72-year-old male with history of gout presents to the emergency department (ED) with worsening knee pain and swelling. He states this pain is worse than previous gout episodes. A diagnostic arthrocentesis is performed. Which of the following synovial fluid results would be most consistent with a diagnosis of septic arthritis?

a. Gram stain negative, positive crystals (needle-shaped), synovial white blood cell (WBC) count 25K, polymorphonuclear (PMN) 70%, turbid appearance

b. Gram stain negative, negative crystals, synovial WBC count 12K, red blood cell (RBC) 1K, fat globules in aspirate

c. Gram stain negative, positive crystals (needle-shaped), synovial WBC count 40K, PMN 92%, turbid appearance

d. Gram stain negative, positive crystals (rhomboid-shaped), synovial WBC count 35K, PMN 60%

Answer: c. A synovial fluid sample with PMN greater than 90% is presumed to be consistent with septic arthritis regardless of total synovial WBC count. Answer choices **a** and **d** may represent septic arthritis in certain scenarios, but neither is diagnostic. These represent acute gout and pseudogout, respectively. Answer **b** is representative of a lipohemarthrosis consistent with intra-articular fracture. A positive Gram stain would also be diagnostic of septic arthritis, although approximately 50% of cases of septic arthritis are associated with a negative Gram stain.

2. A 38-year-old previously healthy female presents to the emergency department with a complaint of worsening elbow pain and swelling. She plays recreational soccer and fell on her elbow one week ago. There was a small laceration over the elbow, but she has been trying to keep it covered with gauze. Over the last two days, the elbow has become more swollen, red, and painful. Joint aspiration confirms septic arthritis with gram stain showing gram positive cocci. What is true about the treatment of this patient with septic arthritis?

a. Intra-articular antibiotic treatment is the gold standard and should be given immediately.

b. Septic arthritis can safely be treated with oral antibiotics as the first line of treatment as long as the patient does not have a fever.

c. Synovial fluid aspiration is rarely helpful in the treatment of septic arthritis. Blood cultures are sufficient.

d. Vancomycin is a good choice for initial antibiotic treatment in this patient with septic arthritis.

Answer: d. A is incorrect because intraarticular antibiotics do not play a role in the treatment of septic arthritis. IV antibiotics are the gold standard. B is incorrect because IV antibiotics are the first line of treatment. Fever is not always present with septic arthritis. C is incorrect because synovial fluid aspiration is extremely important in the treatment of septic arthritis. Drainage of the joint is important, but it can also guide your antibiotic choice. D is correct because vancomycin 15–20 mg/kg/dose every 8 to 12 hours is a good starting antibiotic, with a positive gram stain showing a gram positive organism. Ceftriaxone 2 g IV (or other cephalosporin, see Table 102.8) can be added if gram stain is inconclusive or if results are not readily available.

3. Which of the following statements regarding septic arthritis is true?

a. *Streptococcus pneumonia* is the most common cause of bacterial arthritis

b. Older patients are more likely to present with fever

c. Synovial fluid C-reactive protein (CRP) is more accurate than serum CRP

d. Blood cultures should be obtained when bacterial arthritis is suspected

Answer: d. Blood cultures are recommended in all suspected septic arthritis cases because bacteremia is a relatively common finding. The most common pathogen is *Staphylococcus aureus*. Older patients are less likely to present with a fever. Synovial and serum CRP levels are similar in diagnostic value.

4. A previously healthy 19-year-old female presents with an acutely swollen and painful left knee starting a few days ago. She denies trauma or any previous history of similar episodes. She denies fever or chills but has had minor pelvic discomfort and a vaginal discharge, as well as bilateral wrist pain that resolved spontaneously. In addition to performing diagnostic joint aspiration, what tests should be ordered on this patient?
   a. Oral and genital swabs for polymerase chain reaction (PCR)
   b. C-reactive protein (CRP) and sedimentation rate
   c. Serum uric acid levels
   d. Lyme titers

Answer: a. This patient has gonococcal arthritis. This bacterium is notoriously difficult to find in synovial samples, and it is recommended to obtain oral and genital samples, in addition to samples of any skin lesions that may be present, to maximize diagnostic possibilities. CRP or sedimentation rate may aid in diagnosis of a septic joint but are not helpful in distinguishing the actual causative agent. Uric acid level has limited utility in this setting. Lyme disease is less likely in this patient who also has complaints of vaginal discharge, but this may be consid ered in endemic areas with known or possible tick exposure.

5. A 54-year-old male with poorly controlled hypertension and coronary artery disease on daily aspirin presents with a swollen first metatarsophalangeal (MTP) joint on the right foot. You make the diagnosis of gout. Which of the following is the most appropriate initial treatment for this patient with a presumptive diagnosis of gout?
   a. Naproxen 550 mg twice daily
   b. Naproxen 550 mg twice daily coupled with daily proton pump inhibitor (PPI) therapy
   c. Colchicine
   d. Allopurinol and colchicine
   e. Febuxostat and naproxen

Answer: b. This patient is moderate risk for gastrointestinal (GI) injury due to history of poorly controlled hypertension and daily aspirin use; thus an oral nonsteroidal antiinflammatory drug (NSAID) coupled with daily PPI use is reasonable. If the patient had no cardiac or GI risk factors, use of NSAID alone would be appropriate. Colchicine is an alternative therapy, although given its side effect profile, another NSAID is generally preferred. Allopurinol and febuxostat are medications that may be used to prevent gout attacks but should generally not be started during an acute flair and are not typically initiated in the ED setting.

KEY CONCEPTS

Tendinopathy

- Mechanical overload and repetitive microtrauma are key underlying mechanisms in the development of tendinopathy. Patients most often present with a history of progressively worsening localized pain after repetitive work- or sports-related activities.
- Tendinopathy may also be associated with nonmechanical causes, including systemic manifestations of diseases, infectious etiologies, and the use of fluoroquinolones or statins.
- Most patients with tendinopathy can initially be treated with conservative measures, such as activity modification/protection, icing, medications (e.g., short-term use of nonsteroidal antiinflammatory drugs [NSAIDs] or nitroglycerin patches for certain tendinopathies), bracing/splinting, ergonomic modifications, and graduated exercises.
- Overuse syndromes can take at least 6 to 12 weeks to heal. Advise patients accordingly and provide appropriate referral for follow-up to a musculoskeletal specialist (e.g., a sports medicine, orthopedic, or physical medicine and rehabilitation specialist).
- Emergent imaging may be indicated in the ED when fracture or a condition such as calcific tendinopathy is suspected. The use of point-of-care ultrasound to evaluate tendinopathy can help to identify tendon disruption/rupture.
- Operative treatment may be indicated for selected cases of tendon injury that require primary repair (e.g., rupture of the Achilles tendon) or that have failed to respond to conservative treatment (e.g., rotator cuff tendinopathy).

Bursitis

- Consider the possibility of an infectious cause in all cases of acute bursitis.
- The definitive diagnosis of bursitis is made by aspiration of the bursa and evaluation of the fluid.
- Septic bursitis is most commonly caused by *Staphylococcus aureus*.
- Nonseptic bursitis may be traumatic, rheumatologic (e.g., gout and pseudogout), or idiopathic in nature. It is prudent to consider other conditions, such as septic arthritis, osteomyelitis, or an underlying fracture, in the differential diagnosis of bursitis.
- The management of bursitis includes treatment with appropriate medication (antibiotics for septic bursitis, NSAIDs for nonseptic bursitis), rest, application of ice, compression, elevation, and prompt referral for appropriate follow-up. Hospitalization should be considered for severe local infections, for patients who are immunosuppressed, and in the presence of systemic toxicity.

1. What is the etiology of most tendinopathies?

a. Direct blow to the tendon
b. Recent or previous complete tendon rupture
c. High-grade inflammation of the tendon
d. Mechanical overload and repetitive microtrauma

Answer: D. Tendinopathies are thought to be precipitated by repetitive microtrauma and mechanical overload rather than true inflammation of the tendon

2. Which of the following interventions should routinely be used when treating patients with tendinopathies?

a. Activity modification
b. NSAIDs
c. Intratendinous corticosteroid injection
d. Immobilization of the affected tendon/joint

Answer: A. Patients with tendinopathy should be advised to modify their activities (including ergonomic workplace adjustments, when pertinent) to avoid exacerbating pain and further tendon injury. Some patients may benefit in the near term from a short course (e.g., one week) of NSAIDs, although this can result in impaired tendon healing in the long term. Thus, their use should be dependent on the circumstances of presentation, such as the severity of pain or comorbid conditions. Intratendinous corticosteroid injections and immobilization should not be routinely utilized as they may prove detrimental in the long term.

3. A 38-year-old male presents after feeling a "pop," followed by pain to the back of his leg while playing basketball. You note a defect to his Achilles tendon, decreased resting plantar flexion of the ankle, and a positive Thompson test on examination. What should be the next step in the patient's management?

a. Provide crutches and advise the patient to remain nonweightbearing until follow-up.
b. Splint the affected leg in an equinus position and refer for prompt follow-up.
c. Obtain emergent magnetic resonance imaging (MRI).
d. Perform a steroid injection to the affected tendon.

Answer: B. This patient meets the diagnostic criteria for an Achilles tendon rupture. Although MRI can be a useful adjunct when evaluating tendon injuries, obtaining such imaging in the acute care setting is not essential when the clinical diagnosis is obvious. The same can be stated for point-of-care ultrasound. Treatment consists of splinting the leg in 20° of equinus (plantar flexion) and close follow-up with a musculoskeletal specialist (e.g., sports medicine or orthopaedist) for either operative or nonoperative treatment. Implementing non-weightbearing status without proper splinting is insufficient in most cases.

4. What is the most important predisposing factor in septic bursitis?

a. Chronic obstructive pulmonary disease
b. Diabetes mellitus
c. Human immunodeficiency virus (HIV)
d. Trauma

Answer: D. Trauma is the most common predisposing factor in patients who develop septic bursitis.

5. What is the causative organism in the majority of cases of septic bursitis?

a. Beta-hemolytic streptococci
b. *Pseudomonas aeruginosa*
c. *Staphylococcus aureus*
d. *Prototheca wickerhamii*

Answer: C. *S. aureus* is the most common cause of infectious bursitis.

6. A 42-year-old female without significant past medical history presents with 4 days of edema, mild warmth, and mild erythema in the area of the right olecranon bursa. She is afebrile and has minimal tenderness to palpation of the bursa. Aspiration of the bursa reveals a white blood cell count of 1500/μL.[3] What is the most appropriate next step in the patient's management?

a. Admit for intravenous (IV) antibiotics.
b. Apply a compression dressing and give ibuprofen.
c. Discharge the patient to home with oral antibiotics.
d. Obtain magnetic resonance imaging (MRI) for further evaluation.

Answer: B. The clinical presentation and results of the fluid analysis highly suggest a nonseptic bursitis. Cases of nonseptic bursitis can be managed with a compressive dressing, nonsteroidal antiinflammatory drugs (NSAIDs), avoidance of local trauma, and close follow-up.

KEY CONCEPTS

- Most patients presenting to the emergency department (ED) with back pain have uncomplicated musculoskeletal pain that is self-resolving with conservative therapy and does not require imaging.
- Indications for emergent imaging include "red flags" such as an acute neurologic deficit, bowel or bladder dysfunction, or saddle anesthesia.
- Risk factors for compressive myelopathy include immunocompromised patients with a history of malignancy, injection drug use, fever, chronic steroid or anticoagulant use.
- Back pain due to metastatic disease is more common than primary tumors in the spine, and thoracic metastases are more common than lumbar metastases.
- Epidural abscess or hematoma, cauda equina syndrome (CES), spinal malignancy with compressive symptoms, and spinal osteomyelitis are all indications for emergent surgical consultation or transfer to a center where surgical spine consultation is available.
- Empirical parenteral antibiotics active against staphylococci, streptococci, and gram-negative bacilli should be administered for suspected epidural abscess. Specific antibiotics should be directed against the known pathogen if the culture or Gram stain of the aspirate is positive.
- Corticosteroids given as a single dose in the ED (10 mg dexamethasone) or as a 15-day tapering course after discharge (prednisone 60 mg, 40 mg, 20 mg daily for 5 days each) may improve functional ability but does not improve nerve root pain secondary to disc herniation.

1. A 55-year-old man presents with low back pain for one month. The pain is worse at night and is associated with a 10-pound weight loss. He denies any radicular symptoms. Which of the following is the most likely cause of the back pain?

a. Discitis
b. Muscle spasm
c. Osteosarcoma
d. Prostate cancer
e. Sciatica

Answer: D. The patient's subacute time course of back pain and worrisome finding of weight loss suggest a malignancy. Primary and metastatic bone neoplasms can cause back pain from tumor infiltration into the bone. Primary bone tumors, such as multiple myeloma, chordoma, Ewing sarcoma, and osteosarcoma, are 25 times less frequent than metastatic disease. Of the neoplasms, breast, lung, prostate, thyroid, lymphoma, and kidney are the most likely to metastasize to bone.

2. Which one of the following findings, in conjunction with low back pain, indicates a benign cause of pain?

a. Age of six years
b. Fever and night sweats
c. Negative sitting but positive supine straight leg raise test
d. Post-void residual of 500 mL
e. Saddle anesthesia

Answer: C. A positive supine SLR test but a negative sitting SLR test suggests a nonphysiologic cause for the pain. Low back pain and fever suggest an epidural abscess or osteomyelitis. Saddle anesthesia and post-void residual greater than 100 mL are indicative of cauda equina syndrome (CES). Children complaining of back pain must be investigated. They may have spondylolysis with varying degrees of spondylolisthesis, Scheuermann disease (kyphosis and osteochondritis of the vertebral end plates), infectious diseases, or neoplastic etiologies.

3. To avoid injuring the adult spinal cord, at what level should a lumbar puncture be performed?

a. L1 to L2
b. L3 to L4
c. L5 to S1
d. S2 to S3
e. Coccyx

Answer: B. The spinal cord ends at around L2 in adults, lower in children. When performing a lumbar puncture, to avoid injuring the spinal cord, insert the needle below the conus medullaris at level L3 to L4. Remember that between individuals, there may be significant anatomic variance.

4. A 55-year-old man complains of low back pain when walking downhill that is relieved with walking uphill. His neurovascular examination is unremarkable except for decreased bilateral Achilles reflexes. In addition to analgesia, what is the appropriate management of this patient?

a. Bedrest
b. Emergent spine surgery consultation
c. Lumbosacral plain radiographs
d. Magnetic resonance imaging
e. Surgical referral for pseudoclaudication

Answer: E. This patient presents with typical complaints of spinal stenosis. Patients with spinal stenosis should be managed conservatively with pain medications. In the absence of alarming red flag findings, these patients do not require emergent laboratory or radiographic studies. These patients may be candidates for surgery if they show any of the following conditions: progressive neurologic deficit, progressive reduction in ability to walk secondary to pseudoclaudication, evidence of cauda equina syndrome (CES), or intractable pain.

5. Which of the following findings is most consistent with cauda equina syndrome (CES)?
   a. Absent patellar reflexes
   b. Bilateral buttock pain
   c. Decreased rectal tone
   d. Saddle anesthesia
   e. Urinary retention

Answer: E. The most consistent examination finding in CES is urinary retention. With a high sensitivity of 90%, the patient is unlikely to have this disease process if his or her post-void residual urine volume is less than 100 to 200 mL. Saddle anesthesia (sensory deficit over the buttocks, upper posterior thighs, and perineal area) is frequently an associated finding, with a sensitivity of 75%. In 60% to 80% of cases, the rectal examination reveals a decreased sphincter tone.

6. A 42-year-old man presents to the emergency department (ED) with a four-day history of low back pain after an episode of heavy lifting at work. He reports bilateral low back pain at the level of the iliac crests. He denies sensory or motor symptoms. He also denies bowel or bladder dysfunction. His neurologic examination is normal. What management is indicated?
   a. Erythrocyte sedimentation rate (ESR)
   b. Lumbar spine MRI
   c. NSAIDs and gentle stretching
   d. Oxycodone for three weeks
   e. Strict bedrest for two weeks

Answer: C. The patient most likely suffers from uncomplicated musculoskeletal low back pain. This is also commonly called *acute lumbosacral strain.* Most patients with this injury should not be placed on bed rest and should be allowed to return to normal activity, possibly with some restrictions. The patient has a relatively short history of low back pain with clear onset around an episode of lifting. Given a lack of concerning historical or examination findings, the patient does not require imaging at this time. Blood work would not be of help in evaluating the patient, because he lacks history or examination findings consistent with spinal infection.

7. A 35-year-old woman presents with a 3-day history of severe right lower extremity pain associated with mild low back pain. Her neurologic examination is normal except for a positive straight leg raise (SLR) test on the right and a negative contralateral straight leg raise (CSLR) test on the left. What therapy is indicated?
   a. Anxiolytic medication
   b. High-dose corticosteroids
   c. NSAIDs and gentle stretching
   d. Oxycodone for three weeks
   e. Strict bedrest for two weeks

Answer: C. Patients with herniated lumbar disks often present with radicular leg pain that overshadows the complaint of back pain. The treatment of radicular pain syndromes without neurologic deficit is similar to that of nonspecific uncomplicated back pain: NSAIDs and gentle stretching. It is very common for a patient with lumbar radiculopathy to have no clear motor or sensory deficit but have exacerbation of leg pain with SLR testing. The SLR has high sensitivity but low specificity. In contrast, the CSLR test has high specificity but low sensitivity. Given this, it is common for the patient with lumbar disk herniation to have a positive SLR but negative CSLR. The reverse is very uncommon. Diagnoses (such as spinal epidural abscess and spinal malignancy) usually present with prominent low back pain that is more significant than extremity pain.

8. A 68-year-old man presents with a 5-week history of worsening low back pain. He reports mostly midline spinal pain with occasional radiation into both lower extremities. Two weeks before the onset of his pain, he was discharged from the hospital after an inpatient admission for pneumonia. On examination, he has intact lower extremity motor and sensory function but tenderness to percussion over the lumbar spine. What diagnostic evaluation will confirm the diagnosis?
   a. Complete blood count
   b. Contralateral straight leg raise test
   c. Erythrocyte sedimentation rate (ESR)
   d. Lumbar computed tomography (CT)
   e. Lumbar magnetic resonance imaging (MRI)

Answer: E. The patient's history is suspicious for spinal epidural abscess. He is at higher risk because of his age and recent infection. In addition, the patient has tenderness with percussion of his spine. The optimal diagnostic evaluation to confirm this diagnosis is MRI. Patients with epidural abscess usually have an ESR elevated above 20 mm/hr. However, it is not uncommon for them to have a normal or only mildly elevated WBC count.

9. A 63-year-old man presents with a 9-month history of progressive low back pain with ambulation. He reports significant pain in his buttocks and posterior thighs when he walks distances greater than 25 meters. He says the pain is partially relieved when he flexes forward and completely relieved by recumbency. He reports the pain is not relieved if he stops walking but remains standing. On neurologic examination, he has intact lower extremity strength but diminished Achilles reflexes bilaterally. What is the expected finding on diagnostic imaging?
   a. Diminished Doppler flow in the tibial arteries
   b. Lumbar disc herniation at L5 to S1
   c. Normal lumbar MRI
   d. Post-void bladder volume of 500 mL
   e. Spinal stenosis at L4 through S1

Answer: E. The patient presents with classic findings of spinal stenosis and neurogenic claudication or "pseudoclaudication," including relief with flexing forward and recumbency. Persistence of pain with standing despite having stopped ambulating is also indicative of neurogenic claudication, as are diminished Achilles reflexes. Diminished pulses are indicative of vascular claudication. Pain from vascular claudication is generally relieved if a patient stops walking but remains standing. Unilateral disk herniation does not usually present with bilateral lower extremity symptoms.

KEY CONCEPTS

- Systemic lupus erythematosus (SLE) is a chronic autoimmune disorder that can affect any organ system. A general approach to determine whether SLE is the cause of a nonspecific or single-organ symptom is to search for evidence of other organ involvement or systemic inflammation, which is expected in SLE-mediated presentations.
- Patients with SLE are at significantly higher risk of coronary artery or thromboembolic disease, which should prompt more thorough evaluation for these etiologies even in otherwise low-risk patients, such as young women.
- SLE itself, as well as its treatment, may lead to immunosuppression; thus, it is important to remain vigilant for the possibility of infection in patients with SLE.
- Glucocorticoids are the mainstay for the initial management of the majority of conditions that are associated with increased SLE disease activity, including musculoskeletal, cutaneous, renal, pleural, or pericardial disease.
- Antiphospholipid antibody syndrome (APS) is common in patients with SLE and carries with it an increased risk of venous and arterial thromboembolic disease.
- Rheumatology input may be helpful in diagnostic, management, or disposition decisions for select patients with SLE.

1. Which of the following findings would decrease the likelihood that a patient's pleurisy is related to a lupus flare?
   a. Elevated erythrocyte sedimentation rate
   b. Low complement levels
   c. Negative review of systems
   d. Polyarthritis
   e. PR depression on electrocardiogram

Answer: c. The general approach to assessing whether a symptom is caused by SLE disease activity is to look for signs and symptoms of the disease in other organs or systemically. Patients experiencing an SLE flare are likely to have other signs or symptoms of disease (e.g., arthralgias, fever, malaise, elevated ESR, low complement levels) beyond an isolated chief complaint (in this case, pleurisy). PR depression can be suggestive of pericarditis, which may occur during flares of SLE.

2. A patient with a history of SLE presents actively seizing. In addition to respiratory support, what is the most appropriate initial step in management?
   a. Intravenous ceftriaxone and vancomycin
   b. Intravenous cyclophosphamide
   c. Intravenous lorazepam
   d. Intravenous methylprednisolone
   e. Intravenous tissue plasminogen activator

Answer: c. In nearly all cases (including seizure), the initial resuscitation and stabilization of the patient with SLE are the same as for patients without SLE. While this patient could have meningitis, lupus cerebritis, or a stroke, which could warrant some of the other treatments listed, the patient should be stabilized by treatment of the seizure first.

3. Which of the following symptoms is least likely to be seen in a patient presenting with giant cell arteritis?
   a. Amaurosis fugax
   b. Constitutional symptoms
   c. Dysphagia
   d. Jaw claudication
   e. Scalp tenderness

Answer: c. Patients presenting with giant cell arteritis (GCA) commonly have constitutional symptoms. Other presenting symptoms include headache and scalp tenderness. Jaw claudication is relatively specific for GCA, and visual symptoms (e.g., amaurosis fugax) may herald impending blindness from the disease if not treated. Dysphagia is not a presenting feature of GCA.

4. A patient with a history of SLE presents with acute bilateral arm weakness and decreased sensation below the clavicles starting 90 minutes prior to arrival. A CT of the head is normal, and an MRI of the brain and spine reveals T2 hyperintensity in the cervical spine. Which of the following is the most appropriate next step in management for this patient?
   a. Intravenous ceftriaxone and vancomycin
   b. Intravenous lorazepam
   c. Intravenous methotrexate
   d. Intravenous methylprednisolone
   e. Intravenous tissue plasminogen activator

Answer: d. This patient is presenting with transverse myelitis. The first line of treatment for this condition is high-dose glucocorticoids, such as methylprednisolone, and since this patient has a history of SLE, cyclophosphamide. Refractory cases may be treated with plasma exchange and additional immunosuppressive agents.

5. A patient with a history of SLE presents with shortness of breath. CT angiography reveals two segmental pulmonary emboli. Laboratory testing and bedside echocardiography are unremarkable. Chart review reveals positive anticardiolipin and β2-glycoprotein I titers. Which of the following is the most appropriate next step in therapy for this patient?
   a. Aspirin
   b. Enoxaparin
   c. Rivaroxaban
   d. Tissue plasminogen activator
   e. Warfarin

Answer: b. This patient has SLE and antiphospholipid syndrome (APS) based on the history of thromboembolic disease and positive antiphospholipid (aPL) antibodies. The most appropriate anticoagulation for patients with APS is a vitamin K antagonist (e.g., warfarin) or a heparin analog (e.g., enoxaparin). In the acute setting, however, a heparin analog should be used due to the delayed therapeutic effect of vitamin K antagonists. Aspirin is insufficient to treat acute thromboembolic disease in APS, and direct oral anticoagulants have decreased efficacy when compared to vitamin K antagonists and heparin analogs. This patient does not have an indication for tissue plasminogen activator.

KEY CONCEPTS

- Anaphylaxis is a life-threatening systemic allergic (or nonallergic) reaction of acute onset and multiorgan involvement in which timely recognition and treatment remain essential.
- A history of sudden urticarial rash accompanied by respiratory difficulty, abdominal pain, or hypotension, strongly favors the diagnosis of anaphylaxis.
- The recommended treatment algorithm for anaphylaxis is shown in Box 106.7.
- Epinephrine is the first-line treatment in patients with anaphylaxis and should be given immediately. There are no absolute contraindications to the use of epinephrine in the setting of anaphylaxis.
- Antihistamines and corticosteroids are second- and third-line agents in the management of anaphylaxis and should not replace or precede epinephrine.
- Consider prolonged observation or admission for patients who (1) experience protracted anaphylaxis, hypotension, airway involvement, or unknown trigger; (2) receive IV epinephrine or more than one dose of IM epinephrine; or (3) have poor outpatient social support.
- Patients discharged after an anaphylactic event should be prescribed self-injectable epinephrine devices and instructed on use, encouraged to develop an emergency action plan, and referred to an allergist/immunologist.
- Patients with refractory hypotension may require a continuous IV epinephrine infusion or glucagon in patients with coexisting beta-adrenergic blockade).
- Non-histaminergic angioedema (nonallergic angioedema) does not typically respond to epinephrine and antihistamines, though they should be considered at initial presentation. Newer drugs, including icatibant, ecallantide, and human or recombinant C1 esterase inhibitor, have been approved for use in hereditary angioedema (HAE). Fresh frozen plasma (FFP) has been used with varying success in HAE, acquired C1 esterase inhibitor deficiency (ACID), and angiotensin-converting enzyme (ACE) inhibitor-induced angioedema.

1. Which of the following statements regarding the use of H_2-antihistamines in patients presenting with urticaria and pruritis is true?
   - **a.** A portion of histamine receptors on the skin are H_2 receptors.
   - **b.** Patients with skin manifestations often have concomitant reflux.
   - **c.** H_2-antihistamines have synergistic properties when used with steroids.
   - **d.** Use of H_2-antihistamines can effectively replace H_1-antihistamines and have fewer side effects.
   - **e.** H_2-antihistamines are effective in preventing recurrence of pruritis and urticaria.

Answer: a. Because 15% of the histamine receptors in skin are H_2 receptors, the addition of an H_2-antihistamine (e.g., ranitidine) may be beneficial in the treatment of pruritis and urticaria. There is no proven synergistic effect when used with steroids. Second-generation H_1-antihistamines, such as cetirizine, loratadine, and fexofenadine, are the preferred agents in the treatment of urticaria and pruritis.

2. A 16-year-old female with a known history of hereditary angioedema (HAE) presents with facial swelling. Which HAE medication is not approved for patients in this age group?
   - **a.** Ecallantide
   - **b.** Icatibant
   - **c.** Bertinert
   - **d.** Reconest
   - **e.** Cinryze

Answer: b. Icatibant is approved for treatment in adults aged 18 years or older. Berinert is approved for the treatment of HAE in adult and pediatric patients (no lower age limits). Ruconest approved for use in adults and pediatric patients ages 11 and older. Ecallantide is approved for adult patients and pediatric patients 12 years and older. Cinryze is a prophylactic HAE medication not approved for use in acute attacks.

3. Concomitant use of which of the following classes of medications is associated with increased risk of mortality in patients with drug-induced anaphylaxis?
   - **a.** Penicillins
   - **b.** ACE inhibitors
   - **c.** NSAIDs
   - **d.** Beta-blockers
   - **e.** Fluoroquinolones

Answer: d. Fatal drug-induced anaphylaxis has been associated with hypertension, obesity, male gender, beta-blocker use, and advanced age. Beta-blockers may oppose the actions of adrenergic agents used in anaphylaxis treatment. These patients may require additional treatment with glucagon to counter the effects of beta-blockade.

4. What is the appropriate initial dose and route for a pediatric patient in the treatment of acute anaphylaxis?
   - **a.** 0.3 to 0.5 mg of 1 mg/mL concentration epinephrine concentration, IM in anterolateral thigh
   - **b.** 1 mL of 1:1000 epinephrine IV
   - **c.** 0.01 mg/kg of 1:1000 epinephrine, IM in anterolateral thigh
   - **d.** 2 mL of 1:10,000 epinephrine IV every 5 minutes
   - **e.** 0.01 mg/kg of 1:10,000 epinephrine IV

Answer: c. 0.01 mg/kg (1:1000 concentration) given IM in the anterolateral thigh every 5 to 10 minutes as necessary is recommended for pediatric patients. Alternatively, an EpiPen Jr 0.15 mL can be considered in patients weighing 10 to 25 kg.

5. The majority of the clinical features of anaphylaxis syndrome are produced by which of the following mediators?
   - **a.** Acetylcholine
   - **b.** Histamine
   - **c.** Leukotrienes
   - **d.** Prostaglandins
   - **e.** Tumor necrosis factor

Answer: b. All of the listed mediators contribute, but histamine is the predominant chemomediator.

107 Dermatologic Presentations

KEY CONCEPTS

- Accurate descriptions of dermatologic lesions are essential for diagnosis and management. Primary and secondary lesions are described in Tables 107.1 and 107.2.
- Systemic illnesses with cutaneous findings include systemic infections, autoimmune or connective tissue disorders, malignancies, diabetes mellitus, endocrine disorders, and immunodeficiency states.
- Cutaneous signs of systemic disease include pruritus, urticaria, erythema multiforme, erythema nodosum, pyoderma gangrenosum, or others.
- Most skin and soft tissue infections should be treated with antibiotics to cover methicillin-resistant *Staphylococcus aureus*.
- Cutaneous abscesses are treated with incision and drainage plus antibiotics.
- Tinea capitis requires 4–8 weeks of systemic antifungal treatment.
- Onychomycosis requires long-term systemic treatment.
- Allergic reactions are treated with antihistamines and discontinuation of exposure to the allergen. Nonsedating antihistamines are the preferred agents to control pruritus and histamine-mediated rashes because they allow patients to remain active.
- Suspected infestations should be diagnosed clinically and treated expeditiously, even without definitive proof of the infestation.
- Medication reactions are common and may result from any agent, typically within 4–28 days after use.
- Rashes that are associated with mucosal lesions, blisters, or desquamating skin are often caused by significant soft tissue infections, drug eruptions, or immune disorders.
- Patients with Stevens-Johnson syndrome or toxic epidermal necrolysis require inpatient treatment, preferably in a burn unit.
- Clinicians should be familiar with one or two topical steroid preparations of low, medium, and high potency, and their appropriate therapeutic use.
- The majority of patients with dermatologic complaints are managed as outpatients. Indications for hospitalization include significant fluid and electrolyte abnormalities, disordered thermoregulation, systemic infection or other underlying disorder requiring inpatient management, and inability to care for self or maintain appropriate oral intake.

1. Which of the following statements regarding tinea capitis is TRUE?
   a. It is contagious.
   b. It is not transmitted by household pets.
   c. Tinea capitis presents with alopecia with normal underlying scalp.
   d. Topical treatment is effective.
   e. Treatment should be instituted for 1 to 2 weeks.

Answer: a. Tinea capitis is the dermatophytosis that is contagious. Systemic treatment for 4 to 6 weeks is the minimum. It may be transmitted by pets. The underlying scalp is typically inflamed.

2. A 16-year-old boy presents with cellulitis of his left forearm. What is the appropriate initial antibiotic?
   a. Azithromycin
   b. Ceftriaxone
   c. Clindamycin
   d. Linezolid
   e. Penicillin VK

Answer: c. Clindamycin or trimethoprim-sulfamethoxazole are recommended first-line treatment choices for suspected methicillin-resistant *Streptococcus aureus* (MRSA) infection. Macrolides and penicillins are often ineffective against MRSA. Linezolid, although effective, is expensive and is generally not recommended as a first-line treatment.

3. What is the causative organism of erythema migrans?
   a. *Borrelia burgdorferi*
   b. Group A Streptococcus
   c. Methicillin-resistant *Staphylococcus aureus*
   d. *Neisseria meningitides*
   e. Parvovirus B-19

Answer: a. *Borrelia burgdorferi* is the causative agent of erythema migrans, or Lyme disease. Treatment should be instituted with doxycycline for 10 to 21 days, or as alternates, cefuroxime, ceftriaxone, or penicillin G. Group A *Streptococcus* is the causative organism of scarlet fever. *Neisseria meningitides* is the causative agent of Meningococcemia. *Parvovirus B-19* is the causative agent of erythema infectiosum.

4. A 26-year-old man presents with an erythematous maculopapular eruption of his torso, palms, and soles. He had a painless lesion on his penis 1 month earlier. What is the treatment of choice?
   a. Azithromycin
   b. Benzathine penicillin G
   c. Ceftriaxone
   d. Doxycycline
   e. Trimethoprim-sulfamethoxazole

Answer: b. Secondary syphilis is treated with benzathine penicillin G in a dose of 2.4 million units IM.

5. A 25-year-old female presents with fever, migratory polyarthralgias, and hemorrhagic papules on her fingers and wrists. What is the best treatment?
   a. Ceftriaxone
   b. Ciprofloxacin
   c. Doxycycline
   d. Ofloxacin
   e. Vancomycin

Answer: a. Treatment of disseminated gonococcal infection is with parenteral ceftriaxone, or ceftizoxime or cefotaxime. Patients allergic to β-lactam antibiotics or those with severe penicillin allergies may be treated with spectinomycin. Ciprofloxacin and ofloxacin are not recommended because of increasing resistance patterns. Hospitalization is recommended for patients with disseminated gonococcal infection.

6. Which of the following statements regarding gonococcal dermatitis is TRUE?
   a. Gonococci can usually be seen on gram stain from the lesions.
   b. It affects primarily men.
   c. It occurs in 1% or 2% of patients with gonorrhea.
   d. The lesions have a predilection for the knees and elbows.
   e. The skin lesions are not tender.

Answer: c. Women are affected primarily. The lesions have a predilection for distal joint skin. The lesions are often multiple and have a predilection for periarticular regions of the distal extremities. The lesions begin as erythematous or hemorrhagic papules that evolve into pustules and vesicles with an erythematous halo. They may be tender and may have a gray necrotic or hemorrhagic center. The organism may be cultured from the cutaneous lesions. Gram stain only occasionally reveals the organisms.

7. A 30-year-old man presents with headache, nausea and vomiting, myalgias, fever, and a petechial rash on his extremities and trunk. Lesions are clustered on the palms and soles. What is the best treatment?
   a. Cephalexin.
   b. Doxycycline.
   c. Erythromycin.
   d. Penicillin VK.
   e. Trimethoprim-sulfamethoxazole.

Answer: b. Patients with Rocky Mountain spotted fever present with headache, nausea and vomiting, myalgias, chills, and fever. The disease may last 3 weeks and may be severe with prominent involvement of the central nervous system, cardiac, pulmonary, gastrointestinal and renal systems, disseminated intravascular coagulation, or shock. The rash begins with erythematous macules that blanch on pressure, appearing first on the wrists and ankles. These macules spread up the extremities and to the trunk and face. They may become petechial or hemorrhagic. Lesions on the palms and soles are particularly characteristic. Doxycycline is the antibiotic of choice. Chloramphenicol may be used for patients allergic to tetracyclines and in children younger than 9 years of age. Sulfa drugs should be avoided because they may exacerbate the illness. Rickettsiae are routinely resistant to penicillins, cephalosporins, aminoglycosides, and erythromycin.

KEY CONCEPTS

- Red blood cell transfusion is indicated only to increase oxygen delivery at the tissue level.
- One unit of packed red blood cells (PRBCs) can be expected to raise an adult's hemoglobin level by 1 g/dL. A similar increase is expected in children following the transfusion of 10 mL/kg of PRBCs.
- Controlled trials have supported newer, restrictive, red cell transfusion strategies. Pending further trials, a transfusion trigger of a hemoglobin level below 7 to 8 g/dL is appropriate for most stable hospitalized patients.
- Platelet transfusions are typically used prophylactically for counts less than 10 K/μL in adults without bleeding. For patients undergoing central venous catheter placement, a level of 20 K/μL is recommended. Patients undergoing lumbar puncture and non-neuroaxial surgery should be prophylactically transfused to a level of 50 K/μL.
- Prospective and retrospective reports have suggested a benefit to massive transfusion protocols, with most advocating a 1 : 1 : 1 ratio of fresh frozen plasma (FFP) to platelets to PRBCs.
- When available, low-titer whole blood is safe and effective for transfusion and provides a physiologic mix of blood products.
- When available, prothrombin complex concentrate (PCC) should be given over FFP for reversal of vitamin K antagonism in the setting of a life-threatening bleeding. When PCC is not available, FFP can be used for this purpose, but is considered a second-line therapy.
- Transfusion reactions can vary from minor symptoms to fatal systemic reactions. If any transfusion reaction is suspected, the transfusion should be stopped while the cause and extent of the reaction is investigated.
- An intravascular hemolytic transfusion reaction is usually the result of ABO incompatibility and typically results in immediate symptoms that can include fever, chills, headache, nausea, vomiting, sensation of chest restriction, severe joint or low back pain, burning sensation at the site of the infusion, and feeling of impending doom. Treatment involves stopping the transfusion, fluid resuscitation, and monitoring for the development of renal failure and disseminated intravascular coagulation (DIC).
- Transfusion-related acute lung injury (TRALI) is now the leading cause of reported transfusion-related mortality. Treatment involves stopping the transfusion and providing supportive respiratory care, which may include noninvasive positive-pressure ventilation (NIPPV) or intubation and mechanical ventilation.
- Improved techniques for selecting and testing blood donors has dramatically reduced the risk of viral transmission of disease by transfusion.

1. A 60-year-old man on warfarin presents to the emergency department after a fall. His non-contrast head CT shows a subdural hematoma, and his INR is 4. Which of the following is the recommended agent to reverse his coagulopathy?

a. Cryoprecipitate
b. Fresh frozen plasma
c. Prothrombin complex concentrate
d. Oral vitamin K

Answer: C. Prothrombin complex concentrate (PCC) is the recommended agent for reversal of elevated INR in life-threatening bleeding, including intracranial bleeding. 4 Factor PCC contains clotting factors II, VII, IX, and X making it the ideal agent for patients with elevated INR from warfarin as this medication inhibits synthesis of these vitamin K–dependent clotting factors. Cryoprecipitate contains fibrinogen, factor VIII, factor XII, von Willebrand factors, and fibronectin and is used most often clinically in cases of low fibrinogen

such as massive hemorrhage and DIC. Fresh frozen plasma can be used to reverse warfarin-induced coagulopathy; however, it often requires the administration of significantly more fluid volume and PCC more reliably lowers the INR in a shorter time frame. For these reasons PCC is the preferred treatment. Vitamin K will take hours to days to reverse the coagulopathy, as it relies on the bodies metabolic processes to synthesize clotting factors. It is, however, generally recommended to give IV vitamin K along with PCC, so that once the transfused factors have been consumed the body will have started to produce its own factors again.

2. A 36-year-old female patient is receiving a unit of packed red blood cells to treat her chronic anemia from end-stage renal disease when she develops a temperature of 38.2°C. The rest of her vital signs are normal. Which of the following is the best next step in management?
   a. Administer acetaminophen
   b. Given a bolus of 1 L of NS
   c. Start broad spectrum antibiotics
   d. Stop the blood transfusion

Answer: D. The first step in any suspected transfusion reaction is to stop the blood transfusion. If after investigation and brief observation this appears to be a non-hemolytic febrile transfusion reaction, you may administer acetaminophen and resume the transaction. Fluids resuscitation is the mainstay of treatment for hemolytic transfusion reactions, which can present as a fever. However, as is the case with any transfusion reaction, the first step is stopping the blood transfusion. Antibiotics are indicated in bacterial culture positive transfusion reactions. Typically patients will experience other vital signs abnormalities, such as tachycardia and hypotension, when having one of these reactions.

3. A 76-year-old female is receiving a unit of packed red blood cells, when she suddenly develops shortness of breath. On examination, she has crackles on auscultation of her lungs bilaterally. Which of the following is true regarding transfusion-associated circulatory overload (TACO) and transfusion-related acute lung injury (TRALI)?
   a. Chest x-ray can often distinguish between TACO and TRALI.
   b. Only TACO should be treated with noninvasive positive-pressure ventilation.
   c. TACO typically requires the presence of a significant comorbidity.
   d. TRALI can be treated with aggressive diuresis.

Answer: C. While it is possible to receive enough blood products for a healthy person to become volume overloaded, TACO is much more commonly seen in patients with underlying heart or renal failure. Both TACO and TRALI result in pulmonary edema, meaning they can present with similar chest x-ray findings and physical exam findings. Both entities can be treated with noninvasive positive-pressure ventilation or intubation and mechanical ventilation, as both treatments are effective in treating pulmonary edema. TRALI, however, will not respond to diuretics, because the cause of the pulmonary edema is an inflammatory response within the lungs rather than systemic volume overload.

4. A 15-year-old male is receiving a transfusion of packed red blood cells after a traumatic injury when he complains of itching. On examination, hives are found, and soon the patient develops difficulty breathing, tachypnea, tachycardia, and hypotension. The transfusion is stopped. Which of the following medications should be administered first?
   a. Diphenhydramine
   b. Epinephrine
   c. Furosemide
   d. RhoGAM

Answer: B. Occasionally, anaphylaxis may be caused by an anti-immunoglobulin A (IgA) reaction to IgA found in the donor blood component. The patient is likely to have a congenital IgA deficiency. The presentation is similar to anaphylactic reactions from other causes. Washed red blood cells (RBCs) and plasma products from IgA-deficient individuals can be used to avoid recurrence with subsequent transfusions. As the patient is having an allergic reaction involving more than one organ system, this reaction should be treated as anaphylaxis with epinephrine; while the patient may also receive diphenhydramine, epinephrine is the critical life-saving treatment. Furosemide is indicated to treat transfusion-associated volume overload, which would not present with hives. RhoGAM is a purified anti-D antibody that can be given to Rh− individuals who have received Rh+ PRBC blood transfusions to prevent the patient from developing their own anti-D antibodies. Clinically this is most relevant for female patients to prevent hemolytic disease of the newborn in future pregnancies.

5. A 24-year-old female patient presents with cellulitis. A complete blood count demonstrates a hemoglobin level of 7.5 g/dL. On further questioning, it is discovered that her Ob/Gyn is treating her for menorrhagia, but she has no other comorbidities. She denies any lightheadedness, chest pain, dyspnea, or recent increase in bleeding. Her heart rate is 85 and her blood pressure is 136/84 mm Hg. Which of the following is the most appropriate next step in management of her anemia?
   a. Discharge the patient to follow up with her outpatient provider.
   b. Obtain a type and crossmatch and transfuse 1 unit of type specific packed red blood cells.
   c. Order a STAT hematology consult.
   d. Transfuse 1 unit of O− packed red blood cells.

Answer A. This patient is clinically stable, has a chronic source of blood loss, and her hemoglobin level is above the recommended treatment threshold of 7 g/dL, therefore it is safe to discharge her to follow up with her outpatient provider. Depending on the level of comfort, some physicians might offer a pelvic exam to characterize the degree of bleeding and/or obtain a second hemoglobin level after a brief period of observation. The patient does not warrant an emergent transfusion of either type specific or O− packed red blood cells. O− transfusions should be reserved for those patients with hemodynamic instability who are unlikely to tolerate the 60 to 90 minutes it takes to obtain a type and crossmatch. Similarly, while the patient may benefit from a hematology consult (in particular to be evaluated for von Willebrand disease) it is inappropriate to obtain this in the emergency department for a stable patient.

109 Anemia and Polycythemia

KEY CONCEPTS

- Anemia is caused by three basic mechanisms: loss of red blood cells (RBCs) through bleeding, destruction of RBCs, or decrease in production of RBCs.
- RBC indices along with a peripheral blood smear can help determine the mechanism of anemia.
- Anemia in the elderly often occurs as an exacerbation of pre-existing comorbid diseases.
- Anemia of uncertain etiology should be thoroughly evaluated. If the patient has no adverse hemodynamic consequences, the evaluation can proceed on an outpatient basis or management initiated in an observation setting until the patient is stable.
- Patients with sickle cell disease should be considered to have an acute pain crisis and treated appropriately until proven otherwise.
- Acute chest syndrome is one of the most common causes of death in sickle cell disease and presents with fever, dyspnea, cough, and a new infiltrate on chest radiograph.
- Transfusion therapy is most useful in sickle cell disease associated with acute stroke, acute chest syndrome, or splenic sequestration.
- Primary polycythemia vera is treated with serial phlebotomy to a goal hematocrit of less than 45%.

1. Anemia with an elevated mean corpuscular volume (MCV) is *not* typically seen with which of the following conditions?
- **a.** Folate deficiency
- **b.** Hypothyroidism
- **c.** Iron deficiency
- **d.** Vitamin B_{12} deficiency

Answer: c. Iron deficiency anemia typically presents as a microcytic anemia. Liver disease may present with either a macrocytic anemia or a normocytic anemia reflecting an anemia caused by multiple mechanisms.

2. An elevated mean corpuscular hemoglobin concentration (MCHC) is expected in which of the following conditions?
- **a.** Anemia of chronic disease
- **b.** Iron-deficiency anemia
- **c.** Sideroblastic anemia
- **d.** Spherocytosis

Answer: d. The MCHC index is a measure of the concentration of hemoglobin. Low values represent hypochromia, whereas high values are noted only in patients with decreased cell membrane area relative to volume, such as spherocytosis.

3. A 21-year-old man presents with easy fatigue and lack of energy. During a recent clinic visit, he was found to be anemic with a hemoglobin of 8 g/dL, mean corpuscular hemoglobin concentration (MCHC) of 25%, mean corpuscular volume (MCV) of 61 fL, and a peripheral smear remarkable for target cells and basophilic stippling. Serum iron levels were normal. What is the most likely explanation for his anemia?
- **a.** Iron deficiency
- **b.** Porphyria
- **c.** Sideroblastic anemia
- **d.** Thalassemia

Answer: d. Hypochromic, microcytic anemias with normal serum iron imply thalassemia (a defect in globin chain production). Both alpha- and beta-thalassemia manifest as a hypochromic microcytic anemia with target cells and basophilic stippling. The microcytosis is generally more severe than with iron-deficiency anemia. Sideroblastic anemia generally presents with elevated serum iron.

4. A 73-year-old woman presents with progressive fatigue. Her only past medical history is rheumatoid arthritis for which she takes methotrexate. She has no cardiopulmonary history and does not smoke. Physical examination is remarkable only for bilateral metacarpophalangeal, chronic swelling, and mild splenomegaly. Recent blood tests from the clinic are remarkable for a high normal serum iron, hemoglobin of 10 g/dL, and mean corpuscular volume (MCV) of 69 fL, with peripheral smear showing both microcytes and macrocytes. Which of the following is the most appropriate intervention?
- **a.** A trial of pyridoxine
- **b.** Hematology consultation for bone marrow biopsy
- **c.** Hematology consultation for iron chelation
- **d.** Send blood for vitamin B_{12} and folate levels

Answer: a. Sideroblastic anemia may be acquired or inherited. It is typically a refractory anemia in the elderly characterized by hypochromia and microcytosis but a dimorphic smear also showing normal cells and macrocytes. Some of these patients are pyridoxine deficient and may respond to a course of vitamin B_6. This anemia is a defect in porphyrin synthesis and is associated with rheumatoid arthritis, cancer, and infections. Lead poisoning is a subset. Elevated iron and ferritin levels are seen because the porphyrin defect does not allow iron incorporation and cells hemolyze in the bone marrow.

5. A 43-year-old woman presents with difficulty walking and complaints of depression that have progressively worsened over several weeks. She has no past medical history, takes no medications, and does not drink alcohol or smoke. Physical examination is remarkable for clinical depression, a spastic gait, decreased extremity proprioception, and lower extremity hyporeflexia. What test should be obtained next in this patient's evaluation?
- **a.** Complete blood count with differential diagnosis and RBC indices
- **b.** Lumbar puncture
- **c.** Magnetic resonance imaging (MRI) of the spine
- **d.** Serum potassium and calcium levels

Answer: a. Vitamin B_{12} deficiency presents classically with very low hemoglobin levels, a macrocytic picture, decreased proprioception or vibration, lower extremity spasticity/weakness with hyporeflexia, and often mental status changes, such as depression, forgetfulness, or even paranoia. Irritability and forgetfulness are also seen with folate deficiency.

110 White Blood Cell Disorders

KEY CONCEPTS

- Chronic lymphocytic leukemia (CLL) is the most common leukemia in the elderly, and acute lymphocytic leukemia (ALL) is the most common leukemia in children.
- Splenomegaly is a common finding in leukemias.
- Leukostasis is usually not accompanied by clinical sequelae until the white blood cell (WBC) count is more than 500,000.
- Neutropenia plus a fever should be treated as a potentially life-threatening infection until proven otherwise.
- Infection is the most common cause of neutropenia in children.
- WBC count determination in the emergency department (ED) has poor sensitivity and specificity for any specific disease process.
- Inflammatory markers, absolute neutrophil count, bandemia, or immature granulocyte count may be more useful for identifying infection than the absolute WBC count.

1. Which of the following statements best describes the use of the WBC count in emergency medicine decision making?
   - **a.** The absence of leukocytosis excludes the presence of significant disease.
   - **b.** The absolute neutrophil count, presence of bandemia, and immature granulocyte count may be more helpful than the total WBC count in identifying bacterial infection.
   - **c.** The total WBC count is both specific and sensitive for serious bacterial infection.
   - **d.** The WBC count is highly discerning in screening for illness in the acute care setting.

Answer: B. The WBC count and accompanying differential count are among the most common laboratory tests ordered in the ED. The WBC count has not proved to be a highly sensitive or specific test, and the absence of leukocytosis does not exclude the presence of significant disease. The WBC count has limited screening value in the acute care setting. The absolute neutrophil count or an elevated band count may be more helpful than the total WBC count in identifying bacterial infection. The immature granulocyte count can be used to help identify patients with or without sepsis.

2. Which of the following can affect the "normal" WBC count?
   - **a.** Age and ethnicity
   - **b.** Exercise
   - **c.** Gender
   - **d.** Tobacco use
   - **e.** All of the above

Answer: E. All of the statements are true. One unique problem in WBC disorders is the wide variability in normal values and the multiple factors influencing them. Normal values for the WBC count are listed in Table 110.1. The "normal" WBC count range is age-dependent and may be shifted upward by exercise, female gender, smoking, or pregnancy. Decreases in the total WBC count range by 1000 to 1200 cells/mm^3 have been noted in the Black population.

3. In evaluating a patient with severe neutropenia, which of the following is false?
   - **a.** If the patient is febrile, basic isolation techniques and specific therapies should be initiated after cultures are completed.
   - **b.** If the patient is febrile, urine and blood cultures should be obtained.
   - **c.** The clinician should ask about medications, history of neutropenia, family history, and also review information regarding recent infections.
   - **d.** The classic physical findings associated with infection will be obvious owing to the inflammatory response.
   - **e.** The review of systems focuses on bleeding problems, generalized symptoms such as fatigue, sweats, and weight loss, or autoimmune symptoms.

Answer: D. The classic physical signs of infection may be blunted in severe neutropenia, because the inflammatory or purulent response may be hampered by the limited numbers of neutrophils.

4. A patient with a known history of chronic myeloid leukemia (CML) presents with shortness of breath. The WBC count is 25,000 cells/mm^3. Which of the following conditions is LEAST likely to be the cause of the patient's dyspnea:
   - **a.** Angina
   - **b.** Heart failure
   - **c.** Hyperleukocytosis resulting in pulmonary ventilation-perfusion abnormalities
   - **d.** Renal failure and fluid overload
   - **e.** Severe anemia

Answer: C. Hyperleukocytosis can occur in CML, though the comparatively more mature, "less sticky" cells found in CML rarely cause problems unless the count exceeds 500,000 cells/mm^3. A higher cell count may cause leukostasis and result in deafness, visual impairment, pulmonary ventilation-perfusion abnormalities, or priapism. Treatment involves hydration, leukapheresis, transfusion as necessary, allopurinol to prevent severe hyperuricemia, and specific chemotherapy. The need for urgent therapy in CML is usually related to hyperuricemia and renal injury, or severe anemia and subsequent angina or heart failure.

5. When applied to lymphocytic neoplasms, the terms *acute* and *chronic* describe all of the following except:
   - **a.** Aggressiveness of therapy required in treatment
   - **b.** Cell maturity
   - **c.** Patient survival time
   - **d.** Rapidity of onset

Answer: C. Historically, *acute* and *chronic* were descriptive terms applied to lymphocytic neoplasms with respect to patient survival time. The terms *acute* and *chronic* are currently utilized to describe cell maturity, rapidity of onset, and aggressiveness of therapy required in treatment. Chronic lymphocytic leukemia (CLL) is primarily a B-cell disorder and is the most common form of leukemia in patients 50 years or older. Acute lymphocytic leukemia (ALL) is most commonly diagnosed in children younger than 10 years and is the most frequent malignant neoplasm in children younger than 15 years of age.

KEY CONCEPTS

- Although hemostatic disorders are confirmed through laboratory testing, a careful history and thorough physical examination may provide clues to the diagnosis.
- The use of antithrombotic agents remains widespread, whereas diseases such as hemophilia or disseminated intravascular coagulation (DIC) are encountered infrequently in the emergency department (ED) setting.
- Critical thrombocytopenia increases the risk of bleeding, particularly with trauma or invasive procedures. Platelet dysfunction may occur with platelet levels outside of or within the normal range. Aspirin therapy or renal disease, for instance, can cause platelet dysfunction in the setting of a normal platelet count.
- Patients with suspected new diagnosis of immune thrombocytopenic purpura (ITP) should typically be admitted for further management; glucocorticoid therapy is the mainstay of treatment.
- In patients with possible heparin-induced thrombocytopenia (HIT), clinical scoring systems are helpful in risk stratifying the possibility of HIT, prompting early cessation of heparin, alternative anticoagulation, and hematology consultation. Spontaneous HIT should be considered in patients following major surgery (typically an orthopedic procedure), or with recent serious infection.
- Thrombotic thrombocytopenic purpura (TTP) should be suspected if both thrombocytopenia and microangiopathic hemolytic anemia (MAHA) are identified. Early treatment includes plasma exchange therapy.
- Platelet transfusion is rarely indicated unless platelet counts are below 10,000/mm^3 or severe life-threatening bleeding occurs. Platelet transfusion should be avoided in the setting of thrombotic microangiopathies, such as TTP, hemolytic uremic syndrome (HUS), the "hemolysis, elevated liver function tests, and low platelets" (HELLP) syndrome complicating pregnancy, or HIT.
- Hemophiliacs are often highly informed about their disease. It is imperative that prompt intervention with replacement therapy occurs early when bleeding or the potential for bleeding is suspected. As a general rule, 1 U/kg of factor VIII will increase the circulating factor VIII level by 2%.
- Hemophilia with inhibitors creates a challenge for emergency resuscitation, and a treatment option is recombinant Factor VIIa.

1. What platelet count is generally recommended prior to performance of a lumbar puncture?
   a. 100,000 mm^3
   b. 50,000 mm^3
   c. 40,000 mm^3
   d. 20,000 mm^3
   e. 10,000 mm^3

Answer: B. A platelet count above 50,000 mm^3 is recommended prior to performance of invasive spinal procedures such as lumbar procedure. A platelet count above 20,000 mm^3 is recommended for central line placement. A platelet count above 100,000 mm^3 is recommended prior to neurosurgical procedures or invasive ophthalmologic interventions

2. An isolated elevation in prothrombin time (PT) or INR indicates an abnormality or deficiency in which of the following?
   a. Calcium
   b. Factor VII
   c. Factor XI
   d. Fibrinogen
   e. Platelets

Answer: B. The PT or INR tests the components of the extrinsic pathway, and abnormalities or deficiency in factor VII will cause elevation in PT and INR. Calcium derangements do not typically cause aberrations in PT, INR, or PTT. Factor XI is a component of the intrinsic pathway reflected in the PTT. Fibrinogen function is reflected in both prothrombin time (PT) and PTT values. Platelet deficiencies do not prolong the PT or INR.

3. If initial treatment of ITP with glucocorticoids fails, which of the following is most reasonable additional treatment for severe thrombocytopenia?
   a. Splenectomy
   b. Four factor-PCC
   c. IVIG
   d. Tranexamic acid
   e. Thrombopoietin receptor agonist

Answer: C. IVIG has been shown to improve platelet counts in patients with severe thrombocytopenia secondary to ITP. Splenectomy is considered the last line of treatment. Tranexamic acid prevents fibrinolysis. Thrombopoietin receptor agonists are typically reserved for patients who are not responsive to glucocorticoids and IVIG.

4. A 52 year-old-female presents to the emergency department with confusion, shortness of breath, and fatigue. Laboratory evaluation reveals a platelet count of 14,000 mm^3 and hemoglobin of 7.2. mg/dL, a low MCV, indirect bilirubinemia, and a creatinine of 1.4 mg/dL. Electrocardiogram shows non-specific T wave abnormalities in the lateral leads. She denies chest pain and vitals are normal. What would be the most reasonable next step in management?
   a. Obtain a blood smear and utilize a validated risk score for possible TTP.
   b. Immediate platelet transfusion
   c. Administer tranexamic acid
   d. Start IVIG
   e. Administer heparin

Answer: A. Early identification of TTP is imperative. This patient presents with classic signs and symptoms (neurologic symptoms, fatigue, thrombocytopenia, and microangiopathic hemolytic anemia). A validated score such as the PLASMIC score should be used to rapidly assess the likelihood of TTP, as early plasma exchange is the treatment of choice. A platelet transfusion will more than likely exacerbate symptoms. Tranexamic acid prevents fibrinolysis, and therefore would likely cause harm in the setting of microthrombi in TTP. IVIG has not been shown to be the front-line treatment for TTP. Heparin will further increase this patient's bleeding risk. The nonspecific T wave changes are likely due to TTP.

5. A 22-year-old patient with hemophilia A presents to the ED with a headache. He has a history of migraines but this feels different from his previous headaches. He denies any recent significant closed head injury, though states he fell while skateboarding 1 week prior. He is concerned about the possibility of bleeding. Neurologic examination is unremarkable. Upon further review of available records, the patient has a history of severe hemophilia A with inhibitors. His headache is worsening. What would be the best next step in management?
   a. Obtain imaging of brain and cervical spine.
   b. Consult neurosurgery for possible intracranial bleed.
   c. Administer factor VII as expeditiously as possible.
   d. Administer factor VIII as expeditiously as possible.
   e. Administer a migraine "cocktail" for his headache and reevaluate in 30 minutes.

Answer: C. Patients with known hemophilia should be treated empirically with factor replacement when intracranial bleeding is first *suspected*. An atypical headache, particularly in the setting of trauma (even seemingly minor or incidental trauma), should be treated with factor replacement as expeditiously as possible. Neurosurgery may need to be contacted, though consultation should not delay factor administration. Although this patient has hemophilia A (factor VIII deficiency), the presence of inhibitors (antibodies toward factor VIII) limit the utility of factor VIII replacement. In the setting of hemophilia with known inhibitors and bleeding, recombinant factor VIIa is recommended. Medications for treatment of routine migrainous symptoms may be considered secondary to exclusion of life- or disability-threatening intracranial hemorrhage.

KEY CONCEPTS

- Patients whose absolute neutrophil count is or is expected to soon be 500 cells/mm^3 or lower are considered severely neutropenic. A single temperature of 38.3°C or sustained temperature of 38.0°C for 1 to 2 hours or longer is considered fever.
- Any neutropenic patient with fever or with infectious signs or symptoms (even in the absence of fever) should be evaluated for an infectious source, including drawing of blood cultures, and started on empirical antibiotics. Those with high-risk features (e.g., prolonged or profound neutropenia, pneumonia, hypotension, abdominal pain, neurologic changes, Multinational Association for Supportive Care in Cancer (MASCC) score < 21) should be started on an antipseudomonal beta-lactam (e.g., cefepime, piperacillin-tazobactam, antipseudomonal carbapenem). Those with low-risk features may be appropriate for oral antibiotics. Empirical gram-positive bacterial, antifungal, and antiviral coverage is unnecessary unless the clinical situation dictates otherwise.
- Neutropenic patients with fever should generally be hospitalized, including all high-risk patients. Select low-risk patients may be managed as outpatients.

1. Which of the following statements regarding infections in cancer patients with febrile neutropenia is *true*?
   a. Blood cultures should be obtained from an indwelling catheter if present.
   b. Empirical antifungal coverage should always be immediately initiated.
   c. Head computed tomography (CT) is always indicated.
   d. Lumbar puncture (LP) should be routinely undertaken.
   e. Sinus imaging should be routinely undertaken.

Answer: A. In patients with a preexisting central line, one blood culture should be obtained peripherally while other cultures should be simultaneously drawn off each lumen of the central catheter. Bacterial growth in the catheter-drawn samples more than 2 hours prior to the peripheral samples is suggestive of a catheter-associated infection. Head CT is not routinely indicated. It may be performed if the patient demonstrates focal neurologic deficits, alteration in mental status, or a lumbar puncture is planned. LP and CT scans of the sinuses are not routinely done initially unless signs or symptoms suggest these are involved.

2. A 33-year-old woman with breast cancer presents with fever and neutropenia. She is status post mastectomy and undergoing chemotherapy. She denies respiratory symptoms and chest x-ray is unremarkable. Vital signs are T 38.8°C; HR, 110 beats/min; and BP, 110/60 mm Hg. Examination is otherwise normal. She has no indwelling catheters. Which of the following is the most appropriate empirical antibiotic regimen?
   a. Antibiotics should be withheld pending radiographs and cultures
   b. Cefepime as monotherapy
   c. Ceftazidime and gentamicin
   d. Meropenem and vancomycin
   e. Ticarcillin and amikacin

Answer: B. Monotherapy with imipenem, meropenem, ceftazidime, or cefepime is as effective as traditional dual therapy with an antipseudomonal penicillin and aminoglycoside. However, vancomycin should be included as well if any of the following exist:

- Hypotension or evidence of cardiovascular impairment
- Clinically suspected catheter infection
- Positive blood cultures for gram-positive organisms
- Known colonization with methicillin-resistant *Staphylococcus aureus* or cephalosporin-resistant *Pneumococcus*

3. Which of the following is not a risk factor for acute tumor lysis syndrome (TLS)?
   a. High-burden, rapidly growing and highly chemosensitive leukemias
   b. Hypovolemia
   c. Age older than 60 years
   d. Preexisting renal dysfunction
   e. Rapidly growing solid tumors

Answer: C. TLS is most likely to occur in patients with tumors that are high burden, rapidly growing, and highly chemosensitive, such as Burkitt lymphoma or acute lymphoblastic leukemia (ALL). TLS can also occur in rapidly growing solid tumors, such as breast, testicular, and small cell lung cancers; overall, the diversity of malignancies with potential for TLS is growing as we develop more effective and fast-acting chemotherapeutics. Patient factors predisposing to TLS include preexisting renal failure, hypovolemia, and hyperuricemia. Older age is not a known risk factor for TLS.

4. A 65-year-old male with a history of a large head and neck tumor presents with symptoms of fatigue, weakness, confusion, depression and malaise, nausea, vomiting, constipation, polyuria, and palpitations. His electrocardiogram (ECG) demonstrates a shortened QT interval. Which of the following electrolyte abnormalities is likely to underlie his presentation?
   a. Hypercalcemia
   b. Hyperkalemia
   c. Hyperuricemia
   d. Hypocalcemia
   e. Hypokalemia

Answer: A. This patient's presentation is most consistent with malignancy-associated hypercalcemia. Malignancy-associated hypercalcemia (MAH) signifies advanced disease, with median survival of less than 2 months. Synthesis of PTHrP, classically called *humoral hypercalcemia*, causes about 80% of cases of MAH, and is usually associated with squamous cancers, such as head and neck, lung, esophageal, cervical, ovarian, and endometrial carcinomas. Presenting symptoms often include weakness, lethargy, confusion, abdominal pain, nausea, vomiting, constipation, polyuria, polydipsia, and kidney injury. Electrocardiogram findings may initially exhibit QT interval shortening, progressing to dysrhythmias and heart block as hypercalcemia worsens.

5. Which of the following monoclonal antibodies is involved in the management of severe cytokine release syndrome due to adoptive cell transfer therapy (CAR T-cell therapy)?
   a. Alemtuzumab
   b. Cetuximab
   c. Ofatumumab
   d. Tocilizumab

Answer: D. Tocilizumab (anti-IL-6 receptor antibody) has been shown to effectively mediate the pro-inflammatory effects of cytokine release syndrome. The other options listed are antitumor agents.

113 Acid-Base Disorders

KEY CONCEPTS

- Metabolic acidoses are classified into wide anion gap and normal anion gap acidoses, based on basic metabolic panel (BMP) values. A wide anion gap metabolic acidosis is present when the gap exceeds 15 mmol/L.
- Common causes of a wide anion gap metabolic acidosis are summarized with the mnemonic MUDPILES (Methanol, Uremia, DKA, Paraldehyde, Polyethylene glycol, or Paracetamol (Acetaminophen), Iron, Lactate, Ethylene Glycol, Salicylates).
- When a wide anion gap metabolic acidosis is identified on the BMP, the potential underlying causes can be determined through analysis of the delta gap and the osmolar gap.
- Common causes of a normal anion gap acidosis are summarized with the mnemonic HARDUP (Hyperalimentation/Hospital-acquired administration of saline, Acid infusion, Addison's disease, Carbonic Anhydrase Inhibitors, Renal Tubular Acidosis, Diarrhea, Ureterosigmoidostomy, and Pancreatic drainage/fistula.)
- The optimal use of sodium bicarbonate therapy for metabolic acidosis is not clear and is the subject of ongoing research. A common strategy is to use sodium bicarbonate to increase pH above 7.10 in severe metabolic acidosis or above 7.20 for patients with severe metabolic acidosis and acute kidney injury.
- Metabolic alkalosis is associated with decreased circulating volume. GI-based chloride losses can be corrected with the administration of sodium chloride–containing intravenous fluids.
- Metabolic alkalosis caused by impaired renal excretion of sodium chloride that does not respond to intravenous fluids.
- Respiratory acidosis or alkalosis relates to carbon dioxide (CO_2) clearance, and thus is dependent on minute ventilation. Respiratory acidoses are caused by etiologies that decrease the ability of the body to rid itself of CO_2, such as primary lung disease, chest wall disorders, and entities that decrease respiratory drive.
- Respiratory alkalosis results from disorders that increase CO_2 clearance such as hyperventilation and salicylate toxicity. Values from the blood gas are used primarily to identify the presence of respiratory acid-base disorders.

1. A patient presents to the emergency department with severe metabolic acidosis. In addition to initial stabilization, what should be a priority for the clinician?

a. Infuse sodium bicarbonate to normalize pH to approximately 7.40.
b. Induce a respiratory alkalosis to normalize pH to approximately 7.40.
c. Identify the cause of metabolic acidosis and target therapy at that cause.
d. Infuse saline with the goal of decreasing the strong ion difference.

Answer: C. The key steps to managing acid-base disorders are: (1) identifying the type of acid-base disorder(s) present; (2) identifying the underlying cause of the acid-base disorder; and (3) initiating therapy to reverse the underlying cause. Optimal therapy varies based on the underlying cause. For example, in the case of metabolic acidosis, primary treatment for diabetic ketoacidosis is insulin, for lactic acidosis from sepsis is increasing tissue perfusion with vasopressors and intravenous fluids, and for severe toxic alcohol exposure is a specific reversal agent. In some scenarios with severe metabolic acidosis, sodium bicarbonate infusion is a reasonable adjunctive therapy, especially those caused by loss of bicarbonate or renal failure, while the underlying cause is being addressed.

2. What causes an increase in the anion gap?

a. A decrease in plasma bicarbonate concentration accompanied by an increase in plasma chloride concentration of the same magnitude
b. An increase in the concentration of unmeasured anions in the plasma
c. Saline infusion
d. Hypoventilation

Answer: B. Plasma is maintained as electrically neutral, meaning equal charges from plasma cations and anions. Sodium [Na^+] is the major cation routinely measured in clinical laboratories. Chloride [Cl^-] and bicarbonate [HCO_3^-] are the major anions routinely measured in clinical laboratories. The anion gap is calculated as: [Na^+] − ([Cl^-] + [HCO_3^-]). In normal physiological states, "unmeasured" anions are more plentiful than "unmeasured" cations, and hence the anion gap is positive (typically between about 9 mmol/L and 15 mmol/L). Metabolic acidosis is defined by low [HCO_3^-]. A normal anion gap metabolic acidosis occurs when a decrease in [HCO_3^-] is accompanied by an increase of [Cl^-] by the same amount to maintain electroneutrality. A wide anion gap metabolic acidosis occurs when a decrease is bicarbonate is accompanied by an increase in anions other than chloride (so called "unmeasured anions").

3. A patient in the emergency department has the following laboratory values. Select the term that best characterizes this patient's acid-base status. pH = 7.36; $Paco_2$ = 36 mm Hg; Pao_2 = 135 mm Hg; Na = 142 mmol/L; Cl = 100 mmol/L; HCO3 = 21 mmol/L.

a. Acidemia
b. Alkalemia
c. Respiratory alkalosis with metabolic compensation
d. Metabolic acidosis with respiratory compensation

Answer: D. The patient's pH is in the normal range of 7.35 to 7.45. Therefore, neither acidemia nor alkalemia is present. However, the patient may have an acid-base disorder with compensation into a normal pH range, or multiple primary acid-base disorders that when combined result in a pH in the normal range. To classify an acid-base disorder, initially interpret the basic metabolic panel findings and then the arterial blood gas findings. This patient's bicarbonate is low at 21 mmol/L, indicating a metabolic acidosis. The anion gap is 21 mmol/L, indicating a wide anion gap metabolic acidosis. According to the rule of 15, respiratory compensation for this metabolic acidosis is expected to result in a $Paco_2$ of 36 mm Hg and a pH of 7.36, consistent with the patient's measured values. Looking at the ABG results, the pH is less than 7.40, indicating a primary acidosis, and $Paco_2$ is less than 40 mm Hg, indicating the respiratory findings are a compensatory effect rather than a primary disorder.

4. How can calculation of an osmolar gap be used clinically?
   - **a.** To rule out toxic alcohol ingestion
   - **b.** To assist with understanding potential causes of a wide anion gap metabolic acidosis
   - **c.** To differentiate primary lung pathology from respiratory muscle weakness in a patient with respiratory acidosis
   - **d.** To understand if metabolic compensation has occurred after respiratory alkalosis

Answer: B. Calculation of the osmolar gap is a method to help understand the cause of a wide anion gap metabolic acidosis. It can help identify the presence of a large amount of osmotically active solutes in the blood other than the primary osmotically active solutes that are commonly measured (Na, counter anions to Na, glucose, urea, and ethanol). An elevated osmolar gap occurs when measured osmolality is higher than predicated based on a calculation accounting for the commonly measured osmotically active solutes. An important cause of wide anion gap metabolic acidosis and elevated osmolar gap is toxic alcohol poisoning (e.g., methanol or ethylene glycol). Although an elevated anion gap with an elevated osmolar gap can suggest toxic alcohol poisoning in the right clinical setting, a normal osmolar gap does not rule out toxic alcohol poisoning because the osmolar gap decreases as alcohols are metabolized and calculation of the osmolar gap assumes normal osmolality before the poisoning, which may not be true, especially in chronically ill patients.

5. A patient presents to the emergency department with shortness of breath. Plasma bicarbonate concentration is 44 mmol/L. What is a potential explanation for this laboratory value?
   - **a.** Administration of a large volume of saline
   - **b.** Acute respiratory failure from a spontaneous pneumothorax sustained 2 hours prior to presentation
   - **c.** Chronic respiratory failure from chronic obstructive pulmonary disease
   - **d.** Diabetic ketoacidosis

Answer: C. A plasma bicarbonate concentration of 44 mmol/L indicates either a primary metabolic alkalosis or metabolic compensation for a respiratory acidosis. Metabolic compensation requires alteration of renal excretion of acid, which usually takes at least 2 days to develop. Therefore, metabolic compensation is typically not present with acute respiratory failure but is present with chronic respiratory failure. Administration of a large volume of saline increases plasma chloride concentration and decreases plasma bicarbonate concentration to maintain electroneutrality. Respiratory acidosis from a pneumothorax sustained 2 hours previously would not be associated with metabolic compensation because not enough time would have elapsed for the metabolic compensatory mechanisms to have an effect. Respiratory acidosis from chronic respiratory failure is associated with metabolic compensation and thus an elevated bicarbonate concentration. Diabetic ketoacidosis results in accumulation of "unmeasured" anions (acetoacetic acid and beta-hydroxybutyric acid) and a decrease in plasma bicarbonate to maintain electroneutrality.

ACID-BASE DISORDER SAMPLE CALCULATIONS

Rule of 15 Example 1

Na = 140 mmol/L; Cl = 110 mmol/L; HCO_3 = 10 mmol/L; $PaCo_2$ = 20; pH = 7.32.

- *Check the numbers:* The HCO_3 is low, indicating a metabolic acidosis.
- *Calculate anion gap*: AG = 140 − (110 + 10) = 20
  A calculated value of 20 indicates the presence of wide anion gap metabolic acidosis.
- *Apply the rule of 15*: HCO_3^- + 15 = 10 + 15 = 25.
  Simple respiratory compensation would lead to a $Paco_2$ of 25 mm Hg (±2) and a pH of 7.25 (±0.02). The patient's measured value for $Paco_2$ (20 mm Hg) is lower and the measured value for pH (7.32) is higher than predicted by the rule of 15. These "more alkalotic" values indicate the patient is hyperventilating more than expected to compensate for metabolic acidosis. This patient has a primary respiratory alkalosis in addition to a primary wide anion gap metabolic acidosis.

Rule of 15 Example 2

- *Check the numbers*: Na = 140 mmol/L; Cl = 100 mmol/L; HCO_3 = 20 mmol/L; $Paco_2$ = 35 mm Hg; pH = 7.35.
  A low bicarbonate value indicates metabolic acidosis.
- *Calculate anion gap*: AG = 140 − (100 + 20) = 20.
  A high anion gap indicates a wide anion gap metabolic acidosis.
- *Apply the rule of 15*: HCO_3 + 15 = 20 + 15 = 35.
  Simple respiratory compensation would lead to a $Paco_2$ of 35 mm Hg (±2) and a pH of 7.35 (±0.02). The patient's measured values match these expected values; hence, this patient has a simple wide anion gap primary metabolic acidosis with appropriate respiratory compensation.

Rule of 15 Example 3

- *Check the numbers*: Na = 140 mmol/L; Cl 105 mmol/L; HCO_3 = 10 mmol/L; $Paco_2$ = 32 mm Hg; pH = 7.14.
  A low bicarbonate indicates a metabolic acidosis.
- *Calculate the anion gap*: AG = 140 − (105 + 10) = 25.
  A high anion gap indicates wide anion gap metabolic acidosis.
- *Apply the rule of 15*: HCO_3^- + 15 = 10 + 15 = 25.
  Simple respiratory compensation would lead to a $Paco_2$ of 25 mm Hg (±2) and a pH of 7.25 (±0.02). The patient's measured value for $Paco_2$ (32 mm Hg) is higher and the measured value for pH (7.14) is lower than predicted by the rule of 15. These "more acidotic" values indicate the patient is hypoventilating in the context of a metabolic acidosis. This patient has a primary respiratory acidosis in addition to a primary wide anion gap metabolic acidosis.

Rule of 15 Example 4

- *Check the numbers*: Na = 140 mmol/L; Cl 105 mmol/L; HCO_3 = 10 mmol/L; $Paco_2$ = 32 mm Hg; pH = 7.14.
  A low bicarbonate indicates a metabolic acidosis.
- *Calculate the anion gap*: AG = 140 − (105 + 10) = 25.
  A high anion gap indicates wide anion gap metabolic acidosis.
- *Apply the rule of 15*: HCO_3^- + 15 = 10 + 15 = 25.
  Simple respiratory compensation would lead to a $Paco_2$ of 25 mm Hg (±2) and a pH of 7.25 (±0.02). The patient's measured value for $Paco_2$ (32 mm Hg) is higher and the measured value for pH (7.14) is lower than predicted by the rule of 15. These "more acidotic" values indicate the patient is hypoventilating in the context of a metabolic acidosis. This patient has a primary respiratory acidosis in addition to a primary wide anion gap metabolic acidosis.

Delta Gap Example 1

- *Check the numbers*: Na = 140 mmol/L; Cl 92 mmol/L; HCO_3 = 20 mmol/L; $Paco_2$ = 37 mm Hg; pH = 7.37.
  A low bicarbonate indicates a metabolic acidosis.
- *Calculate the anion gap*: AG = 140 − (92 + 20) = 28.
  A high anion gap indicates wide anion gap metabolic acidosis.
- *Apply the rule of 15*: HCO_3 + 15 = 20 + 15 = 35.
  Simple respiratory compensation would lead to a $Paco_2$ of 35 mm Hg (±2) and a pH of 7.35 (±0.02). The patient's measured value for $Paco_2$ (37 mm Hg) is consistent with the rule of 15, indicating no primary respiratory acid-base disturbance.
- *Calculate the delta gap*: Change in AG = measured AG − 15 = 28 − 15 = 13. Change in HCO_3 = 24 − measured HCO_3 = 24 − 20 = 4.
  The predicted change in bicarbonate concentration is 13 mmol/L, but the measured change is only 4 mmol/L. The measured bicarbonate is higher than predicted by the delta gap calculation, indicating the presence of a primary metabolic alkalosis in addition to the wide anion gap metabolic acidosis.

Delta Gap Example 2

- *Check the numbers*: Na = 140 mmol/L; Cl = 105 mmol/L; HCO_3 = 12 mmol/L; $Paco_2$ = 27 mm Hg; pH = 7.27.
  A low bicarbonate indicates a metabolic acidosis.
- *Calculate the anion gap*: AG = 140 − (105 + 12) = 23.
  A high anion gap indicates wide anion gap metabolic acidosis.
- *Apply the rule of 15*: HCO_3 + 15 = 12 + 15 = 27.
  Simple respiratory compensation would lead to a $Paco_2$ of 27 mm Hg (±2) and a pH of 7.27 (±0.02). The patient's measured value for $Paco_2$ CO_2 (27 mm Hg) is consistent with the rule of 15, indicating no primary respiratory acid-base disturbance.
- *Calculate the delta gap*: Change in AG = AG − 15 = 23 − 15 = 8. Change in HCO_3 = 24 − measured HCO_3 = 24 −12 = 12.
  The predicted change in bicarbonate concentration is 8 mmol/L, but the measured change is 12 mmol/L. The measured bicarbonate is lower than predicted by the delta gap calculation, indicating the presence of a primary normal anion gap metabolic acidosis in addition to a wide anion gap metabolic acidosis.

ABG Determination of Predominant Respiratory or Metabolic Acid-Base Disorder Example 1

ABG: pH = 7.31 / $PaCO_2$ = 53mm Hg / PaO_2 = 98mm Hg

- Determine if the patient is acidemic or alkalemic. The patient's pH is 7.31, indicating acidemia.
- Determine if a predominant respiratory or metabolic acid-base disturbance is present. The patient's $Paco_2$ is high at 53 mm Hg and pH is low at 7.31. This is consistent with a predominant respiratory acidosis.

ABG Determination of Predominant Respiratory or Metabolic Acid-Base Disorder Example 2

ABG: pH = 7.52 / $PaCO_2$ = 60mm Hg / PaO_2 = 84mm Hg

- Determine if the patient is acidemic or alkalemic.
  The patient's pH is 7.52, indicating alkalemia.
- Determine if a predominant respiratory metabolic acid-base disturbance is present. The patient's $Paco_2$ is high at 60 mm Hg and pH is high at 7.52.
  This is consistent with a predominant metabolic alkalosis. The high $Paco_2$ represents respiratory compensation via hypoventilation.

ABG Determination of Predominant and Concomitant/ Compensatory Acid-Base Disorders Example 1

ABG: pH = 7.00 / $PaCO_2$ = 70mm Hg / PaO_2 = 105mm Hg

- Determine if the patient is acidemic or alkalemic.
  The patient's pH is 7.00, indicating acidemia.
- *Determine if a predominant respiratory or metabolic acid-base disturbance is present.* The patient's $PaCO_2$ is high at 70 mm Hg and pH is low at 7.00. This is consistent with a respiratory acidosis.
- Determine if there is concurrent metabolic disturbance.

Change in $PaCO_2$ = 40 – measured $PaCO_2$ = 40 – 70 = –30. Assuming a pure respiratory acidosis with no metabolic acid-base disturbance, predicted pH = 7.40 + [(40 – 70)/10 * 0.08] = 7.40 – 0.24 = 7.16. The measured pH of 7.00 is lower than the predicted pH of 7.16, indicating a concurrent metabolic acidosis.

KEY CONCEPTS

- Electrolyte abnormalities are common in emergency medicine and can vary greatly in importance, severity, and symptoms. Asymptomatic electrolyte abnormalities can be gradually corrected, whereas those causing alterations in consciousness or life-threatening dysrhythmias require immediate therapy to avoid permanent sequelae or death. In some cases, therapy for life-threatening electrolyte disorders may precede laboratory confirmation.
- Asymptomatic electrolyte abnormalities can usually be corrected slowly, but those that cause profound mental status changes or life-threatening arrhythmias require immediate correction to avoid cardiac arrest or seizures.
- IV calcium should be used only for hyperkalemic emergencies, defined as the following: widening QRS; sine wave; bradycardias and cardiac arrest believed to be due to hyperkalemia; or rapidly evolving electrocardiographic changes, from normal to development of tall peaked T waves and loss of the P wave. Acute, rapid rises in serum potassium concentration are rare but may be seen in tumor lysis syndrome, rhabdomyolysis, or massive hemolysis.
- After the critical decision about administration of calcium has been made, a beta$_2$-agonist, insulin and glucose, normal saline, and bicarbonate (if the patient is acidotic) can be given to shift potassium intracellularly.
- When treating hypokalemia, the physician should also replace magnesium sulfate, in addition to potassium, or the patient will excrete most of the infused potassium in the urine.
- Low serum potassium levels reflect a substantial total potassium deficit; correction of large deficits can require several days.
- Hypertonic saline should be reserved for severely hyponatremic patients (typically in the 100 to 110 mEq/L range) who present with coma, seizures, or focal neurologic deficits. Central pontine myelinolysis can occur if serum sodium concentration is raised rapidly by more than 8 mEq/day.

1. A patient with end-stage renal disease misses dialysis and presents feeling weak. His 12-lead electrocardiogram (ECG) shows peaked T waves, absent P waves, and a QRS duration of 240 milliseconds. Which of the following is the next step in emergency department (ED) management?
   a. Calcium
   b. Furosemide
   c. Hemodialysis
   d. Insulin with glucose
   e. Nebulized albuterol

Answer: A. Calcium is the first medication that should be administered in this patient with a widened QRS. The treatment of hyperkalemia is based on the clinical scenario combined with the 12-lead ECG and the laboratory potassium value. Evaluating the ECG of patients at risk for this electrolyte disturbance is critical because results of the serum potassium level can be delayed. Hyperkalemia can be rapidly progressive, and lifesaving interventions should be instituted at the earliest suspicion of toxicity. Intravenous (IV) calcium is used to stabilize the cardiac membrane by restoring the electrical gradient. Calcium increases the depolarization threshold and the calcium gradient across the cardiac membrane, quieting myocyte excitability and increasing cardiac conduction speed, thus narrowing the QRS.

2. For which of the following patients would it be most appropriate to supplement potassium to keep serum levels above 4 mEq/L?
   a. A 23-year-old woman with palpitations and no past medical history
   b. A 42-year-old woman with poorly controlled diabetes mellitus
   c. A 51-year-old man with hypertension, recently started on lisinopril
   d. A 55-year-old man with history of coronary artery disease, chest pain, and premature ventricular complexes
   e. A 68-year-old-man with chronic bronchitis and fever

Answer: D. The 55-year-old patient is at highest risk for life-threatening complications from hypokalemia secondary to his underlying comorbidities. In patients with cardiac ischemia or heart failure, even mild to moderate hypokalemia increases the likelihood of cardiac arrhythmias secondary to potassium's effect on the action potential.

3. A malnourished man with a long history of chronic obstructive pulmonary disease is brought to the emergency department (ED) in respiratory distress. He receives bronchodilators, intravenous (IV) dextrose, and thiamine. He improves significantly, but then begins to complain of severe muscle aches, diffuse weakness, and the feeling that he cannot breathe deeply. Supplementation with which of the following might have prevented his new symptoms?
   a. Calcium gluconate
   b. Diazepam
   c. Normal saline
   d. Potassium phosphate
   e. Sodium bicarbonate

Answer: D. Patients with hypophosphatemia may present with nonspecific complaints, including joint pain, myalgias, irritability, and depression. Severe hypophosphatemia can be manifested as seizures, arrhythmias, cardiomyopathy, insulin resistance, acute tubular necrosis, rhabdomyolysis, and acute respiratory failure. Because hypophosphatemia often presents with hypokalemia, phosphate repletion should be considered in conjunction with potassium administration.

4. A 58-year-old man with history of small cell lung cancer is brought in for weakness and pleuritic chest pain. He is alert, with normal vital signs, nonfocal neurologic examination, and serum sodium concentration of 129 mEq/L. What is the most appropriate management of the hyponatremia?
   a. Free water restriction
   b. Hypertonic saline bolus
   c. Intravenous (IV) furosemide
   d. Normal saline bolus
   e. Sodium chloride tablets

Answer: A. The patient with small cell lung cancer is demonstrating signs and symptoms of hyponatremia. Lung cancers, especially small cell cancer, can lead to the syndrome of inappropriate secretion of antidiuretic hormone (SIADH). The mainstay for treatment of euvolemic hyponatremia is free water restriction. Because the hypoosmolality in SIADH results from a relative abundance of water in the intracellular and extracellular volumes and is maintained by a reduced ability to excrete water, the restriction of free oral water intake is usually the most effective therapy. For relatively asymptomatic patients with sodium values of 115 to 135 mEq/L, a trial of free water restriction to less than 0.8 mL/day to 1.25 L/day should be attempted.

5. A runner is brought in after collapsing during a race. He has been careful to drink at each mile marker. He has altered mental status and is combative even after 10 mg of diazepam. What is the most appropriate treatment after his witnessed tonic-clonic seizure?
   a. 5% Dextrose solution
   b. Fosphenytoin
   c. 3% Hypertonic saline
   d. Lorazepam
   e. Pentobarbital

Answer: C. This runner is demonstrating symptoms concerning for exercise-associated hyponatremia, the occurrence of hyponatremia during or up to 24 hours after prolonged physical activity. Hypertonic saline is indicated in severe cases of hyponatremia when a patient has alterations in mental status, focal neurologic findings, or seizures.

KEY CONCEPTS

- The diagnosis of diabetes can be determined by one or more of four methods—random plasma glucose level above 200 mg/dL, fasting plasma glucose concentration above 126 mg/dL, 2-hour, 75-g post-load oral glucose tolerance test (OGTT) > 200 mg/dL, or HbA_{1c} value above 6.5%.
- Diabetic ketoacidosis (DKA) is diagnosed by the presence of hyperglycemia, anion gap metabolic acidosis, and elevated ketoacid levels.
- The essential treatment of DKA includes administration of insulin, correction of dehydration, correction of potassium level, correction of acidosis, and treatment of the underlying cause.
- The use of sodium bicarbonate to correct acidosis in DKA has not demonstrated any benefit and may be associated with worse outcomes.
- A hyperglycemic hyperosmolar state is usually seen in older adults with multiple comorbid conditions and is distinguished from DKA by the absence of ketoacidosis. In addition to fluid resuscitation and correction of hyperglycemia, treatment should address the underlying cause of the state, which includes infection, myocardial infarction, and cerebrovascular accident.
- Diabetic peripheral neuropathy is common and has multiple treatment options, including gabapentin, pregabalin, and duloxetine.
- Diabetic foot ulcers and other diabetic soft-tissue infections (e.g., gas gangrene, Fournier's gangrene) are frequently polymicrobial and require broad-spectrum antibiotic therapy covering gram-positives, gram-negatives, and anaerobes.
- Hypoglycemia may be associated with significant morbidity and mortality. When the diagnosis is suggested and, if possible, confirmed by laboratory evaluation, therapy should be initiated immediately.
- Hypoglycemia caused by sulfonylurea oral hypoglycemic agents may be prolonged. Patients should be observed for an extended period or hospitalized.

1. Which of the following statements regarding patients with impaired glucose tolerance is true?
   a. Spontaneous reversion to normal glucose tolerance is rare.
   b. The condition is associated with fewer complications than diabetes mellitus.
   c. The rate of decompensation to diabetes mellitus is greater than 10%/year.
   d. There is a predisposition to ketosis.

Answer: b. Patients with impaired glucose tolerance have a glucose level between normal and diabetic. They are at increased risk of cardiovascular disease and development of diabetes (1% to 5%/year), but it is not associated with the same degree of complications as with true diabetes. Many patients spontaneously develop normal glucose levels.

2. What percentage of adults with type 2 diabetes are obese?
   a. 40%
   b. 60%
   c. 80%
   d. 100%

Answer: d. Nonobese patients form a subgroup with a different disease, more similar to type 1 diabetes. Young people with maturity-onset diabetes often have an autosomal dominant inheritance, are nonobese, and have a relatively mild disease course.

3. A 56-year-old man with a 10-year history of type 2 diabetes and poor glucose control (HbA_{1c} = 10.7%) complains of constant burning pain in both feet. Which agent would be most appropriate to start in this patient for the initial management of his symptoms?
   a. Aspirin, 325 mg PO daily
   b. Naproxen, 500 mg bid
   c. Oxycodone/acetaminophen, 5/325 mg qid
   d. Pregabalin, 600 mg daily

Answer: d. Pregabalin in a dose of 600 mg daily gives pain relief in approximately 50% of patients with diabetic neuropathy. Duloxetine, 60 mg daily, achieves similar results. Gabapentin, 300 mg once a day up to tid, and amitriptyline, 25 mg daily, provide pain relief in approximately 33% of patients.

4. A 27-year-old juvenile-onset diabetic is brought by emergency medical services (EMS) for a hypoglycemic coma. Fingerstick glucose level is 30 mg/dL. The paramedics were not able to obtain intravenous (IV) access, and two immediate attempts at IV cannulation failed in the emergency department (ED). What should be the next step in the patient's management?
   a. Albuterol, 2.5 mg nebulized
   b. Central venous catheter placement, then $D_{50}W$ IV
   c. Epinephrine, 1 mg IV
   d. Glucagon, 2 mg intramuscularly

Answer: d. Intravenous dextrose (25 to 75 g for adults, 0.5to 1 g/kg for children) is preferable but, if unable to obtain IV access, administer glucagon, 1 or 2 mg intramuscularly (IM) or subcutaneously (SC; 0.025 to 0.1 mg/kg IM or SC in children). The onset of action is 10 to 20 minutes. It is ineffective in cases of glycogen absence, such as alcohol-induced hypoglycemia.

5. A 33-year-old juvenile-onset, insulin-dependent diabetic suddenly faints without prodrome or warning while walking through the ED. Relatives report that diabetes is his only past history. Which of the following findings is most likely?
   a. Autonomic neuropathy on later orthostatic testing
   b. Fingerstick glucose 27 mg/dL
   c. Hemoccult-positive stool
   d. Positive enzyme-linked immunosorbent assay (ELISA) D-dimer

Answer: b. Hypoglycemia without warning, or hypoglycemia unawareness, is a complication of type 1 diabetes caused by previous hypoglycemic episodes. A single hypoglycemic episode may blunt neurohormonal counterregulatory responses to later hypoglycemic episodes. Risk factors are overaggressive insulin therapy, longer history of diabetes, and autonomic neuropathy, which usually causes orthostasis on first standing or after being upright in a static position. These patients may become abruptly unarousable without warning.

6. A 47-year-old man presents with hypoglycemia. He is a known type 2 diabetic on glyburide. Fingerstick glucose is 27 mg/dL. Twenty minutes after two ampules (50 g) of dextrose, his glucose level is 29 mg/dL. Which of the following agents is indicated?
   a. Adenosine
   b. Epinephrine
   c. Hydrocortisone
   d. Octreotide

Answer: d. A patient with hypoglycemia from sulfonylureas, in addition to standard glucose replacement, frequently requires treatment with an agent to inhibit further insulin release, such as octreotide (a somatostatin analog). Sulfonylureas are insulin secretagogues.

7. What is the most important determinant of mental status in a patient with diabetic ketoacidosis (DKA)?
   a. Acidemia
   b. Calcium level
   c. Glucose level
   d. Osmolarity

Answer: d. The hyperosmolarity produced by dehydration and hyperglycemia is the most important determinant of mental status during an episode of DKA.

KEY CONCEPTS

- Rhabdomyolysis is a process wherein striated muscle cells break down, resulting in the release of creatine kinase, potassium, calcium, phosphate, and uric acid.
- Rhabdomyolysis is associated with a variety of causes including overexertion, extreme body temperature changes, ischemia, infection, immobility, drugs, toxins, endocrine disease, autoimmune reactions, trauma, and genetic conditions.
- The classic presentation of rhabdomyolysis is myalgia, weakness, and dark-colored urine. However, the absence of this triad does not exclude the condition. Clinicians should suspect the disease in patients with risk factors, dark urine, and elevated creatine kinase.
- Diagnosis of rhabdomyolysis is based on elevated creatine kinase more than five times the upper limit of normal (CK > 1000 U/L).
- Although most patients recover without complications, severe rhabdomyolysis is associated with renal failure, electrolyte disorders such as hyperkalemia, compartment syndrome, and disseminated intravascular coagulation.
- Rhabdomyolysis management includes treating the underlying etiology and administering intravenous (IV) hydration with a goal urine output of 2 to 3 mL/kg/hour and euvolemia. Mannitol, sodium bicarbonate, and loop diuretics should not be used routinely in management.
- Although most patients require admission for further evaluation and management, patients with normal vital signs, normal laboratory assessment, normal oral intake, clear etiology, and follow-up may be appropriate for discharge.

1. Which of the following statements regarding muscle cell physiology and rhabdomyolysis is true?
   - **a.** Acute renal failure from rhabdomyolysis is very rare.
   - **b.** Hemoglobin has a higher oxygen affinity than myoglobin.
   - **c.** The final common pathway of injury in rhabdomyolysis is cell membrane damage.
   - **d.** The normal intracellular Na^+ concentration is high.
   - **e.** The normal intracellular Ca^{2+} concentration is low.

Answer: E. Normal intracellular concentrations of Na^+ are low, creating a negative intracellular environment. This allows more facilitated transfer of Ca^{2+} from the intracellular to extracellular space, maintaining low intracellular Ca^{2+} concentrations. The final common pathway of injury is sarcolemma damage with intracellular Ca^{2+} accumulation, proteolytic enzyme inhibition, actin-myosin coupling dysfunction, and cell damage with release of intracellular contents (e.g., myoglobin, PO_4^-, uric acid, lactate dehydrogenase). Myoglobin, which has four times the O_2 affinity of hemoglobin, overwhelms the haptoglobin-binding capacity, is filtered at the glomerulus, and precipitates in renal tubules. Approximately 5% to 15% of cases of acute renal failure in the United States are related to rhabdomyolysis. This myoglobin precipitation is markedly facilitated by acidosis.

2. Which of the following statements regarding exertion (exercise)-related rhabdomyolysis is true?
   - **a.** Eccentric (lengthening) muscle work is more damaging than concentric.
   - **b.** Hypocalcemia increases the risk for this syndrome.
   - **c.** It is the result of voluntary muscle exertion.
   - **d.** It is seen exclusively in untrained athletes.
   - **e.** The mechanism is different than after a crush injury.

Answer: A. Exertional rhabdomyolysis is seen in trained and untrained individuals as well as after exercise or situations of involuntary muscle activity (e.g., psychoses, seizures, tetany, myoclonus). The mechanism (e.g., failure of energy supplies, sarcolemma breakdown, intracellular calcium accumulation, enzyme dysfunction, cellular swelling) is the same. Hypokalemia is a risk factor because a low potassium level limits microvascular dilation and muscle perfusion.

3. Which of the following statements regarding drug-induced rhabdomyolysis is true?
   - **a.** Cocaine myotoxicity is not related to the degree of intoxication.
   - **b.** Colchicine and cyclosporine are myotoxins.
   - **c.** Ethanol myotoxicity is potentiated by high carbohydrate intake.
   - **d.** Statin myotoxicity is unrelated to state of hydration.
   - **e.** The use of the drug ecstasy is not associated with rhabdomyolysis.

Answer: B. Statins, ethanol, cocaine, colchicine, cyclosporine, and many other drugs are myotoxic. Illicit drugs include amphetamine, ecstasy, LSD, and other sympathomimetics. Ethanol is a direct muscle membrane toxin, and this effect is potentiated by starvation, binge drinking, and coexisting electrolyte abnormalities (low K^+, PO_4^-, and Mg^{2+}). Intravenous cocaine use is more myotoxic than inhaled cocaine, and the severity seems to parallel the degree of intoxication.

4. What is the most common viral cause for rhabdomyolysis?
   - **a.** Cytomegalovirus
   - **b.** Epstein-Barr virus
   - **c.** Herpesvirus
   - **d.** Human immunodeficiency virus
   - **e.** Influenza virus

Answer: E. Influenza is the most common viral etiology followed by HIV infection and enteroviral infection.

5. What is the most common bacterial cause for rhabdomyolysis?
   - **a.** *Legionella*
   - **b.** *Pseudomonas*
   - **c.** *Salmonella*
   - **d.** *Staphylococcus*
   - **e.** *Streptococcus*

Answer: A. *Legionella* is the bacterium classically associated with rhabdomyolysis in adult patients. The pathogenesis is believed to be due to direct invasion and toxic degeneration of muscle fibers.

6. Which of the following electrolyte abnormalities has not been associated with rhabdomyolysis?
   - **a.** Hypermagnesemia
   - **b.** Hypernatremia
   - **c.** Hypocalcemia
   - **d.** Hyponatremia
   - **e.** Hypophosphatemia

Answer: A. There are no reported cases of hypermagnesemia-induced rhabdomyolysis to date.

7. A 53-year-old intoxicated alcoholic is brought to the ED by EMS after being found unconscious for an unknown reason. He is now awake but mildly lethargic. He has no complaints of pain or disability. The physical examination is nonfocal. Which of the following statements is true?
   - **a.** A CK-MB fraction of 5% would indicate myocardial damage.
   - **b.** A negative urine myoglobin level would exclude rhabdomyolysis.
   - **c.** His lack of pain complaints would exclude rhabdomyolysis.
   - **d.** Hypocalcemia would be expected in the presence of rhabdomyolysis.
   - **e.** If rhabdomyolysis were found, normal phosphate level would be reassuring.

Answer: D. Hypocalcemia is the most common electrolyte abnormality after rhabdomyolysis. Hypercalcemia may follow later. Only 50% of patients with serum evidence of rhabdomyolysis have complaints of muscle pain. Likewise, the presence of urine myoglobin reflects the glomerular filtration rate, plasma myoglobin concentrations, urine flow, and plasma myoglobin binding. This test result may be negative, especially late in the course of the process. Hyperphosphatemia is expected, and a normal level raises the suspicion that hypophosphatemia was the cause of the rhabdomyolysis. CK-MB levels of 3% to 5% are often seen and reflect skeletal rather than cardiac muscle damage.

8. Which of the following is a proven cornerstone of management for rhabdomyolysis, along with hydration?
   - **a.** Alkalinization
   - **b.** Chelation therapy
   - **c.** Furosemide
   - **d.** Mannitol
   - **e.** None of the above

Answer: E. Furosemide is somewhat contraindicated because of its tendency to acidify the urine. Mannitol and alkalinization are not proven, although they are often used. Chelation therapy is under investigation.

117 Thyroid and Adrenal Disorders

KEY CONCEPTS

Hyperthyroidism

- Hyperthyroidism induces a hypermetabolic state and increases β-adrenergic activity. The resulting clinical manifestations range from vague constitutional symptoms to more organ-specific symptoms (see Box 117.1).
- Hyperthyroidism in elders may be asymptomatic or may manifest with subtle nonspecific symptoms such as weight loss, shortness of breath, and/or dementia.
- The laboratory test of choice for suspected hyperthyroidism is the thyroid stimulating hormone (TSH) concentration with free T_4 and T_3 levels.
- Thyroid storm is a life-threatening thyrotoxic crisis that often presents with fever, extreme tachycardia, and/or altered sensorium. It requires prompt recognition and therapy, as well as identification and treatment of any precipitating cause, such as infection.
- The order of medication administration in thyroid storm is critical. Iodine can precipitate thyroid storm and must be given a minimum of 1 hour after thionamide therapy (PTU or methimazole). As such, the typical order is beta blocker (propranolol), PTU or methimazole, and then iodine (SSKI, Lugol solution).

Hypothyroidism

- Hypothyroidism results from lack of stimulation of the thyroid gland (central or secondary hypothyroidism) or intrinsic gland dysfunction limiting hormone production (primary hypothyroidism).
- Signs and symptoms of hypothyroidism range from asymptomatic to overt organ failure, which can lead to death (see Box 117.5).
- Determination of an elevated TSH level is the most sensitive and single best screening test to confirm the diagnosis of primary hypothyroidism.
- Replacement with levothyroxine (T_4) remains the treatment of choice and resolves physical and psychological signs and symptoms in most patients.
- Myxedema coma is a life-threatening event that presents with altered mental status and hypothermia, along with a concomitant precipitating event (see Box 117.6). It usually occurs in patients with untreated or undertreated hypothyroidism. Treatment with thyroid hormone replacement must be initiated, often based solely on clinical findings.

Adrenal Excess States

- Adrenal excess states run the spectrum from Cushing syndrome, to primary and secondary hyperadrenalism, to pheochromocytoma.
- Symptoms of adrenal excess will vary, depending on the etiology, with chronic, nonspecific symptoms that arise from Cushing syndrome (generalized weakness, fatigue, menstrual irregularities, and weight gain), to simple refractory hypertension with hyperaldosteronism, to acute, refractory hypertension and hyperadrenergic symptoms with pheochromocytoma.
- Diagnosis of hyperaldosteronism is typically clinical, and confirmed by laboratory testing and imaging studies, depending on the etiology. In most cases, acute stabilization of the presenting complaint is paramount, and definitive diagnosis will occur outside the ED. Any adrenal incidentaloma discovered on imaging obtained in the ED, especially in the setting of hypertension, should prompt further evaluation for adrenal excess.

Adrenal Insufficiency

- Clinical manifestations of secondary adrenal insufficiency are often vague and nonspecific, including fatigue, weakness, dizziness, nausea, vomiting, and other nonspecific GI symptoms. Patients with primary adrenal insufficiency characteristically have more pronounced clinical manifestations and skin hyperpigmentation.
- The ACTH stimulation test and measurements of cortisol levels is the most convenient method and is considered the criterion standard to make the diagnosis.
- Refractory hypotension in the acutely ill patient may be the only clue to adrenal insufficiency and is readily treated with IV administration of glucocorticoids (hydrocortisone, 100 mg).

1 What is the most common presenting sign of thyrotoxicosis?

a. Atrial fibrillation
b. Means-Lerman scratch
c. Tachycardia
d. Wide pulse pressure

Answer: C. The most common cardiac sign of thyrotoxicosis is tachycardia. Widened pulse pressure, a Means-Lerman scratch (a systolic friction rub), dilated cardiomyopathy, and pulmonary hypertension are all cardiac findings in patients with thyrotoxicosis, with atrial fibrillation being more common in patients over the age of 65.

2. In thyroid storm, which of the following is the proper sequence of drug administration?

a. All three together (antithyroid drugs, sodium or potassium iodide, steroid) at the same time
b. Antithyroid drugs, sodium or potassium iodide, steroid
c. Sodium or potassium iodide, steroid, antithyroid drugs
d. Steroid, sodium or potassium iodide, antithyroid drugs

Answer: B. Iodine can precipitate thyroid storm and must be given a minimum of 1 hour after thionamide therapy (PTU or methimazole). As such, the typical order is beta blocker (propranolol), PTU, or methimazole, and then iodine (SSKI, Lugol solution). In addition, it is important to identify and treat the precipitating cause of thyroid storm.

3. A 32-year-old female with a history of bipolar mood disorder on lithium presents with asthenia, fatigue, cold intolerance, and dry skin. A TSH test confirms your suspected diagnosis. How should you counsel the patient regarding her lithium use?

a. Instruct the patient to continue the lithium, no other intervention is necessary.
b. Instruct the patient to continue the lithium, while initiating therapy with levothyroxine.
c. Instruct the patient to stop her lithium, while initiating therapy with levothyroxine.
d. Instruct the patient to stop her lithium immediately and follow up with her primary care physician for an alternative medication.

Answer: B This patient presents with symptoms of overt hypothyroidism, confirmed by a high TSH on laboratory testing. Instruct the patient that she may continue her lithium for her chronic bipolar mood disorder, while initiating levothyroxine therapy. A well-known cause of hypothyroidism, lithium causes overt hypothyroidism in 14% to 17% of patients, and subclinical hypothyroidism in 19% to 35% of patients. Most of lithium's effects on the thyroid is due to the inhibition of T_4 and T_3 release. Lithium also increases thyroid autoimmunity if it is present before the initiation of lithium treatment. Treatment with exogenous thyroid hormone is effective, and lithium therapy need not be discontinued.

4. A 52-year-old man is brought in by ambulance from his home with altered mental status and hypotension. EMS reports that he became ill with an influenza-like illness 4 days ago and stopped taking all of his medications secondary to vomiting. Despite multiple fluid boluses, he remains hypotensive. Laboratory results demonstrate severe acidosis, hyponatremia and hyperkalemia. What is the most appropriate treatment for this patient?
   a. Adrenocorticotropic hormone (ACTH)
   b. Continued normal saline (0.9%) fluid boluses
   c. Hydrocortisone
   d. Norepinephrine

Answer: C. This patient presents with adrenal crisis, as evidenced by refractory hypotension and electrolyte abnormalities typical for hypoadrenalism. While refractory hypotension in the acutely ill patient may be the only clue to adrenal insufficiency, the paramedics also report that the patient has been unable to take his usual medications, presumably for primary adrenal insufficiency, and in the setting of acute illness has resulted in adrenal crisis. Additional fluids and pressors will be ineffective until appropriate glucocorticoid treatment is initiated, hydrocortisone, 100 mg IV. ACTH administration is used for evaluation of adrenal insufficiency and is not appropriate for therapeutic administration during the acute crisis.

5. A 37-year-old male presents to the ED with complaints of palpitations, abdominal pain, nausea, vomiting, and diaphoresis. He reports episodes similar to this that have been increasing in frequency and severity over several months. His vital signs are P 127, BP 174/90, RR 28, T 37.1°C, and Sao_2 96%. Once stabilized, what is the most appropriate oral treatment for this patient?
   a. Amlodipine 5 mg PO daily
   b. Metoprolol, 100 mg PO BID
   c. Phenoxybenzamine 10 mg PO BID
   d. Propranolol ER 80 mg PO daily

Answer: C. This patient presents with episodic, or paroxysmal symptoms that are consistent with pheochromocytoma. Following initial stabilization and appropriate workup, the patient will require control of his vital sign abnormalities and symptoms leading up to surgery. The first-line treatment for pheochromocytoma is alpha-blockade, so in this question, phenoxybenzamine is correct. In patients with catecholamine excess, beta blockade alone can result in hypertensive crisis because of unopposed alpha-adrenergic receptor stimulation. Beta blockers can be used, but this should only be done following alpha-blockade; as such, propranolol and metoprolol are incorrect. Calcium channel blockers may be used in mild cases or in patients with undesirable alpha-blockade side effects, but are not considered first line; as such, amlodipine is also incorrect.

KEY CONCEPTS

- All septic patients should be treated with antibiotics as soon as possible, even before a definitive diagnosis is made. Patients with pneumococcemia, meningococcemia, and aggressive soft tissue infections can decompensate rapidly.
- The source of sepsis should be identified as soon as possible, and surgical causes should be addressed. A surgeon should be consulted as soon as possible for patients with sepsis and a débridable source of infection.
- Immunity to diphtheria, tetanus, and pertussis wanes significantly in adults. Pertussis should be considered a cause of persistent cough in adults. A tetanus vaccination history should always be obtained from patients with trauma or infection. When there is doubt about the history, the age-appropriate vaccine according to CDC guidelines should be administered.
- Consider botulism in the differential diagnosis for the infant with failure to thrive, constipation, or decreased muscle tone and for the patient who injects drugs with neurologic symptoms.

1. An 18-year-old male recent immigrant presents with acute-onset sore throat, fever, and weakness. He has no past medical history. Examination is remarkable for a grayish white exudative coat over both tonsils and the posterior pharynx. There is modest neck swelling with cough, low-grade fever, and hoarseness. Rapid strep testing and monospot results are negative. Which of the following statements regarding this patient's disease is *true?*

a. Antibiotics and corticosteroids are the main therapy.
b. Neuromuscular examination is critical.
c. Rapid treatment will obviate the need for intubation.
d. The electrocardiogram (ECG) is a sensitive indicator of myocarditis.

Answer: B. Diphtheria can be a more benign upper respiratory/nasal infection or a "malignant" process with airway obstruction, myocarditis, and neuropathy (palate is involved first; the only cells spared are cortical). Antibiotics stop further organism growth, but diphtheria equine antitoxin is the key to therapy. Death is usually from myocarditis or airway obstruction, with the highest mortality seen in patients with the "bull neck" appearance. The ECG is an insensitive indicator of myocarditis.

2. An 11-year-old child presents with coughing paroxysms for 10 days. This was preceded by a mild upper respiratory tract infection. The paroxysms occur multiple times per day and have occasionally caused vomiting. The child is relatively well between paroxysms. Vaccination history is unknown. Which of the following statements regarding this patient's disease is *true?*

a. Antibiotics will shorten illness duration.
b. Erythromycin should be given to unimmunized contacts.
c. Fever is expected.
d. Thoracic petechiae should prompt a septic evaluation.

Answer: B. Pertussis, caused by a gram-negative bacterium, is a three-phase illness: catarrhal (upper respiratory tract infection), paroxysmal, and convalescence of weeks to months. The coughing phase is characterized primarily by complications related to the paroxysms, such as subconjunctival hemorrhage, pneumomediastinum, headache, rectal prolapse, chest wall petechiae, and even seizures. Fever is rare unless there is secondary infection. Antibiotics only reduce the carrier state. Corticosteroids may help younger children. Antibiotic prophylaxis is indicated for nonimmunized contacts. Nasal culture is diagnostic. Standard cough suppressants are ineffective.

3. After which of the following circumstances has tetanus been reported?

a. Childbirth
b. Chronic skin ulcers
c. Corneal abrasion
d. All of the above

Answer: D. It has also been reported after dental procedures, abortions, and intestinal operations. In these cases, the source of the bacteria is endogenous because up to 10% of humans harbor *Clostridium tetani* in the colon.

4. Which of the following statements concerning the tetanus neurotoxin (tetanospasmin) is *true?*

a. A shorter incubation period portends a better prognosis.
b. It blocks presynaptic release of gamma-aminobutyric acid (GABA).
c. Sensory nerves are blocked, followed by motor nerves.
d. Synaptic binding is reversible and overcome by acetylcholinesterase therapy.

Answer: B. The toxin migrates in the motor nerve from the injury site to the central nervous system (CNS). Presynaptic release of the inhibitory neurotransmitter GABA and glycine is diminished. The unopposed excitatory action results in muscle spasm. Presynaptic inhibition of the autonomic nervous system is also lost, and wide heart rate and blood pressure fluctuations are seen. The toxin binding is irreversible and diminishes only with axon synaptic regeneration. A shorter incubation period is a harbinger of a worse outcome.

5. A 62-year-old male farmer with no medical problems presents with leg pain and muscle spasms. He reports moderate to severe pain and muscle spasms in the calf that have occurred and worsened during 3 days. Two weeks prior, he suffered a puncture wound to his ankle just above his boot top with a piece of metal. Examination is remarkable for a heart rate of 115 beats/minute, blood pressure of 170/110 mm Hg, and a healing clean wound above the medial malleolus with no evidence of active infection. The calf is in active spasm with some increased tone in the peroneal musculature also. Which of the following statements regarding this patient's disease is *true?*

a. A radiograph should be obtained to assess for subcutaneous air.
b. Admission for intravenous (IV) antibiotics is indicated.
c. Outpatient management is indicated.
d. Tetanus immune globulin is indicated.

Answer: D. Localized tetanus reflects a local neuromuscular process with pain and spasm. It is likely due to a partial immunity. Immune globulin is indicated. Although mortality is lower, it can progress to generalized tetanus, and admission is warranted.

KEY CONCEPTS

- Recent outbreaks of vaccine-preventable childhood infections have occurred secondary to unvaccinated individuals and travel to areas where disease is still endemic. Emergency clinicians should recognize the possibility of these once rare diseases.
- Herpes simplex encephalitis is fatal if untreated. Clinicians should suspect this diagnosis when evaluating severely ill patients for suspected meningitis or encephalitis and promptly institute empirical therapy with IV acyclovir while awaiting diagnostic results.
- Primary varicella can be dangerous in select populations, including older children, adults, and pregnant patients. These patients require treatment with acyclovir.
- Zoster patients should be treated with acyclovir if they present within 72 hours of symptoms onset or if they are immunocompromised regardless of duration of illness. Disseminated zoster should be treated with IV acyclovir.
- In healthy patients with influenza infection, the duration of illness can be shortened by almost 1 day if antiviral treatment is administered within 48 hours of symptom onset. Hospitalized patients with influenza infection should be treated with antiviral medication regardless of duration of symptoms, because it may decrease mortality and influenza complications.
- Rabies has the highest case fatality rate of any recognized infectious disease.
- Almost 60,000 deaths per year worldwide are caused by dog-mediated human rabies, and this burden falls disproportionately on children and the poor in rural areas.
- Globally, human rabies results from bites by infected dogs. In North America human rabies results predominantly from wildlife exposures (bat, raccoon, and skunks).
- Treatment is rarely effective once symptoms of human rabies occur.
- Rabies postexposure prophylaxis (PEP) given strictly according to the World Health Organization (WHO) or US Centers for Disease Prevention and Control (CDC) guidelines is extremely effective. Discussion with public health officials is recommended to guide decisions regarding when PEP should be considered. The CDC clinician information line is 877-554-4625 or 800-CDC-INFO.
- Many emerging viral infections, including SARS-CoV2 and Ebola, should be considered in febrile patients. It is important to identify patients at risk by determining travel history and exposure history to individuals with confirmed infection. Once a patient is deemed at risk, the patient should be promptly isolated according to established guidelines while further investigation occurs. It is also important to immediately inform the hospital infection control program and public health agencies.

1. In which of the following patients is antiviral treatment not recommended by the CDC for confirmed influenza infection?
   a. 1-year-old male with 24 hours of symptoms
   b. 22-year-old otherwise healthy female with symptoms for 3 days
   c. 65-year-old male with history of asthma, coronary artery disease, and congestive heart failure with 2 days of mild symptoms
   d. 35-year-old male with no significant past medical history with symptoms for 5 days, intubated with severe hypoxemic respiratory failure, admitted to the ICU

Answer: B. The 22-year-old female with no risk factors for influenza-related complications does not require antiviral treatment because her symptoms have been present for 3 days. The CDC recommends treating all patients as early as possible who are hospitalized, have severe illness, or are at risk for influenza-related complications. The greatest efficacy for antiviral treatment is within 48 hours of symptom onset, but admitted patients with severe disease should be treated regardless of symptom onset.

2. A 19-year-old female presents to the ED with fever, altered mental status, and seizures. She is a college student and lives in a dormitory. Her only past medical history is occasional cold sores. She had been in her usual state of health until 2 days ago when she developed fevers up to 38.5°C and headache. Today she became more lethargic and had two generalized seizures prior to ED arrival. What is the next best course of action?
   a. Administer 1000 mg of intravenous acetaminophen, 2 liters of intravenous crystalloid fluids, and admit to the observation unit with the diagnosis of viral syndrome.
   b. Prescribe 1 g valacyclovir twice daily for 10 days and 650 mg acetaminophen every 4 to 6 hours as needed and discharge home.
   c. Administer 2 g intravenous ceftriaxone, 1 g intravenous vancomycin, intravenous acyclovir 10 mg/kg every 8 hours, order CT scan of head, and then perform lumbar puncture.
   d. Order MRI of the brain, administer 1 g phenytoin, and consult neurology.

Answer: C. This patient has HSV encephalitis. Given the high mortality associated with this condition, antiviral therapy should be started as soon as the diagnosis is suspected. There is significant overlap in the clinical presentation of bacterial meningitis and HSV encephalitis, so patients should empirically be treated for both diagnoses while awaiting CSF results. HSV encephalitis cannot be treated as an outpatient and requires 14 to 21 days of treatment with intravenous acyclovir. An MRI may be necessary eventually to look for temporal lobe involvement but is not necessary at the time of ED presentation and can be negative early in the course of disease.

3. A 32-year-old emergency medicine intern crawls into bed after a 24-hour shift in the medical intensive care unit. Her 36-week pregnant spouse and 18-month-old daughter are asleep as she climbs into their bed. She is awakened from a dead sleep 3 hours later when her wife asks, "What is that brown thing on the ceiling?" Using a tennis racket, the bat is successfully encouraged out of the window. The intern calls you and asks if they need to be vaccinated. Your recommendations should include which of the following?
   a. PEP (postexposure prophylaxis) is contraindicated in the pregnant spouse.
   b. The 18-month-old and intern need PEP.
   c. The spouse needs PEP.
   d. They need better screens on their windows.

Answer: B. In spite of little evidence for a bite or scratch, the intern and 18-month-old fulfill requirements for PEP. Per CDC recommendations, based on developmental staging, the 18-month-old girl cannot know whether an exposure may have happened. The intern does not explicitly meet the requirements for PEP because she is not a child, developmentally challenged, or intoxicated but, after a 24-hour shift, she was likely in a state of deep sleep and would not have been aware of her surroundings. Initiating PEP for her should be seriously considered. Although there is no contraindication for PEP during pregnancy, at 36 weeks, her spouse probably was not sleeping well and was certainly more aware of her surroundings than the intern. PEP should be considered for her but, if she is adamant that no bite or scratch occurred, it may be held.

4. An 18-year-old male who arrived from Sierra Leone 2 weeks ago presents with fever, headache, vomiting, and rash. He has a temperature of 39.5°C and erythematous maculopapular rash over his trunk, back, and arms. He appears ill and severely dehydrated. Which of the following is immediately indicated?
   a. Isolation and contact the public health department
   b. Thin and thick smear to check for malaria
   c. Ribavirin
   d. Intravenous acyclovir for 10 days

Answer: A. This is a classic presentation for Ebola virus disease (EVD). The exposure history is important. Patients should be asked if they have lived or travelled to a country with Ebola or had contact with a confirmed EVD patient in the past 21 days. If the answer is yes, then the next task is to assess for any signs or symptoms compatible with Ebola, including fever, headache, weakness, myalgias, vomiting, diarrhea, abdominal pain or hemorrhage. Once a patient has been screened positive as a potential EVD patient, he or she should be immediately isolated and hospital infection control and the health department should be notified at once.

5. Which of the following viruses is not transmitted by a mosquito vector?
   a. Dengue virus
   b. Chikungunya virus
   c. West Nile virus
   d. Lassa fever virus

Answer: D. The reservoir for the Lassa virus is an African rodent, *Mastomys natalensis*. Humans contract the disease by exposure to urine or feces of *Mastomys natalensis*. Human-to-human transmission can occur via contact with blood or bodily secretions from infected humans. The other viruses are all arboviruses transmitted via a mosquito vector.

KEY CONCEPTS

- Coronaviruses infect humans and animals and have the ability for recombination, generating novel viruses.
- Most human coronaviruses cause mild disease. Three coronaviruses, severe acute respiratory syndrome (SARS) associated virus, Middle East respiratory syndrome (MERS), and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), have caused severe disease in humans.
- The hallmark presentation of severe disease in coronavirus patients is respiratory failure, usually manifesting as acute respiratory distress syndrome (ARDS).
- Public health measures, including social distancing, wearing face masks, limiting social gatherings, and vaccinations, remain integral to controlling the COVID-19 pandemic.
- The vast majority of children have a benign disease course. However, a small number may develop a severe hyperinflammatory illness called multisystem inflammatory syndrome in children (MIS-C).
- Treatment for COVID-19 is rapidly evolving with the available data. Management of nonsevere illness is largely supportive and can occur at home. Treatment of moderate to severe disease occurs in the hospital and focuses on supporting oxygenation and ventilation. These patients may benefit from antiviral and corticosteroid therapies. At this point, the remainder of therapeutics remain mostly investigational.

1. Which of the following viruses are not known to be currently circulating in the wild?
   - **a.** Middle East respiratory syndrome–related coronavirus (MERS-CoV)
   - **b.** Severe acute respiratory syndrome coronavirus (SARS-CoV)
   - **c.** Severe acute respiratory syndrome coronavirus 2 (SARS-CoV2)
   - **d.** Measles

Answer: B. The last reported case of SARS was in 2003. There are still sporadic cases of MERS in the Middle East. Currently there is a global pandemic caused by SARS-CoV-2. SARS carried a high mortality and similarities of SARS-CoV-2 to SARS informed some of the original pandemic response to COVID-19.

2. What are classic CXR findings of SARS-CoV-2 (COVID-19)?
   - **a.** Patchy, ground-glass opacities
   - **b.** Pneumothorax
   - **c.** Lobar consolidation
   - **d.** Pleural effusion

Answer: A. Ground-glass opacities, particularly in the lower lobes, are the most common findings of COVID-19 on CXR and on CT scans. In fact, early diagnosis of COVID-19 in many areas relied on radiographs because the results of PCR testing were often delayed.

3. Which of the following medications has proven to reduce mortality in those with severe COVID-19?
   - **a.** Dexamethasone
   - **b.** Hydroxychloroquine
   - **c.** Lopinavir/ritonavir
   - **d.** Low-molecular-weight heparin

Answer: A. Dexamethasone was the first agent to show a mortality benefit in those who were hospitalized with COVID-19. Hydroxychloroquine and many other medications showed some in vitro activity against the virus, but well controlled studies did not show any benefit.

KEY CONCEPTS

- HIV/AIDS can affect any organ system, and the nonspecific complaints seen with viral illness are common. Consider acute HIV infection in the evaluation of patients with mononucleosis-like syndromes in the presence of risk factors.
- The presenting illness may originate from acute HIV infection, opportunistic infections, medication side effects, inflammation, and immune reconstitution inflammatory syndrome (IRIS).
- Patients with $CD4^+$ count above 500 cells/μL tend to have illnesses similar to those without HIV infection.
- Opportunistic infections are more frequent as the $CD4^+$ count declines but can occur at any stage of HIV infection.
- The patient's immune status should guide formulation of the differential diagnosis, considering the $CD4^+$ count, viral load, current medications, and prior opportunistic infections.
- The current medications used for the treatment of HIV infection (especially ART) can interact with many commonly prescribed drugs.
- The role of the emergency physician has expanded beyond the management of acute HIV-related presentations, but also the facilitation of PreP, PEP, ART initiation, and linkage to care.

1. A 52-year-old man with acquired immunodeficiency syndrome (AIDS) presents with dysphagia. The pain radiates to his chest. He does not appear toxic. There is white plaque on his tongue and oropharynx, which is easily scraped off. His $CD4^+$ count was 52 cells/μL. He was just recently started on highly active antiviral therapy (HAART) and is taking atazanavir-ritonavir, tenofovir, and emtricitabine. What is the most appropriate treatment?

a. Acyclovir
b. Fluconazole
c. Ganciclovir
d. Immediate endoscopy

Answer: C. Odynophagia with a $CD4^+$ count less than 100 cells/μL and oral thrush suggests esophageal candidiasis. If the patient can tolerate oral therapy, fluconazole is first-line treatment. Endoscopy is reserved for treatment failure or those with an atypical presentation. Cytomegalovirus and herpes simplex virus can cause esophagitis but are less common. They are diagnosed with endoscopy, often after treatment failure with fluconazole. Patients with HIV can have esophagitis due to reflux but, in this setting, empirical treatment for candidiasis is the most appropriate. Proton pump inhibitors can decrease the absorption of atazanavir and should not be prescribed without first discussing the case with the patient's human immunodeficiency (HIV) physician.

2. Which of the following is not an AIDS-defining condition?

a. Cellulitis
b. Extrapulmonary *Cryptococcus*
c. Invasive cervical cancer
d. Recurrent pneumonia

Answer: A. All the answers except A are AIDS-defining conditions.

3. A 52-year-old male patient presents with 3 days of confusion, fever, and headache. He was diagnosed with HIV 3 weeks ago and is not currently taking antiretroviral medications. His $CD4^+$ count at that time was 32 cells/μL. He has lived in the United States all his life. A head CT scan with contrast on this visit reveals multiple ring-enhancing lesions. Which of the following is the most likely cause of this central nervous system (CNS) lesion?

a. Cytomegalovirus encephalitis
b. HIV encephalopathy
c. *Mycobacterium* tuberculosis
d. *Toxoplasma gondii*

Answer: D. *Toxoplasma* encephalitis presents over days to weeks with fever, headache, altered mental status, focal neurologic findings, and/or seizures. In 90% of cases, there are ring-enhancing CNS lesions. the $CD4^+$ count is typically less than 100 cells/μL, and often the count is much lower. The main differential diagnosis in developed countries is CNS lymphoma. Patients with ring-enhancing CNS lesions are often empirically treated for *Toxoplasma*, and then a repeat CT scan is performed. A brain biopsy is diagnostic. Tuberculoma must be considered in patients with an exposure to tuberculosis and in those who live in areas highly endemic for tuberculosis.

4. A 36-year-old woman presents with a warm, red, painful lower leg. She has multiple other dermatologic concerns, including flesh-colored, dome-shaped lesions on her face, a new dark pigmented lesion on her arm, cold sores, and facial erythema. Which of her cutaneous findings suggests HIV disease?

a. Cellulitis
b. Facial molluscum
c. Melanoma
d. Oral herpes

Answer: B. HIV has many cutaneous manifestations. In this case, her facial molluscum is highly suggestive of HIV disease.

5. A 35-year-old man with AIDS presents with fever and a productive cough for 1 day. His last known $CD4^+$ count 1 month ago was 538 cells/μL. He has a lobar pneumonia in the left lower lobe on his chest radiograph. There is no evidence of lymphadenopathy. What is the most likely culprit pathogen in this case?

a. *Cryptococcus neoformans* pneumonia
b. *Pneumocystis jiroveci* pneumonia (PCP)
c. Pulmonary tuberculosis
d. *Streptococcus pneumoniae*

Answer: D. It is critical to have an understanding of diseases relative to the absolute $CD4^+$ T cell count. Patients with CD4 counts higher than 500 cells/μL typically develop illnesses similar to the general population. This patient presents with a lobar pneumonia, and *S. pneumoniae* is the most common cause. PCP is the most common opportunistic pathogen in AIDS patients, usually occurring in those with a $CD4^+$ T cell count lower than 200 cells/μL. It would be extremely unusual for PCP to cause a lobar pneumonia in someone who is immune-reconstituted. Tuberculosis can occur at any stage of HIV infection. However, the symptoms tend to be more gradual in onset, and a lobar pneumonia is unlikely.

KEY CONCEPTS

- Parasitic diseases may manifest with almost any constellation of signs and symptoms.
- The combination of presenting signs and symptoms and a history of recent travel to specific geographic regions can lead to early diagnosis and the initiation of pharmacotherapy, decreasing morbidity and mortality and increasing the probability of eradication of the infection.
- Parasitic coinfections are particularly common in patients with HIV infection and AIDS. A travel history is essential because the clinical presentation may be atypical, morbidity and mortality are more severe, and treatment is often prolonged.
- Acute malaria should be suspected in patients with irregular high fevers associated with headache, abdominal pain, or respiratory symptoms. Falciparum malaria, which has a unique morphology easily identifiable on the peripheral blood smear, is the predominant species of malaria that causes coma and death. *P. falciparum* is the most highly resistant to chemotherapy, demanding close observation and clinical follow-up of patients. Patients who are clinically ill or who are suspected of having falciparum malaria should be hospitalized.
- Cysticercosis should be considered in the differential diagnosis of the patient with new-onset seizures, especially in patients who have been living in Central and South America.
- Giardiasis should be suspected in patients with diarrhea who have recently been camping or drinking unfiltered mountain spring water. Patients may have tolerated several weeks of severe bloating, flatulence, eructation, and weight loss without fever before seeking medical attention.
- *Trypanosoma cruzi* infection results in Chagas disease, most notable for the development of acute and chronic myocarditis. Cardiomyopathy can be severe, at times even necessitating heart transplant.

1. A 33-year-old man presents with irregular fevers, shaking chills, intermittent abdominal pain, and fatigue. The fever comes in cycles during approximately 2 or 3 days. He has no medical history and takes no medications. He works as a baggage handler in Miami, Florida. Physical examination reveals a low-grade fever and mildly tender hepatosplenomegaly. Laboratory evaluation is remarkable for hemoglobin 9.6 g/dL, leukocytosis, lactate dehydrogenase 1850 IU/dL, elevated bilirubin, and urine dipstick "blood positive" but no red blood cells or white blood cells. He has had no international travel. Peripheral smear reveals few possible parasites with fragmented red blood cells. What is this patient's most likely infection?

a. Babesiosis
b. Early sepsis
c. Leishmaniasis
d. Malaria

Answer: D. Airport malaria has been reported in people who have never been in endemic areas but who work in or live near an international airport. The infected mosquito is transported from the endemic region and released when the plane arrives. Babesiosis is a parasitic illness with a clinical picture like that of malaria. It is tickborne and is endemic in the northeastern United States.

2. Parasite-induced loss of vision would be suggested by which of the following?

a. Cardiomegaly
b. Edematous and pruritic skin
c. Fever
d. Hepatosplenomegaly
e. Iron deficiency anemia

Answer: B. Onchocerciasis is a major cause of blindness worldwide. Ninety-five percent of cases occur in Africa. The biting flies are found near rivers, and humans are the only host for the parasite. It occupies the skin, resulting in pruritus, edema, and later atrophy with redundant skin folds. The following are other causes of parasite-induced visual loss: *Toxoplasma* can cause retinal hemorrhages, *Toxocara* can cause inflammatory retinal granulomas, and *Acanthamoeba* may cause a keratitis in contact lens wearers.

3. A macrocytic anemia would suggest infection from which parasite?

a. Ancylostoma duodenale
b. Diphyllobothrium latum
c. Falciparum malaria
d. Necator americanus

Answer: B. The fish tapeworm is associated with pernicious anemia. Hookworm and whipworm are associated with gastrointestinal iron loss and microcytic anemia. Malaria causes hemolytic anemia.

4. A 42-year-old man from Ethiopia presents with complaints of skin nodules and skin ulcerations. He has no known past illnesses, exposures, medication use, or systemic symptoms. Examination is remarkable for four 2- or 3-cm cutaneous ulcers on the arms and legs and scattered 1-cm nodules. Vital signs are normal, and physical examination is otherwise unrevealing. Which of the following statements regarding this infection is TRUE?

a. Respiratory tract symptoms would suggest an alternative diagnosis.
b. The lesions always require treatment.
c. The lesions are likely painful to touch.
d. The skin pattern may be confused with leprosy.

Answer: D. Leishmaniasis is transmitted by the sandfly bite. Skin papules and macules develop at bite sites. These may ulcerate into painless ulcers. A microcutaneous variant may be seen, and the inflammatory process may involve the larynx and trachea. Disseminated cutaneous leishmaniasis may resemble lepromatous leprosy.

5. Which of the following is the correct association between the type of parasitic infection and pulmonary symptoms?

a. Hookworm—positive PPD response
b. Leishmaniasis—pulmonary nodules
c. Löffler syndrome—ascariasis
d. Pneumocystis—90% of opportunistic infections in Africa

Answer: C. Ascariasis and hookworm may cause Löffler syndrome of chest pain, fever, rales, wheezing, and eosinophilia. The following are the other correct associations:

- Pneumocystis—less than 10% of pulmonary opportunistic infections in Africa
- *Paragonimus westermani*—positive tuberculin skin test response and chest radiograph resembling tuberculosis
- Echinococcus—anaphylaxis from leakage of cystic contents
- Schistosomiasis—diffuse pulmonary nodules (Katayama fever)

123 Tickborne Illnesses

KEY CONCEPTS

- Tickborne illnesses frequently are misdiagnosed as common viral or bacterial infections. Diagnosis can be facilitated by considering tickborne illnesses in patients who recently have been in endemic areas and routinely asking for a history of recent tick or insect bites in patients with febrile illnesses.
- Lyme disease should be suspected in patients who present with signs of a viral illness, monarticular arthritis, meningitis, multiple neurologic abnormalities, or heart block. Diagnosis can be confirmed with serologic testing of acute and convalescent serum samples.
- Normal physiologic changes from tick bites should not be confused with erythema migrans.
- Relapsing fever should be suspected in patients who present with recurrent viral-like illness associated with high fever. The diagnosis can be confirmed by identifying spirochetes on a blood smear obtained during a period of rising temperature.
- Ulceroglandular tularemia should be suspected in patients with slow-healing extremity ulcers associated with large lesions of regional adenopathy (buboes). The diagnosis can be confirmed with serologic testing.
- Rocky Mountain spotted fever should be considered in patients who present with an unexplained febrile illness, even in the absence of a rash or known tick exposure. Delayed diagnosis and late initiation of specific antirickettsial therapy may lead to a fatal outcome. Treatment never should be delayed pending laboratory diagnosis.

1. What is the most common vector-borne disease in the United States?
- **a.** Babesiosis
- **b.** Cysticercosis
- **c.** Lyme disease
- **d.** Malaria

Answer: c. Lyme disease accounts for 95% of cases.

2. Which of the following statements regarding erythema migrans (EM) is true?
- **a.** The incidence with Lyme disease is 70%.
- **b.** The lesion center may become necrotic.
- **c.** The lesion is tender and warm to touch.
- **d.** The rash does not begin at the tick bite site.

Answer: b. EM occurs in 90% of cases. It begins at the bite site and expands. There may be central clearing, skipped areas, central necrosis, vesicles, and marked erythema. It is warm but not tender. Satellite and secondary lesions may occur due to hematogenous seeding by spirochetes. They are smaller, nonmigratory, and spare the palms and soles. All lesions fade after approximately 1 month.

3. Which of the following clinical pictures is not seen with Lyme disease?
- **a.** Conjunctivitis
- **b.** Hepatitis
- **c.** Meningitis
- **d.** Pleuritis

Answer: e. Clinical pictures consistent with hepatitis, conjunctivitis, and meningitis may be seen with Lyme disease.

4. A 26-year-old woman presents complaining of muscle and joint aches. She has no past history and takes no medications except over-the-counter analgesics. She describes a pattern of migratory and intermittent muscle aches, which she reports as lasting only hours at any single location and then migrating. The physical examination reveals a mild pharyngitis and conjunctivitis and is otherwise normal. Which of the following statements is true?
- **a.** A history of tick bite should be sought.
- **b.** A rheumatoid factor (RF) level should be determined.
- **c.** An erythrocyte sedimentation rate (ESR) would be confirmatory.
- **d.** Creatine phosphokinase (CPK) levels are likely elevated.

Answer: a. The migratory, short-lived, and intermittent nature of the Lyme-related arthralgias is sometimes the best clue to the diagnosis. The ESR, RF, and CPK values will likely be normal. Both rheumatoid arthritis and fibromyalgia present with progressive and usually symmetrical symptoms. Conjunctivitis, pharyngitis, meningitis, and hepatitis pictures may be part of the Lyme presentation.

5. Which of the following statements regarding Lyme carditis is true?
- **a.** Cardiac involvement is uncommon.
- **b.** Electrocardiographic changes are stable and persistent.
- **c.** Onset time from initial illness is 2 or 3 months.
- **d.** The most common manifestation is bundle branch block.

Answer: a. Cardiac involvement is uncommon. The most common manifestation is atrioventricular block, which may fluctuate significantly but often resolves as the cardiac inflammation recedes. Onset time from illness is an average of 3 to 5 weeks.

KEY CONCEPTS

- Early recognition of patients with risk for TB should begin at ED triage. Patients with possible active pulmonary TB should be placed in respiratory isolation promptly.
- TB should be considered in the differential diagnosis of patients who present with fever, cough, and weight loss.
- Risk factors for TB include HIV infection, immunosuppression, age older than 60 years, being from an endemic country, being undomiciled, and close contact with known cases.
- In addition to pulmonary manifestations, a variety of extrapulmonary manifestations may occur, including involvement of lymph nodes, pleura, bones, joints, genitourinary, gastrointestinal, and central nervous systems.
- Therapy should be determined based on consultation with infectious disease specialists. The most commonly used agents are isoniazid (INH), rifampin (RIF), pyrazinamide (PZA), and ethambutol (EMB). Resistant strains, including multidrug-resistant TB (MDR-TB) and extensively drug-resistant TB (XDR), have been increasing in frequency.

1. Which of the following is not a typical vehicle for tuberculosis transmission?
   a. Coughing
   b. Eating utensils
   c. Parenteral transmission
   d. Sneezing

Answer: B. Bedclothes and eating utensils do not require special decontamination. Parenteral transmission is largely a risk of health care workers (e.g., skin ulcers, draining wounds). Talking, coughing, and sneezing all may produce significant droplet spread.

2. Which of the following anatomic areas is not a site of preferential organism spread during stage 2 spread of TB?
   a. Kidneys
   b. Long-bone epiphyses
   c. Lung bases
   d. Lymph nodes

Answer: C. Lung apices are favored rather than the bases. These sites are preferred, possibly because of high oxygen tension.

3. Clinically active TB develops in what percentage of immunocompetent purified protein derivative (PPD) converters?
   a. 5%
   b. 10%
   c. 15%
   d. 20%

Answer: B. TB develops in 10%: 3% to 5% in the first 2 years (acute primary) and 5% later in life (reactivation). Patients with HIV infection develop primary TB at a rate of 37% within 6 months and then develop active TB at a rate of 7% to 10%/year.

4. Which of the following statements regarding chest radiography findings in pulmonary tuberculosis is true?
   a. Adenopathy may distinguish TB from bacterial pneumonia.
   b. Primary TB typically appears as an upper lobe cavity lesion.
   c. The false-negative rate is 10% in immunocompetent patients.
   d. The false-negative rate is 40% in HIV-infected patients.

Answer: A. Hilar adenopathy may be the only way to suspect TB as opposed to a typical bacterial pneumonia. Primary TB is usually a single-lobe infiltrate with a homogenous appearance. It is postprimary TB that has a predilection for the upper lobes, with or without cavitation. The false-negative rate for chest radiography in immunocompetent patients is very low, 1%. The rate increases to 7% to 15% in HIV-infected patients and those with endobronchial disease.

5. Which of the following statements regarding the radiographic appearance of tuberculosis and infectivity is true?
   a. Chronic fibrotic changes are unlikely to be infective.
   b. Normal radiographs are uncommon with HIV infection.
   c. Only the presence of adenopathy can determine active disease.
   d. The presence of cavitation implies high infectivity.

Answer: D. Cavitation suggests high infectivity. Chronic fibrotic changes cannot differentiate old versus active disease, and many of these patients will have positive sputum. Active disease can only be determined radiographically by serial radiographs. Patients with late HIV infection are more likely to have mediastinal adenopathy and less likely to show cavitations. Normal radiographs are common with HIV infection.

KEY CONCEPTS

- Skeletal infection should be considered in the differential diagnosis of all patients who present with bone or joint pain.
- Laboratory evaluation is of little value in the diagnosis of bone and joint infections, with the exception of the ESR and CRP level, which are elevated in approximately 90% of cases of bone and joint infections.
- Joint aspiration is the definitive diagnostic procedure, and intraoperative synovial culture is the only reliable joint fluid test for establishing a diagnosis. When limited fluid is available, it should be sent for a cell count.
- With suspected septic arthritis, joint fluid and blood culture specimens are obtained before IV antibiotics are administered. With suspected osteomyelitis, blood culture specimens are obtained, and IV antibiotics are administered while plans are made for further imaging studies, surgical aspiration, or resection of bone.
- The most important aspect of antibiotic treatment of suspected bone and joint infections is to provide potent bactericidal activity against *S. aureus* with additional empirical antibiotic coverage aimed at suspected organisms on the basis of age, risk factors, and regional variability.

1. Which of the following associations between osteomyelitis and pathogenic organism is true?
- **a.** Dog bite—*Pasteurella*
- **b.** Freshwater wounds—*Pseudomonas*
- **c.** Human bite—*Aeromonas*
- **d.** Intravenous drug use—*Fusobacterium*

Answer: A. The following are correct associations: freshwater—*Aeromonas hydrophila*; intravenous drug use (IVDU)—*Staphylococcus aureus*; sickle cell—*Pseudomonas* and *Salmonella*; cat or dog bite—*Pasteurella multocida*; human bite—mixed with *Fusobacterium*, *Eikenella*, and *Streptococcus anginosus*.

2. Which of the following statements regarding septic arthritis is true?
- **a.** Despite vaccination, *Haemophilus influenzae* remains a frequent pathogen in children.
- **b.** *Pseudomonas aeruginosa* is associated with IVDU-related osteomyelitis.
- **c.** *Pseudomonas aeruginosa* is not associated with prosthetic device joint infection.
- **d.** The most common organism in neonates is *Staphylococcus aureus*.

Answer: B. *Pseudomonas* is associated with IVDU-related cervical osteomyelitis and lumbar osteomyelitis in cases of prolonged urinary catheterization. *H. influenzae* has largely disappeared as a joint pathogen in vaccinated children. The most common neonatal joint pathogens are group B streptococci, *Escherichia coli,* and *Staphylococcus epidermidis.* The most common cause of septic arthritis in people younger than 30 years is gonococcal.

3. Which of the following statements regarding acute hematogenous osteomyelitis in children is ***not*** true?
- **a.** Blood cultures are not usually positive.
- **b.** Skeletal scintigraphy is indicated in neonates.
- **c.** The child usually appears toxic.
- **d.** The most common site is the long-bone epiphysis.

Answer: C. Children may be ill but not usually toxic. There is a 2:1 or 3:1 male predominance, with the most likely site being the distal metaphysis. Blood cultures are positive in 60% of cases. Scintigraphy is not useful in neonates due to a limited inflammatory response. Radiographs are more useful and sensitive early on than in adults.

4. Which of the following statements regarding vertebral osteomyelitis is true?
- **a.** Children are less prone to isolated diskitis.
- **b.** Of the cases of epidural abscesses, 30% are due to osteomyelitis.
- **c.** The diagnostic procedure of choice is a magnetic resonance imaging scan.
- **d.** The most common location is the lumbar spine.

Answer: D. The incidence of associated epidural abscess is 15%. The most common location of vertebral osteomyelitis is the thoracic, lumbar, and cervical spine. The diagnostic procedure of choice is needle biopsy. The disease usually involves two vertebrae and the disk in between. Children are more prone to isolated diskitis, although it may also occur in adults.

5. An 18-year-old woman with known sickle cell disease presents with leg pain of 2 days' duration. Her typical pain syndrome is lower extremity tibial and femur pain. Today's episode is primarily right tibial. She complains bitterly of pain, but there are no gross findings other than trace bilateral anterior tibial swelling, with no discernible warmth or erythema. Vital signs are remarkable for a low-grade fever and heart rate of 110 beats/min. Which of the following statements regarding this patient's condition is true?
- **a.** Bony infection would be expected in the bony metaphysis.
- **b.** Plain radiography can differentiate bony infarction from infection.
- **d.** *Staphylococcus aureus* would be the most likely infectious cause.
- **d.** MRI is not necessary as it will not differentiate infection from infarction.

Answer: A. Sickle cell osteomyelitis is more typically seen in the diaphysis than in non–sickle cell situations, which more often involve the metaphysis. Fever, toxicity, and an elevated ESR suggest infection. The most likely infectious cause is now *Salmonella*, followed by *Staphylococcus aureus*. Often, observation and response to therapy (e.g., analgesics, hydration) ultimately help differentiate the two. When in question, an MRI should be pursued.

KEY CONCEPTS

- Bacterial skin infections such as cellulitis and abscess are common and are rarely life-threatening.
- Necrotizing infection is suggested by pain out of proportion to physical findings, crepitance, gas seen on imaging studies, or clinical instability. Suspected necrotizing infection should be managed with prompt broad-spectrum antibiotics and surgical consultation.
- Emergency clinicians should be familiar with toxic shock syndrome and Rocky Mountain spotted fever, which are rare, life-threatening, systemic infections with skin manifestations.
- Antibiotics reduce treatment failure after surgical drainage of uncomplicated abscesses.
- Current recommendations for the treatment of uncomplicated cellulitis suggest selection of an agent effective against streptococci and methicillin-sensitive *Staphylococcus aureus* (MSSA) (e.g., cephalexin at maximal doses). The addition of community-associated methicillin-resistant *Staphylococcus aureus* (CA-MRSA) coverage does not improve outcomes for uncomplicated cellulitis.
- Clindamycin is no longer recommended for routine treatment of purulent SSTIs due to increasing rates of both MSSA and CA-MRSA resistance.
- White blood cell count should not be routinely measured in patients with uncomplicated skin infections.
- Blood cultures are not necessary to evaluate skin infections, except in cases of sepsis, necrotizing infections, immunocompromised hosts, and multifocal infections suggesting hematogenous seeding.
- Mimics such as venous stasis dermatitis are often misdiagnosed as cellulitis. These conditions are termed *pseudocellulitis* and novel tools are being developed that may assist clinicians improve diagnostic accuracy (e.g., ALT-70 and surface thermal imaging).

1. Which of the following statements regarding cellulitis is true?
   a. Bilateral cellulitis of the lower extremities is a common bacterial infection.
   b. Computed tomography (CT) can rule out necrotizing fasciitis.
   c. Fever and leukocytosis are key to the diagnosis.
   d. Needle aspiration of the uncomplicated cellulitis is unlikely to identify the causative organism.
   e. Ultrasound evaluation cannot rule out a large abscess.

Answer: D. Needle aspiration and even biopsy are rarely able to identify a causative organism in nonpurulent cellulitis. Necrotizing fasciitis is a clinical diagnosis and cannot be ruled out with a CT. Fever and leukocytosis are often absent, and measurement of the white blood cell count is not indicated. Ultrasound evaluation is sensitive and specific and can rule out a large abscess, although distinguishing small abscesses from the cobblestoning of cellulitis can be difficult. Bilateral lower extremity inflammation is usually venous stasis dermatitis, which can be confused with cellulitis. Bilateral cellulitis is rare and suggests hematogenous dissemination.

2. A 53-year-old woman presents with fever and painful swelling of the left side of her face. The physical examination is remarkable for a toxic-appearing woman with a sharply demarcated, raised, bright red, and extremely tender eruption involving the left side of her face. Which of the following statements regarding this patient's condition is true?
   a. Echocardiography is indicated.
   b. Fluoroquinolones are first-line antibiotics.
   c. Penicillin G monotherapy is the correct treatment.
   d. The face is the most commonly involved site.
   e. There is an association with glomerulonephritis.

Answer: E. Erysipelas is an acute cellulitis typically caused by group A streptococci. It presents with an angry red area of inflammation that is well-demarcated from the surrounding skin and has a raised border. Like other group A streptococcal skin infections, erysipelas can give rise to poststreptococcal glomerulonephritis. Penicillin G is probably the ideal therapy, but any toxic-appearing patient should be treated more aggressively, with goal-directed therapy and broad-spectrum antibiotic coverage. *H. influenzae* is a classic cause of facial cellulitis but is more common in children, is rare in the era of HiB vaccination, and typically causes transdermal cellulitis, rather than the more superficial form of cellulitis known as erysipelas. The lower extremities are usually involved.

3. A 5-month-old girl presents with fever and a diffuse dermatitis characterized by bulla formation, with surrounding vesicles leading to the loss of large sheets of epidermis. She has no past medical history and has been on no medications. The areas of desquamation are tender and red. Which of the following statements regarding this patient's condition is true?
   a. Antibiotics are not indicated.
   b. Corticosteroids will help prevent progression.
   c. Culture of the bullae is not indicated.
   d. Mortality is greater than 30%.
   e. Mucous membranes are likely affected.

Answer: C. Staphylococcal scalded skin syndrome is a toxin-mediated process occurring in the very young (6 months–6 years) and older adults. The cornerstone of treatment is resuscitation and prompt administration of antistaphylococcal antibiotics. Culture of bulla fluid is generally negative. Mucous membranes are spared. This condition is rarely fatal.

4. A 7-year-old boy presents with a left leg rash. The mother describes an initial sequence of a patch of small red papules that rapidly became vesicular, then pustular, and then crusted over. The emergency physician observes a 2- × 3-cm area on the left thigh with heavily crusted erythematous macules. There is moderate left inguinal lymphadenopathy. The lesions are not tender, and the child is not toxic-appearing. Which of the following statements regarding this patient's condition is true?
   a. Acute rheumatic fever is a risk.
   b. Corticosteroids are indicated.
   c. Systemic antibiotics are necessary.
   d. The streptozyme test is highly reliable.
   e. Topical mupirocin is indicated.

Answer: E. Impetigo involves a blistering eruption with a honey-colored crust. It may be caused by streptococci or staphylococci. For limited disease, topical mupirocin is the treatment of choice. Oral or systemic antibiotics are only indicated for more severe cases. Corticosteroids are not indicated. No laboratory test is useful. Acute rheumatic fever does not occur after impetigo, but poststreptococcal glomerulonephritis may occur.

5. Which of the following statements about antibiotic therapy for skin infections is true?
   a. Dual coverage of CA-MRSA and *Streptococcus* spp. reduces treatment failure for uncomplicated cellulitis.
   b. Uncomplicated abscesses should always be treated with dual antibiotic therapy covering CA-MRSA and *Streptococcus* spp. after incision and drainage.
   c. Single-dose intravenous antibiotics are recommended prior to discharge for uncomplicated cellulitis to reduce treatment failure.
   d. Clindamycin should be included in the antibiotic regimen for necrotizing infections.
   e. Mammalian bites should routinely receive CA-MRSA coverage.

Answer: D. Multiple trials suggest that dual antibiotic therapy does not improve outcomes for uncomplicated cellulitis. Recent data suggest a margin of benefit for systemic antibiotic after incision and drainage but there is no support for routine dual antibiotic therapy. Single intravenous doses prior to ED discharge have not been demonstrated to reduce outpatient treatment failure. Given its potential to mediate toxin production, clindamycin should be included in the initial antibiotic regimen for necrotizing infections, along with agents providing broad-spectrum gram-positive and gram-negative coverage. CA-MRSA is not considered a common pathogen in mammalian bites and amoxicillin–clavulanic acid is first-line therapy.

6. A young woman presents complaining of a painful itchy rash. She has just returned from a vacation in the Caribbean, where she was snorkeling. Examination reveals streaks of erythema with slight vesiculation along the lateral aspect of the right leg. The lesions are oriented diagonally, not along vascular lines, and are about 15 cm long. What is the most appropriate next step in this patient's management?
   a. Obtain a CT scan to assess for necrotizing fasciitis.
   b. Perform a Tzanck smear on an unroofed vesicle.
   c. Treat with acyclovir.
   d. Treat with corticosteroids.
   e. Treat with trimethoprim-sulfamethoxazole.

Answer: D. This is an example of contact dermatitis, likely from seaweed or jellyfish stings. Nothing in the presentation is suggestive of an infection. Trimethoprim-sulfamethoxazole monotherapy is rarely appropriate because abscesses generally do not require antibiotics, and streptococci are thought to be covered poorly by this antibiotic. A Tzanck smear and acyclovir treatment would be appropriate if herpetic infection were suspected, but neither herpes simplex nor varicella zoster would present with 15-cm streaks.

127 Sepsis Syndrome

KEY CONCEPTS

- Sepsis is a progressive disease due to a dysregulated inflammatory cascade, leading to organ dysfunction and circulatory compromise in severe cases.
- Older adults, immunocompromised and neutropenic patients, and patients with multiple comorbidities are at increased risk for the development of sepsis syndromes.
- A thorough history, physical examination, and laboratory testing should guide the diagnostic evaluation.
- Early treatment should focus on appropriate identification, improvement of tissue perfusion (through the administration of fluids and vasopressor medications), improvement of tissue oxygenation (through administration of oxygen and positive-pressure ventilation), administration of antibiotics, and early identification of infections requiring surgical management.
- Prompt administration of antibiotics is essential and should be based on the suspected source of infection.

1. Which of the following patients meets the criteria for systemic inflammatory response syndrome (SIRS)?
   - **a.** 6-year-old boy with pneumonia, temperature 39.0°C (102.2°F)
   - **b.** 53-year-old man with respirations 30 breaths/min, white blood cell (WBC) count 16,000 cells/mm^3
   - **c.** 74-year-old woman with chest pain, heart rate 130 beats/min
   - **d.** 81-year-old man with WBC count, 2700 cells/mm^3, heart rate, 73 beats/min

Answer: B. SIRS is defined as two or more of the following—tachycardia, tachypnea, temperature higher than 38°C (100.4°F) or lower than 35°C (95°F), high or low WBC count, or bandemia. Sepsis is SIRS with infection. Severe sepsis includes organ dysfunction. Septic shock involves systolic blood pressure below 90 mm Hg.

2. Sepsis is characterized by which of the following?
   - **a.** Depression of tumor necrosis factor levels
   - **b.** Increased endogenous anticoagulant levels
   - **c.** Prolonged suppression of nitric oxide levels
   - **d.** Increased inflammatory cytokine levels

Answer: D. Clinical sepsis is induced by sustained levels of proinflammatory and procoagulant mediators. Cytokines (interleukin-1, interleukin-6, and tumor necrosis factor-α) and prostaglandins are primary mediators. Nitric oxide synthase is upregulated, resulting in sustained elevations of the serum nitric oxide level, with subsequent vasodilations. Sustained suppression of vasopressin adds to this sometimes refractory vasodilated state.

3. Which of the following statements regarding septic shock is true?
   - **a.** Cardiac output is always decreased.
   - **b.** Much of the cardiac decompensation is reversible.
   - **c.** Systemic vascular resistance is high.
   - **d.** The ejection fraction is always increased.

Answer: B. Sepsis affects myocardial function and peripheral vascular tone. The systemic vascular resistance is usually markedly depressed. Cardiac output is generally increased because of a compensatory tachycardia that can at least partially overcome the ventricular dilation and depressed ejection fraction. The myocardial effects are typically reversible.

4. What are the two most common sources of infection in cases of sepsis?
   - **a.** Genitourinary > respiratory
   - **b.** Musculoskeletal > genitourinary
   - **c.** Respiratory > gastrointestinal
   - **d.** Respiratory > genitourinary

Answer: C. Epidemiology studies show that pneumonia is the most common cause of sepsis, followed by an intra-abdominal source. However, a careful investigation to identify the source of infection should occur.

5. In most chemotherapy patients, which neutrophil count should prompt admission, isolation, and empirical antibiotics?
   - **a.** <250 cells/mm^3
   - **b.** <500 cells/mm^3
   - **c.** <750 cells/mm^3
   - **d.** <1000 cells/mm^3

Answer: B. Patients with an ANC <500 cells/mm^3 are at increased risk of infection; thus, a conservative approach should be taken in these patients.

6. Among patients with clinical septic shock, which percentage will have positive blood cultures?
   - **a.** 0%–30%
   - **b.** 30%–60%
   - **c.** 60%–90%
   - **d.** 90%–100%

Answer: B. Although blood cultures are perhaps a gold standard for identification and isolation of bacteria, they may be negative even when the etiology of illness is clearly infectious. Empirical antibiotic treatment in the ED remains a standard approach.

128 Hypothermia, Frostbite, and Nonfreezing Cold Injuries

KEY CONCEPTS

- Patients with hypothermia should be actively rewarmed whenever possible. Specific indications for active rather than passive rewarming include trauma, cardiovascular instability, temperature below 32°C (89.6°F), poor rate of passive rewarming, and endocrine insufficiency.
- Rewarming methods should be chosen to minimize core temperature afterdrop.
- If tachycardia is out of proportion to core temperature then hypoglycemia, hypovolemia, or an overdose should be considered.
- The effects of most medications are temperature dependent. Overmedication to achieve an effect when the patient is cold could cause toxicity during rewarming.
- Laboratory coagulation tests are performed at 37°C (98.6°F). Despite clinically obvious coagulopathy, measures of coagulation will be deceptively normal. Treatment for coagulopathy is to rewarm the patient.
- There are no safe predictors of serum electrolyte levels. Hypothermia enhances the cardiac toxicity of hyperkalemia and obscures premonitory electrocardiographic changes.
- Failure to rewarm despite good technique should suggest infection, endocrine insufficiency, or a futile resuscitation.

1. Emergency medical services (EMS) notifies your emergency department (ED) that an unknown male who was "found down" is being transported. No history is available. Paramedics report that the patient's pulse is 42 beats/min and blood pressure is difficult to obtain. Spontaneous respirations are present at a rate of 10 breaths/min. The electrocardiogram (ECG) showing sinus bradycardia is faxed before the patient's arrival. What treatment should you administer when the patient arrives?
   a. Intravenous epinephrine
   b. Normal saline bolus
   c. Synchronized cardioversion
   d. Warm the patient

Answer: D. Warm the patient. When this hypothermic patient arrives in the ED, after assessment of the airway, the top priority is rapid rewarming. Patients with hypothermia should be actively rewarmed whenever possible. Specific indications for active rather than passive rewarming include trauma, cardiovascular instability, temperature below 32°C (89.6°F), poor rate of passive rewarming, and endocrine insufficiency.

2. Rapid rewarming is the mainstay of treatment for hypothermia. However, rewarming can cause complications. Which of the following complications should be anticipated and prevented when rewarming a hypothermic patient?
   a. Hyperkalemia
   b. Hyponatremia
   c. Hypotension
   d. Rhabdomyolysis

Answer C: Hypotension should be anticipated and prevented when rewarming a hypothermic patient. Patients with moderate or severe hypothermia are usually volume depleted. During rewarming, the total plasma volume is usually high, but the circulatory plasma volume is usually low due to increased peripheral vascular resistance. Rapid volume expansion can be lifesaving, especially in hypothermic neonates. Adult patients with moderate or severe hypothermia should initially receive a 500-mL fluid challenge of warmed normal saline. Avoid lactated Ringers solution because the cold liver metabolizes lactate poorly. Fluids administered via the intravenous (IV) route should be warmed to 40°C to 42°C (104°F to 107.6°F).

3. A 30-year-old woman in cardiac arrest is brought to the ED by emergency medical services. She was intubated in the field, and chest compressions have been continuously performed. Her temperature is 25°C (77°F). When placed on the cardiac monitor, she is noted to be in ventricular fibrillation. A defibrillation attempt is made at the appropriate setting, but she remains in ventricular fibrillation. A nurse resumes providing chest compressions and asks for further instructions. What should be done next?
   a. IV amiodarone
   b. IV lidocaine
   c. IV procainamide
   d. Warm the patient

Answer: D. Warm the patient. This severely hypothermic patient needs further rewarming before subsequent defibrillation attempts are made to treat her ventricular fibrillation. Most hypothermia-induced dysrhythmias convert spontaneously during rewarming. Asystole that develops during rewarming is not as ominous as asystole in normothermic patients. For VF, defibrillation should be attempted at the usual energy level. Successful defibrillation has been reported at 20°C (68°F) but attempted defibrillation is often unsuccessful until the core temperature is above 30°C (86°F). If a defibrillation attempt is unsuccessful, active rewarming should be initiated while continuing CPR. Defibrillation can be attempted occasionally during rewarming. Once the core temperature is above 30°C (86°F), further attempts can be made.

4. A 27-year-old homeless man presents complaining of numb feet. He reports sleeping outdoors overnight. The temperature was consistently below −20°C (−4°F). The feet appear white and waxy. They are cold and hard to the touch. What is the best way to rewarm the feet?
   a. Room temperature air convection
   b. Room temperature water immersion
   c. Warm air convection
   d. Warm water immersion

Answer: D. This patient is suffering from frostbite of the feet. The treatment of choice in this scenario is rapid warm water immersion. Rapidly rewarm completely frozen or partially thawed tissue by immersion in gently circulating water that is carefully maintained at a temperature of 37°C to 39°C (98.6°F to 102.2°F). A whirlpool is ideal, but any large container can be used for the hands or feet. Water warmer than 39°C (102.2°F) does not thaw significantly faster but causes more pain. Tissue can suffer thermal injury when the water temperature exceeds 42°C (107.6°F). Rewarming should be continued until distal erythema is noted. The part should have return of color and feel pliable, which usually requires 15 to 30 minutes of submersion.

5. What is the most common error made when treating frostbite injury?
   a. Débridement of broken blisters
   b. Premature termination of thawing
   c. Use of nonsteroidal antiinflammatory drugs (NSAIDs) for analgesia
   d. Use of thrombolytic agents to restore blood flow

Answer: B. Premature termination of thawing in 37°C to 39°C (98.6°F to 102.2°F) water is a common error. Reperfusion of completely frozen tissue may be painful, requiring parenteral analgesia.

KEY CONCEPTS

- Classic heatstroke is generally diagnosed in older patients with comorbidities during heat waves, whereas exertional heatstroke is more common in young athletic patients or military personnel.
- Heat exhaustion and heatstroke are a continuum of the same pathophysiologic process. Neurologic dysfunction is a hallmark of heatstroke, and cerebral edema is common.
- In heat stroke, morbidity and mortality are directly related to the duration of core temperature elevation. Rapid cooling should be initiated without delay.
- The evaporative technique and ice water immersion are the 2 forms of primary cooling measures. Other cooling measures are secondary and considered adjunct therapy.
- Antipyretics are ineffective and should not be used to control environmental hyperthermia.
- Heatstroke can cause right-sided cardiac dilation and elevated central venous pressure and clinically resemble pulmonary edema but may still require crystalloid resuscitation.

1. An 18-year-old female marathon runner presents to the medical aid station during a hot summer race. She is extremely irritable and diaphoretic. She is complaining of generalized weakness, dizziness, nausea, and headache. The physical examination reveals an oral temperature of 40.5°C (105°F), heart rate of 120 beats/min, muscle twitching, and ataxia. What is the most appropriate management?

a. Assess her volume status and immediately start normal saline to replete volume loss before transfer to a hospital.
b. Encourage her to drink cold water to replace her free water deficit rapidly.
c. Immediately remove her from the hot environment and begin cooling before transfer to a hospital.
d. Prescribe immediate rest, after which she may be allowed to finish the race.

Answer: c. The onset of heatstroke is sudden. Prodromal symptoms lasting minutes to hours can occur that are nonspecific and similar to those of heat exhaustion. Signs and symptoms may include weakness, dizziness, nausea, frontal headaches, confusion, muscle twitching, ataxia and signs of cerebellar dysfunction, and psychiatric symptoms, ranging from anxiety and irritability to psychosis. Heat exhaustion can progress to heatstroke if it is untreated. If the patient is evaluated as this is occurring, differentiation between heat exhaustion and heatstroke is difficult. If heatstroke cannot be excluded, begin cooling the patient immediately.

Whereas rest is part of the treatment for heat exhaustion, it is not the only treatment. She must be removed from the hot environment, not be allowed to finish the race, and assessed for her volume status. Normal saline is used to replete volume if the patient is orthostatic; free water deficits are replaced slowly to avoid cerebral edema.

2. Which of the following statements regarding heat exhaustion is true?

a. It causes body temperatures that often exceed 40.5°C (105°F).
b. It only exists in two discrete forms, either salt depletion or water depletion.
c. It is associated with systemic symptoms.
d. It is characterized by hyponatremia and hyperchloremia.
e. It occurs when muscles are fatigued by heavy work.

Answer: c. Heat exhaustion is a clinical syndrome. Whereas there are typically two types of heat exhaustion, water depletion and salt depletion, pure forms of either type are rare. Most cases of heat exhaustion involve mixed salt and water depletion. In salt depletion heat exhaustion, the syndrome is characterized by hyponatremia, hypochloremia, and low urinary sodium and chloride concentrations. The symptoms and signs associated are variable and nonspecific but usually systemic, such as weakness, fatigue, frontal headache, vertigo, nausea, and vomiting. The body temperature usually remains nearly normal.

3. Despite cooling measures, poor outcomes are seen in heatstroke patients with which of the following?

a. Altered coagulation status
b. History of schizophrenia
c. Need for supplemental oxygen
d. Presentation with acute renal failure
e. Presentation with acute rhabdomyolysis

Answer: a. Factors such as advanced age, hypotension, altered coagulation status, and the necessity for endotracheal intubation on arrival at the emergency department predict a poor outcome, despite successful cooling measures.

4. The usual characteristics of classic heatstroke include which of the following?

a. Diaphoresis
b. Disseminated intravascular coagulation
c. Hypoglycemia
d. Marked lactic acidosis
e. Usual occurrence during heat waves

Answer: e. Usual characteristics of classic heatstroke include predisposing factors or medication, older population, sedentary lifestyle, anhidrosis, normoglycemia, mild coagulopathy, mild elevation in creatine kinase level, oliguria, mild acidosis, and occurrence during heat waves. Diaphoresis, hypoglycemia, disseminated intravascular coagulation, and marked lactic acidosis are characteristics of exertional heatstroke.

5. Which of the following is associated with heat cramps?

a. Excessive salt intake
b. Drinking copious amounts of hypertonic fluids
c. Carpopedal spasms with distal extremity paresthesias
d. Hyponatremia and hypochloremia
e. Rhabdomyolysis

Answer: d. Heat cramps are brief, intermittent, and often severe muscle cramps occurring typically in muscles that are fatigued by heavy work. Heat cramps appear to be related to a salt deficiency. Heat cramps are occasionally confused with hyperventilation tetany. Hyperventilation tetany can be distinguished by the presence of carpopedal spasm and paresthesias in the distal extremities and perioral area. Heat cramps accompanied by systemic symptoms may be part of salt depletion heat exhaustion. Heat cramp victims exhibit hyponatremia and hypochloremia, so serum electrolyte levels should be measured. Rhabdomyolysis or resultant renal damage is not present with isolated heat cramps.

KEY CONCEPTS

- Electrical current follows the path of least resistance, which is often along neurovascular bundles. Deep tissue injuries and organ damage are often more extensive than indicated by examination of the overlying skin.
- Testing is not indicated for victims of low-voltage electrical injuries who are asymptomatic or have minimal local symptoms and physical evidence of burn injury. These patients may be discharged home after emergency department (ED) evaluation.
- Patients exposed to high-voltage sources or lightning strikes who present with syncope, altered mentation, focal neurologic abnormality, significant burns, entry and exit wounds, or persistent symptoms should have testing including electrocardiography, complete blood count, basic chemistry panel, myoglobin and troponin level determination, and urinalysis. Additional testing is directed at suspected areas of injuries.
- Patients with electrocardiographic signs of cardiac injury or dysrhythmias, or with evidence of significant local injury, should be monitored for 12 to 24 hours in the ED, observation unit, or inpatient setting.
- The enormous current of a lightning strike may cause critical injury or death, or the current may be directed superficially over the patient to the ground, resulting in no injury and minor burns.
- Electrocardiography is indicated for all patients evaluated for lightning strike. Additional testing should be based on specific signs and symptoms.
- Lightning strike patients who present without symptoms or signs of injury, or with only minor first-degree burns, and with a normal ECG can be discharged home from the ED after evaluation.
- Lightning strike can cause fixed, dilated pupils in the absence of irreversible brain injury.

1. A 37-year-old woman presents with blurred vision. Her vision has been progressively worsening for approximately 2 months. She denies eye pain or direct eye trauma. She reports that approximately 2 months ago, just before the beginning of her vision problems, she was struck by lightning while playing golf. She states that she went to an emergency department (ED) at that time, had no major injuries, and seems to have recovered well. What is the most likely cause of her decreased vision?

a. Cataracts
b. Glaucoma
c. Iritis
d. Macular degeneration
e. Retinal detachment

Answer: A. Cataracts are a well-known complication of electrical injury, either from artificial electrical sources or from lightning. They may occur immediately or have a delayed presentation. Glaucoma is due to elevated intraocular pressure and is often painful. Iritis and retinal detachment can be caused by electrical injury but are manifested acutely. Macular degeneration is a chronic condition of older adults with central vision loss and is not associated with electrical injury. Macular holes may occur acutely with lightning injury.

2. Which of the following patients presenting to the emergency department should have consultation by the burn surgery service?

a. 22-year-old female who presents after receiving a taser shock who is awake and alert. She has barb wounds in her abdomen that were removed by law enforcement prior to arrival and has mild erythema at the barb sites, but no other injuries.
b. 47-year-old male who presents after a lightning strike, who complains of chest pain and EKG shows sinus tachycardia.
c. 34-year-old male who presents after an occupational accident, in which he received an electrical shock to the right hand from a high-voltage source. He is awake and alert, with normal vital signs and no complaints other than a small area of hyperemia to his right palm and pain to his right arm.
d. 3-year-old female, brought in by parents for concern that she put her finger into an electrical socket. On exam, she is awake and alert, has normal behavior, and has no skin or soft tissue findings.
e. 67-year-old female who presented unresponsive after a lightning strike, but is now awake and alert, but reports hearing difficulties.

Answer: C. Electrical injuries, especially high- voltage injuries, can cause deep tissue burns and damage that may not be readily apparent on the surface, due to decreased resistance of neurovascular bundles. In severe cases, tissue edema and decreased perfusion may require fasciotomy. Often, fluid resuscitation calculations using the Parkland formula are underestimated, as visible injury may be insignificant compared to damage beneath the surface. Patients who receive a taser shock do not require further management beyond local wound care. The victims of lightning strikes may require further evaluation for cardiac and HEENT abnormalities, but skin and soft tissue damage in these patients is less likely as the electrical energy often passes over rather than through them. The 3-year-old does not require further management beyond reassurance.

3. A 42-year-old man suffers an electrical injury while working on power transmission lines near your hospital. Paramedics report that he is unresponsive and initially in ventricular fibrillation but spontaneously converted to sinus tachycardia before treatment. His initial electrocardiogram (ECG) from the field shows ST segment elevation in the inferior leads. An ECG repeated in the ED shows continued inferior ST segment elevation but with decreased magnitude. Creatine kinase (CK) and CK-MB levels are markedly elevated and troponin levels are slightly above normal range. While he does have pulses, he continues to be unresponsive, has no spontaneous respirations, and has bilateral fixed and dilated pupils. What is the appropriate next action?
   a. Brain death testing
   b. Cardiac catheterization
   c. Mannitol infusion
   d. Intubation and supportive care
   e. Thrombolytic administration

Answer: D. Electrical injury can cause a variety of cardiac manifestations, including multiple dysrhythmias, transient ST elevation, and conduction blocks. Myocardial infarction, although reported, is rare. Extensive skeletal muscle damage can be seen in electrical injuries and results in marked elevations of CK and CK-MB levels. Respiratory muscle paralysis and ocular injuries often occur. This patient is probably not brain-dead, nor has he suffered a myocardial infarction. Mannitol is used in cases of impending cerebral herniation, which is not expected in this patient. Supportive care, including mechanical ventilation and treatment of dysrhythmias per ACLS protocols as needed is most appropriate.

4. A 22-month-old female is brought to the emergency department by parents after she was found chewing on an electrical cord. She has minor burns to the lips and oral commissure. Upon discharge, what condition should the parents be warned about as a reason to return to the ED?
   a. Facial compartment syndrome
   b. Cardiac dysrhythmias
   c. Delayed dental eruption
   d. Rhabdomyolysis
   e. Delayed labial artery bleeding

Answer: E. Delayed labial artery bleeding usually occurs two days after the initial injury, when the initial eschar separates. When this occurs, there can be significant bleeding from the labial artery. Traditionally, admission for observation has been advocated for these patients but, if there is good social support, discharge with specific return instructions and close follow-up are appropriate. These types of electrical injury have only local effects, so dysrhythmias, cataracts, and rhabdomyolysis are not complications. Contractures are common and may require reconstructive surgery, but this complication is not life-threatening.

5. A 53-year-old male is brought to the emergency department by law enforcement after they incapacitated him with a taser. There are no visible signs of traumatic injury, but he is not responsive. The most appropriate next step is:
   a. Supportive care as most patients quickly arouse after a taser shock.
   b. Evaluate for burns around the taser barb site.
   c. Perform EKG to evaluate for STEMI.
   d. Evaluate the patient for other etiologies of altered mental status.
   e. Discharge the patient with law enforcement to be taken to jail.

Answer. D. There have been a number of high-profile deaths associated with taser shocks by law enforcement. However, the cause of these deaths has not been linked to the electrical effect of the taser shock itself, and patients who present after a taser shock with altered mental status should be evaluated for other etiologies, including but not limited to infection, toxic/metabolic etiologies, and traumatic injuries. Burns evaluation and EKG alone would not be sufficient evaluation for this patient.

131 Scuba Diving and Dysbarism

KEY CONCEPTS

- The majority of dive injuries are diagnosed on the basis of the focused dive history and physical examination and are best differentiated into disorders of descent, disorders of depth, and disorders of ascent.
- The *U.S. Navy Diving Manual* and the Divers Alert Network (DAN) are valuable resources for the clinician presented with a diving emergency. DAN provides a 24-hour medical emergency hotline at 1-919-684-9111 (see Table 131.4).
- Treatment with 100% oxygen is the initial therapy for *all* diving emergencies until the diagnoses can be determined. It has been demonstrated to reduce the morbidity and mortality related to decompression illness and can be helpful in patients with pneumothorax.
- Diagnostic imaging and laboratory studies are generally not useful for ruling in or out decompression illness and should not delay transfer for recompression therapy.
- Recompression treatment is recommended for patients with decompression sickness (DCS) and arterial gas embolism (AGE).

1. When is a diver most likely to suffer barotrauma?
- **a.** When the diver also suffers from decompression sickness (DCS)
- **b.** When the diver is at extreme depth
- **c.** When the diver is closest the surface
- **d.** When the diver stays at depth for an extended period of time
- **e.** When the diver uses specialized gas mixtures with decreased partial pressure of nitrogen

Answer: C. Barotrauma refers to injuries due to pressure changes. Boyle's law states that pressure and volume are inversely proportional ($P \times V = k$ or $P_1V_1 = P_2V_2$). As the pressure decreases, the volume of gas increases and can damage gas-filled structures (sinuses, inner/middle ears, lungs, and intestines). The incremental changes in pressure (and therefore volume) are greatest at the surface, so barotrauma most commonly occurs near the surface (either at the beginning or at the end of a dive). Long or deep dives are not required for barotrauma but predispose to decompression illness due to the build-up of tissue nitrogen which comes out of solution during ascent.

2. A 55-year-old man presents with the acute onset of left-sided weakness and confusion. Family members report that they are on vacation and had just finished scuba diving when the patient complained of chest pain, became confused, and stopped moving his left arm and leg. This happened 30 minutes ago. The patient has no known medical history and takes no medications. Physical examination reveals a drowsy and confused male who follows commands but does not move his left arm or leg. His blood pressure is 162/98 mm Hg, and his other vital signs are within normal limits. What is the appropriate treatment?
- **a.** Computed tomography (CT) of the brain
- **b.** Endotracheal intubation
- **c.** Intravenous labetalol
- **d.** Intravenous tissue plasminogen activator
- **e.** Recompression in a dive chamber

Answer: E. This patient has suffered an arterial gas embolism (AGE). In AGE, air bubbles gain access to the arterial circulation and can cause mechanical obstruction of the artery. Symptoms can mirror an acute thrombotic or embolic event including myocardial infarction or stroke. Treatment with 100% oxygen should be instituted immediately, but the only definitive treatment is emergent recompression.

3. Which of the following individuals should be advised not to dive?
- **a.** A 10-year-old boy with no known medical problems
- **b.** A 19-year-old woman with no medical problems but who just landed from an intercontinental commercial flight
- **c.** A 20-year-old man with well-controlled asthma
- **d.** A 27-year-old woman who is 20 weeks pregnant
- **e.** A 70-year-old man with asymptomatic coronary artery disease

Answer: D. Upon surfacing, small nitrogen bubbles form in the circulation of all divers. If proper technique is followed, these bubbles are small and asymptomatic. They form in the venous circulation and are eliminated in the lungs. However, if these bubbles are transmitted into the arterial system, embolization with serious consequences can occur. The fetus has a patent foramen ovale (PFO) and ductus arteriosus, both of which would allow air to bypass this filtering effect of the lungs and proceed into the systemic circulation. Asthmatics can dive as long as symptoms are well controlled. There are no definite age recommendations for diving. Flying immediately after diving should be avoided; flying prior to diving is of less concern.

4. A 24-year-old woman presents with complaints of severe chest pain and shortness of breath. The symptoms started approximately 1 hour ago while surfacing from a dive. She denies loss of consciousness or other symptoms. Physical examination reveals decreased breath sounds in the right chest and crepitus in her neck; otherwise, it is normal. Vital signs are normal except for a slight tachypnea. A chest radiograph shows a moderate right-sided pneumothorax and a pneumomediastinum. She has no other symptoms or signs. You place a chest tube on her right side. What is the next course of action?
- **a.** Computed tomography (CT) of the brain
- **b.** CT of the chest
- **c.** Observation and supportive care
- **d.** Pericardiocentesis
- **e.** Recompression in a dive chamber

Answer: C. This patient has suffered from pulmonary barotrauma. Although this injury does predispose one to arterial gas embolism (AGE), the diagnosis of AGE is clinical. The treatment for a pneumothorax from pulmonary barotrauma is the same as for a pneumothorax from another cause—that is, aspiration, catheter insertion, or chest tube placement. Management of pneumomediastinum is supportive. Recompression is only necessary for decompression sickness (DCS) or AGE, not for barotrauma.

5. A 32-year-old man presents with bilateral lower extremity numbness and weakness. The patient reports several days of scuba diving without incident until after his most recent dive, when the symptoms started. Physical examination reveals bilateral lower extremity weakness and decreased sensation to pinprick and light touch. Priapism is also noted. The remainder of the physical examination and all vital signs are within normal limits. You decide that recompression therapy is indicated, but your emergency department does not have a dive chamber and the nearest chamber is 50 miles away. The patient has been accepted in transfer. What is the most appropriate way to transfer this patient?

a. Air ambulance, lying flat
b. Air ambulance, Trendelenburg position
c. Ground ambulance, lying flat
d. Ground ambulance, Trendelenburg position
e. This patient has not yet been stabilized for transport; he should remain at the current hospital.

Answer: C. This patient is suffering from spinal decompression sickness (DCS II). Recompression therapy is the treatment of choice. Unless an inordinate delay to the recompression chamber would result, anything that further decreases ambient pressure should be avoided (e.g., flying). If a patient must be flown, the aircraft should fly at the lowest possible altitude or pressurize the cabin at the highest possible pressure. Traditionally, the Trendelenburg position was advocated because it was believed that it would decrease the incidence of AGE to the brain. This is not the case. In addition, the Trendelenburg position increases cerebral edema and the incidence of coronary artery air embolism. Therefore, the Trendelenburg position should not be used for patients with diving injuries.

6. You decide that a 27-year-old patient with an arterial gas embolism (AGE) following scuba diving requires recompression therapy. Which of the following treatments should be initiated before hyperbaric therapy?

a. Administer 40% oxygen.
b. Administer steroids.
c. Ensure that the endotracheal tube and urinary catheter balloons are filled with water, rather than air.
d. Place prophylactic thoracostomy tubes bilaterally.
e. Place the patient in the Trendelenburg position.

Answer: C. 100% normobaric oxygen is standard of care during evaluation, treatment and transport. Inflate endotracheal tube and urinary catheter balloons with sterile saline. There is no evidence that steroids are effective. The Trendelenburg position increases intracranial pressure and facilitates coronary gas embolization.

KEY CONCEPTS

- All forms of altitude illness can be treated with oxygen and rapid descent.
- The diagnosis of acute mountain sickness requires the presence of headache in the setting of a recent elevation change to greater than 8,000 feet. Additional nonspecific symptoms may include nausea, anorexia, and fatigue. Those with significant symptoms should not ascend further until symptoms improve.
- Patients with mild high-altitude pulmonary edema may be treated in place if experienced providers and treatment options exist. Patients with moderate high-altitude pulmonary edema or high-altitude cerebral edema should descend immediately.
- Dyspnea at rest is an early symptom of high-altitude pulmonary edema. More advanced findings include rest tachypnea, cough productive of frothy sputum, and altered mentation. Treatment involves oxygen and descent.
- Altered consciousness and cerebellar ataxia are early signs of high-altitude cerebral edema. Failure to initiate immediate treatment with oxygen, descent, and dexamethasone can result in permanent disability or death.
- Acute mountain sickness may be prevented by using acetazolamide or dexamethasone. Symptoms can be controlled with analgesia and antiemetics. High-altitude pulmonary edema may be prevented using nifedipine. In patients unable to use nifedipine, consider inhaled salmeterol, and oral phosphodiesterase type 5 inhibitors (sildenafil [50 mg every 8 hours] and tadalafil [10 mg every 12 hours]). Temazepam (7.5 mg qhs) can safely improve sleep quality.

1. Which of the following individuals has the highest risk of developing high-altitude pulmonary edema (HAPE), assuming all are generally healthy, live at the same altitude, and travel to the same altitude?

a. A 16-month-old boy
b. A 19-year-old man
c. A 19-year-old woman
d. A 42-year-old man
e. A 42-year-old woman

Answer: B. Many factors contribute to the development of HAPE including total altitude, sleeping altitude, time at altitude, previous acclimatization, and the presence of comorbidities. However, in general, both females and the elderly have been found to be less likely to develop HAPE. Also, whereas younger males are more at risk, HAPE is extremely rare in individuals younger than 2 years old regardless of gender.

2. Which of the following symptoms must be present to establish a diagnosis of acute mountain sickness (AMS)?

a. Dizziness
b. Fatigue
c. Headache
d. Dyspnea
e. Nausea

Answer: C. To diagnose AMS, a patient must have a recent gain in altitude, be at that altitude for several hours, and have a headache. In addition, at least one of the following must also be present: gastrointestinal upset, fatigue, dizziness or light-headedness. Exertional dyspnea is common at high altitude. Resting dyspnea is concerning for potential HAPE.

3. Which of the following medications used to treat acute mountain sickness (AMS) actually accelerates acclimatization?

a. Acetazolamide
b. Ibuprofen
c. Dexamethasone
d. Oxygen
e. Prochlorperazine

Answer: A. Acetazolamide is a carbonic anhydrase inhibitor and induces a metabolic acidosis that results in increased ventilation. Ibuprofen is an appropriate treatment for the headache associated with AMS. Dexamethasone and oxygen both improve all symptoms of AMS but do not aid acclimatization. Prochlorperazine is appropriate treatment for the gastrointestinal upset caused by AMS and is preferred over promethazine because it does not depress the respiratory centers.

4. Which of the following conditions occurs more commonly at high altitude?

a. Congestive heart failure (CHF)
b. Myocardial infarction
c. Pericarditis
d. Pneumothorax
e. Pulmonary embolism (PE)

Answer: E. Altitude may contribute to hypercoagulability from hyperviscosity due to elevated hematocrit and dehydration. Venous stasis is also more common at altitude due to the relative immobility of sleeping in confined spaces (sleeping bags and small tents) and long flights that tend to precede vacations at altitude. Together, these facts lead to an increased risk of PE. Occasionally, PE may be misdiagnosed as high-altitude pulmonary edema (HAPE) and treated inappropriately.

5. The acute hypoxic stresses of high altitude can cause a range of symptoms suggestive of both cardiovascular and CNS dysfunction. Although high-altitude pulmonary edema (HAPE) and acute mountain sickness (AMS) share the same pathophysiologic trigger (decreased available oxygen), the treatment and prognosis are different. Which of the following features is indicative of HAPE as opposed to AMS?

a. Frequent urination
b. Dyspnea at rest
c. Headache
d. Lower-extremity edema
e. Symptom onset immediately after arrival at altitude.

Answer: B. Dyspnea on exertion is nearly universal at altitude and is not indicative of a pathologic process. Dyspnea at rest is not normal and is an early symptom of HAPE. Cough is common at altitude or any time when one breathes cold dry air. Fluid retention and peripheral edema are common manifestations of AMS, whereas an alkaline diuresis can be a sign of acclimatization. Neither AMS nor HAPE commonly occurs immediately after arrival at altitude. AMS usually occurs within 24 hours and HAPE usually occurs after 2 to 4 days.

6. The definitive treatment for high-altitude pulmonary edema (HAPE) is oxygen and descent to lower altitudes. Which of the following medications is also useful in the treatment of HAPE?

a. Hydrochlorothiazide
b. Metoprolol
c. Nifedipine
d. Nitroglycerin
e. Furosemide

Answer: C. Nifedipine is a nonselective pulmonary vasodilator and is useful in the treatment and prevention of HAPE. Sildenafil and

tadalafil may also be useful in the prevention of HAPE. These phosphodiesterase–5 inhibitors increase cyclic guanosine monophosphate availability and result in pulmonary vasodilation. As a non-cardiac form of pulmonary edema, neither nitroglycerin or furosemide has a role in the treatment of HAPE.

7. Which of the following signs or symptoms is specific for high-altitude cerebral edema (HACE) as opposed to acute mountain sickness (AMS) or high-altitude pulmonary edema (HAPE)?

a. Ataxia
b. Dizziness
c. Dyspnea at rest
d. Headache
e. Vomiting

Answer: A. Ataxia, seizures, slurred speech, focal neurologic deficits, and altered mentation are all concerning to HACE and require immediate descent and treatment. Dizziness, headache, and vomiting can all be seen in AMS. Dyspnea at rest is nearly universal in HAPE.

8. You are the doctor on a group expedition to Denali. Recently, some of the climbers have complained of headaches, dyspnea, nausea, and difficulty sleeping. At breakfast this morning, one of the climbers, a 28-year-old female, is noted to have difficulty speaking and some slurred speech, which she attributes to a restless night's sleep. What treatment is mandated for this patient?

a. 24 hours of rest before further ascent
b. Acetazolamide
c. Descent to lower altitude
d. Nifedipine
e. Ibuprofen

Answer: C. This patient has an acute neurologic emergency, most likely high-altitude cerebral edema (HACE) until proven otherwise. Any hard neurologic finding (ataxia, slurred speech, focal neurologic deficits, seizure, or altered mentation) at altitude is suggestive of HACE, but acute CVA and other malignant etiologies must be considered. Unlike acute mountain sickness (AMS) or high-altitude pulmonary edema (HAPE), which can occasionally be treated without descent, descent is mandatory for all cases of HACE. Dexamethasone should also be given to all patients with HACE.

9. You are the doctor on a group expedition to Denali and are currently at an elevation of 10,000 feet. One of the members of the group experiences headache, lightheadedness, and nausea. You diagnose him with acute mountain sickness (AMS) and treat him with acetazolamide and oxygen. He wants to know when he can continue climbing the mountain. What do you tell him?

a. You may ascend if your oxygen saturation (as measured by pulse oximetry) is higher than 90%.
b. You may ascend when all your symptoms resolve.
c. You may never ascend any higher than this altitude.
d. You may not ascend any higher on this expedition but may try again in 1 or 2 months.
e. You may only sleep 1000 feet higher than your current altitude.

Answer: B. Management of AMS must adhere to the axiom, "After the symptoms of altitude illness occur, further ascent to a higher sleeping altitude is contraindicated."

KEY CONCEPTS

- Drowning is a leading cause of death and loss of years of life with over 90% of cases occurring in lower- and middle-income countries. Cost-effective prevention strategies have been developed for settings where resources limit treatment for drowning victims.
- Significant drowning injuries induce pulmonary injury and hypoxia in proportion to the amount of water aspirated and the duration of submersion.
- Pulmonary and neurologic support is essential to optimize the victim's chance of a favorable outcome.
- Electroencephalography is used in obtunded drowning victims to assess for subclinical seizures.
- No prognostic scale or clinical presentation accurately predicts long-term neurologic outcome after significant drowning injuries. Normal neurologic recovery is documented in patients despite prolonged submersions, persistent coma, cardiovascular instability, and fixed and dilated pupils on presentation.
- Hyperventilation, corticosteroids, diuresis, barbiturate coma, and neuromuscular blockade do not improve outcome following resuscitation of a drowning victim.
- Comatose patients who have been resuscitated after reasonable submersion time regardless of rhythm should not be rewarmed above 34°C.

1. Which of the following is the greatest risk factor for death from drowning?
   a. Ethanol
   b. Male gender
   c. Poor swimming ability
   d. Saltwater drowning
   e. Warm water

Answer: A. Ethanol is a major risk factor for drowning and for death from drowning. Ethanol is a contributing factor in up to 50% of drowning incidents among adolescents and adults. More males drown than females, but male gender alone does not increase the risk of death from drowning. No direct evidence exists relating swimming ability to risk of death from drowning. Although frequently cited in older literature, there is no significant difference between saltwater and freshwater drowning. Although case reports of survival from cold-water drowning exist, overall mortality is greater in cold water than in warm water.

2. A 13-year-old boy is brought to the emergency department (ED) after an apparent drowning. Family and paramedics report the patient was submerged for approximately 4 minutes. Bystanders began cardiopulmonary resuscitation (CPR) immediately. Paramedics intubated the patient and continued CPR, which has now been ongoing for 10 minutes. On arrival in the ED, the patient has a Glasgow Coma Score (GCS) of 3, unreactive pupils, no pulse, and is in asystole on the cardiac monitor. What is the most appropriate next step?
   a. Cessation of treatment/pronounce death
   b. External cardiac pacing
   c. Induction of hypothermia
   d. Intravenous epinephrine
   e. Unsynchronized defibrillation

Answer: D. Although this patient will likely have a poor outcome, asystolic victims of drowning have a higher incidence of full neurologic recovery than patients with asystole from other causes. Poor prognostic indicators are age younger than 3 years old, submersion greater than 5 minutes, delay in initiation of CPR greater than 10 minutes, hypothermia, severe acidosis, unreactive pupils, GCS of 3, and asystole. The decision to cease care must be made on a case-by-case basis, but initial treatment of asystole should be begun in drowning victims following standard advanced cardiac life support (ACLS) protocols and as in this case, administration of intravenous epinephrine. Experiments with induction of hypothermia are ongoing in drowning victims, but the results are not definitive.

3. A 16-year-old boy presents with tachypnea and coughing. Family members state that they were swimming in a nearby lake when they noticed the patient with the previously mentioned symptoms. The patient is awake and alert but in obvious distress. He has a respiratory rate of 35 breaths per minute, with the remainder of his vital signs within normal limits. His oxygen saturation as measured by pulse oximetry is 84% while on high-flow oxygen. Pulmonary examination reveals diffuse rales. There is no evidence of trauma. In addition to intubation, which of the following is indicated?
   a. Administer intravenous diuretics.
   b. Administer intravenous corticosteroids.
   c. Obtain an electrocardiogram (ECG).
   d. Perform maneuvers to remove fluid from the lungs.
   e. Place the patient in cervical spine precautions.

Answer: C. Dysrhythmias are frequently associated with drowning, especially when no obvious cause is found. ECGs should be performed on all drowning patients. Antibiotics, steroids, and diuretics have all been studied in drowning victims, and none show any benefit. Cervical spine precautions are often initiated in all drowning victims but are not needed unless the patient has clinical signs of trauma or a history of motor vehicle crash, fall from height, or diving into the water. Maneuvers to remove fluid from the lungs (Heimlich, Patrick maneuvers) are ineffective and potentially dangerous.

4. An 18-year-old man is brought to the emergency department (ED) after submersion in his swimming pool. Per witnesses, once the patient was brought out from the water, he initially had severe coughing and complained of shortness of breath. On arrival to the ED, the patient denies shortness of breath and is not coughing. Vital signs as well as oxygen saturation are all within normal limits. Electrocardiogram (ECG) and chest radiographs are normal. What is the appropriate disposition of this patient?
   a. Admission to telemetry for 23-hour observation
   b. Admission to the general hospital for 23-hour observation
   c. Admission to the intensive care unit for 23-hour observation
   d. Discharge home after observation in the ED for 8 hours
   e. Discharge home now

Answer: D. Any symptomatic patients or patients with a history of apnea, unconsciousness, or hypoxia should be admitted. Likewise, patients with dysrhythmias or abnormal chest radiographs should be admitted. Some effects of drowning can be delayed, so asymptomatic patients should be observed for 8 hours in the ED but can be safely discharged if they continue to be asymptomatic and can maintain normal room air oxygen saturation. They should be discharged in the care of a responsible family member or friend.

KEY CONCEPTS

- Patients contaminated with radiation pose minimal risk to health care providers when appropriate precautions and decontamination procedures are employed.
- Decontamination should not delay or impede emergency stabilization of patients with radiation exposure.
- Tissues with greater rates of cellular division, particularly the hematopoietic and gastrointestinal systems, are most radiosensitive. Children are more radiosensitive than adults.
- Vomiting, diarrhea, and skin burns occurring shortly following radiation exposure are predictors of severe radiation injury.
- The 48-hour absolute lymphocyte count is the most important prognostic indicator and should be drawn on suspected radiation exposure patients.
- Most therapy is supportive with symptomatic measures except for exposures involving the ingestion or inhalation of radioactive material, when directed therapy with blocking or chelating agents may be indicated.
- Formal consultation at the hospital, regional, and national levels is available 24 hours a day and should be used for assistance when managing patients with radiation injuries (Table 134.4).

1. Your Emergency Department receives a report of a "dirty bomb" explosion with multiple injuries and deaths on scene. Which injuries would you be *most* likely to see?
   a. Cutaneous (local) radiation injuries
   b. Gastrointestinal radiation injuries
   c. Hematopoietic radiation injuries
   d. Neurologic radiation injuries
   e. Soft tissue and orthopedic injuries

Answer: E. "Dirty bomb" is a term for a traditional explosive device to which an amount of radioactive material has been added. The greatest risk of injury is from traditional blast/explosive effects. The risk of radiation injury and contamination from a "dirty bomb" is extremely low. The small amount of radioactive material that survives the explosion is concentrated in the area immediately around the explosion, where victims are likely killed by the blast itself.

2. Which of the following is the proper sequence for caring for patients from a mass casualty incident involving potential radiologic contamination?
   a. Decontamination, assess with radiation meter, emergency stabilization, mass casualty triage
   b. Assess with radiation meter, decontamination, mass casualty triage, emergency stabilization
   c. Assess with radiation meter, decontamination, emergency stabilization, mass casualty triage
   d. Mass casualty triage, emergency stabilization, assess with radiation meter, decontamination
   e. Decontamination, emergency stabilization, mass casualty triage, assess with radiation meter

Answer: D. In radiation incidents, the immediate risk to patients and providers from radiation injury is low. The most immediate risks are from traditional ABCD compromises. Thus, mass casualty triage and emergency stabilization take precedence, followed by an assessment for radiologic contamination, and decontamination if needed. This is in contrast to chemical or biologic incidents, where decontamination takes precedence.

3. A worker in your hospital's blood bank comes into the emergency department (ED) complaining that he has been exposed to radiation on an ongoing basis for approximately 1 month. He reports his job duties include irradiating blood products to be used in the hospital and that he has not been wearing protective clothing. He reports a burning sensation in his hands. His physical examination is normal. What is the appropriate course of action?
   a. Admit for observation
   b. Decontaminate the patient, admit for observation
   c. Decontaminate the patient, discharge home
   d. Decontaminate the patient, place on isolation, admit for observation
   e. Educate patient and arrange appropriate follow-up

Answer: E. This patient has been irradiated but not contaminated. Patients who are exposed to external beam radiation are not made radioactive and are not a hazard to others. Decontamination and isolation are not necessary. With a normal physical examination and minimal symptoms, this patient can be followed as an outpatient. The most important action in this instance is to educate the patient.

4. What is the most important laboratory test to be performed 48 hours after exposure in determining the prognosis of a patient exposed to a significant radiation source?
   a. Absolute lymphocyte count
   b. Absolute neutrophil count
   c. Genetic karyotype
   d. Platelet count
   e. Aspartate aminotransferase

Answer: A. Monitoring the kinetics of lymphocyte depletion during the first 48 hours following a radiation exposure is very useful at estimating the radiation dose received. The 48-hour absolute lymphocyte count is the most important prognostic indicator. Levels greater than 1200/μL indicate a clinically insignificant dose of radiation and an excellent prognosis. Levels less than 500/μL indicate a significant and possibly lethal exposure. None of the other answer choices carry prognostic significance.

5. Which of the following symptoms within the first 24 hours after a significant radiation exposure would indicate a lethal dose of radiation?
   a. Bruising
   b. Gait ataxia
   c. Nausea and vomiting
   d. Skin burns
   e. Shortness of breath

Answer: B. The neurologic system is the most resistant to radiation injury, requiring at least a 10 Gy dose to be affected. The development of neurologic symptoms (such as altered mental status, focal weakness/paralysis, dizziness, ataxia, seizures) within 24 hours of exposure indicates a lethal dose of radiation. The hematopoietic system is the most radiosensitive (can be affected by as little as 0.5 to 1 Gy), followed by the gastrointestinal system (more than 5–7 Gy). The skin will be affected by local doses of radiation greater than 3 to 5 Gy.

6. A 45-year-old man is inadvertently exposed to a radiation field of 850 mSv/hour for 4 hours while working on the clean-up of a nuclear power plant. In addition to general supportive care, which of the following medications may improve his survival?
   a. Acyclovir
   b. Colony-stimulating factor (cytokines)
   c. Diethylenetriaminepentaacetate (DTPA)
   d. Erythropoietin
   e. Potassium iodide

Answer: B. This patient's radiation dose is sufficient to produce the hematopoietic component of acute radiation syndrome (ARS). Cytokines or colony-stimulating factors have shown modest effects in improving survival. Irradiated patients may develop oral herpes, and acyclovir can be used for treatment but does not affect survival. Anemia is present in most irradiated patients but is typically not clinically significant, and erythropoietin has no role. DTPA is a chelating agent that can be used for patients with internal exposure to plutonium or transuranics. Potassium iodine is useful if a patient is exposed to radioactive iodine, because it will compete with the radioactive iodine for uptake into the thyroid gland, resulting in overall less radioactive uptake.

135 Care of the Poisoned Patient

KEY CONCEPTS

- Toxidromes are constellations of signs and symptoms based primarily on vital signs and neuropsychiatric functions that are characteristic manifestations of certain toxic exposures. Recognition of the presence of a toxidrome can suggest a potential intoxicant and guide early interventions and management strategies. Examples of toxidromes include sympathomimetic, antimuscarinic, cholinergic, sedative-hypnotic, and opioid categories.
- Qualitative urine drug assays have limited roles in the clinical setting and are inferior to quantitative serum levels in terms of guiding specific therapy.
- Syrup of ipecac is not indicated in the emergency department (ED) care of a poisoned patient. Gastric lavage is not part of routine care. When given in a timely fashion (1-hour post ingestion), activated charcoal may be indicated for potentially lethal agents in alert, cooperative patients as noted in Figure 135.1. Whole-bowel irrigation is rarely useful for management of poisoned patients but is potentially helpful for specific poisonings, such as metals, illicit drug packets, or sustained-release medications.
- Serum alkalinization enhances urinary drug elimination for certain drugs and is indicated for significant poisoning caused by salicylates, phenobarbital, and methotrexate.
- Hemodialysis is best suited to remove poisons of low molecular weight, low protein binding, and high water solubility; examples include methanol, ethylene glycol, lithium, and salicylates.
- Regional Poison Control Centers (US: 1-800-222-1222) or a medical toxicologist can assist with antidotal therapy and may help facilitate patient disposition.
- If the motivation behind the toxic exposure was self-harm, a psychiatric consultation is warranted. Patients with substance use disorders should be referred to a detoxification center or designated program.

1. A 23-year-old man presents with confusion. His vital signs are blood pressure 105/77 mm Hg, heart rate 69 beats per minute, respiratory rate 10 breaths per minute, temperature 37.0°C, and arterial oxygen saturation (SaO_2) 96% on 2 L NC. His eyes open only to pain, he localizes to pain, but he does not follow commands and his speech is confused. Otherwise, his physical examination is normal. After intravenous (IV) access is obtained and he is placed on continuous cardiac monitoring, what is the most appropriate next step?

a. Check arterial blood gases (ABG).
b. Check bedside serum glucose.
c. Give 50% dextrose 1 amp IV.
d. Give naloxone 2 mg IV.
e. Perform electrocardiogram (ECG).

Answer: B. All patients with altered mental status should have a rapid determination of their glucose. Although this patient has several abnormal vital signs, none necessitate immediate action. His Glasgow Coma Score (GCS) is 11, and it appears that his respiratory status is adequate. Dextrose should not be given unless hypoglycemia is documented. Naloxone is an opioid antagonist, and it could reverse the patient's confusion if it is caused by opioids. This drug could be considered once other more easily reversible causes of confusion (e.g., hypoglycemia) have been ruled out. ABG and ECG are both occasionally needed in patients with altered mental status, but neither is as urgent as glucose testing.

2. The odor on a poisoned patient can provide clues as to the substance causing the poisoning. If a patient smells of garlic, what should you be concerned about?

a. Cyanide
b. Hydrogen sulfide
c. Methyl salicylate
d. Organophosphates
e. Toluene

Answer: D. Arsenic, dimethyl sulfoxide (DMSO), yellow phosphorus, selenium, and tellurium can also have a garlic odor. Odors associated with the other answer choices are as follows: cyanide, bitter almonds; hydrogen sulfide, rotten eggs; methyl salicylate, wintergreen; and toluene, glue. Odors can aid in making the diagnosis but should not be relied on as completely diagnostic. Also, the lack of odors cannot be used to rule out a poisoning resulting from that substance.

3. A 17-year-old girl presents with delirium, tachycardia, dry skin, flushed face, decreased bowel sounds, dilated pupils, and urinary retention. A friend states that the patient had recently ingested some unknown medications in a suicide attempt. Which of the following medications is most likely responsible for the constellation of signs?

a. Alprazolam
b. Clonidine
c. Diphenhydramine
d. Hydrocodone
e. Pseudoephedrine

Answer: C. The patient is expressing the signs of classic antimuscarinic poisoning. Central nervous system (CNS) effects include the delirium and typical "picking movements" of the fingers. Suppression of the cholinergic system results in tachycardia. All secretory functions are inhibited, causing the dry, flushed skin, decreased bowel sounds, and urinary retention. Unopposed sympathetic activity results in dilated pupils. Alprazolam, clonidine, and hydrocodone can all cause the sedative toxidrome. Pseudoephedrine can lead to the sympathomimetic toxidrome, which typically has diaphoretic skin but can otherwise appear similar to an anticholinergic presentation.

4. A 27-year-old individual presents to the emergency department via emergency medical services after being "found down" with pinpoint pupils, slow breathing, and depressed mentation. The patient was given 2 mg of naloxone via the intranasal route and by the time of arrival in the ED is now awake, alert, and cooperative—and asking to leave. There is no evidence of trauma, and the patient has no other complaints. Based on your knowledge of the half-life of naloxone, when do you feel the patient will be safely dischargeable?

a. 30 minutes
b. 1 hour
c. 2 hours
d. 4 hours
e. 6 hours

Answer: D. The exact elimination half-life of naloxone is closer to 1.1 hours (1 hour 6 minutes). This is markedly shorter than the half-life of almost all opioids. For comparison, morphine has a half-life of approximately 2 hours, hydrocodone has a half-life of approximately 4 hours, and methadone has a half-life up to 30 hours. This means that repeat dosing of naloxone is often necessary when treating opioid overdose. A drug similar in action to naloxone, nalmefene, has a half-life of 10 hours and may sometimes be useful.

5. You suspect a patient has a cholinergic syndrome. Which of the following, if found, would make you question your diagnosis?
   a. Confusion
   b. Diaphoresis
   c. Diarrhea
   d. Fasciculations
   e. Mydriasis (dilated pupils)

Answer: E. The typical cholinergic patient is "wet." A common mnemonic used to remember the symptoms is SLUDGE—salivation, lacrimation, urination, defecation, gastrointestinal cramping, and emesis. Confusion can be present but is nonspecific. Frequently, cholinergic syndrome is caused by organophosphate poisoning, so fasciculations are common. Another common sign is miosis, not mydriasis.

6. Gastric decontamination with activated charcoal can decrease the absorption of certain toxins. However, before charcoal can be given, it must be determined that the risk of aspiration is low, the likelihood of reduction of toxicity or improved patient outcome is high, and that the ingestion occurred recently (within approximately 1 hour). Also, it must be determined that the substance ingested is actually adsorbed by charcoal. Which of the following substances is adsorbed by charcoal?
   a. Gasoline
   b. Hydrofluoric acid
   c. Iron
   d. Methanol
   e. Metoprolol

Answer: E. Charcoal does not adsorb hydrocarbons (i.e., gasoline), ionic substances (i.e., strong acids or bases), metals (e.g., iron), or alcohols. It does adsorb most therapeutic drugs with potential major toxicity, such as β-blockers, calcium channel blockers, and cyclic antidepressants.

7. A 6-year-old boy is found playing with liquid pesticide by family members. The family states that he was drinking the solution. Paramedics find the child sitting in a puddle of liquid that smells strongly of pesticide. He is acting normally. During transportation to the hospital, the paramedics report that the child is now sleepier than previously but still easily aroused with a loud voice. His vital signs are within normal limits. They ask what they should do next. What do you advise?
   a. Administer IV atropine
   b. Administer oral activated charcoal
   c. Administer naloxone
   d. Perform rapid sequence intubation
   e. Remove the child's clothing and wash the skin

Answer: E. This child's clothing has been impregnated with poison, and as long as the child is clothed, he will continue to absorb the poison. Ideally, all patients should be decontaminated on scene to expedite decontamination as well as to minimize the risk of contaminating others. Although this child may have ingested some poison, it is likely that far more is being absorbed from his skin than through his gastrointestinal tract. In addition, charcoal can cause complications, whereas removal of contaminated clothing is relatively safe. Although this child may eventually require intubation, he does not need it now. Most pesticides are organophosphates, so this patient may require atropine, but removal from the poison (taking off his clothing) will provide the greater immediate impact on his condition.

8. A patient suffering from serotonin syndrome will experience myriad physiologic abnormalities, including altered mental status, fever, agitation, tremor, and myoclonus. Many of the symptoms are nonspecific. Which of the following diagnoses is often mistaken for serotonin syndrome?
   a. Anticholinergic syndrome
   b. Brown recluse envenomation
   c. Opioid withdrawal
   d. Sepsis
   e. Sympathomimetic overdose

Answer: E. Serotonin syndrome is characterized by altered mental status, fever, agitation, tremor, myoclonus, hyperreflexia, ataxia, incoordination, diaphoresis, shivering, and diarrhea. Similar symptoms are seen with overdoses of sympathomimetics, lithium, and monoamine oxidase inhibitors (MAOIs). Similar symptoms are also seen with neuroleptic malignant syndrome (NMS) and malignant hyperthermia. The antimuscarinic syndrome will give a "dry" patient without diaphoresis. Brown recluse envenomation produces local pain and possibly skin necrosis, as well as systemic symptoms of fever, nausea, and vomiting. Opioid withdrawal produces similar symptoms and will cause myalgias but without other musculoskeletal effects. Sepsis likewise can produce many of the systemic signs and symptoms seen in serotonin syndrome but without the musculoskeletal effects.

KEY CONCEPTS

- The classic finding of an elevated osmolar and anion gap should raise suspicion of methanol or ethylene glycol toxicity but may not be present depending on the timing of ingestion. Early ingestion has a high osmolar gap without acidosis, and late ingestion has acidosis without an osmolar gap.
- Serum osmolarity is calculated by the following equation:

$$\text{Calculated osmolarity (mOsm/kg)} = 2\text{Na}^+ + (\text{BUN}/2.8) + (\text{glucose}/18) + (\text{ethanol}/4.6)$$

The measured osmolar gap is the difference between the measured serum osmolality and calculated serum osmolarity, with a normal range of –15 to +10 mOsm.

- It is important to note that a normal osmolar gap does not exclude toxic alcohol ingestion.
- Initiate therapy based on strong clinical suspicion of exposure to methanol or ethylene glycol. Block alcohol dehydrogenase preferably with fomepizole, but ethanol can be used if fomepizole is unavailable.
- The main priorities in toxic alcohol exposure are correction of acidosis using bicarbonate solution and hemodialysis, inhibition of the production of toxic metabolites, and elimination of the parent alcohol and its toxic metabolites.
- The presence of acidosis indicates the accumulation of the toxic metabolites of methanol (formic acid) and ethylene glycol (glycolic and oxalic acid). Consult nephrology for emergent hemodialysis to correct acid-base disturbances and remove the parent compound as well as its toxic metabolites.
- Severe acidosis is a poor prognostic factor, with high mortality rates in methanol and ethylene glycol ingestions. A comatose state at the time of presentation also is associated with a higher mortality outcome.
- The findings of an elevated osmolar gap with ketonemia or ketonuria and no development of acidosis indicate isopropanol ingestion. Patients can have a prolonged period of inebriation and can be comatose. Alcohol dehydrogenase inhibition is not indicated in these cases.
- Hypotension and gastrointestinal bleeding are poor prognostic factors in isopropanol ingestion.
- Diethylene glycol can result in acidosis and renal failure and should be managed similarly to ethylene glycol poisoning with fomepizole and hemodialysis.

1. A 24-year-old man presents after intentional methanol ingestion. The patient reports that 1 or 2 hours ago he drank approximately 8 ounces of windshield washer fluid in a suicide attempt. He is not sure of the product name. His only current complaint is slight nausea. His vital signs and physical examination are within normal limits. Serum chemistry reveals the following: sodium, 142 mEq/L, potassium, 4.5 mEq/L, chloride, 110 mEq/L, bicarbonate, 22 mEq/L, blood urea nitrogen (BUN), 18 mg/dL, creatinine, 1.5 mg/dL, and glucose, 111 mg/dL. Serum methanol levels are not obtainable at your hospital. The psychiatry service asks if the patient is medically cleared. Which of the following is the most appropriate response?

a. The patient is cleared; he has a normal anion gap, so no significant methanol ingestion occurred.
b. The patient is cleared; ingestion of 8 ounces is below the toxic level regardless of the concentration.
c. The patient is not cleared; he has an elevated anion gap and needs to receive treatment for methanol toxicity.
d. The patient is not cleared; he has evidence of renal failure and needs to receive treatment for methanol toxicity.
e. The patient is not cleared; not enough time has elapsed from the ingestion to determine if significant toxicity has occurred.

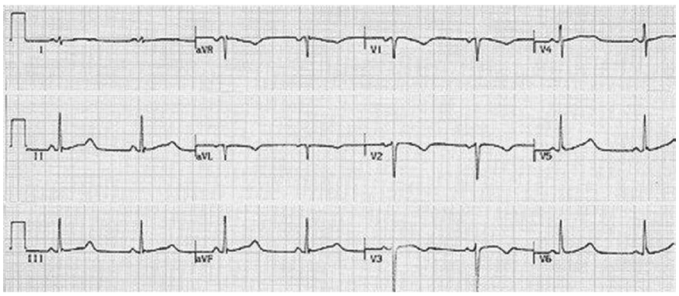
Answer: e. Windshield washer fluid can contain high concentrations of methanol. Methanol itself does not cause toxicity or an anion gap acidosis but its metabolites do, specifically formic acid. It can take 12 to 24 hours for acidosis to develop and even longer if a significant amount of ethanol has also been ingested. This patient should be observed and treated. This patient's anion gap is 10 mEq/L, which is within normal limits. Ingestion of very small amounts of methanol can be fatal or cause permanent neurologic or ophthalmologic damage. As little as 15 mL of 40% methanol can cause death in adults.

2. A 44-year-old woman complains of abdominal pain and headache. Her family reports that the patient drank "something" approximately 8 hours ago in a suicide attempt. The patient is sleepy but arouses with manual stimulation. Her speech is confused, but she follows all simple commands. Her vital signs are blood pressure, 124/82 mm Hg, heart rate, 108 beats/min, respiratory rate, 26 breaths/min, and temperature, 37.0°C (98.6°F). Her physical examination is unremarkable. Her serum chemistry reveals an anion gap of 24 mEq/L. Other laboratory work is pending. Which of the following treatments should be administered initially?

a. Diuresis
b. Hemoperfusion
c. Intravenous flumazenil
d. Intravenous fomepizole
e. Oral activated charcoal

Answer: d. Intravenous fomepizole should be administered. This patient has likely ingested methanol or ethylene glycol, both of which can cause these symptoms, as well as an anion gap metabolic acidosis. The definitive treatment for both ingestions is hemodialysis, which can take some time to arrange. Treatment should be started with fomepizole, even before a definitive diagnosis has been made. Fomepizole and ethanol act by inhibiting the conversion of methanol or ethylene glycol to toxic substances, thereby allowing elimination of the much less toxic parent compounds. Diuresis and hemoperfusion are not effective for methanol or ethylene glycol. Flumazenil will reverse the effect of benzodiazepine but should always be used with caution so as not to induce seizure activity in chronic users. Oral charcoal is not effective for methanol or ethylene glycol, especially after such a long delay.

3. A 20-year-old man presents after ingesting antifreeze. His electrocardiogram (ECG) is shown below. Which electrolyte abnormality is most likely responsible for the findings on the ECG?



a. Hypercalcemia
b. Hyperkalemia
c. Hypocalcemia
d. Hypokalemia
e. Hypomagnesemia

Answer: c. Hypocalcemia is a cause of QT prolongation. Ingestion of ethylene glycol can result in hypocalcemia, which is caused by calcium precipitation with oxalate. Although ethylene glycol ingestion can result in renal failure that can then cause hyperkalemia, it is not associated with QT prolongation. Hypokalemia and hypomagnesemia can both cause QT prolongation; neither is generally associated with ethylene glycol ingestion.

4. An otherwise healthy patient presents after suicidal ethylene glycol ingestion. He is drowsy. His serum pH is 7.1. You have started treatment with fomepizole and have contacted the nephrologist to arrange hemodialysis. In the meantime, what should be done about the patient's acidosis?
   - **a.** Normal saline should be administered to facilitate clearance of the acid.
   - **b.** Nothing should be done; dialysis will correct the acidosis.
   - **c.** Nothing should be done; the fomepizole will correct the acidosis.
   - **d.** Nothing should be done; the patient is not truly acidotic, but ethylene glycol interferes with the laboratory determination of pH.
   - **e.** Sodium bicarbonate should be administered to neutralize the exogenous acid.

Answer: e. Unlike lactic acid, which will be metabolized to bicarbonate, the acidic metabolites of methanol and ethylene glycol cannot be metabolized to bicarbonate and can cause severe acidosis if not treated. Dialysis will correct the acidosis, but the early treatment of the acidosis can reverse some of the adverse effects of methanol or ethylene glycol poisoning. Fomepizole prevents the conversion of methanol or ethylene glycol into toxic metabolites but does nothing to those toxic metabolites already produced. Forced saline diuresis is not beneficial and may increase the incidence of acute respiratory distress syndrome.

5. Which of the following is an indication for hemodialysis after methanol or ethylene glycol ingestion?
   - **a.** Metabolic acidosis
   - **b.** Blood level of 10 mg/dL
   - **c.** Elevated anion gap
   - **d.** Hypocalcemia
   - **e.** Hypokalemia

Answer: a. Metabolic acidosis, renal compromise, visual symptoms, deterioration despite intensive supportive care, and electrolyte abnormalities unresponsive to conventional therapy are all indications for hemodialysis. Although an anion gap is often associated with metabolic acidosis, an anion gap in and of itself is not an indication for hemodialysis. There is debate about the alcohol levels that indicate the need for hemodialysis. Recommendations have been made to dialyze patients with levels of methanol or ethylene glycol between 25 and 50 mg/dL.

6. Which of the following cofactors helps with the elimination of methanol and should be considered in patients with methanol poisoning?
   - **a.** Folinic acid (leucovorin)
   - **b.** Hydroxocobalamin (vitamin B_{12})
   - **c.** Niacin (vitamin B_3)
   - **d.** Pyridoxine (vitamin B_6)
   - **e.** Thiamine (vitamin B_1)

Answer: a. Folinic acid is a cofactor in the degradation of formic acid to carbon dioxide and water, the final step in the metabolism of methanol. Give folinic acid 50 mg IV every 4 hours to adults with methanol poisoning. Thiamine and pyridoxine are indicated for patients with ethylene glycol poisoning and should be considered. Niacin and hydroxocobalamin are antidotes but play no role in the treatment for ethanol or ethylene glycol poisoning.

7. Which of the following statements comparing the effects of isopropyl alcohol and ethylene glycol is true?
   - **a.** Isopropyl alcohol causes less CNS depression and is less toxic than ethylene glycol.
   - **b.** Isopropyl alcohol causes less CNS depression and is more toxic than ethylene glycol.
   - **c.** Isopropyl alcohol causes more CNS depression and is less toxic than ethylene glycol.
   - **d.** Isopropyl alcohol causes more CNS depression and is more toxic than ethylene glycol.
   - **e.** Isopropyl alcohol has the same effects as ethylene glycol.

Answer: c. Isopropyl alcohol causes twice the central nervous system (CNS) depression of ethanol, which causes more depression than ethylene glycol. Isopropyl alcohol is metabolized to acetone, which also causes CNS depression but is relatively nontoxic, except in very high doses.

8. A patient presents with decreased mental status after drinking some homemade alcohol. Serum chemistry reveals sodium, 140 mEq/L, potassium, 4.5 mEq/L, chloride, 108 mEq/L, bicarbonate, 22 mEq/L, BUN, 28 mg/dL, creatinine, 1.0 mg/dL, glucose, 90 mg/dL, and serum osmolality, 320 mOsm/kg. Urinalysis is positive for ketones. Which of the following is the most likely alcohol ingested?
   - **a.** Ethanol
   - **b.** Ethylene glycol
   - **c.** Isopropyl alcohol
   - **d.** Methanol
   - **e.** On the basis of this information, the patient probably did not ingest an alcohol.

Answer: c. This patient has a normal anion gap of 10 but an elevated osmolal gap of 25. The osmolal gap equals the measured serum osmolarity minus the calculated serum osmolarity. The calculated osmolarity is

$$(\text{Sodium} \times 2) + (\text{BUN}/2.8) + (\text{glucose}/18)$$

In this case, it is 295 mOsm/kg. Ethylene glycol and methanol both cause a double gap or elevated anion and osmolal gaps. Ethanol and isopropyl alcohol generally elevate only the osmolal gap. Isopropyl alcohol ingestion causes more CNS depression than ethanol ingestion and results in ketonemia and ketonuria as it is metabolized to acetone.

137 Alcohol-Related Disease

KEY CONCEPTS

- Moderate alcohol consumption is defined as one or two drinks/day for men and one drink/day for women.
- DSM–5 integrates alcohol abuse and alcohol dependence into a single disorder called *alcohol use disorder* (AUD), with mild, moderate, and severe sub-classifications.
- Benzodiazepines are the main treatment of alcohol withdrawal and alcohol withdrawal-induced seizures. Minor alcohol withdrawal occurs as early as 6 h and usually peaks at 24–36 h after the cessation of or significant decrease in alcohol intake.
- Major alcohol withdrawal occurs after 24 h and usually peaks at 50 h (but may take up to 5 days) after the decrease or termination of drinking.
- Delirium tremens is the extreme end of the alcohol withdrawal spectrum; it consists of gross tremors, profound confusion, fever, incontinence, and frightening visual hallucinations.
- Alcohol withdrawal seizures occur 6–48 h after the cessation of drinking, with 60% of patients experiencing multiple seizures within a 6-h period.
- Alcohol withdrawal should be assessed and managed using a validated scale, such as the CIWA-Ar scale.
- Brief intervention and screening (SBIRT—***s***creening, ***b***rief ***i***ntervention, and ***r***eferral to ***t***reatment) can reduce alcohol consumption and is feasible and effective in the emergency department.

1. A 56-year-old man presents with alcohol intoxication. He is drowsy but arousable to painful stimuli. He is confused. Vital signs are within normal limits, and there is no evidence of trauma. Routine testing should include which of the following first?
   a. Basic metabolic panel
   b. Glucose
   c. Serum blood ethanol level
   d. Urine toxicology screen

Answer: b. Chronic alcoholics have decreased glycogen stores and frequently experience hypoglycemia. While a basic metabolic panel could provide you with additional information, the primary electrolyte of interest is glucose. Serum blood ethanol and urine drug screens may be helpful in the undifferentiated patient with altered mental status, but these patients still require a bedside glucose test first.

2. A 42-year-old woman presents with agitation, confusion, and fever. She is noted to experience visual hallucinations during your interview. Her vital signs reveal hypertension, tachycardia, and fever. Physical examination is otherwise unremarkable. Diagnostic studies (including head CT scan and lumbar puncture) are nonspecific. Which diagnosis is most consistent with this patient's presentation?
   a. Acute schizophrenia
   b. Alcohol withdrawal
   c. Anticholinergic poisoning
   d. Thyrotoxicosis

Answer: b. Patients are often confused and agitated and exhibit autonomic instability, resulting in hypertension, tachycardia, and, often, fever. Hallucinations are typically visual. Schizophrenia typically results in auditory hallucinations and, although patients are delusional, they are not typically confused. Patients with anticholinergic poisoning typically present with confusion but also have dry mouth, dry eyes, dry skin, hypoactive bowel sounds, and urinary retention. Thyrotoxicosis patients exhibit lid lag, tremor, and gastrointestinal complaints.

3. In addition to altered mental status (AMS), which of the following is a criterion for diagnosing Wernicke encephalopathy?
   a. Alcohol intoxication
   b. Fever
   c. Oculomotor abnormalities
   d. Seizure

Answer: c. Criteria to diagnoses Wernicke encephalopathy require two of the following: (1) dietary deficiencies; (2) oculomotor abnormalities; (3) cerebellar dysfunction; and (4) AMS or mild memory impairment. Although it is most often diagnosed in alcoholics, alcohol consumption is not required. Treatment is with replacement of dietary deficiencies, particularly thiamine. Magnesium levels should be checked and treated if low. Magnesium is a cofactor for thiamine and is often depleted in chronic alcoholics.

4. Given that brief intervention and screening (SBIRT—***s***creening, ***b***rief ***i***ntervention, and ***r***eferral to ***t***reatment) can reduce alcohol consumption and is feasible and effective in the emergency department, which of the following interventions are particularly effective and preferred for college students?
   a. AA meetings
   b. In-person counseling
   c. Internet-based interventions
   d. Pharmacologic-based interventions

Answer: c. Studies show that providing internet-based interventions is more effective to reduce binge drinking among college students. The Internet could be an economic and acceptable form of delivering brief interventions and is a preferred approach to reach binge drinkers in college. AA Meetings, counseling, and pharmacologic treatment are likely to occur later in the treatment process.

5. A 28-year-old man with alcohol use disorder presents with fever, chills, and a productive cough. The x-ray reveals right lower lobe pneumonia. Which of the following is the most likely etiology of this infection?
   a. *Klebsiella pneumoniae*
   b. *Mycobacterium tuberculosis*
   c. *Mycoplasma pneumoniae*
   d. *Streptococcus pneumoniae*

Answer: d. Pneumococcal pneumonia is the most common type of pneumonia in both healthy individuals and heavy alcohol users. Other infections such as *Klebsiella pneumoniae* and Mycobacterium are also increased in patients with alcohol use disorders and cause disproportionate rates of lung infection, but not the most common. *Klebsiella pneumoniae,* classically associated with alcoholism, is currently more common in patients with cytotoxic chemotherapy, hematologic malignant disease, and transplantation than in the chronic alcoholic.

KEY CONCEPTS

- Acetaminophen concentration should be measured in patients with intentional oral overdoses. Acetaminophen poisoning is relatively silent clinically until serious hepatotoxicity ensues.
- Repeated supratherapeutic dosing of acetaminophen can lead to life-threatening toxicity.
- Use the acetaminophen concentration on the nomogram at 4 hours or later post-ingestion to determine whether *N*-acetylcysteine (NAC) therapy is indicated for acute ingestions.
- IV NAC is preferable to PO NAC. When initiating NAC, continue it until the protocol is completed with adequate clearance of acetaminophen, and there is no evidence of liver injury. If there is evidence of liver injury or acetaminophen concentration remains >10 μg/mL, continue NAC until acetaminophen is undetectable, clinical signs of liver injury have resolved, and liver enzymes are declining (aspartate aminotransferase [AST] <1000 IU/L).
- For maximum benefit, NAC treatment should not be delayed beyond 8 hours after ingestion. If more than 8 hours have passed since ingestion, initiate treatment with ongoing assessment of the amount of ingestion (serial serum acetaminophen levels) and development of hepatotoxicity (elevated transaminases, coagulopathy, and encephalopathy).
- Late or prolonged administration of NAC is beneficial even with low or absent acetaminophen concentrations if hepatotoxicity is evident.
- NAC is safe in pregnancy and is used in the same protocol as for the nonpregnant patient.

1. Which of the following is true regarding pregnant patients with an acute acetaminophen ingestion?

a. All pregnant women should be treated with *N*-acetylcysteine (NAC) after any known ingestion of acetaminophen.
b. Pregnant women in the third trimester should be treated with NAC if their acetaminophen concentration is ever elevated above 50 μg/mL.
c. Pregnant women should be treated with NAC if their acetaminophen concentration is above the treatment line on the nomogram.
d. Pregnant women should be treated with NAC if they ingest >75 mg/kg PO acetaminophen over less than 8 hours.
e. The treatment line on the nomogram is lowered to a line starting at 100 μg/mL at 4 hours after ingestion.

Answer: C. Pregnant patients with an acute ingestion of acetaminophen should have the same initial treatment as nonpregnant patients. After measurement of an acetaminophen concentration, plotting of the acetaminophen concentration on the nomogram treatment with NAC should be given if the concentration is above the treatment line on the nomogram. Both acetaminophen and NAC do cross the placenta, but the fetus is at risk only if the mother becomes ill. Patients with first trimester pregnancies are at risk of miscarriage if the mother becomes ill, and fetuses that are near-term may be at risk for fetal/neonatal hepatotoxicity.

2. A 30-year-old woman presents with drowsiness, nausea, and vomiting. She reports taking several pills in a suicide attempt. Despite your best efforts, she cannot describe what she ingested or how many she ingested. She states she took the pills "last night" and refuses to be more specific. Currently, it is 6 a.m. A family member is present and found the patient vomiting at 4 a.m., at which time she informed the family member she took some pills. The family member further states that the patient was in an argument with a friend at 12 p.m. and may or may not have ingested the pills at that time. The family was with the patient at 8 p.m. and is sure that she had not ingested the pills at that time. What time should be used as the time of ingestion?

a. 2 a.m.—the time that is 4 hours prior to presentation
b. 4 a.m.—the time the patient was found abnormal
c. 6 a.m.—the time of presentation
d. 8 p.m.—the time the patient was last seen and known to be prior to the ingestion
e. 12 p.m.—the time the patient was last seen prior to becoming symptomatic

Answer: D. If the exact time of ingestion cannot be determined, then the "worst case" scenario should be assumed. The longer a patient goes without therapy, the worse the outcome. Therefore, the time of ingestion should be assumed to be the last time the patient was seen normal prior to any possible ingestion.

3. A 20-year-old man is brought to the emergency department by family with a complaint of "overdose." The patient is drowsy but arousable. His vital signs are normal. His physical examination is normal except for appearing to be intoxicated. He has no complaints. The family reports the ingestion of unknown "pills" as well as alcohol and possibly "street drugs." The ingestion occurred approximately 6 hours ago after an argument. Routine supportive care is initiated. What test must be ordered on this patient because it may affect the immediate treatment plan?

a. Acetaminophen concentration
b. Chest radiograph
c. Head computed tomography (CT) scan
d. Serum alcohol concentration
e. Urine drug screen

Answer: A. The antidote should be given within 8 hours of ingestion to prevent development of hepatotoxicity.

4. What is the typical peak serum acetaminophen concentration after a therapeutic oral ingestion?

a. Undetectable
b. 10 μg/mL
c. 30 μg/mL
d. 50 μg/mL
e. 100 μg/mL

Answer: C. This concentration is typically reached approximately 1 hour after ingestion. Four hours after a therapeutic ingestion, the concentration is typically less than 10 μg/mL. Concentrations higher than this should lead one to consider the possibility of chronic ingestion or a person who does not properly metabolize acetaminophen.

5. Two 20-year-old patients present to the emergency department 1 hour after ingesting 15 grams of acetaminophen each in a suicide pact. You are confident of the time of ingestion and that time is confirmed by text messages sent by the patients. Patient A has a 4-hour acetaminophen concentration of 130 mg/dL. Patient B's 4-hour acetaminophen concentration is 170 mg/dL. What is the appropriate treatment for both patients?

a. Neither patient requires treatment because the acetaminophen concentrations are below the treatment cutoff.
b. Neither patient requires treatment because they received activated charcoal (AC).
c. Patient A should receive immediate *N*-acetylcysteine (NAC); patient B should *not* receive NAC.
d. Patient A should *not* receive NAC; patient B should receive immediate NAC treatment.
e. Both patients should receive immediate NAC therapy.

Answer: D. In this case, your confidence of the time of ingestion is high. The treatment line crosses 150 mg/dL at 4 hours. Therefore, if the 4-hour acetaminophen concentration is >150 mg/dL, treatment is indicated. In the case above, patient A does not require NAC therapy, but patient B does require treatment with NAC.

6. What is the antidote for acetaminophen ingestion?

- **a.** Dimercaprol
- **b.** Hydroxycobalamin
- **c.** *N*-acetylcysteine (NAC)
- **d.** *N*-acetyl-*p*-benzoquinone imine (NAPQI)
- **e.** Succimer

Answer: C. Hydroxycobalamin is an antidote for cyanide; succimer is an antidote for lead; and dimercaprol is an antidote for arsenic, lead, and mercury. NAPQI is the toxic metabolite of acetaminophen.

7. A 42-year-old man presents to the emergency department with complaints of nausea and vomiting. He has no other complaints. His symptoms started approximately 12 hours ago. His history is significant for "back pain," which has been worse than normal recently. He states that he finished a bottle of 60 "pain pills" during the past 2 days. His vital signs are normal. Other than right upper abdominal tenderness, his physical examination is normal. Routine laboratory tests including complete blood count, chemistry, and liver panel are normal except an aspartate transaminase (AST) concentration of 1265 IU/L. You are concerned for possible repeated supratherapeutic acetaminophen poisoning. What should you do next?

- **a.** Admit the patient for observation and repeat AST testing; because this is a repeated supratherapeutic ingestion, treatment with *N*-acetylcysteine (NAC) is not beneficial.
- **b.** Initiate treatment with activated charcoal (AC) now.
- **c.** Initiate treatment with NAC now.
- **d.** Obtain two serum acetaminophen concentrations 1 hour apart to determine the drug's elimination half-life to decide if treatment with NAC is indicated.
- **e.** Plot the acetaminophen concentration on the acetaminophen treatment nomogram to determine if treatment with NAC is indicated.

Answer: C. Many prescription and nonprescription analgesics contain acetaminophen, so although a patient may deny use of acetaminophen, suspicion should remain high. A patient with chronic acetaminophen ingestion should have an AST and acetaminophen concentration checked. If the AST is significantly elevated, treatment with NAC should be initiated. Many patients with repeated supratherapeutic ingestion will not have a markedly increased acetaminophen concentration, and the treatment nomogram is not used with repeated supratherapeutic dosing. Regardless of how long ago the ingestion occurred, if a patient displays signs or symptoms of liver damage, NAC should be given because it will still have beneficial effects.

8. A 55-year-old man has taken 10 grams of acetaminophen per day over the last 8 days for a toothache. He arrives to the emergency department with severe right upper quadrant (RUQ) pain, jaundice, and hypoglycemia. His aminotransferases, bilirubin, and acetaminophen concentration are highly elevated, and he is resuscitated and started on intravenous (IV) *N*-acetylcysteine (NAC). Which of the following laboratory findings is a very poor prognostic indicator and indicates that he is a candidate for immediate liver transplant?

- **a.** Aspartate transaminase (AST) >10,000 IU/L
- **b.** Bilirubin >5 mg/dL
- **c.** Lactate >5 mmol/L
- **d.** Acetaminophen concentration >150mg/dL
- **e.** pH <7.4

Answer: C. The patient has severe liver failure due to repeated supratherapeutic dosing of acetaminophen. Prognostic variables that are used to determine immediate hepatic transplant are the Kings College Criteria (a pH <7.3[or lactate >3.5 mmol/L] after resuscitation *or* the combination of Cr >3.3 and PTT >100s [or INR >5] and grade 3 or 4 encephalopathy). Other criteria that indicate high mortality are the APACHE II score >20. If a patient in the emergency department meets these criteria, transfer to a center with liver transplant capabilities is indicated.

9. A 27-year-old woman is brought to the emergency department by emergency medical service (EMS). The patient is lethargic and cannot provide a history. A suicide note indicates that she ingested 50 g of acetaminophen approximately 24 hours ago. The patient is noted to be jaundiced with right upper quadrant (RUQ) tenderness. General supportive care is initiated. Her aspartate transaminase (AST) is 1072 IU. A serum acetaminophen concentration is pending. Which of the following statements is true regarding treatment with *N*-acetylcysteine (NAC)?

- **a.** Treatment with NAC should be delayed until the acetaminophen concentration is obtained.
- **b.** Treatment with NAC should be initiated because an ingestion of 50 g of acetaminophen is potentially fatal.
- **c.** Treatment with NAC should be initiated because the patient has liver injury secondary to acetaminophen poisoning.
- **d.** Treatment with NAC will not alter the outcome; this patient will require a liver transplant.
- **e.** Treatment with NAC will not be beneficial because the ingestion occurred more than 10 hours ago.

Answer: C. Ideally, NAC treatment is initiated within 8 hours of ingestion, but treatment with NAC is still beneficial even after hepatotoxicity has developed. An acetaminophen concentration should be obtained, but there is no reason to delay treatment. Ingestion of 50 g is potentially toxic, but decisions should be based on the patient's current condition. Occasionally, hepatotoxicity from acetaminophen is severe and liver transplant is required, but most cases of hepatotoxicity resolve with NAC treatment.

KEY CONCEPTS

- In the overdose setting, salicylates are profoundly toxic and can be fatal. Salicylate overdose requires vigilant assessment and treatment. The other nonsteroidal antiinflammatory drugs (NSAIDs) generally have self-limited toxicity and respond to supportive measures. There is no specific antidote for any of these drugs.
- Salicylism should be considered in the differential diagnosis of altered mental status, particularly in the elderly.
- Acidemia signifies loss of respiratory compensation and acceleration of toxicity.
- The Done nomogram is no longer used and is not recommended in the evaluation and treatment of salicylate toxicity.
- Salicylate concentrations and blood gas draws should occur every 2 hours until the serum salicylate level is less than 30 mg/dL and is steadily declining at least 10% between assays in the absence of measures to enhance elimination.
- Potassium stores are rapidly depleted in patients with salicylate intoxication and should be repleted with a goal serum level of 4.5 to 5.0 mEq/L (4.5–5.0 mmol/L).
- When possible, mechanical ventilation should be avoided in cases of severe salicylate poisoning. Acidosis may rapidly worsen due to loss of adequate ventilation during the intubation procedure, and it is difficult to maintain ventilation at the level of physiologic hyperventilation.
- If intubation is necessary, a bolus of sodium bicarbonate (50 to 100 mEq) should be given before intubation and post-intubation; minute ventilation should be increased to match pre-intubation respiratory compensation.
- Enhanced elimination through urinary alkalinization with an intravenous sodium bicarbonate drip should be initiated in acute toxicity with a serum level >30 mg/dL.
- Consultation with nephrology and preparation for emergent hemodialysis should occur if the salicylate concentration is above 80 mg/dL or is rising rapidly.
- Hemodialysis is recommended for signs of pulmonary or cerebral edema, coma, seizures, hepatic failure, renal failure, circulatory collapse, or refractory metabolic acidosis along with acute serum levels greater than 100 mg/dL and chronic levels over 40 mg/dL.
- Altered mental status in the setting of salicylate toxicity warrants IV dextrose supplementation.
- Most NSAID overdoses are asymptomatic or cause only minor gastrointestinal symptoms.
- Ibuprofen, along with other propionic acid derivatives, has been associated with sporadic cases of aseptic meningitis.
- The management of NSAID overdose is supportive, and there is no specific antidote. Hemodialysis is reserved for patients with massive overdose (>400–500 mg/kg) and pH <7.1.
- Patients who have ingested a pyrazolone or fenamate require observation for possible seizures for 8 to 12 hours following ingestion.

1. What type of acid-base disorder is caused by salicylate overdose?
   - **a.** Metabolic acidosis and respiratory acidosis
   - **b.** Metabolic acidosis and respiratory alkalosis
   - **c.** Metabolic alkalosis and respiratory acidosis
   - **d.** Metabolic alkalosis and respiratory alkalosis
   - **e.** No acid-base disorder

Answer: B. Early in overdose, salicylates stimulate the respiratory center, causing hyperventilation and a respiratory alkalosis. Later, salicylates inhibit mitochondrial oxidative phosphorylation, resulting in anaerobic metabolism and a metabolic acidosis. This is the classic situation but is later followed by respiratory compensation, resulting in the possibility of a triple acid-base disorder.

2. A 22-year-old man presents 6 hours after an aspirin overdose in an attempt to commit suicide. His only complaint is of tinnitus, and his vital signs and physical examination are normal. His salicylate level is 47 mg/dL. His serum pH is 7.3, with an anion gap of 18 and a serum bicarbonate level of 17 mmol/L. Which of the following is the most appropriate therapy?
   - **a.** Forced diuresis
   - **b.** Hemodialysis
   - **c.** Observation only
   - **d.** Oral activated charcoal due to chronic poisoning
   - **e.** Urinary alkalinization

Answer: E. Salicylates are acidic compounds and therefore readily ionize in an alkali environment. Only in the nonionized state can they traverse cell membranes. Thus, once ionized in the urine, they are effectively "trapped" and can be easily excreted. Urinary alkalinization is indicated in patients with a salicylate level greater than 30 mg/dL, significant acid-base disturbances, or increasing salicylate levels. Forced diuresis does not increase excretion, increases the risk for pulmonary and cerebral edema, and is never indicated. Hemodialysis is the ultimate treatment for salicylate poisoning but is generally reserved for patients with more severe signs or symptoms, more severe acid-base disturbances, or acute salicylate levels greater than 100 mg/dL. Observation alone would be indicated in a patient with a history of salicylate overdose but with no symptoms and normal laboratory values. AC can decrease the amount of salicylate absorbed if given within 1 hour of ingestion. Because large salicylate ingestions tend to form gastric concretions, multiple doses of AC may be indicated. AC is not used in chronic salicylate poisoning because the presentation occurs long after gastrointestinal absorption.

3. Which of the following patients does not require hemodialysis?
   - **a.** A 9-year-old boy with acute salicylate ingestion and an initial salicylate level of 30 mg/dL
   - **b.** A 20-year-old pregnant woman with acute salicylate overdose and an initial salicylate level of 100 mg/dL
   - **c.** A 24-year-old man with acute salicylate overdose and salicylate level of 110 mg/dL despite urinary alkalinization
   - **d.** A 30-year-old woman with chronic salicylate ingestion and salicylate level of 50 mg/dL
   - **e.** A 42-year-old woman with acute salicylate overdose and salicylate level that has risen from 50 to 90 mg/dL despite urinary alkalinization

Answer: A. Indications for hemodialysis in the setting of salicylate toxicity include levels of 100 mg/dl or greater in an acute overdose, a rising level reaching values at or above 80 mg/dl despite urinary alkalinization in an acute overdose, or a level greater than 40 mg/dl in the setting of chronic toxicity. Fetal distress is associated with salicylate toxicity, although recommended salicylate concentration thresholds for hemodialysis are not different than the general population. Rather, discussion with obstetrics regarding emergent delivery should occur if the mother presents in the third trimester of pregnancy.

4. A 41-year-old man presents after an intentional ibuprofen overdose. He complains of epigastric pain, nausea, and one episode of vomiting. He reports that approximately 2 hours ago he took "a handful" of 200-mg ibuprofen tablets in a self-harm attempt. Which of the following is the appropriate next step?

a. Hemodialysis
b. Check serum acetaminophen and salicylate levels
c. Observation for resolution of symptoms
d. Oral activated charcoal (AC)
e. Urinary alkalinization

Answer: B. Oftentimes, patients are unable to differentiate various over-the-counter analgesics. Obtaining salicylate and acetaminophen concentrations in the setting of a reported ibuprofen overdose is essential because occult toxicity due to those compounds may be life-threatening whereas nonsteroidal antiinflammatory drugs (NSAIDs) are generally safe in overdose. Specifically, ibuprofen overdose often results in mild self-limited gastrointestinal (GI) upset. Typically, at least 100 mg/kg of ibuprofen needs to be ingested to cause symptoms. All symptoms are treated with supportive care. Symptoms almost always develop within 4 hours of ingestion. In general, patients may be discharged in cases of small overdoses (<100 mg/kg), once symptoms resolve or after a 4-hour period of observation.

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Anticholinergics

Jason A. Hoppe and Andrew A. Monte

KEY CONCEPTS

- Anticholinergic (antimuscarinic) intoxication is common and can occur as a result of ingestion of a variety of plants and drugs, both prescription and over-the-counter.
- Most patients with antimuscarinic toxicity do well with supportive care and observation.
- Antimuscarinic syndrome varies in clinical presentation with either peripheral or central manifestations predominating.
- Central antimuscarinic delirium is most effectively treated with physostigmine.
- Contraindications to physostigmine administration include narrow angle glaucoma, atrioventricular (AV) blockade, bradycardia, and seizure precipitated by overdose.

PRINCIPLES OF TOXICOLOGY

Overview

Anticholinergic agents cause toxicity through inhibition of muscarinic, nicotinic, parasympathetic, or sympathetic acetylcholine receptors. Nicotinic receptor inhibition and ganglionic acetylcholine inhibition at parasympathetic and sympathetic locations are covered in Chapter 152. This chapter will focus on antimuscarinic effects and toxicity. The terms *anticholinergic* and *antimuscarinic* are used synonymously though the mechanism of toxicity is more accurately described by the term "antimuscarinic," and thus that term will be used in this chapter.

Antimuscarinic effects are due to competitive inhibition of acetylcholine at muscarinic receptors. Muscarinic receptors are found on peripheral postganglionic cholinergic nerves in smooth muscle (intestinal, bronchial, and cardiac), the secretory glands (salivary and sweat), the ciliary body of the eye, and the central nervous system (CNS).

Antimuscarinic agents have been used medicinally from antiquity to the present day. Mandrake plant remains were found in the coffin of Tutankhamen, the Old Testament of the Bible references its use as an aphrodisiac, and antimuscarinic plants were used as anesthetics in Greek and Roman settlements in the 1st century. Atropine, hyoscyamine, and scopolamine are naturally occurring tertiary amine antimuscarinic agents that remain in wide clinical use today. The tertiary amine structure allows the agent to cross the blood-brain barrier; therefore, these agents may precipitate CNS toxicity. Quaternary amine antimuscarinic agents, such as the anti-sialagogue glycopyrrolate, have been developed to mitigate CNS side effects due to their limited ability to cross the blood-brain barrier, though mild delirium may occur in the setting of a large overdose.

CLINICAL FEATURES

Over 600 compounds contain antimuscarinic activity, including prescription drugs, over-the-counter drugs, and plants. The effects of muscarinic receptor blockade are utilized for clinical purposes including pupillary dilation, antispasmodics, sleep aids, treatment of motion sickness, allergic reactions, drying of airway secretions, reactive airway disease, treatment of bradycardia, treatment of Parkinsonism, and the management of urinary incontinence and bladder spasm. The agents that most commonly precipitate antimuscarinic toxicity, such as H1 antihistamines and some antipsychotics, often affect several neurotransmitters and receptor systems in addition to antagonism at muscarinic receptors. This may complicate the clinical presentation and some clinical symptoms may be unique to the specific etiologic agent (Box 140.1).

Antimuscarinic toxicity has both central and peripheral manifestations (Fig. 140.1). Peripheral muscarinic antagonism causes tachycardia, hypertension, hyperthermia, mydriasis, dry mouth, lack of sweating, skin flushing, decreased bowel motility, and urinary retention. Central nervous system blockade of muscarinic receptors may produce delirium characterized by confusion, mumbling speech, agitation, hallucinations, hand picking gestures, myoclonus, tremor, and coma. Manifestations of the toxidrome are frequently incomplete and either peripheral or central components may predominate depending upon which antimuscarinic agent is involved, the dose, and the individual patient (Table 140.1). In one large series of antimuscarinic poisoning, only 28% of patients had all three classic manifestations of tachycardia, dry skin/axilla, and mydriasis. Therefore, most patients will not present with all of these features.[1] Duration of toxicity may be prolonged (18–72 hours) depending on the specific agent, dose, and the pharmacologic effect of delayed gastric emptying.[1]

DIFFERENTIAL DIAGNOSES

The differential diagnosis of altered mental status is broad. Consider antimuscarinic toxicity when there is a history of exposure or if there are physical exam findings consistent with the antimuscarinic toxidrome (Box 140.2).

DIAGNOSTIC TESTING

Laboratory

Patients with mild toxicity, a reliable history of exposure, and symptoms consistent with antimuscarinic toxicity do not require specific laboratory testing. Patients with an unclear history of exposure, other potential etiologies, moderate to severe toxicity or hyperthermia should be evaluated for causes of altered mental status and end-organ toxicity including: serum glucose, electrolytes, cardiac biomarkers, renal function, creatinine kinase to evaluate for rhabdomyolysis, and acid-base status. Patients with an overdose of unclear history should be evaluated for co-ingestion since antimuscarinic agents are often formulated with other potentially toxic agents. Serum acetaminophen and

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CHAPTER 140: QUESTIONS AND ANSWERS

1. A 21-year-old man presents after drinking an "herbal tea" with some friends. He reports visual hallucinations. He has a resting tachycardia and a mildly elevated temperature. On physical examination, he is noted to have dry mucous membranes, dry and flushed skin, and absent bowel sounds. In addition to certain plants, which of the following medications can also cause these symptoms?

a. Amiodarone
b. Clonidine
c. Diphenhydramine
d. Lidocaine
e. Morphine

Answer: C. This patient is experiencing the antimuscarinic toxidrome. In addition to the signs and symptoms described here, patients may also have mydriasis and bladder distention. Mental status can be agitated or depressed. Myoclonus or choreoathetoid movements can also be seen. Amiodarone can cause hypothyroidism or hyperthyroidism and skin discoloration, as well as several other long-term effects. Clonidine can cause dry mouth, drowsiness, bradycardia, and hypotension. Lidocaine can cause headaches, dizziness, confusion, tinnitus, and tremor, as well as bradycardia and hypotension. Morphine can cause respiratory cerebral depression, as well as miosis, bradycardia, and hypotension.

2. Many signs and symptoms of the antimuscarinic syndrome are similar to those of other toxic syndromes, including the sympathomimetic syndrome, serotonin syndrome, and neuroleptic malignant syndrome. Which of the following antimuscarinic findings is most likely to distinguish the antimuscarinic syndrome from the other syndromes listed?

a. Altered mental status
b. Altered movements
c. Dry skin
d. Fever
e. Mydriasis

Answer: C. All the other syndromes often have some degree of diaphoresis. Hyperthermia, altered mental status, and mydriasis can occur in all the named syndromes. Myoclonus can occur in the antimuscarinic syndrome, tremor in the serotonin syndrome, and rigidity in the neuroleptic malignant syndrome.

3. A 31-year-old woman presents with altered mental status after ingesting an unknown quantity of an unknown medication. Her vital signs are significant for tachycardia and hyperthermia. Her physical examination reveals mydriasis, dry mucous membranes, dry skin, decreased bowel sounds, and hypotension. Her electrocardiogram (ECG) reveals a wide QRS complex, at 130 msec, and prolonged QT interval. Which of the following medications is associated with this toxidrome?

a. Amitriptyline
b. Dextroamphetamine
c. Diphenhydramine
d. Fluoxetine
e. Lithium

Answer: A. This patient is experiencing many of the signs and symptoms of antimuscarinic syndrome. However, pure antimuscarinics rarely if ever cause cardiac dysrhythmias (other than sinus tachycardia). Tricyclic antidepressants (TCAs) frequently cause antimuscarinic signs and symptoms but also cause dysrhythmias. Although selective serotonin reuptake inhibitors, stimulants, and lithium can all cause similar signs and symptoms, electrocardiographic abnormalities such as those described here are rare.

4. Which diagnostic test should be performed in almost all patients presenting with the antimuscarinic syndrome?

a. Arterial blood gas analysis
b. Computed tomography (CT) scan of the brain
c. Electrocardiography
d. Electroencephalography
e. Urine drug screen

Answer: C. Patients with a clear presentation and mild symptoms do not necessarily require any diagnostic evaluation. However, patients with more severe symptoms should have measurements of serum electrolytes, renal function, creatine kinase, and glucose concentration performed. Electrocardiography is most helpful because cyclic antidepressants are a common cause of antimuscarinic symptoms and can cause fatal cardiac dysrhythmias. Arterial blood gas analysis might be helpful if the patient has respiratory depression. Head CT might be indicated in patients with altered mental status of unknown cause. Electroencephalography would be indicated only if there is a suspicion of unrecognized seizures. Urine drug screens are almost never helpful in determining treatment, especially in the case of antimuscarinic syndrome because they will not detect most of the medications responsible for this syndrome.

5. What is the best initial treatment of hyperthermia in patients with antimuscarinic syndrome?

a. Acetaminophen
b. Cooling blankets
c. Dantrolene
d. Evaporative cooling
e. Physical restraints

Answer: D. Evaporative cooling is the most effective and noninvasive way to decrease temperature. Death has occurred because of untreated hyperthermia in patients with antimuscarinic syndrome. Antipyretics such as acetaminophen are ineffective at reducing temperature because hyperthermia is not "fever." Dantrolene is useful in malignant hyperthermia but has no role in hyperthermia of other causes. Cooling blankets are ineffective. Physical restraints are likely to worsen the problem and to increase the risk of rhabdomyolysis and myoglobinuric renal failure. If a patient is dangerously agitated from antimuscarinic toxicity, physostigmine is the agent of choice to decrease agitation, muscle activity, and related metabolic activity that contribute to hyperthermia.

CHAPTER 140: QUESTIONS AND ANSWERS—cont'd

6. Which of the following medications crosses the blood-brain barrier and is potentially useful in the treatment of antimuscarinic syndrome?

a. Edrophonium
b. Metoclopramide
c. Neostigmine
d. Physostigmine
e. Pyridostigmine

Answer: D. Metoclopramide is an antiemetic and prokinetic medication that has no role in antimuscarinic syndrome. All of the other agents are acetylcholinesterase inhibitors, but only physostigmine crosses the blood-brain barrier and so it is the only drug that can reverse the central and peripheral effects of antimuscarinic medications. However, physostigmine can cause cardiac dysrhythmias and thus should be used carefully in patients with bradycardia or AV block.

7. Which of the following is a contraindication to physostigmine use in a patient with antimuscarinic syndrome?

a. Altered mental status
b. Bradycardia and atrioventricular (AV) blockade
c. Coexisting myasthenia gravis
d. Hyperthermia
e. Bladder distention and urinary retention

Answer: B. Physostigmine is an acetylcholinesterase inhibitor that is useful to reverse the effects of antimuscarinic medications. However, it is contradicted with narrow angle glaucoma, AV blockade, bradycardia, and seizures due to the causal overdose. The main benefit of physostigmine is to reverse the altered mental status and agitation caused by the antimuscarinic medication. Physostigmine is occasionally used to treat myasthenia gravis. Hyperthermia and urinary retention can occur as symptoms of the antimuscarinic syndrome, and although neither is directly treated with physostigmine, they are not a contraindication to its use.

141 Antidepressants

KEY CONCEPTS

- Although rarely used for depression, MAOIs are used in the treatment of Parkinson disease.
- Because serious symptoms can occur after a lengthy latent period, patients with reported MAOI overdose should be admitted for 24 hours, regardless of symptoms. Toxicity is characterized by tachycardia, hypertension, and CNS changes, and later cardiovascular collapse.
- The primary manifestations of TCA toxicity are seizures, tachycardia, hypotension, and intraventricular conduction delay. IV sodium bicarbonate should be administered for QRS prolongation.
- SSRIs are relatively benign in overdose and generally managed with supportive care alone.
- SNRI ingestions can result in seizures, tachycardia, and occasionally intraventricular conduction delay.
- The hallmark feature of serotonin syndrome is lower extremity rigidity with spontaneous or inducible clonus, especially at the ankles.
- Serotonin syndrome is primarily treated with supportive care, including discontinuation of the offending agent, and benzodiazepines.

1. A 42-year-old woman presents after ingesting an unknown number of pills in a suicide attempt. She does not know the name of the medication she ingested but knows it is an antidepressant. She denies taking any co-ingestants. Currently, she has no complaints; her vital signs, physical examination findings, and electrocardiogram (ECG) are normal. To receive a psychiatric evaluation, she must be transferred off-site. How long should she be observed in the emergency department before transfer to a psychiatric facility?

a. 1 hour
b. 2 hours
c. 6 hours
d. 12 hours

Answer: C. The most dangerous class of antidepressants is the cyclic antidepressants. Typically, peak plasma concentrations and therefore peak effect occur in 2 to 4 hours. However, in overdose, the anticholinergic effects of these agents may result in delayed gastric emptying and delay peak absorption to 6 hours. Most serious signs and symptoms occur within 30 to 60 minutes of ingestion. If the patient has tachycardia or seizure or decreased level of consciousness at 6 hours, she should be admitted for medical observation. If any of these pills were a monoamine oxidase inhibitor (MAOI), however, the patient should be admitted to the hospital for 24 hours.

2. After sinus tachycardia, what is the most common electrocardiographic abnormality seen in cyclic antidepressant overdose?

a. Left bundle branch block
b. PR prolongation
c. QRS prolongation
d. QT prolongation
e. Right bundle branch block

Answer: C. After sinus tachycardia, QRS prolongation of more than 100 milliseconds is the most common specific finding and results from sodium channel–blocking effects of the cyclic antidepressant. Prolonged PR and QT intervals, as well as a right bundle branch block, can also occur but are less common.

3. Which of the following treatment options would be the most appropriate to consider in the awake, asymptomatic patient who presents within 1 hour of a large overdose of a tricyclic antidepressant (TCA)?
   a. Activated charcoal
   b. Endotracheal intubation
   c. Gastric lavage
   d. Physostigmine
   e. Sodium bicarbonate

   Answer: A. Physostigmine should never be prophylactically administered and its use in TCA overdose is generally considered contraindicated, particularly in patients with bradycardia, AV block, and seizures related to the overdose. Sodium bicarbonate should not be prophylactically administered either, but it should be administered as an intravenous (IV) bolus for patients with intraventricular conduction delay and QRS widening. Endotracheal intubation is indicated for patients with significant central nervous system (CNS) depression who are not able to protect their airway. Gastric lavage is not indicated. Activated charcoal can be considered in an awake individual presenting within one hour of ingestion. It should not be "forced" on someone who will not voluntarily drink it (e.g., by placement of a nasogastric tube for the purpose of administering charcoal) because of the increased risk of aspiration and subsequent charcoal pneumonitis.

4. A 23-year-old man presents after an ingestion of a cyclic antidepressant. His initial vital signs are normal. During your initial evaluation, the patient begins to seize. Which agent should be administered first?
   a. Lorazepam
   b. Phenobarbital
   c. Phenytoin
   d. Propofol
   e. Valproic acid

   Answer: A. Lorazepam or any other benzodiazepine is the first-line treatment of toxin-induced seizures. Intravenous phenytoin can increase the incidence of ventricular tachycardia and is not generally effective in controlling toxin-induced seizures. Phenobarbital and propofol are effective but take longer to give and typically are used after benzodiazepines. Valproic acid is not likely to be effective.

5. A 57-year-old woman presents with altered mental status. A friend states that the patient takes antidepressant medications and has recently been complaining of symptoms of an upper respiratory tract infection. The patient is noted to have a temperature of 39.2°C and a pulse of 135 beats/minute. Otherwise, her vital signs are within normal limits. On examination, she has a tremor, myoclonus, and diaphoresis. Which of the following is the most consistent with this presentation?
   a. Anticholinergic syndrome
   b. Cocaine intoxication
   c. Cyclic antidepressant overdose
   d. Neuroleptic malignant syndrome (NMS)
   e. Serotonin syndrome

   Answer: E. Symptoms of serotonin syndrome include altered mental status, agitation, ataxia, diaphoresis, diarrhea, hyperreflexia, hyperthermia, myoclonus, shivering, and tremor. Many of the symptoms are similar to symptoms caused by NMS and sympathomimetic overdoses; however, myoclonus is unique to the serotonin syndrome. Additional historical features consistent with the serotonin syndrome are the fact that the patient is taking an antidepressant, possibly a selective serotonin reuptake inhibitor (SSRI), and has likely added an over-the-counter "cold" medication. Many of these medications contain dextromethorphan, which also decreases serotonin reuptake and can precipitate the serotonin syndrome in patients taking SSRIs.

6. A 24-year-old man presents after taking an overdose of his antidepressant. He does not know the name of the drug. He has no complaints, and his vital signs and physical examination findings are normal. Soon after completing your evaluation, the patient experiences a tonic-clonic seizure. Which of the following antidepressants is most likely to produce seizures without other symptoms of severe toxicity?
   a. Amitriptyline
   b. Bupropion
   c. Fluoxetine
   d. Phenelzine
   e. Trazodone

   Answer: B. Bupropion can induce seizures even at therapeutic levels. Other adverse effects include tachycardia, tremulousness, hallucinations, and QRS prolongation. Cyclic antidepressants (amitriptyline), selective serotonin reuptake inhibitors (SSRIs; fluoxetine), and monoamine oxidase inhibitors (MAOIs; phenelzine) can also cause seizures but less frequently than bupropion and usually with other symptoms of serious intoxication such as central nervous system (CNS) depression or QRS prolongation.

7. A patient is prescribed a new medication that he takes each night. After 4 days, he is drowsy and noted to have orthostatic hypotension, nausea, vomiting, and priapism. What medication is most likely involved?
   a. Amitriptyline
   b. Bupropion
   c. Fluoxetine
   d. Phenelzine
   e. Trazodone

   Answer: E. Trazodone and nefazodone may cause orthostatic hypotension and lethargy. Priapism is a relatively unique complication of trazodone and is more common with therapeutic use rather than in acute overdose.

8. A patient presents after a substantial accidental overdose of her monoamine oxidase inhibitor (MAOI), which she confused with her megavitamin therapy. There is no need for psychiatric evaluation. She is asymptomatic and has normal vital signs and a normal physical examination. What is the appropriate disposition?
   a. Admit to intensive care unit for a minimum of 24 hours of observation
   b. Admit to ward for a minimum of 24 hours of observation
   c. Discharge home
   d. Observe for 6 hours, then discharge home
   e. Observe for 12 hours, then discharge home

   Answer: A. All patients who overdose on MAOIs should be admitted for at least 24 hours of observation because symptom onset is often delayed. In addition, the effects of overdose may be severe and require aggressive therapy.

142 Cardiovascular Drugs

KEY CONCEPTS

- Digoxin toxicity is often occult and should be considered in any patient who is on digoxin and presents with gastrointestinal or visual disturbance and a new dysrhythmia, conduction disturbance or hemodynamic instability, particularly in the setting of ingestion of a natural weight-loss supplement.
- Digitalis Fab is the specific antidote for digoxin toxicity and is dosed based on chronicity of poisoning; most patients require only 1 (chronic poisoning) or 2 (acute poisoning) vials. If full reversal is needed for patients in extremis, dosing is based on total body load of digoxin, not by patient weight.
- Indications for digitalis Fab include progressive and hemodynamically significant bradydysrhythmias and serum potassium >5.0 mEq/L, as summarized in Box 142.4. Of note, Fab therapy should be used before pacing or antidysrhythmic drugs.
- Hyperkalemia in acute digoxin toxicity is best treated with Fab fragments. Conventional treatment as for any other cause of hyperkalemia is also appropriate when Fab fragments are not immediately available. Hyperkalemia in chronic poisoning is likely multifactorial and should be treated with Fab fragments followed by usual hyperkalemia treatments as needed.

1. A 72-year-old man presents with a chief complaint of nausea. He also complains of blurred vision and general weakness. His vital signs and physical examination are normal. He has a past medical history significant for myocardial infarction, congestive heart failure, hypertension, and diabetes. He states that he takes all prescribed medications regularly and just took all of his medications as prescribed approximately 1 hour ago. Reviewing his medication list, you note that one of his medications is digoxin. Which of the following should be done?

a. Administer oral potassium.
b. Give digoxin-specific Fab fragment antibodies now.
c. Give digoxin-specific Fab fragment antibodies only if he has an abnormal electrocardiogram (ECG).
d. Obtain a serum digoxin level in 6 hours.
e. Obtain a serum digoxin level now.

Answer: D. The steady-state (as opposed to peak) digoxin level is most closely correlated with toxicity. Peak level is reached in 1.5 to 2 hours, whereas steady-state levels are not reached until 6 to 8 hours after ingestion. As long as a patient is stable and a chronic ingestion is suspected, digoxin levels taken soon after ingestion are not helpful. Other diagnostic studies should be undertaken while waiting for the digoxin level, including, but not limited to, an electrocardiogram (ECG) and serum electrolyte determination.

2. In a patient with a known digoxin overdose, which of the following is an indication for administration of digitalis antibody fragments?

a. Acute ingestion of 10 mg of digoxin
b. Atrial fibrillation with rapid ventricular response
c. First-degree heart block
d. Ingestion of oleander *(Nerium oleander)*
e. Serum potassium greater than 5 mEq/L

Answer: E. Hyperkalemia is an indication for treatment with digitalis antibody fragments. Other indications include severe ventricular dysrhythmias; hemodynamically significant bradydysrhythmias unresponsive to atropine; rapidly progressive rhythm disturbances or rising potassium level; co-ingestion of cardiotoxic drugs such as beta-blockers, calcium channel blockers, or cyclic antidepressants; or ingestion of a plant known to contain cardiac glycosides in the setting of severe dysrhythmias.

3. A 27-year-old man presents after ingesting 8 mg of digoxin in a suicide attempt. He has ventricular tachycardia on his electrocardiogram (ECG). His blood pressure is 93/54 mm Hg. Laboratory results are not available. You decide to treat with digitalis antibody fragments. How many vials would bind the entire ingestion?

a. 2 vials
b. 4 vials
c. 8 vials
d. 16 vials
e. 20 vials

Answer: D. Each vial contains enough digitalis antibody fragments to neutralize 0.5 mg of digoxin; therefore, the patient should be given 16 vials of Fab fragments to fully treat his ingested dose.

4. A 5-year-old girl presents after taking some of her grandmother's heart medications. She has sinus bradycardia and a slightly low blood pressure. The remainder of her vital signs and physical examination are nonspecific. A bedside glucose test reveals hyperglycemia. The child is not responsive to calcium therapy. The next therapeutic intervention that is likely to treat both the endocrine and cardiovascular pathology would be:
   a. Dobutamine
   b. Epinephrine
   c. High-dose insulin (HDI)
   d. Intravenous fat emulsion (IFE)
   e. Glucagon

Answer: C. Calcium channel blocker toxicity may result in bradycardia and hypotension, much like beta-blocker toxicity. Unlike beta-blocker toxicity, which can cause hypoglycemia, calcium channel blockers can cause hyperglycemia via blockade of calcium channels on beta islet cells. HDI therapy treats calcium channel blocker toxicity via increasing inotropy, as well as normalizing glucose.

5. A 60-year-old man has taken an overdose of his beta-blocker medication. He is awake but lightheaded and has a pulse of 50 beats per minute and a blood pressure of 92/60 mm Hg. Which of the following is the most appropriate initial treatment?
   a. Atropine, crystalloid fluids, calcium salts, glucagon
   b. Atropine, crystalloid fluids, insulin
   c. Atropine, norepinephrine, glucagon
   d. Norepinephrine, crystalloid fluids, insulin
   e. Norepinephrine, glucagon, insulin

Answer: A. Atropine, crystalloid fluids, calcium salts, and glucagon are the initial treatments of choice for beta-blocker toxicity. If these agents are ineffective, high-dose insulin (HDI) is reasonable as next-line therapy for cardiogenic shock. If HDI is ineffective, catecholamines are a reasonable option. The catecholamine choice may vary depending on the type of shock the patient is in, and the dose needed for a response may be greater than for other conditions.

6. A patient presents after an apparent overdose of diltiazem. Which of the following findings would lead you to suspect a co-ingestant in addition to the calcium channel blocker?
   a. Atrioventricular (AV) block on electrocardiogram (ECG)
   b. Hyperglycemia
   c. Metabolic acidosis
   d. Prolonged QRS complex on ECG
   e. Pulmonary edema

Answer: D. Prolonged QRS and QT intervals are generally not seen in calcium channel blocker overdose and should prompt the search for co-ingestants. One exception is the calcium channel blocker bepridil, which can cause QRS or QT prolongation, though bepridil is no longer sold in the United States. All the other listed abnormalities are typical of calcium channel blocker overdose.

7. Treatment for beta-blocker overdose is most similar to the treatment for overdose from which other class of medications?
   a. Anticholinergics
   b. Calcium channel blockers
   c. Cyclic antidepressants
   d. Digoxin
   e. Nitrates

Answer: B. Symptoms from overdose of calcium channel blockers and beta-blockers are similar, as are the therapeutic strategies. Severe anticholinergic toxicity may respond to physostigmine administration, cyclic antidepressants to intravenous (IV) sodium bicarbonate boluses, digoxin poisoning to antibody Fab fragments, and nitrate toxicity to methylene blue therapy.

8. Which of the following drugs is most likely to kill a toddler with ingestion of a single tablet?
   a. Atenolol
   b. Chlorthalidone
   c. Lisinopril
   d. Nitroglycerin
   e. Verapamil

Answer: E. Verapamil and other calcium channel blockers can kill a toddler with the ingestion of a single tablet. Appropriate medical treatment, however, is extremely effective, and very few children die from calcium channel blocker overdose. Although beta-blockers have the potential to cause serious toxicity, propranolol is the beta-blocker most likely to cause serious toxicity. Educating parents and caregivers of young children about the potential effects of these medications is important.

143 Caustics

KEY CONCEPTS

- Health care workers caring for patients with caustic exposures should adhere to universal precautions to prevent additional exposure.
- All symptomatic patients should undergo endoscopy and be should be observed for at least 24 hours.
- Asymptomatic patients can undergo endoscopy in the emergency department or be discharged with close follow-up monitoring.
- Gastric emptying or GI decontamination is not indicated for the majority of caustic ingestions.
- Concentration and pH are the most important characteristics of a substance to predict esophageal and gastric injury.
- Button batteries lodged in the airway or esophagus require endoscopic retrieval.

1. A patient presents after the intentional ingestion of hydrochloric acid (HCl). He complains of mouth, throat, and chest pain, as well as painful swallowing and nausea. His vital signs are normal. Physical examination reveals oral burns without edema. The remainder of the examination is normal. You decide that in addition to psychiatric consultation, the patient should have upper endoscopy. What is the best time for the patient to have the endoscopy?

a. Immediately
b. In 2 to 4 hours
c. In 4 to 12 hours
d. In 12 to 24 hours
e. In 2 or 3 days

Answer: D. The ideal time for endoscopy is 12 to 24 hours. Endoscopy done too soon may miss the extent or depth of injury, whereas endoscopy after 24 hours is actually more likely to cause perforation because the wounds have softened. All patients with signs or symptoms of strong acid ingestion as well as patients with intentional ingestion should have endoscopy performed.

2. A patient presents immediately after the ingestion of bleach. The patient is awake and alert and complaining only of mouth pain. His vital signs and physical examination findings are normal. You consider having the patient drink fluids to dilute the bleach. Which of the following statements regarding this therapy is *true?*

a. Dilution is beneficial only if it is done very soon after ingestion.
b. In cases of alkali ingestions, dilution with a mild acid such as acetic acid is best.
c. Large volumes of fluid should be used.
d. Milk should always be used instead of water.
e. Patients should also be encouraged to eat solids.

Answer: A. Dilution, if it is done at all, should be done early because injuries from caustics occur almost immediately. Water and milk are equally beneficial and are the agents of choice. Weak acids or alkalis should never be used for dilution, because they can cause thermal reactions that worsen the injury. Small volumes up to approximately 500 mL should be used. Solids are not beneficial and can complicate the situation and increase the risk of aspiration.

3. A patient presents after an intentional caustic ingestion. She complains of hoarseness, with mouth, throat, and chest pain. Burns are present on her lips and oral mucosa and she is drooling. Her vital signs are normal, as is the remainder of her physical examination. Which of the following is the most appropriate treatment?

a. Administer 500 mL water orally
b. Administer intravenous Solu-Medrol
c. Endotracheal intubation
d. Obtain electrocardiogram
e. Upper endoscopy

Answer: C. Early intubation is indicated if there is any evidence of airway compromise, such as hoarseness, throat pain, drooling, or edema. Because edema and secretions can both increase rapidly and can make intubation difficult or even impossible, preparations should be made for a difficult airway. Fluids for oral dilution should not be given if the patient has difficulty swallowing. Corticosteroids have been studied to decrease the incidence of stricture formation, but evidence for their benefit is lacking and serious side effects can occur. With the exception of hydrofluoric acid, an electrocardiogram is not routinely needed for caustic ingestions, and endoscopy should be performed 12 to 24 hours after the ingestion and after the airway has been secured.

4. What empirical treatment is indicated to prevent systemic toxicity from hydrofluoric acid ingestions?

a. Calcium chloride
b. Magnesium chloride
c. Potassium chloride
d. Sodium bicarbonate
e. Sodium chloride

Answer: A. Calcium chloride is indicated in significant hydrofluoric acid exposures. Although hydrofluoric acid is a weak acid, the fluoride ion is extremely electronegative and will bind with multiple cations, specifically calcium and magnesium. Profound hypocalcemia is responsible for most deaths from hydrofluoric acid exposure and can occur before a serum calcium concentration can be measured.

5. A 3-year-old boy presents after swallowing a button battery. What is the most appropriate management?
   a. Endoscopic removal
   b. Inpatient observation
   c. Radiograph to assess anatomic location
   d. Surgical removal
   e. Whole-bowel irrigation

Answer: C. Outpatient observation is warranted for button batteries that are located in the stomach or intestines, which can be assessed by plain radiographs. Batteries lodged in the esophagus require endoscopic removal. Examination of the stool for passage of the battery is recommended. If it is not passed in one week, repeated radiographs should be obtained. Inpatient observation is not needed as long as close follow-up can be ensured. Surgical removal and whole-bowel irrigation are not beneficial and are potentially deleterious.

6. A patient presents after a self-harm attempt with 3% household bleach (sodium hypochlorite). The patient has normal vital signs and is asymptomatic. His physical examination is unremarkable. Which is the most appropriate treatment?
   a. Administer 500 mL water orally
   b. Administer intravenous Solu-Medrol
   c. Endotracheal intubation
   d. Obtain electrocardiogram
   e. Upper endoscopy

Answer: E. This patient is high risk for significant esophageal or gastric injury due to the self-harm attempt. Several studies evaluating for the presence or absence of oropharyngeal burns as a predictor of distal esophageal of gastric injury had found these finding to be poorly predictive. Thus, endoscopy should be a standard diagnostic tool used in the management of intentional caustic ingestions.

7. A 45-year-old female presents after an intentional caustic ingestion. She has severe chest pain, and is tachycardic, stridulous, drooling, and hoarse. She has a normal blood pressure. Due to concerns about her airway, she is intubated without difficulty. Which of the following diagnostic imaging modalities is most appropriate for this patient?
   a. Plain chest radiograph
   b. Ultrasound
   c. MRI chest
   d. CT chest
   e. Contrast esophagram

Answer: D. CT scanning is considerably more sensitive than radiography and ultrasound for detecting hollow viscus perforation, for which the patient is at high risk. MRI does not add significant benefit over CT and typically takes longer to obtain.

144 Cocaine and Other Sympathomimetics

KEY CONCEPTS

- Excessive use of stimulants can lead to *sympathomimetic toxicity*, manifesting as tachycardia, hypertension, mydriasis, diaphoresis, hyperthermia, hyperreflexia, and agitation. If untreated, sympathomimetic toxicity can lead to seizures, coma, and death.
- Administration of benzodiazepines is the key therapeutic intervention for sympathomimetic toxicity. Ketamine, olanzapine, and butyrophenones are alternative adjuncts if the cause of severe agitation is not clearly sympathomimetic toxicity.
- Worsening hyperthermia portends imminent death. Reduce body temperature rapidly by external cooling, sedation, and, if needed, paralysis.
- Anti-hypertensives are adjuncts to benzodiazepines. Antihypertensives do not treat as many aspects of sympathomimetic toxicity as do benzodiazepines, however, and short-acting antihypertensive agents (e.g., phentolamine, nitroglycerin, nicardipine, clevidipine) are preferred.
- Wide-complex rhythms secondary to stimulants (cocaine, bupropion) may respond to intravenous sodium bicarbonate administration.
- Cocaine body packers who develop toxicity need emergent surgical intervention to limit bowel necrosis and life-threatening sequelae caused by leaking packets.
- Novel psychoactive substances (e.g., MDMA, bath salts) combine stimulant, hallucinogenic, and psychedelic effects. They generally produce longer-lasting and more intense effects than cocaine or amphetamine.
- Screen for hyponatremia in patients with sympathomimetic or serotonin toxicity.
- If the clinical presentation does not fit the history (e.g., ingested cocaine but hypoxic and somnolent), consider a contaminant or alternate cause. Contaminants change rapidly and vary by geographic region. Consult a medical toxicologist or regional poison control center.

1. Which unique life-threatening electrolyte abnormality is seen with the use of *N*-methyl-3,4-methylenedioxyamphetamine (MDMA)?

a. Hyperkalemia
b. Hypernatremia
c. Hypokalemia
d. Hypomagnesemia
e. Hyponatremia

Answer: e. Hyponatremia can occur with MDMA use and can be severe and life-threatening. MDMA alters the release of vasopressin and will induce a clinical syndrome resembling the syndrome of inappropriate secretion of antidiuretic hormone (SIADH). The patients have concentrated urine with high urine sodium content. Seizures can occur and should be treated with hypertonic saline. In the absence of seizures or other life-threatening conditions, general supportive care with water restriction is adequate treatment.

2. How long does a typical urine drug screen result remain positive after use of cocaine?
   a. 1 day
   b. 3-7 days
   c. 2 weeks
   d. 1 month

Answer: b. Urine drug screens typically detect the cocaine metabolite benzoylecgonine, which is typically present in the urine for 3 to 7 days after the last use.

3. A 24-year-old man is seen by police to be smoking crack cocaine on a street corner. On the way to jail, the patient reports a severe headache, so the police bring him to the emergency department (ED) for medical clearance. The patient is awake with normal mental status but continues to complain of diffuse head pain. His vital signs are blood pressure (BP), 195/99 mm Hg; heart rate (HR), 102 bpm; respiratory rate (RR), 18 rpm; and temperature, 37.2°C. His physical examination is normal. How should his headache be evaluated?
   a. Acetaminophen administration, discharge if headache resolves
   b. Blood pressure reduction, discharge if headache resolves
   c. Head computed tomography (CT) scan, discharge if normal
   d. Head CT scan, lumbar puncture if normal, discharge if both are normal
   e. No evaluation needed; cocaine is not associated with headache

Answer: d. Cocaine is associated with subarachnoid hemorrhage (SAH). Patients complaining of a severe headache after cocaine use should receive a complete evaluation for SAH, including head CT and lumbar puncture. In addition, elevated blood pressure should be lowered while the evaluation is being performed.

4. What is the preferred primary method for controlling a combative patient suffering from a sympathomimetic overdose?
   a. Chlorpromazine
   b. Diazepam
   c. Droperidol
   d. Haloperidol
   e. Physical restraints

Answer: b. Diazepam and other benzodiazepines are the best agents to sedate and establish control of an agitated patient. They cause sedation as well as decrease muscle tone, which can ameliorate hyperthermia. In addition, they can lower the acutely increased blood pressure often seen with sympathomimetic use. In the agitated patient when intravenous (IV) access may be difficult to establish, lorazepam is more predictably absorbed than diazepam. Chlorpromazine can be given intramuscularly to provide rapid sedation but is associated with anticholinergic effects that may exacerbate hyperthermia. Butyrophenone agents (e.g., haloperidol and droperidol) are reserved for more severe agitation and are considered secondary methods of treatment after adequate doses of benzodiazepines. Also, they may have associated dysrhythmic effects that are additive to those of cocaine. Physical restraints may be necessary initially but are not desirable, because they can also increase the risk of hyperthermia as well as increase agitation, which is associated with sudden death.

5. A 27-year-old woman presents with severe chest pain that began soon after use of a large amount of cocaine. She describes the pain as "tearing from my chest to my back." Her vital signs are blood pressure (BP), 210/112 mm Hg; heart rate (HR), 142 bpm; respiratory rate (RR) 22 rpm; and temperature, 38.0°C. A BP measurement taken in the other arm is 147/86 mm Hg. An electrocardiogram (ECG) is normal. You are highly suspicious that the patient is suffering from an aortic dissection and order a transesophageal ultrasound examination while contacting the cardiothoracic surgeon. In the meantime, what should you use to lower her blood pressure?
   a. Labetalol
   b. Nitroglycerine
   c. Nitroprusside
   d. Phentolamine
   e. Treatment should be withheld until a definitive diagnosis is made.

Answer: d. Phentolamine is a direct alpha-adrenergic antagonist and is the drug of choice for sympathomimetic-induced hypertension with end-organ damage. Labetalol and beta-blockers are not recommended, because they have little clinical benefit in this setting. Nitroprusside and nitroglycerine are acceptable agents if phentolamine is not available. Treatment should not be withheld because rapid blood pressure control could be lifesaving.

6. A 35-year-old man presents with chest pain that started approximately 2 hours ago, soon after he smoked crack cocaine. The pain occurred with exertion and has not resolved. He has no previous medical history. His vital signs are blood pressure (BP), 182/99 mm Hg; heart rate (HR), 122 bpm; respiratory rate (RR) 18 rpm; and temperature, 38.2°C. His electrocardiogram (ECG) shows sinus tachycardia with ST depression in the anterior leads. The chest radiograph is normal. Laboratory results are significant for elevations in troponin I. What is the likely explanation for his elevated serum cardiac markers?
   a. He has a completely occluded coronary artery causing ischemia.
   b. He has coronary artery spasm without ischemia.
   c. He has coronary stenosis causing ischemia.
   d. He has fever and rhabdomyolysis but no cardiac disease.
   e. His elevated blood pressure is causing cardiac strain but not ischemia.

Answer: c. Although cocaine can cause coronary spasm with resultant chest pain, patients with positive serum markers are likely to have significant angiographic stenosis. Cocaine users who have complete coronary occlusion typically develop ST elevation just as non-cocaine users do. His elevated blood pressure and fever are certainly increasing the workload on the heart but are not causing his elevated cardiac markers. Cocaine patients may suffer from rhabdomyolysis, and although the resultant renal failure can slow the clearance of the cardiac markers, it does not cause the elevation.

7. A 24-year-old man is brought to the emergency department (ED) in police custody after he admitted to swallowing multiple packets of cocaine to smuggle them through an airport. The patient has no complaints. His physical examination is normal. An abdominal radiograph shows multiple slightly radiopaque areas consistent with packets of cocaine. An electrocardiogram (ECG) is normal. What is the most appropriate management of this patient?
   a. Endoscopic removal of packets
   b. No therapy needed; the patient may be discharged
   c. Observation alone to watch for elimination of packets
   d. Surgical removal of packets
   e. Whole bowel irrigation to remove packets

Answer: e. Whole bowel irrigation with polyethylene glycol facilitates passage of the packets and is safe and effective. Endoscopic removal should be avoided, because there is increased risk of packet rupture. Emergent surgical removal is indicated if there is evidence of packet leak and the patient becomes symptomatic. The patient needs to be observed with cardiac monitoring at a center capable of emergent surgery until all packets have passed.

KEY CONCEPTS

- Hallucinogens include many types of drugs and chemicals with different associated effects, including action at serotonin receptors, dopamine receptors, and glutamate *N*-methyl-D-aspartate receptors.
- Diagnosis and management are based primarily on the history and physical examination, with hallmarks of therapy including supportive care, a calm quiet environment, and sedation with benzodiazepines such as diazepam or lorazepam. Severely agitated patients may benefit from butyrophenone antipsychotic agents such as haloperidol and droperidol.
- Screening tests for hallucinogenic drugs of abuse are of limited value in the acute management of intoxicated patients.
- Novel synthetic hallucinogens continue to emerge and may have effects from hallucinogenic, serotonergic, and dissociative toxidromes. These drugs are rarely detected by screening tests, and cases of toxicity may occur in regional outbreaks.
- Patients with phencyclidine (PCP) toxicity can have unpredictable, violent behavior, and may sustain traumatic injuries. Extreme agitation, although possible, is less common with abuse of ketamine and methoxetamine.
- Extremely agitated, violent PCP-intoxicated patients may require rapid sedation to decrease danger to the patient and health care providers. For hyperthermic patients, sufficient sedation to decrease neuromuscular hyperactivity may require intubation, paralytics, and active external cooling to decrease the risk of multiorgan failure and mortality.
- The care of patients intoxicated from cannabis and synthetic cannabinoids consists of prevention of injury and reassurance for those who have panic reactions. An extremely agitated patient can be sedated with oral or parenteral administration of benzodiazepines or antipsychotics. High doses of antiemetics may be necessary to treat the nausea and vomiting associated with synthetic cannabinoids and heavy daily cannabis use, referred to as the "cannabinoid hyperemesis syndrome."
- The central nervous system and physiologic effects of mescaline use are similar to those of lysergic acid diethylamide (LSD) derivatives, but more vivid hallucinations can occur. Nausea and vomiting are pronounced and almost always precede the hallucinogenic effects.

1. Intoxication with morning glory seeds will mimic intoxication with which of the following hallucinogens?
   a. Ecstasy
   b. Ketamine
   c. LSD
   d. Cannabis
   e. PCP

Answer: C. Several plants contain alkaloids similar to LSD, including morning glory and Hawaiian baby wood rose.

2. A 14-year-old boy presents with altered mental status. On examination he has mydriasis, nystagmus, and lethargy. There is a history of exposure to "skittles." His urine drug screen for drugs of abuse is presumptively positive for PCP. Which toxicant did he ingest?
   a. Dextromethorphan
   b. Ecstasy
   c. LSD
   d. Psilocybin
   e. Synthetic cannabinoids

Answer: A. Dextromethorphan presents with more of a dissociative clinical pattern than opioids. "Skittles, red hots, and triple Cs" are street names for over-the-counter tablets of dextromethorphan. Most immunoassays for PCP cross-react with dextromethorphan to give a false-positive PCP report. Ketamine can also give a similar clinical picture, so the history is also necessary to determine which is the most likely intoxicant in the differential.

3. Which of the following is likely to be detected on a typical urine immunoassay drug screen?
   a. LSD
   b. Kratom
   c. Salvia
   d. Cocaine
   e. Synthetic cannabinoids

Answer: D. The immunoassay screen for cocaine detects its metabolite, benzoylecgonine. Unlike immunoassays for amphetamines, benzodiazepines, phencyclidine, and THC, which may be highly unreliable, inconsistent, or nonspecific, the immunoassay for benzoylecgonine does not typically result in false-positives or false-negatives. Detection of benzoylecgonine is a specific indication of cocaine use in the past 3 days. LSD, kratom, salvia, and synthetic cannabinoids do not typically cross-react with immunoassays that are commonly used.

4. Which of the following mushroom species has hepatotoxic but not hallucinogenic properties?
   a. *Amanita muscaria*
   b. *Amanita pantherina*
   c. *Amanita phalloides*
   d. *Conocybe smithii*
   e. *Psilocybe cubensis*

Answer: C. *A. phalloides* mushrooms do not have hallucinogenic properties. Their toxicity is mainly hepatic due to a number of cyclopeptide toxins contained in all parts of the mushroom. *C. smithii* and *P. cubensis* contain psilocybin and psilocin tryptamines. *A. muscaria* and *A. pantherina* contain the isoxazole compounds ibotenic acid and muscimol, which are analogues of glutamic acid (excitatory) and γ-aminobutyric acid (GABA; inhibitory) neurotransmitters, respectively.

5. Which of the following agents have been used to treat acute withdrawal syndromes and chronic drug dependence?
   a. Kratom
   b. Ibogaine
   c. Buprenorphine
   d. Clonidine
   e. All of the above

Answer: E. Kratom, ibogaine, buprenorphine, and clonidine have all been used to treat acute opioid withdrawal and chronic drug dependency with varying effects and long-term outcomes.

146 Iron and Heavy Metals

KEY CONCEPTS

- Asymptomatic patients seeking emergency department (ED) care for an abnormal metal test need follow-up evaluation arranged with a medical toxicologist.
- Metal testing in the ED should be ordered in consultation with a medical toxicologist or regional poison center.
- Acute ingestion of the salts of most metals causes rapid severe gastrointestinal pain and emesis.
- Abnormal neurologic signs in a patient with any metal exposure warrants admission for further evaluation and chelation therapy.
- Acute iron poisoning can result in gastrointestinal symptoms, metabolic acidosis, and hepatoxicity. Serum iron levels at 3 and 6 hours after ingestion determine toxicity and need for therapy.
- The chelation agent of choice for severe iron poisoning is deferoxamine, which is indicated for peak serum iron concentrations greater than 500 μg/dL and in patients with severe signs and symptoms regardless of the iron level.
- The most important intervention for lead poisoning is removal from the source of exposure.
- The gastrointestinal decontamination method of choice for iron and lead toxicity with radiographic presence of pills or paint chips is whole bowel irrigation (WBI).
- The chelation agent of choice for acute arsenic poisoning is intramuscular British antilewisite (BAL) or oral succimer.
- Elemental mercury is nontoxic to the gastrointestinal tract but may cause pulmonary and central nervous system (CNS) toxicity from inhalation of volatilized vapors.

1. An 18-month-old female presents to the emergency department with her mother after they were informed of an elevated venous blood lead level of 38 mg/dL by their pediatrician. The child is asymptomatic and has a normal physical exam. She is an only child, does not attend daycare, and spends her time in the family home which was built in 1942. Which of the following would be the most appropriate initial management?

a. Admit for parenteral chelation
b. Start oral chelation and discharge home
c. Admit or secure alternative housing to limit exposure
d. Discharge home to follow-up with their pediatrician
e. Obtain an abdominal x-ray to determine if leaded foreign bodies are present

Answer: E. The most important treatment for all heavy metal poisonings is to limit further exposure. The source of lead in this case is likely from leaded paint in the home. Evaluating if the child currently has paint chips or other leaded foreign bodies in her gastrointestinal tract is important. If present, she will require decontamination to prevent absorption. After confirming the absence or presence of leaded foreign bodies, environmental control is paramount. To limit any further exposure, parents should ensure alternative housing (i.e., staying with a relative) or the child should be admitted to the hospital until the local public health entities can inspect and remediate the home. Discharging home with primary care follow-up is not appropriate because the child will continue to be exposed. Parenteral chelation is required for children with a lead level of 69 mg/dL and above or for severely symptomatic patients. Oral chelation is for children with blood lead levels above 44 mg/dL. This asymptomatic child does not meet criteria for chelation with a level of 38 mg/dL.

2. A 20-year-old man presents after an iron overdose. An abdominal radiograph shows many radiopaque objects in his stomach, consistent with iron tablets. You decide to try to decrease the gastrointestinal absorption of the iron. Which of the following methods is most effective?

a. Activated charcoal
b. Gastric lavage
c. Surgical removal
d. Syrup of ipecac
e. Whole bowel irrigation (WBI)

Answer: E. WBI is the preferred method to minimize iron absorption. WBI should not be routinely used in small overdoses or with mild symptoms, but it should be considered when a significant ingestion is suspected or when multiple tablets are identified by radiography. Iron

is not adsorbed to oral activated charcoal. Gastric lavage and syrup of ipecac do not remove significant amounts of iron. Open surgical removal of tablets has been used in the past and is effective, but it is invasive and has a higher rate of adverse outcomes than WBI.

3. A healthy 26-year-old female was helping her great aunt clean out her home when she spilled a large tray of old mercury thermometers on the floor. She then attempted to vacuum up the mercury. She is now presenting to the emergency department 3 hours later with cough, fever, and shortness of breath. Her oxygen saturation is 93% on room air and her respiratory rate is 24 breaths per minute. A chest x-ray shows bilateral patchy infiltrates. Which of the following would be the most appropriate initial management?
   a. Provide supplemental oxygen and intravenous fluid resuscitation and admit for further monitoring.
   b. Give oral activated charcoal.
   c. Consult nephrology for potential acute kidney injury.
   d. Administer antibiotics and admit for further monitoring.
   e. Call the department of public health to evaluate the home for hazardous material clean-up.

Answer: A. This patient is exhibiting acute inhalation elemental mercury toxicity. After an inhalation exposure where mercury is volatilized and inhaled, patients can have fever, shortness of breath, and cough that can rapidly progress to ARDS. This may occur hours after exposure. Later, as the mercury is absorbed patients are at risk for inorganic mercury poisoning resulting in nephrotoxicity. The best approach to these patients is supportive care and close monitoring of their respiratory status. There are no roles for activated charcoal or antibiotics as activated charcoal does not bind heavy metals and this is a pneumonitis. Nephrology may ultimately need to be consulted but now the focus should be on the patient's cardiorespiratory status. Answer choice E is also reasonable to prevent further exposure at the home given the large amount of elemental mercury spilled, however it is not the most emergent management step.

4. A 5-year-old boy presents with abdominal pain, nausea, vomiting, and bloody diarrhea. The boy was found rummaging through his grandmother's medicine cabinet and says he swallowed some of her medications. His vital signs are significant for hypotension and tachycardia. An electrocardiogram (ECG) shows sinus tachycardia. Laboratory tests are significant for an anion gap acidosis, hyperglycemia, and moderate leukocytosis. Which of the following medications did he most likely ingest?
   a. Digoxin
   b. Diltiazem
   c. Iron
   d. Metformin
   e. Metoprolol

Answer: C. Iron toxicity typically presents with gastrointestinal symptoms including occasional bleeding soon after ingestion. The previously mentioned laboratory findings are also typical. Digoxin toxicity can also be manifested with gastrointestinal symptoms, but there is usually no gastrointestinal bleeding and there are also typically ECG changes. Beta-blocker toxicity, such as with metoprolol, is manifested with hypotension and bradycardia, as well as with hypoglycemia. Calcium channel blocker toxicity, such as with diltiazem, can be manifested with gastrointestinal symptoms, acidosis, and hyperglycemia. However, patients typically have bradycardia, and gastrointestinal bleeding is not expected. Metformin toxicity is manifested with gastrointestinal upset, typically without bleeding. Lactic acidosis can result, but the vital sign abnormalities and other laboratory results are not expected.

5. A 57-year-old woman with hypothyroidism presents to the emergency department with painful, burning circumferential bilateral lower leg pain. She states that over the last 2 weeks the pain has gotten progressively worse so that she feels as though she cannot even wear socks. She lives in a rural community, works part-time at the public library, and denies alcohol use or taking any herbal supplements. Her only medication is levothyroxine. She denies fevers or chills, polyuria, chest pain or shortness of breath. Her exam is remarkable for hyperalgesia in the bilateral lower extremities. Looking through her chart you notice that she was in the ED 2 weeks ago for gastroenteritis, was given IV fluids and then discharged home. As you leave her room, she tells you she is convinced her husband is poisoning her. You then decide to order a complete blood count, a basic metabolic panel, and a:
   a. Neurology consult
   b. Venous blood lead level
   c. Vitamin B_{12}
   d. 24-hour urine collection for arsenic testing
   e. Ethylene glycol level

Answer: D. Acute inorganic arsenic poisoning is characterized by cholera-like diarrhea followed weeks to months later by peripheral neuropathy and nail and skin changes. Arsenic was commonly used as a rodenticide until the mid-19th century when it was no longer produced. It can still be found in rural areas, stored in old barns. A 24-hour speciated urine arsenic test to determine the amount of inorganic and organic arsenic is the gold standard for determining the need for chelation. Although lead poisoning can cause peripheral neuropathy, this is typically motor in nature and not preceded by nausea, vomiting, and diarrhea. Constipation and colicky abdominal pain are characteristic for lead poisoning. Vitamin B_{12} deficiency can lead to a peripheral neuropathy however in this patient it is not the most likely diagnosis.

KEY CONCEPTS

- Aspiration is the major toxic risk of hydrocarbon poisoning.
- Hydrocarbons may cause systemic toxicity, burns, cardiac dysrhythmias, altered mentation, and seizures depending upon the specific agent, dose, and duration of exposure.
- Gastrointestinal decontamination is potentially harmful in cases of hydrocarbon ingestion and is contraindicated.
- Hydrocarbon inhalant abuse can cause central nervous system (CNS) and cardiotoxic effects.
- In most cases of hydrocarbon ingestion or inhalation, symptomatic care with observation and monitoring are the cornerstones of management. There are no specific antidotes for hydrocarbons. Patients with pulmonary symptoms should have a chest radiograph.
- Symptoms of toxicity, especially aspiration, can be delayed, so asymptomatic patients should be observed for 6 hours and given instructions to return if symptoms develop after discharge.

1. Hydrocarbons can affect many organ systems, but derangements in which organ system most commonly lead to death after hydrocarbon exposure?
- **a.** Cardiac
- **b.** Gastrointestinal
- **c.** Nervous
- **d.** Pulmonary
- **e.** Renal

Answer: D. Most fatalities from hydrocarbon ingestion occur because of aspiration. Hydrocarbons cause direct lung injury, as well as displace oxygen and disrupt surfactant. Hydrocarbons can sensitize the myocardium to catecholamines, which can result in ventricular dysrhythmias and sudden death. Although an important consideration, death is more frequently caused by pulmonary complications. Gastrointestinal complications from hydrocarbon exposure are rare unless related to aspiration potential. Chronic hydrocarbon abuse or exposure causes nervous system dysfunction, including peripheral neuropathy, cerebellar degeneration, and neuropsychiatric disorders, but typically does not result in death. Several hydrocarbons cause renal failure, particularly toluene, but rarely result in death.

2. A healthy 20-year-old man presents after accidentally ingesting furniture polish that was stored in a plastic container that he mistook for a beverage. The ingestion occurred approximately 1 hour ago. He has no complaints. His vital signs and physical examination are normal. Which is the most appropriate course of action?
- **a.** Carbon monoxide level
- **b.** Careful gastric decontamination using nasogastric tube
- **c.** Drug screen
- **d.** Observe for 6 hours and discharge if condition does not change
- **e.** Serum electrolytes

Answer: D. Furniture polish is typically a mixture of nontoxic chemicals with a hydrocarbon base. The primary toxicity from hydrocarbon is from aspiration. Gastric decontamination is not recommended. Serum electrolytes and drug screens are rarely helpful in acute asymptomatic hydrocarbon exposures. Wood strippers often contain methylene chloride, which can cause carbon monoxide poisoning, but this chemical is not found in furniture polish.

3. A 4-year-old boy with a history of asthma is brought to the emergency department (ED) following an accidental hydrocarbon ingestion 2 hours ago. He initially was noted to be coughing. His vital signs are normal. His physical examination reveals mild bilateral wheezing but with good air movement. His oxygen saturation is 99% on room air. A chest radiograph is normal. The child is placed on a cardiac monitor and intravenous (IV) access is obtained. What is the most appropriate treatment plan?
- **a.** Admission and observation
- **b.** Endotracheal intubation
- **c.** Intravenous sodium bicarbonate
- **d.** Racemic epinephrine
- **e.** Oral activated charcoal

Answer: A. Patient is symptomatic after a hydrocarbon exposure. Hydrocarbon pneumonitis can worsen over the next 24hours. Typically, they will cause few symptoms when ingested. Any therapy that risks inducing vomiting should be avoided (i.e., gastric lavage and oral activated charcoal). Patient is not ill enough to get intubated. Racemic epinephrine may be helpful if stridor or upper airway swelling is evident. Sodium bicarbonate has no role in hydrocarbon toxicity.

4. A 16-year-old male presents by emergency medical services (EMS) paramedics with sudden cardiopulmonary arrest after being found unresponsive by friends at a high school beach party. In the emergency department (ED), he is unresponsive with no gag reflex. When you attempt to orally intubate the patient, you note gold metallic colorations inside his oral cavity. What else are you immediately concerned about clinically?
- **a.** Airway obstruction from anaphylactic shock
- **b.** Cardiac arrest from ventricular dysrhythmias
- **c.** Sedative hypnotic toxicity and respiratory arrest
- **d.** Toxic alcohol poisoning and acute renal failure
- **e.** Traumatic intracranial hemorrhage

Answer: B. This adolescent has likely abused a hydrocarbon (such as toluene from gold spray paint) for recreational and euphoric reasons. Cardiac arrest could occur from primary asphyxia if they huffed using a bag over their head or possible direct cardiotoxicity from the inhaled hydrocarbon resulting in ventricular dysrhythmias.

148 Inhaled Toxins

KEY CONCEPTS

- An asphyxiant is any gas that displaces sufficient oxygen from the breathable air. Treatment consists of removal from exposure, supplemental oxygen, and supportive care.
- Highly water-soluble gases produce rapid irritation and predominantly upper respiratory tract effects, such as airway irritation. Poorly water-soluble gases, like phosgene, often produce delayed lower respiratory tract findings, such as bronchospasm or acute respiratory distress syndrome (ARDS).
- Carbon monoxide (CO) poisoning is confirmed by co-oximetry measurement. Cyanide poisoning is treated empirically when cardiovascular instability (e.g., hypotension), altered mental status, or a serum lactate greater than 10 mmol/L are present in a patient with a concerning history, such as a fire victim.
- Hydroxocobalamin is the preferred antidote for most cyanide-poisoned patients due to its efficacy, ease of use, and safety in patient with concomitant CO poisoning. Sodium thiosulfate may be administered concomitantly and may provide additional benefits.
- Patients with hydrogen sulfide poisoning generally respond to removal from exposure and ventilatory support.
- Normobaric oxygen therapy is sufficient to many patients with CO poisoning, but we recommend consultation with a hyperbaric (HBO) facility, poison control center, or medical toxicologist for consideration of HBO therapy under specific conditions. Indications for consultation and HBO treatment include patients with a carboxyhemoglobin (COHb) greater than 25% in the absence of clinical findings, a COHb greater than 15% or signs of fetal distress in pregnancy, or an elevated COHb level with one or more of the following: syncope, coma, altered mental status, abnormal cerebellar function, or a prolonged CO exposure with minor clinical findings.

1. A laboratory worker is brought to the emergency department (ED) after being found unconscious. His colleague reports that the patient was found in his vehicle with its windows closed which he was using to transport dry ice. The patient is now awake but reports feeling tired and confused. He has no other complaints. His vital signs and physical examination are normal. What toxin-specific diagnostic test should be ordered for this patient?

a. Carboxyhemoglobin (COHb)
b. Chest radiograph
c. Electrocardiogram
d. Methemoglobin (MetHb)
e. No tests are indicated

Answer: e. Carbon dioxide is primarily a simple asphyxiant, meaning that its major consequential adverse effects stem from its displacement of oxygen in the lungs. Once patients are removed from the source, they generally recover completely. Patients should be observed until this time. COHb measurement would be indicated if there is a suspicion of carbon monoxide (CO) exposure. Chest radiographs should be ordered if patients have pulmonary complaints after an unknown exposure. Electrocardiograms should be ordered if patients are exposed to known cardiac toxins. MetHb levels should be checked when there is suspicion for oxidative stress on the red blood cells.

2. A 32-year-old woman presents following exposure to an irritant gas at her job site. She reports cough, burning eyes, and shortness of breath. She has mild tachypnea, with the remainder of her vital signs within normal limits. Her oxygen saturation is 96% on room air. She is noted to have stridor on physical examination. What is the preferred method to evaluate her upper airway symptoms?

a. Arterial blood gas (ABG)
b. Chest radiograph
c. Computed tomography of the neck
d. Fiberoptic laryngoscopy
e. Soft tissue neck radiograph

Answer: d. Fiberoptic or direct laryngoscopy is the preferred method to evaluate upper airway symptoms after exposures to irritant gases. Radiographs and laboratory tests have no role and should not influence the decision to provide a definitive airway. Symptoms can progress rapidly, so patients with upper airway symptoms require either placement of a definitive airway or close observation with frequent serial examinations.

3. A 52-year-old man is brought to the emergency department (ED) after being rescued from a house fire. He has not suffered any cutaneous burns. He complains of a sore throat, hoarse voice, and cough. Vital signs are normal. Physical examination reveals soot in his oropharynx and carbonaceous sputum. What therapy should be instituted first?

a. Endotracheal intubation
b. Intravenous (IV) methylprednisolone
c. Nebulized albuterol
d. Nebulized sodium bicarbonate
e. Saline bronchoalveolar lavage

Answer: a. Endotracheal intubation should be performed early in patients with signs and symptoms of significant airway burns (as with this patient). Corticosteroids are not beneficial and can worsen associated injuries. Inhaled beta-agonists are commonly used, but there is no evidence of improved outcome. Inhaled sodium bicarbonate plays no role in the management of smoke inhalation. Bronchoalveolar lavage can be performed if there is suspicion of inhaled debris or toxins, but the airway should first be secured.

4. A 47-year-old woman is brought to the emergency department (ED) after being rescued from a house fire. She was found unconscious at the scene and intubated before arrival. Her vital signs are significant for hypotension and tachycardia. Physical examination is significant for soot in the oropharynx. No cutaneous burns are noted. You suspect that she is suffering from cyanide poisoning. What is the most appropriate immediate therapy?
   a. Hyperbaric oxygen (HBO)
   b. Intravenous (IV) hydroxocobalamin
   c. IV methylene blue
   d. IV sodium nitrite
   e. Observation and supportive care

Answer: b. One of the two major treatments for cyanide poisoning is hydroxocobalamin (the other is the traditional cyanide antidote kit). The nitrite compounds in the cyanide antidote kit convert hemoglobin to methemoglobin (MetHb), which in turn binds to cyanide. However, nitrites produce hypotension and the MetHb prevents proper oxygen deliver, which may compound the reduction in oxygen delivery associated with carbon monoxide (CO) poisoning. HBO has no role in acute cyanide poisoning. Methylene blue has been used for cyanide poisoning in the past but is not as useful as the cyanide kit. Its primary use is in treating methemoglobinemia. General supportive care is not appropriate because there is an effective antidote for this patient's poisoning.

5. A 22-year-old man is brought to the emergency department (ED) after being found unconscious in a car with an intentionally prominent suicide note visible in the window. By the time he arrives in the ED, he has regained consciousness and is complaining of headache and nausea. Paramedics report that the car engine was not running when the patient was discovered. His vital signs and physical examination are normal. Which of the following therapies should be instituted?
   a. Hyperbaric oxygen (HBO)
   b. Intravenous (IV) methylene blue
   c. IV sodium nitrite
   d. IV sodium thiosulfate
   e. Observation and supportive care

Answer: e. This patient has been exposed to hydrogen sulfide (a common form of suicide in some parts of the world), which has similar effects on the mitochondria as cyanide. However, hydrogen sulfide is rapidly removed from the body; and as long as patients are recovering, removal from the source is usually all that is necessary. HBO and methylene blue have no role in hydrogen sulfide poisoning. Sodium nitrite can be used in patients who are not recovering once removed from the source or for severe exposures. Sodium thiosulfate is not necessary because hydrogen sulfide is detoxified by a different pathway than cyanide and does not need a sulfur donor.

6. A worker at a chemical plant was exposed to a cloud of chlorine gas (Cl_2). On examination in the emergency department you note a normal oxygen saturation, increased respiratory rate, and wheezing on lung auscultation. In addition to administration of inhaled beta-adrenergic agonists, which of the following can be administered for symptomatic relief?
   a. Nebulized hypertonic saline
   b. Systemic corticosteroids
   c. Nebulized 2% sodium bicarbonate solution
   d. Antibiotics
   e. Intravenous magnesium sulfate

Answer: c. This patient was exposed to chlorine gas which is an intermediate water-soluble gas that combines with water in the respiratory tract to form hydrochloric acid. Patients may have improvement of their symptoms with the administration of nebulized 2% sodium bicarbonate solution. Systemic corticosteroids have not shown a benefit in these patients. Antibiotics should not be administered routinely in these patients unless an overlying infectious process is suspected.

7. What is the major benefit of hyperbaric oxygen (HBO) therapy for patients suffering from carbon monoxide (CO) poisoning?
   a. Decreased rate of hospitalization
   b. Improvement of 24-hour mortality
   c. Improvement of 30-day mortality
   d. Prevention of delayed cardiovascular complications
   e. Prevention of delayed neuropsychiatric complications

Answer: e. There is controversy regarding the role of HBO therapy for patients with CO poisoning, but the best evidence suggests that it can significantly decrease the incidence of delayed neuropsychiatric complications. There is no change in rate of hospitalization, nor on overall mortality, either short term or long term. There are no delayed cardiovascular symptoms associated with CO poisoning.

8. Assuming that all patients have similar vital signs and complaints of headache and nausea, which of the following patients suffering from carbon monoxide (CO) poisoning should be considered highest priority for hyperbaric oxygen (HBO) therapy?
   a. A 22-year-old otherwise healthy man with a carboxyhemoglobin (COHb) level of 30%
   b. A 25-year-old otherwise healthy pregnant woman with a COHb level of 25%
   c. A 30-year-old otherwise healthy man also suffering from cyanide poisoning with a COHb level of 15%
   d. A 35-year-old otherwise healthy woman with second-degree burns to 20% of her body and with a COHb level of 20%
   e. A 67-year-old asymptomatic woman with coronary artery disease and with a COHb level of 25%

Answer: b. Pregnant patients should be considered for HBO therapy. CO binds more strongly to fetal hemoglobin than to adult hemoglobin and can cause severe hypoxia to the fetus. There is controversy about an absolute level of COHb that requires HBO therapy. HBO does not benefit cyanide victims, nor is it indicated in uncomplicated burn patients or those with stable comorbidities.

9. You successfully treat a patient for cyanide poisoning. Several hours after improvement of the patient's hemodynamics, you are alerted that the patient has developed a bright red skin discoloration. Which of the following is the most likely explanation for this new physical exam finding?
   a. Administration of hydroxocobalamin
   b. Concomitant carbon monoxide toxicity
   c. An anaphylactic reaction
   d. Development of hyperthermia
   e. Peripheral vasodilation

Answer: a. Patient presented with cyanide toxicity for which the antidote, hydroxocobalamin, was given. Hydroxocobalamin has only mild adverse effects, although it can cause hypertension in those without cyanide toxicity and it can cause a bright red discoloration of the patient's skin. This discoloration can often be mistaken as an allergic reaction and it can interfere with spectrophotometric laboratory assays including COHb level. Cherry red skin discoloration in the setting of CN or CO poisoning is a postmortem finding.

149 Lithium

KEY CONCEPTS

- The clinical pattern of acute and chronic lithium toxicity is different. Gastrointestinal symptoms occur early, and neurologic toxicity manifests late in acute toxicity. Neurologic findings (e.g., tremors, altered mental status and seizures) often are presenting signs of chronic lithium toxicity, whereas gastrointestinal symptoms often are absent.
- Neither activated charcoal nor whole bowel irrigation (WBI) is indicated in the routine management of acute or chronic lithium toxicity.
- Serial lithium concentrations should be obtained every 2 to 4 hours initially to determine the peak level and the need for hemodialysis.
- Fluid hydration with crystalloid is essential to enhance renal excretion of lithium. Diuretics are contraindicated.
- We recommend hemodialysis for acute lithium concentrations greater than 5 mEq/L or for patients with signs of severe neurologic lithium toxicity, particularly tremor, clonus, altered mental status, and seizure, regardless of the serum concentration.

1. What are the two most common electrolyte abnormalities seen in chronic lithium toxicity?

a. Hyperkalemia and hypernatremia
b. Hypernatremia and hypercalcemia
c. Hypokalemia and hypercalcemia
d. Hypomagnesemia and hyponatremia
e. Hyponatremia and hypercalcemia

Answer: b. Chronic lithium use can cause hypernatremia due to the development of diabetes insipidus. Prolonged therapy with lithium alters cellular calcium sensing and leads to elevated parathyroid hormone levels and hypercalcemia. Magnesium and potassium levels are typically normal, and abnormalities are not caused by lithium. Hyponatremia and hypercalcemia are frequently seen in cases of acute lithium overdose.

2. Which of the following conditions is most frequently seen in patients with chronic lithium use?

a. Anticholinergic syndrome
b. Diabetes insipidus
c. Hyperthyroidism
d. Hypoparathyroidism
e. Syndrome of inappropriate antidiuretic hormone

Answer: b. Diabetes insipidus commonly occurs in patients on chronic lithium therapy or with chronic overdose. Diabetes insipidus can cause dehydration and a further increase in lithium concentration and is a frequent contributory cause to chronic lithium toxicity. Rarely, hypothyroidism can also develop. Lithium use is also associated with neuroleptic malignant syndrome and serotonin syndrome but none of the other listed conditions.

3. Which of the following drugs should be used with caution in patients taking lithium?

a. Acetaminophen
b. Hydrochlorothiazide
c. Metformin
d. Metoprolol
e. Penicillin

Answer: b. Hydrochlorothiazide, other diuretics, angiotensin-converting enzyme (ACE) inhibitors, and nonsteroidal antiinflammatory drugs (NSAIDs) can increase lithium levels and consequently cause chronic lithium toxicity by interfering with renal elimination. Lithium has also been implicated in serotonin syndrome when combined with other serotonergic drugs, such as monoamine oxidase (MAO) inhibitors, selective serotonin reuptake inhibitors (SSRIs), and dextromethorphan. None of the other drugs listed causes significant interactions with lithium.

4. A 26-year-old woman presents following a witnessed seizure. She was found with an empty bottle of lithium and a suicide note. Her vital signs reveal hypotension and tachycardia. She is lethargic and her only response is to withdraw from painful stimuli. The remainder of her physical examination is normal. Laboratory tests including a serum lithium level are pending. Which of the following treatments is the most appropriate?

a. Activated charcoal
b. Forced diuresis
c. Hemodialysis
d. Urinary alkalinization
e. Whole bowel irrigation (WBI)

Answer: c. Hemodialysis is the most effective way to remove lithium. It can remove lithium at a rate five to seven times the rate of typical renal elimination. Common indications for dialysis include decreased level of consciousness and seizures. Activated charcoal does not adsorb lithium. Lithium overdose patients are often dehydrated and occasionally hypernatremic and should be fluid resuscitated, but once dehydration is corrected, forced diuresis is of no benefit and can be detrimental. Urinary alkalinization has no role in lithium toxicity. Because this patient has already experienced a seizure, the administration of WBI is not advised. In addition, the seizure suggests that the brain concentration of lithium is already toxic and this is not improved by WBI.

KEY CONCEPTS

- Antipsychotics are commonly categorized into typical, or first-generation antipsychotics (FGAs) with primary antagonism to dopamine receptors, and atypical, or second-generation antipsychotics (SGAs) which include serotonin receptors as a target. Aripiprazole is an example of a third type of antipsychotic that acts as a partial agonist at dopamine sites.
- Extrapyramidal symptoms are common side effects of antipsychotics. First line treatment is benztropine or diphenhydramine. Lorazepam may be used in refractory cases.
- The most common presentation of an antipsychotic overdose is central nervous system (CNS) depression. Treatment focuses on supportive care, airway management, and cardiac monitoring.
- QT prolongation and *torsades de pointes* are potential complications of antipsychotic overdose but may also occur with therapeutic use.
- Clozapine is associated with potentially life-threatening agranulocytosis. Treatment includes cessation of the medication, treating potential infections, and supportive care.
- Neuroleptic malignant syndrome (NMS) is characterized by altered mental status, hyperthermia, muscle rigidity, and autonomic instability. Treatment includes supportive care with airway management, benzodiazepines, muscular rigidity management, and evaporative cooling methods for hyperthermia.

1. Which of the following is a second-generation atypical antipsychotic?
   a. Aripiprazole
   b. Haloperidol
   c. Lumateperone
   d. Olanzapine

Answer: d. Olanzapine is an atypical second-generation antipsychotic. Aripiprazole is an atypical psychotic but would be considered a third generation. Haloperidol is a first-generation, high-potency antipsychotic. Lumateperone is a newer unique antipsychotic that affects multiple systems including serotonergic, dopaminergic, and glutamatergic.

2. A 19-year-old woman presents with urinary frequency, urgency, and dysuria. She had a urinalysis which demonstrated 20 to 40 white blood cells per high-power field. She has schizophrenia for which she takes haloperidol. You diagnose her with an uncomplicated urinary tract infection. Which of the following antibiotics could cause a life-threatening arrhythmia if administered to this patient?
   a. Amoxicillin/clavulanic acid
   b. Ciprofloxacin
   c. Cephalexin
   d. Nitrofurantoin

Answer: b. Antipsychotics can cause QT prolongation, so other drugs that cause QT prolongation should be avoided. Macrolides, fluoroquinolones, and trimethoprim-sulfamethoxazole are common antibiotics that can all cause QT prolongation.

3. A 32-year-old man presents complaining of neck pain. His neck is turned to the right and he states that he has difficulty moving it. He denies any trauma and has not had a fever. He has no previous similar episodes. He has schizophrenia but cannot recall the names of his medications. His vital signs are normal and his examination reveals a palpable spasm of the right trapezius and sternocleidomastoid muscles. What is the most appropriate treatment?
   a. Bromocriptine
   b. Cyclobenzaprine
   c. Diphenhydramine
   d. Morphine

Answer: c. This patient has an acute dystonic reaction to an antipsychotic medication best treated with an anticholinergic medication, such as diphenhydramine or benztropine. Bromocriptine is a dopamine agonist used to treat pituitary disorders and can worsen psychosis. Cyclobenzaprine can be used for typical muscle spasms but will not significantly improve acute dystonia. Morphine may help but does not resolve the underlying problem.

4. A 52-year-old man presents to the emergency department with a cough. He is tachycardic with a temperature of 38.6°C. His chest x-ray has a left lower lobe infiltrate. His lab work is significant for a leukopenia with an absolute neutrophil count of 250 cells/μL. His past medical history is significant for psychiatric illness. Which of the following medications is he most likely to be taking?
   a. Clonidine
   b. Clozapine
   c. Fluoxetine
   d. Lithium

Answer: b. Clozapine is an antipsychotic associated with agranulocytosis and neutropenia. Clonidine is an alpha-adrenergic agonist with no known association with this condition. Fluoxetine is a selective serotonin reuptake inhibitor commonly used in the treatment of depression and is not known to cause neutropenia. Lithium is used as a mood stabilizing medication such as in bipolar disorder and not known to cause an alteration in cell counts.

5. A 23-year-old woman is brought to the emergency department by her boyfriend with a change in mental status. He reports that the patient had her psychiatric medications adjusted 3 days ago and has been confused since then. The patient's vital signs are: blood pressure: 162/100 mm Hg; heart rate: 140 beats/min; respiratory rate: 22 breaths per minute; and temperature: 40.2°C. On physical examination, the patient is noted to have muscle rigidity. Laboratory tests are pending. Intravenous (IV) access is obtained, and balanced crystalloid fluids are started. Cool mist and fans are applied to the patient. What therapy is indicated next?
   a. Acetaminophen
   b. Cyproheptadine
   c. Dantrolene
   d. Lorazepam

Answer: d. The patient has neuroleptic malignant syndrome (NMS) and the best first line treatment option is a benzodiazepine, such as lorazepam. Acetaminophen is ineffective in treating the hyperthermia, which should be treated with active cooling measures. Cyproheptadine can be used in serotonin syndrome but does not improve NMS. Dantrolene can be used in malignant hyperthermia and for refractory, severe muscle rigidity but has no mechanistic indication in NMS, and evidence for its use is of low quality. Diphenhydramine is used for acute dystonia and does not affect the course of NMS.

151 Opioids

KEY CONCEPTS

- The opioid toxidrome includes three prominent findings—central nervous system depression, miosis, and, most importantly, respiratory depression—but presentations may be variable.
- A negative urine screen is often unreliable, and absence of detection should not deter a diagnosis of opioid intoxication when clinical findings support it.
- Airway protection, oxygenation, ventilation, and early administration of the reversal agent naloxone are the cornerstones for management of patients with opioid toxicity.
- The duration of action of many opioids, especially after overdose, is significantly longer than that of naloxone. Patients responsive to naloxone should be observed for recurrence of respiratory depression because they may require additional doses of naloxone.
- Naloxone distribution, prescription drug monitoring programs, and initiation of buprenorphine with a referral to addiction treatment programs are ways in which the medical profession is trying to combat the epidemic of opioid-related deaths.
- Opioid withdrawal syndrome does not include altered cognition. Patients with known or suspected opioid withdrawal who also have altered mental status should be evaluated for an alternative cause of altered cognition.

1. A 26-year-old female is brought to the emergency department (ED) from the local airport by law enforcement. She is sleepy, with small pupils, and mumbles incoherently in a foreign language. Vital signs include the following: blood pressure, 104/66; respiratory rate, 14 breaths/min; and temperature, 98.6°F. Which of the following tests might identify the cause of this patient's symptoms?

a. Abdominal radiograph
b. Electrocardiogram (ECG)
c. Electroencephalogram (EEG)
d. Head computed tomography (CT)
e. Urine drug screen

Answer: a. An abdominal radiograph would likely reveal multiple packets of illicit opioid in the gastrointestinal tract of this body packer. One or more of the packets has leaked, producing the opioid toxicity. A urine drug screen may not identify an opioid but would not identify the internal packets. A head computed tomography (CT) scan would not be helpful unless associated head trauma is suspected. An EEG and ECG would not provide specific information to identify the internal packets.

CHAPTER 151: QUESTIONS AND ANSWERS

2. Most opioids cause mild hypotension related to histamine release and bradycardia. Which of the following opioids can also cause sodium channel blockade and QRS widening?
   a. Hydromorphone
   b. Hydrocodone
   c. Morphine
   d. Oxycodone
   e. Propoxyphene

Answer: e. Propoxyphene and its metabolite norpropoxyphene can cause QRS widening. None of the other listed opioids has significant effects on the cardiac conduction system.

3. Which of the following laboratory abnormalities is most commonly seen in opioid overdose?
   a. Hypocalcemia
   b. Hypochloremia
   c. Hypoglycemia
   d. Hypokalemia
   e. Hyponatremia

Answer: c. Hypoglycemia is the only consistent laboratory abnormality found in opioid overdose. It is generally mild but can contribute to the decreased level of consciousness seen in opioid overdose.

4. A 32-year-old man presents with confusion, nausea, vomiting, diarrhea, and abdominal pain. His friends report that he is withdrawing from heroin. Vital signs reveal mild hypertension, tachycardia, and tachypnea. Physical examination is significant for confusion, mydriasis, diaphoresis, lacrimation, piloerection, and mild diffuse abdominal tenderness. Which of the following signs and symptoms makes you concerned that this may not be a simple opioid withdrawal case?
   a. Confusion
   b. Diarrhea
   c. Mydriasis
   d. Piloerection
   e. Tachycardia

Answer: a. Opioid withdrawal almost always causes restlessness, agitation, and anxiety. Cognition and mental status are not affected in simple opioid withdrawal and, if present, should prompt the clinician to search for other causes instead of, or in addition to, withdrawal.

5. A 20-year-old woman is brought to the emergency department (ED) after being found with decreased mental status at a club. Vital signs indicate mild hypotension and bradycardia. She is drowsy but arousable, and she has an otherwise normal physical examination. Upon receiving naloxone 2 mg IV, her mental status immediately improves and soon thereafter she vomits. She now reports nausea but has no other complaints. She states that she took some "pain pills" to get high but does not know what they were. What diagnostic tests should be performed?
   a. Acetaminophen
   b. Arterial blood gas
   c. Chest radiograph
   d. Lactate
   e. Urine drug screen

Answer: a. Because many prescription opioid medications are combinations of an opioid and acetaminophen, ibuprofen, or salicylate, concentrations of acetaminophen and salicylate should also be ordered. Acetaminophen overdose might otherwise remain undiagnosed but, if identified, can be treated with an existing antidote, *N* acetylcysteine. The chest radiograph is not indicated unless a pulmonary complication is suspected. Lactate and a urine drug screen would not change patient management.

6. Which of the following medications can be used to treat opioid withdrawal?
   a. Clonidine
   b. Dextromethorphan
   c. Diphenhydramine
   d. Nalmefene
   e. Valproic acid

Answer: a. Clonidine suppresses the sympathetic hyperactivity of opioid withdrawal. Dextromethorphan is an opioid derivative used as a cough suppressant, but it does not treat the symptoms of opioid withdrawal. Diphenhydramine is not the most commonly used antihistamine for a patient in withdrawal. Valproic acid has no role in opioid withdrawal. Nalmefene is an opioid antagonist similar to naloxone but with a longer duration of action. Administration of nalmefene is longer-acting than naloxone and would worsen opioid withdrawal symptoms.

7. A 14-month-old child is brought to the emergency department (ED) 4 hours after he was found with his grandmother's antidiarrheal medication bottle. A pill count identifies that only one Lomotil tablet is missing. The child is playful, has a normal respiratory rate and pattern, and has a soft abdomen with normal bowel sounds. Appropriate management includes which of the following?
   a. Administration of activated charcoal
   b. Administration of naloxone
   c. Admission to a monitored unit
   d. Discharge home
   e. Gastric lavage

Answer: c. Activated charcoal and gastric lavage are means of gastrointestinal decontamination and are not routinely recommended in opioid toxicity. Opioid intoxicated patients with central nervous system (CNS) and respiratory depression should be treated with naloxone, but asymptomatic patients do not require antidote administration. Asymptomatic patients with known or suspected Lomotil (diphenoxylate/atropine) ingestion should be observed in a monitored setting for delayed onset of toxicity from the metabolite of diphenoxylate.

8. An 18-year-old male is driven to the emergency department (ED) by friends and carried into the triage area. He has agonal respirations and is cyanotic. Immediate resuscitative measures include bag-valve-mask (BVM) ventilation, establishment of an intravenous (IV) line, and administration of 0.4 mg of naloxone. The patient's respiratory status improves and, although sleepy, he is able to answer some questions. During subsequent monitoring, the patient's respiratory status again declines, and he requires two additional doses of naloxone. Additional treatment should include which of the following?
   a. Nalmefene
   b. Naloxone infusion
   c. Hemodialysis
   d. Naltrexone
   e. Narcan nasal spray

Answer: b. This patient likely has toxicity from a long-acting opioid agent, and a continuous infusion of naloxone will be necessary for ongoing reversal of toxicity. Nalmefene and Naltrexone are a longer-acting opioid antagonists but are not preferred over naloxone infusion because naloxone allows for dose titration. Once multiple doses of IV narcan are given, there is no role for narcan nasal spray. Opioids are not dialyzable due to large volumes of distribution.

152 Pesticides

KEY CONCEPTS

- Organophosphates cause symptoms by accumulation of acetylcholine.
  - Treat cholinergic symptoms with atropine.
  - Reverse the inhibition of acetylcholinesterase with oximes.
- Aging, which results in prolonged toxicity, occurs with organophosphate poisoning, but not with carbamates.
- Chlorinated hydrocarbons can present with seizures and cardiac toxicity.
- Substituted phenols are found in weight loss products and exert their toxicity by uncoupling oxidative phosphorylation. These products can cause cardiac, liver, and renal injury.
- Chlorophenoxy compounds cause muscular injury. Measure creatinine kinase; assess for acute rhabdomyolysis, kidney injury, and liver injury.
- Bipyridyl compounds (paraquat and diquat) cause pulmonary and renal injury.
  - Paraquat concentrates in lungs; supplemental oxygen therapy should be limited, since this will exacerbate pulmonary toxicity.
  - Diquat causes renal injury.
- Pyrethrins and pyrethroids cause local dermatologic symptoms.
- Acute toxicity with glyphosate is likely related to the surfactant included in the product.
- DEET (N, N-Diethyl-meta-toluamide) should not be used in infants younger than 2 months old. DEET in concentrations of more than 30% should not be used in children and may result in neurotoxicity and self-limited seizure activity if used in excessive amounts.
- Most rodenticide exposures will be superwarfarin compounds.
  - For large exposures, International Normalized Ratio (INR) should be checked at a minimum of 2 days after ingestion.
  - Vitamin K should be used for reversal; blood products should be used for active bleeding.

1. A 37-year-old man arrives at the emergency department (ED) after exposure to an organophosphate. He is severely symptomatic, and atropine is given. When should atropine treatment be discontinued?

a. After 100 mg has been given
b. When fasciculations stop
c. When mydriasis occurs
d. When secretions have stopped
e. When tachycardia occurs

Answer: d. Patients with organophosphate poisoning may require very large doses of atropine (up to 500 mg). Proper dosing is 1 or 2 mg intravenously, with doubling of the dose every 5 minutes until the drying of secretions. Atropine has no effect at the neuromuscular junction. Mydriasis may occur before secretions have dried. Tachycardia is likely, but it is not a contraindication for continued atropine treatment. The tachycardia often improves as the pulmonary status improves.

2. A 30-year-old agricultural worker arrives at the emergency department (ED) with confusion, abdominal pain, nausea, vomiting, and shortness of breath. His coworkers report that he was spraying plants with an insecticide. The patient's vital signs are significant for hypotension, bradycardia, and tachypnea. Physical examination reveals miosis, wheezing, vomiting, diarrhea, and urinary incontinence. Which clinical finding is associated with higher mortality?

a. Hyperglycemia
b. Hypoglycemia
c. Glasgow Coma Scale (GCS) ≤3
d. Acidosis
e. Poison Severity Score (PSS) 4

Answer: d. This patient has symptoms of the cholinergic syndrome (SLUDGE symptoms) and was likely exposed to an organophosphate or carbamate pesticide. Patients poisoned with organophosphate or carbamate pesticides that are acidotic have a higher mortality than those with normal acid-base status. Elevated and decreased glucose concentrations have both been associated with increased morbidity and mortality. As for PSS and GCS, these vary by type of poisoning and cannot reliably be used to determine risk of fatality.

3. A 4-year-old boy arrives at the emergency department (ED) after drinking a medication that was being used to treat his sister's head lice. An unknown amount was consumed, and the bottle is not available. The patient's only complaint is of nausea. Vital signs and physical examination findings are normal. Which of the following symptoms should be anticipated?

a. Gastrointestinal hemorrhage
b. Hallucinations
c. Hypotension
d. Paralysis
e. Seizure

Answer: e. Lindane is a chlorinated hydrocarbon insecticide that is used for the topical treatment of head lice and scabies. It is rapidly absorbed and can result in difficult-to-control seizures requiring high doses of benzodiazepines or barbiturates and that may require sedation, paralysis, and intubation. Because lindane is a hydrocarbon, it can sensitize the cardiac membrane and predispose to ventricular dysrhythmias. It can also cause pulmonary compromise if it is aspirated.

4. An 18-year-old woman arrives at the emergency department (ED) complaining of feeling generally weak. She reports reading online about a pesticide that can help in weight loss, and she has recently tried this. Her vital signs are blood pressure, 110/70 mm Hg; heart rate, 121 bpm; respiratory rate, 22 rpm; and temperature, 104°F (40.0°C). Her physical examination reveals dry mucous membranes and yellow staining on her abdomen. Which of the following laboratory findings can you anticipate?

a. Hypoglycemia
b. Hypokalemia
c. Hyponatremia
d. Hypoxia
e. Methemoglobinemia

Answer: a. Substituted phenols, such as dinitrophenol, are pesticides that uncouple oxidative phosphorylation. This causes an increased metabolism, which in turn consumes glucose and generates heat, often causing an increased body temperature. For this reason, they have been used as diet aids. They can be applied topically, and the yellow skin staining is considered pathognomonic for nitrogen compounds. Treatment is supportive and should be aimed at stopping further exposure; providing the needed substrates of oxygen, water, and glucose; and applying active cooling measures.

5. What organ system is most affected by paraquat ingestion?

a. Cardiac
b. Gastrointestinal
c. Nervous
d. Pulmonary
e. Renal

Answer: d. Paraquat is concentrated in the lungs and directly damages the alveolar capillary membrane. This results in surfactant loss, adult respiratory distress syndrome, pulmonary fibrosis, and respiratory failure and is accelerated with supplemental oxygen. All the other organ systems listed are affected but to a much lesser degree.

6. A patient arrives at the emergency department (ED) after an intentional paraquat ingestion complaining of severe mouth, throat, and chest pain. What potentially fatal complication of paraquat ingestion should be suspected?

a. Aortic dissection
b. Esophageal rupture
c. Myocardial infarction
d. Pneumothorax
e. Pulmonary embolism

Answer: b. Paraquat is extremely corrosive and can cause severe burns to the oropharynx, as well as to the esophagus. Frequently, esophageal rupture occurs, leading to mediastinitis and death. None of the other listed conditions occurs with any frequency in paraquat poisoning.

7. How do pyrethrins cause toxicity?

a. Sensitivity or allergic reactions
b. Bone marrow suppression
c. Cardiovascular instability
d. Inhibition of coagulation
e. Uncoupling of oxidative phosphorylation

Answer: a. Pyrethrins are naturally occurring substances from the yellow chrysanthemum and commonly cause allergic reactions in humans. These reactions can be mild or life-threatening, with bronchoconstriction and laryngeal edema. They also affect gamma-aminobutyric acid (GABA)-mediated chloride channels in the nervous system, but this typically results in only a mild headache and paresthesias.

8. DEET is a commonly used insect repellant that can cause contact dermatitis and more severe neurologic complications, including seizures with high doses that are absorbed through the skin. Which of the following methods will minimize DEET absorption through the skin?

a. Apply to skin at night
b. Cover skin with clothing
c. Expose skin to direct sunlight
d. Keep skin dry
e. Remove from skin with oil-based products

Answer: d. DEET absorption and toxicity increase with repeated applications, with increased ambient temperatures, with sweating, when it is applied to abraded or thin skin, and when it is covered with tight-fitting clothing. Oils or lipophilic substances applied to the skin also increase absorption of DEET.

9. A patient has long-standing depression that was exacerbated by the recent COVID-19 pandemic. During the isolation of the pandemic, the patient was cleaning their attic and found an old box of rat poison. During a particularly challenging time, the patient attempted suicide by consuming the contents of the rat poison container. The patient woke up the next day and was asymptomatic. Two days later, the patient noted bruising and hematuria. Which class of pesticides likely represents the overdose in this situation?

a. Bipyridal compounds
b. Organophosphates
c. Pyrethrins
d. Superwarfarins
e. Chlorophenoxy compounds

Answer: d. Superwarfarins were developed from warfarin for rodenticides, after increasing numbers of rats developed resistance to warfarin. They act by inhibiting vitamin K, which is a cofactor in the synthesis of clotting factors II, VII, IX, and X. The therapeutic and toxic effects require significant reduction of these clotting factor concentrations, thus taking at least 15 hours to present. While warfarin has a half-life of 35 hours, the duration of action can be up to 5 days. Superwarfarins are 100 times more potent than warfarin, leading to increased and prolonged toxicity, which can manifest as superficial, as well as internal, bleeding (such as bruising and hematuria). Bipyridal compounds, such as paraquat and diquat, are pesticides that are not used in the United States. Organophosphates are insecticides that are highly regulated in the United States. Pyrethrins are insecticides derived from chrysanthemums, commonly used to treat scabies. Chlorophenoxy compounds are herbicides, and not used as rodenticides.

153 Plants, Herbal Medications, and Mushrooms

KEY CONCEPTS

- Most botanical exposures result in minimal toxicity and management is largely supportive.
- Most serious toxicities result from exposure to plants with anticholinergic, antimitotic, cardiovascular or convulsive properties.
- Most cases of mushroom ingestion in which gastrointestinal (GI) symptoms begin within the first 2 h will prove to involve a non–life-threatening substance.
- Delayed GI symptoms with an onset of more than 6–8 h after exposure suggest a potentially life-threatening ingestion, such as the cyclopeptide and gyromitrin mushroom groups.
- Regional poison control centers, mycologists, and botanists can assist in identifying potentially toxic plants and mushrooms. We recommend smart phone digital photography with expert consultation.
- Herbal medications are largely unregulated and may have inherent toxicity, herb-drug interactions, or contaminants. Clinicians should advise against the routine use of herbal medications.

1. A 52-year-old man presents after ingestion of water hemlock complaining of nausea but no other symptoms. He reports that he ate the water hemlock in a suicide attempt but has since changed his mind. His vital signs and physical examination are normal. What is the most appropriate initial treatment of this patient?
   a. General seizure precautions
   b. Hemodialysis
   c. Neutralization with milk
   d. Supportive care only
   e. Whole bowel irrigation

Answer: a. Water hemlock is very toxic, with fatality rates as high as 70%. Because this patient is at risk for seizures, oral activated charcoal and gastric lavage are contraindicated. Hemodialysis, whole bowel irrigation, and milk are not beneficial. Supportive care is not adequate with water hemlock ingestion. Most deaths are caused by intractable seizures, so this complication should be anticipated.

2. A family of four presents to the emergency department feeling ill after consuming a homemade stew containing Datura stramonium (Jimsonweed). Which of the following can be expected upon physical examination?
   a. Bradycardia
   b. Confusion
   c. Diaphoresis
   d. Miosis
   e. Respiratory depression

Answer: b. Jimsonweed contains atropine, hyoscyamine, and scopolamine, all of which have anticholinergic properties. Anticholinergic signs include confusion, dry skin and mucous membranes, mydriasis, tachycardia, urinary retention, confusion. In severe cases, physostigmine can be used to reverse effects.

3. Oleander contains cardiac glycosides similar to digoxin. Which of the following should be considered when treating oleander ingestion?
   a. Activated charcoal should be avoided.
   b. Serum digoxin levels correlate with degree of toxicity.
   c. Treatment is supportive.
   d. Treatment with digoxin-specific Fab fragments is beneficial, but a higher dose may be necessary.
   e. Treatment with digoxin-specific Fab fragments is not beneficial.

Answer: d. Treatment is with digoxin-specific Fab fragments, but much higher doses are necessary compared with those needed to treat digoxin toxicity. Activated charcoal is useful if the ingestion was recent. A serum digoxin level should be measured, but it is not a reliable indicator of level of toxicity. An elevated level confirms oleander ingestion, but a negative level cannot rule out ingestion.

4. A husband and wife present to the emergency department (ED) complaining of nausea and vomiting after eating mushrooms that they picked while hiking. They report eating the mushrooms on a salad approximately 1 hour ago. Each is complaining of severe nausea, vomiting, and diffuse abdominal pain. There is no hematemesis. Vitals signs and physical examination are normal. You inform them that they will be symptomatically treated and observed, but there is likely nothing to worry about. Why are you not concerned about their ingestion?
   a. Early onset of gastrointestinal symptoms
   b. No hematemesis
   c. No right upper quadrant tenderness
   d. Normal mental status
   e. Normal vital signs

Answer: a. The vast majority of severely toxic mushrooms have delayed onset of symptoms (>6 hours from ingestion). Occasionally, people will ingest two different types of mushrooms—one benign and causing early onset of symptoms and one more toxic. Therefore, patients should be observed and rechecked.

5. What is the most common cause of death after mushroom ingestion?
   a. Gastrointestinal hemorrhage
   b. Heart failure
   c. Liver failure
   d. Renal failure
   e. Respiratory failure

Answer: c. The *Amanita* species of mushrooms are very toxic and cause fulminant liver failure often necessitating liver transplant. Hepatotoxic mushrooms account for about 90% of all mushroom fatalities. Although certain mushrooms (*Cortinarius* spp. and *Amanita smithiana*) can cause renal failure, death is uncommon with these mushrooms. Mushrooms generally do not affect the other listed organ systems.

6. A 46-year-old man presents after waking in the night with headache and severe nausea and vomiting. For dinner the night before, he consumed some mushrooms that he had gathered from a nearby forest. He felt fine immediately after dinner. His vital signs are normal. While conducting your physical examination, he begins to seize. Benzodiazepines are administered without improvement. What treatment is indicated?
   a. Phenobarbital
   b. Phenytoin
   c. Propofol
   d. Pyridoxine
   e. Vecuronium

Answer: d. Most mushrooms do not cause seizures, but gyromitrin-containing mushrooms are an exception. They are commonly mistaken for edible morels because they look quite similar. They contain an isoniazid-like toxin that causes seizures. Traditional seizure medications can be used (because they are typically more readily available), but seizures can be intractable until pyridoxine is given.

7. A 54-year-old male with a history of depression presents to the emergency department with confusion. He is prescribed nortriptyline only, but his wife reports that he has recently started taking an herbal supplement. His vital signs are temperature 100.3°F, blood pressure 130/75, pulse 124, respiratory rate 20, oxygen saturation 96%, You note spontaneous clonus to his lower extremities and diagnose him with serotonin syndrome. Which of the following herbal supplements did he most likely start taking?
   a. Ephedra
   b. *Hypericum perforatum*
   c. *Gingko biloba*
   d. *Piper methisticum*
   e. Saw palmetto

Answer: b. *Hypericum peforatum* (St. John's wort) is a non-selective serotonin reuptake inhibitor and is thought to also upregulate post-synaptic serotonin receptors. Combination therapy with other anti-depressants that increase serotonin has been reported to cause serotonin syndrome. Ephedra is no longer commercially available as it was banned by the FDA for its association with cardiovascular disease. *Gingko biloba* may have antioxidant properties and inhibit platelet aggregation. *Piper methisticum* can increases GABA-mediated CNS depression when combined with alcohol. Saw palmetto is a remedy for benign prostatic hypertrophy and is thought to work by inhibiting 5-α-reductase.

154 Sedative-Hypnotics

KEY CONCEPTS

- Supportive care, with a focus on respiratory depression, is the foundation of management of all sedative-hypnotic ingestions.
- Benzodiazepines are commonly used medications both medically and recreationally. Coingestions with other sedative-hypnotic agents can potentiate their neurologic and respiratory effects.
- Barbiturate medications are less commonly prescribed, and intoxications are infrequent. Most patients recover with supportive care alone.
- A positive urine toxicology screen for benzodiazepines or barbiturates does not prove a causal linkage between the drug and the current clinical condition.
- We do not recommend routine use of flumazenil for benzodiazepine toxicity, particularly in chronic benzodiazepine users in whom flumazenil can precipitate seizures. Due to the short duration of action of flumazenil, when administered, patients should be monitored closely for recurrence of respiratory depression.
- Chloral hydrate toxicity may result in sedation and cardiotoxicity, principally in the form of supraventricular tachycardias, which are best treated with a short-acting beta blocker.
- Withdrawal from sedative-hypnotic use, including benzodiazepines, barbiturates and gamma hydroxybutyrate (GHB), can be life threatening. Management often requires high doses of benzodiazepines or barbiturates.

1. All of the following patients have depressed mental status, decreased respiratory drive, and mild hypotension. Supportive care has been initiated. All of the patients have a confirmed benzodiazepine overdose by a reliable historian. For which patient is it most reasonable to consider the use of flumazenil?

a. A 5-year-old boy with no known medical problems who took his mother's medication
b. A 22-year-old woman with no known medical problems
c. A 30-year-old man who is known to abuse cocaine
d. A 43-year-old man with depression and anxiety
e. A 79-year-old woman with congestive heart failure, coronary artery disease, and diabetes who takes benzodiazepines for insomnia

Answer: a. Flumazenil variably reverses the effects of benzodiazepines but can cause dysrhythmias and intractable seizures. Contraindications for flumazenil include history or suspected chronic benzodiazepine use (which are the vast majority of benzodiazepine overdose patients), coingestants that lower the seizure threshold, tricyclic antidepressants, cocaine, history of seizure disorder, chronic alcoholism, and head trauma. Its use should be for a known non-habituated patient (e.g., the pediatric patient described here). Because flumazenil's duration of action is short compared with the benzodiazepine and it has variability to reverse respiratory depression, patients require close, monitored observation, repeated dosing or continuous infusion, and preparations to manage the airway. Because benzodiazepine overdose is rarely lethal and patients recover with supportive care, using an antidote requires a careful benefit/risk analysis.

2. How do barbiturates affect peripheral and cerebral blood flow?
   a. Decrease peripheral pressure and have no effect on intracranial pressure
   b. Decrease peripheral pressure and increase intracranial pressure
   c. Decrease peripheral and intracranial pressure
   d. Increase peripheral and intracranial pressure
   e. Increase peripheral pressure and decrease intracranial pressure

Answer: c. Barbiturates cause direct cardiac depression with decreased cardiac output. In addition, they cause venous pooling and blunt the normal compensatory increase in system vascular resistance that would occur with decreased cardiac output. They also decrease cerebral blood flow and lower intracranial pressure. These effects are particularly pronounced in individuals with an already low cardiac output, such as in congestive heart failure or hypovolemic shock.

3. Other than general supportive care, which of the following is most useful in the management of a severe phenobarbital overdose with hemodynamic instability?
   a. Flumazenil
   b. Hemodialysis
   c. Multidose activated charcoal
   d. Urinary alkalinization
   e. Whole bowel irrigation

Answer: b. Although rarely indicated, hemodialysis is effective for acute phenobarbital toxicity (persistent serum levels over 100 mcg/mL) in the presence of prolonged coma, refractory hypotension, renal or cardiac failure, metabolic acidosis, or inadequate response to conventional measures and ventilatory support. These patients often require mechanical ventilation, so the patient's airway should be secured prior. Although there is evidence that multi-dose activated charcoal (MDAC) increases clearance of phenobarbital and may shorten duration of clinical toxicity, there is no evidence it results in improved outcomes over supportive care alone. Flumazenil will reverse the effect of benzodiazepines but has no effect on barbiturates. Urinary alkalinization has traditionally been used to "trap" the acidic barbiturates in the urine to increase excretion, However, a recent comprehensive review concluded that there is little clinical role for urine alkalinization in acute barbiturate poisoning. Whole bowel irrigation is not beneficial.

4. Which of the following sedative-hypnotics is most likely to cause fatal cardiac dysrhythmias in overdose?
   a. Buspirone
   b. Chloral hydrate
   c. Eszopiclone
   d. Flunitrazepam
   e. Zolpidem

Answer: b. Chloral hydrate sensitizes the myocardium to catecholamines and also independently induces dysrhythmias. Treatment of choice of dysrhythmias from chloral hydrate is beta-blockers. None of the other listed medications cause significant cardiac toxicity.

5. A 53-year-old woman presents with decreased mental status and mild hypotension. She is arousable to pain. Her husband reports that approximately 4 hours ago she had been complaining of insomnia and said she was going to take something to help her sleep. He then found her in her present state with an empty bottle of "sleeping pills" on the counter. He does not remember the name of the sleeping pills. Supportive care is initiated. Which of the following laboratory tests would be the most helpful to manage this patient?
   a. Acetaminophen level
   b. Arterial blood gas
   c. Serum chemistry
   d. Serum myoglobin
   e. Urine drug screen

Answer: a. Many over-the-counter (OTC) sleep aids contain acetaminophen or salicylates. The active ingredient in OTC sleep aids is an antihistamine, typically diphenhydramine or doxylamine. Both have anticholinergic effects in addition to causing sedation. Toxicity is typically mild and can be successfully managed with supportive care. It is rare that an antihistamine overdose results in rhabdomyolysis, and a serum creatine phosphokinase level, urinalysis, and urinary myoglobin level can be checked if there is concern. Myoglobin is cleared rapidly from the serum and is not a useful diagnostic test. Arterial blood gases, serum chemistries, and urine drug screens can be ordered if there is a complicated clinical course—but only rarely alter treatment.

6. A 21-year-old woman is brought to the emergency department (ED) by her friends after they found her unconscious in the bathroom of a local bar. The friends report that she had only had one or two alcoholic drinks. The patient's vital signs show bradycardia, hypotension, and bradypnea. Physical examination reveals short periods of apnea with hypoxia. She wakens briefly to noxious stimuli. You suspect that she may be suffering from gamma-hydroxybutyrate (GHB) poisoning. What is the most appropriate next step in management?
   a. Atropine
   b. Electrocardiogram
   c. Endotracheal intubation
   d. Flumazenil
   e. Naloxone

Answer: c. GHB frequently causes emesis and aspiration because patients are unable to protect their airways. Patients with significant GHB overdose should be intubated, but supportive care alone is all that is required for most GHB overdoses. The half-life is short, and many patients will return to baseline mental status in several hours. Bradycardia is common, usually mild, and does not typically require atropine, but this can be used if bradycardia is severe. GHB does not generally cause dysrhythmias. Neither flumazenil nor naloxone has any effect on GHB.

7. A 30-year-old man presents with anxiety and insomnia. He is agitated and has a tremor on physical examination. His vital signs are as follows: blood pressure, 200/124 mm Hg; heart rate, 123 beats/min; respiratory rate, 22 breaths/min; and temperature, 37.5°C. On further questioning, he reports that he chronically uses gamma-hydroxybutyrate (GHB) for muscle building but has not been able to take any for approximately 24 hours. Lorazepam is given in two escalating doses without response. Which of the following should be given next?
   a. A barbiturate
   b. Additional lorazepam
   c. Clonidine
   d. GHB
   e. Labetalol

Answer: a. GHB withdrawal is severe, and deaths have been reported. GHB acts as a central nervous system (CNS) depressant, so withdrawal causes CNS excitement and autonomic instability. Chronic GHB abuse can lead to GABA depletion in the CNS. Because benzodiazepines require the presence of GABA, they may not be effective. Barbiturates act independently of GABA and are therefore more effective in GHB withdrawal. Clonidine can be used in opioid withdrawal but is not useful in GHB withdrawal. Labetalol would improve the vital signs but would have no effect on the underlying cause of the autonomic instability.

155 Care of the Pediatric Patient

KEY CONCEPTS

- Patterns of illness and injury vary by age, and a number of anatomic and physiologic characteristics affect the presentation and management of pediatric emergencies.
- A basic understanding of normal development will aid the emergency clinician in assessment of the pediatric patient.
- The pediatric assessment triangle (PAT) can be used as a tool for rapid evaluation of the patient's overall status.
- Tachypnea in children should be evaluated relative to age norms and is often a sign of increased metabolic demands. A child with tachypnea despite normothermia should be evaluated for respiratory and nonrespiratory causes (e.g., hypoperfusion, acidemia).
- Maintenance of a neutral thermal environment is necessary for critically ill infants.
- Child abuse should be considered when injuries are inconsistent with history, when details of the history change, or with certain injury patterns.
- A joint guideline of the American College of Emergency Physicians, American Academy of Pediatrics, and Emergency Nurses Association summarizes the role of pediatric emergency care coordinators, development of pediatric policies, and recommended equipment, supplies, and medications for emergency departments (EDs).
- EDs that comply with the national guidelines for pediatric readiness have improved outcomes for care of critically ill children.
- The family's presence should be encouraged for pediatric procedures and resuscitations.
- A variety of pharmacologic and nonpharmacologic techniques are available to decrease procedural pain and anxiety.

1. Which of the following respiratory signs is characteristic of a child in compensated or decompensated shock?
   a. Grunting with abnormal breath sounds
   b. Tachypnea with nasal flaring
   c. Tachypnea with clear breath sounds on auscultation
   d. Tripoding with stridor
   e. Wheezing with retractions

   Answer: c. Effortless tachypnea, or rapid respirations with clear breath sounds, is characteristic of a child in compensated or decompensated shock. Grunting, nasal flaring, tripoding, stridor, wheezing, and retractions are all signs of increased work of breathing and would be more indicative of pulmonary or airway disease.

2. As the pediatric physician coordinator for your emergency department, you decide to institute new policies to prevent pediatric medication errors. Which of the following strategies will be most effective in decreasing risk of dosing errors?
   a. Pharmacy review of medication orders
   b. Use of length-based resuscitation tapes
   c. Use of resuscitation calculators
   d. Weighing and recording the weight in kilograms

   Answer: d. Weights should be measured and recorded in kilograms, not pounds, to avoid inadvertent dosing calculations using pounds instead of kilograms. The use of length-based resuscitation tapes and resuscitation calculators and pharmacy review of medication orders have also been suggested to reduce medication dosing errors, but the most effective measure is to weigh and record the weight in kilograms.

3. You have written an order for a blood draw and placement of an intravenous line in a nervous 3-year-old boy. Which of the following is least likely to be helpful in decreasing the patient's procedure-related distress?
   a. Application of a lidocaine-epinephrine-tetracaine mixture
   b. Having the patient blow bubbles prior to the needle stick
   c. Needle-free jet injection of local anesthetic
   d. Use of a vapocoolant

   Answer: a. The formulation of lidocaine-epinephrine-tetracaine only works on broken skin (i.e., lacerations). Unlike eutectic mixture of local anesthetic (EMLA) and 4% liposomal lidocaine preparations which are effective on intact skin, it will not work on intact skin. The use of vapocoolants and a needle-free jet infection of local anesthetic may decrease the patient's pain. Distraction techniques such as blowing bubbles, singing a song, or watching a video may relieve procedure-related anxiety.

KEY CONCEPTS

- Pediatric advanced airway management is a relatively rare skill to perform in most emergency departments (EDs), and skill maintenance is difficult based solely on clinical practice.
- There are several anatomic differences that impact pediatric airway management, and these occur mostly in the very young child (<2 years of age). Infants have a large occiput and a high, anterior airway, which impacts positioning during intubation. The narrowest portion of the pediatric airway is at the level of the cricothyroid membrane which means a foreign body could be lodged below the cords. They are also more dependent on diaphragmatic excursion for ventilation, thus gastric insufflation can result in difficulty with rescue ventilation.
- Children are prone to desaturation due to their high metabolic rate and their lungs' small functional residual capacity, making preoxygenation and maintenance of oxygenation during intubation attempts crucial.
- The cognitive burden inherent in dealing with the large age/size spectrum in pediatrics can be overcome with reference aids that organize equipment selection and drug dosing based on length/age/size. Formulas have been developed to aid in selection of the correct endotracheal tube (ETT) size and determine appropriate depth of ETT insertion. For estimation of uncuffed tube sizes in children older than 1 year old: ETT size = 4 + (age in years/4). Subtract 0.5 in size for cuffed tubes. To estimate the depth of ETT insertion (the so-called "lip to tip" distance), multiply the ETT size × 3 (e.g., a 5.0 ETT would be inserted to 15 cm at the lip).
- Rapid sequence intubation (RSI) is the preferred method of airway management in the vast majority of pediatric cases in the ED.
- Compared to adults, children are more prone to desaturation over the time it takes for a neuromuscular blocking agent (NMBA) to take effect. Use of high-flow nasal cannula during the apneic period of RSI has not been well studied in children in the emergency setting, but we recommend its use at 1 to 2 L/min/year of age to a maximum of 15 Lpm. Because children desaturate more rapidly than adults, we recommend that assisted ventilation (coordinated with the child's respiratory efforts if not yet fully paralyzed) be initiated if oxygen saturation drops below 95%.
- Video laryngoscopy is an evolving technology for use in pediatrics and assists in visualization of the airway but may prolong time to intubation.
- Surgical airway techniques differ in infants and young children, necessitating a needle technique that is different from the older child or adult. This technique provides a mechanism to oxygenate the "can't intubate, can't ventilate" child, but should not be relied on as a definitive airway.

1. Which of the following is a correct formula for use in airway management of children?
   - **a.** Cuffed ETT size (1 to 10 years old) = 4 + (age in years/2)
   - **b.** ETT depth = 3 × ETT size
   - **c.** ETT depth = 4 × ETT size
   - **d.** Uncuffed ETT size (1 to 10 years old) = (16/age in years) + 4

Answer: B. Correct formulas are as follows:
Uncuffed ETT size = 4 + (age in years/4)
Cuffed ETT size = subtract 0.5 from formula for uncuffed ETT; (4 + (age in years/4)) − 0.5
ETT depth = 3 × uncuffed tube size

2. Which of the following should be used for proper positioning of the pediatric patient during endotracheal intubation?
   - **a.** The chin should be tilted and the head lifted.
   - **b.** The infant younger than 6 months old should have a towel roll placed under the occiput to align the airway axes.
   - **c.** The neck should be placed in slight flexion and the shoulders extended.
   - **d.** The patient should be positioned so that a line drawn through the external auditory canal and the anterior shoulder is parallel to the bed.

Answer: D. The relatively large head and occiput of the infant results in slight flexion at the neck when supine, impeding the ability to visualize the glottis. The infant is correctly positioned so that a line drawn through the external auditory canal and the anterior shoulder is horizontal and parallel to the bed. In the infant (<6 months old), this is accomplished by placing a towel roll under the patient's shoulders, elevating the body and overcoming the flexion associated with their large occiput. A head tilt–chin lift may open the airway.

3. Relative to rapid sequence intubation (RSI) in adults, children undergoing RSI:
   - **a.** Are less affected by stomach pressures.
   - **b.** Are more likely to develop bradycardia in response to hypoxia.
   - **c.** Have less missed intubations.
   - **d.** Have less pliable airways.
   - **e.** Maintain their oxygen saturations longer after paralytics are administered.

Answer: B. Children have a higher metabolic rate and thus a higher oxygen demand. In the absence of respiratory support, children will drop their oxygen saturations more quickly than adults due to their relatively greater consumption of oxygen. Hypoxia is the most common reason for bradycardia during intubation in children. The more pliable pediatric thorax combined with children's lower ventilatory reserve make stominflation a significant impediment to effective respiratory support. Pediatric intubations are infrequent and multiple pediatric intubation attempts are more common in children than in adults.

4. Which of the following does *not* characterize the pediatric airway relative to adults?
   - **a.** A straight blade is used to pick up the floppy epiglottis.
   - **b.** Infants may require a shoulder roll to align airway axes.
   - **c.** The airway is more pliable.
   - **d.** The airway is more superior and posterior.
   - **e.** The epiglottis is larger and omega-shaped.

Answer: D. Small infants have a large occiput and a high, anterior airway, which impacts positioning during intubation, often requiring a shoulder roll to align airway axes. The airway is also more pliable and the membranes thinner, necessitating needle cricothyroidotomy for surgical airways in children younger than 10 years old. Blade choice is age dependent, and a straight blade is recommended in young children; this blade lifts the relatively larger epiglottis to visualize the vocal cords.

5. Following the administration of succinylcholine in a 7-year-old boy with respiratory distress, severe masseter spasm is noted. Which of the following medications should be administered to terminate this spasm?
   - **a.** Diazepam
   - **b.** Fentanyl
   - **c.** Repeated dose of succinylcholine
   - **d.** Rocuronium
   - **e.** Thiopental

Answer: D. Masseter muscle spasm is a rare side effect of succinylcholine administration. It primarily occurs in pediatric patients and is typically terminated by the administration of a competitive neuromuscular blocking agent (NMBA), such as rocuronium, vecuronium, or pancuronium. Failure to respond to such therapy should prompt consideration of malignant hyperthermia.

157 Pediatric Sedation and Analgesia

KEY CONCEPTS

- Patients of all ages experience pain, including infants, neonates, and premature babies.
- Oligoanalgesia, the inadequate treatment of pain, has many short-term and long-term consequences: worse patient outcomes, increase in patient's pain threshold, and development of chronic pain.
- Pain management may include a combination of techniques: analgesics, topical anesthetics, local anesthetic injections, oral sucrose in infants, and nonpharmacologic interventions.
- Nonpharmacologic interventions to decrease pain or anxiety include parental presence; physical measures, such as heat or cold therapy and splinting for musculoskeletal injuries; and behavioral or cognitive measures, such as distraction and play therapy.
- Topical anesthetics are recommended to decrease the pain of minor procedures, such as venipuncture or IV cannulation.
- Techniques for decreasing the pain of intradermal injections include topical agent prior to the intradermal injection; slowly injecting warmed, buffered local anesthetic solution from within the wound with the smallest gauge needle possible; and limiting the number of needle punctures.
- When using large amounts of local anesthetics in small children or infants, calculate the drug dose to avoid toxicity; a 1% solution = 1 g/100 mL or 10 mg/mL.
- Procedural sedation and analgesia (PSA) requires pre-sedation evaluation; sufficient monitoring (during and after the procedure) by qualified individuals capable of dealing with any adverse events that may occur; age-appropriate equipment (including airway equipment) and medications (including reversal agents and advance life support drugs); and discharge criteria for when the patient is fully awake, returns to baseline with normal vital signs, and is able to be discharged in the care of a responsible adult.
- Overall, preprocedural fasting is not necessary for most emergency patients, because large studies show no clinically significant differences with airway complications, emesis, or other adverse effects between groups of patients stratified by their preprocedural fasting status.
- Choice of sedative and analgesic for PSA depends on many variables including patient factors and the procedure to be done. Slow titration of medications can achieve the desired level of sedation and analgesia while minimizing risk of adverse events.

1. When do most adverse events associated with emergency department (ED) procedural sedation occur?

a. During the manipulation or intervention
b. 5 to 20 minutes after the last sedative dose
c. 20 to 30 minutes after the last sedative dose
d. 30 to 60 minutes after the last sedative dose
e. 60 to 90 minutes after the last sedative dose

Answer: b. High-risk times are 5 to 20 minutes after the last medication administration and at the completion of the procedure when there is no longer a painful stimulus, but the patient remains sedated.

2. Which of the following modalities has proven most effective for monitoring patients undergoing procedural sedation?

a. Capnometry or capnography
b. Cardiac rhythm monitoring
c. Continual direct visual observation of qualitative clinical signs
d. Documented respiratory rate
e. Pulse oximetry

Answer: c. The patient's ability to follow commands in response to varied levels of stimulation and direct observation of the ventilatory status have been the most reliably documented methods of assessing the level of consciousness during procedural sedation. Pulse oximetry is a reliable adjunct, but it identifies hypoventilation late, especially when used with supplemental oxygen. Cardiac monitoring has been shown to be helpful in older patients or in those with a history of cardiac disease, but there is no evidence that it is of any benefit in young healthy patients. End-tidal carbon dioxide monitoring has been shown to be useful to detect inadequate ventilation earlier than oximetry, especially when direct observation of the patient is difficult, but no studies have demonstrated an effect on clinical outcome to date. Recently, the American Society of Anesthesiologists (ASA) updated its procedural sedation standards to include capnography during moderate or deep sedation, in addition to the continual observation of qualitative clinical signs. Respiratory rate alone is an insensitive indicator of adequacy of ventilation.

3. Which of the following agents is matched with the correct associated side effect?
   a. Etomidate—longer (>30-minute) duration of sedation
   b. Ketamine—laryngospasm
   c. Methohexital—venoirritation
   d. Pentobarbital—seizures
   e. Propofol—myoclonus

Answer: b. Ketamine has been associated with laryngospasm in children younger than 3 months old and those with a respiratory infection. The following are the other correct associations:

- Etomidate—myoclonus
- Methohexital—seizures
- Pentobarbital—longer (>30-minute) duration
- Propofol—venoirritation

4. Which of the following statements regarding the use of ketamine is false?
   a. Benzodiazepine administration may be useful for emergence phenomenon in children that are not improved by removing stimulation and providing calming interventions.
   b. Despite increased secretions, airway reflexes are generally well maintained.
   c. Hypotension is common.
   d. Profound analgesic and sedative effects occur with minimal respiratory depression.
   e. Repeat doses are well tolerated in longer procedures.

Answer: c. Ketamine increases the release of catecholamines upon administration and supports blood pressure well. It also decreases smooth muscle tone in the bronchial tree and may have a benefit in patients with reactive airways disease. Several studies have failed to show benefit with the concurrent administration of low-to-moderate dosages of benzodiazepines in preventing emergence phenomenon in children. These studies have shown a slightly increased risk of side effects. Their routine use is discouraged and should be reserved for the actual treatment of severe emergency phenomenon that are not improved by removing stimulation and providing calming interventions.

5. Which of the following statements regarding the use of propofol is true?
   a. Propofol has a long duration of action and provides significant analgesia.
   b. Propofol has significant antiemetic properties.
   c. Propofol can be easily reversed with a reversal agent.
   d. Propofol is well tolerated in volume-depleted patients.
   e. The use of "ketofol" (ketamine in combination with propofol) is clinically superior to the use of propofol alone.

Answer: b. Propofol is an ultra–short-acting, sedative-hypnotic, cerebroprotective agent with no analgesic but profound antiemetic properties. Its adverse effects include dose-dependent respiratory depression, apnea, hypotension, and pain on injection. Preload-dependent patients are particularly susceptible to hypotension. Its combined use with ketamine is common. The two agents are felt to have synergistic effects that balance each other's deficits. The combined use has been shown to improve provider satisfaction, sedation quality, and decrease emesis but has not been shown to be clinically superior to either agent used alone regarding respiratory depression, airway complications, or improved recovery times.

6. Which of the following statements is true regarding the need for fasting before procedural sedation?
   a. A 6-hour period of fasting is required after the ingestion of liquids or solids before procedural sedation.
   b. Preprocedural fasting is required in all circumstances.
   c. The recommendation for preprocedural fasting is based on controlled trials involving patients undergoing procedural sedation.
   d. The risk of vomiting and the loss of the airway protective reflexes is an extremely rare occurrence during procedural sedation.
   e. There is an increased risk of aspiration during procedural sedation after a liquid or solid meal.

Answer: d. The American College of Emergency Physicians organized a multidisciplinary Consensus Statement on Unscheduled Sedation that recommends the assessment of the timing and nature of recent oral intake. The urgency of the procedure will dictate the necessity of providing sedation without delay, regardless of fasting status. For patients with established risk factors for aspiration (e.g., serious underlying illness, obstructive sleep apnea, obesity, age less than 12 months, upper endoscopy as the procedure, or bowel obstruction), consider the risks versus benefits of delaying procedural sedation after recent ingestion of a substantial meal. American Society of Anesthesiologists (ASA) currently recommends a period of 2 hours after ingestion of clear liquids, a period of 4 hours after ingestion of breast milk, and a period of 6 hours after ingestion of other liquids or solids before the performance of procedural sedation. This recommendation is based on expert consensus and extrapolated from data on patients receiving general anesthesia and manipulation of the airway during intubation and extubation. There are no published studies showing increased risk of aspiration after a liquid or solid meal, nor benefits of fasting before procedural sedation. There are large studies showing no clinically significant differences with airway complications, emesis, or other adverse effects between groups of patients stratified by their preprocedural fasting status. Adherence to the ASA preoperative fasting guidelines for procedures is not necessary in emergency department (ED) patients undergoing procedural sedation and analgesia (PSA).

KEY CONCEPTS

- Unlike adults, most cardiac arrests in children arise from respiratory etiologies. Therefore, emphasis is first on oxygenation and ventilation.
- Detection of a child's pulse may be difficult; if a brachial pulse is not definitively present after 10 seconds of palpation, initiate cardiopulmonary resuscitation (CPR).
- Hypotension is a late finding in pediatric shock and requires immediate intervention to prevent cardiac arrest. Progression from tachycardia to bradycardia in a child is often a harbinger of imminent arrest.
- Although few studies support recommendations for exact rate, depths, and ratios of compressions to ventilations in pediatric CPR, emergency clinicians should perform rapid compressions over the lower sternum with minimal interruptions. The hand encircling technique is recommended for infants requiring chest compressions and the two-hand approach in mid sternum for older children. In the emergency department (ED), we recommend 15 compressions to 2 ventilations for children less than 8 years and 30 compressions to 2 ventilations for older children.
- Verbal and quantitative feedback improves compressions and ventilations.
- Advanced airway management (i.e., endotracheal intubation) may be harmful for children in arrest in the ED. If attempted, physicians should strive for minimal interruptions in compressions and to ventilate at 8 to 10 breaths/min. High quality bag-mask ventilation is a reasonable alternative to advanced airway maneuvers.
- If ventricular fibrillation or pulseless ventricular tachycardia arise, defibrillate at 2 J/kg as soon as possible. While preparing the equipment and charging, continue to perform high quality compressions. Administer subsequent defibrillations at escalating energy doses of 4 J/kg and then 10 J/kg.
- Early administration of epinephrine for non-perfusing rhythms may improve survival.
- Empiric administration of medications to children in arrest worsens outcomes. Reserve medications for specific indications, for example, bicarbonate administration for hyperkalemia.
- Knowledge of the child's approximate weight is necessary to administer the appropriate medication doses and to utilize the appropriate equipment. Determine actual weight in kilogram and estimate weight based on parental report or the child's length.
- Prompt vascular access is critical in a resuscitation. Intraosseous access tends to be the easiest and fastest, but care should be taken that the needle is not residing in the subcutaneous tissue. Central venous access is resource intensive and not necessary in the first few hours of a resuscitation.
- Except in specific circumstances, resuscitations greater than 30 minutes are unlikely to yield favorable outcomes for pediatric arrest.
- After return of circulation, maintain normothermia and avoid hypotension.
- Current definitions of sepsis are based on consensus and primarily for research purposes.
- Systemic inflammatory response syndrome (SIRS) criteria are sensitive but not specific for identifying sepsis in children.
- Hypotension should prompt administration of 20 to 60 mL/kg fluids and likely vasopressors.
- Most children with septic shock have cold shock and therefore epinephrine (0.05 mcg/kg/min) is the first line vasopressor.
- Protocols bundling intravenous fluids, antibiotics, and blood culture acquisition in septic children improve outcomes.
- Brief resolved unexplained events (BRUEs) involve alterations in breathing, tone, or behavior that last less than 1 minute before resolving spontaneously. Children appear well on presentation to the ED, and there are no elements of the history that suggest a particular etiology for the event.
- Low-risk BRUEs occur in children that were born greater than 32 weeks, are greater than 60 days of age, have had no prior BRUEs, and CPR was not required during the event. These children require an assessment and brief observation in the ED, little or no diagnostic testing, and may be discharged with close follow-up.

1. You are treating a 4-month-old with bronchiolitis. While being observed in the ED, the child becomes bradycardic and then loses pulses. What should be the primary focus of your initial minutes of the resuscitation?
   - **a.** Preparing to insert an endotracheal tube to improve oxygenation and ventilation
   - **b.** Correction of potential acidosis with bicarbonate
   - **c.** Vascular access to administer epinephrine
   - **d.** Uninterrupted compressions with ventilations via bag-mask device

Answer: d. At the beginning of the arrest, the most important elements are restoration of perfusion and ventilations with compressions and ventilations. Early insertion of an endotracheal tube is likely to impede compressions, and initial ventilations should occur with a bag valve mask and oral airway. Vascular access will be needed but is not as time sensitive. Bicarbonate should be given only if there is a clear indication.

2. After return of circulation, which action is most critical to improve a child's prognosis?
   - **a.** Empiric amiodarone to prevent arrhythmias
   - **b.** Avoidance of hypotension
   - **c.** Induced hypothermia to 33°
   - **d.** Strict glucose control less than 110 mg/dL with an insulin drip
   - **e.** Maintenance of oxygen saturation at 100%

Answer: b. A single episode of hypotension increases mortality and therefore efforts should be made to avoid it. Induced hypothermia does not appear to improve outcomes in pediatric arrest. Strict glucose control, hyperoxemia, and empiric amiodarone are not beneficial and may induce harm.

3. A 6-year-old male with Trisomy 21 presents with decreased responsiveness and not eating. His temperature is 39°C, heart rate is 150, blood pressure is 80/55, respiratory rate is 24, oxygen saturation is 98%, and glucose is 90 mg/dL. He receives cefepime and 20 mL/kg of lactated Ringers but his vital sign abnormalities persist. On examination, he is lethargic with a neck that is supple, lungs that are clear to auscultation, cool extremities, and capillary refill of 3 seconds. What is the next best step in the management?
   - **a.** Hydrocortisone 50 mg/m^2
   - **b.** Vancomycin 50 mg/kg
   - **c.** Epinephrine 0.05 mcg/kg/min
   - **d.** Lactated Ringers 20 mL/kg

Answer: D. This child is in septic shock. He received an initial fluid bolus but has ongoing signs of decreased perfusion. In the absence of signs of pulmonary edema, the next step should be to administer another bolus of isotonic fluids. He may benefit from epinephrine and vancomycin but after a second bolus is initiated.

4. A 4-month-old girl presents after changing color and tone at home. She was feeding and started to choke. She turned red-purple in her face and upper body while her arms stiffened. After 30 seconds, she spit up some formula, and the episode resolved. She was born full-term and has no other medical issues. In the ED she is asymptomatic and has a normal physical examination. What is your next step in the management of this patient?
   - **a.** Discharge after 2 hours of observation
   - **b.** Classify the event as a high-risk BRUE and admit to the hospital
   - **c.** Obtain a pH probe to assess for GERD
   - **d.** Obtain an ECG and blood glucose

Answer: a. Choking in the setting of feeding with complete resolution of the symptoms after vomiting suggests reflux. As a potential etiology is apparent based on the history, this event should not be classified as a BRUE. Diagnostic testing is not necessary; pH probes are not specific for GERD and are rarely helpful in the ED.

5. Which of the following medications have shown a mortality benefit for children in arrest without a clear etiology?
   - **a.** Calcium chloride 20 mg/kg
   - **b.** Sodium bicarbonate 1 mEq/kg
   - **c.** Epinephrine 0.01 mg/kg
   - **d.** Normal Saline 20 mL/kg

Answer: c. There are no randomized control trials of medications in pediatric arrest. However, observational studies suggest that epinephrine is beneficial. Other medications are only beneficial for specific indications, such as glucose for hypoglycemia.

159 Neonatal Resuscitation

KEY CONCEPTS

- Resuscitation should be anticipated for all deliveries; 10% of newborns will require some resuscitation, and 1% will require advanced life support interventions after birth.
- Predictable indications for resuscitation include hypoxia, hypothermia, hypoglycemia, hypovolemia, prematurity, maternal infection, and adverse effects of maternal medication.
- Drying, warming, positioning, and stimulating the infant are sufficient resuscitative measures for most deliveries.
- Adequate ventilation will reverse most bradycardia, while oxygenation with more than 21% fraction of inspired oxygen (FiO_2) is not indicated for most neonatal resuscitations.
- The Neonatal Resuscitation Program (NRP) resuscitation algorithm provides a proven guide for management and its implementation has shown to improve short- and long-term outcomes, including neurodevelopment.
- Routine tracheal suctioning of vigorous and nonvigorous infants born through meconium-stained amniotic fluid is no longer recommended.
- Weight-based epinephrine and volume expanders are rarely required.
- Significant hypovolemia is rare in neonates. Hemorrhage is one of the few predictable situations in which volume expansion improves newborn outcome.
- Preterm infants and those born to mothers with suspected infection, including chorioamnionitis, should receive empirical antibiotic therapy. An acceptable regimen includes dual therapy with ampicillin and gentamicin.
- Any neonate with persistent cyanosis or signs of respiratory distress (e.g., grunting, nasal flaring, and tachypnea) should be assisted by continuous positive airway pressure (CPAP) or positive pressure ventilation (PPV). Endotracheal intubation should be performed in several situations, such as when bag-mask ventilation is ineffective or prolonged, chest compressions are performed, an extremely low birth weight infant is born, and tracheal suctioning for meconium in infants results in failure to improve, despite effective PPV.
- Chest compressions are rarely required, because bradycardia generally responds to effective ventilation. However, compressions should be started for a heart rate (HR) less than 60 beats/min despite oxygen and adequate ventilation for 30 seconds.
- The umbilical vein is the preferred route of immediate vascular access, followed by peripheral veins, peripherally inserted central catheter lines, and the femoral vein. Intraosseous (IO) line placement can be problematic in neonates.
- No reliable and widely adopted set of parameters has been identified for newborns who should not receive resuscitative efforts. Unless there is clear family, parent, and health care provider agreement, resuscitation efforts should ensue.
- Infants receiving appropriate resuscitation efforts nonetheless showing no signs of life after 10 minutes may have further efforts withheld, particularly when this decision is in accord with parental preference.
- All newborns requiring intravenous (IV) line placement, medication administration, chest compressions, or endotracheal intubation should be transferred to an appropriate neonatal intensive care unit.

1. With most neonatal deliveries, which resuscitative measures are usually sufficient?

a. Administer fluids.
b. Bag-mask ventilation.
c. Intubate.
d. Warm, dry, stimulate, and position.

Answer: d. Drying, warming, positioning, and stimulating the infant are usually sufficient resuscitative measures in most deliveries.

2. For a term newborn with cyanosis, respiratory distress, and a heart rate more than 100 beats/min, which of the following is not initially indicated?

a. Apply 100% oxygen.
b. Position airway.
c. Suction.
d. Warm, dry, and stimulate.

Answer: a. 100% oxygen is not indicated for initial resuscitation in all neonates born at 35 weeks' gestation or later; avoiding unnecessary supplemental oxygen is thought to minimize free radical creation in the brain and decreases the incidence of retinopathy of prematurity. Initial resuscitation with room air is recommended. For select newborns less than 35 weeks' gestation, initial FiO_2 of 21% to 30% may be required to achieve normal oxygen saturations.

3. In a typical neonatal resuscitation, what is the preferred compression-to-ventilation ratio?

a. 3:1
b. 5:1
c. 10:2
d. 15:2
e. 30:2

Answer: a. Unlike pediatric or adult cardiopulmonary resuscitation (CPR), neonatal CPR is performed at a ratio of three compressions to one breath, with a goal of approximately 90 compressions with 30 synchronized breaths (120 "events") per minute. If the cause of the bradycardia is known to be cardiac, a ratio of 15:2 is acceptable.

4. A nonvigorous and crying newborn is delivered with copious meconium-stained fluid. What is the correct recommended resuscitative measure?

a. Bag-mask ventilate.
b. Intubate.
c. Suction at maternal perineum before cutting umbilical cord.
d. Gentle mouth suctioning if needed, followed by warming, drying, and stimulation.

Answer: d. For infants born with meconium-stained amniotic fluid, routine intubation and endotracheal tube suctioning are no longer recommended because they have shown no consistent benefit. Vigorous and nonvigorous infants born through even thick meconium should instead have gentle mouth suctioning, if needed, followed by warming, drying, and stimulation.

5. After drying, stimulating, and bag-mask ventilation, what is the next step in resuscitation of a newborn that appears floppy and apneic and with a heart rate of 50 beats/min?

a. Give a normal saline bolus of 20 mL/kg.
b. Give epinephrine (0.1 mg/mL) intravenous (IV) at a dose of 0.1 mg/kg.
c. Intubate.
d. Start with a chest compression-to-ventilation ratio of 3:1.

Answer: d. With a heart rate less than 60 beats/min in a neonate, intubation may be considered, but compressions should be started. If the low heart rate persists, IV epinephrine (0.1 mg/mL) may be considered at a dose of 0.01 mg/kg.

KEY CONCEPTS

- Trauma is the leading cause of death in children in the United States.
- Avoid hypoxia and hypotension by early administration of oxygen and assisted ventilation, and fluid resuscitation with crystalloid at 20 mL/kg increments. Initiate transfusion of 10 mL/kg of packed red blood cells (pRBCs) if hypotensive or signs of hypovolemic shock after 40 mL/kg of crystalloid is infused.
- Key pediatric anatomic and physiologic differences include:
  - Children are smaller, so bodily force is more widely distributed, making multi-system injuries more likely.
  - The infant's head-to-body ratio is greater, creating a relatively higher center of gravity. This, combined with a less myelinated brain and thinner cranial bones, predisposes infants to more serious head injury.
  - Children have a higher anatomic fulcrum in the cervical spine (C2 to C3 in children <8 years old), resulting in higher cervical spine injuries.
  - Children have greater laxity of the cervical column ligaments, leading to a greater risk of spinal cord injury without radiographic abnormalities (SCIWORA).
  - Children have more horizontally positioned ribs, resulting in a more upward movement during inspiration; this leads to a limited ability to increase tidal volume and risk for respiratory failure with chest or diaphragmatic injury.
  - Children have more elastic ribs, allowing for pulmonary injury without skeletal injury.
  - Children have thinner abdominal walls and a more anterior location of the liver and spleen; this results in a greater chance of injury to those organs.
  - Despite large intravascular volume loss, children are able to substantially increase their systemic vascular resistance to remain normotensive; hypotension is a very late sign.
- Most minor head trauma may be managed with observation and without computed tomography (CT) imaging. When applied, clinical decision rules may reduce imaging and radiation exposure.
- In major trauma patients, indications for intubation include respiratory failure or a Glasgow Coma Scale ≤8.
- The diagnostic test of choice for the evaluation of intra-abdominal injury in a stable patient with high suspicion for injury is abdominal CT.
- Solid organ injuries are generally treated nonoperatively in children.

1. A 12-year-old male fell while climbing over a 12-foot barbed-wire fence and sustained a deep 10-cm laceration to his medial left thigh. Parents bring him to the ED directly from the scene. There is active oozing from the laceration. What is the first step in the management of this patient?

a. Apply a tourniquet to the leg.
b. Begin with a primary survey and assess the patient's airway and breathing.
c. Obtain intravenous (IV) access and begin blood transfusion immediately.
d. Pack the wound to decrease hemorrhage.

Answer: b. The primary survey should quickly assess the airway, breathing, and circulation (ABCs). Approaching trauma patients in a systemic fashion will ensure that large, obvious injuries do not distract from the detection of other injuries. Oftentimes multiple interventions would be done simultaneously. Jaw thrust to open an airway and then another health care providers quickly applying pressure on the wound for hemorrhage control. The child in the question fell from a fence and may have other fall-related injuries that should be identified during the secondary survey.

2. Which of the following statements regarding imaging of a multitrauma pediatric patient is correct?

a. A negative computed tomography (CT) scan of the cervical spine rules out spinal cord injury, and if normal, immobilization can be discontinued.
b. A negative focused assessment with sonography in trauma (FAST) examination rules out traumatic intra-abdominal injury, making a CT scan unnecessary.
c. In a hemodynamically stable pediatric trauma patient, CT imaging should be completed before transfer to a pediatric trauma facility, even if it delays transfer.
d. In a hemodynamically stable pediatric patient with a high level of concern for intra-abdominal trauma, CT scan is the imaging test of choice.

Answer: d. CT scan is the diagnostic test of choice for evaluation of intra-abdominal trauma in children. Spinal cord injury without radiologic abnormality (SCIWORA) is more common in pediatric patients, and a normal cervical spine CT scan does not rule out ligamentous injuries or spinal cord edema. Although often a useful adjunct, a FAST examination does not rule out intra-abdominal injury. In a hemodynamically stable patient, CT imaging does not need to be completed before transfer to a pediatric trauma center and definitely should not delay transfer.

3. A 5-year-old male is struck by a car while riding a bicycle with no helmet, presents with heart rate of 92 beats/min, blood pressure 60/30 mm Hg, cervical spine immobilized with a collar, and quadriplegia. The patient has no evidence of hemorrhage on examination or diagnostic testing. You suspect neurogenic shock and attempt to correct the hypotension with fluids but are not successful. Which of the following should be your first choice for a vasopressor?
   a. Dobutamine
   b. Dopamine
   c. Epinephrine
   d. Norepinephrine

Answer: d. Patients with neurogenic shock lose their sympathetic tone, manifested as the triad of hypotension, bradycardia and peripheral vasodilation. Treatment includes fluid administration, parasympathetic receptor blocking agents (e.g., atropine), and vasopressors with chronotropic, vasoactive, and inotropic characteristics (e.g., norepinephrine).

4. A 6-month-old infant is brought to your emergency department (ED) by his mother after he reportedly fell 3 feet from his changing table approximately 45 minutes ago. His mother reports that the child did not lose consciousness but is fussy and has a large scalp hematoma on the right side of head. On exam, the Glasgow Coma Scale (GCS) is 14 (spontaneous eye and motor, irritable cries). The patient has a right sided 4 cm scalp hematoma in the temporal area but otherwise the examination is normal. What is your most important intervention at this point?
   a. Consulting a social worker to help screen for child abuse
   b. Cranial computed tomography (CT) scan
   c. Ice packs for the scalp hematoma
   d. Observation in the ED for 4 hours
   e. Skull radiographs

Answer: b. A 6-month-old infant with a GCS 14 has a 4% risk of clinically important traumatic brain injury (TBI) by the Pediatric Emergency Care Applied Research Network rules. Additionally, the temporal scalp hematoma is associated with a 1.6% risk of clinically important TBI while a severe mechanism of injury carries a 0.5% risk of clinically important TBI. Solely on the GCS 14 score, the infant needs a non-contrast cranial CT to rule out TBI. If the infant had a GCS 15, and was acting normally, based on the severe mechanism alone, shared decision-making with the parents to observe the infant in the ED is a reasonable alternative to an immediate cranial CT scan.

5. Which of the following is the most commonly injured solid organ in pediatric patients with abdominal trauma?
   a. Bladder
   b. Duodenum
   c. Kidney
   d. Liver
   e. Spleen

Answer: e. Injuries to the spleen are the most common injuries in pediatric abdominal trauma. Children involved in motor vehicle collisions (MVCs), sudden deceleration injuries, and contact sports–related injuries may sustain splenic trauma. Clinical exam findings include left upper quadrant abdominal pain that may radiate to the left shoulder. Splenic injuries are best identified and characterized by abdominal computed tomography (CT) scans. Treatment includes fluid resuscitation and blood transfusion as needed. Splenic salvage is important for immunocompetence in children and operative intervention is avoided as long as the patient can be stabilized with fluid resuscitation.

6. A 4-year-old female is brought to your emergency department (ED) from the scene of a motor vehicle collision. Paramedics intubated the child in the field because she was unresponsive at the scene. On arrival, her vital signs are within normal limits. Before transport to the computed tomography (CT) scanner, however, you note that she is becoming mildly bradycardic and hypertensive. Her left pupil becomes dilated and nonreactive. You elevate the head of bed 30 degrees and maintain $ETCO_2$ at 35 mm Hg with controlled ventilation. Which of the following should be your next immediate action?
   a. Mannitol or 3% hypertonic saline IV
   b. Nicardipine infusion
   c. Phenytoin administration
   d. Proceed immediately to the CT scanner

Answer: a. The sudden onset of a dilated, nonreactive pupil, along with bradycardia and hypertension, is indicative of acute brain herniation from raised intracranial pressure. Controlled ventilation to keep $ETCO_2$ greater than 30 mm Hg, a hyperosmolar agent, such as mannitol or hypertonic saline given intravenously, and raising the head of the bed 30 degrees are recommended strategies. Phenytoin can be considered for seizure prophylaxis in this patient, but not before the other measures are instituted. Nicardipine and other antihypertensive agents have no place in the management of a head-injured child. Hypertension is a physiologic response to brain herniation, where increasing the mean arterial pressure (MAP) preserves cerebral perfusion pressure (CPP).

7. A 7-year-old girl is brought into the emergency department (ED) by emergency medical service (EMS) personnel from the scene of a motor vehicle collision (MVC). On your primary survey, she is noted to have a patent airway, decreased breath sounds in the right lung field, and subcutaneous emphysema. She has hypotension and weak distal pulses. What is your next step in stabilizing this patient?
   a. Airway control with endotracheal intubation (ETI)
   b. Bag-mask ventilation (BMV)
   c. Immediate needle thoracostomy in the second midclavicular space on the right
   d. Portable chest radiograph to confirm diagnosis

Answer: c. This patient is presenting with signs of tension pneumothorax. In this case, immediate decompression with a needle thoracostomy, followed by the placement of an appropriately sized chest tube or immediate chest tube placement is required to avoid cardiovascular collapse. A portable chest radiograph should be performed to confirm chest tube placement.

8. Which of the following statements regarding chest injuries in children is correct?
   a. Aortic transection is more likely in a pediatric patient than in an adult patient.
   b. Multiple rib fractures without significant underlying lung injury are common in children.
   c. Penetrating chest trauma is more common than blunt chest trauma in pediatric patients.
   d. Significant pulmonary contusions may be present in the absence of rib fractures in children.

Answer: d. The pediatric rib cage has more compliance than an adult rib cage. Thus children are predisposed to pulmonary injury in the absence of rib fractures. Blunt chest trauma is more common than penetrating chest trauma in pediatric patients and concurrent chest and abdominal injuries are common. Aortic transection is more common in adults.

KEY CONCEPTS

- Fever is the most common complaint among pediatric patients presenting to the emergency department (ED). Although rates of bacterial illness are lower since the advent of universal vaccination for *Haemophilus influenzae* type b and *Streptococcus pneumoniae*, serious bacterial infection (SBI) should be considered in the under-vaccinated or unvaccinated child.
- Viruses cause the vast majority of childhood febrile illnesses and are generally self-limited and benign.
- SBI is the growth of pathogenic bacteria in a previously sterile site, such as urinary tract infection (UTI), bacteremia, meningitis, osteomyelitis, bacterial gastroenteritis, bacterial pneumonia, cellulitis, or septic arthritis.
- The rate of SBI in infants younger than 3 months old with fever is between 6% and 10%.
- Infants 28 days old and younger are at much higher risk for bacterial illness with fever because of their immature immune systems and incomplete vaccination status.
- Empirical treatment of a febrile neonate is indicated, and appropriate antibiotic regimens include ampicillin plus either gentamicin or cefotaxime, which cover the bacterial organisms likely in this age group.
- Empiric treatment for herpes simplex virus (HSV) in neonates with a maternal history of genital herpes, ill appearance, fever and seizure, cutaneous vesicles on physical examination, transaminitis, or evidence of coagulopathy is recommended.
- The most common cause of SBI in children continues to be UTI. In infants who are not toilet-trained, bladder catheterization is the best method for collecting uncontaminated urine. However, for select patients, bagged urine may be obtained, and if results do not indicate infection, no further studies are needed. Suprapubic aspiration may be considered when a urine sample cannot otherwise be obtained.
- Bacterial meningitis can occur at any age, but most commonly presents in a relatively small proportion of febrile infants younger than 3 months old (3/1000).
- Respiratory syncytial virus (RSV) and influenza are common viral causes of fever and respiratory distress in infants, but the presence of viral infection does not lower the risk of concomitant SBI in children younger than 28 days old.
- In older infants and children, the documented presence of RSV or influenza significantly reduces the incidence of SBI and can be used to modify the evaluation. Because UTI is still common in this population, a urinalysis should be obtained.
- There are numerous risk-stratification strategies (e.g., Boston, Rochester, and Philadelphia criteria) reported in the literature that have similar performance characteristics. All involve a laboratory evaluation designed to identify a subset of febrile infants younger than 3 months old that can safely be managed as outpatients with or without antibiotics.
- Standardization and adoption of a clinical practice guideline for the evaluation of the febrile infant have been shown to reduce variation and cost.
- Due to universal vaccination against pneumococcus, the evaluation of highly febrile children 3 to 36 months old has evolved from one of universal screening for occult bacteremia to one where clinical assessment determines the need for bloodwork.
- Inflammatory markers, such as C-reactive protein (CRP) and procalcitonin, have been shown to predict bacterial illness in febrile children more accurately than the white blood cell (WBC) count but does not solely rule out SBI.
- Traumatic lumbar punctures occur relatively commonly in young infants and can make interpretation of cell counts difficult. The use of various formulas to account for the protein and WBCs in the cerebrospinal fluid (CSF) after a traumatic tap should be used with caution.
- As the risk of meningitis is exceedingly low, we do not recommend lumbar puncture in well-appearing children with a simple febrile seizure.
- Children presenting with fever and petechiae are at risk for infection with meningococcus; blood should include a complete blood count (CBC) and culture and, if available, CRP and procalcitonin (>0.5 ng/mL). Children with an abnormal CRP, WBC count (<5000/mm^3 or >15,000/mm^3), or bandemia should be treated with parenteral antibiotics and admitted. Lower risk, well-appearing children with normal laboratory parameters can be considered for close outpatient follow-up.
- Children with fever who also are receiving cytotoxic chemotherapy for cancer are at high risk for bacteremia and sepsis and should receive prompt broad-spectrum antibiotic therapy after appropriate diagnostic evaluation (at minimum, a CBC and blood culture).
- Patients with fever and a history of sickle cell disease are at risk for bacteremia from encapsulated organisms due to functional asplenia and should be considered high risk, admitted, and treated.

1. Which of the following are appropriate methods to obtain urine as part of a fever evaluation in a non-toilet-trained child?

a. Catheterized specimen
b. Clean catch, midstream
c. Suprapubic aspiration
d. A and C

Answer: d. The only reliable method to obtain sterile urine in a non-toilet-trained child is bladder catheterization or suprapubic aspiration if a catheter specimen cannot otherwise be obtained. Bag collection has a high rate of false-positive results and is not preferred in a child who is not toilet trained.

2. A 38-day old male infant presents to the emergency department (ED) after parents noted a fever at home today. In the ED, the temperature was noted to be 40.0°C. The child was born at 35 weeks gestation and 1 week ago was diagnosed with otitis media by his primary care physician and has been taking amoxicillin as treatment. Which of the following historical aspects places this child at higher risk for invasive bacterial infection?

a. Fever responsive to acetaminophen
b. Temperature of 40.0°C
c. Prior treatment with antibiotics
d. Prematurity
e. B, C, and D

Answer: e. Risk for serious bacterial infection (SBI) varies with age. Children less than 3 months are at high risk for SBI, with those less than 28 days of age at highest risk. Hyperpyrexia, prior treatment with antibiotics, and prematurity have also been identified as risk factors for SBI. Response to antipyretics has not been linked to SBI risk.

3. Which of the following statements regarding occult bacteremia in children younger than 36 months old is *true?*

a. Children with no obvious source of fever and a temperature higher than 102.2°F (39°C) have an incidence of bacteremia of 5%.

b. Most patients appear toxic.

c. The most common pathogen is *Neisseria meningitidis.*

d. There has been a marked decrease in the incidence of occult bacteremia since the advent of universal vaccination against pneumococcus and *Haemophilus influenzae* type B.

e. With pneumococcal bacteremia, most patients remain febrile until antibiotic therapy is initiated.

Answer: d. Children younger than 36 months old with a fever higher than 102.2°F (39°C) and no obvious source have an incidence of occult bacteremia of less than 1%. Most patients appear nontoxic. The most common pathogen in positive cultures is *Streptococcus pneumoniae,* and the incidence of infection has dropped dramatically since the advent of universal vaccination. With pneumococcal bacteremia, most patients become afebrile in 3 or 4 days with or without antibiotic therapy.

4. A 3-year-old boy presents with a fever of 103°F. His mother reports that the fever started approximately 5 days ago, and he has an associated maculopapular rash. On examination, you find the patient also has bilateral conjunctival injection, a strawberry tongue, and swelling of his hands and feet. Which of the following medications should be included in the treatment of this patient?

a. Aspirin

b. Decadron

c. Penicillin G

d. Ceftriaxone

Answer: a. This patient has Kawasaki disease. Goals of treatment include amelioration of symptoms and prevention of coronary aneurysms, which are normally accomplished with aspirin and immunoglobulin. There are no indications for antibiotics or steroids.

5. A 2-year-old presents with a high fever and vomiting. On examination, you find an irritable child with a rectal temperature of 102°F rectal and a stiff neck. The patient's past medical history is significant for hydrocephalus with a ventriculoperitoneal shunt placement. You suspect the patient has a ventriculoperitoneal shunt infection. Which of the following is the most likely bacterial pathogen?

a. *Haemophilus influenzae*

b. *Neisseria meningitidis*

c. *Staphylococcus aureus*

d. *Staphylococcus epidermidis*

Answer**: d.** Patients with ventriculoperitoneal shunts and fever should be evaluated for shunt infection. The most common bacterial pathogen is *S. epidermidis*.

162 Pediatric Upper Airway Obstruction and Infections

KEY CONCEPTS

Respiratory arrest precedes most pediatric cardiac arrests. Quick recognition of an airway problem and intervention in potentially life-threatening upper airway obstruction in children are critical.

Retropharyngeal Abscess

- This is a potentially life-threatening emergency in young children with signs of upper airway obstruction or meningismus; a retropharyngeal abscess is often related to oral trauma.
- Retropharyngeal abscess is most frequently caused by *Staphylococcus aureus*, group A streptococci, and anaerobes. Treatment is admission, IV antibiotics, and for more severe cases, surgical drainage.

Epiglottitis

- Epiglottitis may be caused by many bacteria or local injury. In the post–*Haemophilus influenzae* type b vaccine era, the incidence of pediatric epiglottitis has decreased, and epiglottitis is now more likely in older patients.
- Clinical features of epiglottitis are often subtle, such as in the older adolescent, (e.g., sore throat out of proportion to physical findings, anterior neck tenderness), but may also be dramatic, as in infants and young children (i.e., drooling, stridor, toxicity, severe respiratory distress).

Croup

- Viral croup is the most common infection of the upper airway in young children.
- Glucocorticoids (usually given as a single oral dose of dexamethasone) reduce symptoms, hospitalizations, and length of stay in the emergency department (ED).
- Treatment of moderate to severe croup includes vaporized epinephrine in addition to glucocorticoids. Patients can be discharged from the ED after a posttreatment observation period. We recommend discharge after a period of observation, if the child is free of resting stridor and distress and has access to follow-up care.

Bacterial Tracheitis

- Suspect bacterial tracheitis when an upper respiratory infection (URI) progresses to acute toxicity and marked respiratory distress and stridor. Standard treatment for croup may be tried, but usually does not significantly improve the patient's symptoms. Antibiotic therapy should include a cephalosporin plus coverage for *S. aureus*, which is the most common cause of this infection.
- Bronchoscopy is diagnostic and therapeutic and should be emergently performed.

Airway Foreign Body

- Complete obstruction due to an airway foreign body requires emergent life support procedures for removal of the foreign body.
- Plain films may be negative in aspirated foreign bodies. Bronchoscopy should be performed with a clinical suspicion of aspiration.
- Emergency cricothyroidotomy may be required for obstructed patients who cannot be intubated or ventilated as a lifesaving temporizing measure; needle cricothyroidotomy is preferred for infants and young children because of the challenges in identifying landmarks and associated complications of surgical cricothyrotomy.

1. Which of the following is the most common cause of upper respiratory obstruction in childhood?
- **a.** Airway foreign body
- **b.** Bacterial tracheitis
- **c.** Croup
- **d.** Epiglottitis

Answer: c. Although all the choices may lead to symptoms of airway obstruction in children, the most common cause of upper airway obstruction is viral croup.

2. A 3-year-old girl presents at 2 AM with complaints of a barky cough, which started abruptly overnight. Vital signs are heart rate, 140 beats/min, respiratory rate, 40 breaths/min, and temperature, 100.1°F (38°C). She has no history of asthma or wheezing. She appears to be in moderate distress and has audible stridor. Indications for admission include which of the following?
- **a.** Low-grade fever
- **b.** Prior history of croup
- **c.** Stridor at rest
- **d.** Severe dehydration

Answer: d. Indications for admission of patients with croup include severe respiratory distress or failure, unusual symptoms (hypoxia and hyperpyrexia), anything but mild dehydration, persistence of stridor at rest after aerosolized epinephrine and steroids, persistence of tachycardia or tachypnea, and complex medical history (e.g., prematurity, pulmonary or cardiac disease).

3. A 5-year-old immunized boy presents with severe stridor at rest, low-grade fever, and nasal congestion. His family reports a barky-sounding cough. After initiating vaporized epinephrine, he appears well and is in no distress. The parents are asking about discharge. Which of the following would be the most appropriate next step in management?
- **a.** Administer dexamethasone, observe the patient for 2 to 3 hours, and discharge if well.
- **b.** Admit the patient for overnight observation.
- **c.** Allow the patient to go home.
- **d.** Allow the patient to go home with a prescription of steroids.

Answer: a. The child has croup and, after being treated with nebulized epinephrine, should be observed for recurrent stridor after 2 hours. If the child is well, is in no respiratory distress, has a good hydration status, and is able to access emergency and follow-up care, he or she can then be safely discharged.

4. What is the ideal head position to assess a pediatric soft tissue radiograph of the neck for upper airway pathology?
- **a.** Extension during inspiration
- **b.** Flexion
- **c.** Flexion during inspiration
- **d.** Neutral

Answer: a. Gentle extension of the head gives the most accurate images, avoiding the artificial soft tissue widening that can be seen in flexion. Inspiration, if possible, allows maximal distention of the pharynx and the best viewing of soft tissue structures defined by an air–soft tissue interface.

5. Which of the following factors is least consistent with the diagnosis of peritonsillar abscess?
- **a.** Muffled, hot potato voice
- **b.** Pain radiating to the ear
- **c.** Patient 3 years of age
- **d.** Patient 13 years old
- **e.** Trismus

Answer: c. Peritonsillar abscess more commonly occurs in older children and teenagers, whereas retropharyngeal abscess is more common in a younger population. All the other signs or symptoms listed are consistent with peritonsillar abscess, along with deviation of the uvula away from the abscess side.

6. A 3-year-old immunized girl presents after a brief viral illness with progressive dyspnea, ill appearance, and high fever. The child is relatively still, appearing as if she is trying not to cough. Stridor is heard, and she does not respond to croup therapy. You notify the operating room, where the patient undergoes bronchoscopy, with suctioning and airway placement. Culture results are most likely to grow which of the following organisms?

a. *Bacteroides fragilis*
b. *Candida albicans*
c. Parainfluenza
d. *Staphylococcus aureus*
e. *Streptococcus pneumoniae*

Answer: d. The case described is consistent with bacterial tracheitis. Although *Candida,* parainfluenza, and *Streptococcus* have all been reported, *S. aureus* is most common. Broad-spectrum antibiotics are appropriate, with an emphasis on covering *S. aureus*.

163 Pediatric Lower Airway Obstruction

KEY CONCEPTS

- No single asthma score has been universally adopted to assess the degree of illness or treatment responses. However, most scores include some combination of respiratory rate, degree of wheezing, inspiratory-to-expiratory ratio, use of accessory muscles, and oxygen saturation.
- Chest x-ray (CXR) is not required for wheezing children, even for those who are febrile, are wheezing for the first time, or require hospitalization. CXR is indicated for those with a history of choking, focal chest findings, extreme distress, subcutaneous emphysema, diagnostic uncertainty relative to respiratory illness, or with clinical findings suggestive of a cardiac etiology.
- Albuterol delivered by metered-dose inhalers with spacers (MDI-S) is as effective as that delivered by nebulizers for children with acute asthma. The mode of delivery is largely chosen on the basis of cost and ability to achieve the goal of three treatments within the first hour of care. Per the 2007 National Heart Lung and Blood Institute guidelines, a high dose (4–12 puffs) of a short-acting beta-agonists (SABAs) metered-dose inhaler (MDI) with a spacer has "equivalent bronchodilation" to nebulized treatment.
- Levalbuterol does not lead to better emergency department (ED) outcomes compared with racemic albuterol. Racemic albuterol, at a substantially lower cost, should remain the drug of choice for children with acute asthma exacerbations.
- Dexamethasone is as effective as prednisone in the ED treatment of acute asthma. Dexamethasone is associated with fewer doses, less vomiting, and greater compliance.
- Continuously nebulized albuterol, corticosteroids, magnesium sulfate, and parenteral SABA are cornerstones of therapy for moderately to severely ill children with asthma.

1. A 19-month-old girl with a history of asthma presents with severe respiratory distress marked by wheezing, tachypnea, deep retractions, and an oxygen saturation of 92% in room air. She fails to improve 10 minutes after nebulized short-acting β_2-agonist (SABA) therapy and repeatedly tries to pull off her face mask. Which strategy is likely to produce the most rapid clinical benefits in this setting?
   a. Administering IM terbutaline
   b. Administering IV methylprednisolone
   c. Administering the SABA via metered-dose inhaler (MDI) instead of by nebulization
   d. Doubling the dose of SABA in the nebulizer reservoir
   e. Taking off her face mask and holding it close to her face to reduce her agitation

Answer: a. This patient's degree of bronchospasm and lack of cooperation makes it difficult to deliver aerosolized medication effectively. In this setting, IM terbutaline is most likely to result in rapid bronchodilation, making subsequent aerosolized therapy more effective.

2. A 6-month-old male infant presents to the ED with his father, who reports that the patient has been wheezing. His vital signs are a temperature of 39.7°C, respiratory rate 62 breaths/min, and oxygen saturation of 90%. On examination, you find diffuse wheezing and copious nasal secretions. While in the ED, the patient does not have a wet diaper; his father reports that he has had decreased oral intake for the past 2 days. Which of the following findings with bronchiolitis is not associated with the need for admission?
   a. Age
   b. Decreased oral intake
   c. Oxygen saturation
   d. Respiratory rate
   e. Temperature

Answer: a. Patients often in need of admission for bronchiolitis include those with respiratory rates more than 70 breaths/min, O_2 saturation of 95% or less, age younger than 3 months, and poor feeding.

3. Which of the following side effects are seen with the use of intravenous magnesium in the treatment of asthma in children?
   a. Change in serum pH
   b. Development of a prolonged QT interval on the electrocardiogram
   c. Hypercalcemia
   d. Hyperkalemia
   e. Hypotension

Answer: e. Most patients who receive a magnesium infusion at the recommended dose will experience a clinically insignificant decrease in blood pressure. This may be minimized by infusing the medication over 20 minutes and by concurrently administering normal saline solution.

4. You are about to treat a child with acute asthma with albuterol. Which of the following factors is most important when considering the method of aerosolized drug delivery?
   a. About 20% to 30% of nebulized drug reaches the alveoli.
   b. Children receiving β-agonists by nebulizer (NEB) have similar outcomes compared with those using a metered-dose inhaler (MDI-S).
   c. Children weighing >20 kg should receive a maximum of four puffs of albuterol via MDI-S.
   d. MDI-S therapy is more costly than NEB therapy.
   e. The use of MDI-S for acute asthma is not supported by national guidelines.

Answer: b. Clinical trials and systematic reviews have repeatedly shown these two forms of therapy to be equivalent. With NEB treatment, less than 10% of nebulized drug reaches the alveoli. A recommended MDI-S dose for older children is eight puffs. Most studies demonstrate that MDI-S therapy is more cost-effective because of a slightly reduced need for hospitalization. National guidelines support the use of MDI-S to deliver albuterol to children with acute asthma.

5. A 7-month-old male infant was diagnosed 2 days ago with bronchiolitis. He continues to have wheezing and increased work of breathing, prompting his parents to bring him into the ED because of a new-onset fever. Which of the following is the most likely secondary bacterial infection in this infant with bronchiolitis?
   a. Pneumonia
   b. Acute otitis media
   c. Meningitis
   d. Urinary tract infection

Answer: b. Bacterial acute otitis media (AOM) is the most common condition associated with bronchiolitis, with a prevalence of up to 60%. The bacterial pathogens are similar to those recovered in other children with AOM; thus, it should be treated according to standard recommendations. Other concurrent bacterial infections are rare.

KEY CONCEPTS

- Determining the causative agent of pneumonia by clinical presentation and radiographic findings is not reliable; empirical treatment is based on guideline recommendations and likely pathogens.
- Causative agents vary by age; viral agents predominate, especially in younger children, and *Streptococcus pneumonia* is the leading bacterial cause outside of the neonatal period.
- Infants and younger children with pneumonia may have subtle or nonspecific symptoms and signs on presentation, and fever may be the only sign of disease.
- First line therapy for the treatment of bacterial pneumonia in children is amoxicillin for an outpatient and ceftriaxone or ampicillin for an inpatient.
- Pertussis should be considered in a young infant with a staccato cough or episodes of cyanosis.
- In patients with cystic fibrosis, defects in chloride transport across the airway epithelium result in reduced ciliary clearance of thickened mucus, which leads to an increased likelihood for pneumonia, especially that caused by *Pseudomonas aeruginosa.*
- Cystic fibrosis may respond favorably to bronchodilator therapy and mucolytics, such as inhaled *N*-acetylcysteine.
- Patients with bronchopulmonary dysplasia have increased airway resistance, decreased lung compliance, and obstructive lung disease; reactive airway disease and pneumonia are common in these patients.

1. A 5-month-old boy presents with a cough. His parents report that for the past several weeks he has had mild respiratory tract symptoms and cough; however, during the past day, he has developed a severe paroxysm of staccato cough followed by posttussive emesis. What is the most appropriate antibiotic choice for this patient?
   a. Amoxicillin
   b. Ampicillin
   c. Azithromycin
   d. Trimethoprim-sulfamethoxazole

Answer: C. This patient likely has pneumonia caused by *Bordetella pertussis*. All children younger than 6 months with presumed pertussis should be observed in the hospital for monitoring and supportive care. First line treatment is with azithromycin, and erythromycin is a possible alternative. Amoxicillin and ceftriaxone are used to treat other types of pneumonia.

2. Which of the following findings is an indication for admission to the hospital in children with a diagnosis of pneumonia?
   a. Abnormal chest radiograph showing pulmonary infiltrates
   b. Decreased breath sounds in a lower lung field
   c. Dehydration and vomiting
   d. Diffuse rales and mild tachypnea

Answer: C. Many children with pneumonia can be managed as an outpatient with appropriate therapy. Clinical dehydration or inability to orally maintain hydration should prompt admission for ongoing treatment and monitoring. Abnormal lung auscultation or findings on a chest radiograph by themselves would not be contraindications to outpatient treatment.

3. Which of the following statements best describes the epidemiology of pneumonia in children?
   a. B*ordetella pertussis* is the most common cause in infants.
   b. *Haemophilus influenzae* type b is still an important pathogen.
   c. *Listeria monocytogenes* may cause illness in children younger than the age of 5 years.
   d. Viral agents are the most common cause of pneumonia in children overall.

Answer: D. Viral agents are the most common cause of pneumonia in children. Bacteria predominate in neonates but are less common causative agents in toddlers and older children. Although *L. monocytogenes* may cause pneumonia in infants, it is rare after 3 months of age. *H. influenzae* type B has decreased by 90% since the onset of immunization of infants and young children.

4. What is the most common viral agent causing pneumonia in infants younger than 1 year?
   a. Adenovirus
   b. Enterovirus
   c. Epstein-Barr virus
   d. Respiratory syncytial virus (RSV)

Answer: D. RSV and parainfluenza are the most common viral agents in infants younger than 1 year. Viruses that may be responsible for the other agents may also cause viral pneumonia but are not as common as RSV.

5. A 2-year-old presents with fever, cough, and rales. Chest radiograph reveals right middle lobe pneumonia. Which of the following antibiotics would be recommended for outpatient treatment for this toddler who is not penicillin allergic?
   a. Amoxicillin
   b. Azithromycin
   c. Ceftriaxone
   d. Cephalexin

Answer: A. High-dose amoxicillin is the treatment of choice for children with pneumonia younger than 5 years. Once the child reaches school age, azithromycin is suggested as empirical therapy because of the increased risk of *Mycoplasma pneumoniae* infection.

KEY CONCEPTS

- Consider a congenital heart defect in an infant with central cyanosis who does not respond to 100% supplemental oxygen (hyperoxia challenge).
- Neonates with ductal-dependent cardiac lesions typically present within the first 2 to 3 weeks of life with either acute cyanosis or shock. Prostaglandin E_1 (alprostadil, 0.05 to 0.1 μg/kg/min) can maintain a patent ductus arteriosus to supply mixed blood and temporize the patient.
- Treatment of a hypoxic "tet spell" first includes the placement of an infant in the knee-to-chest position (or an older child in a squatting position) to increase systemic vascular resistance (SVR) and the provision of supplemental oxygen. Sedative agents can be used to decrease hyperpnea. Various medications can be used as adjunctive treatment to increase the SVR and thereby decrease the degree of right-to-left shunting across the ventricular septal defect (VSD).
- Prompt recognition of the clinical findings and symptoms of Kawasaki disease along with the rapid initiation of high-dose aspirin and intravenous immune globulin (IVIG) infusion may prevent the formation of coronary aneurysms.
- Acute bacterial endocarditis should always be considered in a child with a known congenital heart defect or an acquired cardiac defect who presents with fever of unknown origin, acute neurologic deficits, new-onset microscopic hematuria, myalgias, splenomegaly, petechiae, or other signs of systemic embolization.
- Oxygen, positive-pressure ventilation (noninvasive or invasive), diuretics, and possibly inotropes are the main emergency department (ED) treatment of infants and children who present with congestive heart failure (CHF). Nitroglycerin can cause profound hypotension in children and is not a first-line therapy.
- If vagal maneuvers fail to convert stable paroxysmal supraventricular tachycardia in children, rapid adenosine administration (0.1 mg/kg for the first dose, followed by 0.2 mg/kg on repeated doses) is the treatment of choice. Verapamil should be avoided in children because of its profound hypotensive effects.
- Young athletes with a positive family history of sudden unexplained death or exertion-induced symptoms (such as, chest pain, dyspnea, palpitations, and syncope) should be evaluated by a cardiologist before their resumption of vigorous activity.

1. Which of the following increases the systemic vascular resistance (SVR), thus producing a left-to-right shunt through the ventricular septal defect (VSD) associated with Tetralogy of Fallot?
   a. Acute hypovolemia
   b. Crying
   c. Defecation
   d. Squatting

Answer: D. Squatting or knee-to-chest positions increase the SVR, thus improving tet spells by producing the left-to-right shunt. Acute hypovolemia, crying, defecation, and tachycardia are all events that suddenly lower the SVR and produce large right-to-left shunts across the VSD, beginning the vicious cycle of a hypoxic (tet) spell.

2. A 14-year-old girl presents to the emergency department (ED) with altered level of consciousness and trouble breathing. Her vital signs are blood pressure (BP), 72/39 mm Hg; heart rate, 240 beats/min; temperature, 99.6° F; respiratory rate, 60 breaths per minute; and oxygen saturation, 80%. An electrocardiogram (ECG) is performed, which shows supraventricular tachycardia. The patient weights 50 kg. Which of the following is the most appropriate initial treatment?
   a. Adenosine 5 mg IV
   b. Cardioversion with 50 J
   c. Cardioversion with 200 J
   d. Diltiazem 12.5 mg IV

Answer: B. This patient should be considered unstable supraventricular tachycardia (SVT), with trouble breathing, low O_2 saturation, and hypotension. In stable patients, vagal maneuvers may be appropriate to try first, followed by adenosine. Diltiazem may be used for rate control in patients with atrial flutter or atrial fibrillation. Cardioversion is the treatment of choice in patients with hemodynamic instability and SVT. The dose is 0.5 to 1 J/kg.

3. What is the most common cause of bradycardia in infants?
   a. Complete heart block
   b. Hypothermia
   c. Hypothyroidism
   d. Hypoxia

Answer: D. Although all of these may cause bradycardia in infants and children, the most common cause is hypoxia.

4. A 16-year-old boy presents with complaints of syncope during a basketball game. His mother reports a family history of sudden death in young adults, and you are concerned that the patient may have hypertrophic cardiomyopathy. Which of the following increases the murmur associated with this condition?
   a. Hand grip
   b. Methoxamine
   c. Squatting
   d. Valsalva

Answer: D. During the Valsalva maneuver, the venous blood return to the heart is decreased, which in turn transiently reduces the left ventricular size. This transient reduction will increase the degree of obstruction and thus cause an increase in the intensity of the murmur. Hand grip, methoxamine, and squatting will increase return of blood to the heart and therefore decrease the murmur associated with hypertrophic cardiomyopathy.

5. A 5-month-old girl presents with fever for a week, rash, and fussiness; reportedly yesterday her rash was faint throughout her body and has since resolved. On examination, she is febrile with otherwise reassuring vital signs; she is fussy and has conjunctival injection in both eyes. Her parents think she got sick from their other children. Which of the following statements regarding this patient's most likely disease is *true?*
   a. If present, other systemic signs (such as, nausea, vomiting, and diarrhea) suggest an alternative diagnosis (such as, acute gastroenteritis).
   b. Laboratory investigation in the emergency department (ED) will assist in her risk stratification.
   d. Older children are at the highest risk for aneurysm formation.

Answer: B. Kawasaki disease is a systemic vasculitis, and gastrointestinal findings (nausea, vomiting, abdominal pain, diarrhea) and neurologic findings (irritability, positive pleocytosis on lumbar puncture if done) may mislead the clinician. Although she only has one criterion currently, signs of Kawasaki disease may occur in series or simultaneously over the course of the disease. With two or more signs in a child who is febrile for more than 5 days, markers of systemic inflammation (such as, C-reactive protein [CRP] and erythrocyte sedimentation rate [ESR]) are indicated to determine the need for further laboratory testing (complete blood count, liver function tests [LFTs], albumin, urinalysis), imaging (echocardiography), or next day follow-up (with repeat laboratory tests). This child is at high risk because infants are more prone to vascular complications (such as, giant aneurysm formation); tragically they are also more likely to present with incomplete Kawasaki disease and may be overlooked. Accordingly, consensus guidelines recommend that all infants younger than 6 months old with fever of 1 week or greater (regardless of other findings) should undergo laboratory testing for markers of inflammation and, if positive, should have an echocardiogram performed emergently.

KEY CONCEPTS

- Physiologic jaundice of the newborn and breast milk jaundice are the most common causes of jaundice in the neonatal period.
- Direct hyperbilirubinemia in infants is always pathologic and requires a detailed evaluation.
- Hypertrophic pyloric stenosis is associated with gradually progressive nonbilious emesis that becomes projectile.
- Hypochloremic-hypokalemic metabolic alkalosis is the classic electrolyte derangement associated with hypertrophic pyloric stenosis.
- Bilious vomiting in the neonate is an ominous sign and should initiate a diagnostic evaluation for possible malrotation with volvulus or other intestinal obstructive pathology.
- Infants with bilious emesis should receive an emergent surgical consultation, particularly if ill-appearing.
- Necrotizing enterocolitis (NEC) occurs more commonly in premature infants, but 10% of affected infants are full term. Pneumatosis intestinalis in neonates—intramural air seen on x-ray—is pathognomonic for NEC.
- Gastroesophageal reflux (GERD) is very common in infants and is usually benign and self-limited. Occasionally, GERD may cause more severe symptoms, including irritability, respiratory distress, and failure to thrive. GERD usually responds to conservative measures (e.g., positioning, thickening of formula, smaller and more frequent feedings); pharmacologic interventions are seldom needed.
- The classic clinical triad of intussusception includes colicky, intermittent abdominal pain, a palpable sausage-shaped abdominal mass, and bloody "currant jelly" stools; however, this triad occurs in less than one-third of patients.
- Children with intussusception may present atypically, with an altered level of consciousness (e.g., lethargy) rather than abdominal pain.
- Hirschsprung disease is a pathologic cause of constipation in the neonate and usually manifests as delayed passage of meconium. Occasionally, children may present later in life with symptoms of chronic constipation. Toxic megacolon is the most serious complication.
- Meckel diverticulum classically manifests in children younger than 5 years with massive, painless, "brick red" colored rectal bleeding.
- More than 90% of GI foreign bodies pass without complications.
- Lithium button batteries lodged in the esophagus may cause serious burns, erosions, and perforations within as little as two hours. Of all foreign bodies, button batteries require the most expeditious removal, usually by endoscopy.
- Appendicitis is the most common surgical disease in children. Diagnosis depends on a combination of clinical factors, including history, physical examination, laboratory values, and imaging studies.
- Effective imaging strategies for children with suspected appendicitis include initial ultrasound examination followed by CT scanning of the abdomen for those with equivocal findings.
- Causes of pancreatitis in children include viruses, trauma, drugs, and toxins.
- Biliary disease in children is more commonly caused by cholestasis rather than biliary obstruction.
- Pigment gallstones are more common than cholesterol stones in children. Biliary tract disease is usually diagnosed with right upper quadrant ultrasound imaging; management strategies are similar to those for adults.

1. What is the most common cause of jaundice in the newborn?
- **a.** Breast milk jaundice
- **b.** Crigler-Najjar syndrome
- **c.** Gilbert syndrome
- **d.** Physiologic jaundice of the newborn

Answer: D. Although each of these may be a cause of hyperbilirubinemia in the newborn, the most common cause of jaundice is physiologic jaundice of the newborn.

2. A 4-week-old white infant presents with projectile vomiting. The mother denies that the patient has a history of fevers, irritability, or signs suggestive of abdominal pain. On physical examination, you palpate an olive in the patient's right epigastrium. Which of the following laboratory abnormalities would you expect to find?
- **a.** Hyperchloremia and hypokalemia
- **b.** Hyperglycemia and hypokalemia
- **c.** Hypernatremia and hyperkalemia
- **d.** Hypochloremia and hypokalemia

Answer: D. This patient likely has pyloric stenosis. As vomiting continues, the infant loses hydrogen and chloride ions through emesis of gastric juices. As this metabolic derangement worsens, the kidney attempts to retain hydrogen ions by substituting potassium, resulting in a hypochloremic-hypokalemic metabolic alkalosis.

3. An 11-month-old infant presents with vomiting. The patient's mother reports that he has been crying out in pain intermittently throughout the day, at which times he brings his knees to his abdomen. In between these episodes, the patient acts normally and plays. He has not had a fever, but the mother complains that his stool earlier looked like currant jelly. On examination, you find a playful afebrile patient, with a soft nontender abdomen. Which of the following may be used as an initial screening examination?
- **a.** Air-contrast enema
- **b.** Barium enema
- **c.** Computed tomography scan of the abdomen-pelvis
- **d.** Ultrasound

Answer: D. The initial screening examination for intussusception is abdominal ultrasonography. Although each of the other imaging modalities may be useful to exclude other diagnoses or identify intussusception, ultrasonography has high sensitivity and specificity and is the screening modality of choice.

KEY CONCEPTS

Identification of Pathogen

- Stool studies are not indicated in most uncomplicated cases of acute gastroenteritis (AGE). Exceptions are those cases in which specific treatment, specific prophylaxis, or health precautions are required, or in which the patient has systemic involvement, underlying medical complications, or dysenteric features.
- Antibiotics are not required for most cases of uncomplicated acute bacterial enteritis. Antibiotics are recommended routinely for *C. difficile, Giardia intestinalis,* and *E. histolytica.* Antibiotics can be considered for *Campylobacter*, *Cryptosporidium,* traveler's diarrhea, and *Shigella* (because antibiotics have been shown to decrease diarrhea and eradicate organisms in the stool).
- Patients with Shiga toxin–producing *E. coli* (STEC) should not empirically receive antibiotics, because they may increase the risk of hemolytic-uremic syndrome (HUS).
- Testing for fecal leukocytes is a useful initial test because it may support a diagnosis of invasive disease. This test should be considered in children with diarrhea who are febrile or have mucus or blood in the stool. If the test result is positive, stool culture is indicated to further guide management.

Oral Rehydration

- Most patients with mild to moderate dehydration can be treated with oral rehydration therapy (ORT). Resumption of feeding with age-appropriate diets should begin as soon as vomiting subsides. Routine fasting with infectious diarrhea is not recommended.

Dehydration Assessment

- The degree of volume depletion is estimated from the history and physical examination findings. The desired volume of oral rehydration solution is calculated as 30 to 50 mL/kg for mild dehydration and 60 to 80 mL/kg for moderate dehydration; 25% of the volume of oral rehydration solution is to be replaced every hour (100% over 4 hours). Continue to replace ongoing losses with 10 mL/kg for each diarrheal stool and 2 mL/kg for each vomiting episode. Patients who fail an oral rehydration trial of 4 to 8 hours in the emergency department (ED) should be admitted for intravenous hydration.

Severe Dehydration

- In severe dehydration, 20 mL/kg of 0.9% saline (or other appropriate isotonic crystalloid solution) given intravenously or intraosseously should reverse signs of shock within 5 to 15 minutes. Repeated boluses of 20 mL/kg are indicated until clinical improvement occurs, but volume requirements greater than 60 mL/kg without signs of improvement suggest other conditions, such as septic shock, hemorrhage, capillary leak with third-space fluid sequestration, and adrenal insufficiency. Rapid correction of sodium derangements in dehydration can lead to central nervous system complications.

1. For which of the following patients should the emergency clinician not obtain diagnostic testing of the stool?
   - **a.** 1-month-old with fever to 38°C with bloody diarrhea
   - **b.** 8-month-old with sickle cell disease with fever and diarrhea
   - **c.** 12-month-old, well-appearing child with 2 days of watery diarrhea
   - **d.** 2-year-old with nephrotic syndrome on prednisone with diarrhea and fever for 3 days
   - **e.** Community outbreak of diarrhea

   Answer: C. Most children with uncomplicated gastroenteritis or acute diarrhea do not need laboratory studies. Stool cultures are useful and important to obtain in febrile infants and children with blood in their stools, in community outbreaks, and in the immunosuppressed.
2. A 4-year-old child presents with moderate dehydration from gastroenteritis. He is still actively vomiting. Which of the following statements regarding this patient's rehydration is correct?
   - **a.** A nasogastric tube should be placed for rehydration.
   - **b.** Oral rehydration should not be started.
   - **c.** Oral rehydration success can be determined in 2 or 3 hours.
   - **d.** Oral rehydration target volume is 80 mL/kg.
   - **e.** Target oral rehydration volume should be administered over 2 hours.

   Answer: D. Oral rehydration can be effective even in the face of vomiting in cases of mild or moderate dehydration. Target oral volumes are 60 and 80 mL/kg, respectively. Nasogastric tubes may occasionally be useful but should not be used unless oral rehydration therapy (ORT) fails. The target oral volume should be given in doses of 25% of total replacement per hour over 4 hours.
3. A 9-year-old child with a history of sickle cell disease presents for evaluation of fevers, diarrhea, and vomiting. The parents say the patient has not had a sickle cell crisis in almost a year. On examination, the vital signs are stable and the child is nontoxic-appearing with mild dehydration. His abdominal examination is benign. In addition to rehydration, which of the following would be the most appropriate management and disposition for this child?
   - **a.** Obtain blood cultures and admit for observation.
   - **b.** Obtain cultures (stool and blood), begin antibiotics, continue hydration, and admit for observation.
   - **c.** Obtain electrolytes; and if he has a normal bicarbonate level, then discharge home.
   - **d.** Perform a stat Rotazyme assay; and if negative and patient tolerates oral intake, then discharge home.
   - **e.** Provide oral rehydration therapy (ORT); and if successful, then discharge home.

   Answer: B. Patients with sickle cell disease are more susceptible to *Salmonella* infections and are at increased risk for complications. It has been shown that giving antibiotics to patients with nontyphoidal *Salmonella* has been ineffective in shortening the duration of symptoms and prolonging the carrier state. Therefore, antibiotics are generally not recommended for asymptomatic cases or for uncomplicated cases. However, antibiotic treatment is indicated in infants younger than 3 months old or those with complications, such as failure to improve within 5 to 7 days; bacteremia; focal infection in the central nervous system (CNS), bone, joint, kidney, or pericardium; or those with immunosuppressive conditions, hemoglobinopathies, malignant neoplasms, human immunodeficiency virus (HIV), or chronic gastrointestinal disease. The recommended antibiotics are for unknown susceptibility or in areas of high resistance to use ceftriaxone or cefotaxime until susceptibility is known. Ampicillin and trimethoprim-sulfamethoxazole (TMP-SMX) may also be effective, but they should be used only in high-risk individuals once susceptibility is known.
4. A 5-month-old infant presents with persistent vomiting and diarrhea for 3 days. The parents came to the emergency department because their child suddenly seemed to be passing out. Parents have noted significant weight loss and poor breastfeeding. Exam shows an unconscious infant with doughy skin and dry mucous membranes. IV access is obtained, the airway is secured, and a normal saline infusion is started. What central nervous system complication of dehydration is most likely in this patient?
   - **a.** Cerebral edema
   - **b.** Central pontine myelinolysis
   - **c.** Hypoxic ischemic encephalopathy
   - **d.** Primary epilepsy
   - **e.** Subdural hematoma

   Answer E: The clinical picture suggests hypernatremic dehydration (infant with poor breastfeeding, doughy skin, significant weight loss). Hypernatremia leads to dehydration and shrinking of the brain, which increases the risk of injury to bridging veins, subdural hematoma, and CNS thrombi. Cerebral edema is a complication of hyponatremic dehydration (more often in older children with skin tenting on examination). Central pontine myelinolysis is a complication of rapid sodium correction in hyponatremic dehydration. HIE and epilepsy are not complications of dehydration alone.
5. A 4-year-old girl presents with acute gastroenteritis. On history and examination the child has lost 12% of her weight from her last primary care appointment 3 weeks ago. She appears lethargic and has mottled skin with a capillary refill of 5 seconds. Multiple attempts at IV access are unsuccessful. What is the next best approach for rehydration?
   - **a.** Hypodermoclysis (subcutaneous rehydration)
   - **b.** Obtain intraosseous access
   - **c.** Oral ondansetron
   - **d.** Nasogastric tube insertion
   - **e.** Transfer to a pediatric tertiary hospital

   Answer B: Patients with severe dehydration (>10 % body weight loss) or evidence of decreased perfusion (poor capillary refill, lethargy) should have emergent infusion of isotonic crystalloid at a rapid rate to restore perfusion. In this patient, the fastest method to achieve this is with intraosseous access. In a conscious patient, the line can be primed with lidocaine for 1 to 2 minutes before the infusion to help with the pain of the infusion. Hypodermoclysis, nasogastric tube rehydration, and oral rehydration are all appropriate management options for mild-moderate dehydration. Transferring the patient before IV rehydration would delay her care and increase her mortality risk.

KEY CONCEPTS

- Ischemic priapism is a urologic emergency due to compartment syndrome of the penis and is managed with local analgesia with or without sedation, cavernosal aspiration, irrigation, and possibly injection with phenylephrine.

1. A 12-year-old male presents due to right-sided scrotal pain for the last 2 days. On examination you note a swollen right testicle with normal cremasteric reflex that has a "bag of worms" consistency on examination. What is your next step in this patient's workup?
   a. Further imaging of the abdomen
   b. Further imaging of the testicles
   c. Mumps titers
   d. Urine to assess for *Neisseria gonorrhea* and *Chlamydia trachomatis*

   Answer: A. Further imaging of the abdomen. The patient described has a right-sided varicocele. Most patients have a left-sided varicocele due to the angle of drainage of the left spermatic vein to the left renal vein; however, a right-sided and abrupt onset varicocele warrants abdominal imaging to investigate for an intra-abdominal mass. Imaging of the testicle would be appropriate to investigate for testicular torsion, but abdominal imaging is still necessary to rule out mass in this presentation. Mumps may present with orchitis, but not usually with varicocele. STIs do not usually present as varicoceles and in absence of other symptoms testing is not necessary at this time.
2. A family brings their 10-day-old girl into the ED for a newly developed bulge in the right inguinal area which appeared about an hour ago and is not reducible. Patient has otherwise been feeding well and afebrile prior to this. On examination, the infant is fussy but consolable and has a palpable lump in the right inguinal region which is not reducible and has no overlying erythema. What is your next step in care?
   a. Full septic workup with CBC, UA, blood culture, and CSF studies
   b. Pain control and outpatient follow up
   c. Surgery consult
   d. Ultrasound with Doppler

   Answer: D. Ultrasound with Doppler. Inguinal hernia in females may contain ovary in the hernia sack due to failure of closure of the canal of Nuck, whereas hernias in males are more likely to have intestines. All females with inguinal hernia should have an ultrasound with Doppler to assess contents and assess for ovarian viability because torsion could occur. These cases do require surgical consultation after ultrasound as ovarian hernia will require more urgent surgical management. Sending the patient home with incarceration would not be appropriate in this case. Full septic workup is not indicated in an afebrile otherwise well child without signs of infection.
3. You are caring for a 3-year-old boy who was brought in by his mother for a possible abdominal mass palpated during his bath tonight. As part of your workup you obtain a renal bladder ultrasound that demonstrated a right-sided renal mass. What is the most likely etiology of his renal mass?
   a. Congenital mesoblastic nephroma
   b. Neuroblastoma
   c. Renal cell carcinoma
   d. Wilms tumor

   Answer: D. Wilms tumor. The most common renal cancer in children is Wilms tumor. The second most common cancer in patients <12 months is congenital mesoblastic nephroma and <5 years is renal cell carcinoma.
4. Patients with nephrotic syndrome are at risk for which complication?
   a. Henoch-Schönlein purpura
   b. Supraventricular tachycardia
   c. Thrombosis
   d. Vascular malformations

   Answer: C. Thrombosis. Due to increased permeability of the basement membrane, patients lose antithrombin 3 molecules which predisposes them to thrombosis. Patients are also at risk for infections, fluid overload, and electrolyte abnormalities. In general, patients do not have increased risk of Henoch-Schönlein purpura, supraventricular tachycardia, or vascular malformations.
5. A 5-year-old male presents with 5 days of diarrhea followed by 1 day of bloody diarrhea, fever, and abdominal pain. Labs reveal acute kidney injury, thrombocytopenia, and schistocytes on smear. Further workup most likely reveals which pathogen?
   a. *Campylobacter*
   b. *E. coli* 0157
   c. *Salmonella*
   d. *Shigella*

   Answer: B. *E. Coli* 0157. Most cases of HUS occur secondary to Shiga toxin–producing *E. coli* 0157 and less frequently due to *Shigella* or *Salmonella*. Though *Campylobacter* can result in bloody diarrhea, it does not usually cause hemolytic uremic syndrome as described with acute kidney injury, thrombocytopenia, and microangiopathic hemolytic anemia.

169 Pediatric Neurologic Disorders

KEY CONCEPTS

- Altered mental status in children has a varied spectrum of clinical presentations, and may include any of the following: altered level of consciousness, excessive sleepiness, irritability, lethargy, and abnormal behavior.
- A careful and detailed history is instrumental in determining whether an event was a seizure.
- Status epilepticus constitutes a neurologic emergency that carries high morbidity and mortality rates. Initial treatment is typically with IV benzodiazepines, followed by fosphenytoin or levetiracetam. If the seizure continues to be refractory after a second-line agent, the patient may require airway management.
- A simple febrile seizure is generalized, lasts less than 15 minutes, and occurs in a neurologically and developmentally normal child between 6 months and 60 months of age.
- Breath-holding spells occur in children 6 months to 6 years of age, and are triggered by pain or emotional upset. After a trigger, the child becomes pale or cyanotic and may lose consciousness, sometimes with a brief period of clonic movements or opisthotonos that may mimic a seizure.
- Warning signs of secondary headaches include sudden onset, occurrence with straining or exertion, association with neurologic symptoms, change in headache pattern, nocturnal awakening, worsening in a recumbent position, and bilateral occipital headaches.
- If there are red flags on the history or physical exam, radiologic evaluation by computed tomography (CT), magnetic resonance imaging (MRI), or both may be necessary to rule out secondary causes of headache, such as intracranial hemorrhage, subarachnoid hemorrhage, brain tumor, or brain abscess.
- A toxicology screen is the test with the highest diagnostic yield for acute-onset ataxia in children.
- In children, 40% of ataxia cases are caused by acute cerebellar ataxia.
- Approximately 45% to 60% of all childhood brain tumors arise in the brainstem or cerebellum and can manifest with slowly progressive ataxia.
- When assessing an infant or child with motor weakness, it is important to distinguish presentations consistent with upper motor neuron pathology from lower motor neuron processes.
- Strokes represent a pediatric neurologic emergency and may be hemorrhagic or ischemic in nature. Imaging with CT or MRI can help confirm the diagnosis of stroke. Children with stroke may present with less specific signs such as headache, seizure, or alteration level of consciousness.
- Children presenting with suspected or confirmed strokes should be emergently transferred to a pediatric stroke center for timely consideration of therapies.
- Spinal cord compression is a medical emergency and requires prompt diagnosis and treatment. It may arise from trauma, infection and inflammation, or malignancy.
- The diagnosis of Guillain-Barré syndrome (GBS) is largely clinical, although lumbar puncture (LP) may be helpful in confirming the diagnosis. Patients with GBS are at risk for respiratory compromise and should be admitted to the hospital for observation and supportive care.
- The diagnosis of infant botulism is largely clinical. If there is high clinical suspicion, treatment should be initiated promptly, without awaiting laboratory confirmation. Given the risk of respiratory compromise, infants with botulism should be admitted to the hospital for observation and supportive care.
- Diagnosis of myasthenia gravis is often not confirmed in the ED. The disorder can often be treated on an outpatient basis, but patients with truncal involvement and concern for respiratory compromise should be admitted to the hospital for observation and supportive care.

1. Which of the following clinical findings of bacterial meningitis is more common in newborns compared with infants and children?
   a. Hypothermia
   b. Neck ache
   c. Rash
   d. Shaking chills
   e. Tachycardia

Answer: A. Hypothermia is more commonly found in newborn infants with bacterial meningitis compared with older infants and children with a similar disease.

2. A 15-month-old child is seen with a history of one 10-minute tonic-clonic event, which was nonfocal and accompanied by a temperature of 39°C (102.2°F) and period of illness preceding this event. Two hours after the event, the child is still unresponsive to external stimulation. Which of the following is the next step in the management of this child?
   a. Admit to the inpatient service for antibiotics after blood cultures are drawn.
   b. Admit to the inpatient service for continued observation.
   c. Carry out imaging by obtaining an MRI of the brain.
   d. Obtain a complete blood count (CBC) and urinalysis if the child does not improve in 1 hour.
   e. Order neuroimaging, a lumbar puncture, complete blood count, and blood and urine cultures to evaluate for infectious causes.

Answer: E. Additional diagnostic testing should be obtained because this child has exhibited a prolonged postictal period and has not returned to baseline.

3. What is the appropriate treatment for a 12-year-old boy with a history of epilepsy and brought from home by emergency medical services to the emergency department (ED) after multiple seizures within the 15 minutes. The child is still unconscious and seizing. Which of the following is the next step in the management of this child on arrival at the ED?
   a. Benzodiazepine
   b. Electroencephalography (EEG) and CT scan of the head
   c. Intubation using etomidate and vecuronium
   d. Methohexital
   e. Propofol

Answer: A. If the airway cannot otherwise be maintained, there is respiratory failure, or there is evidence of increased intracranial pressure, the patient should be intubated. Do not paralyze the patient unless absolutely necessary. If needed, consider short-acting neuromuscular blockers such as succinylcholine and vecuronium. The three most commonly used agents to treat convulsive status epilepticus are benzodiazepines, fosphenytoin, and levetiricam.

4. A 3-year-old boy is seen for seizure-like activity at home. He has no prior history of epilepsy and no family history of seizures. His fever has been as high as 39°C (102.2°F), and his parents have been watching him at home. Seizure activity has ceased and the history and physical examination are normal; the child is fully immunized for age. Which of the following statements is *not* consistent with appropriate counseling of parents regarding their 3-year-old?
   a. Although acetaminophen may make him feel better, it will not prevent recurrence.
   b. He is at a substantially higher risk for developing epilepsy later in life.
   c. It is safe for him to go home with the parents.
   d. Seizures with fever are common in young children and do not lead to brain damage.
   e. There is no need to start any antiepileptic medication.

Answer: B. In general, a child between 6 months and 5 years who has a normal neurologic examination and has had a brief seizure in the setting of fever can be assumed to have a simple febrile seizure. Future epilepsy is not predicted and is unlikely in children with simple febrile seizures.

5. Which one of the following statements best describes the signs and symptoms associated with pediatric brain tumors?
   a. At the time of diagnosis, most patients with brain tumors have associated symptoms, such as nausea, vomiting, visual effects, problems with walking, weakness, changes in personality or school performance, or speech changes.
   b. Headaches are never the first symptom of a pediatric brain tumor.
   c. Most pediatric brain tumors are diagnosed within the first month after symptom onset.
   d. Papilledema is one of the least common neurologic findings in children with brain tumors.

Answer: A. Of patients with brain tumors, most have some of the associated signs or symptoms noted in choice A. Headaches may be the first symptom of a pediatric brain tumor, although the frequency of this presentation increases with age. Papilledema is one of the more common neurologic findings, along with abnormal eye movements, ataxia, abnormal tendon reflexes, and abnormalities on the visual examination. CT scanning is usually not performed with the first sign of a headache unless clinically indicated.

6. A 10-year-old boy is brought in by his mother for gait imbalance. He is alert and oriented to person, place, and time, and he has a negative drug and alcohol screen. He has a persistent wide-based unsteady gait. What should be the next step in management?
   a. Head imaging, admission, and neurologic evaluation
   b. Discharge home if CT scan is negative
   c. Perform a lumbar puncture
   d. Reassurance and primary care referral
   e. Start sepsis evaluation

Answer: A. Most children with ataxia are seen in the first few days after onset, usually because of a refusal to walk, unsteadiness of arm movements, or the sudden development of a wide-based "drunken" gait. History should include recent infection, injury, inadvertent drug ingestion, or other family members with the same problem. Children with ataxia usually need admission for a complete evaluation. Consultation with a pediatric neurologist should be sought for patients in whom the cause of the ataxia is not evident on ED evaluation.

7. A 14-year-old girl with a history of migraines presents to the ED with the complaint of being unable to move her left side following a severe right-sided headache. Which of the following statements best describes her management?
   a. Imaging, analgesics, and admit to the hospital because symptoms may last for days.
   b. Immediate evaluation with neuroimaging for a stroke
   c. Inpatient vasculitis evaluation is indicated.
   d. She can be safely discharged home, and the symptoms will resolve in an hour.

Answer: B. Hemiplegic migraine is characterized by the sudden onset of hemiparesis or hemisensory loss followed by headache in the contralateral hemisphere. This type of migraine is seen more frequently in children than adults, but imaging should be obtained to rule out the diagnosis of stroke. Although symptoms usually last for hours or days, affected individuals are rarely left with permanent deficits.

8. A mother brings in her 4-year-old son for periods of lip smacking that occur multiple times per week and last several minutes. During these episodes, he may turn his head to her when she calls, but he does not speak. You suspect which of the following conditions?
   a. Complex partial seizure
   b. Generalized seizure
   c. Infantile spasm
   d. Lennox-Gastaut syndrome
   e. Simple partial seizure

Answer: A. In complex partial seizures, the patient has a change in level of awareness and may exhibit bizarre behaviors including staring, lip smacking, wandering, or picking at clothing. In simple partial seizures, the patient has no change in mentation. Lennox-Gastaut syndrome is characterized by mental retardation, multiple seizure types, and a classic EEG pattern of slow spike and wave. Infantile spasms manifest during the first year of life and consist of rapid, jackknife flexor or extensor spasms that appear in clusters.

KEY CONCEPTS

- The pediatric physis is the weakest part of the bone and more likely to separate before adjacent tendon or ligament tears, occurring more frequently during periods of rapid growth.
- Displaced supracondylar fractures are at greater risk for neurovascular injury and compartment syndrome. A lateral elbow radiograph with an elevated fat pad is suspicious for occult fracture.
- Transient synovitis has a peak presentation between 3 and 6 years of age. History, physical examination, radiographs, and laboratory values can help distinguish from septic arthritis and Lyme arthritis.
- Slipped capital femoral epiphysis (SCFE) is the posterior and inferior slippage of the proximal femoral epiphysis on the metaphysis, occurring through the epiphyseal plate. This condition affects boys at twice the rate of girls, occurring more commonly between 8 and 15 years of age. This prevalence is changing due to the increasing obesity rate.
- Lyme arthritis typically presents as mono-arthritis in two-thirds of cases, involving the knees 90% of the time. The knee tends to be swollen with limited range of motion, but pain varies and patients can usually still ambulate with a limp. Approximately 50% to 60% of patients not treated for early stages of Lyme will go on to develop Lyme arthritis, which can be their presenting symptom of the disease.
- Little League elbow describes a group of elbow injuries, including apophysitis, medical epicondylitis, and osteochondritis dissecans of the radial head and capitellum. The pediatric elbow is vulnerable to overuse injury, because it has multiple muscle and ligamentous attachments as well as six ossification centers that close at different ages of skeletal maturity.
- Apophyseal injuries of the hip occur at the multiple sites of muscle origination or insertion including on the pelvis. Athletes most at risk are dancers, distance runners, and those participating in kicking sports.
- Gymnast wrist is a chronic wrist pain affecting almost 80% of pediatric gymnasts at some point. Compressive loading and shearing forces cause physeal microfractures at the hypertrophic zone, causing physeal widening and metaphyseal irregularity in almost three-fourths of patients.

1. An 8-year-old boy presents with left wrist pain after fall off a trampoline. Wrist radiograph shows a fracture at the distal left radius extending through the epiphysis, physis, and metaphysis. According to the Salter-Harris classification, what fracture pattern is described?

a. Salter-Harris type I
b. Salter-Harris type II
c. Salter-Harris type III
d. Salter-Harris type IV
e. Salter-Harris type V

Answer: D. Salter-Harris type IV fractures involve the growth plate extending from the physis into both the epiphysis and metaphysis. The risk of growth plate damage is higher in type IV and requires orthopedic consult and commonly open reduction to stabilize.

2. A 13-year-old overweight male presents with a complaint of insidious left hip pain that now radiates down to his knee. No trauma is noted on history and the patient is not athletic. On examination the patient walks with a limp and is unable to fully range his left hip due to pain but has full range of motion of the knee. Radiograph of the left hip is likely to show what finding?

a. Inferiorly and posteriorly displaced femoral epiphysis relative to the metaphysis on a cross-table lateral view of the hip
b. Femoral neck fracture
c. Avulsion fracture at the ischial tuberosity
d. Femoral head fragmentation

Answer: A. The patient's history and examination are consistent with a slipped capital femoral epiphysis (SCFE). The diagnosis of SCFE is made with AP and lateral radiographs of both hips as it can be present in 18% to 50% of patients. Femoral head fragmentation is consistent with Legg-Calvé-Perthes. Medial joint space widening can be seen with a joint effusion in the setting of septic arthritis, transient synovitis, or Lyme arthritis.

3. Which of the following statements regarding Lyme arthritis is true?

a. Lyme arthritis is an early localized sign of Lyme disease.
b. Synovial fluid WBC count ranges between 50,000 and 100,000 cells per microliter.
c. Azithromycin is the treatment of choice.
d. Monoarthritis is noted in two-thirds of cases, involving the knee 90% of time.

Answer: D. Lyme disease is caused by the spirochete *Borrelia burgdorferi* and is now considered the most common vector-borne illness in North America and Europe. The primary vector is the *Ixodes* tick. Lyme arthritis is a finding in late Lyme disease. Synovial fluid WBC count ranges are typically 10,000 to 50,000. Treatment recommendations depend on prior treatment failures and additional Lyme manifestations.

4. Which of the following statements regarding gymnast wrist are true?

a. Initial radiographs of the wrist demonstrate a distal radius Salter-Harris II fracture.
b. Gymnast wrist only affects about 10% of gymnasts.
c. Radiographic evidence is seen in almost three-fourths of patient's with the clinical diagnosis.
d. The weight load to the athlete's wrist is generally greater than 30 times their body weight.

Answer: C. Distal radial epiphysitis, also known as "gymnast wrist," is a chronic wrist pain affecting almost 80% of pediatric gymnasts at some point. Radiographs demonstrate widening of the distal radial physis. This condition is commonly seen in gymnasts who bear weight through their upper extremities, increasing the weight load to their wrist from 2 to 16 times their body weight. Treatment is generally conservative with cessation from weightbearing activities.

KEY CONCEPTS

- Awareness of differences in pediatric pharmacokinetics and specific drug toxicities is of critical significance for the safe and effective use of medications in children.
- Avoid prescription and over-the-counter (OTC) cough and cold medications in children because these agents have limited efficacy data and may cause harm.
- Counsel parents about the management of fever and appropriate indications and proper use of antipyretics.
- Perform a risk assessment (for prescription drug abuse and diversion) prior to prescribing opioid analgesics and, when indicated, limit prescribing to the lowest duration and amount possible.
- A multifaceted approach using clinical support systems and readily available reference tools is essential for the delivery of optimal emergent pediatric care.

1. A 12-year-old girl is being discharged from the emergency department (ED) after having her displaced forearm fracture reduced. Which one of the following actions would be considered potentially dangerous?

a. Prescribe codeine.
b. Consult state prescription drug monitoring programs (PDMP) and prior medical and pharmacy records.
c. Use a screening tool, such as CRAFFT, to evaluate for high-risk behaviors.
d. Prescribe a minimum number of days to decrease risk for misuse and diversion.

Answer: A. Codeine is no longer recommended for pain in pediatric patients. This is due to the variation in pharmacogenomics (CYP 2D6) and metabolism to morphine, which can lead to either poor pain control or unintentional opioid toxicity. In adolescent patients, it is recommended that the patient be evaluated for high-risk factors that may lead to opioid use, misuse or diversion. This includes checking the state's PDMP and screening to evaluate the risk for drug abuse and addiction. There are currently no pediatric-specific guidelines for prescribing opioids, but it is generally recommended not to prescribe opioids for acute pain more than 3 to 7 days in duration. Providers should also emphasize a nonopioid, multimodal approach to pain control, including nonpharmacologic methods and adjunct medications such as nonsteroidal antiinflammatory drugs, acetaminophen, or neuroactive agents.

2. What is a unique pharmacokinetic property observed in the pediatric population?

a. Infants have decreased absorption of topically applied medications.
b. Neonates and infants have higher total body water and larger volumes of distribution and extracellular fluid.
c. All drug metabolizing enzymes are at full capacity at birth.
d. All maternal drugs should be considered not safe when breast feeding.

Answer: B. Young children can have significant differences in the pharmacokinetic properties based on their physiology. Absorption can be impaired and delayed based on gut pH and motility. Metabolizing enzymes mature at different rates during the first year of life. Young children have a higher total body water and larger volumes of distribution and extracellular fluid. There are several factors to consider when assessing the risk/benefit profile of taking medications while breast-feeding, including limited human data, maternal comorbidities, therapeutic alternatives, properties of the drug, and the benefits of continued breast-feeding, when possible. LactMed is an excellent resource to determine the safety of the drug during breast-feeding.

3. A healthy, fully vaccinated 11-month-old male infant presents to your department for evaluation of a fever of 38.4°C (101.1°F) this morning that responded to a dose of acetaminophen. The physical examination reveals a well-appearing, afebrile infant with clear lungs. The patient has an unremarkable evaluation and, during the discharge process, his parents ask for advice regarding fever management. Which of the following statements is correct?

a. Aspirin should not be used in children younger than 15 years because of the risk of Reye syndrome.
b. Cool water baths and creams should be used to supplement antipyretics, even if they cause some discomfort to the patient.
c. Ibuprofen cannot be used in this age group because of ongoing renal development.
d. Over-the-counter (OTC) antipyretics are standardized, contain similar products and formulations, and are thus interchangeable.

Answer: A. Counsel parents and caregivers about the management of fever and appropriate indications for and proper use of antipyretics. There is no need to cause discomfort with external cooling methods for fever control. Ibuprofen should not be used in children younger than 6 months because of ongoing renal development. The formulations and dosing of OTC antipyretics are varied and cannot be used interchangeably. The correct answer is A because of the risk of Reye syndrome with

aspirin administration during a viral illness in children younger than 15 years.

4. Which of the following statements regarding pediatric pharmacokinetics is correct?
   - **a.** A thinner stratum corneum and increased body surface area contribute to a greater risk for systemic toxicity from dermally administered drugs.
   - **b.** Ceftriaxone administration in neonates results in increased bilirubin production.
   - **c.** Because of minimal differences in the volume of distribution and renal development, weight-based dosing of gentamicin without attention to age is sufficient.
   - **d.** Quicker gastric emptying and decreased gastric pH in neonates increase systemic absorption of enterally administered medications.

Answer: A. Awareness of differences in pediatric pharmacokinetics and specific drug toxicities is of critical significance for the safe and effective use of medications in children. The dosing of gentamicin needs to account for age-based differences in renal development in addition to weight-based differences in distribution volume. Decreased gastric emptying times and an increased pH can prolong exposure to medications before they pass the pylorus. An immature blood-brain barrier can result in kernicterus from bilirubin displacement by ceftriaxone. The correct answer is that a thinner stratum corneum and increased body surface area contribute to a greater risk for systemic toxicity from dermally administered drugs.

5. Which of the following steps can be taken to reduce pediatric dosing errors?
   - **A.** Adoption of electronic health records with clinical support tools to decrease weight-based dosing errors
   - **B.** Calculation of weight-based dosing for all emergent medications administered in code situations as opposed to using a validated quick reference guide
   - **C.** Limiting hospital pharmacist presence in the emergency department to avoid delays in bedside care
   - **D.** Medication reconciliation should occur with just the patient present to limit primary caregiver influence

Answer: A. A multifaceted approach using clinical support systems and readily available reference tools are essential for the delivery of optimal emergent pediatric care. Caregivers are an integral part of medication reconciliation. ED pharmacists have been proven to increase departmental accuracy in pediatric medication management. Validated quick reference guides for code drug administration have been shown to decrease errors and improve efficiency.

KEY CONCEPTS

- The ultimate determination of whether abuse has occurred can take days or weeks. Emergency clinicians should focus on recognizing possible abuse, treating medical injuries, and establishing a safe disposition for the child.
- Completely undress infants and preverbal children for the physical examination; pay particular attention to the skin, ears, mouth and oral cavity, scalp, fontanel, and genitalia.
- Consider abuse routinely for sentinel injuries in young children without an independently witnessed traumatic mechanism. Sentinel injuries include the following: bruising in children younger than 6 months old; bruising on the torso, ears, neck, jaw, cheek, or eyelid; oral injuries; patterned cutaneous injuries; subdural hematoma; long-bone fractures in infants; intra-abdominal injuries; and rib fractures.
- Consider abuse for children when family violence (child abuse, intimate partner violence, elder abuse, or animal abuse) is recognized in the child's home.
- The goal of diagnostic testing is to identify additional clinically or forensically significant injuries or medical entities that may present similarly to abuse.
- Use objective, nonaccusatory, matter-of-fact statements to communicate concern for abuse.
- Emergency clinicians in the United States, Canada, and many other countries are legally mandated to report reasonable concerns for abuse to public child protective services (CPS) agencies.

1. A 12-month-old girl is referred to the emergency department (ED) because her daycare noticed that she cries whenever she is picked up and there was crepitus in the left side of the chest. They deny any witnessed trauma in the daycare facility. The mother arrives and appears concerned. On physical examination, you find some circular contusions on the inner aspects of both her upper arms. The mother states this occurred from a fall a week ago. Which of the following should be the next step(s) in the patient's management?
   a. Conduct a skeletal survey, and report to child protective services (CPS).
   b. Determine whether the arm contusions are of the same age.
   c. Perform a dedicated retinal examination to determine if the child has retinal hemorrhages and, if she does, perform a head computed tomography (CT) scan.
   d. Perform a humerus x-ray to determine if there is a humerus fracture.

Answer: A. Contusions on the inner aspect of the upper arm are atypical injuries in light of the mother's claim of a fall for the child. This pattern of injury is often related to being held tightly around the arm. Chest wall pain in a child could indicate a rib fracture. Chest x-ray is insensitive for rib fractures, which are very uncommon at this age unless caused by inflicted trauma. This should raise your suspicion of child physical abuse. In general, children younger than 2 or 3 years old with suspected inflicted injuries should be evaluated with a skeletal survey.

2. A 6-year-old, pre-pubertal girl presents to the emergency department (ED) at midnight after her parents noticed malodorous vaginal discharge over the last 3 days after visiting her grandfather 2 weeks ago. The patient is tired and irritable at the time of presentation. Which of the following are indicated in the ED?
   a. Empiric treatment for gonorrhea and chlamydia
   b. Obtain a forensic interview.
   c. Post-exposure prophylaxis for HIV.
   d. Report to child protective services (CPS)

Answer: D. Genital discharge can be caused by both sexually transmitted and non-sexually transmitted infections, as well as other inflammatory processes. Several days of discharge implies a subacute timeframe for a potential abusive exposure. For subacute sexual abuse, ED management should include fulfilling the mandate to report to CPS and ensuring that the child has a safe disposition. A genital examination may be attempted in a cooperative patient and can assist in assuring the child that they are normal and healthy, but the examination should not be forced on the child if they are reluctant; it is unlikely to identify transient evidence of sexual contact. The detailed forensic interview should be conducted by a trained interviewer and may be deferred until the patient is rested. HIV PEP is unlikely to be effective 2 weeks after the assault. Empiric treatment for gonorrhea or chlamydia should not be given to pre-pubertal children until a sexually transmitted illness is definitively confirmed.

3. Which of the following implies multiple episodes of trauma at different times?
   a. Bruises that are different colors.
   b. Mixed density subdural hematoma on head CT.
   c. Rib fractures with robust periosteal reaction are identified the day after a child presents with sudden coma and subdural hematoma.
   d. Too numerous to count intraretinal hemorrhages and liver laceration with elevated hepatic transaminases

Answer: C. Periosteal reaction and other signs of fracture healing are not visualized until approximately 1 week after injury, and persist for weeks or months, whereas sudden coma implies an acute, traumatic event. Mixed-density subdural hematomas can be observed in the acute, subacute, and chronic phases. The color of bruises is affected by several factors in addition to the age of a bruise, including the child's complexion, depth of the bruise, natural variation, and even the perceptions of the observer. Too numerous to count intraretinal hemorrhages do not persist beyond a few days, but this could be consistent with a hepatic laceration.

4. Which of the following is very concerning for sexual abuse in a pre-pubertal child?
   a. A new history of constipation and dilated anal sphincter (>1 cm)
   b. An 8-year-old girl whose anogenital exam is completely normal reports that her camp counselor "put his wee-wee in my butt."
   c. Erythema of the hymen and posterior fourchette.
   d. Thin, fragile epithelium surrounding the labia with petechiae and subcutaneous bleeding.

Answer: B. A child's report of sexual abuse is highly concerning and is often the strongest evidence of abuse. A normal exam neither confirms nor excludes the possibility of sexual abuse, even with penetration. Dilation of the anal sphincter is nonspecific, especially in the setting of constipation. Like many mucosal tissues, erythema can occur for many reasons, including trauma, foreign bodies, lack of hygiene or even normal variation. "Cigarette-paper" skin with petechiae or hemorrhage suggests the presence of *lichen sclerosis et atrophicus*.

5. In a 2-week-old infant that is not yet crawling, which of the following is most highly concerning for abuse?
   a. Clavicle fracture with callous
   b. Lateral rib fracture with callous
   c. Parietal skull fracture without callous
   d. Torn lingual frenulum with scant bleeding

Answer: D. Lingual frenulum tears are very concerning for abuse in pre-mobile infants. These injuries heal quickly, and the presence of bleeding excludes the possibility of birth injury. Several self-limited injuries can result from the birth process, even in deliveries that are not recognized as particularly traumatic. These injuries can be clinically subtle and are not-uncommonly missed in the newborn nursery. Rib fractures are uncommonly seen from birth and are more common in larger infants or difficult deliveries. The presence of obvious callous on the rib fractures is consistent with a fracture that is at least 7 to 14 days old. Clavicle and skull fractures are not uncommon parturitional injuries. Unlike other fractures, skull fractures do not develop callous and their age cannot be estimated.

173 Complications of Pregnancy

KEY CONCEPTS

- Miscarriage is defined as the spontaneous termination of pregnancy before 20 weeks of gestational age.
- Approximately 25% of women experience some bleeding during pregnancy.
- Most miscarriages are due to uterine malformations or chromosomal abnormalities.
- Beta hCG discriminatory levels for detecting intrauterine pregnancy are usually considered to be 6500 mIU/mL for transabdominal ultrasonography and 1000 to 2000 mIU/mL for transvaginal ultrasonography.
- Treatment of the patient with incomplete miscarriage includes expectant management, medical management, or surgical evacuation.
- All patients with pregnancy loss require patient education, return precautions, and support through the grieving process.

1. An 18-year-old G1P0 at 8 weeks of gestation presents with abdominal pain and vaginal bleeding for 1 day. Her serum human chorionic gonadotropin (hCG) level is 3700 IU/L. Transvaginal ultrasonography does not reveal an intrauterine pregnancy (IUP) or mass. The pelvic examination is remarkable for a closed cervical os and a small amount of blood. Which of the following should be the next step?
   a. Coagulation panel
   b. Gynecologic consultation
   c. RhoGAM, 300 μg intramuscularly
   d. Repeat hCG in 2 days
   e. Serum progesterone level

Answer: B. Ultrasonographic detection of an IUP is likely at hCG levels higher than 1500 to 2000 IU/L. A negative ultrasound, with an hCG level of 3700 IU/L, is concerning for an ectopic pregnancy or miscarriage. β-hCG levels should peak at the 7- to 10-week range, with mean values of 50,000 IU/L. A persistently low hCG level is even more suspicious for ectopic pregnancy, and gynecologic consultation is warranted.

2. A 36-year-old G3P2 presents with painless vaginal bleeding during the past hour. She is at 33 weeks of gestation, and her pregnancy has been uncomplicated. Her bleeding lasted approximately 20 minutes. Which of the following should not be included in the management of this patient?
   a. Baseline hemoglobin level and platelet count
   b. Immediate complete pelvic examination
   c. Immediate obstetric consultation
   d. Intravenous fluid resuscitation
   e. Transvaginal ultrasound

Answer: B. Painless late pregnancy bleeding is placenta previa until proven otherwise. Digital or instrumental probing of the cervix should be avoided until the diagnosis is excluded via ultrasound. An injudicious vaginal examination can precipitate severe hemorrhage in the patient with an asymptomatic or minimally symptomatic placenta previa.

3. Which of the following sexually transmitted diseases does not require treatment during pregnancy?
   a. *Chlamydia trachomatis*
   b. Herpes simplex virus
   c. *Neisseria gonorrhoeae*
   d. Pelvic inflammatory disease (PID)
   e. *Trichomonas vaginalis*

Answer: E. Although trichomoniasis has been associated with increased prematurity, treatment with metronidazole has not been shown to improve fetal outcomes, so emergency clinicians should counsel patients and consider deferring treatment in asymptomatic pregnant women until after 37 weeks' gestation.

4. A 28-year-old G3P0 at 34 weeks of gestation presents with new-onset seizures. The patient has multiple seizures in the ED and is noted to be hypertensive at a blood pressure of 164/87 mm Hg. Which of the following should be administered next?
   a. Benzodiazepines
   b. Calcium gluconate
   c. Labetalol
   d. Magnesium sulfate
   e. Pralidoxime

Answer: D. Magnesium sulfate has little antihypertensive effect but is the most effective anticonvulsant, preventing recurrent seizures while maintaining uterine and fetal blood flow. A loading dose of 4 g intravenous (IV) magnesium, followed by 2 g IV/hr, is recommended.

5. A 36-year-old G5P4 at 8 weeks of gestation presents to the ED with painless vaginal bleeding. Ultrasonography shows products of conception in the uterus. The pelvic examination shows a dilated cervix and a moderate amount of blood in the vaginal vault. Which of the following is appropriate management?
   a. Surgical evacuation
   b. Expectant management
   c. Medical management with misoprostol
   d. None of these
   e. All of these

Answer: E. All of these are appropriate management options for an incomplete miscarriage.

6. Which of these is the most common surgical emergency in pregnancy?
   a. Abruptio placentae
   b. Appendicitis
   c. Cholecystitis
   d. Ovarian torsion
   e. Ruptured peptic ulcer

Answer: B. Appendicitis is the most common surgical emergency in pregnancy. Clinical presentations may be atypical, particularly during the second half of pregnancy.

7. For the treatment of nausea and vomiting in pregnancy, which of the following has been found to have some associated risk of fetal abnormalities?
   a. Diclegis (delayed-release combination of doxylamine and pyridoxine)
   b. Ginger
   c. Metoclopramide
   d. Ondansetron
   e. Promethazine

Answer: D. Ondansetron has been shown in studies to be linked with fetal cardiac abnormalities, as well as cleft lip and palate. Diclegis is first-line pharmacologic treatment of nausea and vomiting in pregnancy; metoclopramide and promethazine are alternatives if it fails. Ginger, a nonpharmacologic treatment, has been shown to be safe and effective at a dose of 250 mg QID.

KEY CONCEPTS

Asthma

- The treatment goal for a pregnant woman with an acute asthma exacerbation is to prevent fetal hypoxia by keeping maternal oxygen saturation above 95%. Inhaled beta-agonists and corticosteroids are first-line emergency department treatment and are considered safe for use in pregnancy.

Cardiac Disease

- Hypertensive emergency in pregnancy is defined as acute-onset persistent hypertension with systolic blood pressure greater than 160 mm Hg or diastolic blood pressure greater than 110 mm Hg that is persistent for greater than 15 minutes. In these cases, antihypertensive therapy should be administered as soon as reasonably possible, and no later than 30 to 60 minutes after diagnosis with a target blood pressure of 140 to 150 mm Hg systolic and 90 to 100 mm Hg diastolic. The drugs of choice are oral nifedipine, IV hydralazine, and IV labetalol.
- The risk for acute coronary syndrome and acute myocardial infarction is increased in pregnant women compared to age-matched controls. The most common cause of AMI in the pregnant women is spontaneous coronary artery dissection. Treatment is similar to the nonpregnant agent, although P2Y12 receptor inhibitors should be avoided and fibrinolytic agent use should be carefully considered in women close to term.

Anemia

- Anemia in pregnancy is defined as a hemoglobin less than 11. Serum ferritin is the most accurate lab value for diagnosing iron deficiency anemia in pregnancy. Women with mild iron deficiency anemia can be started on daily iron supplementation while those with severe iron deficiency in the second and third trimester can be referred for IV iron infusion.
- Sickle cell disease causes maternal complications including more frequent pain crises, and increased risk of venous thromboembolism and preeclampsia. Treatment of pain crises is the same as in nonpregnant patients with the exception that hydroxyurea is contraindicated due to known teratogenicity.

Epilepsy

- Gravid patients with epilepsy have a tenfold risk of death compared to pregnant women without epilepsy. Many antiepileptic drugs (AEDs) have known teratogenicity and levetiracetam and lamotrigine are considered the safest agents during pregnancy. Treatment of status epilepticus is with benzodiazepines followed by a phenytoin as first-line AED and levetiracetam as second-line AED.

Endocrine

- Pregnant patients with type 1 diabetes mellitus (T1DM) are recommended to transition to insulin during gestation to achieve HgbA1C values <6%.
- Hypoglycemia is most common in the first trimester, and up to 40% of pregnancies are marked with at least 1 episode of severe hypoglycemia.
- Comorbid obesity increases the risk of cesarean section and venous thromboembolism.
- Graves disease commonly rebounds in the immediate postpartum period with thyrotoxicosis.
- Radioiodine is strongly contraindicated for the treatment of hyperthyroidism in pregnancy.
- All gravid patients with new-onset nephrolithiasis should be screened for hypercalcemia. Treatment of hypercalcemia with bisphosphonates is contraindicated in pregnancy.

Psychiatric Disorders

- Prenatal discontinuation of methamphetamines and other stimulants, although desirable, can cause depression and psychosis.
- Maternal and neonatal outcomes in women on opioid agnostic therapy show decreased rates of neonatal abstinence syndrome (neonatal opioid withdrawal syndrome).

Inflammatory Disorders

- Antiphospholipid syndrome (APS) in lupus patients is characterized by deep vessel clotting, pregnancy-related morbidity, and positive anticoagulant serum markers. Catastrophic APS has rapid-onset small vessel thrombosis, multiorgan dysfunction, and a high maternal mortality rate.

Renal Disease

- Management of chronic kidney disease (CKD) in pregnancy can be treated with intensified hemodialysis (increased length of treatment time or increased frequency) to improve fetal outcome.
- Patients post renal transplantation have fertility rates that return to normal within 6 months.

Infectious Disorders

- Moderate to severe anemia in pregnancy in an HIV-infected mother should prompt a workup for tuberculosis.
- During pregnancy, penicillin is the only known effective treatment for congenital syphilis, and pregnant patients with penicillin allergy should be desensitized and treated with penicillin.
- Lamivudine given in late pregnancy to women with high viral loads of HBV DNA reduces viral transmission when given in conjunction with HBV vaccine and immune globulin.
- Obstetric hemorrhagic complications, including DIC and shock and subsequent need for transfusion, are more common with HCV infection.

1. A 28-year-old G2P1 woman at 26 weeks of gestation presents with a recurrent asthma flare. Vital signs are temperature 36°C, heart rate 110 beats/min, blood pressure 120/60 mm Hg, respiratory rate 28 breaths per minute and O_2 saturation 96%. She has diffuse expiratory wheezes. Arterial blood gas reveals Po_2 90 mm Hg, Pco_2 40 mm Hg, and pH 7.34. Which of the following statements best describes the issues in management with this patient?
   a. Corticosteroids are contraindicated.
   b. Most patients with asthma improve during pregnancy.
   c. She has a metabolic acidosis.
   d. Treatment and discharge are likely.
   e. Inhaled beta-agonists are first-line therapy.

Answer: E. Beta-agonists followed by corticosteroids are the mainstay of asthma therapy during pregnancy. During pregnancy, one third of asthmatics worsen, one third improve, and one third stay the same. Blood gas interpretation must take into account the "normal" alkemia of pregnancy with a Pco_2 of 30 to 32 mm Hg and a compensatory HCO_3 level of 18 to 20 mEq/L. This patient has a relative "hypoventilation" and may indeed need admission for close observation.

2. A pregnant woman in the second trimester presents to the ED with a complaint of a 5/10 headache and has a blood pressure of 165/100. She reports regular prenatal follow-up. She reports she has a history of mild hypertension dating to before her pregnancy and her blood pressures have been in the 150/90 range during this pregnancy and that she is not on any anti-hypertensive medications. A repeat blood pressure after 15 minutes is 165/100. Which of the following best describes the issues in management of this patient?
   a. The patient should have been placed on an anti-hypertensive medication by her OBGYN for her blood pressures of 150/90.
   b. A full workup should be completed before treating the patient's hypertension.
   c. An IV must be established quickly in order to administer IV anti-hypertensives.
   d. Oral nifedipine can be used to lower the patient's blood pressure.
   e. ACE inhibitors can be used to lower the patient's blood pressure.

Answer: D. ACOG states that mild chronic hypertension in pregnancy defined as less than a systolic blood pressure of 160 or a diastolic blood pressure of 110 does not need to be treated. Hypertensive emergency in pregnancy is defined as acute-onset persistent hypertension with systolic blood pressure greater than 160 mm Hg or diastolic blood pressure greater than 110 mm Hg that is persistent for greater than 15 minutes. In these cases, antihypertensive therapy should be administered as soon as reasonably possible, and no later than 30 to 60 minutes after diagnosis with a goal blood pressure of 140 to 150 mm Hg systolic and 90 to 100 mm Hg diastolic. Oral nifedipine, IV hydralazine, and IV labetalol are the drugs of choice in this situation, and IV access attempts should not delay treatment as oral nifedipine can be given. ACE inhibitors are contraindicated in pregnancy.

3. An HIV-positive woman G3P2 at 15 weeks gestation presents with increasing fatigue. She is mildly tachycardia but otherwise with normal vital signs and has normal fetal heart tones. Her labs show a WBC of 10.1, Hgb 6.2, Hct 18, and platelets of 60,000. You should consider what other comorbid condition?
   a. Antiphospholipid syndrome
   b. Hepatitis B infection
   c. Thyrotoxicosis
   d. Tuberculosis

Answer: D. The diagnosis of new infection or activation of latent tuberculosis in pregnancy can be challenging, given that common symptoms can be mimicked in normal pregnancy including tachypnea and fatigue. Moderate to severe anemia in an HIV-infected mother should prompt a workup for TB.

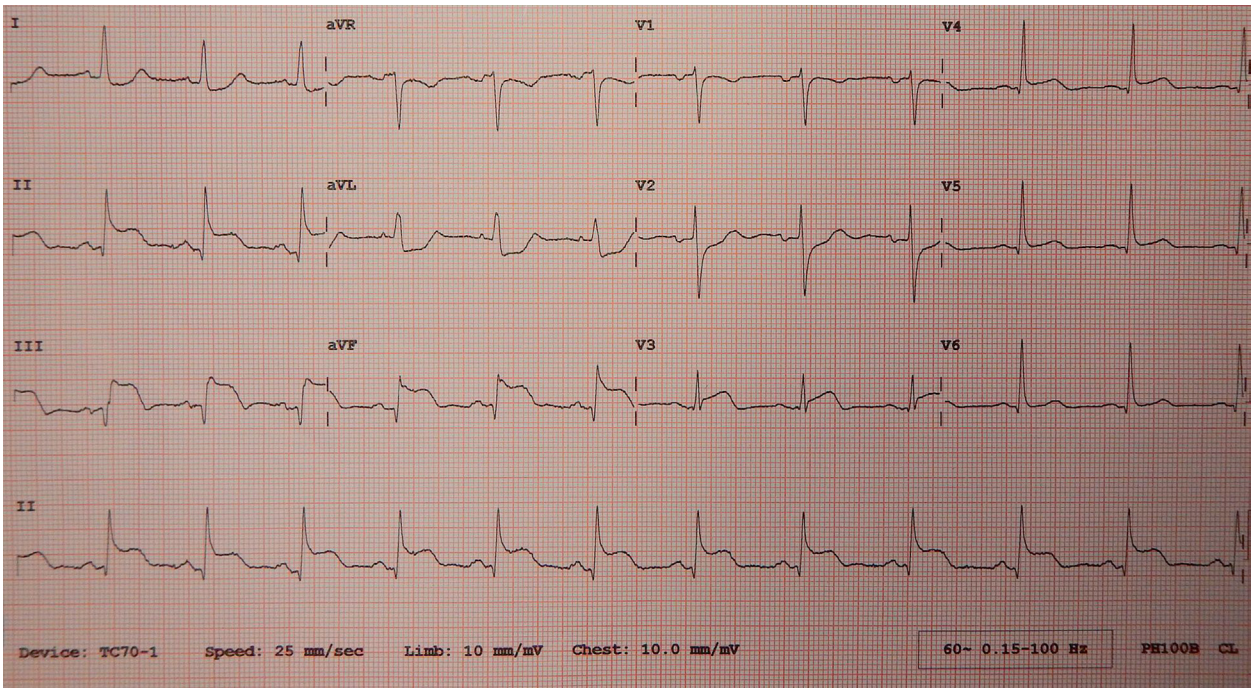
4. A 24-year-old G1P0 woman at 8 weeks' gestation presents with anxiety, sweating, and palpitations. She has a history of Graves disease. She is tachycardic and hypertensive. Her TSH is <0.02 and her free T4 level is 9.2 ng/dl (normal 0.8 to 2.8 ng/dL). Which of the following treatments is most appropriate?
   a. Alpha-adrenergic blockade
   b. Methimazole
   c. Propylthiouracil
   d. Radioiodine

Answer: **C.** Radioiodine is strongly contraindicated in pregnancy because it will destroy the fetal thyroid gland. PTU is the treatment of choice in the first trimester because of the risk of adverse congenital drug effects of methimazole during organogenesis. Beta blockade remains the treatment of choice for abnormal vital signs, not alpha-adrenergic blockade.

5. A 20-year-old G2P1 woman at 20 weeks' gestation with a known history of SLE presents with acute onset of R hemiplegia and dysarthria. One hour post onset of symptoms, her head CT is normal. Her symptoms are most likely caused by which of the following?
   a. Catastrophic antiphospholipid syndrome (CAPS)
   b. Cerebritis
   c. Eclampsia
   d. Stroke

Answer: D. The most common arterial presentation of neurologic symptoms in lupus patients is stroke; positive aPL results are found in up to 20% of ischemic stroke patients younger than 50 years of age. CAPS is characterized by rapid-onset, small vessel thrombosis, multiorgan dysfunction or failure (renal, hepatic), and a high maternal mortality rate, but not as a large vessel occlusion. Patients with lupus are not at increased risk of cerebritis during pregnancy, and eclampsia does not present as a focal hemiplegic deficit.

6. A 41-year-old G4P3 woman with known hypertension presents at 36 weeks with chest pain and the ECG that follows. Which of the conditions listed below is the most likely diagnosis?
   a. Hypertensive emergency
   b. Normal physiologic ECG changes of pregnancy
   c. Spontaneous coronary artery dissection
   d. Stanford type B dissection



English: 12 lead ECG showing inferior and right ventricular infarct

Date 19 May 2014, 06:20:17
Source Own work
Author James Heilman, MD

Answer: C. ECG changes in normal pregnancy include T wave flattening, T wave inversion (mainly in lead III), and nonspecific ST changes during pregnancy, as well as ST depression during labor induction for cesarean section. This ECG demonstrates an inferior MI. Although aortic dissection is more common than ACS in pregnant patients, a type B dissection would not involve the coronary outflow tract. The most common cause of ACS in pregnancy is spontaneous coronary artery dissection.

7. Perinatal maternal complications in women with sickle cell disease include which of the following?
   - **a.** Deep vein thrombosis
   - **b.** Maternal infections
   - **c.** Pain crises
   - **d.** Preeclampsia
   - **e.** All of the above.

Answer: E. Maternal complications are common in patients with SCD; these include preterm labor, premature rupture of membranes, maternal infections, more frequent pain crises, thrombosis, preeclampsia, and increased need for cesarean delivery.[38,39] Due to these complications, pregnant women with SCD have a sixfold increased risk of maternal death compared with controls.

8. A patient with poorly controlled epilepsy is at highest risk of breakthrough seizures during which time period of pregnancy?
   - **a.** 1 to 2 weeks postpartum
   - **b.** During delivery
   - **c.** Post 32 weeks
   - **d.** She is not at increased risk of seizing

Answer: B. Delivery and the first 24 hours postpartum are the most likely times for a seizure to occur, with a nine fold greater incidence of seizure than during pregnancy in general.

9. Which of the following autoimmune disorders most consistently worsens during pregnancy?
   - **a.** Addison disease
   - **b.** Graves disease
   - **c.** Multiple sclerosis
   - **d.** Rheumatoid arthritis

Answer: A. Pregnancy has an immunosuppressive effect. Because of this, pregnancy can have a variable effect on autoimmune disorders. Rheumatoid arthritis consistently improves, Graves disease has a transient worsening in the first trimester but improves later in pregnancy. Patients with MS see a decrease in the frequency and severity of exacerbations. Addison disease is exacerbated, especially in the third trimester, because of the biggest need for cortisol replacement.

175 Drug Therapy in Pregnancy

KEY CONCEPTS

- Chemically induced birth defects are responsible for approximately 1% to 3% of anomalous births.
- Gestational age is crucial in determination of the impact of any given exposure, especially during organogenesis (days 21–56 of fetal life), when major body organs are formed.
- Human data on teratogenicity and fetal toxicity of medications are often limited, and causal associations are difficult to determine, especially with newer medications.
- In general, the health of the fetus is directly related to the health of the mother, and drugs should be given when the maternal benefits outweigh the risks to the fetus.
- Certain medications should be avoided during pregnancy because they are known teratogens or cause potential toxic effects in the newborn; these include anticonvulsants, warfarin derivatives, NSAIDs, sulfonamides, fluoroquinolones, and ACE inhibitors. If there are no alternatives to these agents, it is recommended to use the lowest dose for the shortest duration possible.

1. Which of the following statements is true regarding the teratogenic risk of medications administered during pregnancy?

a. A drug should be given when maternal benefits outweigh the risks to the fetus.
b. The FDA assigns one of five letters (A, B, C, D, and X) to classify teratogenic risk.
c. The fetus is at greatest risk of anatomic defects during the third trimester.
d. Most drugs do not enter the breast milk due to protein binding.
e. Most drugs have epidemiologic studies confirming the risks in pregnancy.

Answer: A. In general, the health of the fetus is directly correlated with the health of the mother and necessary medications should not be withheld for fear of harming the fetus. The FDA no longer uses letter categories to describe teratogenic risk, which was a flawed system. The Pregnancy and Lactation Labeling Rule (PLLR), effective in 2015, changed the content and format for prescription drug labeling to better summarize the risks of drugs in pregnancy, lactation, and reproduction. The fetus is at greatest risk during organogenesis in the first trimester and functional development of the CNS in the 10th to 17th weeks of pregnancy. Unfortunately, current studies are insufficient to determine reliable and accurate risks in the use of medications in pregnancy. Most drugs passively diffuse into the breast milk.

2. Which of the following drugs is a known teratogen?

a. Albuterol
b. Hydroxycobalamin
c. Penicillin VK
d. Pyridoxine
e. Valproic acid

Answer: E Several antiepileptic drugs (AEDs) are known teratogens, including valproic acid, carbamazepine, ethosuximide, phenobarbital, phenytoin, and topiramate. Valproic acid is associated with the most frequent serious adverse effects on the pregnancy and fetus compared with other AEDs. Newer-generation AEDs, such as lamotrigine and levetiracetam, have not been associated with increased risk of congenital malformations compared to controls but the data are limited. Maternal mortality is higher in women with a seizure disorder and control of seizures is felt to be important. Use of monotherapy in the lowest effective anticonvulsant dose is optimal. The other drugs are safe in pregnancy.

3. A 28-year-old woman who is 7 months pregnant was well until she experienced sudden onset of shortness of breath and left-sided sharp chest pain one hour prior to arrival. The pain is worse on inspiration and is not related to exertion, although her dyspnea becomes worse with any activity. She is alert and appears in mild distress. Her pulse is 112 beats/min, RR is 24 breaths/min, and BP is 110/68 mm Hg. Her temperature is 100.8°F (32°C), and O_2 saturation is 92%. Lung examination reveals diminished breath sounds on the left base with scattered wheezes. Cardiac examination reveals tachycardia. Chest radiograph reveals elevated right hemidiaphragm but no infiltrates. ECG reveals generalized T wave inversions. Which of the following treatments is most appropriate?

a. Administer a low-molecular-weight heparin subcutaneously and start oral warfarin.
b. Administer intravenous ceftriaxone and azithromycin.
c. Administer nebulized bronchodilators and start prednisone.
d. Administer oxygen and start low-molecular weight heparin
e. Place a central line, and start early goal-directed therapy for sepsis.

Answer: D On the basis of the information provided, the patient most likely has a pulmonary embolus. The patient is hypoxic and requires oxygen administration and anticoagulation. This may be accomplished with heparin, low-molecular-weight heparin (LMWH), or warfarin. Warfarin, a vitamin K antagonist, is teratogenic and is contraindicated in pregnancy. The treatment of pulmonary embolism is LMWH preferentially or heparin. Heparin is preferred if the patient is hemodynamically unstable due to its rapid onset and short half-life. The patient has no symptoms of an infectious process; choices B and E are therefore not appropriate. Bronchodilators and steroids are indicated for asthma but not for thromboembolic disease. Choice C is therefore not appropriate.

4. Which of the following drugs may be associated with complications in the newborn when used at term?
   a. All of these
   b. Nitrofurantoin
   c. Nonsteroidal antiinflammatory drugs
   d. Propranolol
   e. Sulfonamides

Answer: A. All the choices have been associated with complications in the newborn when used at term. Sulfonamides compete with bilirubin for protein-binding sites, leaving large amounts of free bilirubin to diffuse freely into the brain. This results in bilirubin deposition in the infant's brain, thus causing kernicterus. Nitrofurantoin at term has been associated with hemolytic disease of the newborn. Nonsteroidal antiinflammatory drugs are associated with premature closure of the ductus arteriosus. Propranolol at term has been associated with neonatal hypoglycemia, respiratory depression, and neonatal jaundice.

5. A 7-week pregnant (by LMP) patient arrives in the emergency department. She is complaining of severe nausea and vomiting. She is unable to keep any solids or liquids down. Her heart rate is 115, BP 110/75, Temp 98.8°F, RR 15, and glucose is 98. The patient is given IV fluids and antiemetics with improvement of her symptoms. Which of the following medications is the most appropriate for the outpatient management of this patient once her symptoms are controlled and she is tolerating fluids well?
   a. Doxylamine 10 mg and pyridoxine 10 mg once per day
   b. Haloperidol 2 mg twice a day
   c. Omeprazole 20 mg every morning
   d. Ondansetron orally disintegrating tablets 4 mg every 8 hours
   e. Prochlorperazine 10 mg three times per day

Answer: A. This patient's symptoms likely represent nausea and vomiting of pregnancy or possible hyperemesis gravidarum. The preferred initial home treatment is Diclegis, which is a combination drug of doxylamine and pyridoxine (10 mg/10 mg). The non-combination version is: half-tab (21.5mg) of OTC Unisom and 10 mg pyridoxine (Vitamin B_6). There are some inconsistent studies suggesting a possible link with long term use of Zofran (Ondansetron) and cardiac abnormalities and cleft palates. Omeprazole is not first-line treatment for hyperemesis gravidarum, it is for the treatment of GERD. Prochlorperazine is teratogenic in animals and is not considered first-line treatment. Haloperidol is an antipsychotic and not indicated in this patient.

6. In determining a causal link between a specific drug and congenital malformations, which of the following may be viewed as confounding factors?
   a. Genetic background of the fetus
   b. Maternal illicit drug use
   c. Presence of maternal illness
   d. Presence of nutritional deficits
   e. All of the above.

Answer: E. The process of establishing teratogenicity of a substance is often flawed. Much of our current knowledge on teratogenicity has been derived from case reports, case-controlled studies, and cohort studies, which are inherently weak in establishing a causal link. These reports are often complicated by a number of confounding factors, which make a causal link difficult to establish. The presence of any of the listed choices may confound results. In the presence of maternal illness, for example, the outcome of pregnancy may be related to the medical condition and not the medication.

KEY CONCEPT

- Most ED deliveries require only basic equipment to cut and clamp the umbilical cord and dry and suction the infant. However, the ED should have additional equipment and trained staff should be available to care for a newborn requiring further resuscitation.
- Women in labor who present to the ED are generally best cared for in the obstetric suite. Women with the urge to push or with the head of the infant crowning are at imminent risk of delivery, which should take place in the ED.
- The Braxton Hicks contractions of false labor do not escalate in frequency or duration like the contractions of true labor. When in doubt, external electrical monitoring of uterine activity can rule out true labor.
- Preterm labor is defined as uterine contractions with cervical changes before 37 weeks of gestation. Treatment includes tocolytics and fetal maturation therapy combined with bed rest and hydration.
- Premature rupture of membranes (PROM) occurs after 37 weeks' gestation. Its management depends on several factors, including gestational age and fetal maturity, presence of active labor, presence of infection or placental abruption, and degree of fetal well-being or distress.
- Preterm PROM should be treated with antibiotics to prevent infection (chorioamnionitis).
- The first stage of labor averages 8 hours in nulliparous women and 5 hours in multiparous women. Throughout labor, ongoing assessment of fetal well-being is important, and continuous external electrical monitoring helps identify fetal distress.
- Ultrasonography provides crucial information regarding pending delivery, including fetal viability, lie, and presentation.
- The fourth stage of labor refers to the first hour after delivery of the placenta and is a critical period during which postpartum hemorrhage is most likely to occur.
- Deliveries complicated by dystocia, malpresentation, or multiple gestations are life-threatening emergencies. The emergency clinician should develop strategies to treat each of these potential complications of delivery. Please see the following link for a video demonstration of maneuvers used to treat shoulder dystocia: https://www.hopkinsmedicine.org/gynecology_obstetrics/education/training/shoulder-dystocia.
- When a prolapsed cord occurs with a viable infant, cesarean section is the delivery method of choice. If surgical delivery is available, maneuvers to preserve umbilical circulation should be instituted immediately. The mother is placed in the knee-chest position, with the bed in the Trendelenburg position, and instructed to refrain from pushing to avoid further compression of the cord. The presenting part is then manually elevated off the cord. Elevation is maintained until the baby can be delivered surgically.
- Uterine atony accounts for 75% to 90% of cases of postpartum hemorrhage. Administration of a uterotonic, such as oxytocin in conjunction with massage usually provide enough stimuli to control bleeding.
- Approximately 10% of postpartum hemorrhage cases are due to retained placental tissue. Treatment requires manual removal of the remnant placental tissue.
- Pelvic bleeding postpartum can be difficult to control without hysterectomy. When available, embolization of bleeding vessels by an interventional radiologist has reported success rates of 95% to 100%.
- Maternal complications of labor and delivery include obstetric trauma, uterine inversion and rupture, amniotic fluid embolism, coagulation disorders, and infections. Many of these problems can initially be managed in the ED while awaiting obstetric consultation.

1. After episiotomy, which is the first maneuver that a clinician should attempt when treating shoulder dystocia?
   a. McRoberts maneuver
   b. Fracture the clavicle
   c. Deliver the posterior shoulder
   d. Cesarean section

Answer: A. The most important first step is to use McRoberts maneuver (see Fig. 176.10). Maternal leg flexion to a knee-chest position may disengage the anterior shoulder. This maneuver "walks" the pubic symphysis over the anterior shoulder and flattens the sacrum, helping the fetus pass through the birth canal, one shoulder at a time.

2. Vaginal delivery of a frank breech is best accomplished by which of the following actions?
   a. Pulling hard on the presenting part
   b. Pushing the presenting part back through the cervix
   c. Supporting the presenting part and giving the cervix time to dilate.
   d. Administering oxytocin

Answer: C. The head is the widest part of the fetus. Pulling on the presenting part causes the head to become trapped behind an incompletely dilated cervix.

3. Which of the following medications is effective at reducing postpartum uterine bleeding?
   a. Tranexamic acid
   b. Terbutaline
   c. Magnesium sulfate
   d. Gentamycin

Answer: A. Tranexamic acid (1 gr IV) is effective at reducing bleeding. Oxytocin and fundal massage should also be used.

4. Which antibiotic combination provides the best initial coverage for postpartum endometritis?
   a. Cephalexin and vancomycin
   b. Ciprofloxacin and metronidazole
   c. Clindamycin and gentamicin
   d. Amoxicillin and nitrofurantoin

Answer: C. Antibiotics should include anaerobic coverage because endometritis is usually caused by genital flora.

5. The first stage of labor ends with which of the following?
   a. Delivery
   b. Ruptured membranes
   c. A fully dilated cervix
   d. Placental delivery

Answer: C. The first stage of labor is the cervical stage, ending with a completely dilated, fully effaced cervix.

KEY CONCEPTS

- Female trauma patients of reproductive potential should be screened for pregnancy and assumed to be pregnant until proven otherwise.
- Management of life- and limb-threatening injury in the mother comes first. Saving the mother provides the best chance of saving the baby.
- Even in the stable pregnant trauma patient, the fetus is at increased risk of morbidity and mortality; therefore, the fetus should be monitored by cardiotocography continuously for a minimum of 4 hours after any trauma.
- The fetus is considered viable between 22 to 24 weeks' gestation (>500 grams). This usually corresponds to when the fundus is at or above the umbilicus.
- Alterations in anatomy and physiology that occur during pregnancy alter the pattern of injuries in trauma and the clinical findings related to blood loss, which may mask injuries, making a systematic approach essential.
- Keeping the mother tilted 15 to 30 degrees to the left, or manually displacing the uterus leftward, in any pregnancy of 20 weeks' gestation or greater is recommended to alleviate hypotension due to aortocaval compression in order to improve maternal and fetal perfusion.
- Resuscitative hysterotomy should be initiated for a potentially viable fetus (fundus above the umbilicus) within 4 minutes and completed in 5 minutes after the onset of cardiopulmonary arrest and no return of spontaneous circulation (ROSC) for optimal maternal and fetal benefits.
- The use of ionizing radiation to the pregnant patient, including CT and plain radiography, should be minimized, but imaging should not be withheld if it may provide significant diagnostic information. In certain circumstances ultrasound, MRI, or a period of observation can preclude the need for ionizing radiation.

1. A 30-year-old female who is 28 weeks' gestation presents to the emergency department after being a restrained passenger in a low-speed MVC. Which of the following is the best indicator of fetal outcome in this scenario?

a. Abdominal tenderness
b. Cardiotocographic monitoring for 4 hours
c. Maternal blood count and arterial blood gas results
d. Maternal vital signs
e. Ultrasonography

Answer: B. For women with mild blunt trauma, fetal outcome is not predicted by maternal vital signs, abdominal tenderness, blood tests, or ultrasound results. Only cardiotocographic monitoring for a minimum of 4 hours is useful to predict fetal outcome. Fetal monitoring can identify fetal distress and the need for further intervention.

2. Which of the following factors is most concerning in the presentation of a pregnant trauma patient?

a. Diastasis of the symphysis pubis
b. Electrocardiogram (ECG) findings of Q waves in III and aV_F
c. Hematocrit 34% in third trimester
d. Hypotension in the third trimester
e. Respiratory alkalosis in the third trimester

Answer: D. Blood pressure declines in the first trimester, levels out in the second trimester, and then returns to nonpregnant levels during the third trimester. In pregnancy, minute ventilation increases, leading to hypocapnia. Therefore, a partial pressure of arterial carbon dioxide ($Paco_2$) of 35 to 40 mm Hg may indicate inadequate ventilation and impending respiratory decompensation. The physiologic anemia of pregnancy, resulting from a 48% to 58% increase in plasma volume and only an 18% increase in red blood cells, results in hematocrits of 32% to 34% by gestational age of 32 to 34 weeks. Electrocardiographic changes include a left-axis shift averaging 15 degrees, caused by diaphragm elevation. Consequently, flattened T waves or Q waves in leads III and aV_F may be seen.

3. A 26-week gravid woman presents to the emergency department (ED) after a moderate-speed motor vehicle collision (MVC). The patient is without complaints, and her vital signs are as follows: blood pressure, 100/60 mm Hg; heart rate, 100 beats per minute; and respiratory rate, 18 breaths per minute. Ultrasound examination shows good fetal movement, with a fetal heart rate of 150 beats per minute. What is the appropriate disposition for this patient?

a. Consult obstetrics for a minimum of 4 hours of cardiotocographic monitoring.
b. Perform a FAST examination and if negative discharge home.
c. Consult trauma and obstetrics for admission and serial examinations.
d. Consult trauma surgery for exploratory laparotomy.
e. Discharge the patient with close follow-up with obstetrics.

Answer: A. Placental abruption results when the inelastic placenta shears away from the elastic uterus during sudden deformation of the uterus. Because deceleration forces can be as damaging to the placenta as direct uterine trauma, abruption can occur with little or no external sign of injury to the abdominal wall. For the viable fetus (more than 22 to 24 weeks' gestation), monitoring is the next step.

4. A 26-year-old, 30-week gestation woman presents unresponsive with cardiopulmonary resuscitation (CPR) in progress after a high-speed motor vehicle collision (MVC). The patient lost her vital signs 3 minutes before arrival in the emergency department (ED). What is the most appropriate next step in the management of this patient?
   - **a.** Transfer to the operating room for emergency cesarean section by obstetrics
   - **b.** Resuscitative hysterotomy with vertical midline incision
   - **c.** Thoracotomy with cardiac massage
   - **d.** Administer epinephrine and continue ACLS guidelines for cardiac arrest
   - **e.** Defibrillate immediately at 300 J

Answer: B. In the event of maternal cardiopulmonary arrest, resuscitative hysterotomy is indicated in any pregnancy of greater than 20 weeks' gestation (above the umbilicus) to improve both maternal and fetal survival. It should ideally be performed within 4 minutes of loss of spontaneous circulation and completed by 5 minutes of no ROSC. The most experienced physician available should perform the procedure.

5. A 28-week pregnant patient comes in after a gunshot wound to the epigastrium. Her blood pressure is 95/30 and her heart rate is 125 after receiving 1 liter IV fluids by EMS en route. What is the optimal management for this patient?
   - **a.** Explore the wound locally for entrance and exit wounds.
   - **b.** Expectant management with serial abdominal examinations.
   - **c.** Intraoperative exploratory laparotomy or laparoscopy by a trauma surgeon.
   - **d.** Fetal monitoring for a minimum of 4 hours

Answer: C. The gravid uterus affects the injury pattern seen with penetrating trauma to the upper abdomen with the probability of harm to the bowel, liver, or spleen at almost 100%. The gravid uterus tends to protect abdominal organs from trauma but substantially increases the likelihood of bowel injury from penetrating trauma to the upper abdomen. A trauma surgeon/trauma service should be activated or consulted. With persistent hypotension after an initial 1-liter IV fluid bolus, the massive transfusion protocol should be initiated in anticipation of the expected course of action. The patient will need to go to the operating room for an exploratory laparotomy.

178 Care of the Geriatric Patient

KEY CONCEPTS

- The population is aging, with 10,000 Americans turning 65 every day, resulting in an increasing proportion of ED patients that are elderly. Older ED patients have the greatest resource use, longest lengths of stay, and highest admission rates of any age group.
- Delirium, especially hypoactive delirium, is underrecognized by emergency clinicians; the diagnosis of delirium should prompt investigation for life-threatening emergencies, including infection, metabolic abnormalities, and acute coronary syndrome (ACS).
- Nonspecific complaints, such as generalized weakness, are among the top ten presenting complaints for older ED patients, and are often the harbinger of serious underlying illness.
- Atypical presentations of disease result in diagnostic challenges and can lead to misdiagnosis.
- Many older patients with ACS present without chest pain, especially females and those older than 85 years. ACS in older adults is more often complicated by acute heart failure due to age-related decreases in left ventricular compliance. Current recommendations regarding medical and revascularization therapy in those with non-STEMI and STEMI have no age limitations.
- Almost one-third of older ED patients presenting with abdominal pain are ultimately found to have a surgical condition causing the pain; therefore, there should be a low threshold for diagnostic imaging.
- Mortality from sepsis approaches 40% for patients older than 85 years, with respiratory and genitourinary infections being the most common sources. Older adults with infection are less likely to present with fever or leukocytosis and may have SIRS-negative sepsis.

1. An 80-year-old female with history of COPD and arthritis comes to the ED following an episode of loss of consciousness while seated at the table after eating lunch. The patient and her husband, who witnessed the event, deny any symptoms prior to the event and deny any recent illness or symptoms. Her vitals are normal, and her physical exam is unremarkable. Which of the following is the most likely etiology of the syncopal episode?
   a. Cardiac event
   b. Cerebrovascular accident
   c. Infectious process
   d. Hypoglycemic reaction

Answer: A. Syncope is more likely to have a serious etiology in older adults, with 40% ultimately found to have a cardiac etiology. Hypoglycemia is highly unlikely because she does not have diabetes and just ate lunch. Cerebrovascular accident and infection are potential causes of syncope but are much less likely given that she has been well, asymptomatic, and her examination is normal.

2. Which of the following is a key feature of delirium that helps differentiate it from dementia?
   a. Use of inappropriate language with swearing
   b. Inability to correctly name the current date
   c. Physical agitation with combativeness
   d. Inability to focus or maintain attention

Answer D. Delirium and dementia can be difficult to diagnose and can exist together, with dementia being a predisposing factor to the development of delirium. A key feature of delirium is inattention with the inability to maintain focus. Confusion with the inability to name the current date, agitated behavior with combativeness, and the use of inappropriate language with swearing can all be seen in both dementia and delirium and therefore do not help with differentiation.

3. The presentation of infection in older patients differs from the presentation in younger patients. Older patients are more likely to have which of the following?
   a. Elevated inflammatory markers, such as erythrocyte sedimentation rate and C-reactive protein
   b. Fever
   c. Left shift of the white blood cells
   d. Normal white blood cell count

Answer: D. Older patients are more likely to have a normal white blood cell count, as well as a normal differential. They are less likely to have fever and signs or symptoms from the infection process. They are more likely to have complications and suffer long-term morbidity or mortality.

4. Which of the following statements is true regarding acute myocardial infarction (AMI) in older adults?
   a. Atypical presentations portend a more benign course.
   b. Atypical presentations increase with age.
   c. Painless AMIs occur more commonly in men.
   d. The absence of chest pain is rare.

Answer: B. Atypical presentations of AMI increase with age and are of equal severity as typical cases. Women have painless presentations more than men, and present with abdominal pain almost one-third of the time. The absence of chest pain is not uncommon.

5. A 74-year-old man with a history of hypertension and hyperlipidemia presents to the ED with acute onset of severe, nontraumatic, left flank pain. His vital signs include a heart rate of 72 beats/min and blood pressure of 120/70 mm Hg. Which approach to his care would be most appropriate?
   a. Arrange for him to be in the observation unit overnight for serial abdominal examinations to help clarify his diagnosis.
   b. Consider abdominal aortic aneurysm only after the patient's urinalysis and renal ultrasound are negative for nephrolithiasis.
   c. Diagnose him with nephrolithiasis without any further testing.
   d. Evaluate for a possible abdominal aortic aneurysm.

Answer: D. Vascular disease prevalence increases with age. Vascular emergencies remain some of the most time-sensitive and highly morbid causes of abdominal pain in the older patient and should always be considered first, particularly for abrupt-onset symptoms.

KEY CONCEPTS

- Do not let a low-impact mechanism, patient cognitive impairment, or vital signs within the range of normal reduce your pretest probability of significant injury in an older patient.
- Age-specific trauma alert criteria improve the care of injured older adults.
- Vital signs, including tachycardia and hypotension, are unreliable to detect hemodynamic instability in older adults. Ultrasound is a helpful tool to assess volume status in the older trauma patient.
- Older patients are at high risk of hypothermia and develop pressure ulcers more rapidly than younger patients. Unnecessary spinal immobilization (cervical collars and backboards) causes pressure ulcers, respiratory distress, and delirium in this population.
- Clinical decision tools for radiographic imaging have generally excluded older patients. A low threshold for imaging should be used for older adults with trauma, and computed tomography (CT) should be used as the primary modality, except for extremity imaging.
- Falls are the leading cause of injury-related death in older adults, and ground-level falls can result in major injuries. Assessing the patient's future fall risk, home safety, and home resources is important prior to leaving the ED or hospital.
- Rib fractures and pulmonary contusions are associated with poor outcomes in older patients. ICU care should be considered for those with two or more rib fractures or pulmonary contusions.
- Older adults with hip fractures have improved survival on a dedicated orthogeriatric service. Consider transfer of patients to hospitals where these services are available.
- Routinely screen for elder abuse. A valid screening question is: "Has anyone close to you tried to hurt you or harm you recently?" Another query is: "Does anyone at home scare you or threaten you?"
- All older adults with fractures should be assessed for osteoporosis/osteopenia, and malnutrition as leaving this untreated reduces healing and increases morbidity and mortality.

1. A 76-year-old man is brought to the emergency department (ED) after a fall at home. The paramedics tell you that he slipped in the shower and struck his head on a tile floor. The patient is awake and alert. He is unsure if he lost consciousness and complains of headache. When you obtain a computed tomography (CT) scan of his head, it reveals evidence of a subdural hematoma. The paramedic tells you that he has a history of severe congestive heart failure (CHF) and takes warfarin and furosemide daily. When he returns from CT, he seems slightly confused. Laboratory tests are pending. What is the best initial treatment?

a. Administer prothrombin complex concentrate (PCC)
b. Administer tranexamic acid
c. Administer protamine sulfate
d. Transfuse 2 units fresh frozen plasma

Answer: A. Prompt reversal of anticoagulation is important. Specific considerations for reversing coagulation abnormalities in older trauma patients are the volume of reversal agents required and corresponding risk of fluid overload. PCCs require minimal volume compared with fresh-frozen plasma (FFP) but are costly. To reverse anticoagulation fully, 1 to 2 L of FFP may be required, presenting a limitation to rapid reversal in older patients at risk of fluid overload. Platelets, tranexamic acid, and DDAVP will not reverse the elevated international normalized ratio caused by warfarin.

2. Which of the following is the most common cervical spine fracture in geriatric trauma patients?

a. Compression fracture
b. Jefferson fracture
c. Spinous process fracture
d. Type 2 odontoid fracture

Answer: D. Because of the relative immobility of the cervical spine related to degenerative joint disease, the most common level of cervical spine injury in older adults is C1 to C3, a higher level than in younger patients. Among these upper cervical spine fractures, the most common is a type 2 odontoid fracture.

3. An 80-year-old woman presents after a motor vehicle accident. Her vital signs on arrival are: heart rate of 112 beats/min, blood pressure of 88/65 mm Hg, respiratory rate of 18 breaths/min, and oxygen saturation measured by pulse oximetry (Spo_2) of 98%. The paramedic tells you that she has a history of CHF and takes furosemide daily. FAST does not reveal a source of bleeding. There are no obvious external injuries or bleeding. The IVC collapses more than 50% with inspiration. Which of the following treatments is most appropriate to address the patient's tachycardia and hypotension while searching for its cause?

a. Give a bolus of 500 mL warmed normal saline.
b. Give 2 units of non–crossmatched type O blood.
c. Run an infusion of warm normal saline wide open.
d. Start norepinephrine at a low dose.

Answer: A. If the patient has an obvious source of hemorrhage, use of non–crossmatched type O blood would be appropriate as a first step, starting with 1 unit. However, this patient has no signs of bleeding on FAST or external exam. The most prudent approach in this hypotensive older trauma patient is controlled boluses of warmed isotonic fluids, with frequent assessment of physical examination, vital signs, pulse oximetry, lactate, and IVC status. Resuscitation with crystalloid may correct the patient's hypotension, obviating the need for transfusion. Hypotension should be immediately addressed, and treatment should not be delayed for type-specific blood. Given her history of CHF, a wide open infusion of normal saline without an endpoint should not be administered. Vasopressors such as norepinephrine should not be given to any trauma patient, except in extreme circumstances of persistent hypotension despite adequate volume resuscitation.

4. A 76-year-old man presents after slipping and falling down several stairs at home. He notes his doctor just started him on a medication for dizziness (meclizine). The patient is awake and alert and denies loss of consciousness after the fall. He complains of chest pain and has left chest wall tenderness on examination. When you obtain a CT scan of his chest, it reveals three contiguous lateral rib fractures. The patient's pain is only moderately relieved by 0.15 mg/kg of morphine sulfate IV. What is the most appropriate disposition for this patient?

a. Admission for pain control with IV opioids
b. Admission with anesthesia consultation for pain management
c. Continued observation in the ED for 6 additional hours of IV opioid management, followed by reassessment
d. Discharge home after a bupivacaine rib block for pain.

Answer: B. Geriatric trauma patients with severe pain from rib fractures often require hospitalization to allow adequate and safe pain management, and those with flail segments or larger numbers of rib fractures may require intensive care unit admission. Pain control is of particular importance because rib fractures will lead to splinting and atelectasis and increase the risk of pneumonia. Analgesia can be administered with IV opioids via patient-controlled analgesia or via rib block with anesthetics. An anesthesia consultation is appropriate for the older rib fracture patient without pain relief from parenteral opioids.

5. Late in the evening, an 82-year-old woman is transferred to your ED for a burn to the hand. Your ED is the local burn center. Her vital signs on arrival are as follows: heart rate of 94 beats/min, blood pressure of 110/70 mm Hg, respiratory rate of 12 breaths/min, and oxygen saturation measured by pulse oximetry (Spo_2) of 97%. You note circumferential first and second degree burns to the entire left hand with some sparing over the flexor areas of the 2-5th digits. She has no respiratory distress and there are no other injuries on exam. The adult grandson who cares for her says she was cooking and stuck her hand into boiling soup to grab a spoon. He also notes her dementia is worsening and it has been harder to care for her. She has no known history of neuropathy or self-injury. In addition to local wound care, what is the next step in the management of this patient?
   a. Pain control prescription, and discharge to home with family.
   b. Pain control prescription, and discharge to home with family. Put in a call to Adult Protective Services to follow up on the patient.
   c. Analgesics, IV fluid resuscitation per the Parkland formula, and admission.
   d. Analgesics, and admission for further investigation of elder abuse vs caregiver fatigue.

Answer: D. Older adults are at risk for inadvertent burns while cooking, but a circumferential burn with signs of a clenched fist (sparing over the flexor areas of the fingers) suggests the hand was forced into the liquid. This woman is not safe for discharge both due to her injuries (circumferential hand burn) and her safety risk. The patient should be admitted for continued medical care and a full social work assessment and collaboration with Adult Protective Services. Most police agencies prefer to be notified at the time of injury as well if there is concern for criminal abuse, just as you would for a younger victim of interpersonal violence. As isolated hand burns are less than 10% BSA, there is no indication to start Parkland formula resuscitation. The main intervention is keeping her safe from possible abuse.

6. A 67-year-old construction manager is the restrained driver in a head on collision with a roadside barrier at 45mph. Airbags deployed and he complains of right knee pain, chest pain, and back and neck pain. Vital signs are heart rate of 85 beats/min, blood pressure of 130/67 mm Hg, respiratory rate of 12 breaths/min, and oxygen saturation measured by pulse oximetry (Spo_2) of 97%. He is oriented to self, time, and place but does not remember everything about the accident or how he veered off the road. He has a history of diabetes, hypertension, and cirrhosis and takes a daily aspirin, lisinopril, hydrochlorothiazide, metformin, and glipizide. Which laboratory tests should be ordered in addition to the basic trauma evaluation (coagulation studies, a type and cross, alcohol and drug testing, a complete blood count and a basic metabolic panel)?
   a. Erythrocyte sedimentation rate
   b. Troponin, hepatic function panel, ammonia, and urinalysis
   c. Creatinine kinase.
   d. Venous blood gas.

Answer: **B.** This patient was in a high speed MVA without a clear inciting cause. He requires both a trauma evaluation for any injuries but also an evaluation for the cause of the collision. The initial workup therefore includes a bedside glucose (he is on two oral antihyperglycemic medications), a troponin to assist in evaluating whether his chest pain is from a cardiac cause or chest trauma, an ammonia and hepatic function level to evaluate for cognitive difficulties from cirrhosis, and a urinalysis to check for infection, volume depletion, and other causes. An erythrocyte sedimentation rate is not be specific for injury. A venous blood gas is likely unnecessary as he is oxygenating well and has no history of pulmonary disease. A creatinine kinase to evaluate for rhabdomyolysis is unnecessary as the patient was not immobilized and so is not at risk for this.

7. A 72-year-old woman who runs a dance instruction studio presents for 3 days of L hip pain. She slipped and fell on the hip and initially required assistance to get up, but has been ambulating since the injury. She presents to your ED because the pain is not improving with acetaminophen and time. On exam, there is no angulation, bruising, deformity or limb length discrepancy. She tolerates passive range of motion of the L hip but is tender over the greater trochanter and with pelvic compression. The initial x ray demonstrates osteoporosis but no fractures. An MRI shows a stable pelvic ring fracture that does not require operative management. What is the best recommendation for pain management?
   a. Prescribe two weeks of ibuprofen 600mg tabs to use every 6-8 hours.
   b. Prescribe tramadol and a strong muscle relaxer such as Valium to prevent spasms.
   c. Prescribe a walker to decrease her weight bearing on that limb, a referral to early physical therapy, and topical nonsteroidal anti inflammatory cream.
   d. Prescribe two weeks of oxycodone 5mg tablets to use every four hours.

Answer: **C**. A multimodal approach that also assists with ambulation and healing is appropriate. Non-steroidal anti-inflammatories can cause renal injury or peptic ulcer disease if used at high doses or for a longer time period. Try to limit use to lower doses (200-400mg) and under a week at a time. Tramadol and skeletal muscle relaxers are not indicated for use in older adults, and this combination is especially concerning for increasing the patient's risk of another fall and injury. Oxycodone may be needed if her pain is severe, but should not be prescribed without a bowel regimen.

KEY CONCEPTS

- Those with a chronologic age of 65 years or older are commonly referred to as older adults (or the elderly), but physiologic age is more indicative of a drug's therapeutic or toxicologic effect. Besides age, overall patient assessment should include organ function, comorbidity, and functional status to guide drug dosing.
- Pharmacokinetic and pharmacodynamic changes that occur with age need to be considered to optimize drug dosing and minimize toxicity in older adults. In most cases, a "start low, go slow" approach is recommended. Multiple or repeated dosing is more likely to lead to drug accumulation compared to single doses in the emergency department (ED).
- Polypharmacy is common in older adults, predisposing them to adverse drug effects, drug interactions, and functional and cognitive impairment. Some of these medications do not have legitimate indications or may be inappropriate.
- Published lists of potentially inappropriate medications, such as the Beers list and the STOPP and START criteria, can help to identify potentially problematic medications; however, there are limited studies to enable extrapolation to the ED setting.
- Anticoagulation-related hemorrhagic complications are common in elderly patients, particularly if drug accumulation occurs in patients taking renally eliminated direct-acting oral anticoagulants.
- Older adults often present to the ED with altered mental status. Drug-related causes such as anticholinergic medication burden should be considered in the differential diagnosis.
- Geriatric patients with pain-related complaints are less likely to receive analgesics in the ED compared to younger adults, placing them at risk for poor pain control. Dosing of opioids should be cautious, with frequent monitoring and titration. Given the availability of alternative opioids, the use of meperidine should be avoided.
- A growing number of institutions have pharmacists practicing in the ED. In geriatric EDs, there is a great opportunity to integrate and consult with pharmacists, given the myriad drug therapy issues that can lead to suboptimal care.

1. Which of the following pharmacokinetic change in older adults is associated with increased plasma concentration of drug?

a. Decrease in absorptive capacity in an orally administered medication
b. Decreased glomerular filtration rate in a renally eliminated medication
c. Increase in gastric pH in a medication that is a weak base
d. Increased liver blood flow in a high–extraction ratio medication
e. Increased total body water in hydrophilic medications

Answer: B. Decrease in glomerular filtration rate is common with age. This leads to decreased elimination and increased plasma concentration of a drug. All other options listed would lead to decreased plasma drug concentrations. Liver blood flow, glomerular filtration rate, total body water, and absorptive capacity decrease with age. However, gastric pH increases.

2. Which of the following is the most appropriate strategy for analgesic provision in an older patient with renal insufficiency and severe abdominal pain in the emergency department?

a. Hydromorphone 0.5 mg IV, followed by the same dose in 15 minutes, if needed
b. Ketorolac, 30 mg IV qid
c. Meperidine, 50 μg IV every 2 hours
d. Morphine, 4 mg IV every 2 hours
e. Oxycodone/acetaminophen, 5/235 mg orally every 2–4 hours

Answer: A. One strategy that has been used successfully in older adults is a two-step hydromorphone protocol. Patients are given 0.5 mg IV hydromorphone, which is repeated in 15 minutes if the patient desires another dose when asked, "Do you want more pain medication?" Hydromorphone does not have active metabolites that would accumulate with renal insufficiency, so it would be an appropriate option for severe pain. Ketorolac should be avoided in older adults with renal insufficiency because the inhibition of prostaglandins can decrease renal blood flow. Meperidine use is no longer recommended because of a neurotoxic metabolite and accumulation in renal insufficiency. Morphine has an active metabolite that also accumulates in renal insufficiency. Oxycodone-acetaminophen given orally is not appropriate for severe acute pain due to delayed onset of effect. Bioavailability of oral medications may further be delayed in older adults.

3. Which of the following agents should be avoided for outpatient use in an older patient being discharged from the emergency department?
   - **a.** Diphenhydramine
   - **b.** Loratadine
   - **c.** Acetaminophen
   - **d.** Clindamycin
   - **e.** Metronidazole

Answer: A. Diphenhydramine and other first-generation antihistamines should be avoided in older patients given the potential for anticholinergic toxicity. The use of diphenhydramine in the emergency department for severe allergic reactions is appropriate; however, longer term outpatient use is not recommended. This effect is much less common with second-generation antihistamines such as loratadine. Acetaminophen is generally considered safe in older patients. Antibiotics such as clindamycin and metronidazole have the potential to cause drug-drug interactions with other home medications; however, by themselves they are not typically problematic.

4. Which of the following medications prescribed on discharge is most likely to interact in an older patient taking warfarin?
   - **a.** Albuterol inhaler for asthma
   - **b.** Ciprofloxacin for urinary tract infection
   - **c.** Insulin NPH for hyperglycemia
   - **d.** Oxycodone for fracture pain
   - **e.** Promethazine for nausea

Answer: B. Studies of harmful drug interactions in older adults have shown that the initiation of antibiotics in patients taking warfarin can lead to bleeding. Ciprofloxacin can lead to an elevation in the international normalized ratio (INR) due to disruption of vitamin K synthesis in patients taking warfarin. Thus, more frequent monitoring is required when ciprofloxacin is initiated. The other options listed do not interact with warfarin. However, oxycodone and promethazine can be harmful because of central nervous system effects, potentially leading to falls.

5. Which of the following statements is true regarding neurologic emergencies in older adults?
   - **a.** Ampicillin is added for the treatment of bacterial meningitis for additional coverage against *Enterococcus faecalis.*
   - **b.** Anticholinergic medication use is not associated with delirium.
   - **c.** Haloperidol, 10 mg IV, is recommended for acute agitation due to underlying psychiatric illness.
   - **d.** In patients with ischemic stroke who are older than 80 years, alteplase should not be administered if the patient presents 4 hours after the onset of symptoms, according to guidelines.
   - **e.** Large doses of haloperidol given via the IV route have been associated with QTc interval prolongation

Answer: E. There is a US Food and Drug Administration (FDA) black box warning regarding the use of haloperidol and QTc interval prolongation. This risk is increased with large doses given intravenously. Polypharmacy in older adults also puts them at risk for drug interactions due to concomitant QTc-prolonging agents. Doses of haloperidol should be less than 5 mg in most older adults. Ampicillin is added for bacterial meningitis to cover *Listeria monocytogenes.* Although recent evidence has extended the 3-hour time window to 4.5 hours for receiving alteplase after the onset of stroke in a small subset of patients, those older than 80 years are not eligible, according to guidelines. Anticholinergic medications are known to be associated with delirium and sedation.

KEY CONCEPTS

- Elder mistreatment, which includes physical abuse, sexual abuse, neglect, emotional/psychological abuse, abandonment, financial/material exploitation, and self-neglect, is common and may have serious medical and social consequences.
- Elder mistreatment is under-recognized by emergency clinicians and underreported to the authorities.
- Signs suggestive of potential elder abuse and neglect that should be recognized by emergency clinicians may exist in the medical history, physical examination, and medical/laboratory markers.
- Emergency clinicians should be vigilant in assessing for the possibility of elder abuse or neglect and routinely ask elderly patients about mistreatment, even in the absence of signs and symptoms. Screening protocols may be helpful.
- Using a team-based approach including social workers and other emergency department (ED)-based professionals may improve elder abuse detection, and Emergency Medical Services can play a critical role.
- ED management of elder abuse should include the following: treating acute medical and psychological issues, ensuring patient safety, and proper reporting to the authorities. Trauma-informed care should be provided.
- Emergency clinicians should hospitalize elderly patients who are in immediate danger or implement a care plan that prevents them from having any contact with the suspected abuser(s) but must respect the wishes of an older adult with decision-making capacity who refuses interventions and desires to return to an abusive situation. Trauma-informed care should be provided.
- Emergency clinicians should document completely and accurately the history and all physical findings in cases of suspected elder abuse or neglect, as this documentation may be critical to ensure justice for the victim.

1. A 70-year-old man who lives alone but has 12 hours per day of a home attendant is brought to the emergency department (ED) by his daughter, who found him in his caregiver's home with severe bruising. On examination, he had multiple bruises at varying stages of healing on the chest and arms, as well as a possible pattern injury—a bruise in the shape of a rectangle—across his left chest. What should you do?

a. Ask the daughter to make a report to adult protective services (APS) because she is the one who found him.
b. Contact the long-term care ombudsman.
c. Do not make a report because he has a long history of falls.
d. File a report with APS or law enforcement if your suspicion for abuse is high.

Answer: D. The injuries noted are of significant concern for elder abuse. The patient should be asked questions in a nonthreatening atmosphere, separate from the caregiver. If the emergency clinician still has high suspicion, a report should be made to APS and/or law enforcement. The physician has the obligation to make the report. The long-term care ombudsman should be contacted only if the suspected abuse occurs in a facility, such as a nursing home.

2. A 72-year-old female is brought to the emergency department (ED) by paramedics for "not eating for 5 days," according to her family. The paramedics noted that she was covered in urine and feces and that the house was filthy, and she was found lying on the floor. On examination, she is noted to have a stage 4 pressure ulcer on her sacrum, which appears infected. Which of the following are concerning for neglect?

a. Delay in seeking care
b. Evidence of dehydration
c. Poor hygiene
d. Pressure ulcer
e. All of the above

Answer: E. All of the above signs are potential markers for neglect.

3. A 69-year-old man with dementia presents to the emergency department (ED) with pneumonia and delirium. He is very confused, and his cognition is markedly worse than his baseline. You are called to the bedside because his adult daughter is at the bedside asking him to sign a legal form giving her the deed to his house. What are the next steps in advocating for your patient?

a. Perform an assessment of cognition and capacity and document it on the chart.
b. Request that the hospital lawyer and notary be present.
c. Tell her that there must be two witnesses present for the form to be legal.
d. Tell the daughter that if there are other children, they must be present for the form to be valid.

Answer: A. When an elderly patient has delirium or is confused because of another medical condition, he or she may lack the capacity to enter into a legal agreement or sign a legal document. The patient described is clearly too confused to understand all of the ramifications of signing his or her home over to another person. Confusion resulting from delirium or dementia is a risk factor for elder financial abuse.

4. An 85-year-old female with dementia, who lives in a skilled nursing facility, presents with bruises on her bilateral inner thighs. The facility staff states that she sustained these while being lifted from her bed to a chair. Who should you notify?

a. Adult protective services (APS)
b. Law enforcement
c. Sexual assault response team
d. The long-term care ombudsman
e. All of the above

Answer: E. The physical findings are concerning for sexual abuse, therefore law enforcement and APS should be called, as well as the sexual assault response team to collect forensic evidence. Because the patient lives in a facility, the long-term care ombudsman should also be called to investigate the possible abuse.

182 The Immunocompromised Patient

KEY CONCEPTS

- Immunocompromised patients who present with acute infections, especially those that are neutropenic, may appear deceptively benign initially. Their symptoms and signs often mimic noninfectious complications.
- Broad-spectrum antibiotics are indicated after obtaining appropriate cultures of all potential sites of infection, especially if the patient is neutropenic.
- Immunocompromised patients can have serious local or systemic infections without fever.
- Symptoms, signs, and findings of infection may include tachypnea, tachycardia, mental status change, metabolic acidosis, increased volume requirements, rapid changes in serum glucose or sodium concentration, or acute abdominal pain.
- The incidence and severity of febrile neutropenia are inversely proportional to the absolute neutrophil count and directly proportional to the duration of neutropenia.
- In neutropenic patients, the temperature should be measured orally or tympanically, not rectally, due to theoretical risk of bacterial translocation and subsequent bacteremia.
- Febrile neutropenia is more common in hematologic malignancy (compared with solid malignancy) and is most likely to occur 7 to 10 days after chemotherapy.
- Gram-positive organisms are responsible for most serious infections in neutropenic cancer patients, but infections due to gram-negative organisms are more rapidly lethal.
- Neutropenic cancer patients with chemotherapy-induced oral mucositis can develop rapid onset of fever with shock. Viridans streptococci is a common pathogen and requires Vancomycin.
- When pneumonia develops in patients with febrile neutropenia, purulent sputum may be absent, and the initial chest radiograph may not show an infiltrate.
- Some low-risk febrile neutropenic patients may not require admission to the hospital. After calculating a Multinational Association for Supportive Care in Cancer (MASCC) risk index score and consulting with their oncologist, discharge may be reasonable.
- Diabetic patients have a high incidence of MRSA infection, osteomyelitis, and wound infections and are at higher risk of bacteremia. Severe infections may be more insidious in presentation.
- Patients with cell-mediated immune deficiency, including those on high-dose corticosteroids, may develop life-threatening infections with intracellular bacteria (*Listeria*, *Salmonella*, tuberculosis), fungi (*Cryptococcus*, *Coccidioides*, *Histoplasma*), herpes simplex virus, and varicella-zoster virus.
- Guidelines no longer support empiric antibiotic treatment of aspiration in alcoholic patients.
- In patients with cirrhosis, empirical treatment of suspected spontaneous bacterial peritonitis (SBP) with antibiotics should be started regardless of ascitic cell count.
- Patients who require hemodialysis for end-stage renal disease have high mortality if they develop pneumonia, *C. difficile* disease, or infections of the dialysis access site.
- Functional or surgical asplenia predisposes to fulminant infection with pneumococci and other encapsulated organisms (*H. influenzae*, *N. meningitidis*, and *Capnocytophaga canimorsus* after dog bites) and, when seen early, may be misdiagnosed as a viral illness, gastroenteritis, or food poisoning.

1. What is the most common site of infection in febrile neutropenic patients?
   a. Gastrointestinal tract
   b. Perineum and anorectal area
   c. Respiratory tract
   d. Skin and soft tissue

Answer: C. The respiratory tract is the most common site of infection, with 25% of infections in the lung and another 25% in the mouth or pharynx (and an additional 5% in the nose or sinuses). Still, all neutropenic patients with fever need to have a thorough physical examination because an undiagnosed infection can cause severe morbidity or mortality.

2. A 49-year-old man with diabetes who has been receiving chemotherapy for non-Hodgkin lymphoma presents with fever and facial pain. Six weeks ago, he was treated with broad-spectrum antibiotics for fever but had negative cultures of blood and urine. He is taking ciprofloxacin as prophylaxis against infection. Examination is normal except for a purple-black lesion on his hard palate that looks like dried blood. A complete blood count shows severe neutropenia. In addition to blood cultures and chest radiography, what action should be taken immediately?
   a. Administer intravenous clindamycin or ampicillin-sulbactam.
   b. Administer intravenous levofloxacin.
   c. Observe in the emergency department (ED) because he may be a candidate for home antibiotic therapy if other laboratory tests are normal, if he appears stable after an initial dose of antibiotics, and if his oncologist agrees.
   d. Obtain a computed tomography (CT) scan of the paranasal sinuses and initiation of intravenous antifungal therapy in the hospital.

Answer: D. This patient has invasive aspergillosis or mucormycosis until proven otherwise. These fungi invade tissues and produce life-threatening necrotizing infections in cancer patients with neutropenia, as well as in diabetic patients, especially those who have received broad-spectrum antibiotics. CT scan will often show deep extension of the infection into the sinuses, orbit, or brain. The initial lesion in the palate or nose is often mistaken for a benign process. The antibiotics listed in A and B are incorrect. The most appropriate empirical antibiotic regimen is cefepime or piperacillin-tazobactam, with or without an aminoglycoside, in addition to antifungal therapy. He is not a candidate for home therapy.

3. A 27-year-old man presents to the emergency department (ED) with acute onset of fever, chills, headache, myalgias, vomiting, mild abdominal cramping, and diarrhea for 8 hours. A splenectomy was performed 15 years earlier when he was treated for lymphoma, which has been in remission since then. He is not taking any medications and has been well. Vital signs are as follows: pulse, 125 beats/min; blood pressure, 110/60 mm Hg; respiratory rate, 20 breaths/min; and temperature, 39.5°C. His mental status is normal and he has mild generalized abdominal tenderness. What is the most appropriate treatment for this patient at this time?
   a. Blood cultures followed by immediate administration of ceftriaxone or cefotaxime, with or without vancomycin
   b. Hydration, antipyretic, antiemetic, and observation in the ED
   c. Immediate hospital admission with observation and frequent abdominal examinations
   d. Lumbar puncture

Answer: A. This patient is at high risk for overwhelming postsplenectomy sepsis, usually caused by *Streptococcus pneumoniae.* Persons who have undergone splenectomy for a hematologic disorder or lymphoma are at much higher risk for overwhelming postsplenectomy infection than are those undergoing splenectomy for trauma. The initial prodromal symptoms may be misdiagnosed as a viral illness, gastroenteritis, or food poisoning before there is abrupt deterioration with development of septic shock with disseminated intravascular coagulation, purpura, and multiorgan dysfunction. After blood cultures are obtained, he should immediately receive antimicrobials active against pneumococci, meningococci, and *Haemophilus influenzae.* He can be investigated for other possible etiologies of his symptoms after this initial critical action is taken.

4. Which answer is correct regarding the diagnosis and management of patients on long-term high-dose corticosteroid therapy?
   a. Acute, short-term administration of high-dose corticosteroids, but not long-term administration, can make the diagnosis of peritonitis particularly difficult.
   b. After laceration repair, sutures should be left in place for 50% to 100% longer than usual.
   c. Complications of chronic corticosteroid use include pancreatitis, pseudotumor cerebri, avascular necrosis of bone, cataracts, myopathy, spontaneous vertebral fractures, psychosis, and hypoglycemia.
   d. Long-term corticosteroid use is frequently accompanied by peripheral blood neutrophilia.

Answer: B. Corticosteroids interfere with wound healing, so sutures need to remain in place longer than is usual for the type of laceration. Despite the profound defect in CMI that occurs with long-term corticosteroid use, infections with organisms associated with defective CMI are unusual. Most serious infections in these patients are caused by pyogenic bacteria, such as *Staphylococcus aureus,* streptococci, and gram-negative bacilli. Both short-term and long-term administration of corticosteroids interfere with the diagnosis of peritonitis. These patients will have poorly localized abdominal discomfort with minimal findings on examination. All the conditions listed in B are associated with chronic corticosteroid use—except hypoglycemia. Corticosteroid use is associated with hyperglycemia, hyperosmolar nonketotic diabetic coma, and diabetic ketoacidosis.

5. A 34-year-old woman on chemotherapy for lymphoma presents to the emergency department with severe oral pain and difficulty swallowing. She has no fevers, cough, abdominal pain, or dysuria. On physical exam, her oral mucosa is erythematous, friable, and edematous. Her labs reveal an ANC of 600 cells/μL. What is the most important next step?
   a. Administer 30 mL/kg of crystalloid for sepsis.
   b. Draw blood cultures and start empirical vancomycin and cefepime.
   c. Start IV analgesics to allow oral rehydration.
   d. Draw blood cultures and start empiric metronidazole and cefepime.

Answer: B. This patient is a neutropenic patient presenting with mucositis. Mucositis is a frequent prelude to viridans streptococcal bacteremia, which can produce sudden onset of acute respiratory distress syndrome, a toxic shock–like syndrome, rash, and pneumonia. Importantly, mucositis predisposes the patient to gram-positive infections, and vancomycin is appropriate in addition to a broad-spectrum gram-negative agent.

KEY CONCEPTS

- The solid organ transplant recipient's altered anatomy, denervated allograft, and immunosuppression frequently result in atypical disease presentations both related and unrelated to the transplanted organ.
- An understanding of the solid organ transplant recipient's altered anatomy, including vascular and nonvascular anastomoses, is critical to evaluating early post-transplantation complications.
- Rejection can manifest at any point post-transplantation with constitutional symptoms and signs of allograft insufficiency, requiring prompt recognition and augmented immunosuppression to salvage the transplanted organ.
- Timing post-transplantation, the net state of immunosuppression, and ongoing antimicrobial prophylaxis should be incorporated into the evaluation of a solid organ transplant recipient with fever and other concerns for infection.
- In addition to affecting a specific arm of the immune system, each antirejection agent is associated with unique toxicities which may be independently responsible for a solid organ transplant recipient's clinical condition.
- Cardiac allograft vasculopathy, a form of chronic rejection and similar to coronary artery disease in presentation, has emerged as a common complication of orthotopic heart transplantation as transplant recipients live longer.
- Especially in lung transplant recipients with underlying cystic fibrosis or bronchiectasis, prior imaging and culture data should help guide initial management of new respiratory symptoms.
- Recurrent allograft pyelonephritis post-transplantation merits thorough evaluation for ureteral stones and strictures, perinephric abscesses, and urinary retention to identify potentially actionable contributors to infection.
- Hepatic artery thrombosis is an uncommon but devastating early complication of liver transplantation resulting in allograft dysfunction, biliary necrosis, and sepsis if left undetected and untreated.
- Allograft rejection, bacterial or viral enteritis, and altered intestinal transit time can all contribute to diarrhea in the intestinal and multivisceral transplant recipient.
- As chronic corticosteroids are incorporated into immunosuppression regimens for the majority of solid organ transplant recipients, adrenal insufficiency should be considered in the differential of transplant recipients presenting with hypotension and fever.
- In addition to the organ-specific complications described in the sections above, solid organ transplant recipients experience graft-versus-host disease, malignancy, trauma, and psychosocial distress post-transplantation requiring heightened awareness and assembly of multidisciplinary teams.

1. A 42-year-old man presents from home with a 3-day history of fever, cough, and shortness of breath. He underwent orthotopic heart transplantation 5 years ago, has never experienced rejection, and is maintained on minimal doses of prednisone and tacrolimus. He takes trimethoprim-sulfamethoxazole for antimicrobial prophylaxis. Chest radiograph reveals a diffuse interstitial pattern. What is the most likely cause of his pneumonia?
   a. Community-acquired respiratory virus
   b. Cytomegalovirus
   c. *Pneumocystis jirovecii*
   d. *Nocardia*

Answer: A. Not all transplant recipients are equivalently immunocompromised. Timing post-transplantation, the net state of immunosuppression, and antimicrobial prophylaxis should all be considered in developing a differential diagnosis for a transplant recipient with suspected infection. His 5-year rejection-free post-transplant course and minimal immunosuppression make opportunistic infections much less likely than community-acquired respiratory viral infections. Given his ongoing use of trimethoprim-sulfamethoxazole, *Pneumocystis jirovecii* pneumonia is unlikely.

2. A 57-year-old deceased donor kidney transplant recipient is sent to the ED from her primary care physician's office 3 months postoperatively because routine blood work revealed a creatinine of 3.45 (baseline creatinine 1.0). Which of the following medications may be responsible for her acute kidney injury?
   a. Albuterol
   b. Tacrolimus
   c. Pancrelipase
   d. Prednisone

Answer: B. Tacrolimus is associated with dose-dependent nephrotoxicity. A serum level of tacrolimus should be obtained in any solid organ transplant recipient maintained on this immunosuppressive agent presenting with acute kidney injury. It is important to recognize that tacrolimus can both cause and be affected by renal insufficiency, so it may not have been the initial trigger but may have ultimately worsened the renal injury. Alternatively, a new drug (fluconazole, azithromycin, etc.) may have been initiated which resulted in higher serum concentrations of tacrolimus and resultant renal injury.

3. A 35-year-old woman with cystic fibrosis status post bilateral lung transplantation 2 weeks ago is brought to the ED by her husband. He reports that she was discharged several days prior to presentation with confusion attributed to intensive care unit delirium. Instead, she has developed worsening episodes of alternating agitation and lethargy. She is afebrile and normotensive on presentation, with a 2-liter oxygen requirement to maintain saturations above 90%. No leukocytosis or abnormalities of liver or kidney function are noted. A blood gas test does not demonstrate hypercarbia. What is the next best test?
   a. Lumbar puncture
   b. Magnetic resonance imaging
   c. Serum ammonia
   d. Electroencephalography

Answer: C. Hyperammonemia syndrome affects 1% to 4% of lung transplant recipients, resulting from systemic infection with *Mycoplasma hominis* or *Ureaplasma urealyticum*. These urease-splitting organisms metabolize urea as an energy source and produce ammonia as a by-product, which can accumulate in tissues such as the trachea and lung to cause unexplained agitation, lethargy, and new-onset seizures. If ammonia levels are elevated, whole blood or plasma and bronchoalveolar samples should be obtained and tested for *Mycoplasma hominis* and *Ureaplasma urealyticum* polymerase chain reaction (PCR). While awaiting these results, empirical antibiotic therapy against these organisms, such as doxycycline or ciprofloxacin, should be initiated. Hemodialysis, intravenous sodium benzoate, and sodium phenylacetate can help to clear ammonia from the blood and tissues but will be incompletely effective if the underlying infection is not addressed.

4. A 30-year-old liver transplant recipient presents to the ED 10 days post-transplantation with fevers, right upper quadrant pain, and jaundice. Laboratory evaluation reveals leukocytosis, transaminase elevations, and hyperbilirubinemia, which have all worsened since hospital discharge. Which of the conditions responsible for this presentation is most important to diagnose emergently?
   a. Rejection
   b. Cholangitis
   c. Donor-derived hepatitis B infection
   d. Hepatic artery thrombosis

Answer: D. The hepatic artery serves as the sole blood supply to the liver allograft and thus hepatic artery thrombosis has devastating consequences to allograft survival. This complication develops within days to weeks post-transplantation and requires immediate recognition and management in close collaboration with transplant surgeons because thrombectomy or urgent re-transplantation may be required. Diagnosis is with Doppler ultrasound. Consideration should be given to assessing the hepatic blood flow for any liver transplant recipient presenting to the ER in the first weeks after transplantation.

5. A 53-year-old intestinal transplant recipient with a history of Crohn disease status post multiple bowel resections resulting in short gut syndrome has been maintained on tacrolimus and prednisone for the last 2 years since transplantation without previous significant infection complications. He presents to the ED with 5 days of nausea, vomiting, cramping abdominal pain, and diarrhea, and states that his 2-year-old son had been ill with similar symptoms that have since resolved. The recipient has not been able to consistently take his medications for the last 3 days. He is febrile and hypotensive with blood pressures in the 80s/50s. His lab work is significant for acute kidney injury and hyponatremia. Which of the following is unlikely to contribute to this patient's clinical presentation?

a. Community-acquired gastroenteritis
b. Acute rejection
c. Adrenal insufficiency
d. CMV enteritis

Answer: D. Intestinal transplant recipients may develop diarrhea from a wide variety of causes and, in fact, dehydration is the most common reason for recurrent presentations to the ED and readmission in this population. While rejection, bacterial overgrowth, and CMV infection are common reasons for diarrhea in the first year after transplant, later causes for diarrhea are more likely to be community-acquired. This patient's history of an exposure to a child with a like illness suggests a new infection with a typical pathogen. Acute rejection can be precipitated by the inability to ingest and absorb immunosuppressive drugs and could now be a part of this patient's disease process. Given the use of chronic corticosteroids in this population, endogenous steroid production by the adrenal glands is often impaired and seemingly minor insults such as viral infections and hypovolemia may lead to signs of adrenal insufficiency. CMV is unlikely to emerge as a new infection this late post-transplant.

184 The Morbidly Obese Patient

KEY CONCEPTS

- Obesity has reached epidemic proportions in the United States (US), as well as much of the world, with nearly 40% of American adults classified as obese.
- Obesity results in several physiologic changes including aberrations to lung physiology and predisposes patients to obstructive sleep apnea and obesity hypoventilation syndrome.
- Drug dosages can be challenging in obese patients because some medications are lipophilic and will need to be dosed based on total body weight while hydrophilic medications are dosed closer to ideal body weight. Many medications require specific scalers for appropriate dosing.
- Many drugs lack large-scale trials in the obese population to guide appropriate dosing.
- Obese trauma patients suffer greater rates of chest and abdominal injuries from blunt trauma than healthy-weight patients.
- Bariatric surgery is the only weight loss strategy that consistently results in long-term significant weight loss.
- Laparoscopic gastric banding, sleeve gastrectomy, and Roux-en-Y gastric bypass are the most common weight loss surgeries in the United States, and each presents its own short-term and long-term complications.
- Procedures common in the emergency department (ED), such as lumbar puncture, venous access, CPR, and intubation, may be more difficult in the obese patient.
- The increased body mass of the obese patient presents several challenges to obtaining interpretable radiographic images.

1. Rates of obesity in the United States and around the world have been:
   - **a.** Increasing in the US and decreasing around the world.
   - **b.** Increasing in the US and staying constant around the world.
   - **c.** Staying about the same worldwide.
   - **d.** Increasing in the US and around the world.

Answer: D. Increasing in the United States and around the world.

2. When lying supine, obese patients tend to:
   - **a.** Breathe normally because they have more developed chest musculature.
   - **b.** Have trouble breathing due to stomach distention.
   - **c.** Have trouble breathing due to changes in lung mechanics.
   - **d.** Have trouble breathing due to chronic back pain.

Answer: C. Increased chest wall mass in conjunction with substantial abdominal fat mass leads to reduced lung compliance and collapse of small airways resulting in increased airway resistance. These changes in turn result in a decrease in functional residual capacity (FRC) leading to increased atelectasis.

3. Challenges to obtaining high-quality CT scan images in the obese patient include which of the following?
   - **a.** Difficulty lying flat for the scan
   - **b.** Weight limits of the gantry table
   - **c.** Girth greater than aperture size
   - **d.** Claustrophobia during the scan
   - **e.** Area of interest laying outside the CT field of view (FOV)

Answer: A, B, C, and E are all correct. Claustrophobia is often a difficulty during MRI, not CT.

4. Drug dosing of obese patients is based on which of the following factors?
   - **a.** Ideal body weight for the vast majority of drugs
   - **b.** Total body weight for the vast majority of drugs
   - **c.** A complex decision based on the properties of the drug and the clinical picture of the patient
   - **d.** Total body weight if weight is less that 150 kg and ideal body weight if above that.

Answer: C. Drug dosing is a complex decision based on the properties of the drug and the clinical picture of the patient. The clinician faces several challenges regarding proper medication dosing in the obese patient. The volume of distribution (V_d) of a drug is the principal factor involved in the loading dose whereas subsequent maintenance dosing will primarily be governed by total body clearance (Cl).[14] The V_d is affected by many factors including drug lipophilicity, plasma binding, regional blood flow, body composition, molecular size, and degree of ionization.

5. Which of the following statements regarding internal hernias following Roux-en-Y gastric bypass is true?
   - **a.** They can be intermittent and may require exploratory surgery for diagnosis.
   - **b.** They are easy to diagnose based on history and physical examination.
   - **c.** They are almost always visualized on CT studies.
   - **d.** They are an exceedingly rare complication.

Answer: A. They can be intermittent and may require exploratory surgery for diagnosis Weight loss will predispose patients to internal hernias as potential spaces open in the small bowel mesenteries. They can be particularly difficult to diagnose because they can be intermittent and CT scans are often falsely negative. Patients may complain of intermittent pain without clear etiology which may localize to the left flank. Exploratory laparoscopy may be required to confirm the diagnosis.

KEY CONCEPTS

- The emergency department (ED) should develop a written plan of action to deal with violence that integrates the roles and activities of ED staff, hospital administration, security, and local authorities.
- ED staff should be trained to recognize potentially violent individuals and to intervene with verbal de-escalation techniques prior to the use of physical or chemical restraint when possible.
- The emergency clinician should be familiar with the use of physical and chemical restraints as well as the breadth of options for chemical sedation and circumstances that may guide selection of particular medications.
- For the undifferentiated, severely agitated patient requiring rapid tranquilization, we recommend a benzodiazepine (such as lorazepam) either alone or with a first-generation antipsychotic (such as haloperidol).
- The possibility of an organic (medical) cause of aggressive behavior should be considered in all violent patients, even those with known psychiatric disease.
- Difficult patient encounters may result in undesirable implications for both patients and their ED caregivers, including compromised patient care, compassion fatigue, or professional burnout.
- Management of the difficult patient can be optimized by understanding the multiple issues contributing to the impaired clinician-patient relationship, including factors of the ED setting (such as time constraints and lack of privacy), individual clinician influences (such as personal bias or poor communication), and patient contributions to the interaction, including behavioral, social, or substance use issues.
- Pejorative stereotypes of difficult patients should be avoided. To aid in clinician strategies for challenging encounters, instead aim to characterize the patient's primary difficult behaviors.
- Understanding one's own biases and reactions and optimizing communication are helpful strategies in dealing with a suboptimal clinician-patient relationship.

185.1. Which of the following is a positive predictor of violent behavior in patients?

a. Age
b. Ethnicity
c. Gender
d. Level of education
e. Marital status

Answer: C. Identification of potentially violent patients is difficult, with male gender, prior history of violence, and drug or ethanol abuse the only positive predictors. Ethnicity, diagnosis, age, marital status, and education are not reliable identifiers.

185.2. Which of the following medications is associated with the highest incidence of sedation, hypotension, and anticholinergic symptoms?

a. Droperidol
b. Haloperidol
c. Loxapine
d. Molindone
e. Thioridazine

Answer: E. Antipsychotics can be divided into low potency (such as chlorpromazine and thioridazine), midrange potency (such as loxapine and molindone), and high potency (such as haloperidol and droperidol). The incidence of sedation, hypotension, and anticholinergic symptoms is highest in the low potency group.

185.3. Which of the following antipsychotic medications is associated with the highest incidence of extrapyramidal symptoms?

a. Chlorpromazine
b. Haloperidol
c. Loxapine
d. Molindone
e. Thioridazine

Answer: B. Antipsychotics can be divided into low potency (such as chlorpromazine and thioridazine), midrange potency (such as loxapine and molindone), and high potency (such as haloperidol and droperidol). The incidence of extrapyramidal symptoms is greatest in the high potency group.

185.4. Which of the following disorders is best defined as a pattern of instability of interpersonal relationships, self-image, affect, and marked impulsiveness that begins by early adulthood and presents in a variety of contexts?

a. Antisocial personality disorder
b. Borderline personality disorder
c. Dependent personality disorder
d. Histrionic personality disorder
e. Paranoid personality disorder

Answer: B. Borderline personality disorder is a pattern of instability of interpersonal relationships, self-image, affect, and marked impulsiveness beginning by early adulthood and present in a variety of contexts. It may be indicated by five or more of the following:

1. Frantic efforts to avoid real or imagined abandonment
2. A pattern of unstable and intense interpersonal relationships characterized by alternating extremes of idealization and devaluation
3. Identity disturbance: Markedly and persistently unstable self-image or sense of self
4. Impulsiveness in at least two areas that are potentially self-damaging (e.g., spending, sex, substance abuse, reckless driving, or binge eating)
5. Recurrent suicidal behavior, gestures, threats, or self-mutilating behavior
6. Affective instability caused by a marked reactivity of mood (e.g., intense episodic dysphoria, irritability, or anxiety usually lasting a few hours and rarely more than a few days)
7. Chronic feelings of emptiness
8. Inappropriate, intense anger, or difficulty controlling anger (e.g., frequent displays of temper, constant anger, or recurrent physical fights)
9. Transient, stress-related paranoid ideation or severe dissociative symptoms

185.5. A patient demonstrates inappropriate sexually seductive behavior while being interviewed by her physician. This characteristic is most consistent with which of the following personality disorders?

a. Antisocial personality disorder
b. Borderline personality disorder
c. Dependent personality disorder
d. Histrionic personality disorder
e. Paranoid personality disorder

Answer: D. Histrionic personality disorder is a pattern of excessive emotionality and attention seeking beginning by early adulthood as indicated by five or more of the following:

1. Discomfort in situations in which the center of attention is someone else
2. Interaction with others often characterized by inappropriate sexually seductive or provocative behavior
3. Displays rapidly shifting and shallow expression of emotions
4. Consistently uses physical appearance to draw attention to self
5. Has style of speech that is excessively impressionistic and lacking in detail
6. Shows self-dramatization, theatricality, and exaggerated expression of emotion
7. Is suggestible (i.e., easily influenced by others or by circumstances)
8. Considers relationships to be more intimate than they actuallyare

Multiculturalism, Diversity, and Care Delivery 186

KEY CONCEPTS

- Changing demography and an evolving culture in the United States are changing emergency medicine practice as disparities in health and health care delivery continue, despite efforts to improve care.
- Emergency department interventions should focus on social determinants of health, health literacy, and empowerment of patients to participate in their care.
- Treatment plans created with patients and based on what matters to them have the greatest opportunity for success.
- Emergency clinicians will improve the quality of care provided if they can meet federal standards for culturally and linguistically appropriate care, recognizing that patients with limited English proficiency have a right to medical interpretation.
- Culturally sensitive, patient-centered, and trauma-informed care will improve patient clinician communication and satisfaction, decrease medical errors, and promote patient follow-through with recommendations.

1. Which of the following factors has not been identified as a health care disparity for minorities?
   - **a.** High rates of no insurance or underinsurance
   - **b.** Lack of prenatal care
   - **c.** Likelihood of living in an area with adequate physician concentration
   - **d.** Lower life expectancy when compared to white populations
   - **e.** Relatively low number of minority providers

Answer: C. Minority patients are likely to live in impoverished areas with limited access to both primary and specialty medical care. As a result, they may present to the ED with previously undiagnosed medical problems underlying their symptoms.

2. Patients' cultural perceptions can have a significant impact on how they perceive their provider and the care they are receiving. Which of the following methods can be used by providers to approach different cultural perceptions so as to enhance the provider-patient relationship?
   - **a.** Being aware of one's own cultural values
   - **b.** Developing assumptions of cultural perceptions base on patients' ethnicity or race
   - **c.** Limiting participation of alternative therapies and folk medicine in the patient's care plan
   - **d.** Maintaining the role of the physician as the decision maker, even when it may conflict with the patient's autonomy

Answer: A. Being aware of one's own preconceived notions and biases is the first step a health care provider can take on the path toward developing cultural sensitivity toward patients from minority groups.

3. Which method of communicating is adequate to utilize with a deaf or hard-of-hearing patient?
   - **a.** Certified interpreter
   - **b.** Family member
   - **c.** Lip reading
   - **d.** Written communication on notepad

Answer: A. Using a certified interpreter for the hard-of-hearing is the only reliable way of communicating with such patients.

4. A helpful language translator can be essential in communication with a patient who is facing a language barrier. Which of the below methods are recommended for using an interpreter during a medical examination?
   - **a.** Asking the patient to have friends or family translate discharge instructions for them
   - **b.** Only obtaining an interpreter if the patient requests one
   - **c.** Using a family member is appropriate as long as they are older than 18 years
   - **d.** Using a trained medical interpreter
   - **e.** Using health care staff members who are fluent in the patient's language

Answer: D. Using a certified interpreter is the only consistently reliable way of communicating with patients with language barriers.

5. Which of the following organizations have suggested guidelines to address cultural competence in educational standards?
   - **a.** American Association of Medical Colleges
   - **b.** Liaison Committee on Medical Education
   - **c.** Accreditation Council on Graduate Education
   - **d.** All of the above

Answer: D. These are just some of the medical organizations that have developed guidelines to assist medical schools and residency programs in teaching physicians-in-training to become culturally competent in their approach to patients from minority groups.

KEY CONCEPTS

- Labor and sex trafficking involve the exploitation of a person for labor or commercial sex, respectively, affecting up to 25 million persons worldwide.
- Human smuggling, a crime in which a person contracts a smuggler to facilitate their illegal entry into a country, can evolve into trafficking during transit or at the destination and under such circumstances the person is considered a victim of human trafficking.
- Child victims of sex trafficking are considered victims of child abuse and neglect under the law and thereby call into relevance state mandated reporting statutes.
- Populations at greater risk for trafficking include persons with histories of child abuse, family dysfunction, diverse sexual orientation or gender identity, intellectual disability, homelessness, financial insecurity, and migration.
- Human trafficking often involves the use of abusive and violent tactics, including forced substance use and psychological coercion to entrap and exert control over trafficked persons, with profound implications for survivors' physical, reproductive, and mental health.
- Trafficked persons seek health care services during their exploitation for a wide range of health conditions and the emergency department (ED) is the most common access point for this patient population where indicators of abuse, control, and the physical and psychosocial red flags of trafficking can assist in the recognition of trafficked victims.
- Inquiry about forms of interpersonal violence, including trafficking, is fundamentally different from screening for medical issues with the goal of providing a safe environment in which patients feel empowered to share as much or as little as they choose, and where strengths and resilience are recognized.
- Trafficked people may return to exploitative situations repeatedly before exiting permanently; therefore, safety planning is critical for the discharged trafficked patient.
- Trauma due to trafficking results in neurobiologic changes such that commonly occurring smells, sounds, sights, and procedures of the ED environment may be perceived as threats by those with trauma histories.
- The six principles of trauma-informed care include physical and psychological safety; trustworthiness and transparency; peer support; collaboration and mutuality; empowerment, voice, and choice; and cultural, historical, and gender acknowledgment.
- Law enforcement involvement should be limited to patient request, in state-specific mandated reporting scenarios, or when clinicians suspect imminent danger to staff or the patient.

1. Which of the following statements regarding human trafficking inquiry is true?
   a. It is imperative that trafficked victims recount their history as often as possible to ensure consistency and to support law enforcement in the investigation of trafficking networks.
   b. The primary goal of inquiry is disclosure.
   c. Inquiry is a series of protocol-based standard questions designed to efficiently obtain information from traumatized victims.
   d. The goal of inquiry is to provide a safe environment for patients to share as much or as little as they choose.

Answer: D. Rather than disclosure, the goal of inquiry is to provide a safe environment in which patients feel empowered to share as much or as little as they choose where strengths and resilience are recognized. Inquiry-based assessment is an active process that includes open-ended questions and dialogue.

2. Which of the following statements regarding health impacts on trafficked victims is false?
   a. Traffickers employ abusive tactics that can cause physical, reproductive, and mental health problems.
   b. Schizophrenia is not a diagnosis commonly associated with the psychological and psychiatric sequelae of victims of trafficking.
   c. Commonly occurring smells, sounds, sights, and procedures in the emergency department may be perceived as threats by trafficking victims who have experienced trauma but is not caused by neurobiologic changes.
   d. Substance abuse and drug use rates are common in trafficked victims under 18 years of age.

Answer : C. Trauma due to trafficking results in neurobiologic changes such that commonly occurring smells, sounds, sights, and procedures of the emergency department environment may be perceived as threats by those with trauma histories.

3. Which of the following statements regarding the indicators and red flags used to identify victims of trafficking in the ED are true?
   a. Hypervigilance and hyperstartle reflexes are pathognomonic of human trafficking.
   b. Accompanying individuals who are reluctant to leave the room, insist on answering questions or attempting to control encounters, or communications that may seem rehearsed may all be indicators that a victim is being trafficked.
   c. Physical red flags of human trafficking often include immediate presentation for care for easily preventable injuries.
   d. Evidence of psychological distress including limited eye contact, low trigger threshold, hypervigilance, and hyperstartle reflex are uncommon presenting characteristics of trafficked victims and are not reliable enough indicators in emergency departments to raise suspicion.

Answer: B. There is no single characteristic that is pathognomonic of human trafficking. Physical red flags of human trafficking include delayed presentation for care of injuries or infections; signs of abuse or neglect (e.g., malnourishment, poor oral health); work injuries or exposures that would be easily prevented by personal protective equipment (PPE). Other potential indicators include patient communications that seem rehearsed or scripted, patient reluctance or inability to answer simple questions about their living or working situation, conflicting information, reported mechanism of injury inconsistent with the physical evidence, and evidence of psychological distress (e.g., limited eye contact, low trigger threshold, hypervigilance, hyperstartle reflex).

4. Which of the following statements regarding the evaluation and treatment of trafficked victims is true?
   a. Empiric STI treatment and emergency contraception is indicated only for sex trafficked victims and not labor trafficked victims.
   b. Emergency department evaluation should include addressing acute medical issues and focus on forensic examination but should exclude the evaluation of chronic medical conditions.
   c. Law enforcement involvement should be limited to patient request or when clinicians appreciate imminent danger to staff or patients.
   d. When a suspected human trafficking case is identified, forensic examination should be performed by the original provider because including a sexual assault examination by forensic nurse examiners can worsen victim traumatization.

Answer: C. For both labor and sex trafficking, empirical STI treatment and emergency contraception may be indicated. ED evaluation may include addressing acute medical issues, evaluation of possible untreated chronic medical problems, documentation of acute and remote injuries, and STI testing and treatment, and consideration of a sexual assault medical forensic examination and evidence collection. Research shows that the use of sexual assault/forensics nurse examiners results in better patient outcomes in legal and emotional support.

5. Which of the following statements regarding the trauma-informed care of trafficking victims is false?
   a. Mitigation of conscious and unconscious biases is important in contributing to the traumatization of trafficked victims.
   b. Preventing retraumatization is an essential feature of trauma-informed care in the emergency department.
   c. The PEARR Tool provides a trauma-informed framework for emergency clinician assessment but is only validated for victims of human trafficking.
   d. If a patient denies victimization or declines assistance, the clinician may offer information about resources that can assist in the event of an emergency.

Answer: C. The Privacy, Educate, Ask, Respect, and Respond (PEARR) tool provides a trauma-informed framework for the emergency clinician assessing for any form of interpersonal violence, including human trafficking.

KEY CONCEPTS

- Sexual orientation, gender identity, and gender expression occur along a continuum and may be completely separate and discordant from one another.
- Multiple health disparities exist in sexual minority populations, but many of the contributing factors are likely unknown due to lack of studies.
- ED provider education about sexual minorities is often inadequate, limited, or absent.
- Transgender individuals may utilize a variety of medical and surgical interventions in their transition to affirm their gender identity. Emergency clinicians should be aware of complications related to these practices that may require treatment in the ED.
- Sexual minority patients at extremes of age, or who share another minority identity, have additional disparities that require consideration to deliver equitable and safe care.
- Open, inclusive, and nonjudgmental patient-centered communication using appropriate terminology is fundamental to fostering provider trust and minimizing health disparities in sexual minorities.

1. Which of the following statements regarding sexual minority populations is true?
   - **a.** Sexual orientation, gender identity, and gender expression define psychological, cultural, and behavioral characteristics of either being male or female.
   - **b.** Lesbian, gay, bisexual, and homosexual are all acceptable terms when addressing the sexual minority community.
   - **c.** The demographics of sexual minority populations indicate that a majority of vast transgender individuals were born after 1994 (Generation Z).
   - **d.** In sexual minority populations, it is important to separate sexual and gender identities from sexual behavior.

Answer: D. Sexual orientation, gender identity, and gender expression occur along a continuum and may be completely separate and discordant from one another whereas gender refers to the psychological, behavioral, and cultural characteristics of being either male or female. The term *homosexuality* is associated with past bias and discrimination and is not a preferred term when addressing sexual minority populations. Sexual minority individuals cross every demographic, and multiple cultural and societal influences can affect their visibility and disclosure of sexual and gender identity.

2. Which of the following factors does not affect equitable care in sexual minority populations?
   - **a.** Parental barriers
   - **b.** Same-sex marriage
   - **c.** Access to care
   - **d.** Provider education and knowledge
   - **e.** None of the above

Answer: E. Same-sex couples with adopted children can face difficulty in legal recognition of parenthood, resulting in barriers in medical decision making and obtaining appropriate health care. Although marriage equality provided the opportunity for stronger and automatic protections previously denied to same-sex couples, only 51% of same-sex couples' households are legally married compared to over 90% of heterosexual households. Sexual minority patients have less access to health care and are less likely to have health insurance compared to majority counterparts. ED provider education about sexual minorities is often inadequate, limited or absent, leading to lack of understanding and poorly informed decision making or advocacy on behalf of sexual minority populations.

3. Which of the following statements regarding medical gender affirming therapy is true?
   - **a.** Commonly used medications include hormone supplementation with androgen blocking medications in conjunction with estrogen.
   - **b.** The most common gender affirming medical treatment for the transgender male is spironolactone in addition to 5-alpha reductase inhibitors and aromatase inhibitors.
   - **c.** Transgender women who went through puberty and had exposure to testosterone will usually have complete elimination of male patterned facial or body hair and change in pitch in voice although little change in skeletal bone structure would occur with hormone therapy alone.
   - **d.** Finasteride is a 5-alpha reductase inhibitor used only in transgender males for gender affirming therapy.
   - **e.** Both A and B are true.

Answer: A. The most common gender affirming medical treatment utilized by transgender men is testosterone. The most commonly used androgen blocker in the United States is spironolactone. 5-Alpha reductase inhibitors (finasteride, dutasteride) are alternative androgen blocking medications. Transgender women who went through puberty and had exposure to testosterone will not have complete elimination of male patterned facial or body hair, change in skeletal bone structure, or change in pitch of their voice with just hormone therapy. These require specific interventions such as plastic surgery, laser hair removal or electrolysis, and voice therapy.

4. Which of the following statements regarding surgical gender affirming therapy is false?
   - **a.** Genital procedures in gender affirming surgeries for transgender women also include the removal of the prostate gland as well as orchiectomy and vaginoplasty.
   - **b.** Complications of metoidioplasty and phalloplasty include spraying or dribbling of urine and the development of urethral strictures and fistulas.
   - **c.** In transgender males, pregnancy is not a consideration.
   - **d.** Both a and c are false.
   - **e.** None of the above are true.

Answer: D. Genital procedures in gender affirming surgeries for transgender women do not include the removal of the prostate gland, so the risk for prostate cancer persists necessitating screening. Pregnancy should always be considered in any patient who still has a uterus and ovaries.

5. Which of the following statements is false?
   - **a.** There is an increased risk for mental health disorders among the sexual minority population.
   - **b.** The population of older Americans who identify as LGBTQ is likely to shrink by the year 2030 and experience less depression.
   - **c.** African American and Latino sexual minorities experience greater stigma relative to white counterparts.
   - **d.** Rates of HIV have risen among the Asian and Latino males who have sex with males over the last decade.

Answer: B. There are approximately 3 million Americans who identify as LGBTQ over the age of 55, and by 2030 it is estimated there will be 6 million, with increasing burdens of mental health and lifetime physical disabilities. Although overall rates of HIV have been stable in the last decade, rates rose 52% among Asian and 18% among Latino MSM. The HIV infection rate among Black MSM was stable and the infection rate among white MSM decreased 16%.

189 Social Determinants

KEY CONCEPTS

- Challenges with food, housing, transportation, and other social determinants of health account for approximately 50% of poor health outcomes in the United States.
- Emergency providers can connect patients to resources that can help address challenges with different social determinants of health.
- Legal services attorneys provide low-cost or free services that can assist with a number of health-harming legal needs.
- Health care providers and systems need to work with nonprofits, community groups, community members, and community leaders to expand the capacity to address social determinants of health.

1. Which of the following statements regarding social determinants of health (SDOH) is true?
   - **a.** Less than 20% of health outcomes can be attributed to SDOH.
   - **b.** SDOH equally impacts all segments of the US population regardless of income.
   - **c.** Immigration status is not considered a SDOH.
   - **d.** Health literacy is one of the most significant aspects of SDOH.

Answer: D. Patients may have limited health literacy, which is defined as limited capacity to obtain, process, and understand basic health information and services needed to make appropriate health decisions may have poorer adherence to care plans and poorer health outcomes. An estimated 20% of a patient's health outcome is influenced by medical care, whereas social and economic factors account for 50% of such outcomes. Social concerns disproportionately affect the poor and impede health. Immigration as an experience has profound effects on individuals and their health. Immigration status has an important effect on patients for two reasons—immigration status is linked to ability to work and to access public benefits, including health care.

2. Which of the following factors impact health outcomes?
   - **a.** Access to transportation
   - **b.** Homelessness and housing
   - **c.** Food insecurity
   - **d.** Income and insurance
   - **e.** All of the above

Answer: E. Access to transportation affects an individual's ability to access care, adhere to appointments, and carry out treatment plans, having a direct impact on health. These individuals have worse health outcomes and mortality compared with those who are housed. Delayed care is a common issue that arises in the ED as uninsured and underinsured patients often are unable to access care prior to an ED visit. Lack of access to healthy foods leads to increased rates of mental health problems, low birth weight, depression, hypertension, diabetes, and hyperlipidemia.

3. Which of the following ED-based strategies can counteract SDOH?
   - **a.** Systematic screening in the emergency department can be a significant step in recognizing patients with significant SDOH.
   - **b.** Addressing SDOH must occur only within the health care system for the strategy to be successful.
   - **c.** Addressing concerns of health literacy first will substantively impact health outcomes by improving patients' failure to comply with follow-up care instructions and access to community resources.
   - **d.** Violence prevention is not a reasonable strategy in combating poor outcomes due to SDOH.

Answer: A. Most departments do not currently screen for social needs. Instead, social needs arise during the course of the patient encounter. Options for systematic screening exist but require implementing a framework to address social needs.

4. Which of following health-related benefits available in the United States can impact access to care and health outcomes?
   - **a.** Children's Health Insurance Program (CHIP) is a specific program focused on providing access to care for children up to age 26.
   - **b.** The Affordable Care Act requires plans and issuers that offer dependent child coverage to make coverage available until the child reaches the age of 21.
   - **c.** The Hill-Burton Act of 1946 mandates charity care obligations for facilities that benefited from the funding.
   - **d.** Part A Medicare benefits are essential to supporting prescription drug coverage.

Answer: C. Facilities built with assistance from the Hill-Burton Act of 1946 have legally mandated charity care obligations. There are specific programs focused on providing access to care for children under the age of 18, including the Children's Health Insurance Program (CHIP). The Affordable Care Act requires plans and issuers that offer dependent child coverage to make coverage available until the child reaches the age of 26. Medicare is administered and funded by the federal government and contains several parts including inpatient hospital coverage (Part A), outpatient medical coverage (Part B), alternative coverage options including Medicare Advantage (Part C), and prescription drug coverage (Part D).

5. Which of the following statements regarding legal services available to ED patients is true?
   - **a.** Legal services are typically unhelpful in increasing access to health care coverage but are invaluable in supporting immigration status adjustments.
   - **b.** A public charge applies to anyone seeking health care services, so referral to legal services is essential to evaluate individual cases and public benefits.
   - **c.** In managing family and domestic concerns, referral to legal services should only occur if there is evidence of intimate partner violence or elder abuse.
   - **d.** Heath-harming legal needs are defined as a group of legal factors that can adversely impact health outcomes.

Answer: D. Legal services are typically helpful in a wide variety of circumstances and address what may be considered health-harming legal needs. The idea of becoming a "public charge," or one who is reliant on public benefits, has caused fear and hesitation among immigrant communities to seek care. While applying for and receiving certain public benefit programs *may* make one a public charge, seeking and receiving health care services never does. However, for the undocumented individual, eligibility to receive public benefits is impermissible unless the specific exceptions are met. Family dynamics and relationships can be a source of stress and lead to emergency department visits and poor health. Addressing challenges can improve health for patients. Patients may face challenges with marriage and children. If there is no intimate partner violence, then a referral to the local legal services agency, courthouse, and/or bar association can help patients access resources to resolve these concerns.

KEY CONCEPTS

- It is important for prevention efforts, including those in hospitals and emergency departments, to consider unjust social conditions disproportionately experienced by minority communities, and especially by African Americans; these include generational poverty, residential segregation, and other forms of racism that limit opportunities to grow up in healthy and violence-free environments.
- Interventions should focus on altering the quality of life course for survivors of violence through targeting their total wellness and thriving through employment, education, financial stability and economic mobility, independence, and stable emotional health.
- Hospital based violence intervention programs (HVIPs) that incorporate at least some brief intervention and case management demonstrate HVIP cost-effectiveness with decreases in injury recidivism.

1. Which of the following statements correctly describes a difference between uninsured African American (AA) patients and uninsured white patients with depression?
   a. AA patients are less likely to be prescribed antidepressants.
   b. AA patients are less likely to be diagnosed with depression.
   c. AA patients are more likely to think about suicide.
   d. AA patients are less likely to need ongoing therapy.

Answer: A. A study observed that African American parents were much less likely to seek mental health care services for their children if they themselves held high levels of stigma towards mental health.[40] Furthermore, uninsured African Americans with depression are less likely to be prescribed antidepressants compared to those who are insured, and those who are insured are still less likely to be prescribed medications compared to their white counterparts.[41]

2. Which of the following statements is true regarding individuals who are survivors of interpersonal violence and are exposed to community violence?
   a. They are more at risk for reinjury/trauma recidivism.
   b. They are less at risk for retaliation.
   c. They are more likely to receive follow-up care.
   d. They are less likely to seek psychological support.

Answer: A and D. Victims of interpersonal violence may experience traumatic physical injuries and may seek treatment from hospitals/emergency departments/trauma centers. The vast majority of these centers focus exclusively on stabilizing patients and clinicians in these settings may not recognize the important of assessing psychological and mental health issues victims may be experiencing post-injury. Most trauma centers are not equipped with the resources, staff, and systematic approach to identify and address the social factors that contribute to persistent community violence among the patient population they serve.[42] Individuals who are victims of interpersonal violence and have exposure to community violence are more at risk for reinjury/trauma recidivism.[43] Furthermore, discrimination and limited access to mental health facilities increase disparities in mental health for communities of color.[38]

3. Which of the following young populations is most at risk for poor physical and mental health outcomes from exposure to physical violence (e.g., homicide and aggravated assault)?
   a. Men
   b. Women
   c. African Americans
   d. Latino Americans

Answer: C. African American youth in the United States disproportionately carry the largest burden of violence experiences in general. The homicide rate for African Americans in the United States is eight times higher than that of their white counterparts[3] and African Americans are 500 times more likely to die this way.[5] African American young adults and adolescents only make up about 2.4% of the US population, but they are disproportionately victims of injuries from assault (26%) and homicide (20.7%).[3,6]

4. According to Rosenblatt et al.,[42] several fundamental principles of HVIPs include all of the following except:
   a. Violence is not preventable, but it can be managed.
   b. Hospitals should use public health–based principles to approach violence prevention, including working to address poor education, lack of job opportunities, injury and criminal recidivism, socioeconomically deprived neighborhoods, substance abuse, complex posttraumatic stress disorder, and lack of positive role models.
   c. There is a "teachable" moment during which a patient may be more receptive to psychological first aid and prevention principles following a traumatic injury.
   d. Intervention will begin in the hospital but must follow the patient and family into their community. Case management principles are implemented, ideally including professionals with therapeutic skill sets and "credible messengers" with knowledge of the patient population.

Answer: A. Several fundamental principles of HVIPs include:
"1. Violence is preventable,
2. Hospitals should use public health-based principles to approach violence prevention, including working to address the following in the patients and communities they serve: poor education, lack of job opportunities, injury and criminal recidivism, socioeconomically deprived neighborhoods, substance abuse, complex posttraumatic stress disorder, lack of positive role models,
3. There is a 'teachable' moment during which a patient may be more receptive to psychological first aid and prevention principles following a traumatic injury, and
4. Intervention will begin in the hospital but must follow the patient and family into their community. Case management principles are implemented, ideally including professionals with therapeutic skill sets and 'credible messengers' with knowledge of the patient population."[42]

5. Racism is not correlated with which of the following factors?
   a. Poor housing
   b. Inferior education
   c. Longer life
   d. Access to health services

Answer: C. African Americans in the United States are exposed to constant racism that is systemic (organized socially and culturally) through exclusion, prejudice, and discrimination.[21] Racism is correlated with second-rate employment, housing, education, income, and access to health services. It is well known that racism is a factor in health disparities and inequities that is not explained by other demographic factors such as age, gender, and educational attainment.[21] Also, there are many risks for individuals who are racial or ethnic minorities such as occupational hazards, exposures to toxic substances and allergens in the home, low-quality schooling, and lack of availability of healthy foods.[20] While these risk factors are not clinical in nature, they make up the social determinants of health, which includes factors such as race/racism, poverty, education, housing, access to healthy food, environmental exposures, violence, and gender.

191 Sexual Assault

KEY CONCEPTS

- Sexual assault is most common in women but can happen to gay and heterosexual men and to lesbian, gay, bisexual, transgender, and gender-nonconforming individuals.
- Sexual assault often results in no physical signs of injury.
- Optimal care includes creating a safe confidential environment while incorporating the principles of trauma-informed care. The patient should be included in decision making and ultimately decide treatment. Options include injury evaluation, treatment to prevent pregnancy and sexually transmitted infections (STIs), support and trauma counseling, evidence collection, and comprehensive toxicology testing if appropriate.
- The sexual assault evidence collection examination is an intensive, protocol-driven, multistep process, best performed by a certified sexual assault examiner.
- Adult sexual assault patients should be treated empirically according to Center for Disease Control (CDC) guidelines to prevent STIs (e.g., gonorrhea, syphilis, chlamydia, trichomonas, human immunodeficiency virus [HIV], human papillomavirus [HPV], and hepatitis B). Children and adolescents should be tested and, if symptoms develop, treated for STIs.
- All adolescent and adult female sexual assault patients should be offered pregnancy prophylaxis.
- HIV postexposure prophylaxis should be offered if the assailant is known to be HIV-positive, multiple assailants are involved, or if the HIV status of the assailant is unknown.
- Alcohol and drugs may have been ingested voluntarily or involuntarily by the patient. If the patient consents, comprehensive toxicology testing may be performed.
- A strangulation attempt with loss of consciousness, bowel and bladder incontinence, persistent voice changes, difficulty swallowing, or shortness of breath should be comprehensively evaluated in the emergency department (ED). Evaluation options include a chest x-ray, flexible laryngoscopy, and CTA or MRI of the neck. We recommend prolonged observation or hospital admission for persistent symptoms.
- Many victims will not have obvious physical injuries; this does not imply consent or refute a sexual assault.
- The emergency clinician should record observations, statements, and findings that were gathered during the course of ED treatment objectively.

1. Which of the following statements best describes hepatitis B infection prevention for victims of sexual assault?
   - **a.** Give HBIG and hepatitis B vaccine if the patient has not been immunized.
   - **b.** Give HBIG only if the patient has not been immunized.
   - **c.** Give hepatitis B vaccination if the patient is unimmunized or uncertain.
   - **d.** Give hepatitis B vaccine only if serologic testing shows that the patient is not adequately immunized.
   - **e.** Serologic testing is always required, followed by hepatitis B immunoglobulin (HBIG).

Answer: C. Give hepatitis B vaccination if the patient is unimmunized or uncertain. Follow-up doses should be given at 1 to 2 months and 4 to 6 months (total of three doses). This strategy, which avoids the need for serologic testing, has been shown to be effective. HBIG is not recommended by the CDC after sexual assault unless the assailant is known to be HBsAg-positive and the victim is unimmunized.

2. Which of the following empirical antibiotic regimens is indicated for sexual assault patients to prevent sexually transmitted infections?
   - **a.** Cefixime 400 mg PO
   - **b.** Cefixime 400 mg PO once plus doxycycline 100 mg PO bid for 10 days
   - **c.** Ceftriaxone 1 g IM (intramuscularly)
   - **d.** Ceftriaxone 1 g IM plus azithromycin 2 g orally (PO)
   - **e.** Ceftriaxone 250 mg IM plus metronidazole 2 g PO plus azithromycin 1 g PO

Answer: E. Ceftriaxone is given to cover gonorrhea, azithromycin to cover chlamydia, and metronidazole (Flagyl) to cover *Trichomonas*. Ceftriaxone is preferred over oral cefixime to cover incubating syphilis and due to increasing gonorrhea resistance. Many providers opt to give the metronidazole to take at home because it increases the risk of nausea and vomiting, a common side effect of many of the medications (including emergency contraception and HIV postexposure prophylaxis).

3. Which of the following statements best describes sexual assault in males?
   - **a.** Ejaculation should not occur in the victim during male sexual assault.
   - **b.** Males are more likely to overreport sexual assault.
   - **c.** Males are more likely to require sexually transmitted infection (STI) prophylaxis.
   - **d.** Males do not require referral to rape crisis centers.
   - **e.** Males may require anoscopy to detect anogenital injuries.

Answer: E. Males may actually suffer more anogenital injuries than women; injury detection can be aided or enhanced by using an anoscope. Males underreport the crime, do not seek medical attention, and absolutely need referral to a rape crisis center for post-rape care and counseling. Males are not more likely to require STI prophylaxis because the risk of transmission per act does not change based on gender. Ejaculation may occur during sexual assault due to prostatic stimulation and fear arousal. This should not be taken to infer that the assault was consensual.

4. Which of the following factors reduces the likelihood of finding genital injury during the sexual assault examination?
   - **a.** Digital penetration
   - **b.** Increased time since sexual assault occurred
   - **c.** Penile penetration
   - **d.** Use of foreign object during the assault
   - **e.** Victim sexual immaturity

Answer: B. The genital structures heal quickly, so the longer the time since the sexual assault occurred, the less likelihood of finding evidence of injury on examination. All the other factors increase the likelihood of finding genital injury at the time of the sexual assault examination.

5. A 25-year-old woman presents 4 days after vaginal penetration. Her body mass index (BMI) is 33. Which of the following is true about emergency contraception (EC)?

- **a.** A pregnancy test is mandatory prior to offering EC.
- **b.** She should be offered levonorgestrel because it is more effective in this situation.
- **c.** She should be offered ulipristal.
- **d.** She should have an intrauterine device (IUD) inserted because this is the most effective form of EC for her.
- **e.** She should not receive EC because it will likely be less effective due to her BMI.

Answer: C. Emergency contraception should be offered up to 5 days after vaginal assault. Ulipristal, levonorgestrel, and high-dose birth control pills are options. Ulipristal is more effective after 72 hours and in women with a BMI greater than 26. At a BMI above 35, both forms of oral EC are less effective, but should still be administered if there is no alternative. IUD placement is the most effective form of EC; it can be placed up to 5 days after assault. IUD placement allows for ongoing birth control in situations where there is likely to be loss of reproductive control (intimate partner assault) but is often less desirable after assault. IUD placement is most often not available in a timely manner. A pregnancy test is not mandatory before giving EC because it will not harm an existing pregnancy. A pregnancy test is suggested prior to ulipristal administration, given the lack of large studies in pregnant women.

KEY CONCEPTS

- Intimate partner violence encompasses a pattern of controlling behaviors, including intentional physical assault, sexual assault, psychological violence, and financial control.
- Intimate partner violence (IPV) patients are best served by a coordinated system response plan that includes staff training, social work, victim advocates, and a close relationship with area IPV service provider groups and law enforcement.
- The emergency clinician's role in caring for patients affected by intimate partner violence involves 4 basic steps.
  1. Identification (asking the patient)
  2. Treatment (supporting messages and medical issues)
  3. Documentation (regarding violent actions and threats)
  4. Referral (e.g., community, social services, legal)
- Sequelae of IPV include chronic pain, mental health issues (e.g., depression, posttraumatic stress syndrome [PTSD], substance abuse), sexually transmitted infections (STIs) and unintended pregnancy, and worsening of medical illnesses.

1. Intimate partner violence occurs in which of the following population of patients?
   a. Restricted to lower socioeconomic classes and is never seen in wealthy highly educated individuals.
   b. Found only amongst women in heterosexual relationships and not in other individuals.
   c. Patient has injuries from a weapon used in the assault.
   d. Similar prevalence in same-sex and gender nonconforming individuals as in heterosexual couples.

Answer: D. IPV occurs with at least as high a prevalence in same-sex and gender nonconforming individuals as in heterosexual couples. It also is found commonly in wealthy and highly educated individuals. Most patients who present to the ED with conditions related to IPV do not have injuries at the time of the visit. Finally, IPV violence consists of acts or threats of physical, psychological, or sexual violence between intimate or formerly intimate partners who need not ever have been married.

2. The emergency clinician after identifying a patient suffering from IPV should employ which of the following strategies?
   a. Call the police and have the perpetrator immediately arrested.
   b. Validate the disclosure of abuse and communicate the importance of leaving the situation immediately to ensure personal safety.
   c. Provide kind and supportive messages and information.
   d. Report to the public health department.

Answer: B. Emergency clinicians should validate the disclosure of abuse, emphasize that the victim is not at fault, and encourage future discussions with IPV community agencies or other health care providers. Immediate safety should be assessed, but most patients will not want to leave the abuser immediately; however, a positive initial conversation may begin the process of ending the abusive relationship. Reporting to law enforcement or health department should be done only with victim consent or if mandated by law. A templated list will allow the ED staff to create a basic safety plan with the patient; an individualized plan is best done in conjunction with trained domestic violence advocates, typically in follow-up. Referral to couples counseling is inappropriate and ineffective in violent relationships and potentially dangerous.

3. The best practice for inquiring about IPV is which of the following?
   a. Use open-ended questions that describe actions and feelings.
   b. Use the term "abuse" when questioning the patient.
   c. Ise the term "victim" to ensure the patient knows the clinician is concerned.
   d. Question the patient's partner about what happened.

Answer: A. Partners should be asked to leave in a routine manner before the patient is asked about IPV. Patients will rarely admit to abusive acts unless they are guaranteed privacy, and initiation of the topic is potentially dangerous in the presence of an abusive partner who should not be questioned by providers. Providers should query the patient with any reasonable options that provide an empathetic and open approach facilitating improved screening rates. Providers should ask about specific actions and avoid use of the terms "victim" and "abuse" because patients may not yes see the actions as abuse or themselves as a victims.

4. Which of the following is most suspicious for intentional injury from IPV?
   a. Ecchymosis of lower extremity
   b. Paronychia
   c. Pattern injuries
   d. Infected earring

Answer: C. Signs of an intentional injury include a central location (i.e., trunk and breasts), bilateral injuries (both arms or both legs), defensive injuries (i.e., ecchymoses on the back of the hand as a result of protecting the face), and patterned injuries (such as carved words on the back, a cigarette burn or other pattern of an object,).

5. By the state law where Dr. Jane works, emergency clinicians practicing in the state are mandated reporters for suspicious injuries from assaultive or abusive conduct. After identifying a patient with such suspicious injuries from IPV Dr. Jane should do which of the following regarding reporting?
   a. Not report because the Health Insurance Portability and Accountability Act (HIPAA) prohibits reporting of this patient's injuries.
   b. Report without telling the patient so the patient cannot object.
   c. Report as required by state law because it supersedes patient's HIPAA consent rights.
   d. Obtain a signed consent to make the report.
   e. Call the legal department before reporting.

Answer: C. Some states have laws that require injury reporting to local authorities. Patients should be informed about state-specific reporting requirements that may accompany disclosure of abusive injuries. Mandatory reporting of health conditions required by local laws are exempted from HIPAA regulations. In general, IPV should not be reported to police without the consent of the survivor unless mandated by law (as in this case) or in cases of coexisting child, elder, or disabled abuse or based on state-specific reporting statutes (e.g., burns or injuries inflicted by weapons). If reporting is mandated, the provider should make every effort to involve the patient. However, concerns for HIPAA violations do not apply in this circumstance; the privacy rule contains a provision allowing disclosure of protected health information to law enforcement in the case of reporting required by law.

KEY CONCEPTS

- World Health Assembly resolution 72.16, "Emergency care systems for universal health coverage: ensuring timely care for the acutely ill and injured," was passed in 2019 and calls for the strengthening of emergency care delivery through a detailed list of initiatives across the prehospital, facility, public health, and disaster preparedness fields.
- In low- and middle-income countries (LMICs), substantial excess mortality amenable to emergency health care exists, and the development of high-quality emergency care can be a significant systemic contribution to improving mortality.
- Global emergency medicine (GEM) is an academic subspecialty of emergency medicine focused on the strengthening of the specialty worldwide and on improving international humanitarian response.
- GEM can be divided into development and humanitarian work. Development work is typically focused on strengthening local systems in a sustainable manner, while humanitarian work involves an acute response to large-scale crises in which local resources are overwhelmed.
- The model curriculum for a GEM fellowship includes experiences in emergency medicine systems, humanitarian relief, disaster medicine, public health, travel and field medicine, program administration, academic skills, and clinical skills.

1. Research projects based in settings outside of the GEM physician's home country require which of the following special considerations with regard to research ethics?
   - **a.** Ethical approval must be obtained only at the GEM physician's home institution.
   - **b.** United States-based IRBs do not cover international settings, and as such the GEM physician does not need to seek ethical approval.
   - **c.** Ethical approval is required from both the GEM physician's home institution and any partner institutions or research sites.
   - **d.** If the research is planned in a country that does not have an IRB, no formal ethical approval is required.

Answer: C. Ethical approval for any research project must be obtained from the GEM physician's home institution, any partner institutions, and any research sites. In the event that a formal IRB does not exist for a certain research site, efforts must be made to obtain and document ethical approval from an appropriate site representative.

2. Which of the following is required when planning a clinical experience abroad in which the GEM physician will be providing direct clinical care without supervision?
   - **a.** Current medical licensure from the country within which care will be delivered.
   - **b.** Current medical licensure from the GEM physician's home country.
   - **c.** Written agreement between the GEM physician and a licensed physician in the country within which care will be delivered.
   - **d.** No local medical license is required as long as the GEM physician is practicing within the same scope as allowed by home licensure.

Answer: A. The direct delivery of unsupervised medical care should only be provided by a GEM physician with a current and valid license issued by the appropriate regulatory body of the country within which care will be delivered. Additional requirements, such as hospital credentialing, are also generally required.

3. The practice of global emergency medicine is an academic subspecialty that focuses on which areas of medicine and public health?
   - **a.** All areas of medicine are represented except for surgical conditions, as these are addressed by global surgeons.
   - **b.** All areas of medicine for all age groups are represented, inclusive of trauma or surgical conditions, infectious and noninfectious conditions, and pregnancy.
   - **c.** All adult areas of medicine are represented, but not pediatrics as this field is addressed by global pediatricians.
   - **d.** GEM focuses only on emergency conditions, excluding chronic disease as this generally does not contribute to the need for GEM.

Answer: B. Recognizing that patients present to the emergency department (ED) without a diagnosis, but with a complaint, GEM focuses on presentation-based solutions for conditions for all age groups, both trauma and medical, infectious and noninfectious, and pregnancy. GEM seeks to ensure early access to quality emergency care for all people, regardless of gender, ethnicity, age, or ability to pay.

4. What is the relationship between humanitarian work and development work as it pertains to the GEM physician?
   - **a.** Humanitarian work and development work are best performed separately to avoid confusion around roles, tasks, and funding commitments.
   - **b.** GEM physicians provide only development work and do not focus on humanitarian response, which requires separate training and expertise.
   - **c.** Humanitarian and development work are the same thing, and the difference in terminology is a function of who is performing the actual work.
   - **d.** Humanitarian work and development work complement and strengthen each other.

Answer: D. While GEM is broadly divided into humanitarian and development work, these two subfields are closely related and complement each other. An emergency care system strengthened through development work should be able to surge during crises to enhance humanitarian efforts, and an effective humanitarian response will include strengthening the local system to provide resilience against future crises.

e2 Humanitarian Aid in Disaster and Conflict

KEY CONCEPTS

- A key feature of humanitarian crises is the mass displacement of large numbers of people from their homes. Displaced populations are designated either as internally displaced persons (IDPs) or refugees.
- The international response community has developed practice guidelines that stem from international law, research, and expert consensus. Responders should be familiar with these international standards and laws to ensure that intervention is appropriate and ethical.
- The *Sphere Handbook,* a collaborative project of humanitarian response experts, sets international standards for the provision of humanitarian aid and is widely considered to define the gold standard of global humanitarian response.
- Demand for humanitarian responders is likely to rise in the future as global urbanization and an increase in climate-related disasters conspire to create more frequent and severe crises.

1. Which of the following scenarios often creates ethical dilemmas for those providing humanitarian assistance?
   a. Availability of water
   b. Creating housing for patients
   c. Local government policies
   d. Triage of patients by local custom

Answer: D. Although all these issues are important, humanitarian workers often have ethical concerns about triage based on local customs versus severity of illness

2. Which of the following best describes the *Sphere Handbook*?
   a. A global compilation of nongovernmental agencies
   b. A description of legal decisions in international courts
   c. An educational guide to global relief efforts
   d. A guide that sets internal standards for humanitarian aid

Answer: D. The *Sphere Handbook*, a collaborative project of humanitarian response experts, sets international standards for the provision of humanitarian aid and is widely considered to define the gold standard of global humanitarian response.

3. What is the distinction between internally displaced persons and refugees?
   a. Internally displaced persons may remain in country or reside in another country.
   b. Refugees are persons who are displaced and reside in another country.
   c. Internally displaced persons have more rights and benefits than refugees.
   d. The refugee is dependent on the number of displaced persons.

Answer: B. Displaced populations are designated either as internally displaced persons (IDPs) or refugees. An IDP is someone who has been forced from his or her home but who remains within the country of origin. If an IDP crosses an international border to seek help in another country, she or he becomes a refugee. Although refugees enjoy protections and rights guaranteed by international treaty, IDPs must depend (in the absence of international aid) on their own government for help, even though the actions—or inaction—of their government often caused their displacement,

KEY CONCEPTS

- Over the past 20 years, emergency ultrasound (EUS) has become an integral part of emergency medical care in the United States and has become standard in the evaluation of emergency medical conditions.
- EUS answers specific, often binary questions, and is therefore neither sufficient nor intended to diagnose all of the broad range of pathologic processes encountered in emergency medicine.
- During cardiac arrest, ultrasound can be used to rapidly detect ventricular motion in asystole and pulseless electrical activity and confirm cardiac standstill.
- The subcostal four-chamber view, as in the focused assessment of sonography in trauma, is ideal for assessment of pericardial effusion and useful during cardiac arrest because it does not interfere with chest compressions.
- Although cardiac tamponade is a clinical diagnosis, there are several suggestive echocardiographic features, including diastolic collapse of the RV, loss of respiratory variation of the inferior vena cava (IVC), and transvalvular flow velocity paradoxus.
- EUS is more sensitive and specific compared with supine chest radiography for the detection of pneumothorax, approaching that of CT.
- Pneumonia on EUS appears as dynamic air bronchograms, hyperechoic areas within bronchi that move with respiration usually visualized within the consolidated lung, and are highly specific for alveolar consolidation.
- Color Doppler can help differentiate mild hydronephrosis from renal vasculature, and possibly accentuate any renal stones by producing the renal twinkle artifact.
- EUS has reduced the morbidity of ectopic pregnancy by shortening the time to diagnosis and operating room treatment.
- In the ED, ultrasound-guided internal jugular cannulation is associated with decreased time to vessel penetration and improved success in the difficult access patient, improved overall and first-attempt success rates, reduced time to insertion, and reduced complication rate.

1 Which frequency range defines US?
- **a.** Less than 0.2 Hz
- **b.** Over 2 Hz
- **c.** Over 200 Hz
- **d.** Over 20,000 Hz

Answer: D. US is defined as sound with frequency over 20,000 Hz.

2 Which of the following performance characteristics is true for higher-frequency probes?
- **a.** Deeper penetration, higher resolution
- **b.** Deeper penetration, poorer resolution
- **c.** Superficial penetration, higher resolution
- **d.** Superficial penetration, poorer resolution

Answer: C. Higher-frequency probes sacrifice penetration (depth of scan) for improved resolution (image detail).

3 Which type of US has the best resolution when the probe is perpendicular to the object of interest?
- **a.** Doppler US
- **b.** Grayscale US

Answer: B. Grayscale or B-mode US produces best imaging when perpendicular to the area of interest, whereas Doppler US is best performed when the angle of insonation is <60 degrees.

4 Sensitivity is typically greater than specificity for FAST examination.
- **a.** True
- **b.** False

Answer: B. The FAST examination is typically more specific than sensitive.

5 Which of the following is the correct order of imaging for increasing the sensitivity of detecting a pneumothorax?
- **a.** Chest x-ray (CXR) > CT > US
- **b.** CT > CXR > US
- **c.** CT > US > CXR
- **d.** US > CT > CXR

Answer: C. CT is only slightly more sensitive than US at detecting pneumothorax, but US is significantly more sensitive than supine CXR.

KEY CONCEPTS

- The rapid increase in the number of older adults in the United States (US) and around the world, as well as the unsustainable costs of the current US health care system, mandates improved emergency care systems for these vulnerable patients.
- Older adults are at high risk of experiencing harm from busy, crowded emergency departments (ED). While in the ED, they may experience prolonged lengths of stay, iatrogenic infections, misdiagnosis, delirium, bedsores and other adverse consequences.
- Older adults who are discharged are at high risk of ED return visit, hospitalization, or death within three months. Because older adults admitted to the hospital are at high risk of adverse events, care transition programs are needed to reduce the risk of inpatient stays and improve post-ED care.
- The Geriatric ED Guidelines are an excellent resource to improve the physical and process design to better address the needs of older adults.
- The Geriatric ED Collaborative and Geriatric ED Accreditation programs are innovative programs that facilitate the implementation of better emergency care for older adults.

1. For which of the following outcomes are hospitalized elders at increased risk?
   a. Infections
   b. Delirium
   c. Functional decline
   d. Iatrogenic injuries
   e. All of the above

Answer: E. Geriatric patients are particularly vulnerable to the deleterious consequences of hospitalization, including nosocomial infections, functional decline, increased rates of delirium, and iatrogenic complications.

2. The Geriatric ED (GED) Guidelines developed by ACEP, SAEM, AGS, and ENA make recommendations in all of the following areas except:
   a. Education and patient care protocols
   b. Proper technique for central line placement
   c. Quality improvement measures
   d. Staffing, supplies, and administration
   e. Follow-up and transitions of care

Answer: B. The GED guidelines consist of 40 specific recommendations in six general categories: (1) staffing and administration; (2) equipment and supplies; (3) education; (4) policies, procedures, and protocols; (5) follow-up and transitions of care; and (6) quality-improvement measures.

3. The GED guidelines recommend elder-specific educational content in all of the following areas except:
   a. Atypical presentations of disease
   b. Cognitive and behavioral disorders
   c. Psychotherapy for adjustment disorders
   d. Pain management and palliative care
   e. Medication management

Answer: C. See Box e4.1.

4. ED consultation from which of the following interdisciplinary services has been shown to decrease the likelihood of an elder returning to the ED with a fall within 30 days?
   a. Pharmacy
   b. Social work
   c. Case management
   d. Physical therapy
   e. Occupational therapy

Answer: D. Physical therapy. Although all of the above interdisciplinary services likely improve care for older adults in the ED, being evaluated by a physical therapist in the ED after a fall is associated with lower likelihood of returning to an ED with another fall within 30 days.

5. Which of the following core content areas is not a component of geriatric emergency medicine education?
   a. Elderly trauma, including falls and hip fracture
   b. Myocardial infarction
   c. Cognitive and behavioral disorders
   d. Medication management
   e. Transitions of care

Answer: B. Although myocardial infarction is a condition commonly suffered by older adults, knowledge of myocardial infarction is not in and of itself geriatric emergency medicine education. Geriatric emergency medicine education involves geriatric syndromes, conditions underlying the acute presentation of disease such as polypharmacy or frailty, and management of conditions that are different for older adults than younger adults.

e5 End of Life

KEY CONCEPTS

- Palliative care is physical, spiritual, psychological, and social support provided to patients and families at any stage of serious illness.
- Palliative care teams are specialized interdisciplinary teams that should be involved early in the course of illness.
- Hospice care is a care system for patients and families with a prognosis of 6 months or less if the disease runs its usual course; referral from the emergency department (ED) may be indicated.
- Patients at the end of life often follow one of four terminal illness trajectories: sudden death, organ failure, cancer, or frailty. Awareness about these trajectories facilitates decision making.
- Outcomes are very poor for patients with advanced age or illness and cardiac arrest or respiratory failure requiring mechanical ventilation. Decision support tools should be incorporated into shared decision making.
- Goals-of-care conversations that outline patient and family hopes and expectations of interventions should be attempted in the ED.
- Emergency clinicians should make recommendations for care plans, interventions, and treatment courses based on prognosis, goals of care, and expected benefits of meeting the identified goals of care.
- Interpretation of an existing advance care plan or receipt of a prehospital order (e.g., POLST—***P***hysician ***O***rders for ***L***ife-***S***ustaining ***T***herapy) should factor into ED treatment plans in the context of the patient's illness trajectory.
- Symptom management (e.g., dyspnea, pain, delirium, agitation, secretions) is critical at the end of life.

1. Mr. Samuels is a 46-year-old man with severe breathlessness and anxiety due to advanced chronic obstructive pulmonary disease. His vital signs are temperature, 36.8°C (98.2°F); blood pressure, 110/60 mm Hg; heart rate, 115 beats/min; respiratory rate, 34 breaths/min; oxygen saturation, 98% on room air. He is visibly anxious and reports dizziness and dyspnea. Which of the following is the best initial treatment?

a. IV fluids
b. Lorazepam
c. Morphine
d. Oxygen

Answer: C. In the absence of hypoxia, morphine has been shown to relieve the sensation of breathlessness. Benzodiazepines have not been shown to relieve breathlessness and may increase side effects.

2. The primary focus of palliative care is to
   a. transition dying patients to hospice.
   b. administer morphine to terminally ill patients.
   c. improve the quality of life of patients living with serious illnesses.
   d. facilitate the withdrawal of life-sustaining treatments in terminally ill patients.

Answer: C.

3. Emergency providers commonly discuss the issue of code status with their patients. In terms of prognosis after a cardiac arrest,
   a. a recent study of nursing home patients experiencing cardiac arrest found high rates of survival to hospital discharge.
   b. there are no good data, so providers should rely on their gestalt when counseling patients and families about code status determination.
   c. a clinical decision rule called GO-FAR has been validated and can give providers predictions for neurologically intact survival in individual patients using variables from their clinical data.
   d. older patients and those with a terminal illness fare surprisingly well.

Answer: C.

4. An 80-year-old patient with stage IV pancreatic cancer, enrolled in hospice, presents to the ED in respiratory distress. His vital signs are HR 120, RR 30, O_2 saturation 85% on RA and 100% on NRB, BP 100/50. He lives at home with his family and has been relatively stable and asymptomatic until today. The emergency provider should do all of the following, EXCEPT
   a. assume that because the patient called 911, his goals are to focus on life prolongation and that he would want to be intubated.
   b. notify the patient's hospice provider that the patient is in the ED and obtain more history about recent events.
   c. gather clinical data, review advance directives, and then engage the patient and/or family in a conversation about goals of care.
   d. consider administering low-dose morphine to alleviate the patient's dyspnea.

Answer: A.

5. Which of the following is expected to be the least likely cause of death for adults in the United States?
   a. Cancer
   b. Cardiovascular disease
   c. Heart disease
   d. Trauma

Answer: D. According to the Center for Disease Control, accidents are the third leading cause of death in the United States. (http://www.cdc.gov/nchs/fastats/leading-causes-of-death.htm.)

6. In a nondeath disclosure setting, deliverance of bad news should start with which of the following approaches?
   a. Allowing the patient to react to the news
   b. Asking the patient if he or she wants to know
   c. Making sure you keep the conversation going to provide comfort and thereby dull the emotional impact of the news
   d. Your knowledge of the complete medical implications and ramifications of the news

Answer: B. Respectful disclosure of difficult news includes assessing the readiness and willingness of the patient to receive the news.

7. Mr. Jones is a 54-year-old man currently receiving hospice care at home for advanced colon cancer. His wife calls 911 when he becomes unresponsive. On presentation to the ED, his vital signs are blood pressure, 60/40 mm Hg; heart rate, 140 beats/min; temperature, 39.6°C (103.3°F); respiratory rate, 12 breaths/min. He is moaning with a facial grimace. His wife, who is the legal decision maker, is at the bedside. Which of the following represents the best course of action?
   a. Initiate sepsis protocol, begin treatment for pain, and call the palliative care service.
   b. Initiate sepsis protocol, page the hospice nurse, and speak to his wife regarding goals of care.
   c. Page the hospice nurse, begin treatment for pain, and speak to his wife regarding the goals of care.
   d. Page the hospice nurse and speak to his wife regarding the goals of care.

Answer: C. Under the hospice benefit, the hospice agency is the care manager. The patient has lost capacity and his wife is his decision maker and she should be consulted regarding goals. Pain and non-pain symptoms should be addressed while other actions are being taken.

8. Mrs. Smith is a 68-year-old woman with advanced heart failure who presents in cardiogenic shock with multiple previous admissions that state that she is currently optimized with all available therapies. You decide to initiate hospice care from the ED. Which of the following do you do next?
   a. Call the hospice liaison on call for the ED to speak with her family regarding their options.
   b. Call the palliative care service to discuss hospice care with the patient and her family.
   c. Conduct a goals-of-care discussion with the patient and her family.
   d. Discuss the benefits of hospice care that would allow the patient to stay at home with her family.

Answer: C. Central to the discussion of hospice care is an assessment of the goals of care focused on care that addresses the distress of the illness and not the reversal of the primary illness.

9. In performing a death disclosure, which of the following statements represents best practice when communicating news of death?
   a. "Your loved one didn't make it."
   b. "Your loved one has died."
   c. "Your loved one has gone to a better place."
   d. "Your loved one has passed on."

Answer: B. It is not recommended to use euphemisms or unclear statements to communicate news of death because this can cause confusion and ambiguity for the receiver of the news.

10. Mr. Stone is a 27-year-old man with glioblastoma multiforme. He has completed radiation treatment and is on oral chemotherapy. Over the last week, he has become progressively weaker and is now in bed. He is no longer speaking and is dropping things. He is taking sips of clear liquids and coughs after swallowing. Which of the following is the strongest predictor of his death?
   a. Aspiration risk
   b. CT scan that shows progression
   c. Deterioration of speech
   d. Functional status

Answer: D. The palliative performance scale is the most studied prognostication tool for patients with terminal illness. It incorporates the patient's ability to perform activities of daily living (ADLs), level of consciousness, and oral intake[7] (see Table e5.1). In cohorts of patients with a terminal illness, a score of < 50% correlates with a prognosis of weeks, not months.

KEY CONCEPTS

- Bioethics is a method of using values and moral principles to come to defendable decisions for ethical dilemmas. In emergency medicine, many ethical dilemmas often go unrecognized.
- Ethical dilemmas arise from conflicts between multiple good options or multiple bad options, not between good and bad choices.
- Values come from a variety of sources: community, culture, religion, and family. The values driving medical decisions should be the patient's values.
- Basic ethical principles include autonomy, beneficence, nonmaleficence, justice, and confidentiality.
- Consent requires decision-making capacity or appropriate surrogate decision makers.
- Not only religious but also family, cultural, and other values contribute to patients' medical care decisions. Without asking, it is impossible to know what decision a specific person would make.
- Patient autonomy recognizes an adult person's right to accept or to reject recommendations for medical care, even to the extent of refusing all care, if that person has appropriate decision-making capacity.

1. Which of the following statements regarding patient confidentiality is FALSE?
   - **a.** It describes the patient's right to enough physical and auditory isolation so that they cannot be seen or heard by others during interactions with medical personnel.
   - **b.** It imposes a duty on health care workers.
   - **c.** It may conflict with the law, especially public health statutes.
   - **d.** It presumes that what the patient tells the physician will not be revealed to any other person or institution without the patient's permission.

Answer: A. Although isolation of patients for confidentiality is a good goal, the limited space and hectic nature of an ED would sometimes prevent timely and quality care.

2. Which of the following can emergency clinicians always rely on for appropriate guidance when faced with ethical dilemmas?
   - **a.** State laws and medical board policies
   - **b.** The American College of Emergency Physicians' (ACEP) Code of Ethics
   - **c.** The Hippocratic Oath
   - **d.** None of the above

Answer: D. All these examples are great for guidance on ethical dilemmas, but none of them can "always" have guidance for every situation that can occur.

3. To have adequate decision-making capacity in any circumstance, a person must understand all of the following EXCEPT:
   - **a.** The available options.
   - **b.** The consequences of acting on those options.
   - **c.** The costs and benefits of consequences in relation to a relatively stable framework of personal values and priorities.
   - **d.** The court's decision about their competency.

Answer: D. To be deemed competent an individual must understand A, B, and C. Answer D is a legal issue and is not required for competency.

4. Which of the following statements regarding the withholding of treatment in the ED is TRUE?
   - **a.** It legally differs from withdrawing treatment.
   - **b.** It morally differs from withdrawing treatment.
   - **c.** It requires clinical information that is often unavailable immediately.
   - **d.** It should never be done.

Answer: C. In an emergency not all relevant information is available at the time important decisions must be made.

5. When using the "rapid approach to ethical problems in the emergency department (ED)" to decide on a course of action:
   - **a.** Always consult with the bioethics committee/consultant before acting.
   - **b.** Assume that each ethical problem in the ED requires a unique solution.
   - **c.** Test your chosen action against your religious values.
   - **d.** When practicable and safe for the patient, buy time to consult on possible options.

Answer: D. Ethical problems can often take some time to work out, but a rapid approach rarely has enough time for a formal ethics consult. Buying some time so a reasonable solution can be derived is often the best that can be done.

e7 Emergency Medical Treatment and Labor Act and Medicolegal Issues

KEY CONCEPTS

- The Emergency Medical Treatment and Labor Act (EMTALA) governs virtually every aspect of hospital-based emergency services.
  - Centers for Medicare and Medicaid Services (CMS) deems anyone on hospital property to have "come to the emergency department." Hospital property consists of the entire main hospital campus, including parking lots, sidewalks, and driveways, and any ambulance owned and operated by the hospital, even if the ambulance is not on hospital grounds.
  - Any person who comes to an ED requesting examination or treatment for a medical condition must be provided with an appropriate medical screening examination (MSE). The purpose of the MSE is to determine whether the patient has an emergency medical condition (EMC).
  - Presentation to the ED or on hospital property is not sufficient to trigger the hospital's duty to provide an MSE; a request for examination or treatment of a medical condition is also necessary. The request can be made by the patient or by anyone on behalf of the patient (e.g., EMS personnel, a police officer, or a babysitter, family member, or legal guardian).
  - CMS regulations require that the screening examination be done by "qualified medical personnel." Typically, the designation applies to physicians or advanced practice providers (nurse practitioners [NPs] or physician assistants [PAs]).
  - An appropriate MSE must be reasonably calculated to identify critical medical conditions and must be uniformly provided to all patients who present with substantially similar complaints.
  - Hospitals need to have a policy and practice for left-without-being-seen (LWBS) patients that adequately documents pertinent findings and protects the hospital from liability.
  - Medicare-participating hospitals that have specialized capabilities or facilities are required by EMTALA to accept appropriate transfers of patients who require such capabilities or facilities if the hospital has the capacity to treat the patient.
  - Once a hospital admits the patient in good faith as an inpatient for further treatment, the hospital's obligation under EMTALA ends.
- Emergency Consent for Procedures
  - The emergency clinician who proposes to undertake the procedure must be the one to obtain the patient's informed consent. The duty to obtain consent cannot be delegated.
  - Whenever emergency clinicians are in doubt about the legality of a situation related to consent, they should do and document what they believe to be in the patient's best interest. Delaying treatment in an emergency to obtain informed consent is a much more serious and common medicolegal problem than failure to obtain proper informed consent.
  - Consent for minors in the ED is a nonissue. Consent is a creature of state law; EMTALA preempts state law, and it requires the ED to provide an MSE and stabilizing treatment to all minors presenting to the ED.
  - A parent's right to freedom of religion (e.g., Jehovah's Witness) does not include the right to deny life-sustaining medical intervention for that person's children.
- Leave Against Medical Advice
  - Emergency clinicians should be cautious when patients the demonstrate intention to leave the ED against medical advice. The capacity to make reasonable medical decisions can easily be affected by alcohol, drugs, pain, or any number of medical conditions. Careful evaluation and documentation of the patient's decision-making capacity supports the decision to allow the patient to leave or to retain the patient against their will. The courts (and public opinion) tend to support clinicians who act in the best interests of the health and safety of their patients.

1. Which of the following is a common source of liability for failure to comply with the Emergency Medical Treatment and Labor Act (EMTALA)?
   - **a.** Failure to comply with written hospital policies and procedures
   - **b.** Failure to diagnose an emergency medical condition (EMC)
   - **c.** Failure to properly treat an admitted patient who is boarded in the emergency department (ED) while waiting for an intensive care unit (ICU) bed to become available
   - **d.** Failure to stabilize a patient after admission to an inpatient unit

Answer: A. Hospital policies become the hospital's standard under EMTALA. Thus, failure to comply with written policies, a failure to follow your own rules, becomes an EMTALA violation when the policies are related to the medical screening and stabilization of patients in the ED.

2. Which of the following violates the Emergency Medical Treatment and Labor Act (EMTALA)?
   - **a.** A 7-hour wait after triage to be medically screened by the emergency clinician
   - **b.** Boarding an admitted patient in the emergency department (ED) for 72 hours after the patient's physician has accepted the patient for admission
   - **c.** Delaying the patient's access to an ED bed by declining to promptly accept the patient from an arriving emergency medical service (EMS) unit or by failing to immediately triage or evaluate the patient to determine if the patient can wait on a stretcher with the EMS folks until a bed becomes available
   - **d.** Delaying the patient's medical screening examination (MSE) in the ED to triage the patient, and then for the emergency clinician to see the patients in the order based on the triage nurse's perception of the patients' acuity

Answer: C. Centers for Medicare and Medicaid Services (CMS) guidelines require the hospital to immediately triage ambulance patients arriving at the hospital's ED.

3. A 73-year-old woman is brought by private ambulance to the emergency department (ED) for shortness of breath. Due to ED overcrowding, there are several ambulance stretchers in line ahead of her at triage. At what point does the hospital accept responsibility for this patient?
   - **a.** When she was placed in the ambulance
   - **b.** When the ambulance arrived on hospital property
   - **c.** When the patient is seen by the emergency clinician
   - **d.** When the patient is seen by triage personnel

Answer: B. The Emergency Medical Treatment and Labor Act (EMTALA) defines the time of acceptance of responsibility as when the patient comes to the ED. In its EMTALA regulations, the Centers for Medicare and Medicaid (CMS) defines the term *comes to the ED* to include anywhere on hospital property, so the hospital's duty and responsibility for the patient are triggered when the ambulance arrives on hospital property.

4. A physician on-call for the hospital's emergency department (ED) has a duty under the Emergency Medical Treatment and Labor Act (EMTALA) to come to the ED under which of the following circumstances?
   - **a.** If asked to help stabilize a patient in the ED who has an emergency medical condition (EMC)
   - **b.** To help transfer a stable patient to another acute care hospital that has a specialist available that is not available at the transferring hospital
   - **c.** When asked to consult on a patient who has been stabilized in the ED and is being admitted by the patient's primary care physician
   - **d.** Whenever requested to come to the ED by the emergency clinician on-duty

Answer: A. Under EMTALA, the on-call physician only has a duty to come to the ED if needed to help determine whether the patient has an EMC or to stabilize an EMC. Any other requirements concerning when the on-call physician must come to the ED are governed by state law and/or by the hospital's medical staff bylaws, not by EMTALA.

5. According to the Centers for Medicare and Medicaid Services (CMS), which of the following patients may a hospital with specialized capabilities and available capacity refuse to accept in transfer solely because the patient is uninsured?
   - **a.** A patient with an unstable emergency condition admitted to observation at another hospital that is unable to stabilize the patient's condition
   - **b.** Any patient the hospital chooses to reject
   - **c.** An emergency department (ED) patient with an unstable emergency condition who is at another hospital that is unable to stabilize the patient
   - **d.** An inpatient with an unstable emergency condition at another hospital that is unable to stabilize the patient

Answer: D. CMS guidelines state that hospitals do not have to accept an inpatient, as defined by CMS, in transfer from another hospital under any circumstances, even if its decision to decline to accept the patient in transfer will result in the patient's death.

KEY CONCEPTS

- The work of health care occurs within a complex socio-technical system so that a change in even one component has an impact on other parts of the system and ultimately on safety.
- Patient safety and mitigation of risk emerges from multifactorial interactions within the clinical work system. Incidents of patient or staff harm are most often the result of system failures, not individual human error.
- Process improvements to clinical care often have unintended consequences elsewhere within the work system. Partnerships with information technology; cognitive, behavioral, and social scientists; and engineers can help reduce the likelihood of these events and improve adoption by workers.
- Don't set up reporting systems without the resources to fully analyze the reports.

1. Which of the following triage assessments are important contributors to error in emergency medicine?
   - **a.** Triage by age
   - **b.** Triage by nursing personnel
   - **c.** Inappropriate triage to a treatment area
   - **d.** Triage based on comorbidities
   - **e.** In-room patient triage and registration

Answer: C. An inappropriate triage to a specific treatment area may create a bias in the minds of the treating physician and staff, leading to delays in care or misdiagnoses. This may be more pronounced with undertriage versus overtriage.

2. A senior attending asks a resident to discharge a young patient with abdominal pain. The resident has noted lab abnormalities she believes merits further work-up, but assumes the attending has seen these results and defers to the attending's judgment. These abnormalities had in fact been overlooked by the attending and the patient gets sent home without appropriate work-up. This is an example of which potentially negative process?
   - **a.** Authority gradient
   - **b.** Anchoring
   - **c.** Countertransference
   - **d.** Fatigue and shift work
   - **e.** Visceral bias

Answer: A. The hierarchical nature of medicine can lead to authority gradients which may contribute to errors. If a more junior member of the team does not feel empowered to voice a concern regarding patient safety, it can potentially have adverse consequences.

3. Which of the following may ameliorate the effect of shift work?
   - **a.** Five consecutive night shifts
   - **b.** Assigning more senior staff to nights only
   - **c.** Sedatives
   - **d.** Circadian schedule

Answer: D. A circadian schedule with forward rotating shifts has the potential to improve sleep habits for those with a rotating shift schedule. Excessive amounts of consecutive nights, sedatives, and increased night shifts for senior attendings are all poor strategies to deal with the burden of shift work.

4. Which of the following are more likely to cause error in emergency medicine?
   - **a.** Low patient volumes, poor teamwork, electronic health record
   - **b.** Inadequate supervision, high communication load, multiple transitions of care
   - **c.** Overstaffing, multiple distractions, low numbers of transitions in care
   - **d.** Physician salaries, language barriers, quiet work environment

Answer: B. Many factors affect an ED's performance, making it more at risk for a medical error (see Table e8.1). Although each choice contains at least one of these performance-affecting factors, only choice B contains three of the factors.

Patient Experience in the Emergency Department

KEY CONCEPTS

- Factors that can improve the patient experience in the emergency department include:
  - Communication that expresses empathy
  - Working as a high-functioning team
  - Setting realistic expectations regarding wait time
- Clear discharge instructions
- Improvement in the patient experience has been shown to impact composite outcomes such as health-related quality of life to patients, reduced malpractice risk, and improved staff satisfaction.

1. Which of the following outcomes are impacted by patient experience?
   a. Health-related quality of life
   b. ED length of stay
   c. ED return visits ("bounce backs")
   d. Staff salaries

Answer: A. Patient-related outcomes (health-related quality of life, various pain scores, anxiety, depression, and specific disease conditions such as asthma, blood pressure, diabetes, osteoarthritis, and weight loss) and staff-related outcomes (improve experience, satisfaction, and joy at work and decrease burnout, depersonalization, and emotional exhaustion).

2. What is a key driver of patient experience?
   a. Communication
   b. Years of staff experience
   c. ED boarding
   d. Nursing ratios

Answer: A. Key drivers of patient experience are communication, empathy, expectation setting, and wait times.

3. What is an evidence-based practice you can use to improve your patient's experience?
   a. Using a computer on wheels
   b. Sitting in the room
   c. Bedside rounding
   d. Direct bedding

Answer: B. Sitting is a straightforward technique that has been shown to increase patients' perception of the provider's skill at communication and listening, makes instructions easier to understand, and leaves patients with a more positive evaluation of the experience. It has also been shown to result in patients' perception that emergency clinicians have spent more time at the bedside, without having actually spent more time, than those that stand. None of the other mentioned strategies have been studied in relationship to patient experience in the ED.

KEY CONCEPTS

- Physician Wellness: A quality of life that includes the presence of positive physical, mental, social, and spiritual well-being experienced in connection with activities and environments that allow physicians to develop their full potentials across personal and work-life domains.[1]
- Physician Burnout: A work-related syndrome characterized by emotional exhaustion, depersonalization, and a sense of reduced personal accomplishment. It is frequently associated with chronic stress and emotionally intense work demands for which resources are inadequate. Physicians exhibit a higher rate of burnout than the general population, with emergency physicians experiencing some of the highest rates in the medical profession.[2-6]
- Stress: A nonspecific response of the body to any demand that can have both positive and negative effects. Without any demand, performance suffers. With too much stress, anxiety and exhaustion lead to poor performance.[1] Chronic stress is associated with burnout.
- Compassion Fatigue: Resulting from exposure to a traumatized individual, compassion fatigue has been described as the convergence of secondary traumatic stress and burnout. It can lead to a reduced capacity and interest in being empathetic toward future suffering.[7]
- Resilience: The ability of a person, community, or system to withstand, adapt, recover, rebound, or even grow from adversity, stress, or trauma. When considering individual resilience, personality traits such as optimism and altruism are found in resilient people, but behaviors such as mindfulness and reflective practices also aid resilience. External factors such as social support and good self-care are also important.[8-10]
- Impaired Physician: A physician who is unable to practice medicine with reasonable skill and safety due to mental or physical illness; due to a condition that adversely affects cognitive, motor, or perceptive skills; or due to substance abuse. Substance abuse has been described as a direct consequence of work stress and burnout for physicians.[1,11]
- A Systems Model of Physician Burnout: The process of physician burnout is complex and occurs within a multilevel system that includes the individual, the front-line care delivery team, the health care organization, and the external environment (governmental agencies, regulatory bodies, societal norms, etc.). To effectively address burnout, interventions must be directed at all levels of this complex system.[10]

1. Which of the following statements regarding factors that contribute to occupational stress of the emergency clinician are true?
   - **a.** Emergency clinicians with experience are not significantly impacted by end of life issues in the emergency department.
   - **b.** Personal safety is usually not a concern for the emergency physician as there is often sufficient security to guard against the impact of violence or verbal abuse in the ED.
   - **c.** The expectation of error-free clinical practice is a major stressor for emergency clinicians.
   - **d.** Most emergency departments are not significantly stressful environments because of the predictability of patient acuity and patient arrival volumes.

Answer: C. Personal safety is also a major issue while working in the ED environment, where exposure to acts of violence, verbal abuse, and risk of exposure to potentially life-threatening diseases is higher than in other practice settings. In addition to the aforementioned stresses is the expectation of error-free practice and diagnostic certainty, with the constant possibility of human error, a missed diagnosis, and the attendant risk of medical negligence as an ever-present stress for emergency clinicians. The ED is a highly demanding environment because of 24/7 patient care in settings where life-threatening illness and injury occur with great frequency, and where arrival volumes, patient acuity, and nature of emergencies rapidly and unpredictably shift.

2. What specific strategies exist to impact individual wellness for emergency clinicians?
   - **a.** A focus on individual well-being alone is usually sufficient to prevent burnout.
   - **b.** Vigorous daily exercise has been shown to improve the ability to cope with difficult situations but only improves mood less than 10% of the time.
   - **c.** Peer groups that allow for reflection and sharing on the meaning inherent to the practice of medicine increase physician engagement and decrease depersonalization.
   - **d.** Health care organizational engagement has little impact in individual physician wellness but can improve efficiency of care in the work environment.
   - **e.** All of the above.

Answer: C. The current consensus suggests the importance of physicians focusing on their own well-being, but this alone is not enough to prevent burnout. Physical exercise has been shown to improve the mood of physicians and to help them cope with difficult situations. A strong business case exists for organizations to invest in and promote engagement in combatting and reducing physician burnout, including strategies of supporting executive leadership dedicated to improving and sustaining professional well-being across organizations.

3. Which of the following statements best describes physician burnout?
   - **a.** A non-work-related syndrome characterized by emotional exhaustion, depersonalization, and a sense of reduced personal accomplishment.
   - **b.** Physician burnout is frequently associated with chronic stress and emotionally intense work demands for which resources are inadequate.
   - **c.** Emergency physicians exhibit a lower rate of burnout compared to physicians in other specialties and to the general population.
   - **d.** Physician burnout can often result in impairment that results in an inability to practice medicine and can lead to substance abuse.
   - **e.** Both B and D are true.

Answer: E. Physician burnout is a work-related syndrome characterized by emotional exhaustion, depersonalization, and a sense of reduced personal accomplishment. It is frequently associated with chronic stress and emotionally intense work demands for which resources are inadequate. Physicians exhibit a higher rate of burnout than the general population, with emergency physicians experiencing some of the highest rates in the medical profession.

4. A systems model of physician burnout includes all of the following features, except:
   - **a.** The external environment
   - **b.** Organizational support and buy-in
   - **c.** Governmental agencies and regulatory bodies including licensing boards
   - **d.** Religious factors

Answer: D. The process of physician burnout is complex and occurs within a multilevel system that includes the individual, the front-line care delivery team, the health care organization, and the external environment (governmental agencies, regulatory bodies, societal norms, etc.) (see Fig. e10.3). To effectively address burnout, interventions must be directed at all levels of this complex system

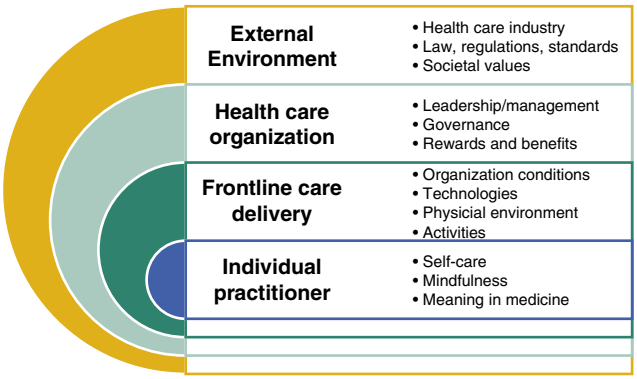


Fig. e10.3 The four levels of the systems model of physician burnout. (Modified from: National Academies of Sciences, Engineering, and Medicine. *Taking Action Against Clinician Burnout: A Systems Approach to Professional Well-Being*. Washington DC: The National Academies Press; 2019.)

KEY CONCEPTS

- Knowledge of wound mechanics, production, and appearance can provide practicing emergency clinicians with important clues regarding the forensic interpretations of injuries.
- Wounds and injuries should be diagrammed and photographed.
- The medical record should accurately document objective findings associated with a patient's wounds. Emergency clinicians should not speculate about their mechanism or cause.
- For the chain of custody to be preserved, any evidence collected during the course of treatment must be documented in the medical record, including to whom the evidence was given.

1. When documenting a gunshot wound, it is important to do which of the following?
- **a.** Describe the appearance, physical characteristics, and location of wounds.
- **b.** Describe the bullet trajectory.
- **c.** Determine the type of bullet used.
- **d.** Estimate the caliber of the bullet.
- **e.** Interpret the wounds as "entrance" or "exit."

Answer: A. Emergency clinicians should not identify wounds as "entrance" or "exit." Instead, their charting should include a detailed description, using appropriate forensic terminology, of the wound's characteristics and location without speculating about its function or the caliber or type of bullet (projectile) used to create it.

2. A young man is brought to the emergency department following an accidental rifle injury to his right leg. He has an obvious deformity to the extremity on examination with intact sensation and motor function, but pulses are diminished. Plain radiographs of the extremity show comminuted tibial and fibular fractures. Which of the following is/are indicated in his management?
- **a.** Superficial débridement and wound closure
- **b.** CT angiography, prophylactic antibiotics, and operative management
- **c.** Splinting and urgent orthopedic follow-up
- **d.** Emergent fasciotomy
- **e.** Urgent surgical consultation

Answer: B. High-energy gunshot wounds with associated fractures are managed as open fractures and are generally treated with prophylactic antibiotics because contaminants can be sucked into the wound due to cavitation. Operative intervention is required for patients with unstable fractures, fractures with exposed bone, compartment syndrome, or vascular injuries requiring repair. The latter is likely present in this patient given his diminished pulses and should be further evaluated by CT angiography. These patients are also at risk for compartment syndrome and should be closely monitored.

3. An abrasion collar can be associated with which of the following types of wounds?
- **a.** Close-range wound
- **b.** Contact wound
- **c.** Long-range or indeterminate-range wound
- **d.** Intermediate-range wound
- **e.** Shotgun wound

Answer: C. An abrasion collar (also referred to as an abrasion margin, rim, or ring) results from skin indentation as the bullet penetrates the surface. This collar is an abraded area of tissue that surrounds an entry wound as the result of friction between the bullet and epithelium. These usually result from distant or long-range wounds, which are inflicted from far enough away that only the bullet makes contact with the skin. When an abrasion collar is the only visible superficial clinical finding present, the term *indeterminate range* describes the range of fire. Most entrance wounds will have an abrasion collar aside from those on the palms and soles, which usually appear slit-like.

4. A 23-year-old woman presents after having been physically and sexually assaulted by an unknown assailant. In addition to multiple abrasions and brown contusions, she has a bite mark on her left shoulder. Which of the following is not appropriate in your evidence collection procedures and evaluation of this patient?
- **a.** Estimating the age of the contusions based on their color
- **b.** Measuring and documenting each visible injury on a diagram
- **c.** Preserving the condition of the bite mark until after a forensic examination
- **d.** Swabbing the bite mark with a sterile cotton-tipped applicator moistened with sterile water or saline
- **e.** Taking photographs of the bite mark and each injury

Answer: A. When an acute bite mark is identified, take care that critical evidence is not washed away. The skin surface should first be swabbed with a sterile cotton-tipped applicator moistened with sterile saline. Swabbing the area may reveal the assailant's DNA present in dried saliva. Remember, a contusion's color is never a predictor of its age. The emergency clinician may be asked to render an opinion regarding the age of a contusion. Because contusions develop as a result of multiple variables, there is no reproducible standard for dating them.

5. When managing a gunshot wound victim, which of the following is not important for forensic evidence collection?
- **a.** Do not clean the patient's the hands with alcohol or Betadine.
- **b.** If soot is noted on the hands of the victim, cover with paper bags.
- **c.** Maintain a chain of evidence.
- **d.** Place each article of clothing in a separate paper bag.
- **e.** Use metal forceps to handle a bullet to prevent contamination.

Answer: E. Never remove a bullet with metal hemostats or pickups because metal tools can obliterate the microscopic markings (the telltale fingerprint) of the gun from which it was shot. A victim's clothing may hold the answer to critical questions, such as "How far away was the assailant who fired the weapon?" and "Which hole is the entrance and which the exit?" Articles of clothing removed from a wounded patient should be placed in separate paper bags to preserve the trace evidence on them and to avoid accidentally transferring evidence from one article to another (cross-contamination). Always cover a patient's hands with paper bags when the presence of soot is suspected. A gunshot residue test may determine whether a victim or suspect has been in close proximity to a weapon that has been fired. Factors that decrease the sensitivity of gunshot residue tests include washing the skin with alcohol or Betadine.

e12 Emergency Medical Services: Overview and Ground Transport

KEY CONCEPTS

- Published in 1966, *Accidental Death and Disability: The Neglected Disease of Modern Society* by the National Academy of Sciences–National Research Council was instrumental in emergency medical service (EMS) maturation in the United States.
- There are multiple models for EMS systems, including public and private services, those operating at basic and advanced levels of care, and those including single or multiple tiers of response capability.
- There are four levels of prehospital providers recognized nationally—emergency medical responder (EMR), emergency medical technician (EMT), advanced emergency medical technician (AEMT), and paramedic, which is the most advanced level.
- The community paramedic provider focusing on population health issues such as access, chronic disease, and decreasing utilization and readmission is now being considered by many communities.
- Advances in emergency medical dispatching and positioning resources at locations and during specified times where expected call volumes are prevalent are innovations that are being implemented to decrease response times and improve outcomes.
- Regulatory oversight for EMS systems lies at the individual state level, and medical direction for individual public or private services resides at the local level.
- Direct medical oversight involves real-time interaction with the prehospital providers via face-to-face or radio communications. Indirect medical oversight involves off-line processes such as protocol development, quality improvement, and education.
- Advances in prehospital care of medical patients have included analgesics and anxiolytics that can be administered intranasally, negating intravenous routes, noninvasive measures to support patients in respiratory distress, alternative adjuncts in place of endotracheal intubation for managing airways, and tourniquets for controlling hemorrhage.
- More advanced diagnostics (such as 12-lead electrocardiography and use of stroke screens) have assisted in transporting patients to appropriate facilities based on their illness and acuity.

1. Which of the following describes an EMS system that is contracted to provide services?
   a. Public utility
   b. Third-service municipal
   c. Private
   d. Fire-based
   e. Hospital-based

Answer A. A public utility model is a hybrid between private and public design that allows local government to contract with a private or public provider.

2. Which of the following best defines direct medical oversight in the prehospital setting?
   a. Concurrent direction of EMTs providing patient care
   b. Development of prehospital care policies
   c. Implement quality improvement program
   d. Use of prehospital patient care protocols
   e. Use of standing field treatment protocols

Answer: A. Direct medical oversight is the concurrent direction of EMTs providing patient care, which can be at the scene or online over radio or cell phone.

3. Which of the following recognized the need to formally develop EMS in the United States?
   a. Highway Safety Act of 1973
   b. The EMS Agenda for the Future
   c. *Accidental Death and Disability: The Neglected Disease of Modern Society*
   d. 1965 President's Commission on Highway Safety
   e. The Future of Emergency Care in United States Health System

Answer: C. The National Academy of Sciences–National Research Council published this document as a result of the growing concern for EMS and hospitals not being prepared to respond to the growing number of injuries resulting from traffic accidents on the nation's highways.

4. Which of the following prehospital interventions has been shown to improve patient outcome?
   a. Defibrillation
   b. Endotracheal intubation for children
   c. Endotracheal intubation in severe head trauma
   d. Intraosseous needle placement
   e. Needle cricothyrotomy

Answer: A. Rapid defibrillation has been shown to improve outcomes for patients in cardiopulmonary arrest; other interventions have not shown proven benefit in the prehospital setting.

5. The EMS Agenda 2050 describes a vision for EMS that is defined by which of the following?
   a. Availability of ambulance resources
   b. Centralized resources that can be deployed
   c. Community-based
   d. People-centered

Answer: D. The EMS Agenda 2050 describes a vision for EMS that is people-centered along with six guiding qualities: inherently safe and effective, integrated and seamless, socially equitable, reliable and prepared, sustainable and efficient, adaptable and innovative.

KEY CONCEPTS

- Air medical transport (AMT) is a critical component of a comprehensive health care system and a vital link for rural communities and critical access hospitals to distant emergency care.
- Boyle's law and Dalton's law have the greatest impact and explain the development of hypoxia and most common altitude-related symptoms. Other stresses of flight that can affect the patient or crew include temperature fluctuations, dehydration, noise, and vibration.
- Although most flight programs do both primary (scene flights) and secondary (interfacility) response, ground ambulance remains the primary means of out-of-hospital and interfacility patient transport.
- The helicopter offers several advantages over other transport vehicles, including reduction in travel time by up to 75%, ability to avoid common ground delays (traffic, obstacles, and so on), and ability to fly into locations that may be inaccessible to other modes of travel.
- All air medical services require involvement of a medical director responsible for supervising, evaluating, and ensuring the quality of medical care.
- As a general rule, helicopters are less useful in urban settings because of the proximity of health care facilities.
- Helicopter emergency medical services (HEMS) represents the only modality by which nearly 28% of United States residents have timely access (within 1 hour) to level I or level II trauma centers.
- HEMS may benefit in other time-critical situations, including ST-elevation myocardial infarction (STEMI) patients going to a catheterization laboratory and acute stroke patients going to regional stroke centers.
- Safety is the predominant concern of air medical operations. *Helicopter shopping*, the practice of a requesting EMS agency or hospital calling numerous HEMS programs after other programs have declined the flight because of bad weather, must be avoided.

1. Which of the following statements is true regarding air medical transport (AMT)?
- **a.** AMT does little to improve outcomes when used for STEMI transport from rural areas.
- **b.** AMT scene response for acute stroke is limited by emergency medical service (EMS) providers' inability to effectively triage and identify patients who would likely benefit from HEMS scene response.
- **c.** Faster times to a trauma center are not required for outcomes benefit from AMT.
- **d.** The emphasis for AMT of neonates is often more on the *mode* of transport than on the transport team.

Answer: C.

2. Safety is a critical consideration for all air medical operations. Which of the following statements is true?
- **a.** If the patient's condition is severe, it is prudent to request service from several AMT providers in the face of marginal weather.
- **b.** Spectators should be kept at least 200 feet away from the landing zone.
- **c.** The use of a hospital helipad for emergency medical service (EMS) rendezvous triggers the Emergency Medical Treatment and Active Labor Act (EMTALA).
- **d.** Vehicles and personnel should be kept at least 50 feet away from the landing zone.

Answer: B.

3. A working knowledge of aviation physiology is vital to understanding the effects of AMT on pilots, medical personnel, and patients. Which of the following statements is correct regarding the aviation physiology gas laws?
- **a.** Boyle's law explains the physiologic effects of expansion and contraction of gases within the closed spaces of the body that may occur with altitude change.
- **b.** Charles' law can be shown as Pt = P1 + P2 + P3…Pn.
- **c.** Dalton's law accounts for the gas changes that result in decompression sickness.
- **d.** Henry's law explains why the ambient temperature decreases with increased altitude.

Answer: A.

4. Which of the following statements is correct with regard to landing zone safety?
- **a.** During night operations, spotlights should be toward the approaching aircraft.
- **b.** If the aircraft is parked on a slope, approach and depart on the downhill side.
- **c.** In a rear-loading helicopter, it is safe to approach or depart from the rear of the helicopter.
- **d.** Vehicles and personnel should be kept at least 50 ft from the landing zone.

Answer: B.

5. Determining which patient(s) might benefit from AMT is a critical decision for prehospital and interfacility personnel. Which of the following would represent a safe and appropriate request for AMT?
- **a.** Any burn victim being transferred to a regional burn center
- **b.** Based on distance for pediatric trauma patients, regardless of the type and severity of injury
- **c.** The preferred HEMS program has turned down the flight due to weather conditions but there are other programs that can be called
- **d.** When the distance to the closest appropriate facility is too great for safe and timely transport by ground ambulance

Answer: D.

KEY CONCEPTS

- Comprehensive emergency management consists of four phases: mitigation, preparedness, response, and recovery.
- Mass casualty planning should account for the breakdown of traditional transportation and communications systems during a disaster.
- Field personnel should be specifically trained in mass casualty triage and stabilization because austere field conditions change management strategies.
- All plans must protect caregivers and rescue personnel.
- Critical incident stress management is highly desirable after an event and should be planned for in advance. Psychological triage tools, such as Psy-START, may help.
- Planners should establish and exercise a hospital-based incident management system.
- Disaster planning needs to include policies to address the needs of vulnerable populations such as children, disabled, and the elderly.
- Multiple United States government agencies support disaster response, including the National Disaster Medical System, the Department of Defense, the Department of Veterans Affairs, and the Centers for Disease Control and Prevention.

1. How can one characterize application of the Secondary Assessment of Victim Endpoint (SAVE) triage guidelines?
   - **a.** Triage decisions are based on patient outcome expectations.
   - **b.** It is applicable to day-to-day hospital operations.
   - **c.** Patients are sorted immediately after the incident.
   - **d.** The guidelines provide detailed management of psychogenic cases.

Answer: A. The SAVE system is designed specifically for care in an austere environment. It triages patients into (1) those who will die regardless of treatment, (2) those who will live regardless of treatment, and (3) those who would benefit from austere intervention. The triage decisions are based on field outcome expectations from existing survival and morbidity statistics.

2. Which of the following best describes the National Disaster Medical System (NDMS)?
   - **a.** It does not include private sector institutions.
   - **b.** It does not involve the Department of Veterans Affairs (VA).
   - **c.** The medical response element consists of Department of Defense (DoD) personnel.
   - **d.** The Department of Health and Human Services (HHS) has oversight.

Answer: D. The NDMS is a federally coordinated initiative that is a cooperative program between the DoD, VA, DHHS, and Department of Homeland Security (DHS) with oversight provided by DHHS. The NDMS provides a system of coordinated mutual aid agreements among federal, state, local, and private institutional entities for resource and personnel provision in times of disaster. The medical response element includes dozens of volunteer civilian medical teams that supplement the local medical infrastructure.

3. What is accurate advice that one would give to those planning a disaster response?
   - **a.** Critical incident stress debriefing is best conducted 1 or 2 weeks after the event.
   - **b.** Medical resupply systems are the most vulnerable component of a disaster plan.
   - **c.** One of the Department of Veterans Affairs (VA) system's four legally mandated missions is emergency management.
   - **d.** The Federal Emergency Management Agency (FEMA) has a coordinating responsibility for the entire spectrum of disasters.

Answer: C. One of the VA health system's four mandated missions is emergency management. Although controversy exists, critical incident stress debriefing, if used, should be implemented as early as possible. The Department of Homeland Security (DHS) has full-spectrum disaster coordinating responsibility. Communications systems are likely the most vulnerable systems.

4. Which of the following statements is *not* a feature of comprehensive emergency management?
   - **a.** An incident management system consistent with NIMS should be implemented.
   - **b.** Health care facilities should perform a hazard vulnerability analysis to assess community risks.
   - **c.** Most incident management systems are characterized by five functional elements: incident command, operations, logistics, planning, and finance.
   - **d.** The four phases of comprehensive emergency management (CEM) are pre-event, event, post-event, and recovery.

Answer: D. The four phases of CEM are mitigation, preparedness, response, and recovery. These phases represent a continuum over time, and more than one phase may be in effect at the same time (e.g., even during the response phases, recovery actions may take place.)

5. After an earthquake, what resource would be expected to provide the most rapid and effective search and rescue activity finding the most victims who would most likely survive?
   - **a.** Urban search and rescue teams
   - **b.** Private citizens
   - **c.** Disaster medical assistance teams
   - **d.** National Guard Civil Support Teams

Answer: B. Uninjured survivors perform the most rapid and effective search and rescue activity after an earthquake. These private citizens will recover most victims who ultimately survive. Although formal urban search and rescue teams will eventually arrive, they do not find many victims and it is unknown how many survive. National Guard Civilian Support Teams and disaster medical assistance teams do not perform search and rescue activities.

KEY CONCEPTS

- Emergency department (ED) preparedness for a radiation incident should address decontamination (an external freestanding decontamination unit is best), triage, staff safety, personal protective equipment (PPE), and diagnostic procedures that emphasize radiation monitoring. It is important that emergency personnel know their radiation safety officer.
- External decontamination of radioactive particles can be achieved by removing clothing and washing the skin with soap and water.
- Management of acute, life-threatening conditions (such as major traumatic injuries) takes priority over radiation-associated issues.
- Aerosol dispersal is a likely route that terrorists may use to deploy biologic weapons, so victims will present primarily with respiratory complaints.
- In addition to influenza-like illness (ILI) symptoms, anthrax typically causes mediastinal widening, pulmonary consolidation, and pleural effusions best seen on chest computed tomography (CT) scans.
- Smallpox can spread in a hospital environment; thus, patients thought to have smallpox should be admitted to locations isolated from the rest of the hospital.
- Decontamination is a key activity in the management of patients exposed to chemical agents and frequently first occurs on emergency department arrival.
- Nerve agents are organophosphates; patients exposed to these agents are treated with large doses of atropine (repeated frequently), pralidoxime, and benzodiazepines.
- Patients with blast injuries can have subtle and delayed presentations of life-threatening conditions.

1. Three patients arrive at triage simultaneously: One has received a 4 Gy work-related irradiation exposure from a food sterilizer but no other injury; one is experiencing an acute ST elevation myocardial infarction (MI); and one likely has urosepsis but with a stable blood pressure and heart rate of 105 beats/min. Which of these actions is your first priority?
   a. Activate the cardiac catheterization team.
   b. Decontaminate the irradiated victim before placing him in a room.
   c. Initiate intravenous (IV) fluid bolus and prepare for intubation of the radiation-exposed patient.
   d. Place a central venous catheter in the urosepsis patient.

Answer: A. Even patients who have received a lethal radiation dose do not die quickly as a consequence of the ionizing exposure. Patients should be triaged according to severity of the medical conditions and/or vital sign derangements. Here, the patient with the MI is most acute and should be the highest priority necessitating activation of the cardiac catheterization team. The radiation-exposed patient was not contaminated, just irradiated. As such, decontamination and isolation are unnecessary. Although the victim has received the LD_{50} dose of ionizing radiation, no significant injury will result just after exposure, so this patient does not take priority over the MI patient at this point and intubation is not indicated. The septic patient is not critically ill and should not be treated ahead of the MI patient.

2. Which of the following statements best describes issues in management of anthrax infection?
   a. Antibiotics do not change the mortality of cutaneous disease.
   b. Cutaneous anthrax lesions are nontender.
   c. Doxycycline or ciprofloxacin are both options for single-agent treatment of symptomatic inhalational anthrax.
   d. Sputum culture and Gram stain obtained early in the disease help differentiate inhalational anthrax.

Answer: B. Cutaneous anthrax causes a severe, although nontender, peripheral vesicle that progresses to an eschar accompanied by regional adenopathy. Antibiotics lower the mortality from cutaneous anthrax twenty-fold. For inhalational, gastrointestinal, and cutaneous anthrax with toxicity, intravenous (IV) treatment is with ciprofloxacin or doxycycline plus at least two other agents. Regarding inhalational anthrax, mechanical ventilation may not improve mortality and sputum cultures and Gram stains are not helpful until late in the disease.

3. Several children ages 5 to 8 years old have definitely been exposed to anthrax spores. Health department officials have brought these children to the emergency department (ED). They are all ambulatory with normal vital signs and without symptoms. Which of the following is the most appropriate management?
   a. Admission for parenteral penicillin G
   b. Doxycycline for 5 days
   c. Outpatient ciprofloxacin for 60 days or until the children have received three doses of vaccine
   d. Penicillin VK for 7 to 10 days

Answer: C. See Box e15.5. For children without toxicity, ciprofloxacin, doxycycline, or amoxicillin orally is indicated for a minimum of 60 days or until the child has received three doses of vaccine. The vaccine has not been approved for children but may be indicated to reduce the long-term exposure to antibiotics. Note that weaponized anthrax may be penicillin-resistant.

4. An individual is exposed to sarin vapor. She presents complaining of difficulty with vision, salivation, vomiting, and the urge to defecate. The most appropriate treatment for this patient is which of the following?
   a. 5 mg diazepam intravenous (IV)
   B. 6 mg atropine IV and 1 g 2-PAM IV every hour for a total dose of 3 g
   C. 10 mg diazepam intramuscular (IM) via autoinjector
   D. One or two Mark 1 or the DuoDote autoinjector kits IM

Answer: D. The victim described would be characterized as a moderate exposure to sarin vapor. Treatment is indicated with one or two Mark 1 or DuoDote autoinjectors IM. Observation would not be appropriate. Diazepam is not indicated for moderate exposures. If an IV is established, the initial treatment is atropine 2 to 4 mg IV and 2-PAM 1 g IV.

5. Which of the following statements regarding blast injury is true?
   a. Classically, there are three different types of blast injury.
   b. Primary blast injury is caused by flying debris and bomb fragments.
   c. Exacerbation of asthma is an example of tertiary blast injury.
   d. Secondary blast injury is the most common cause of morbidity and mortality following a blast event.

Answer: D. Of the four types of blast injury, secondary blast injury is the most common cause of morbidity and mortality. These injuries are caused by flying debris generated by the explosion. An asthma exacerbation could be considered a quaternary injury in the correct setting. Tympanic membrane rupture is not an accurate predictor of internal injuries in blast events.

e16 Tactical Emergency Medical Support and Urban Search and Rescue

KEY CONCEPTS

- Tactical emergency medical support (TEMS) facilitates the overall success and safety of law enforcement missions during all phases of a tactical operation through the delivery of preventive, urgent, and emergency medical care.
- A fundamental principle in tactical medicine is that the medical mission may be subordinate to the overall law enforcement mission.
- Tactical combat casualty care (TCCC) has adapted civilian advanced life support principles to provide medical care during a hostile force encounter. Its goals are to treat the casualty, prevent additional casualties, and complete the mission.
- TCCC is divided into three phases of care: care under fire (CUF), tactical field care (TFC), and combat casualty evacuation care (CASEVAC).
- Tactical emergency casualty care (TECC) addresses casualty management during high-threat civilian tactical and rescue operations and is divided into three phases; direct threat, indirect threat, and evacuation care.
- Urban search and rescue (USAR) encompasses responding to, locating, reaching, medically treating, and safely extricating victims entrapped by collapsed structures. The primary role of the emergency clinician is support of the health and welfare of the team members, including canines.
- USAR teams often treat crush syndrome, particulate inhalation, hazardous materials exposures, and blast injuries.
- In crush syndrome, treatment with fluids begins prior to extrication to avoid life-threatening complications once the patient is extricated.

1. Which of the following is a priority for the tactical emergency medical support (TEMS) provider in caring for a victim shot during a hostage incident?
   a. Bag-valve-mask ventilation
   b. Cervical spine stabilization
   c. Intravenous access and fluid resuscitation
   d. Tourniquet application to bleeding extremity

Answer: D. This scenario describes care rendered by TEMS in the hot zone, or care under fire; thus, conventional care sequences do not apply. Priorities include mitigation of threats, tourniquet use for serious bleeding, and evacuation to a safer location.

2. A 41-year-old man has been trapped under a piece of large crane that overturned during a partial construction site collapse. The patient has been trapped for over 7 hours and the rescue team is actively working to extricate him. His right leg is both mangled and pinned up to the level of his pelvis. He is conscious, in pain, has a systolic blood pressure of 140, a heart rate 112, SpO_2 97%, RR 20. He received pain medication and fluid resuscitation. The monitor shows a widening QRS interval. Which of the following treatments should be initiated?
   a. Blood products
   b. Calcium
   c. Epinephrine
   d. Hypertonic saline

Answer: B. Prolonged entrapment raises the concern for crush injury, which increases after 4 hours of immobilization. Continuous cardiac monitoring during extrication is indicated to monitor the development of arrhythmias which may occur in the setting of hyperkalemia. Evidence of hyperkalemia should be treated and includes calcium. There is no clinical evidence to suggest hemorrhagic shock or uncontrolled bleeding to warrant blood product administration. Fluid resuscitation may help prevent the hypotension associated with release of the entrapped limb(s) and should be initiated before extrication.

3. The principles of TEMS have largely developed from which type of incidents?
   a. Civilian multicasualty events
   b. Military conflicts
   c. Natural disasters
   d. Pandemics

Answer: B. The principles of TEMS have largely developed from lessons learned during military conflicts and most closely emulate the medical support structure of military special operations units.

4. Which of the following reflects an important distinction between tactical medicine and stand EMS practice?
   a. Airway compromise is treated before hemorrhage.
   b. Medical care may be delayed in order to accomplish the law enforcement mission.
   c. Care under fire requires the immediate movement of patients from the scene.
   d. Tourniquet use is secondary to other forms of hemorrhage control.

Answer: B. A fundamental principle in tactical medicine is that the medical mission may be subordinate to the overall law enforcement mission. In contrast to conventional EMS and hospital practices, in which the sole priority is usually the health and welfare of the patient, the essential priority in a tactical mission is the success of the law enforcement objective. When a casualty occurs during a tactical operation, medical providers may be directed to delay or modify medical care until the tactical commander determines that rendering care will not jeopardize the overall mission. Bleeding control is emphasized with rapid use of tourniquets. In care under fire patients should not be moved unless the danger is alleviated.

5. Which of the following is the primary responsibility of the USAR team emergency clinician?
   a. Ensuring a public health framework exists for the region
   b. Ensuring that entering a collapsed building is safe
   c. Medical treatment of entrapped victims
   d. Supporting the health and welfare of team members

Answer: D. The USAR team emergency clinician must be familiar with the types of medical conditions that may be encountered by the team. The USAR team emergency clinician has a primary responsibility to the team members should they require medical attention. Secondarily, treatment of entrapped patients can be done but is not their primary mission. Ensuring broader elements of a public framework in a disaster-affected area is typically beyond the conventional role of the USAR clinician, though this individual may liaison with other health and medical personnel in the same geographic area of operations.